

CSCM Tool

Controller Status Code Management Tool

Full documentation

The complete documentation package

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Controller Status Code Management Tool (CSCM Tool)

Software Delivery Documentation Package

Document Set Version: 4.0 **Date:** 2026-10-23 **Status:** Implemented baseline (v4 — the v3 field model plus the improvement pack I-01 … I-36 of `21-improvement-ideas.md`: concurrency control, import toolkit, audience- and profile-scoped publication with signed manifests, software-version binding, SLAs and delegation, step-up re-authentication, hash-chained audit log, verified backups, German UI, and operations tooling)

This package is the complete, self-contained requirements and design basis for building the CSCM Tool without additional requirements gathering. It is intended to be read as a set; cross-references between documents use document numbers (e.g. "see 07-database-design.md"). **Version history:** v1.0 -> v2.0 replaced Controller Type (Main/Hub) numbering with Functional System Group numbering, introduced the `StCd-XXXXX` identifier format, the Chief Engineer role, the dual-review-then-approve workflow, Turbine Platform multi-select, sandbox/test status codes, and on-premises-only deployment. v2.0 -> v3.0 (2026-09-17) grounded the entire attribute set in a real 897-row export of the live status code library (correcting Status Category, Available Group, Alarm, delay fields, and replacing the single Access Rights enum with eight independent audience fields), made Functional Subgroup mandatory, added three Hub Controller Functional System Groups (11000–13999) alongside the nine Main Controller ones, added CSV/XLSX library import (fully governed, not a bypass), added Administrator-created subgroups and whole new Functional System Groups, added XLSX to export formats, and adopted the OneTool shared design system for the UI shell while keeping CSCM Tool's own authentication.

v4 (2026-10-23): all 36 improvement ideas are implemented and specified: requirements FR-090 … FR-155, NFR-009 … NFR-014, BR-016 … BR-022 and decisions D-01 … D-08 in 03-srs.md; security controls SEC-026 … SEC-030 and audit requirements AUD-072 … AUD-075; schema v4 (07-database-design.md §12, `artifacts/schema.sql`) and API v4.0.0 (`artifacts/openapi.yaml`, 161 operations). **v4.2 (2026-10-23):** status codes can be imported from the generated `.st` programs, several at a time, with the legacy -> new mapping (FR-048o … FR-048w, BR-023, SEC-031; schema v4.2). `21-improvement-ideas.md` now records what was built and lists the next improvements.

Open items still pending your input: (1) the exact meaning of `Covert` across several audience-access vocabularies is unconfirmed (06-data-dictionary.md §9d) — the system treats it as not visible meanwhile (decision D-02); (2) the descriptive labels for Available Group codes and the eight numeric Program Reference fields are not available from the source export and need Controls Engineering input (06-data-dictionary.md §2c, §9c). The single-file `StCd\\\_4960.st` example is superseded — the 2026-09-17 update grounds the entire data dictionary in a real 897-row export of the live library instead, which is more complete than one example file would have been. `10000?10999`, `14000?14999`, `15000?15999`, and `16000?16999` are confirmed free/reserved capacity for future Functional System Groups — no longer an open item.

Field-model correction (2026-09-17): the attribute set, Functional Subgroup names, and several field types in this package were corrected against a real export of the live status code library and a legacy-tool screenshot. Functional Subgroup is now mandatory (never optional); see 16-mvp-roadmap.md §8 "Corrections Log" and 06-data-dictionary.md's provenance note for the full list. CSV/XLSX library import is now in scope (reversing the earlier "no bulk import" position, still fully governed per row — 05-governance-handbook.md §6b), and the UI adopts the shared OneTool design system for its visual shell while keeping its own authentication (10-ui-ux-specification.md §0).

Document Set

|#|Document|File| |-|-|-| |01|Executive Summary|01-executive-summary.md| |02|Product Requirements Document (PRD)|02-prd.md| |03|Software Requirements Specification (SRS)|03-srs.md| |04|Domain Model Specification|04-domain-model.md| |05|Status Code Governance Handbook|05-governance-handbook.md| |06|Data Dictionary|06-data-dictionary.md| |07|Database Design Document|07-database-design.md| |08|System Architecture Document|08-system-architecture.md| |09|API Specification|09-api-specification.md| |10|UI/UX Specification|10-ui-ux-specification.md| |11|Security Architecture|11-security-architecture.md| |12|Audit & Compliance Specification|12-audit-compliance.md| |13|Test Strategy|13-test-strategy.md| |14|Deployment Architecture|14-deployment-architecture.md| |15|Development Standards| 15-development-standards.md| |16|MVP Roadmap|16-mvp-roadmap.md| |17|Implementation Backlog| 17-implementation-backlog.md| |18|Claude Code Implementation Guide| 18-claude-code-implementation-guide.md| |19|Glossary (living document — update with every change)| 19-glossary.md| |20|Traceability Matrix — spec vs. code vs. tests, no-cloud verification record| 20-traceability-matrix.md| |21|Improvement Ideas — prioritised (HIGH / MEDIUM / LOW), mapped to epics|

21-improvement-ideas.md| |22|Software Description|22-software-description.md| |23|User Manual| 23-user-manual.md| |24|Test Report (generated by `scripts/generate_test_report.py`)|24-test-report.md| |25|Concept — integrating Hub Controller codes into the 4xxx / 9xxx scheme (proposal)| 25-hc-integration-concept.md| |26|Translations — concept, review guide for native speakers and the generated catalogue of every translation|26-translations.md|

Supplementary Artifacts

|Artifact|File|Referenced By| |-|-|-| |SQLite DDL (schema v4)|artifacts/schema.sql|07-database-design.md| |Optional FTS5 search index DDL|artifacts/schema-fts5.sql|07-database-design.md §12| |OpenAPI 3.0 Specification (v4.0.0)|artifacts/openapi.yaml|09-api-specification.md| |Test run history (appended by the report generator)|artifacts/test-run-history.json|24-test-report.md| |HC numbering proposal (mapping of all Hub Controller codes)|artifacts/hc-numbering-proposal.xlsx|25-hc-integration-concept.md|

Naming & Terminology Baseline

Full definitions live in 19-glossary.md; this is the quick-reference subset used throughout the package.

- **Status Code** — the master governed record, identified by its Status Code Identifier.
- **Status Code Identifier (StCd)** — `StCd-XXXXX`, a five-digit zero-padded number equal to the code's position within its Functional System Group's numeric range, e.g. `StCd-01232`. Not assigned until Draft -> Review.
- **Status Code Revision** — a specific versioned instance of a Status Code's attributes, carrying its own lifecycle state.
- **Functional System Group** — the classification and numbering axis (replaces the former Controller Type field). Nine groups; see 05-governance-handbook.md §2.
- **Functional Subgroup** — sub-division of a Functional System Group owning a numeric sub-band, **mandatory on every Status Code, never optional**; real taxonomy seeded, with deliberate reserved blocks for future additions and Administrator-created new subgroups (06-data-dictionary.md §9a).
- **Turbine Platform** — multi-select classification (minimum `2XM`, `3XM`, `4XM`).
- **Change Request (CR)** — a proposal to create a new Status Code or revise an existing one, routed through the dual-review-then-approval workflow.
- **Release** — a named, dated, immutable bundle of Released status code revisions exported as a catalogue; it may target one or more controller software versions (v4).
- **Lifecycle State** — one of `Draft`, `Review`, `PendingApproval`, `Approved`, `Released`, `Deprecated`, `Archived` (see 05-governance-handbook.md).
- **Sandbox Status Code** — Administrator-only test record, deletable, never part of a Release.

Roles

|Role|Organizational Equivalent| |-|-| |Administrator|Product Owner Controls Software / Component Owner Controls Software (highest level)| |StCd Owner|Owner of the status code library: reads everything, moves users between Viewer, Engineer and Reviewer (v4.3)| |Engineer|Controls engineer or cross-functional engineer (tagged with an Engineering Domain)| |Reviewer|Sub-PO for Status Codes| |Chief Engineer|Final approval authority| |Viewer|Read-only consumer|

Functional System Group Numbering Table

|Range|Functional System Group|Prefix| |-|-|-| |1000–1999|Converter & Grid Interface|WCNV| |2000–2999|Generator System|WGEN| |3000–3999|Meteorology & Nacelle Environment|WNAC| |4000–4999|Rotor & Pitch System|WROT| |5000–5999|Tower & Structure|WTOW| |6000–6999|Transformer & MV System|WTRF| |7000–7999|Drivetrain & Gearbox|WTRM| |8000–8999|Yaw System|WYAW| |9000–9999|Turbine Control & Operation|WTUR| |10000–10999|(free — reserved for a new Functional System Group)|—| |11000–11999|Hub Controller Control System|WTUR (reused — different range from 9000–9999)| |12000–12999|Hub Controller Rotor & Pitch|WROT (reused — different range from 4000–4999)| |13000–13999|Wind Farm / Plant Dispatch|WPPD| |14000–14999|(free — reserved for a new Functional System Group)|—| |15000–15999|(free — reserved for a new Functional System Group)|—| |16000–16999|(free — reserved for a new Functional System Group)|—|

Each group subdivides into real Functional Subgroups owning numeric sub-bands, with deliberate reserved (unassigned) blocks for future growth — see 05-governance-handbook.md §2 and 06-data-dictionary.md §9a for

the full subgroup list (38 subgroups across 12 groups as of 2026-09-17). `code` (the prefix) is unique together with `range\\\_start`, not on its own — `WTUR` and `WROT` each identify two different groups, one Main Controller and one Hub Controller.

Requirement Identifier Conventions

|Prefix|Meaning|Defined In| |-|-|-| |`FR-###`|Functional Requirement|03-srs.md| |`NFR-###`|Non-Functional Requirement|03-srs.md| |`BR-###`|Business Rule|03-srs.md, 05-governance-handbook.md| |`SEC-###`|Security Control|11-security-architecture.md| |`AUD-###`|Audit/Compliance Requirement|12-audit-compliance.md| |`D-##`|Decision taken for the improvement pack|03-srs.md §6| |`I-##`|Improvement idea|21-improvement-ideas.md| |`TC-###`|Test Case|13-test-strategy.md| |`EPIC-##` / `US-###`|Backlog Epic / User Story|17-implementation-backlog.md|

Deployment Principle

The system runs entirely on local, on-premises servers. No public cloud (SaaS/PaaS/IaaS) component of any kind is used anywhere in this design, including for future authentication (on-premises Active Directory / ADFS, not a cloud identity provider) — see 08-system-architecture.md §7 and 14-deployment-architecture.md §1.

01. Executive Summary — CSCM Tool

1. Business Problem

Wind turbine Main Controller and Hub Controller software generate large, growing catalogues of status codes (faults, warnings, operational states, informational messages). Today these codes are typically authored inside engineering notes, spreadsheets, or directly inside technical manuals and specification documents. This creates five compounding problems: **(1) No single source of truth** — the same code number can exist with conflicting descriptions in different manuals, service bulletins, or SCADA configuration exports, because each document is edited independently. **(2) No governance** — status codes are added or changed by whoever edits the document next, with no mandatory review, approval, or sign-off before a code becomes authoritative. **(3) No traceability** — there is no reliable way to answer "who changed this code, when, why, and what did it say before," a requirement for engineering change control and for any future functional-safety or quality audit. **(4) No release discipline** — manuals are regenerated ad hoc; there is no concept of a versioned, approved status code catalogue that a given software build, manual edition, or SCADA integration can be pinned to. **(5) Duplication and drift** — because status codes are copy-pasted between documents rather than referenced from one place, duplicate or near-duplicate codes accumulate over time, and downstream documents silently drift out of sync with engineering intent.

2. Business Value

CSCM Tool inverts the current document-first workflow: status codes are created, cross-domain-completed, dual-reviewed, Chief-Engineer-approved, versioned, and released inside CSCM Tool first, and every downstream artifact (technical manuals, specifications, SCADA tag lists, future software repositories) is derived from CSCM Tool, not the other way around. It delivers authoritative data (one record per code, classified by Functional System Group, with one current approved state at any time), controlled change (every change to a status code goes through a seven-state lifecycle — Draft, Review, PendingApproval, Approved, Released, Deprecated, Archived — with named accountability at each of three distinct decision points: Reviewer, Administrator, and Chief Engineer), full history (every revision of every status code is retained and queryable), repeatable trustworthy exports (Approved/Released catalogues exported on demand, with full-database export reserved to the Administrator for disaster-recovery and migration purposes), and elimination of duplicate/conflicting codes through a numbering scheme tied directly to the turbine's functional subsystems rather than an arbitrary sequence.

3. Expected Outcomes

Outcome	Description
Single system of record	All Main Controller and Hub Controller status codes exist in exactly one authoritative place, classified by Functional System Group instead of the former Controller Type split.
Governed lifecycle with real accountability	No status code reaches Released status without cross-domain-complete authorship, independent Reviewer and Administrator sign-off, and a separate Chief Engineer approval.
Full audit trail	Every create, edit, sign-off, approval, rejection, release, deprecation, and sandbox test event is logged immutably with actor and timestamp.
Faster manual production	Technical writers pull an approved, versioned export instead of manually re-collating status codes from multiple sources.
Safe test capability	Administrators can create and permanently delete sandbox status codes for testing and training without ever touching production numbering or manuals.
Foundation for automation	REST API and structured export formats enable future, strictly on-premises integration with documentation generation, software repositories, and SCADA systems, without re-architecting the data model.
Fully on-premises operation	No public cloud component anywhere in the system, including authentication — a hard deployment constraint, not a default.
Trustworthy hand-over (v4)	Every export carries a manifest with SHA-256 hashes (optionally signed), can be limited to an audience, a language and a controller software version, and can be verified later without the database; release notes show what changed since the previous Release.
Complete manuals per release (v4.10)	A Release can be exported as the complete baseline — every status code valid as of it, about 900 for the Main and 350 for the Hub Controller — narrowed to a platform and a controller, and as one package of every language and access level with a manifest. Everybody may download what is released; only the StCd Owner and the Administrator what is not.
Operable by the owning team (v4)	Verified nightly backups, a tamper-evident audit log, readiness and metrics endpoints, SLA ageing with reminders, delegation for absences, and one-script installation keep the single-file system safe to run on-premises.

4. Success Metrics

Metric	Target (post-MVP, first 2 quarters of production use)
% of status codes with unknown/undocumented owner	0%
Duplicate identifiers created	0 (enforced by uniqueness constraint and atomic allocation; see 07-database-design.md)
Median time from Change Request submission to Approved (through all three decision points)	<= 7 business days
% of Released status codes with complete audit trail from Draft, including both sign-offs and the Chief Engineer approval	100%
Manual technical-writing rework caused by status code inconsistency	Reduced by >= 80% vs. baseline
Export catalogue adoption	100% of new manual editions sourced from a CSCM Tool export within 2 releases of go-live
Cross-Domain Sign-Off turnaround (when required)	<= 3 business days
System availability (production)	>= 99.5% monthly, per NFR-002 (03-srs.md)

Since v4 the first four process metrics (unknown owners, approval time, cross-domain turnaround, audit-trail completeness) are **measured by the application itself** on the Success Metrics page (FR-123) from timestamps it already stores, instead of by a separate report.

5. Scope Boundary (Summary)

In scope for MVP: status code lifecycle management with Functional System Group/Subgroup-based numbering, Turbine Platform multi-select classification, Cross-Domain Sign-Off, the three-decision-point approval workflow, revision history, release management, sandbox test codes, catalogue export (role-scoped) and full-database export (Administrator-only), role-based access control across five roles, full audit logging, and an internal-network-only REST API — all running on local, on-premises infrastructure with no cloud component. Out of scope for MVP: on-premises Active Directory/ADFS SSO integration itself (the hook is built, not wired up), automated push integration to documentation generation tools/software repositories/SCADA systems, multi-tenant/multi-OEM support. See 02-prd.md §6 for the complete scope statement.

v4 addition to scope (2026-10-23): the improvement pack of 21-improvement-ideas.md is part of the implemented baseline: governance extensions (change sets, delegation, step-up, field comments, review checklists), publication (profiles, audiences, manifests, software versions, translations, offline bundle), import toolkit, insight (health report, fleet feedback, saved searches), integration (API tokens, change feed, webhooks, Hub-to-Main consolidation) and operations (backups, audit chain, readiness, metrics, installers, CI). Requirements: 03-srs.md §1.10–§1.15.

02. Product Requirements Document (PRD) — CSCM Tool

1. Product Vision

CSCM Tool is the master system of record for wind turbine Main Controller and Hub Controller status codes. It replaces document-first authoring with a governed, auditable, database-first workflow: engineers propose codes (completing cross-domain fields where their own specialty cannot cover them), a Reviewer and an Administrator each independently sign off, a Chief Engineer gives final approval, and every other document or system that needs status code data consumes it from CSCM Tool, never the reverse. The system runs entirely on local, on-premises infrastructure.

2. Goals

1. Provide one authoritative, versioned record per status code, classified and numbered by Functional System Group and Functional Subgroup — both mandatory — rather than an arbitrary series.
2. Enforce a mandatory, three-decision-point review/approval workflow before any code becomes Released.
3. Preserve complete, immutable revision history for every status code.
4. Provide release management so a specific, named, dated snapshot of approved codes can be exported and referenced by manuals, SCADA configuration, and future integrations.
5. Provide a REST API, reachable only from the internal network, suitable for future on-premises integration with documentation generation tools, software repositories, and SCADA systems.
6. Provide role-appropriate access so contribution (Engineer), cross-domain completion, gatekeeping (Reviewer, Administrator, Chief Engineer), and consumption (Viewer) are cleanly separated.
7. Provide a safe, isolated sandbox mechanism for testing and training that can never contaminate production numbering, catalogues, or manuals.

3. Personas

3.1 Engineer (primary contributor)

A controls engineer or cross-functional engineer (electrical, mechanical, grid, safety), tagged with one or more Engineering Domains. Creates new status codes and change requests. Needs a fast, structured entry form with strong validation, clear visibility into which domain-owned fields still need a second engineer's input, and visibility into where their submissions sit across the three decision points.

3.2 Reviewer (Sub-PO for Status Codes)

Responsible for one of the two required Review sign-offs: verifying a proposed status code or change is technically correct, non-duplicate, and consistent with existing conventions. Needs a focused sign-off queue, side-by-side diff of proposed vs. current revision, and mandatory-comment rejection.

3.3 Administrator (Product Owner Controls Software / Component Owner Controls Software)

Owns system configuration, user/role/Engineering-Domain administration, controlled vocabularies (including the Functional System Group/Subgroup and Turbine Platform lookups), release process, sandbox status codes, and full-database export. Also provides the second of the two required Review sign-offs, during which any Engineer-proposed restricted field (e.g. Access Rights) is confirmed or overridden.

3.4 Chief Engineer (final approval authority)

Performs the final Approval decision once both Review sign-offs are complete. Cannot approve a submission they authored or already signed off on as Reviewer or Administrator. Needs an Approval Queue and the same diff/comment tooling as the Reviewer/Administrator sign-off screens.

3.5 Viewer (consumer)

Technical writer, service technician, or other stakeholder who looks up current or historical status code definitions but never edits data. Needs fast search/filter and read-only export of Approved/Released catalogues.

4. Stakeholders

Stakeholder	Interest
Controls Engineering	Accurate, unambiguous status code definitions tied to firmware behaviour, and the final real Functional Subgroup taxonomy
Technical Publications	Reliable source data for manuals and specifications, on a predictable release cadence
Quality / Compliance	Auditability, traceability, and change control evidence across all three decision points
Service & Field Operations	Correct, current status code meaning for troubleshooting
SCADA/Software Integration teams (future)	Machine-readable, versioned status code data via an internal-network-only API
IT/Platform Operations	Operable, securable, maintainable, fully on-premises system

5. User Journeys

5.1 Engineer creates a new status code

1. Engineer logs in, navigates to "New Status Code," selects Functional System Group (e.g. Converter & Grid Interface) and Functional Subgroup (e.g. Converter) — both mandatory — plus at least one Turbine Platform. No identifier is shown yet.
2. Engineer completes all mandatory attributes; validation blocks submission until the record is complete and internally consistent.
3. If fields owned by a domain the Engineer does not hold are still unset, the system flags them and lets the Engineer request sign-off from a colleague holding that domain.
4. Once complete, Engineer submits for Review; the system allocates a fixed `StCd-XXXXX` identifier at that moment, not before.
5. Engineer tracks sign-off/approval progress and sees Reviewer/Administrator/Chief Engineer comments if rejected at any stage.

5.2 Reviewer and Administrator sign off

1. Reviewer opens their queue, opens a submitted code, reviews the field-level diff, checks for duplicate-meaning codes, and signs off or rejects with a comment.
2. Independently, the Administrator does the same, additionally confirming or overriding the Engineer-proposed Access Rights value.
3. Once both sign-offs are recorded (in either order), the system automatically advances the code to Pending Approval.

5.3 Chief Engineer approves

1. Chief Engineer opens the Approval Queue, reviews the same diff plus both sign-offs' comments, and approves or rejects with a comment.
2. On approval, the code becomes Approved and enters the Release candidate pool.

5.4 Administrator builds and publishes a Release

1. Administrator creates a Release scoped by Functional System Group/Subgroup, reviews the Approved (non-sandbox) candidate list, confirms the set, and publishes.
2. Publishing atomically transitions included revisions to Released, stamps Effective Date, and locks the Release's contents.
3. Administrator exports the Release catalogue (CSV/JSON/XLSX/PDF) for Technical Publications. (v4) The Administrator may first set the controller software versions the Release targets, sees warnings for items whose planned effective date is still in the future, and confirms the publish with their password (step-up). Each export can use an export profile, an audience, a language and a software version, ships with a manifest of SHA-256

hashes, and the Release has generated release notes against the previous Release.

5.5 Viewer looks up a status code

Search, view current Released definition and history, export a filtered read-only catalogue. (v4) Search is ranked and tolerates typos; filters and visible columns can be saved and shared as a link; the interface can be switched to German; state and role tooltips and a per-role quick start explain the process.

5.6 Administrator deprecates an obsolete code

Deprecation is routed through the full three-decision-point cycle when initiated as an Engineer Change Request.

5.7 Administrator uses a sandbox status code

1. Administrator creates a sandbox status code for testing an export template or training a new Reviewer, drawing an isolated `StCd-T#####` identifier that can never enter a real Functional System Group range.
2. Administrator exercises it through the lifecycle as needed; it can never be added to a real Release.
3. Administrator permanently deletes it when done; the deletion itself remains in the audit trail forever, even though the record does not.

5.8 Administrator migrates the legacy library (v4)

1. Downloads the import template, fills it, and runs a **dry run**: every problem is listed per row and column with a fix hint and nothing is created.
2. Imports the corrected file (large files run in the background with a progress bar); similar titles and clashing legacy numbers are reported.
3. Re-uploads corrected rows in **update mode**: open Drafts are revised in place, Released codes get a new revision.
4. Opens the **reconciliation report** to see how many legacy rows are imported, in review, released and still missing.

5.9 Reviewer works the queue (v4)

Each request shows how long it has waited (amber/red). The Reviewer reads the field diff, comments on single fields, ticks the Administrator-defined review checklist, and signs off with their password. Out of office, they name a delegate for the dates; the delegate acts as themselves and the audit trail shows both.

5.10 Engineer prepares a software release (v4)

Groups the change requests that ship together into a **change set**, attaches evidence (test reports, safety assessment), checks reference warnings and the Where-used list, optionally stamps planned dates, and states the software-version range the code applies to.

5.11 Service technician looks up a code at the turbine (v4)

Opens the **offline bundle** generated from a Release — one HTML file with search, needing no server — and finds the code, its remedy text and affected component.

5.12 Technical Publications and integrators consume releases (v4)

Download an audience-scoped, language-specific profile export with its manifest and verify it later without the database; poll the change feed or receive a signed webhook on publish using a scoped read-only API token.

5.13 Technical Publications builds the manuals of a Release (v4.10)

Open a published Release, see how many codes it holds in total (the complete baseline, by controller and platform), export the baseline for one platform and controller in one language and access level, or build the package: one ZIP with every language for every access level, the release notes and a manifest that lists each file with its SHA-256. Anybody signed in may do this for a released Release; the working set and unpublished Releases are for the StCd Owner and the Administrator only.

6. Functional Scope

In scope (MVP): status code CRUD via the governed three-decision-point Change Request workflow; Functional System Group **and mandatory** Subgroup classification and numbering (`StCd-XXXXX`, allocated at Draft->Review, not before); Administrator-created new subgroups, non-overlap validated; Turbine Platform multi-select; Cross-Domain Sign-Off; full attribute set per 06-data-dictionary.md (corrected 2026-09-17 against a real library export); seven-state lifecycle with enforced valid transitions; revision history and field-level diffing; role-based access control across five roles; sandbox status codes (create/delete, Administrator-only); CSV/XLSX library import producing governed Drafts (05-governance-handbook.md §6b); Release management (build, publish, lock, export); search/filter/sort; full audit trail; session-based authentication and internal user management; catalogue export (CSV/JSON/XLSX/PDF, role-scoped) and full-database export (Administrator-only); REST API covering all core resources, internal-network-only; OneTool shared design system for the visual shell (10-ui-ux-specification.md §0), with CSCM Tool retaining its own authentication.

In scope since v4 (2026-10-23): the improvement pack I-01 … I-36 (03-srs.md §1.10–§1.15): optimistic concurrency, reference warnings and Where-used, duplicate detection, platform variants, the import toolkit, export profiles / audiences / manifests / release notes / offline bundle, software-version binding, planned dates, translations (English source, German), SLA ageing and reminders, KPIs, change sets, delegation, step-up, field comments and checklists, evidence, remedy fields, catalogue health and capacity, saved searches, fleet feedback, API tokens / change feed / webhooks, Hub-to-Main consolidation, ranked search, backups, audit hash chain, readiness and metrics, German UI and accessibility, installers and CI.

Out of scope (MVP and v4): on-premises Active Directory/ADFS SSO integration itself (the `AuthProvider` hook is built; the integration is not); live/automated push integration into documentation generation tools, code repositories or SCADA systems (v4 offers internal pull — tokens, change feed, signed webhooks — instead); content in languages other than English and German; multi-OEM/multi-tenant data partitioning; mobile native applications; configurable/multi-stage approval chains beyond the fixed Reviewer+Administrator+Chief-Engineer model; any public-cloud component of any kind, at any phase.

7. Out-of-Scope Items (Explicit Exclusions)

Item	Reason
Cloud-hosted SSO (e.g. Azure AD)	Explicitly excluded by the no-cloud-infrastructure requirement; the future SSO path is on-premises Active Directory/ADFS only (16-mvp-roadmap.md, 11-security-architecture.md §2)
Automated SCADA tag sync	Requires SCADA-side integration work outside CSCM Tool's control; v4 provides the pull interfaces (API tokens, change feed, webhooks, fleet-feedback import) on the internal network only
Configurable/multi-stage approval chains	The fixed Reviewer->Administrator (parallel)->Chief Engineer model is sufficient for current governance need
(removed 2026-09-17) Bulk import was previously excluded here on principle	The business explicitly requested CSV/XLSX library import; it is now in scope (§6), on the condition that every imported row still passes the full governance cycle before Release (05-governance-handbook.md §6b) — the "codes are governed in CSCM Tool, not imported pre-approved" principle survives, the "no import mechanism at all" position does not
Real-time collaborative editing	Not required; optimistic concurrency control (FR-090) is sufficient
Public-cloud infrastructure of any kind	Hard deployment constraint (00-INDEX.md "Deployment Principle"); every environment is on-premises

8. Acceptance Criteria (Product-Level)

1. A user cannot create two Status Codes with the same `status_code_identifier` (enforced at the database and API layer).
2. A Status Code Draft never displays a fixed identifier; one is allocated exactly once, atomically, at Draft->Review.
3. A Status Code cannot reach Released without passing through both required Review sign-offs (Reviewer and Administrator, neither the CR's author) and a separate Chief Engineer approval (also not the author or either sign-off).
4. Submission for Review is blocked while any Engineering-Domain-owned field required by the code's classification is not yet signed off by a qualified second Engineer, when the originating Engineer does not hold that domain.
5. Every mandatory attribute in 06-data-dictionary.md is enforced as required at both the UI and API layer before

a revision can leave Draft.

6. Every state transition, field change, sign-off, approval, and role assignment produces an immutable audit log entry with actor, timestamp, before/after values.
7. A published Release is immutable, and a sandbox-flagged revision can never be included in one.
8. Role-based access control prevents Viewers from performing any write operation, prevents Engineers from self-signing-off or self-approving their own Change Requests at any of the three decision points, and restricts full-database export to the Administrator role alone.
9. Search returns results across all major attributes (identifier, title, category, Functional System Group, lifecycle status, owner) within NFR-001 performance targets.
10. An exported catalogue contains only Approved or Released, non-sandbox records, correctly scoped to the export filter and reproducible.
11. A sandbox status code can be permanently deleted only by an Administrator, and only while flagged sandbox; the deletion is itself permanently audited with a full before-state snapshot.
12. (v4) A save based on an outdated Draft is refused with 409 and nothing is overwritten; the user's entries are shown beside the current values.
13. (v4) Reviewer and Administrator sign-off, Chief Engineer approval, publishing and the full-database export are refused without the actor's password; delegates and change-set decisions never bypass segregation of duties.
14. (v4) An audience export contains no code whose confirmed value for that audience is not visible, and none of the other audiences' columns; every export has a manifest whose hashes verify, and a changed file is detected.
15. (v4) An edited, removed, reordered or truncated audit entry is detected by the verification; backups are proven restorable and the readiness probe reports their age.
16. (v4) A dry run creates nothing; an update-mode re-import never creates a duplicate; the reconciliation report names the legacy numbers still missing.
17. (v4) Every page is operable in English and German and meets the WCAG 2.1 AA checks of NFR-010; all requirements of 03-srs.md are cited by a test or listed as not testable in the test report.

9. KPIs (Product Health, Ongoing)

KPI	Target
Active Engineer/Reviewer/Administrator/Chief Engineer adoption (weekly active / licensed users in those roles)	>= 90% by end of Phase 1
Change Requests rejected due to preventable data-entry errors	< 10% of all submissions after month 2
Mean combined Reviewer+Administrator sign-off turnaround	<= 3 business days
Mean Chief Engineer approval turnaround	<= 2 business days
API error rate (5xx)	< 0.1% of requests
Audit log completeness (state-changing actions with a corresponding log entry)	100%
Cloud-infrastructure findings in any architecture review	0 (standing check, 11-security-architecture.md §12)
Verified backup younger than 36 h (readiness probe)	100% of days
Change requests past their red SLA threshold	< 5% of open requests
Released codes with an approved German translation (once German manuals are in scope)	tracked on the translation worklist

03. Software Requirements Specification (SRS) — CSCM Tool

> **v4 (2026-10-23):** the improvement pack of `21-improvement-ideas.md` (I-01 … I-36) is specified in §1.10–§1.15 (FR-090 … FR-155, plus FR-048i–n), NFR-009 … NFR-014, BR-016 … BR-022 and the decisions D-01 … D-08 in §6; **v4.2 (2026-10-23)** adds §1.16 (status code import from the generated `.st` programs and the legacy -> new mapping: FR-048o … FR-048w, BR-023, SEC-031). Security and audit requirements added with it (SEC-026 … SEC-031, AUD-072 … AUD-075) are in `11-security-architecture.md` and `12-audit-compliance.md`. Requirement IDs are never reused; the test report (`24-test-report.md` §6a) lists any requirement that no test cites.

> **v4.7 (2026-10-24):** the e-mail address of a user is fixed (FR-162), the Windows sign-in with its developer simulation (FR-163, SEC-033) the translation review inside the application (FR-164), the PDF in every script (FR-165), the technical nomenclature (FR-166) the complete interface catalogues of all languages (FR-167) the fields of the generated programs (FR-168), the menu (FR-169), the beta mark of languages (FR-170), the languages a Viewer may see (FR-172), the language of the words around an export (FR-171), the complete baseline of a Release (FR-173) and the package of all languages and access levels (FR-174) the rule who may download what (FR-175), the separate approval of new and changed translations (FR-176) the confidence check of the translations (FR-177) the language impact of a change of the vocabulary (FR-178) the connections between status codes (FR-179) and the bulk workflow with its sample inspection (FR-180) are specified in §1.17.

> **v4.17 (2026-10-27):** the legacy -> new **mapping** is kept in the database and changed only with the approval of the StCd Owner, the Chief Engineer and an Administrator (FR-191 … FR-198, BR-027); the `.st` import takes the title, the type, the up/down counter, the access levels, cause and directive from it (FR-192, FR-193) and no longer lets the uploader choose a category (FR-181 is superseded).

> **v4.3 (2026-10-23):** the sixth role **StCd Owner**, role colours, and nineteen languages with flags are specified in §1.17 (FR-156, FR-157), BR-024 and SEC-032 (`11-security-architecture.md`).

1. Functional Requirements

1.1 Authentication & Session Management

- **FR-001** The system shall authenticate users via username/email and password using session-based authentication (server-side session, HttpOnly + Secure + SameSite=Strict cookie).
- **FR-002** The system shall enforce session expiry after 30 minutes of inactivity (configurable) and require re-authentication thereafter.
- **FR-003** The system shall allow a user to explicitly log out, invalidating the server-side session immediately.
- **FR-004** The system shall lock a user account for 15 minutes after 5 consecutive failed login attempts.
- **FR-005** The system shall provide an architecture hook (pluggable auth provider) for a future **on-premises Active Directory (LDAP/Kerberos) or ADFS/SAML** integration without changing the session/authorization model downstream, and shall never integrate with a cloud-hosted identity provider (see 08-system-architecture.md §7, 11-security-architecture.md §2).

1.2 User & Role Management

- **FR-010** The system shall allow an Administrator to create, edit, deactivate, and reactivate user accounts.
- **FR-011** The system shall allow an Administrator to assign exactly one of six roles (Administrator, StCd Owner, Engineer, Reviewer, Chief Engineer, Viewer) per user; only an Administrator creates users, and an StCd Owner may move a user between Viewer, Engineer and Reviewer (FR-156, BR-024).
- **FR-011a** The system shall allow an Administrator to tag an Engineer-role user with one or more Engineering Domains (06-data-dictionary.md §10), used to route Cross-Domain Sign-Off (§1.3a).
- **FR-012** The system shall prevent deactivation of the last active Administrator account.
- **FR-013** The system shall allow a user to change their own password, enforcing the password policy in 11-security-architecture.md §4.
- **FR-014** The system shall allow an Administrator to force a password reset for any user.

1.3 Status Code Creation & Editing

- **FR-020** The system shall allow an Engineer or Administrator to create a new Status Code by specifying a Functional System Group and a Functional Subgroup — **both mandatory, corrected 2026-09-17: Functional Subgroup is never optional** — and all mandatory attributes defined in 06-data-dictionary.md.
- **FR-021** The system shall NOT assign a fixed Status Code Identifier while a code is in Draft. It may offer a non-binding preview of the likely next identifier for display purposes only (`GET /status-codes/next-available`); the identifier is allocated exactly once, atomically, at the Draft->Review transition (05-governance-handbook.md §2, 07-database-design.md §7a).
- **FR-021a** The system shall allow an Administrator to override the automatically allocated identifier for a documented migration/reservation case, with the reason logged.
- **FR-022** The system shall reject creation or save of a Status Code Revision if any mandatory attribute is missing, malformed, or fails cross-field validation rules (06-data-dictionary.md "Validation" column).
- **FR-023** The system shall prevent creation of a duplicate `status_code_identifier` (unique constraint; see 07-database-design.md).
- **FR-023a** The system shall require at least one Turbine Platform to be selected before a revision can be submitted for Review.
- **FR-024** The system shall allow an Engineer to save an in-progress Status Code as Draft an unlimited number of times without triggering workflow transitions.
- **FR-025** The system shall NOT allow direct in-place editing of a Status Code Revision that is in Review, PendingApproval, Approved, Released, Deprecated, or Archived state; any change requires creating a new Draft revision from the current one (copy-on-write).
- **FR-025a** The system shall treat each of the eight audience-access fields (corrected 2026-09-17: Development, Sales, TCC, Service, Turbine Operator Package, Grid Operator, Service Partner, Customer — replaces the single `access_rights_id`) as a restricted field: an Engineer may propose values while the revision is in Draft, but none is authoritative until confirmed or overridden by the Administrator's Review sign-off (06-data-dictionary.md §2d, SEC-013).
- **FR-026** The system shall allow an Engineer to withdraw/cancel a Draft, Review, or PendingApproval Change Request before it reaches Approved.

1.3a Cross-Domain Sign-Off

- **FR-027** The system shall determine, from the field-to-Engineering-Domain ownership mapping (06-data-dictionary.md §2a), which domains are required on a given revision, and shall flag any domain not covered by the originating Engineer's own domain tags as pending.
- **FR-028** The system shall allow the originating Engineer to request sign-off from an Engineer holding a required, pending domain, and shall surface that request on the target Engineer's Dashboard and Change Request list.
- **FR-029** The system shall block Submit-for-Review (422) while any required domain lacks a recorded `DomainSignoff`, and shall only accept a `DomainSignoff` from a user who actually holds the named Engineering Domain.

1.4 Change Request & Review Workflow

- **FR-030** The system shall require every creation of a new Status Code and every revision of an existing one to be represented as a Change Request (CR) that carries the proposed revision through the lifecycle.
- **FR-031** The system shall allow an Engineer to submit a Draft CR for Review, which allocates the Status Code Identifier (FR-021), transitions the associated revision from Draft to Review, and notifies eligible Reviewers and the Administrator.
- **FR-032** The system shall present Reviewers and the Administrator with a field-level diff between the proposed revision and the current Approved/Released revision of the same Status Code (or "new code" if none exists).
- **FR-033** The system shall allow a Reviewer to record an independent sign-off (approve or reject) on a CR in Review; rejection requires a mandatory comment and returns the revision to Draft, clearing both sign-off columns.
- **FR-034** The system shall allow an Administrator to record an independent sign-off (approve or reject) on a CR in Review, following the same rules as FR-033, and shall confirm or override any restricted field (FR-025a) at the moment of approval.
- **FR-035** The system shall automatically transition a CR from Review to PendingApproval, atomically with whichever sign-off action is recorded second, once both the Reviewer's and the Administrator's sign-offs are present (in either order).

- **FR-036** The system shall allow a Chief Engineer to record the final Approval decision (approve or reject) on a CR in PendingApproval; rejection requires a mandatory comment and returns the revision to Draft, clearing both prior sign-offs.
- **FR-037** The system shall prevent the Engineer who authored a CR from signing off or approving their own CR at any of the three decision points (Reviewer sign-off, Administrator sign-off, Chief Engineer approval), and shall additionally prevent the Chief Engineer decision from being made by whichever individuals already provided the Reviewer or Administrator sign-off on that same CR (BR-005).
- **FR-038** The system shall allow a Reviewer, Administrator, or Chief Engineer to add comments to a CR at any stage without changing its state.
- **FR-039** The system shall allow an Administrator to reassign or unassign a CR between Reviewers.

1.5 Release Management

- **FR-040** The system shall allow an Administrator to create a Release container with a name, target version label, description, and scope filter (e.g. Functional System Group, Functional Subgroup).
- **FR-041** The system shall list all non-sandbox Status Code Revisions in Approved state matching a Release's scope filter as candidates for inclusion.
- **FR-042** The system shall allow an Administrator to include/exclude individual candidate revisions before publishing.
- **FR-043** The system shall, upon publishing a Release, transition all included revisions from Approved to Released, stamp Effective Date, and mark the Release as immutable.
- **FR-044** The system shall prevent modification of a published Release's contents; a correction requires a new Release.
- **FR-045** The system shall allow export of any Release, or of the current set of all Released status codes, in CSV, JSON, XLSX, and PDF formats (XLSX added 2026-09-17), all as flat, table-shaped output (one row per status code), per the role-scoped catalogue-export matrix (11-security-architecture.md §3). Since v4 an export may additionally be shaped by an export profile, an audience, a language and a software version, and ships with a manifest (FR-100–FR-104, FR-107, FR-111). Since v4.10 a Release export is its complete baseline (FR-173).
- **FR-046** The system shall allow export of a Draft/ad-hoc filtered working set, distinctly labeled "Not for distribution — unapproved data," restricted to the StCd Owner and the Administrator since v4.10 (FR-175; before: Engineer/Reviewer/Administrator/Chief Engineer).
- **FR-047** The system shall allow an Administrator, and only an Administrator, to generate a full raw database export/backup, distinct from catalogue export (08-system-architecture.md §6, SEC-015). Since v4 it is requested with `POST` and the Administrator's password (FR-126).
- **FR-048** The system shall prevent a sandbox-flagged revision from ever being added to a Release candidate list or included in a published Release, enforced independently at the service layer and the database layer (07-database-design.md §5).

1.6 Deprecation & Archival

- **FR-050** The system shall allow an Administrator (or Engineer via CR, subject to the full three-decision-point cycle) to initiate deprecation of a Released Status Code, requiring a mandatory reason and an optional superseding identifier reference.
- **FR-051** The system shall allow an Administrator to archive a Deprecated Status Code after the configured minimum retention period has elapsed (05-governance-handbook.md §11); Archived codes become read-only.
- **FR-052** The system shall prevent deletion of any non-sandbox Status Code or Revision at any lifecycle state; removal from active use is only possible via Deprecated/Archived states.

1.5a Library Import (new 2026-09-17 — reverses the prior "no bulk import" position)

- **FR-048a** The system shall allow an Engineer or Administrator to upload a CSV or XLSX file of status codes for bulk import.
- **FR-048b** The system shall validate every imported row against the same rules as manual entry (06-data-dictionary.md), creating a Draft `StatusCode`/`StatusCodeRevision`/`ChangeRequest` for every row that passes and recording every row that fails in a downloadable error report (FR-048m); a failing row creates nothing.
- **FR-048c** The system shall require every imported row to pass through the full governance cycle (Cross-Domain Sign-Off where applicable, both Review sign-offs, Chief Engineer approval) before it can be Released — import is a faster data-entry path, never a governance bypass (confirmed explicitly by the business;

05-governance-handbook.md §6b).

- **FR-048d** The system shall preserve any legacy/source numbering supplied in the import file as `legacy_reference_number` for traceability, distinct from and never confused with the governed `StCd` identifier.

1.5b Subgroup Administration (new 2026-09-17)

- **FR-048e** The system shall allow an Administrator to create a new Functional Subgroup within any Functional System Group, specifying a name and a numeric sub-range.
- **FR-048f** The system shall reject a proposed subgroup sub-range that overlaps any existing subgroup in the same Functional System Group (409), naming the conflicting subgroup, enforced at both the service layer and the database layer (07-database-design.md §5).
- **FR-048g** (new 2026-09-17) The system shall allow an Administrator to create a brand-new Functional System Group, specifying a code (prefix), label, and a numeric range, into any currently free range (confirmed reserved: `10000?10999`, `14000?14999`, `15000?15999`, `16000?16999` — 05-governance-handbook.md §2), without requiring a schema change.
- **FR-048h** (new 2026-09-17) The system shall reject a proposed Functional System Group range that overlaps any existing group's range (409), naming the conflicting group, enforced at both the service layer and the database layer (07-database-design.md §5).

1.6a Sandbox Status Codes

- **FR-053** The system shall allow an Administrator, and only an Administrator, to create a Status Code flagged `is_sandbox = true`, drawing its identifier from an isolated `StCd-T#####` band that never overlaps a real Functional System Group range.
- **FR-054** The system shall exclude sandbox-flagged records from default search results, catalogue exports, and Release candidate lists by construction, not by convention (FR-048).
- **FR-055** The system shall allow an Administrator, and only an Administrator, to permanently delete a sandbox-flagged Status Code, and shall reject any deletion attempt against a non-sandbox record regardless of role, enforced at both the service layer and a database trigger.
- **FR-056** The system shall record a `SANDBOX_DELETE` audit entry containing a full pre-delete snapshot of the deleted record, in the same transaction as the delete, so the fact of creation and deletion remains permanently traceable even though the record itself no longer exists.

1.7 Search, Filter, and Browse

- **FR-060** The system shall provide full-text search across Status Code Identifier, Title, and Description (ranked and typo-tolerant since v4, FR-148).
- **FR-061** The system shall provide filtering by Functional System Group, Functional Subgroup, Turbine Platform, Lifecycle State, Status Category, Owner, Software Version, and sandbox flag (default excluded), individually or combined; a filter combination can be saved and shared as a link (FR-142).
- **FR-062** The system shall provide sorting on any listed column, ascending or descending.
- **FR-063** The system shall paginate list results with a configurable page size (default 50, max 200).
- **FR-064** The system shall allow a Viewer or higher role to view the complete revision history of any non-Draft/Review/PendingApproval Status Code, including diffs between any two revisions; Draft/Review/PendingApproval-state data is visible to Engineer, Reviewer, Administrator, and Chief Engineer roles only (BR-009).

1.8 Audit Trail

- **FR-070** The system shall record an immutable audit log entry for every create, update, state transition, sign-off, approval, rejection, release, deprecation, archival, sandbox create/delete, login, failed login, step-up confirmation, role change, delegation, settings change, evidence, translation, token and webhook change, and export action, capturing actor, timestamp (UTC), entity type, entity ID, action, and before/after field values where applicable (12-audit-compliance.md).
- **FR-071** The system shall allow Administrators (and Reviewers/Chief Engineers, read-only, for records they are entitled to act on) to search and filter the audit log by actor, entity, action type, and date range.
- **FR-072** The system shall prevent any user, including Administrators, from editing or deleting audit log entries through the application.
- **FR-073** The system shall retain the audit trail for a deleted sandbox record indefinitely, even though the application provides no means of retrieving the record itself once deleted (FR-056).

1.9 API

- **FR-080** The system shall expose a versioned REST API (`/api/v1/...`) covering Status Codes, Revisions, Change Requests (incl. sign-off/approval/domain-sign-off endpoints), Releases, Users, Lookups, and Audit Log read access, per 09-api-specification.md (171 operations in v4.10; `docs/artifacts/openapi.yaml` is the contract and a test fails if the code and the contract differ).
- **FR-081** The system shall enforce the same role-based authorization on the API as on the UI; the API is not a privilege-escalation path.
- **FR-082** The system shall support API pagination, filtering, and sorting equivalent to the UI list capabilities (FR-061–FR-063).
- **FR-083** The system shall be reachable only from the internal corporate network in every deployment; no endpoint shall be exposed to the public internet, and no request shall be proxied through or dependent on a public-cloud service.

1.10 Concurrency & Data Integrity (v4 — I-01, I-04, I-12, I-25; EPIC-17)

- **FR-090** The system shall detect conflicting edits of a Draft: every revision carries a `row_version` that increases on each save and an `updated_at` stamp; an edit that names a stale `expected_row_version` (or `expected_updated_at`) is rejected with 409 and changes nothing, and the edit form re-shows the user's entries beside the current values (NFR-007a). An edit that names neither is accepted (backwards compatible).
- **FR-091** The system shall warn, never block, when `status_code_for_repeated_error`, `related_status_codes` or `superseded_by_status_code_id` point at a code that is unknown, Deprecated or Archived, or at the code itself. The warnings are returned with the saved revision, shown after saving, and listed on the code's detail page and on the change request.
- **FR-092** The system shall show, on every code, a **Where-used** list of the other codes whose current revision refers to it (repeated-error pointer, related codes, supersession, platform variant), and shall require the requester of a deprecation to acknowledge the live dependants (`acknowledge_dependants`) before the request is created.
- **FR-093** The system shall show existing codes of the same Functional Subgroup with an identical or similar title when a code is created or imported ("did you mean to revise StCd-...?"). The check is a warning; it never blocks a save or an import row.
- **FR-094** The system shall allow an Engineer or Administrator to create a **platform variant** (e.g. 4XM from a 3XM code): a new Draft that copies the latest decided revision for another turbine platform, linked through `platform_variant_of_status_code_id`, with the family shown on both detail pages.

1.11 Library Import Toolkit (v4 — I-05, I-32; extends 1.5a)

- **FR-048i** The system shall offer the import template for download (CSV: the exact headers; XLSX: headers, instructions and the lookup codes), and the unmodified template shall import cleanly.
- **FR-048j** The system shall offer a **dry run** that validates a file with the full import rules and reports every problem per row and column (value, problem, how to fix) and every warning, while creating nothing — not even an import job.
- **FR-048k** The system shall offer an **update mode** keyed on `legacy_reference_number` (unique within a Functional System Group): a changed row revises the already-imported open Draft in place, or opens a new revision of an already-Released code; an unchanged row is skipped; a row without a match is created; a code that is in review, has another owner, or would change group or subgroup is reported as an error on that row.
- **FR-048l** The system shall provide a **reconciliation report** — legacy rows imported, in review, approved, released, retired and, given an expected total or the uploaded legacy master list, still missing (with the missing legacy numbers named).
- **FR-048m** The system shall provide each import job's error report as CSV and as XLSX, with one line per problem (row, column, value, problem, how to fix); values that start with a formula character are neutralised (SEC-024).
- **FR-048n** The system shall process files above a configurable row threshold (default 300) in a background worker and show progress (`processed_count`, percentage) on the job; the upload returns 202 and the job ends COMPLETE or FAILED, never silently stuck.

1.12 Publication, Release Integrity & Software Versions (v4 — I-02, I-03, I-06, I-07, I-15, I-24, I-27; EPIC-18, EPIC-20, EPIC-22)

- **FR-100** The system shall let an Administrator define **export profiles** — named column sets with header labels and an optional default audience and language (seeded: Service manual, SCADA tag list, Customer list) — as

data, and shall let any export, release export or bundle use one.

- **FR-101** The system shall never truncate cell text in the PDF export: cells wrap over several lines, the header row repeats on every page, and a title page shows the release, version, effective date, filters, row count and the content digest.
- **FR-102** The system shall allow an export to be scoped to one of the eight audiences: only codes whose Administrator-confirmed value for that audience is visible are included, and the other audiences' columns are omitted. Visibility is data on each access vocabulary value (`audience_visible`); a code with no value for the audience is hidden (decision D-02, BR-016).
- **FR-103** The system shall accompany every export of a Release with a **manifest** (release, scope, row count, SHA-256 of every file, content digest, generator, timestamp, applied profile/audience/language/version), signed with HMAC-SHA-256 when `MANIFEST_SIGNING_KEY` is configured, and shall record each export with its hash. The export can ship as a ZIP bundle containing the data file, `MANIFEST.json` and the release notes.
- **FR-104** The system shall verify a file against a manifest without the database — in the UI and API (`/exports/verify`) and on the command line (`flask verify-export`) — reporting the SHA-256, the matching recorded exports, whether the file is listed and whether the signature is valid, invalid or absent.
- **FR-105** The system shall generate **release notes** for every published Release against the previous published Release: codes added, codes changed (with the field-level diff of FR-032) and codes deprecated, as Markdown, CSV and PDF.
- **FR-106** The system shall maintain an ordered vocabulary of **controller software versions**; each revision may declare an inclusive from/to version range (either end open) and each Release targets one or more versions, which an Administrator sets before publishing (decision D-01).
- **FR-107** The system shall answer "which Released codes apply to version X" and "what was added, removed and changed between versions X and Y" (applicability matrix), and shall allow any export or bundle to be limited to one software version.
- **FR-108** The system shall generate, from any Release, an **offline lookup bundle**: one self-contained HTML file with the codes embedded and a search box, loading nothing from the network, listed in the manifest.
- **FR-109** The system shall let an Administrator stamp a **planned effective date** and a **planned deprecation date** on the current revision (audited); the release builder and the publish response warn — never block — when an included revision's planned effective date is still in the future.
- **FR-110** The system shall hold translations of title and description per language (English is the source; German is seeded) with a status of MISSING, DRAFT or APPROVED. A translation is saved only against an Approved or Released source revision, approved by a Reviewer or Administrator who is not the translator (BR-018), sent back to DRAFT by any edit, and carried over from the previous revision as DRAFT on request.
- **FR-111** The system shall export in a chosen language using approved translations only, falling back to the source text unless the export is strict, and shall list the codes still lacking an approved translation (worklist).

1.13 Workflow Aids & Governance Extensions (v4 — I-08, I-10, I-11, I-13, I-14, I-16, I-17, I-23, I-30; EPIC-04, EPIC-19, EPIC-23)

- **FR-120** The system shall show how long every change request has waited at its current stage, in business days (Monday–Friday, no holiday calendar), and shall mark it amber or red against Administrator-configured thresholds (defaults: review 3/5 days, approval 2/3 days).
- **FR-121** The system shall create **in-app notices** for whoever holds the next decision (including anyone covering for them, FR-125) on a stalled request, at most once per user, request, stage, level and day, through `flask send-reminders`; if an SMTP relay is configured (decision D-04) the same text is also mailed. A mail failure never breaks the triggering action.
- **FR-122** The system shall give Administrators a list of stalled change requests with one-click reassignment (FR-039).
- **FR-123** The system shall show Administrators and Chief Engineers the business-case KPIs: median business days from submission to approval (target <= 7), median cross-domain turnaround (target <= 3), share of Released revisions with a complete audit trail (target 100 %), the age distribution of open requests and the codes whose owner is no longer an active user.
- **FR-124** The system shall let Administrators, Engineers and Chief Engineers group change requests into a **change set** for one controller software release and let reviewers decide several of its requests in one sitting. Each request is still decided by its own service call with its own segregation-of-duties check, comment and audit entry; a request the actor may not decide is reported as skipped and never blocks the others (BR-005 is unchanged).
- **FR-125** The system shall let a Reviewer, Administrator or Chief Engineer name a **delegate** for a date range (an

Administrator may do so for anybody). The delegate acts as themselves; the audit entry names both; delegation grants the decision actions of the delegated role only, never user administration; segregation of duties applies to the delegate personally (BR-017, decision D-08).

- **FR-126** The system shall require the actor's password again (**step-up**) for Reviewer and Administrator sign-off, Chief Engineer approval, publishing a Release and the full-database export; a missing password returns 403 `step_up_required`, a wrong one 403 `step_up_failed`; both outcomes are audited and the same password prompt covers a whole change-set decision (decision D-07, SEC-026).
- **FR-127** The system shall let a Reviewer, Administrator or Chief Engineer comment on one specific field of a change request in Review or Pending Approval; the author sees the open comments inline and marks them resolved.
- **FR-128** The system shall let an Administrator maintain a **review checklist** per Functional System Group (or for all groups); an approving Reviewer must confirm every active item that applies, and the confirmed wording is stored with the sign-off.
- **FR-129** The system shall show the open reviews per Reviewer and shall assign the least-loaded available Reviewer on request, or automatically at submit when the Administrator enables it; a Reviewer who is out of office (FR-125) is skipped.
- **FR-130** The system shall let the author or an Administrator attach **evidence** to an open change request (PDF, PNG, JPG, TXT, LOG, MD, CSV, XLSX, DOCX; at most 10 MB and 20 files each), checked against the claimed type, stored outside the web root, downloadable only as an attachment by the working roles and removable only while the request is a Draft (SEC-024).
- **FR-131** The system shall carry the optional fields remedy text, affected component and standard references on a revision; they have an "Any" owner (06-data-dictionary.md §2a), so they never gate Cross-Domain Sign-Off, and they appear in the matching export profiles and the offline bundle.

1.14 Insight, Integration & Hub Consolidation (v4 — I-19, I-26, I-28, I-29, I-31, I-36; EPIC-16, EPIC-24)

- **FR-140** The system shall provide a **catalogue health report**: expected-but-missing optional data (remedy text, component, software-version range, audience access), inconsistent combinations (INFO with an alarm; repeat count without escalation code or the reverse; repeat window without count; planned deprecation date passed), unused numbers inside a band, and the upcoming planned dates.
- **FR-141** The system shall show how full every subgroup's number band is and alert Administrators, Reviewers and Chief Engineers when at most N percent or N numbers remain (defaults 15 % / 20), and shall preview the remaining capacity on the New Status Code form. On the health page the rows of Main Controller groups are blue and those of Hub Controller groups light blue, each with an MC or HC chip, so the two numberings can be told apart at a glance.
- **FR-142** The system shall let users save a search (filters and visible columns) under a name, choose the columns of the list, and share the same view as a URL.
- **FR-143** The system shall let an Administrator or Chief Engineer import anonymised occurrence counts per status code and period (CSV/XLSX with `status_code_identifier` and `occurrences`) and report the Released codes that never fired and the most frequent ones, flagging those without remedy text.
- **FR-144** The system shall let an Administrator issue **scoped, expiring API tokens** (`catalogue:read`, `changes:read`; at most 365 days; stored only as a hash; shown once; revocable). A token authenticates `Authorization: Bearer cscm_?` on the read-only endpoints that declare its scope, with Viewer-level visibility, and can never write (SEC-027).
- **FR-145** The system shall provide a **change feed** (`GET /changes`) of releases, deprecations, reinstatements and archivals in audit order, paged with a `since_id` cursor and supporting `ETag/If-None-Match`.
- **FR-146** The system shall let an Administrator register **webhooks** on `release.published`; a delivery carries an HMAC-SHA-256 signature, goes only to a private or loopback address, does not follow redirects, is recorded with its outcome, and can never block or fail the publish (SEC-028).
- **FR-147** The system shall support the Hub -> Main Controller consolidation of assumption A-008: create a Main Controller Draft from an Approved or Released Hub Controller code (new MC number at submit, the HC identifier kept as the legacy reference, mapping recorded), report every Hub code with the state of its replacement, and open deprecation requests for all Hub codes of a group whose replacement is Released.
- **FR-148** The system shall rank search results by relevance (a title hit outranks a description hit) and tolerate small spelling errors ("convertor" finds "converter"), falling back to substring search where the SQLite build lacks FTS5.

1.15 Operations & Usability (v4 — I-09, I-18, I-21, I-22, I-33, I-34, I-35; EPIC-25)

- **FR-150** The system shall back the database up with SQLite's online-backup API (`flask backup-db`), keep a configurable number of generations, and **prove** each backup is usable (integrity check, restore into a scratch database, row-count comparison, audit-chain check), writing the result to a status file that the readiness probe and the Administrator view read.
- **FR-151** The system shall store with each audit entry the hash of the previous one (HMAC-keyed when `AUDIT_HMAC_KEY` is set), verify the chain on demand (`flask audit-verify`, UI, API) and detect an edited, removed, reordered or truncated row, and shall archive — never delete — entries older than the retention period (`flask audit-archive`) (AUD-074).
- **FR-152** The system shall expose an unauthenticated **readiness probe** (`GET /health/ready`) reporting database reachability, schema at the expected Alembic head, the newest audit rows' integrity and the age of the last verified backup: 200 when ready (status `ok` or `degraded`), 503 when not.
- **FR-153** The system shall expose Prometheus-style **metrics** (`GET /metrics`: request counts, latency and write-lock-wait histograms, queue sizes, backup state) to the internal network only (allow-listed networks, optional bearer token) and shall optionally write the JSON request log to a rotating file.
- **FR-154** The system shall explain itself in the application: tooltips for lifecycle states and roles, a "what happens next" hint on each change-request state, and a per-role quick start (`/help`).
- **FR-155** The system shall offer the interface in English and German (per-user preference, else the browser language) and meet WCAG 2.1 AA for contrast, keyboard operation, landmarks, labelled controls and table headers (NFR-010).

1.16 Status Code Import from Generated Programs (v4.2 — extends 1.5a; EPIC-14)

The generated controller programs (`PROGRAM StCd_`, IEC 61131-3 Structured Text, written by the status code generator) already carry every status code of the controller: its number, name, delays, repeat rule, alarm and availability settings, subsystem programs and user-group levels. These requirements let the business bring new codes into the catalogue **from the** `.st` **files alone**, several at a time, without bypassing governance (FR-048c). The reader never executes or compiles anything; it scans the text (`app/importers/stcd.py`).

- **FR-048o** The system shall let an Engineer or Administrator upload several `StCd_.st` files at once, a whole folder, or one ZIP archive of them, read every `PROGRAM StCd_` in them (UTF-8, falling back to the generator's CP1252) and report files that hold no program or exceed the size limit instead of failing the upload (SEC-031).
- **FR-048p** The system shall show a **preview** before anything is created: one line per program with its new identifier, legacy number, controller, name, the group and subgroup its number lies in, a state (ready, already in the catalogue, blocked), every problem and every warning, and shall let the user tick any subset (select all, filter by text) for creation. The preview writes nothing except the user's staged copy of the upload.
- **FR-048q** The system shall **reserve** the number a program carries for the Draft created from it (`status_code.reserved_identifier`): the identifier issued at Submit for Review is exactly that number, the sequence allocator skips reserved numbers, an Administrator override is refused for a reserved code, and the database refuses a second reservation or the issue of a reserved number to another code (BR-023).
- **FR-048r** The system shall place each imported code in the Functional System Group and Subgroup whose band holds its number (`NumberingService.band_of`); a number no active subgroup covers, or a number in a Hub band for a Main Controller program (or the reverse), blocks that program with a message naming the missing band.
- **FR-048s** The system shall treat the number in a Hub Controller file as the **legacy** Hub number and take the new number from the **mapping** (FR-191): the row of the Hub family with that legacy number gives the new number, in the Hub range (five digits, 11000 and up). A legacy number missing from the mapping blocks the program until the row is requested and approved (FR-194). Moving the Hub codes into the Main range is a later, systematic change of the mapping, requested and approved like any other (I-75). A file is a Hub file when it is uploaded as such or, by default, when its number is 10000 or higher.
- **FR-048t** The system shall create every ticked, still-ready program as an ordinary Draft status code with its change request (`cr_type = IMPORT`), filling the attributes the catalogue has (title, availability group, alarm, set and reset delay, repeat rule, snapshot, zero-load operation, the eight subsystem programs it has columns for, the programs the set condition refers to as related codes), and shall put **all Drafts of one confirmation into one new change set** (I-11) so the batch is reviewed and decided together; every Draft still needs Cross-Domain Sign-Off, both Review sign-offs and Chief Engineer approval (FR-048c, BR-015). The status category is not in a program: the uploader chooses one for the batch.
- **FR-048u** The system shall provide the **legacy -> new mapping**: for every catalogue code that carries a legacy number, the legacy number (Main or Hub), the identifier it was issued or has reserved, its title and state, with

filters and a CSV/XLSX export, for internal roles (not Viewers). The mapping runs from legacy to new only: the new library holds more codes than the old one, so a code without a legacy number is normal and counted as n/a, and no reverse lookup is offered. The mapping is the code's `legacy_reference_number` read with its controller type; there is no second copy to keep in step.

- **FR-048v** The system shall keep, for every imported code, its **provenance** — source file name, SHA-256 of the file, program number and line, and every attribute parsed from it as JSON, including those the catalogue has no column for yet (user-group levels, the remaining subsystem programs) — and shall store the program's `PrgIdx_GUi` when it is free (`status_code.prg_idx`, 06-data-dictionary.md §1), warning otherwise.
- **FR-048w** The system shall re-check every ticked program when the user confirms (the catalogue may have changed since the preview): a program that is meanwhile present, or conflicts with a number, a legacy number or another program of the upload, is refused with a message and nothing is created for it; the rest proceeds. Files above the background threshold (FR-048n) are processed in a worker with progress.
- **FR-048x** The system shall, in the preview of an import from program files, also recognise programs whose number is already in the catalogue and say what they would do: **unchanged** (nothing), **changed** — the fields in which the file differs are listed, and the program can **update the open Draft** in place or open a **new revision** of an Approved or Released code — and **withdrawn**, a code whose change request was withdrawn and which can be **restored** as a new Draft on the same number. A changed or withdrawn program is never ticked by default; the user chooses. A code in Review, Deprecated or Archived is reported as blocked. The result joins the batch change set like a new program.

1.17 Roles and Languages (v4.3 — EPIC-04, EPIC-25)

- **FR-156** The system shall have six fixed roles. The sixth, **StCd Owner** (the owner of the status code library), reads the library and the workflow (change requests, change sets, the Hub mapping) but takes no part in any decision, and may change the role of a user **only between Viewer, Engineer and Reviewer** — never the role of an Administrator, a Chief Engineer or another StCd Owner, never to a role above Reviewer, and nothing else about a person (BR-024). Creating users, the password reset and every other role change stay with the Administrator. Every role is shown as a **coloured badge** whose colour is its level — red for the highest — in the user list and in the top bar: Administrator 100 (red), StCd Owner 80 (purple), Chief Engineer 60 (amber), Reviewer 40 (blue), Engineer 30 (green), Viewer 10 (grey).
- **FR-157** The system shall offer nineteen languages — English, German, French, Hindi, Arabic and, after Arabic in this order, Bengali, Marathi, Telugu, Tamil, Gujarati, Urdu, Kannada, Odia, Malayalam, Punjabi, Spanish, Italian, Danish and, last, Georgian — both as languages of the translation lookup and as interface languages chosen per user like FR-155. Every language name reads **country letters on the left, the name, and the national flag on the right** (`GB English GB`, `DE Deutsch DE`, `SA ??????? SA`, `IN ????? IN`, `PK ???? PK`, `ES Español ES`, `DK Dansk DK` …); in the language menu of the top bar the flag is drawn as an image, because Windows shows flag emoji as two letters, and the menu scrolls. The Indian languages carry the flag of India, Urdu that of Pakistan, Arabic that of Saudi Arabia, Spanish, Italian and Danish those of Spain, Italy and Denmark, Georgian that of Georgia. Arabic and Urdu are written right to left: the page is mirrored (`dir="rtl"`). French, Hindi, Arabic, Spanish, Italian, Danish and Georgian cover the most visible text (top bar, navigation, login, dashboard, status code list and the role names); the ten languages after Arabic cover about sixty strings (top bar, navigation, login, role names); all other text falls back to English, and none of the translations has been reviewed by native speakers. What is translated and what never is — identifiers, numbers, stored codes, audit entries, the content of a status code outside the approval workflow — is fixed in `10-ui-ux-specification.md` §10a.
- **FR-158** The system shall let an Administrator **view** a user (profile, role, domains, last sign-in and what the account has done in the tool), **edit** the user, **deactivate** and **activate** the account with a button, and **delete** an account — in the user list and on the user's page. A deactivated account cannot sign in and an open session ends with its next request; nothing it did is lost. Deletion needs the Administrator's own password (SEC-026), is refused for the Administrator's own account and for the last active Administrator, and works for every account (BR-025): an account that nothing else points at is removed, one with history is erased — its name, address and password go and it leaves every list, but the row stays so that everything it did keeps its author. The audit log is never changed. The StCd Owner sees the list and each user's page but has none of these buttons except the role change of FR-156.
- **FR-159** The system shall let an Engineer or Administrator **select several change requests and withdraw them in one action** (the list My Change Requests and `POST /change-requests/withdraw`). Every request goes through the single-request rules — only its author, only while it is a Draft, in Review or waiting for approval — and has its own audit entry; one that cannot be withdrawn is reported and does not stop the others.
- **FR-160** The system shall make clear what a withdrawal does. The withdrawn revision and its change request are deleted and their content is kept in the audit entry (FR-026). A code that is left with no revision and never

received its identifier gives back what it held — the number an import reserved, its legacy number and its PrgIdx_GUi — so the same program can be imported again; a code that already has its identifier keeps it (BR-001, BR-026) and can be restored from its program file (FR-048x).

- **FR-161** The system shall keep every interface translation checkable by a native speaker: a generated document (26-translations.md) lists every string of the interface with a stable id, the screens that show it, the English source and each language's translation, per language with its coverage and a fingerprint of the catalogue; a generator (`scripts/gen_translations_doc.py`) writes one review sheet per language (XLSX: current translations with a verdict column of OK / CHANGE / UNCLEAR, a proposal and a comment, the strings still missing, and a checklist); the review itself is done in the application (FR-164), which records a completed review with the fingerprint of the version reviewed and shows a language as not reviewed, in progress, reviewed, or changed since the review. A test keeps the document equal to what the generator produces from the code. How the status code manuals reach every designated language, and what is still open, is the concept of that document (sections 4 and 8).
- **FR-162** The system shall keep a user's **e-mail address fixed** once the account exists. The address is the stable identity of the person (it is what the Windows sign-in matches, FR-163): it is entered when an Administrator creates the user (lower-cased, checked for form and for duplicates) and can never be edited — not in the user form, not through `PUT /users/{id}`, which answers 422 for a body that carries an `email` at all, and not in the database, where the trigger `trg_user_email_immutable` aborts any update of `user.email` that changes it. A person who needs another address gets a new account; the old one is deactivated (BR-025).
- **FR-163** The system shall support **sign-in with the Windows account** and let a developer simulate it. `AUTH_MODE` is `password` (default), `windows` or `both`. In Windows mode the organisation's gateway (IIS or ARR with Windows authentication, Apache with Kerberos, nginx with spnego — any on-premises proxy that authenticates against Active Directory) passes the user in the header `WINDOWS_AUTH_HEADER` (default `X-Remote-User`); the application believes the header only from an address in `WINDOWS_AUTH_TRUSTED_PROXIES` and ignores it from anywhere else. The identity is matched with an active account **by e-mail address** (`name@corp.example`; `CORP\name` becomes `name@WINDOWS_AUTH_EMAIL_DOMAIN`, else the user name); nobody is created on the fly, an unknown or deactivated Windows user is refused with a clear message, and every sign-in and refusal is audited like a password sign-in. In `windows` mode the password form, the password change and the self-service password pages are switched off (API: 403 `password_login_disabled`). **Simulation for development:** with `WINDOWS_AUTH_SIMULATE` (on in the development configuration) the login page offers Sign in as Windows user, where the developer types the identity the gateway would have supplied; it runs through the same code as a real header, so what works there works behind the gateway. The production configuration refuses to start with the simulation switched on or with an unknown `AUTH_MODE`. Endpoints: `GET /login/windows`, `POST /login/windows/simulate` (development only), `POST /api/v1/auth/windows`.
- **FR-164** The system shall let native speakers **review the interface translations in the application itself**, in a section of its own (Translations, S42 to S44). An Administrator or StCd Owner **assigns reviewers** to a language (any active account; an external native speaker gets a Viewer account that sees only this section). A reviewer sees one row per string — screens, English, current translation — and answers OK, CHANGE with a proposal (at most 1,000 characters, the `%(name)s` placeholders of the English text kept) or UNCLEAR with a comment; strings without a translation can be translated on the way; answers are saved page by page, a completed XLSX sheet can be uploaded instead, and the review is **completed** only when every translated string has a verdict. The completion is recorded with the date and the fingerprint of the version reviewed. An Administrator or StCd Owner **decides** on each proposal — accept (optionally correcting the text), reject or note — never on a proposal of their own; a decision is final. An accepted proposal is an **override** that the application shows at once for that language, without a release; an override can be removed, which restores the shipped text. Assigning, completing, deciding and removing an override are audited (AUD-077).
- **FR-165** The system shall print **every script of its languages in the PDF export**: Latin (with Greek and Cyrillic), Devanagari (Hindi, Marathi), Bengali, Gurmukhi (Punjabi), Gujarati, Odia, Tamil, Telugu, Kannada, Malayalam, Georgian, and Arabic-script text (Arabic in Naskh, Urdu in Nastaliq). The PDF text is Unicode: Noto Sans is the base font, a cell that contains another script also gets the Noto font of that script, the text is shaped with HarfBuzz (conjuncts, vowel signs, joining forms), and Arabic and Urdu cells are laid out right to left and right-aligned; the title page, notice, column headings and footer follow the same rules. The fonts (SIL Open Font License, with the licence text) are bundled in `app/static/fonts/` and the shaping library `uharfbuzz` is a requirement, so a deployment needs no font installation and no internet access. A character that no bundled font draws is replaced by an ASCII stand-in or `?` and never leaves an empty gap; plain Latin text is not shaped and stays fast (NFR-001). The export's language (FR-111) selects Nastaliq or Naskh for Arabic-script text, otherwise the letters decide.
- **FR-166** The system shall keep the **technical nomenclature** — the words of the controller and of the workflow (Status Code, Hub Controller, Set delay, Draft, the roles, the audiences ...) — in a vocabulary of its own, apart

from the interface translations (FR-164), in a section Terminology (S45 to S47). Every term has a stable code, the English text, a category (product, data, state, process, role, audience or never translated), a definition for a translator who does not know the product, and one translation per language. A translation is a **Draft until a second person approves it** (the segregation of BR-018: nobody approves their own proposal; Send back needs a note and reopens an approved translation). Everybody who is signed in reads; an Administrator, StCd Owner, Chief Engineer, Engineer or Reviewer proposes; an Administrator, StCd Owner, Chief Engineer or Reviewer approves; an Administrator, StCd Owner or Chief Engineer adds and changes terms, and a changed English text reopens the approved translations of the term. Being the reviewer of an interface language gives no right in the terminology. A term marked never translated has no translation. The tool ships 78 terms with a draft translation in each of the 18 languages besides English, taken from the interface catalogue where both contain the word. A consistency check lists the interface strings that contain a term whose translation does not appear in the translation of the string (a hint, not a verdict, because languages that inflect words change the ending), and the vocabulary is exported per language and for all languages as XLSX and CSV (a formula in a cell is neutralised). Every change is audited (`TERM_CREATE`, `TERM_UPDATE`, `TERM_TRANSLATE`, `TERM_APPROVE`, `TERM_REJECT`).

- **FR-167** The system shall translate the **whole interface inventory** into every language it offers: every text a template wraps in `_()`, the role names, the help texts and the sentences of the service layer that a person reads in a flash message (found by reading the source, so none can be missed), 1,093 strings in each of the 18 languages besides English. Only a unit or a code (`d`, `WNEW`, `TCC`) stays as it is, and every placeholder survives in every translation. The catalogues are data files (`app/translations/.json`) that `app/i18n_generated.py` merges: an entry written by hand wins over the file and an accepted override (FR-164) wins over both. The messages of the API stay English (they are part of the contract). The translations of the Indian languages, Arabic, Urdu, Georgian, Spanish, Italian and Danish are **proposals written by the maintainer's tooling and not yet checked by native speakers**; the review of FR-164 is how they become reviewed.
- **FR-168** The system shall store and show **everything the generated controller programs carry** (schema v4.1; 06-data-dictionary.md §2c.1, §2d.1, §2f): the 13 further subsystem programs of the Main Controller and the three Hub Controller programs (pitch error, pitch warning, rotor brake), the operating-mode mask (a bit string: 20 bits on the Main Controller, 6 on the Hub Controller), the task call frequency, the flags zero-load operation, availability calculation and the five Hub flags (each yes, no or not captured), and the numeric read and write levels of the 16 user-group slots. They are fields of the revision like any other: entered in the form (the sections Further Program References, Operation Parameters and User-Group Levels), shown on the detail page for the controller of the code, copied into every new revision and deprecation request, listed in the diff, the exports and the API (`user_groups`, `operating_mode_mask`, `operating_mode_bits`, `controller_type`), and never part of an audience export. A field that does not apply to the controller of the code (§2f) is refused with 422 and, in a program import, left out with a warning; a caller that does not send these fields (an older API client, a spreadsheet import) leaves them unchanged. The program import fills all of them from the file, except the five Hub flags, which stay not captured because the bit order of the Hub mask is not verified, and reads `loadless_spinning_permitted` from `ZeroLoadPrg_ArBo` (before v4.9 it wrongly took `ZeroLoadOperation_ArBo`). A named audience value brings the levels it held on every code that carried it (`default_read_level`, `default_write_level`, §9d.9) when the slot has none yet, and the CSV/XLSX import has columns for all of these fields (`user_group_levels` as `slot:read/write`, `operating_mode_mask` as a bit string); an empty or missing column leaves the stored value alone.
- **FR-169** The system shall show a **menu on the left** that is organised by the work of the user, collapsible and counted: the groups Catalogue, Workflow, Import and Migration, Insight, Languages and Administration (with the blocks People, Configuration and Operations), each link shown only to the roles that may use it (SEC-010), a + New Status Code button for the roles that create codes, a badge on the links and a total on the group for what waits for the user (the review queue, the approval queue, stalled requests, withdrawn codes; one's own open change requests at their link), groups that stay open or closed as the user left them (the group of the current page always shows) and, below 768 px, where the menu used to be hidden altogether, a Menu button that opens it. The menu is defined as data (`app/ui/nav.py`), so that what each role sees is tested.
- **FR-170** The system shall mark a language **beta** in the language menu while no native speaker has reviewed it (FR-164): the mark stays until a review of the language is completed and for as long as a string awaits approval (FR-176). English, the source, never gets it. An Administrator can switch the mark off in the settings (`languages.beta_marker`, on by default). The mark is a hint to the people who read the interface, not a barrier.
- **FR-171** The system shall print the words around an export in the language of the export: the title page (heading, labels and values such as not yet published and all), the note on the digest, the column headings of the default PDF, the page footer and the banner of the working set. English, or no language, keeps the English text; the data rows, the field names of CSV, JSON and XLSX, the file names and the manifest are never translated. The words belong to the interface inventory (FR-167) and exist in all languages.
- **FR-172** The system shall show **only the reviewed languages to Viewers and to people who are not signed in**.

While the mark of FR-170 is on, a language of which no review has been completed is hidden from the language menu, the login page and the account page of a Viewer and of a guest, cannot be chosen by them, and reads as English if a preference was stored earlier; a Viewer who was assigned a language to review (FR-164) still sees that one. Engineers, Reviewers, Chief Engineers, Administrators and the StCd Owner see every language, the unreviewed ones marked beta. Switching the mark off in the settings (`languages.beta_marker`) shows every language to everybody.

- **FR-173** The system shall export a Release as its **complete baseline**: every status code that is valid as of the Release - the latest published revision of each code in this Release or any earlier one - and not only the revisions the Release carries (roughly 900 codes of the Main Controller and 350 of the Hub Controller, so the manual of a Release is the whole catalogue). The delta stays available as Changes of this release only (`content=delta`). Every Release export, the offline lookup page and the package can be narrowed to a **turbine platform** (2XM, 3XM, 4XM) and to a **controller** (MC, HC); the baseline of a published Release never changes, and an unpublished Release shows the baseline it would have. The title page and the manifest name the scope. The Release page shows the counts (baseline, by controller, by platform) and, for working roles, how many codes are translated per language and how many have no confirmed access levels (information, never a block).
- **FR-174** The system shall build a **package** of a published Release in one step: a ZIP with the complete baseline in **every language** and for **every access level** (the eight audiences and the unfiltered file), in the chosen formats (PDF, XLSX, CSV, JSON), the release notes and a `MANIFEST.json` that lists every file with its SHA-256, row count, language and audience and the translation coverage per language. It can be narrowed to a platform and a controller. Everybody who may see the published Release may build and download it (FR-175); it is built in the background with a visible progress, recorded as an export with the SHA-256 of the ZIP, and audited (`PACKAGE_REQUEST`, `PACKAGE_EXPORT`). A code without an approved translation appears in the source language, marked in `content_language`.
- **FR-175** The system shall let **everybody who is signed in download what has been released** — the Release exports, the offline page, the release notes and the package of FR-174, Viewers included — and shall let **only the StCd Owner and the Administrator download what is not released yet**: the working set (FR-046) and the files of a Release that is not published. For anybody else an unpublished Release does not exist (404); the readiness check of FR-173 stays with the working roles.
- **FR-176** The system shall let **new and changed translations be approved separately**. What a native speaker approved stays approved: an OK answer, an accepted proposal (its final text), a rejected proposal or a noted question count as approval of that text, and the newest answer of a completed review wins. A string that is new, or whose text changed after it was approved (a shipped change, an accepted proposal that is edited, an override), is awaiting approval: the language keeps its beta mark for the working roles while any string awaits approval, a **follow-up review** lists only those strings (with All translations one click away) and is completed when each of them has a verdict, and a Viewer or a guest reads those strings in English until they are approved - the rest of the language stays in the language they chose. A language that no review has ever completed stays hidden from Viewers and guests (FR-172). The overview shows the number of strings awaiting approval per language.
- **FR-177** The system shall **check the confidence of the translations** whenever the text or the technical nomenclature changes, and show the result. Every string is checked mechanically (placeholders, missing or untranslated text, terms that differ from the nomenclature), a sample of 160 strings per language is read for meaning, register and consistency, and the result is a percentage per language: the share of strings a native speaker is expected to accept unchanged. A change of a string or of a term must be read one by one and recorded (`scripts/translation_precheck.py --check`, `--delta`); a test fails while a change is not covered. The overview (S42) and the catalogue document show the percentage on a colour from red (50 % or less) through yellow to green (above 95 %). The percentage is an estimate by a checker, never a review: it does not remove the beta mark and does not approve anything.
- **FR-178** The system shall **test every change of the vocabulary automatically and tell the StCd Owner which languages it touches**. A change that ships with a version (new or changed interface text, a changed term of the seeded nomenclature) is caught by the test of FR-177 in the build, and when the server starts - and with `flask translation-impact` after a deployment - the StCd Owner receives an in-app notice with the number of changed strings per language, whether the nomenclature changed, and the reviewed languages whose strings now await a follow-up review (FR-176). A change made in the application (a term added or changed, the translation of a term approved) is checked against every language at once - no approved translation, interface strings that do not use the term, reviewed languages affected - and the notice goes to the StCd Owner and to the reviewers assigned to an affected language. A notice is sent once per change and person, and a failing notice never undoes the change it reports.
- **FR-179** The system shall show **how status codes are connected**, as far as the generated programs say so. From every imported program three kinds of connection are read, as text and never by interpreting the controller

logic: a code is suppressed by another when its set condition holds the other's active flag behind a NOT (the other is the cause, this one a consequence that is silenced while the cause is present); a code is only set while another is active when the flag appears without a NOT; a code escalates to its repeat code (`RepeatStCdArUi`), which mostly lies in a pool outside the catalogue and is then marked external. The code page shows, for working roles, what the code depends on, what depends on it and the chain of causes above it; the page Connections shows the causes with the codes they suppress as a tree, the escalation targets with the codes that escalate to them, and what looks wrong (a cause that is not in the catalogue, a deprecated or archived cause or escalation target, a loop of suppressions). The connections are derived from the newest imported program of each code, so a re-import changes them; an imported Draft is shown under the number it has reserved. The API gives both views (`/status-codes/{id}/connections`, `/status-codes/connections`). Viewers do not see them (as the where-used panel).

- **FR-180** The system shall let the people who hold each step work on a **whole batch of codes at once without weakening the step**, so that the first migration of ~900 Main and ~350 Hub Controller codes can pass the governance. On the workbench of a change set: the readiness of every Draft (what still stops it: no platform, the domain sign-offs pending); **bulk edit** of the same value on many Drafts (status category, platforms, available group, the eight access levels), only by the author or an Administrator; **bulk submit** of the Drafts of the author; **bulk domain sign-off** by a holder of the domain; and the bulk decisions of the Reviewer, the Administrator and the Chief Engineer. Every code still goes through its own service call: its own segregation-of-duties check, its own audit entry and, for a decision, the password of the person (entered once for the sitting); a code the person wrote or has signed already is skipped and reported. A bulk **decision or sign-off on more than a handful of codes needs a sample inspection first**: the system draws a random sample of the batch, stratified by subgroup so that every part of the controller is looked at - 10 % of the batch but at least 8 codes by default, none for a batch of 5 or fewer (settings `bulk.sample_percent`, `bulk.sample_minimum`, `bulk.sample_above`) - and **the person who decides inspects it**, marking every sampled code OK or DEFECT (a defect needs a note). The decision is possible only with a sample that is complete, has no defect, was drawn by the same person for the same step and batch as it is now (the batch changed: a new sample), is not older than 24 hours and has not served a decision yet. One defect blocks the batch: the defective codes are sent back with the note as the reason (Reviewer, Administrator and Chief Engineer; a defective Draft is for its author) and a new sample is drawn from what is left. The sample, its seed, the person, every verdict and the decision are kept, the comment of every code decided with it names the sample, and the **dossier** of the change set (XLSX: the codes with their decisions, the samples with their verdicts, the audit entries) puts it in one document for an auditor. A decision of the Reviewer cannot be repeated by the same Reviewer on a code already signed.
- **FR-181** (v4.15; superseded in v4.17 by FR-192) The system let the user give each program of a `.st` import its own status category - a default, one per alarm type, or one per program in the preview. The program files carry no category and the uploader's choice was an assumption of the kind FR-189 forbids; the **type of the mapping row** replaces it, and the upload form, the preview and the API (`default_category`, `alarm_type_categories`, `categories`) no longer take a category.
- **FR-182** The system shall let an Engineer or Administrator **delete several of their own Drafts at once from the Status Codes list**: a tick box per own Draft, a select-all box in the header, a Delete selected Drafts button and, when the list is filtered to Drafts, Delete all my Drafts of this filter across every page. Each Draft goes through the single withdrawal of FR-026 (only its author, only while open, one audit entry each), so one that cannot be deleted is reported and never stops the others; a number that was already issued is kept (BR-001), one that was only reserved is given back (BR-026).
- **FR-183** The system shall offer a **dark colour theme** next to the light one, chosen by the user in the top bar (and on the login page) as Automatic (system), Light or Dark. Automatic follows the operating system's setting and changes with it; the choice is kept on the device the user works on (it is not part of the account) and is applied before the page is first drawn, so the wrong theme never flashes. Both themes meet the contrast rules of NFR-010 (4.5 : 1) for every text and fill pair, status and role colours stay recognisable, a printout and every export (PDF, XLSX, CSV, ZIP) keep the light colours, and the theme needs no inline script (SEC-025 stays unchanged). Every colour of the interface lives in a named token, so a new screen cannot bypass the theme (guarded by a test).
- **FR-184** The system shall **show its version**, so that a user, an administrator and a monitoring tool can tell which build is running. The number (for example `4.14.1`) is written once in `app/version.py` and shown at the foot of the menu, at the foot of the login page, and as `version` in the answers of `GET /api/v1/health` and `GET /api/v1/health/ready`; `pyproject.toml` is kept equal to it by a test. The version is raised with every release together with the git tag; the REST contract keeps its own version in `openapi.yaml`.
- **FR-185** The system shall produce, for every release, **four PDF documents with the version in the file name**: the full documentation (`CSCM-Tool_v_Full-Documentation.pdf`: every document of `docs/` with a contents list and bookmarks, the quick deployment guide, and the artefacts - the OpenAPI contract and the database schema -

as appendices), the software description (`..._Software-Description.pdf`), the user manual (`..._User-Manual.pdf`) and the test report (`..._Test-Report.pdf`). `python scripts/build_docs_pdf.py` makes them from the Markdown sources into `releases/v/` (committed with the release), without a browser or an office program, in the Noto fonts of the application; `python scripts/ci.py --pdf` does so after a green pipeline, so the test report is the one of that run. Every PDF carries a title page with the version, the date, the commit and a digest of its sources, and a footer with the version and the page number. The version is the one of FR-184. The catalogue of interface texts (`26-translations.md`, several MB) is left out unless asked for.

- **FR-186** The system shall produce, for every release, **two PowerPoint decks of its design** in English, built from real screenshots of the application with fictitious data: a short deck **for management** (what the tool gives, the three independent decisions, the migration, trust, languages and themes, releases, what was delivered) and a **walkthrough for the team** (design principles, login, the shell, every main screen, languages and right to left, light and dark, phone, the rules for new screens). `python scripts/build_showcase.py` builds both - `CSCM-Tool_v_Design-Showcase-Management.pptx` and `..._Design-Showcase-Team.pptx`, into `releases/v/` - from a throw-away copy of the application (its own database, never a real one), photographed with the headless Edge or Chrome of the machine; figures (tests, coverage, requirements, languages, what was delivered) are read from the repository, so the decks follow every release. A text that does not fit its box stops the build; `--previews` shows every slide as the build sees it; `--pdf` (always given by `ci.py --showcase`) writes a PDF copy of each deck beside it, so a release holds each deck as `.pptx` and as `.pdf`.
- **FR-187** The system shall carry its **identity** the same way everywhere. The full name is **Controller Status Code Management Tool** (short: CSCM Tool) and appears on the title page of every release document and deck; **on top of every title page** stand the Software Owner and Administrator, Bastian Ritter, with the e-mail address bastian.ritter@senvion.in; and a **logo** - a wind turbine whose hub is a status ring, between code brackets, above three status lights - heads the title pages and appears in the menu bar, on the login page and as the icon of the browser tab. The names are written once (`app/branding.py`); the Git account and the repository are not named anywhere (the address of the repository is given by the Software Owner or the Administrator).
- **FR-188** The system shall be handed over to a team repository by a **repeatable, audited procedure** (`scripts/handover.py`, docs/15 section 9). The first release of a team repository is a clean start: the tree of the release tag becomes one commit on `main` with the author the owner names, plus the annotated tag; the history of the development machine is not carried over. Before anything is exported the tree is audited and any finding stops the hand-over: the Git account or repository name, keys and tokens, databases, `.env` files, paths of a personal machine, e-mail addresses other than the owner's, files over 5 MB, a version that disagrees with the tag, missing release files, a test report that is not green. The push needs the address of the repository, `--push` and `--yes` together and is never forced.
- **FR-189** The `.st` import shall **assume no value**. A field of the Draft holds a value the program gives, a value the user chose, or - when the program gives a variable instead of a value - the **name of that variable** (a delay stays `StCd14000SetDelayGpTi`, never `0ms`); when the program has no timer at all the delay says `not in the program`. What the program does not give is **listed in the comment of the Draft** (its description, before the set condition so that a long condition cannot cut it off): the variables and parameters whose value is not in the program, the flags and program references that were not in the file and had to be entered as No or 0, and the up/down counter when the mapping has none. The preview warns with the name of every such parameter. **What the files do not carry is not assumed either**: the type, the title, the up/down counter and the access levels come from the mapping (FR-192), and a program whose row is missing or has no type is blocked. A code that already exists and whose row is gone keeps what it has. A delay accepts a time with a unit, the name of a parameter, or `not in the program`; the parameter has to be looked up before the code is released.
- **FR-190** The `.st` import shall give the eight audiences their **named access levels from the mapping** and check them against the numbers of the program. The program gives a read and a write number for each of its user-group slots (slots 1 to 8 are the audiences, docs/06 section 2d.1); the mapping row names the level (`Development = See only`). Where the named level has default numbers of its own (docs/06 section 9d.9) and the program gives other ones, the difference is warned about in the preview and listed in the comment of the Draft; the mapping names the level, and the numbers of the program are kept with the Draft all the same. An audience the mapping leaves empty stays empty and is listed in the comment (FR-189). (v4.16.1 guessed the name from the numbers when exactly one level fitted; v4.17 takes it from the mapping.)
- **FR-191** The system shall keep the **legacy -> new mapping in its database** (`status_code_mapping`), one row per new number: the family (Main or Hub Controller), the legacy number (none for a code without a predecessor), the new number, the **English title**, the functional group and logical node, the **type** (Error, Warning, Info), the up/down counter, the value of the parameter that sets a delay, the **named access level of each of the eight audiences**, cause, directive and a note, and a status - `PROPOSED`, `FROM_PROGRAM`, `APPROVED`, `ISSUED` or `REJECTED`. The mapping holds **only what the generated programs do not carry**; the trigger snapshot, the alarm, the repeats, the programs and the user-group numbers stay in the programs and are never copied. Main and Hub

Controller numbers are never in conflict (a new number belongs to one row), the 2XM and 3XM platforms share one legacy numbering, and the Hub rows lie in the Hub range. Its initial content is the workbook `seed_data/legacy_mapping.xlsx`, loaded by `flask mapping-seed` (the installer and `first_start.py` run it; an Administrator can also load it from the empty Mapping page, and only into an empty table); a row takes the status written in the workbook - every row of the draft workbook is `PROPOSED`, so nothing is approved by loading, and the audit entry `MAPPING_SEEDED` counts the rows per status.

- **FR-192** The `.st` import shall take from the mapping, and from nowhere else: the **title** (the English description of the row; the program name is German and goes into the comment), the **status category** (the type of the row), the **up/down counter**, the **named access levels**, the **remedy text** (cause and directive) and - when the program sets a delay with a parameter and gives no time - the **value of the parameter** (`*5s`; a time in the program wins). A Main Controller program is found by its number, a Hub Controller program by its legacy number. A program whose number is not in the mapping, whose row is rejected, belongs to the other family or has no type is **blocked**, with the message that says which row to request; a row that is only `PROPOSED` or `FROM_PROGRAM` is used with a warning that the number is not approved yet. When a number is imported the row becomes `ISSUED` and its numbers can no longer change. The upload form and the preview show no category control.
- **FR-193** The import shall **check every program against its mapping row**: the legacy number in the header of a Main Controller program must be the one of the row (a difference blocks the program, a missing one on either side is warned about); the logical node of the row is compared with the group of the number range; and the English title is compared with the German program name through the **glossary** (`app/importers/title_glossary.csv`, from the glossary sheet of the workbook) - for each technical term in the German name one of its English renderings has to occur in the title, otherwise the preview warns. The glossary check never blocks and never changes a title.
- **FR-194** The mapping shall change **only through a request** (`mapping_change`) that the **StCd Owner, the Chief Engineer and one Administrator all approve**: three different people, none of them the person who made the request, each deciding once with the password confirmed again (I-17). One rejection - which needs a reason - closes the request; the third approval applies it, in one transaction, and the rows become `APPROVED` (a row that is `ISSUED` stays so). Nothing changes before. A request can be withdrawn by the person who made it while it is open. Delegation does not apply: the three approvals are given by the holders of the three roles. A row that changed after the request was made makes the request unusable (it is withdrawn and made again).
- **FR-195** A request shall be made **for one row, for the ticked rows, or for a whole workbook**, with a reason that the three approvers read first. The actions are ADD (a new row, starting at a new number), CHANGE (fields of a row), CONFIRM (ask for the approval of the number as it is - the way a loaded `PROPOSED` row becomes `APPROVED`) and REJECT (the number is never used). Every row is checked before the request exists, and one wrong row refuses the whole request with all the messages: the family, whole numbers, the new number in 1...99999 and in a numbering band of the family's controller, a title of at most 120 characters, a type and an up/down counter for a row that is to be approved, a delay that is a time or the name of a parameter, access levels that are active named levels of their audience, a new number and a legacy number that stay unique, and no change of the numbers of a row that is `ISSUED`.
- **FR-196** The system shall show the mapping on a page of its own (Mapping, filter by controller, status and text, 100 rows a page, the same for everyone who may read it) and the requests on another (Mapping changes): each request with its reason, its rows (the fields that change, before and after), and the **three places for the three approvals** with name, date and comment. The person who may decide sees the decision form, the others see why not. The menu counts the requests that wait for the user's decision. The list shows 100 rows a page and one click on Show all puts every row of the filter on one page, so that select all covers them. The decision of a request is **below its rows**, after the status and the changes have been read. A database that has not been upgraded gets a page that says which two commands to run (`flask db upgrade`, `flask mapping-seed`), not an error. The roles that work with the library and the StCd Owner read the mapping and make requests; Viewers see neither.
- **FR-197** The mapping shall be **exchanged as a workbook**: the export writes the sheets MC and HC with the columns the upload reads (a cell that would start a formula is written as text), and the upload compares a workbook with the mapping and opens **one request** with every new and every differing row (unchanged proposed rows only when asked). Rows the workbook does not mention stay as they are. A wrong row names its sheet and row and nothing is requested; a file that is not a workbook, or is larger than 5 MB or 3,000 rows, is refused.
- **FR-198** Every step of the mapping shall be **audited** (`MAPPING_SEEDED`, `MAPPING_CHANGE_REQUESTED`, `MAPPING_APPROVAL`, `MAPPING_REJECTION`, `MAPPING_CHANGE_REJECTED`, `MAPPING_CHANGE_APPLIED`, `MAPPING_CHANGE_WITHDRAWN`, `MAPPING_ROW_ISSUED`) and reachable through the API (`/mapping`, `/mapping-changes`, docs/09 section on the mapping).

2. Non-Functional Requirements

2.1 Performance — NFR-001

Targets (13-test-strategy.md §5): 500ms p95 list/search at 20,000 revisions, 200ms p95 single-record read, 5s CSV/JSON export of 5,000 records, 15s PDF export of 5,000 records — now additionally validated for identifier-allocation contention within a single popular Functional System Group under concurrent submission (TC-025).

2.2 Availability — NFR-002

99.5% monthly availability during business hours, <= 4 hours/month scheduled maintenance.

2.3 Security — NFR-003

See 11-security-architecture.md. Summary: TLS 1.2+ only, on the internal network exclusively; no plaintext password storage/logging; CSRF protection on all state-changing endpoints; input validation and output encoding against injection; **no dependency on any public-cloud service, including for authentication, storage, or logging (NFR-003a, new in v2)**.

2.4 Scalability — NFR-004

The architecture supports >= 50 concurrent named users and 20,000 revisions without breaching NFR-001; repository pattern preserves a PostgreSQL migration path.

2.5 Maintainability — NFR-005

Layered structure per 15-development-standards.md, business logic in the service layer, >= 80% line coverage on service-layer and validation code.

2.6 Auditability — NFR-006

100% of state-changing actions (now including three decision points and sandbox create/delete) produce a corresponding audit log entry; retention >= 7 years, with the explicit sandbox-delete exception documented in 12-audit-compliance.md §3.

2.7 Usability — NFR-007

Core workflows completable by a trained user without external documentation; desktop-primary responsive layout, now `max-width: 1280px` per the OneTool design system (10-ui-ux-specification.md §0). Two requirements added 2026-09-17, direct responses to documented legacy-tool complaints: **NFR-007a** no form in the application may discard already-entered data as a result of a validation failure, network retry, or session event — a failed submission returns field-level errors only, never a reset form (10-ui-ux-specification.md §4 S5, §9). **NFR-007b** every lookup-backed field must surface its selected value's description as inline help text, so a user is never left choosing between unexplained options (10-ui-ux-specification.md §5).

2.8 Deployment Independence — NFR-008 (new in v2)

The system shall be deployable and fully operable on an isolated internal network with no outbound internet connectivity whatsoever — no feature in MVP scope may require reaching an external service to function. This is stricter than "no cloud dependency" alone: it is a testable claim (14-deployment-architecture.md §1) that the application continues to operate correctly with egress blocked entirely.

2.9 Multi-Process Readiness — NFR-009 (new in v4, I-20)

Several worker processes may serve one SQLite file. Everything that must be consistent across them lives in the database, not in process memory: the login rate limit (10 attempts per IP per 5 minutes, 5 per username per 15 minutes) is counted in `login_attempt` inside the login request's write transaction, so it is shared between processes and survives a restart. Concurrent writers queue on SQLite's write lock (`BEGIN IMMEDIATE`, 15 s busy timeout) instead of failing; the lock wait is measured (FR-153). Proven by running two real server processes against one file (13-test-strategy.md §8a).

2.10 Accessibility — NFR-010 (new in v4, I-22)

The interface meets WCAG 2.1 level AA: text and badge colours reach a contrast ratio of at least 4.5 : 1, every page has one `main` landmark, a skip link, a language attribute and a single heading hierarchy, every form control has a label or accessible name, table headers carry `scope`, state is never conveyed by colour alone, and the style sheet honours `prefers-reduced-motion`. Checked on every page in both languages by an automated test (13-test-strategy.md §8a); a manual screen-reader pass remains part of the pre-go-live checklist.

2.11 Observability — NFR-011 (new in v4, I-21)

Every request is logged as one JSON line with a request id (stderr, optionally a rotating file); `GET /health` (liveness), `GET /health/ready` (readiness) and `GET /metrics` (internal network only) let the on-premises monitoring see rates, latencies, write-lock waits, queue sizes and the state of the last verified backup. No telemetry leaves the network (NFR-008).

2.12 Recoverability — NFR-012 (new in v4, I-09)

A verified backup is taken at least nightly (RPO <= 24 h, <= 1 h with WAL shipping), the shipped schedule keeps 35 generations (the command's own default is 14), and the restore procedure is rehearsed quarterly (14-deployment-architecture.md §6–§7). The readiness probe reports `degraded` when no verified backup is younger than 36 hours (configurable).

2.13 Packaging & Continuous Verification — NFR-013 (new in v4, I-33)

The application installs from a checkout with one script per platform (Windows Server scheduled tasks; Linux systemd units), upgrades idempotently, never overwrites configuration, data or backups, and takes a verified backup before migrating an existing database. A CI definition for the on-premises Git server runs the whole verification (tests in a rotating order, coverage, black, ruff, mypy, bandit, pip-audit) on every push; the install scripts are syntax-checked and dry-run in the test suite.

2.14 Database Portability — NFR-014 (new in v4, I-34)

SQLite-specific constructs (triggers, `BEGIN IMMEDIATE`, `GLOB`, FTS5, the online-backup API, `sqlite3` calls) are inventoried by `scripts/pg_readiness.py`, each with its PostgreSQL porting note, and counted against a committed baseline: CI fails when a change adds SQLite-specific code, and when one is removed until the baseline is lowered. Engine set-up is already guarded by dialect. The application has **not** been run on PostgreSQL; that is the Phase 3 port this requirement keeps cheap (07-database-design.md §12).

3. Business Rules

- **BR-001** A Status Code's identity is immutable once its identifier is allocated; it cannot be changed by editing, only superseded via deprecation.
- **BR-002** A Status Code has at most one revision flagged `is_current = true` at a time.
- **BR-003** Only one open (Draft, Review, or PendingApproval) revision may exist per Status Code at a time.
- **BR-004** A revision cannot transition to Released except via inclusion in a published Release, cannot reach Approved except via explicit Chief Engineer approval, and cannot reach PendingApproval except via both the Reviewer's and the Administrator's independent sign-off.
- **BR-005** No individual may both author and decide on the same Change Request at any of the three decision points (Reviewer sign-off, Administrator sign-off, Chief Engineer approval); the Chief Engineer additionally may not be whoever provided either of the two prior sign-offs on that CR.
- **BR-006** Deprecating a Status Code requires a reason; archiving requires the configured minimum retention period (default 180 days) to have elapsed since deprecation.
- **BR-007** A published Release's member revisions are immutable; correcting a released status code requires a new Draft revision, a new full three-decision-point cycle, and a new Release.
- **BR-008** Only Administrators may create, publish, or delete (pre-publish only) Releases.
- **BR-009** Viewers have read-only access to Approved, Released, Deprecated, and Archived data; Draft, Review, and PendingApproval-state data is visible only to Engineer, Reviewer, Administrator, and Chief Engineer roles.
- **BR-010** A sandbox-flagged Status Code may never be included in a Release, and is the only category of record that may be permanently deleted, and then only by an Administrator.
- **BR-011** Submit-for-Review is blocked while any Engineering-Domain-owned field lacks coverage from either the

originating Engineer's own domain tags or a recorded `DomainSignoff`.

- **BR-012** Each of the eight audience-access fields (corrected 2026-09-17; replaces the single `access_rights_id`) is not authoritative until confirmed by the Administrator's Review sign-off, regardless of what value the originating Engineer proposed.
- **BR-013** A Status Code must have exactly one Functional Subgroup at all times; Functional Subgroup is never optional (corrected 2026-09-17).
- **BR-014** A new Functional Subgroup's sub-range must not overlap any existing subgroup within the same Functional System Group.
- **BR-015** Every row imported via the library import feature must pass through the full Change Request lifecycle before Release; import creates Drafts only, never Approved or Released records directly. This holds for every import mode, including update mode (FR-048k) and Hub-to-Main carry-over (FR-147).
- **BR-016** (v4) An audience sees a code only if the Administrator-confirmed value of that audience's access field is marked visible; a missing value hides the code, and `Covert` is not visible until the business defines it otherwise (decision D-02). A Draft's proposed values never count (BR-012).
- **BR-017** (v4) A delegate acts as themselves. Delegation grants the decision actions of the delegated role only; the segregation-of-duties rules (BR-005) apply to the delegate personally, the two Review sign-offs must still come from two different people, and the audit entry names the delegate and the person they cover for.
- **BR-018** (v4) A translation is approved by one Reviewer or Administrator who is not the translator; it is not subject to the three-decision cycle of the source text, and editing an approved translation returns it to DRAFT.
- **BR-019** (v4) A save of a Draft that is based on an outdated `row_version` is refused and changes nothing (FR-090); a Released revision is never edited in place (BR-007).
- **BR-020** (v4) Audit entries are never deleted: archiving writes a copy and an `AUDIT_ARCHIVE` entry (FR-151), and any gap, edit or reordering in the hash chain is a reportable integrity failure.
- **BR-021** (v4) A Release may target one or more controller software versions, set by an Administrator before publishing. When it has targets, its candidate list holds only Approved revisions whose version range includes at least one target (an open end includes everything); with no targets every Approved revision is a candidate. The target versions of a published Release are immutable like its items (BR-007), and a revision's range is otherwise informational — it never blocks a lifecycle transition.
- **BR-022** (v4) A sensitive action — Reviewer or Administrator sign-off, Chief Engineer decision, publish, full-database export — needs a fresh password confirmation from the actor, unless the deployment deliberately turns step-up off (`STEP_UP_REQUIRED`, a documented deviation).
- **BR-023** (v4.2) A number that an imported program already carries is reserved for exactly one status code and is issued to that code unchanged at Submit for Review: no other code can be allocated or overridden to it, the code itself cannot be given another number, and the mapping from its legacy number to the new one is never inverted (FR-048q, FR-048u).
- **BR-024** (v4.3) Only an Administrator creates users and gives or takes away the roles Administrator, StCd Owner and Chief Engineer. An StCd Owner may change a user's role only when the user holds Viewer, Engineer or Reviewer now and is to hold one of those three afterwards; the service layer refuses everything else with 403, and every change is audited as `ROLE_CHANGE` with the actor. The last active Administrator can never lose the role (FR-012).
- **BR-025** (v4.6, changed in v4.9) An Administrator can delete every account, and the audit log is never modified. An account that nothing else points at - no change request, review, release, audit entry or any other record - is **removed**, with its own rows (domain tags, password history, saved searches, notices). An account with history is **erased**: its user name becomes `deleted-`, its address `deleted-@deleted.invalid`, its name Deleted user, its password is destroyed, its own rows and reviewer assignments go, `user.deleted_at` is set, and it leaves every list and every call (404). The row stays, so every record it created still has an author and every audit entry still resolves its actor; the old user name and address are free again. The deletion is audited as `USER_DELETE` with the snapshot of what the account held and the mode (`removed` or `erased`). Deactivating, which keeps the name, remains the gentler choice.
- **BR-026** (v4.6) Withdrawing a change request never reuses an issued number: a code that already received its identifier keeps it. A code that never did gives back the number an import reserved for it, its legacy number and its PrgIdx_GUi, because nothing was issued and the same program has to be importable again. A code whose last revision was withdrawn is an empty shell (`status_code.withdrawn_at`) that keeps its number, legacy number and PrgIdx_GUi and so blocks the same program or row from being imported again (v4.9). An Administrator can **delete such a code** on the page Withdrawn Codes (S48), with the password again. The deletion is done in place (`status_code.deleted_at`): the row stays, so no id is ever reused and every audit entry keeps pointing at one code, but the code gives up its identifier, reserved number, legacy number and PrgIdx_GUi, and the program or row can be imported again. It is audited as `WITHDRAWN_CODE_DELETE` with what

the code held. The database trigger refuses every delete of a non-sandbox code, so a code that has ever had a revision (and so may have been Released) is never touched. A number freed this way can be allocated again if it was the highest of its range; restoring the code from its program file instead clears the withdrawal mark.

- **BR-027** (v4.17) A number of the mapping counts as approved only after the **StCd Owner, the Chief Engineer and an Administrator** have each approved it. They are three different people and none of them made the request; a delegate cannot give the approval of a role. This is the one decision the StCd Owner takes part in - everywhere else the owner of the library has no say in approvals (BR-024) - because the owner of the library is the one who must stand behind a number before it is used.

4. Constraints

- **C-001** MVP persistence is SQLite (single-file, file-based); this bounds MVP concurrent write throughput and requires the repository-pattern abstraction (NFR-004) to enable a later migration. SQLite is not intended as a permanent production datastore; see 16-mvp-roadmap.md for the Phase 3 migration trigger.
- **C-002** MVP authentication is session-based, internal user store only; on-premises AD/ADFS integration is out of scope for MVP (FR-005).
- **C-003** Backend framework is fixed to Python/Flask; frontend is server-rendered HTML/CSS/vanilla JavaScript, no external CDN dependency (11-security-architecture.md §12).
- **C-004** The application must run in the target on-premises deployment environment described in 14-deployment-architecture.md.
- **C-004a** No public cloud (SaaS/PaaS/IaaS) component may be introduced at any phase without an explicit, separate decision to relax C-004 (new in v2 — see 00-INDEX.md "Deployment Principle").
- **C-005** (superseded 2026-09-17 — retained for history) The original position was that no bulk import feature would exist in MVP. The business has since explicitly requested CSV/XLSX library import (FR-048a–d) to support the legacy migration; the constraint now is that import must never bypass governance (FR-048c), not that import must not exist.

5. Assumptions

- **A-001** The initial user population is small (tens, not thousands, of named users across all six roles), consistent with NFR-004's 50-concurrent-user target.
- **A-002** Engineers, Reviewers, Administrators, Chief Engineers, and Viewers are employees or contractors with existing corporate accounts; user provisioning is manual (Administrator-driven) until on-prem AD/ADFS integration lands.
- **A-003** Status code catalogues across all nine Functional System Groups remain in the low tens of thousands of revisions over the system's MVP lifetime, consistent with NFR-001's sizing.
- **A-004** Technical Publications and any future SCADA/documentation-generation consumers will integrate against the versioned REST API or scheduled exports, over the internal network only; no direct database access will be granted to external systems.
- **A-005** The organization has an existing password/credential policy that CSCM Tool's password rules (11-security-architecture.md §4) can align with.
- **A-006** The real Functional Subgroup taxonomy, supplied 2026-09-15 and seeded directly (06-data-dictionary.md §9a), is treated as authoritative; any further changes Controls Engineering makes before go-live remain a data-only update. The `10000?10999`, `14000?14999`, `15000?15999`, and `16000?16999` ranges are confirmed free/reserved capacity, not an outstanding classification gap.
- **A-008** (new 2026-10-22) The Hub Controller (HC) may be removed from the hardware setup in future. If so, the status codes of the functions it carried move to the Main Controller (MC) side and are **defined there anew** — they are new MC codes in MC groups/subgroups, not renumbered HC codes. The design must therefore keep Hub groups as ordinary data (deprecatable, no code paths that depend on the 11000–13999 range) and make it easy to create new MC groups/subgroups; see I-36 in 21-improvement-ideas.md.
- **A-007** Chief Engineer is a distinct, sufficiently-staffed role in the organization such that the added approval step does not become a single-person bottleneck; this should be validated against actual staffing before go-live (16-mvp-roadmap.md §6 risk).
- **A-009** (v4) Technical Publications can use an internal SMTP relay for reminders if the business wants mail; until that is confirmed, in-app notices are the whole mechanism (decision D-04). Mail is never required for any workflow step.
- **A-010** (v4) The meaning of `Covert` in the audience vocabularies is still unconfirmed; the system treats it as not visible (D-02) so that a wrong guess can only hide a code, never disclose one.

6. Decisions (v4)

These resolve the "decisions needed from the business" of `21-improvement-ideas.md` with the safest workable default. Each is data or configuration where possible, so it can be revisited without a release.

ID	Decision	Rationale	Where it lives
D-01	The software version is a range per revision (applies_from/applies_to, inclusive, either end open) plus the set of versions a Release targets.	The legacy tool filters by version; a single value cannot express "from 4.1 on".	lookup_controller_software_version, revision columns, release_target_version (FR-106, BR-021)
D-02	Covert means not visible. Visibility is the flag audience_visible on each access vocabulary value; no value for an audience also means hidden.	Disclosure-safe default until the business defines Covert.	seed data + Lookup Administration (FR-102, BR-016, A-010)
D-03	Languages: English is the source, German is the first translation. Status per language is MISSING / DRAFT / APPROVED; the approver is not the translator.	Cheapest model that serves multilingual manuals without a second approval cycle.	lookup_language, revision_translation (FR-110, BR-018)
D-04	In-app notices are the mechanism; SMTP is optional, through the organisation's own relay only.	No external mail provider (NFR-008); mail may be absent.	notification, SMTP_* (FR-121, A-009)
D-05	Manifests are signed with an HMAC-SHA-256 key (MANIFEST_SIGNING_KEY) and the audit chain is keyed with AUDIT_HMAC_KEY. Both optional; unsigned manifests still prove file integrity.	Symmetric keys need no PKI; asymmetric signatures are listed as a further improvement.	config (FR-103, FR-151)
D-06	controller_type MAIN | HUB is a column of the Functional System Group (not of the subgroup).	Hub groups stay ordinary data (A-008); the range test range_start >= 11000 seeded it.	lookup_functional_system_group (FR-147)
D-07	Step-up means re-entering the actor's own password per sensitive action; one prompt covers a whole change-set decision.	Smallest change that makes approvals deliberate; no second factor infrastructure on-premises.	AuthService.step_up, STEP_UP_REQUIRED (FR-126, SEC-026, BR-022)
D-08	Delegation covers decision actions only (Reviewer / Administrator sign-off, Chief Engineer decision and their queues), never user administration; segregation of duties applies to the delegate.	Prevents a delegation from becoming a privilege escalation.	delegation (FR-125, BR-017)

04. Domain Model Specification — CSCM Tool

1. Entity Overview

Entity	Description	Ownership
User	A named system account, tagged with a Role and (for Engineers) one or more Engineering Domains	Administration
Role	Fixed enumeration: Administrator, StCd Owner, Engineer, Reviewer, Chief Engineer, Viewer	Administration
EngineeringDomain	Lookup of engineering specialties (Controls, Electrical, Mechanical, Grid, Safety, ...)	Administration
FunctionalSystemGroup	Lookup of the nine numbering bands (05-governance-handbook.md §2)	Governance
FunctionalSubgroup	Lookup of sub-bands within a Functional System Group (real taxonomy seeded; mandatory on every StatusCode, never optional)	Governance
TurbinePlatform	Lookup of platforms a code applies to (multi-select)	Governance
StatusCode	Immutable identity for a governed status code (StCd identifier, allocated at Draft->Review)	Engineering
StatusCodeRevision	A specific versioned set of attribute values for a StatusCode, with its own lifecycle state	Engineering / Governance
StatusCodeRevisionPlatform	Join: which Turbine Platform(s) a revision applies to	Engineering
PlatformVariantLink	Cross-reference between independent Status Codes that represent the same condition on different platforms	Engineering
ChangeRequest	The workflow object that carries a StatusCodeRevision through Draft->Review->PendingApproval->Approved	Engineering / Governance
ReviewComment	A comment/decision attached to a ChangeRequest, tagged with which of the three decision roles it represents	Governance
DomainSignoff	Record of a second Engineer completing their Engineering Domain's fields on a Change Request	Engineering
Release	A named, versioned, immutable bundle of Released revisions	Governance
ReleaseItem	Join entity: which revisions belong to which Release	Governance
AvailableGroup	Soft-validated lookup of the numeric availability codes observed in production; labels TBD	Governance
Eight *Access lookups	Independent audience-visibility vocabularies — Development, Sales, TCC, Service, Turbine Operator Package, Grid Operator, Service Partner, Customer (06-data-dictionary.md §2d, §9d)	Governance
ImportJob	Record of a CSV/XLSX library import; produces Draft Change Requests, never bypasses governance	System / Engineering
AuditLogEntry	Immutable record of a state-changing action	System
ExportJob	Record of a generated export (catalogue, release notes, offline bundle or full database): formats CSV/JSON/XLSX/PDF/HTML/MD/ZIP, with SHA-256, size, row count, profile, audience, language, software version and manifest (v4)	System
ControllerSoftwareVersion (v4)	Ordered lookup of controller software versions; a revision declares a from/to range, a Release targets versions	Governance
Language / RevisionTranslation (v4)	Content languages (English source, German) and the per-revision translation with its own status	Governance / Technical Publications
ExportProfile (v4)	Administrator-managed column set with header labels, default audience and language	Governance
ChangeSet (v4)	Group of change requests that ship with one controller software release	Engineering / Governance
Delegation (v4)	Out-of-office cover for a decision role for a date range	Governance
Notification (v4)	In-app notice (optionally mailed) for the holder of the next decision	System
CrAttachment (v4)	Evidence file on a change request	Engineering
FieldComment, ReviewChecklistItem, CrChecklistConfirmation (v4)	Field-level review comments; configurable review checklist and the confirmations stored with a sign-off	Governance
SavedSearch (v4)	A user's named filter-and-column view	User
ApiToken, Webhook, WebhookDelivery (v4)	Machine access: scoped read-only tokens; signed release webhooks and their delivery log	Administration

Entity	Description	Ownership
OccurrenceImport, CodeOccurrence (v4)	Fleet-feedback occurrence counts per code and period	Administration / Chief Engineer
HubMcMapping (v4)	Link from a Hub Controller code to its Main Controller replacement	Engineering
AppSetting (v4)	Administrator-managed settings (SLA thresholds, auto-assignment, capacity alert)	Administration
LoginAttempt (v4)	Rate-limit counters shared by all worker processes	System

2. Entity Definitions

2.1 User

Attributes: `id`, `username`, `email`, `password_hash`, `full_name`, `role_id`, `is_active`, `created_at`, `last_login_at`, `failed_login_count`, `locked_until`. Relationships: one `User` has one `Role`. An Engineer-role `User` has zero or more `EngineeringDomain` tags (M:N via `user_engineering_domain`). One `User` authors many `StatusCodeRevision` (as `created_by`), records many `ReviewComment` decisions, owns many `StatusCode` (as `owner`).

2.2 Role

Fixed lookup: `ADMINISTRATOR`, `ENGINEER`, `REVIEWER`, `CHIEF_ENGINEER`, `VIEWER`. Not user-extensible in MVP. See 00-INDEX.md "Roles" and 05-governance-handbook.md §7 for organizational mapping and responsibilities.

2.3 EngineeringDomain

Lookup: `CONTROLS`, `ELECTRICAL`, `MECHANICAL`, `GRID`, `SAFETY` (seed set, Administrator-extensible). Used to route Cross-Domain Sign-Off (05-governance-handbook.md §6a) and to tag which fields on a `StatusCodeRevision` a given domain is responsible for.

2.4 FunctionalSystemGroup

Attributes: `id`, `code` (`WCNV`, `WGEN`, ...), `label`, `range_start`, `range_end`, `is_active`. Twelve seed rows per 00-INDEX.md's numbering table (nine Main Controller groups plus three Hub Controller groups added 2026-09-17). This lookup **replaces** the former `controller_type` field — it is now both a classification and the source of numeric allocation for `StatusCode.status_code_identifier`. `code` **is unique together with** `range_start`**, not on its own**: `WTUR` and `WROT` are each intentionally reused across a Main Controller group and a numerically distinct Hub Controller group (same thematic prefix, confirmed by the business), so any lookup by prefix alone is ambiguous — resolve by `(code, range_start)` or `id`.

2.5 FunctionalSubgroup

Attributes: `id`, `functional_system_group_id` (FK), `code`, `label`, `sub_range_start`, `sub_range_end`, `is_active`. Seeded with the real taxonomy supplied 2026-09-15 (06-data-dictionary.md §9a): each subgroup owns a 100-number sub-band aligned to the hundreds digit of the identifier (e.g. `010xx` = 01000–01099). Not every hundred-block in a group is currently assigned — unassigned blocks are deliberate reserved capacity for future subgroups, addable as a pure data change. Selecting a Functional Subgroup at creation time narrows numeric allocation to its sub-band; selecting only the Functional System Group (no subgroup) allocates from any number in the group's range not already claimed by a subgroup-scoped allocation, including reserved blocks.

2.6 TurbinePlatform

Attributes: `id`, `code` (`2XM`, `3XM`, `4XM`, ...), `label`, `is_active`. Administrator-managed, extensible (the business explicitly stated "minimum" 2XM/3XM/4XM, more will follow).

2.7 StatusCode

Represents the durable identity of a governed code. Attributes: `id`, `status_code_identifier` (`StCd-XXXXX`, nullable until first Review submission — see §4 lifecycle note), `functional_system_group_id` (FK), `functional_subgroup_id` (FK, **mandatory — corrected 2026-09-17, was nullable**), `is_sandbox` (bool, default false), `platform_variant_of_status_code_id` (nullable, see `PlatformVariantLink` §2.10), `legacy_reference_number` (nullable text, populated on library import — §2.18), `owner_id`, `created_at`. **Cardinality:** one `StatusCode` has many `StatusCodeRevision` (1:N); exactly one revision at a time is flagged `is_current = true` (BR-002). **Uniqueness:**

`status_code_identifier` is unique once assigned. **Sandbox exception:** if `is_sandbox = true`, the identifier is drawn from an isolated `StCd-T#####` counter (05-governance-handbook.md §12) and the record is eligible for hard delete by an Administrator — the only entity in this schema with that property.

2.8 StatusCodeRevision

The versioned, governed record. Full field-by-field detail lives in 06-data-dictionary.md §2–§2e (grounded in a real 897-row export of the live library, 2026-09-17) — this section summarizes the categories rather than repeating every field, to avoid the two documents drifting apart:

- Identity/linkage: `id`, `status_code_id` (FK), `revision_number`, `is_current`.
- Descriptive/classification: `title`, `description`, `software_version` (nullable free text, kept for legacy rows; since v4 the governed form is the range `applies_from_version_id`/`applies_to_version_id`, decision D-01), `status_category` (`ERROR`/`WARNING`/`INFO`), `available_group` (numeric, soft-validated — §2.15), `alarm_flag` (boolean, not a behaviour enum), `set_delay`/`reset_delay` (free text with unit).
- Behavioural flags: `up_down_counter`, `trigger_snapshot`, `loadless_spinning_permitted`.
- Repeated-error escalation (all optional together): `number_of_times_repeated`, `repeated_over_the_course`, `status_code_for_repeated_error` (opaque numeric reference to a separate, unrelated code pool — confirmed 2026-09-17, **not** a foreign key to `StatusCode`).
- Program references (numeric, externally defined by controller firmware, soft-validated — §2.15): `yaw_program`, `manual_reset_program`, `auto_reset_program`, `converter_reactive_power_program`, `brake_program`, `umrichterprogramm_gsc`, `umrichterprogramm_msc`, `yaw_bearing_lubrication_program`.
- Audience/access (governance-restricted, eight independent fields — see §2.9): `development_access`, `sales_access`, `tcc_access`, `service_access`, `turbine_operator_package_access`, `grid_operator_access`, `service_partner_access`, `customer_access`.
- v4 content and control fields: `remedy_text`, `affected_component`, `related_status_codes`, `standard_references` (all optional, "Any" owner), `planned_effective_date`, `planned_deprecation_date`, and the concurrency pair `row_version` (increases on every save) / `updated_at` (FR-090).
- Technical addressing (derived, read-only, not stored — §2e of 06-data-dictionary.md): `logical_node`, `node_value`.
- Governance/workflow: `owner_id`, `lifecycle_status` (7-state enum, 05-governance-handbook.md §4), `created_by`, `created_at`, `reviewer_signoff_by/_at`, `admin_signoff_by/_at`, `chief_engineer_approved_by/_date`, `effective_date`, `deprecated_reason`, `superseded_by_status_code_id`.
- **Cardinality:** many `StatusCodeRevision` : one `StatusCode` (N:1). One `StatusCodeRevision` : one `ChangeRequest` (1:1). One `StatusCodeRevision` : many `TurbinePlatform` (M:N via `status_code_revision_platform`). One `StatusCodeRevision` : zero-or-one `ReleaseItem`.
- **No subtype hierarchy**: Functional System Group on the parent `StatusCode` remains the sole classification axis; no per-type subtables.

2.9 Restricted Fields

The eight audience/access fields (§2.8) may each be set by the authoring Engineer as a **proposed** value but are not authoritative until the Administrator's mandatory Review sign-off (§2.12) confirms or overrides them — enforced at the service layer, not merely a UI convention (11-security-architecture.md §3). This replaces the single `access_rights_id` restricted field from the prior revision with eight independently-restricted fields, following the same propose-by-Engineer/confirm-by-Administrator-at-sign-off pattern.

2.10 PlatformVariantLink

Represented as the nullable self-referential `platform_variant_of_status_code_id` on `StatusCode` (not a separate join table, since it is a simple pointer, not M:N): when a code is renamed or functionally regenerated for a different Turbine Platform rather than simply extended to cover it, the new `StatusCode` is created as an **independent record** with its own `StCd` identifier, optionally pointing back to the originating code via this field, purely for navigability — it carries no lifecycle or approval semantics of its own (05-governance-handbook.md §2a). Since v4 the link is created by Create platform variant (FR-094): a Draft copy of the latest decided revision for another platform, with the family shown on both detail pages.

2.11 ChangeRequest

Attributes: `id`, `status_code_revision_id` (FK, 1:1), `cr_type` (`NEW` | `REVISION` | `DEPRECATION`), `requested_by`, `justification`, `state` (mirrors the revision's lifecycle for Draft/Review/PendingApproval/Approved), `submitted_at`, `chief_engineer_decided_at`, `chief_engineer_decided_by`, `assigned_reviewer_id` (nullable), `change_set_id` (nullable, v4), `import_job_id` (nullable). **Cardinality:** 1:1 with `StatusCodeRevision`; 1:N with `ReviewComment`; 1:N with `DomainSignoff`.

2.12 ReviewComment

Attributes: `id`, `change_request_id`, `author_id`, `comment_text`, `decision` (`REVIEWER_APPROVE` | `REVIEWER_REJECT` | `ADMIN_APPROVE` | `ADMIN_REJECT` | `CHIEF_ENGINEER_APPROVE` | `CHIEF_ENGINEER_REJECT` | `NULL` for a plain comment), `created_at`. The `decision` enum's role-tagging is what lets the system verify both required Review sign-offs came from the correct, distinct roles (05-governance-handbook.md §7) rather than the same role acting twice.

2.13 DomainSignoff

Attributes: `id`, `change_request_id` (FK), `engineering_domain_id` (FK), `signed_off_by` (FK -> User, must hold that domain), `signed_off_at`. A Change Request may only be submitted for Review once every domain tagged as required on its revision (derived from which domain-owned fields are still at their unset/default value — 06-data-dictionary.md §2a) has a corresponding `DomainSignoff` row (05-governance-handbook.md §6a).

2.14 Release / ReleaseItem

Scoped by Functional System Group / Subgroup in `scope_filter` (07-database-design.md). Since v4 a `Release` also has zero or more `ReleaseTargetVersion` rows (the controller software versions it is published for; immutable after publish) and its candidate list is limited to revisions whose version range includes a target (BR-021).

2.15 Lookup & Reference Entities

Corrected 2026-09-17 against real data — replaces the prior revision's `StatusCategory/AvailabilityGroup/BrakeProgram/ResetProgram/OperationalState/AccessRights/AlarmBehaviour` set entirely:

- `AvailableGroup` — soft-validated lookup of the numeric availability codes observed in production (currently 23 codes); labels not yet available from source data (06-data-dictionary.md §9c).
- Eight independent `*Access` lookups (Development, Sales, TCC, Service, Turbine Operator Package, Grid Operator, Service Partner, Customer) — real observed text values, replacing the single `AccessRights` enum (06-data-dictionary.md §9d).
- `status_category` and `alarm_flag` are plain `CHECK`-constrained columns (three values; boolean), not FK'd lookup tables — the real data doesn't support the richer enums assumed previously.
- The eight program-reference fields (`brake_program`, `yaw_program`, etc.) are **not** lookup-backed at all — they are numeric passthroughs to a controller-firmware-defined table CSCM Tool does not own (06-data-dictionary.md §2c).

2.16 AuditLogEntry

Since v4 every row also carries `prev_hash` and `entry_hash`, forming a chain in insertion order (AUD-074, 12-audit-compliance.md §6). Notable addition: a `SANDBOX_DELETE` action captures the full pre-delete `StatusCode/StatusCodeRevision` snapshot in `before_value`, since this is the one case where the audited entity itself ceases to exist (12-audit-compliance.md §2).

2.17 ExportJob

Attributes as before, plus (v4) `sha256`, `size_bytes`, `row_count`, `profile_code`, `audience`, `language_code`, `software_version_code` and the JSON `manifest`; `export_scope_type` is `CATALOGUE` | `FULL_DATABASE` | `RELEASE_NOTES` | `OFFLINE_BUNDLE` — `FULL_DATABASE` exports are restricted to the Administrator role (08-system-architecture.md §6, FR-047). `export_type` now also accepts `XLSX` alongside `CSV/JSON/PDF/SQLITE_BACKUP` (new 2026-09-17).

2.18 ImportJob

New entity (2026-09-17). Attributes: `id`, `requested_by`, `source_format` (`CSV|XLSX`), `source_file_reference`, `row_count`, `status`, `created_count`, `error_count`, `error_report_reference`, `error_report_xlsx_reference`, `created_at`, `completed_at`; v4 adds `processed_count`, `warning_count` and `import_mode` (`CREATE | UPDATE`) and the status values `PENDING` / `VALIDATING` / `COMPLETE` / `FAILED` for background processing. **Cardinality:** one `ImportJob` produces zero-or-more `ChangeRequest` rows (via `change_request.import_job_id`, tagged `cr_type = IMPORT`), each following the identical Draft-through-Approval path as a manually authored Change Request (05-governance-handbook.md §6b) — import creates work for the normal governance pipeline, it does not create a parallel one.

3. UML-Style Textual Class Diagram



```
| AuditLogEntry (append-only, no FK)       |
| entity_type/entity_id, action, actor_id, |
| occurred_at, before_value, after_value   |
+------------------------------------------+

+------------------+
|   ExportJob      |
| export_scope_type: CATALOGUE | FULL_DATABASE (Administrator only)
+------------------+
```

4. Lifecycle Note (see 05-governance-handbook.md §4–§5 for full detail)

`Draft -> Review -> PendingApproval -> Approved -> Released -> Deprecated -> Archived`. The `status_code_identifier` is `NULL` throughout `Draft` and is allocated exactly once, atomically, at the Draft->Review transition (never before, never re-allocated).

5. Cardinality Summary

Relationship	Cardinality
User -> StatusCode (owner)	1 : N
User -> EngineeringDomain (tags)	M : N
FunctionalSystemGroup -> FunctionalSubgroup	1 : N
FunctionalSystemGroup -> StatusCode	1 : N
FunctionalSubgroup -> StatusCode	1 : N (mandatory — every StatusCode has exactly one)
ImportJob -> ChangeRequest	1 : N (via cr_type = IMPORT)
StatusCode -> StatusCodeRevision	1 : N
StatusCodeRevision -> ChangeRequest	1 : 1
StatusCodeRevision -> TurbinePlatform	M : N
StatusCode -> StatusCode (platform_variant_of)	0..1 : N (self-referential)
ChangeRequest -> ReviewComment	1 : N
ChangeRequest -> DomainSignoff	1 : N
Release -> ReleaseItem	1 : N
Release -> ControllerSoftwareVersion (targets)	M : N (v4, release_target_version)
StatusCodeRevision -> ControllerSoftwareVersion (from / to)	N : 0..1 each (v4)
StatusCodeRevision -> RevisionTranslation	1 : N, at most one per language (v4)
ChangeSet -> ChangeRequest	1 : N (v4)
ChangeRequest -> CrAttachment / FieldComment / CrChecklistConfirmation	1 : N each (v4)
User -> Delegation (as delegator / as delegate)	1 : N each (v4)
User -> Notification / SavedSearch / ApiToken	1 : N each (v4)
Webhook -> WebhookDelivery	1 : N (v4)
OccurrenceImport -> CodeOccurrence	1 : N (v4)
StatusCode (Hub) -> StatusCode (Main) via HubMcMapping	1 : 0..1 (v4)

6. Business Ownership Matrix

Entity	Primary Business Owner	Secondary
StatusCode / StatusCodeRevision	Controls Engineering (Engineer role)	Governance (Reviewer/Administrator/Chief Engineer decide)
FunctionalSystemGroup / FunctionalSubgroup / TurbinePlatform / EngineeringDomain	Governance (Administrator)	Controls Engineering (proposes changes)
ChangeRequest / ReviewComment / DomainSignoff	Shared: Engineering initiates and cross-signs, Governance decides	—
Release / ReleaseItem	Governance (Administrator)	Technical Publications (consumer)
User / Role	Administrator	—
AuditLogEntry	System (immutable, with the sandbox-delete exception noted in §2.16)	Quality/Compliance (consumer)

Entity	Primary Business Owner	Secondary
ExportJob	System	Administrator (full-database), all roles per matrix (catalogue)
ImportJob	Engineering (initiates)	Governance (every resulting Draft still goes through the full review pipeline)
ExportProfile / ControllerSoftwareVersion / Language / ReviewChecklistItem / AppSetting	Governance (Administrator)	Technical Publications (profiles, languages)
RevisionTranslation	Technical Publications / Engineering (translate)	Reviewer or Administrator (approve)
ChangeSet / CrAttachment / FieldComment	Engineering	Governance
Delegation	The delegating decision maker	Administrator (may create or revoke for anyone)
ApiToken / Webhook	Administrator	Integration teams (consumers)
OccurrenceImport / CodeOccurrence	Chief Engineer / Administrator	Operations / SCADA (data source)
HubMcMapping	Engineering	Governance (the resulting MC Draft is an ordinary CR)

05. Status Code Governance Handbook — CSCM Tool

1. Purpose

This handbook is the authoritative governance policy the application must enforce in software. Where this document and any other document disagree on a process rule, this document wins.

2. Code Numbering Strategy

The Status Code Identifier is `StCd-XXXXX`: a five-digit, zero-padded number, e.g. `StCd-01232`. Its numeric value is not arbitrary — it equals the code's position within its Functional System Group's fixed numeric range, so the identifier itself encodes classification. There is no separate prefix letter scheme in the identifier (the former `MC-/HC-` split and its Controller Type basis are retired entirely); the four-letter group codes (`WCNV`, `WGEN`, …) are displayed alongside the identifier for cross-reference but are not part of it.

Range	Functional System Group	Prefix (display only)
1000–1999	Converter & Grid Interface	WCNV
2000–2999	Generator System	WGEN
3000–3999	Meteorology & Nacelle Environment	WNAC
4000–4999	Rotor & Pitch System	WROT
5000–5999	Tower & Structure	WTOW
6000–6999	Transformer & MV System	WTRF
7000–7999	Drivetrain & Gearbox	WTRM
8000–8999	Yaw System	WYAW
9000–9999	Turbine Control & Operation	WTUR
10000–10999	free — reserved for a new group	—
11000–11999	Hub Controller Control System	WTUR
12000–12999	Hub Controller Rotor & Pitch	WROT
13000–13999	Wind Farm / Plant Dispatch	WPPD
14000–14999	free — reserved for a new group	—
15000–15999	free — reserved for a new group	—
16000–16999	free — reserved for a new group	—

Prefix reuse (added 2026-09-17): `WTUR` and `WROT` each identify two numerically distinct groups — one Main Controller, one Hub Controller — confirmed explicitly by the business as intentional (same thematic domain, separate status code range per controller's software). The prefix is a display/cross-reference label only; the numeric range is always the authoritative identifier of which group a code belongs to (06-data-dictionary.md §9).

Reserved capacity (confirmed 2026-09-17): `10000?10999`, `14000?14999`, `15000?15999`, and `16000?16999` are explicitly free — no Functional System Group occupies them yet, and none is required to. This includes the `15000?15999` range previously found in the live 4XM codebase, which turned out to be exactly this kind of reserved space, not a code-classification gap needing resolution. An Administrator can assign a whole new Functional System Group (and its subgroups) into any of these ranges later, following the same non-overlap validation as adding a subgroup (§9a.1 of 06-data-dictionary.md) — a pure data change, never a schema migration.

Subgroups (mandatory — corrected 2026-09-17): every Functional System Group is divided into Functional Subgroups, and every Status Code must belong to exactly one — Functional Subgroup is **never optional**, confirmed against a real export where all 897 live status codes carry one. Most subgroups own a 100-number sub-band aligned to the hundreds digit of the identifier; the **first subgroup of each Main Controller group owns the first 200 numbers** (e.g. `StCd-010xx`, 01000–01199, is the Converter subgroup within Converter & Grid Interface; Tower is 05000–05199); the Hub Controller groups follow the same alignment and the same naming (`123xx` is 12300–12399, `110xx` is 11000–11099). **A subgroup code is the digits of the first number of its band, with `x` for the free positions, and never carries a dash or a letter:** a band shorter than a hundred is named by its first ten with a digit in place of the second `x` (`1110x` is 11100–11149, `1115x` is 11150–11199, `1205x` begins at 12050); within Turbine Control & Operation, six subgroups share the `099xx` hundred-block in finer ten-number sub-bands and are called `0990x` (9900–9929), `0993x` (9930–9959), `0996x`, `0997x`, `0998x` and `0999x`. The range shown next to a code is always the true band. The real taxonomy, corrected 2026-09-17 against the authoritative row-level export (superseding an earlier hand-typed summary that used the wrong subgroup names — the exact issue flagged by the business), is seeded

in full: 06-data-dictionary.md §9a, `artifacts/schema.sql`.

Not every block within a group is currently assigned to a named subgroup — e.g. `011xx` and `045xx–049xx` within Rotor & Pitch System are unassigned. This is deliberate reserved capacity, not a gap to close: an Administrator can create a new Functional Subgroup at any time, in any Functional System Group, choosing any sub-range that does not overlap an existing one (validated by the service layer and, as a backstop, a database trigger — 07-database-design.md §5). No hundreds-digit alignment is enforced — the WTUR ten-number sub-bands prove narrower ranges are legitimate — but the UI suggests the next hundred-aligned free block as a sane default. Adding a subgroup, like adding a Functional System Group (subject to resolving the open `15000?15999` item above), is a pure lookup-table data change, never a schema migration or a renumbering of existing codes. A code that already exists inside a currently-unassigned block (e.g. legacy `StCd-04960`, found in the live 4XM codebase, sitting in the unassigned `049xx` block) predates this taxonomy and needs formal reclassification once a covering subgroup exists — the mandatory-subgroup rule applies to newly created codes; it does not retroactively invalidate legacy data.

Allocation rule: the identifier is **not** assigned at Draft creation. An Engineer creating a new Status Code selects a Functional System Group and a Functional Subgroup via dropdown — **both required, the subgroup is never optional** — and the Draft is worked on with no fixed number (displayed as "TBD — assigned on submission"). Only at the moment the Change Request is submitted for Review (Draft -> Review) does the system atomically allocate the next free number within the selected Subgroup's sub-band. This satisfies the requirement that in-progress codes never carry a fixed number.

A Change Request that is rejected back to Draft and later resubmitted, and every later revision of an already-numbered code, **keeps the identifier it already has** — allocation happens exactly once per code (BR-001). Numbers are never reused, even if a code is later Archived. Gaps are expected wherever a Draft is abandoned before submission (its number was never allocated in the first place — it never had one). An Administrator may allocate an out-of-sequence number only for a documented migration/reservation case, recorded and audited as a manual override.

Sandbox exception: Administrator-created test/sandbox Status Codes never draw from the ranges above. They receive an identifier from an isolated `StCd-T#####` counter, are flagged `is_sandbox = true`, are excluded from every Release, catalogue export, and manual, and are the only records in the system an Administrator may permanently delete (§12).

Identifier + Functional System Group assignment is immutable once allocated (BR-001). If a code was genuinely created in error before any Review activity, it is deprecated, not renumbered or reassigned to a different group.

2a. Platform Variants

A condition that differs between Turbine Platforms is represented in one of two ways. If the same code simply also applies to another platform, the platform is added to the code's Turbine Platform multi-select — one code, one identifier. If the code is renamed or functionally regenerated for another platform, it becomes an **independent Status Code** with its own `StCd` identifier and its own full cycle, optionally pointing back to the originating code through `platform_variant_of_status_code_id` purely for navigability; the link carries no lifecycle or approval meaning (04-domain-model.md §2.10). Since v4 the Create platform variant action copies the latest decided revision into a new Draft for the other platform and sets the link (FR-094).

3. Code Ownership

Every Status Code has exactly one Owner at all times, captured on the current revision (`owner_id`). Default Owner is the Engineer who authored the first revision; reassignable by an Administrator without altering revision content or requiring re-approval. Controlled vocabularies (Functional System Group, Functional Subgroup, Turbine Platform, Engineering Domain, Status Category, Availability Group, Brake Program, Reset Program, Operational State, Access Rights, Alarm Behaviour) are owned by Governance (Administrator role); Engineers may propose new values, only an Administrator may activate one.

4. Lifecycle States

Seven states (extended from the original six to represent the dual-review-then-Chief-Engineer-approval sequence explicitly, rather than collapsing it into a single "Review" step):

State	Meaning	Mutable?
Draft	Being authored; no fixed StCd identifier yet	Yes, freely

State	Meaning	Mutable?
Review	Submitted; StCd identifier now fixed; awaiting the two required sign-offs (Reviewer + Administrator)	No (read-only to the author)
PendingApproval	Both required Review sign-offs recorded; awaiting Chief Engineer decision	No
Approved	Chief Engineer has approved; awaiting inclusion in a Release	No
Released	Published as part of a formal Release; authoritative and citable	No
Deprecated	No longer active; retained for historical reference	No (metadata-only: reason, superseding code)
Archived	Long-term retention state after the retention period	No

5. Valid State Transitions

From	To	Trigger	Actor
(none)	Draft	Create new Status Code (select Functional System Group [+ Subgroup])	Engineer, Administrator
Draft	Review	Submit Change Request — identifier allocated here (§2); requires all Cross-Domain Sign-Offs complete (§6a)	Engineer (author), Administrator
Review	PendingApproval	Both required sign-offs recorded (Reviewer AND Administrator — order-independent)	System-triggered on the second sign-off
Review	Draft	Either required reviewer rejects (mandatory comment)	Reviewer, Administrator
PendingApproval	Approved	Chief Engineer approves	Chief Engineer
PendingApproval	Draft	Chief Engineer rejects (mandatory comment)	Chief Engineer
Approved	Released	Included in a published Release	Administrator (system-executed at publish)
Approved	Draft	Withdraw before release (exceptional; Administrator + reason)	Administrator
Released	Deprecated	Deprecation action (mandatory reason)	Administrator, or Engineer CR + full dual-review-and-approval cycle
Deprecated	Archived	Archival, only after retention period elapsed	Administrator
Deprecated	Released	Reinstatement (exceptional; Administrator + reason)	Administrator

Any transition not listed is prohibited by the system.

6. Status Code Creation Process

1. Engineer (or Administrator) initiates "New Status Code," selects Functional System Group and Functional Subgroup via dropdown — both required (§2). No identifier is shown yet.
2. System creates a `StatusCode` (identifier `NULL`, `functional_system_group_id/functional_subgroup_id` set) + first `StatusCodeRevision` + a `ChangeRequest` (`cr_type = NEW`), all in **Draft**.
3. Engineer completes mandatory attributes; system validates continuously.
4. If the originating Engineer's tagged Engineering Domain(s) do not cover every domain-owned field on the revision, the Draft is flagged "awaiting domain input" and cannot be submitted until each missing domain is signed off (§6a).
5. Engineer submits for Review; the system allocates the StCd identifier atomically and transitions Draft -> Review.
6. Reviewer and Administrator each independently evaluate and sign off (§7); on the second sign-off, the system auto-transitions Review -> PendingApproval.
7. Chief Engineer evaluates and approves or rejects (§8); on approval, PendingApproval -> Approved.
8. On inclusion in a published Release, Approved -> Released, Effective Date stamped.

(v4) Around this process: saving a Draft that someone else changed meanwhile is refused (§9); the form warns about unknown or retired references and similar titles (§6c); the Engineer may attach evidence, group the request into a change set and state the software-version range; submitting may auto-assign the least-loaded Reviewer (§14); each decision asks for the actor's password (§7a).

6a. Cross-Domain Sign-Off

Some Functional System Groups require input from more than one Engineering Domain (e.g. a Generator-group code created by an Electrical engineer may still need Controls-domain parameters — delay/alarm/reset behavior — that the originating engineer cannot authoritatively provide). Each field in 06-data-dictionary.md is tagged with an owning Engineering Domain. When a Draft's originating Engineer lacks a domain tag required by one or more of its fields, the system marks those domains as "pending" and blocks Submit-for-Review until a `DomainSignoff` exists for each pending domain, recorded by an Engineer who holds that domain. This is a completeness gate, not a content judgment — it does not replace Review.

6b. Library Import

New capability (2026-09-17): an Engineer or Administrator may import a CSV or XLSX file of status codes (e.g. a legacy library such as the 897-row export used to ground this taxonomy) instead of creating codes one at a time through the UI. **Import is a faster data-entry path, not a governance bypass, confirmed explicitly by the business:** every row that passes field validation becomes an ordinary `StatusCode`/`StatusCodeRevision`/`ChangeRequest` triple in **Draft**, exactly as if an Engineer had typed it in — it still requires Cross-Domain Sign-Off where applicable, both Review sign-offs, and Chief Engineer approval before it can ever be Released. A row that fails validation is skipped and reported in the import job's error report (06-data-dictionary.md §5); it never partially creates a record. The source file's legacy numbering (e.g. the old "Status Code Number" column) is preserved on each created code as `legacy_reference_number` for traceability, and is never mistaken for the governed `StCd` identifier, which is still allocated only at Draft->Review (§2) — including for imported rows.

Status codes from the generated programs (v4.2). The controller programs `StCd_.st` already carry the codes, so an Engineer or Administrator may import them directly (FR-048o … FR-048w) — several files, a folder or a ZIP at a time, first as a **preview** in which any subset is ticked, then confirmed. The same rule holds as for the CSV import: every program becomes an ordinary Draft that still needs Cross-Domain Sign-Off, both Review sign-offs and Chief Engineer approval; all Drafts of one confirmation join one change set so the batch is reviewed together. What is specific to this source: **the number is already decided by the program**, so it is reserved for the Draft and issued unchanged at Submit for Review (BR-023) — no sequence allocation, no Administrator override, and the allocator skips it; the subgroup follows from the band the number lies in; the legacy number (Main Controller: the old number in the program header; Hub Controller: the number in the file, which is the legacy Hub number and is translated through the numbering table) is stored as `legacy_reference_number`, which makes the **legacy -> new mapping** — one direction only, because the new library holds more codes than the old one. The file carries no status category, so the uploader picks one for the batch and each Draft's author refines it.

6c. Soft Checks While Authoring (v4)

These never block a save; they make a mistake visible while it is cheap to fix (FR-091, FR-092, FR-093):

- **References.** A repeated-error pointer, a related code or a superseding code that is unknown, Deprecated, Archived or the code itself produces a warning — the pointer may legitimately name an external escalation number, so it stays a warning.
- **Similar titles.** An identical or very similar title in the same Functional Subgroup produces "did you mean to revise StCd-…?".
- **Where-used.** Every code shows which other codes refer to it. Deprecating a code that live codes still refer to requires the requester to acknowledge the dependants.
- **Platform variants.** Create platform variant copies the latest decided revision into a new Draft for another turbine platform and links the two; it is an ordinary Draft that still passes the whole cycle (FR-094).
- **Planned dates.** An Administrator may stamp a planned effective and planned deprecation date on the current revision; they warn at release time and appear in the health report, but never change a lifecycle state by themselves (FR-109).

7. Review Sign-Off Rules

Review requires **two independent sign-offs**, from two different people holding two different roles:

1. **Reviewer** (organizationally the Sub-PO for Status Codes).
2. **Administrator** (organizationally the Product Owner / Component Owner Controls Software).

Both are required; they may occur in either order and do not depend on each other. Either one rejecting (with a mandatory comment) returns the revision to Draft — the other sign-off, if already given, is discarded and must be

re-given after resubmission. Neither sign-off may be given by the Change Request's author (segregation of duties, BR-005). The Administrator's sign-off is also the point at which any Engineer-proposed value in a restricted field (e.g. Access Rights — 04-domain-model.md §2.9) is confirmed or overridden as authoritative.

Reviewer checklist (procedural, not fully machine-checkable):

1. No duplicate or overlapping-meaning code exists (system-assisted search, human judgment).
2. Classification fields are internally consistent (hard incompatibilities are system-enforced per 06-data-dictionary.md; soft conventions are reviewer judgment).
3. Delay Parameters are sane for the selected Operational State.
4. Description is unambiguous, not a restatement of the Title.
5. Functional System Group / Subgroup assignment matches the condition's actual subsystem.

(v4) The Administrator can turn this checklist into **configured review-checklist items** per Functional System Group (or for all groups). An approving Reviewer must confirm every active item that applies; the confirmed wording is stored with the sign-off, so later edits never rewrite history (FR-128). A Reviewer, Administrator or Chief Engineer may also comment on **one specific field** while the request is in Review or Pending Approval; the author sees open field comments inline and marks them resolved (FR-127).

7a. Password Confirmation and Delegation (v4)

Step-up (SEC-026, BR-022). Reviewer and Administrator sign-off, Chief Engineer decision, publishing a Release and the full-database export require the actor to enter their own password again, so each is a deliberate, attributable act and an unattended session cannot approve anything. A missing password gives `step_up_required`, a wrong one `step_up_failed`; both are audited.

Delegation (FR-125, BR-017). A Reviewer, Administrator or Chief Engineer who will be absent names a delegate for a date range (an Administrator may arrange it for anybody). While the delegation is active the delegate may take the decision actions of that role — never user administration or other Administrator functions. The delegate acts **as themselves**: the audit entry names them and the person they cover for, segregation of duties applies to them personally (they cannot decide a request they wrote or already signed, and the two Review sign-offs must still come from two different people), and the Reviewer auto-assignment skips an absent Reviewer. The delegator or an Administrator can revoke it.

8. Chief Engineer Approval

Once both Review sign-offs are recorded (PendingApproval), a Chief Engineer performs the final Approval decision — approve or reject with a mandatory comment on rejection. The Chief Engineer may not be the Change Request's author, the Reviewer who signed off, or the Administrator who signed off on the same Change Request (segregation of duties extends across all three decision points). (v4) A Chief Engineer may work through a **change set** — the requests that ship with one controller software release — in one sitting: each request is still decided by its own single-request service call with its own segregation-of-duties check, comment and audit entry, the password is asked once, and a request the actor may not decide is reported as skipped without blocking the others (FR-124). Approval is the final governance gate before a revision is eligible for Release; it is a distinct accountability step from Review and is always performed by a named individual, never inferred from role membership alone.

9. Revision Numbering & Concurrency

Each new Draft opened against an existing `StatusCode` increments `revision_number`; a rejection (Review or PendingApproval -> Draft) reuses the same revision row; only one open (Draft/Review/PendingApproval) revision is permitted per `StatusCode` at a time (BR-003). (v4) Every Draft carries a `row_version` that increases on each save; a save based on an older version is refused with 409 and the user's entries are kept for comparison, so two people editing the same Draft can no longer overwrite each other silently (FR-090, BR-019).

10. Release Rules

Only an Administrator may create/publish/pre-publish-delete a Release (candidate pool is Approved-state revisions matching the scope filter, now expressed in terms of Functional System Group/Subgroup rather than Controller Type; publish is atomic; a published Release is permanently immutable). (v4) Before publishing, the Administrator may set the controller **software versions** the Release targets, which limits the candidate list to revisions whose version range includes a target (BR-021); publishing asks for the password (§7a) and warns, without blocking, about included revisions whose planned effective date is in the future. Every export of a Release carries a manifest of

SHA-256 hashes (signed when a key is configured), the Release has generated **release notes** against the previous published Release, and exports can be scoped to an audience (only values the Administrator confirmed at sign-off count; `Covert` is treated as not visible, BR-016), a language (approved translations only, §16) and a software version (FR-100 … FR-111).

11. Deprecation, Retention & Archival Rules

Deprecation requires a reason and optional superseding reference; minimum 180 days in Deprecated before Archival; 7-year minimum retention for Archived records and all associated Change Requests/comments/sign-offs/audit entries, never hard-deleted — **except** sandbox Status Codes (§12). (v4) The deprecation request lists the live dependants (Where-used) and the requester acknowledges them; audit entries older than the retention period are archived to compressed files with a hash and are never deleted (§18).

12. Sandbox / Test Status Codes

An Administrator may create a Status Code flagged `is_sandbox = true` for testing purposes (e.g. verifying a new export template, training a new Reviewer). Sandbox codes:

- Draw their identifier from an isolated `StCd-T#####` band, never from a real Functional System Group range, so they can never collide with or consume a production number.
- Are excluded from every Release candidate list, catalogue export, and manual by construction (a hard filter, not a convention).
- May proceed through the same lifecycle states for realistic testing, but can never actually be included in a real Release (the system blocks adding a sandbox-flagged revision to any Release, regardless of state).
- Are the **only** records in the system an Administrator may permanently delete. Deletion is logged as an immutable `SANDBOX_DELETE` audit entry that captures the full record snapshot in `before_value` — the audit trail entry survives even though the record itself does not (12-audit-compliance.md §2).
- This is a narrow, explicit, audited exception to the otherwise absolute no-hard-delete rule (§11) and must never be generalized to non-sandbox records.

13. Compliance Rules

Every Released Status Code must be traceable, without gaps, through Draft -> Review (both sign-offs, with identity and timestamp) -> PendingApproval -> Approved (Chief Engineer, with identity and timestamp) -> Released (Release identity, Effective Date). Segregation of duties is enforced server-side at all three decision points (Reviewer, Administrator, Chief Engineer), never only in the UI.

14. Service Levels and Reminders (v4)

A request waiting for a decision is aged in **business days** (Monday–Friday; no holiday calendar) from the moment it entered its stage and shown amber or red against Administrator-configured thresholds (defaults: Review 3 / 5, Approval 2 / 3 days — settings `sla.review.*`, `sla.approval.*`). `flask send-reminders`, run by the scheduler, creates one in-app notice per decision holder (including delegates), request, stage, level and day; if the organisation's own SMTP relay is configured the same text is mailed. Administrators see the **stalled requests** list with one-click reassignment (FR-039) and the open reviews per Reviewer; with `reviewer.auto_assign` enabled the least-loaded available Reviewer is assigned at submit. The business-case KPIs (median approval time <= 7 days, cross-domain turnaround <= 3 days, audit-trail completeness 100 %, queue ages, codes without an active owner) are computed on the Success Metrics page (FR-120 … FR-123, FR-129).

15. Evidence, Remedy Content and Change Sets (v4)

The author (or an Administrator) may attach **evidence** — test reports, simulation results, design notes, a safety assessment — to an open request (allow-listed types, checked against their content, at most 10 MB and 20 files each, stored outside the web root, downloadable by the working roles only, removable only while the request is a Draft; FR-130). Revisions may carry remedy text, affected component, related codes and standard references for service use (FR-131). Requests that ship together are grouped in a **change set** (FR-124); grouping changes presentation and queueing only, never the decisions required.

16. Translations (v4, decision D-03)

English is the source language of every code. A translation of title, description and remedy text is saved as DRAFT against an Approved or Released revision by an Engineer or Administrator, and approved by one Reviewer or Administrator **other than the translator** — a light review, deliberately not the three-decision cycle (BR-018). Editing an approved translation returns it to DRAFT; translations can be carried over from the previous revision as DRAFTs. Exports in a language use approved translations only (falling back to English unless the export is strict), and the translation worklist lists the current codes still lacking an approved translation (FR-110, FR-111).

17. Hub -> Main Controller Consolidation (v4, assumption A-008)

If the Hub Controller leaves the hardware setup, its codes are defined anew on the Main Controller side without renumbering anything: Create MC draft on an Approved or Released Hub code copies the descriptive fields into a new Draft in a chosen Main Controller group and subgroup (the new MC number is allocated at submit, the Hub identifier is kept as the legacy reference, a mapping row is recorded). The mapping report shows every Hub code with the state of its replacement; once a replacement is Released, an Administrator opens deprecation requests for all such Hub codes of a group at once — each an ordinary deprecation request needing review and approval (FR-147, BR-015). Hub groups stay ordinary data (`controller_type = HUB`, decision D-06).

18. Integrity, Backup and Retention (v4)

Each audit entry stores the hash of the previous one; an Administrator verifies the chain on demand and a scheduled job verifies it weekly, so an edited, removed, reordered or truncated entry is found (FR-151). Verified backups are taken nightly and each is proven restorable (FR-150); the readiness probe reports `degraded` when the newest verified backup is older than 36 hours. Audit entries older than seven years are archived, never deleted (BR-020). The restore drill is rehearsed quarterly (14-deployment-architecture.md §7).

06. Data Dictionary — CSCM Tool

Conventions: **Required** = enforced not-null at both UI and API before the entity can leave Draft (for revision fields) or at all times (for structural fields). Datatypes are given in SQLite-native terms with the logical type in parentheses where useful (see 07-database-design.md for full DDL).

Resolved in v4 — software version (decision D-01): the legacy tool scopes the whole table by a software version (`SW_1_1_1_4.2MW`). The governed model is now the ordered vocabulary `lookup_controller_software_version`, an inclusive from/to **range per revision** (`applies_from_version_id`, `applies_to_version_id`, either end open) and the set of versions a **Release targets** (`release_target_version`); see §2 and §6. The free-text `software_version` column stays for legacy rows and is informational only.

v4 (2026-10-23): the tables and fields added by the improvement pack are marked (v4); §13 lists every new table. `docs/artifacts/schema.sql` is the executable definition (65 tables, 14 triggers) and a test fails if a fresh database and an upgraded one differ.

v4.1 (2026-10-23) — fields read from the generated controller programs: 14 generated `StCd_*.st` programs (Main Controller 1000–1010, Hub Controller 14000, 14005 and 14025) were compared field by field with the status code workbooks (Main and Hub, Customer and Service exports, the 4XM "Rise0B" workbook and the Hub numbering workbook). Everything the workbooks contain matched the generated programs for all 14 codes, with one exception (`ROTOR_BREAK`, §2f). They also showed what the workbooks do not carry: the block identification `PrgIdx_GUi` (§1), further subsystem programs and Hub programs (§2c.1), the operating-mode mask and other per-code parameters (§2f), and the numeric user-group levels (§2d). Fields added from this comparison are marked (v4.1); migration `c9e2a4b6d815` brings an older database up to them, and the existing library has no values for them until they are imported from the generated programs. **Since v4.9 the application stores, shows, copies, compares and exports all of them and the program import fills them (FR-168); the controller rule of §2f is enforced for the new fields.**

Provenance note (2026-09-17): the field set, real value vocabularies, and Functional Subgroup names in this revision are grounded in an actual export of the live status code library (`New Statuscode for sharing.xlsx`, 897 rows across 9 sheets, one per Functional System Group) and the legacy tool's screenshot supplied by the user. Fields whose exact business meaning could not be determined from the export alone (the numeric "program" reference fields, and the leading `*` marker on delay values) are flagged **semantics TBD** rather than guessed.

1. status_code (StatusCode identity)

Field	Description	Datatype	Required	Default	Validation	Allowed Values	Example
id	Surrogate primary key	INTEGER (PK)	Yes	auto	system-generated	—	1042
status_code_identifier	Human-readable governed identifier (StCd)	TEXT	No while Draft; Yes from Review onward	NULL	^StCd-\d{5}$ (or ^StCd-T\d{5}$ for sandbox); unique once non-null; allocated atomically at Draft->Review (05-governance-handbook.md §2)	—	StCd-01232
functional_system_group_id	Classification + numbering band	INTEGER (FK -> lookup_functional_system_group)	Yes	—	must be is_active = true	see §9	WCNV (1000–1999)
functional_subgroup_id	Sub-classification + numbering sub-band	INTEGER (FK -> lookup_functional_subgroup)	Yes — never optional (corrected 2026-09-17; was mistakenly nullable in the prior revision)	—	must belong to the selected functional_system_group_id; must be is_active = true	see §9a	Converter (010xx)
is_sandbox	Test/sandbox flag	INTEGER (bool)	Yes	0	Administrator-only to set true	0, 1	0
platform_variant_of_status_code_id	Optional pointer to the originating code this one was split from across Turbine Platforms	INTEGER (FK -> status_code.id, nullable)	No	NULL	must reference a different Status Code	—	988

Field	Description	Datatype	Required	Default	Validation	Allowed Values	Example
legacy_reference_number	Free-text pointer to the pre-migration status code number, for traceability during library import (06-data-dictionary.md §12)	TEXT	No	NULL	—	—	3000 (legacy "Status Code Number" from the source system)
prg_idx (v4.1)	PrgIdx_GUi: the block identification the generated program writes to the controller's global variable on entry and restores on exit, so the controller knows which block was running. Entered manually by an engineer; the tool records, validates and audits it and never allocates it	INTEGER	No while Draft or not yet imported; Yes from Review onward	NULL	1–65535 (the generated program declares it UINT); unique among non-sandbox codes (ux_status_code_prg_idx); frozen once a revision of the code is Released, Deprecated or Archived, except the first assignment NULL -> value (trg_prevent_prg_idx_change_after_release); every set or change is audited with its before and after value	—	14066 (Main 1000), 18926 (Hub 14000)
reserved_identifier (v4.2)	The number an imported StCd_.st program already carries, held for this Draft: the identifier issued at Submit for Review is exactly this one (FR-048q, BR-023). Set only by the status code import, never through the API payload	TEXT	No	NULL	StCd-NNNNN; unique; may not equal the identifier of another code (service and triggers)	—	StCd-01000
withdrawn_at (v4.9)	Set when the last revision of the code was withdrawn (BR-026); only such a code can be deleted (erased in place, deleted_at), by an Administrator, and a restore from the program file clears it	TEXT (UTC timestamp)	No	NULL	set by the withdrawal, never through the API		
deleted_at (v4.9.2)	Set when an Administrator deletes a withdrawn code: the row stays (no id is reused, the audit entries keep one meaning) but the identifier, the reserved number, the legacy number and prg_idx are cleared	TEXT (UTC timestamp)	No	NULL	set by the deletion, never through the API		
owner_id	Accountable engineer	INTEGER (FK -> user.id)	Yes	creator's id	active user, Engineer or Administrator role	—	14
created_at	Row creation timestamp (UTC)	TEXT (ISO-8601)	Yes	now()	system-generated	—	2026-03-02T09:14:00Z

Why Functional Subgroup is mandatory: every one of the 897 rows in the real export carries a `Functional Group Name` (subgroup) value — there is no "group-only, no subgroup" case in production data, and the business explicitly confirmed a code must never be created without one. An Engineer selecting only a Functional System Group at creation time is therefore an incomplete Draft, not a valid one; the Functional Subgroup dropdown blocks progression exactly like any other mandatory field (10-ui-ux-specification.md §4).

PrgIdx_GUi (v4.1): the value exists only in the generated programs, not in any workbook, so the existing library holds `NULL` until it is imported from them. "Required from Review" is enforced by the service layer (not yet wired in the application) and backed up in the database by `trg_require_prg_idx_for_review`, which stays inert until the setting `status_code.require_prg_idx` is `1`; set it once the library carries its values and the form collects the field. Observed values: Main 14066–14253 (codes 1000–1010), Hub 18926, 18931 and 18993 (codes 14000, 14005, 14025). The two ranges do not overlap, so uniqueness is checked across both controllers; whether Main and Hub share one ID space or are separate PLC projects is open, and other programs of the controller also write `PrgIdx_GUi`, so the tool can only guarantee uniqueness among status codes. All 14 values fit `base + (n ? first) + 21 × [(n ? first) / 10]`, where n is the code's legacy number (3000… for Main, 14000… for Hub); that may serve as a suggestion or a warning, never as a rule, because the number is entered by hand.

2. status_code_revision — descriptive, classification, and timing fields

Field	Description	Datatype	Required	Default	Validation	Allowed Values	Example
id	Surrogate primary key	INTEGER (PK)	Yes	auto	system-generated	—	5031
status_code_id	Parent Status Code	INTEGER (FK)	Yes	—	must reference existing status_code	—	1042
revision_number	Sequential revision counter per Status Code	INTEGER	Yes	1	monotonic, never reused	>= 1	2
is_current	Flags the single "current" revision per code	INTEGER (bool)	Yes	1	exactly one true per status_code_id	0, 1	1
title	Short human-readable name	TEXT	Yes	—	1–120 chars; must not equal Description	—	Frequency converter not ready
description	Full technical description of the condition (maps to the real export's Description column, which in production doubles as the short title — CSCM Tool splits it into title + a longer description for clarity; see open item in 00-INDEX.md)	TEXT	Yes	—	10–4000 chars	—	Frequency converter is not in a ready state; the converter has not completed its pre-charge and self-check sequence.
status_category	Nature of the status (corrected 2026-09-17: real data uses exactly three values, not the six-value FAULT/WARNING/SAFETY/OPERATIONAL/MAINTENANCE/INFORMATION set previously assumed)	TEXT (enum)	Yes	—	—	ERROR, WARNING, INFO	ERROR
software_version	Firmware/software version this revision is associated with	TEXT	No (see note)	NULL	^\d+\.\d+(\.\d+)?$ when present	—	1.1.1.4.2MW
available_group	Availability-accounting code (corrected 2026-09-17: real data is a numeric code, not a four-letter A–D group)	INTEGER	Yes	—	must appear in lookup_available_group's currently-observed set (soft-validated — see note below)	observed: 1,2,4,5,6,9,11–16,18,20–25,27,28,30,32 (labels TBD, see §9c)	27
alarm_flag	Whether the condition raises an operator-facing alarm (corrected 2026-09-17: real data is a plain boolean, not a four-value Alarm Behaviour enum)	INTEGER (bool)	Yes	0	—	0, 1	1
set_delay	Debounce delay before the condition is set/raised, as free text with unit (corrected 2026-09-17: real data is not a plain seconds integer); (v4.16.1) it may also be the name of the parameter that sets it, or not in the program, when an imported program gives no time (FR-189) - the time must be looked up before release	TEXT	Yes	—	^*?\d+(\.\d+)?\s?(ms|s|min|MIN|h|d)$	—	10ms, *500ms, 2min
reset_delay	Minimum time the condition must be clear before it resets	TEXT	Yes	—	same as set_delay	—	2min, *0 ms
up_down_counter	Whether this status uses an up/down occurrence counter rather than a simple set/clear (new field, 2026-09-17)	INTEGER (bool)	Yes	0	—	0, 1	0
trigger_snapshot	Whether occurrence of this status triggers a diagnostic data snapshot (new field, 2026-09-17)	INTEGER (bool)	Yes	0	—	0, 1	1
loadless_spinning_permitted	Whether loadless rotor spinning is permitted while this status is active (new field, 2026-09-17)	INTEGER (bool)	Yes	0	—	0, 1	1
owner_id	Accountable engineer for this revision	INTEGER (FK -> user.id)	Yes	inherited	active user, Engineer/Administrator	—	14

Field	Description	Datatype	Required	Default	Validation	Allowed Values	Example
lifecycle_status	Current governance state	TEXT (enum)	Yes	DRAFT	must follow valid transitions	DRAFT,REVIEW,PENDING_APPROVAL,APPROVED,RELEASED,DEPRECATED,ARCHIVED	PENDING_APPROVAL
created_by	Engineer who authored this revision	INTEGER (FK -> user.id)	Yes	current user	—	—	14
created_at	Revision creation timestamp	TEXT (ISO-8601)	Yes	now()	—	—	2026-03-02T09:14:00Z
reviewer_signoff_by / _at	Reviewer (Sub-PO for StCd) sign-off	INTEGER (FK -> user.id) / TEXT	No	NULL	!= created_by	—	22
admin_signoff_by / _at	Administrator (Product Owner) sign-off	INTEGER (FK -> user.id) / TEXT	No	NULL	!= created_by	—	2
chief_engineer_approved_by / _date	Chief Engineer final approval	INTEGER (FK -> user.id) / TEXT	No	NULL	!= created_by, reviewer_signoff_by, admin_signoff_by	—	31
effective_date	Date the revision became Released	TEXT (ISO-8601)	No	NULL	set only at Release publish	—	2026-03-15T00:00:00Z
deprecated_reason	Why the code was deprecated	TEXT	Required if lifecycle_status = DEPRECATED	NULL	10–2000 chars when present	—	—
superseded_by_status_code_id	Optional pointer to replacement code	INTEGER (FK -> status_code.id)	No	NULL	must reference a different Status Code	—	1090
row_version (v4)	Optimistic-concurrency counter; increases on every save (FR-090)	INTEGER	Yes	1	a save naming an older expected_row_version is refused (409)	>= 1	3
updated_at (v4)	Time of the last save	TEXT (ISO-8601)	No	NULL	system-generated; an older expected_updated_at is refused	—	2026-10-20T08:00:00.000Z
remedy_text (v4)	Recommended action for service staff (FR-131)	TEXT	No	NULL	<= 4000 chars	—	Reset the converter, then check the DC-link fuse.
affected_component (v4)	Subsystem or component concerned	TEXT	No	NULL	<= 200 chars	—	DC link
related_status_codes (v4)	Related codes, comma separated; soft-checked (FR-091)	TEXT	No	NULL	<= 500 chars; each StCd-NNNNN	—	StCd-01040, StCd-01041
standard_references (v4)	Applicable standards	TEXT	No	NULL	<= 1000 chars	—	IEC 61400-25
applies_from_version_id / applies_to_version_id (v4)	Inclusive software-version range the code applies to; either end open (D-01)	INTEGER (FK -> lookup_controller_software_version)	No	NULL	the order is the vocabulary's sort_order	—	4.1 … 4.3
planned_effective_date / planned_deprecation_date (v4)	Planned dates, informational (FR-109, BR-021)	TEXT (YYYY-MM-DD)	No	NULL	ISO date; set by an Administrator, audited	—	2027-03-31

2a. Field-to-Engineering-Domain Ownership

Principle (05-governance-handbook.md §6a); the v4 content fields (`remedy_text`, `affected_component`, `related_status_codes`, `standard_references`, the version range and the planned dates) have an "Any" owner and never gate Submit-for-Review. `title`, `description`, `status_category`, `available_group` are domain-agnostic ("Any"); timing fields (`set_delay`, `reset_delay`, `up_down_counter`, `trigger_snapshot`), the repeated-error fields (§2b), and the program reference fields (§2c) are owned by `CONTROLS`. The eight audience/access fields (§2d) are Governance-confirmed, following the same propose-by-Engineer/confirm-by-Administrator pattern as the old restricted `access_rights_id` field. (v4.1) `prg_idx`, the further program fields, the operation flags, the operating-mode mask, the task call frequency and the availability-calculation flag are owned by `CONTROLS` as well; the user-group levels (§2d) follow the audience fields.

2b. Repeated-Error Escalation Fields

Field	Description	Datatype	Required	Default	Validation	Example
number_of_times_repeated	How many occurrences within the window (below) before escalation	INTEGER	No	NULL	>= 1 when set	10
repeated_over_the_course	The window duration, free text with unit	TEXT	No	NULL	^\d+(\.\d+)?\s?(s|min|h|d)$	1d
status_code_for_repeated_error	The escalation code raised once the repeat threshold is hit	INTEGER	No	NULL	Not a foreign key to status_code — confirmed 2026-09-17 to reference a separate, unrelated numeric code pool outside StCd governance (e.g. a fixed system/SCADA alarm code). Stored and displayed as an opaque reference number; CSCM Tool does not validate it against any internal table.	455

All three are optional together — a status code with no repeat-escalation behavior leaves all three `NULL`.

2c. Program Reference Fields (numeric, externally defined — semantics TBD)

These fields reference controller-firmware program/subroutine numbers maintained **outside** CSCM Tool (in the Main/Hub Controller codebase itself), not a CSCM Tool-governed vocabulary. CSCM Tool stores and soft-validates them against the currently-observed set below (a warning, not a hard block, since the firmware-side table can gain new program numbers CSCM Tool doesn't yet know about) rather than enforcing a closed lookup. **The descriptive meaning of each numeric code is not available from the source export and must be confirmed with Controls Engineering before these can be upgraded to labeled lookups.**

Field	Description	Datatype	Required	Default	Observed values
yaw_program	Yaw subprogram reference	INTEGER	Yes	0	0,1,3,5,10,13,15,20,30,50,70,240
manual_reset_program	Manual reset subprogram reference	INTEGER	Yes	0	0,5,10,20
auto_reset_program	Automatic reset subprogram reference	INTEGER	Yes	0	0,10,20,30
converter_reactive_power_program	Converter reactive-power subprogram reference	INTEGER	Yes	0	0,10,20
brake_program	Brake subprogram reference (replaces the invented BP-NONE/BP-1...4 enum)	INTEGER	Yes	0	0,20,50,52,60,75,170,180,190,...
umrichterprogramm_gsc	Converter program, grid-side converter (GSC)	INTEGER	Yes	0	0,10,20,30
umrichterprogramm_msc	Converter program, machine-side converter (MSC)	INTEGER	Yes	0	0,6,10,15,20
yaw_bearing_lubrication_program	Yaw bearing lubrication subprogram reference	INTEGER	Yes	0	0,10

This replaces the prior revision's single `reset_program_id`/`brake_program_id` enum FKs entirely — real data has **two** distinct reset-program fields (manual and automatic), not one, and brake/converter/yaw programs are numeric passthroughs, not a small named enum.

2c.1 All program fields and where they come from (v4.1)

The generated programs hold one array per subsystem (`?PrgArUs[Idx]`); the 4XM workbook renamed them (sheet "Naming Conventions", `?Lev`). Each is a numeric reference into the controller firmware, 0 meaning the code does not act on that subsystem. **Applies to:** M = Main Controller, H = Hub Controller. Not-applicable fields stay 0 (§2f).

Column	4XM name	Generated array	Applies to	Notes
manual_reset_program	Manual Reset Program	ManResetPrgArUs	M, H	0, 5, 10, 20
auto_reset_program	Auto Reset Program	AutoResetPrgArUs	M, H	10 = reset after the reset delay whatever the condition; 20 = reset once the set condition is gone; 30 = reset on the code's own reset condition; 0 = never. A reset is also held back while the brake program check (CheckBP) fails and, for codes with a repeat limit, once the limit is reached

Column	4XM name	Generated array	Applies to	Notes
brake_program	BrkLev (Main), "Stop Program" (Hub workbooks)	BrakePrgArUs	M, H	One field: the Hub "Stop Program" (0, 40, 80) matched BrakePrgArUs in all 3 Hub programs
yaw_program	YwLev	AzimutPrgArUs	M	
converter_reactive_power_program	ConvQLev	QConverterPrgArUs	M	
umrichterprogramm_msc	ConvMscLev	ConverterMSCPrgArUs	M	machine-side converter
umrichterprogramm_gsc	ConvGscLev	ConverterGSCPrgArUs	M	grid-side converter
yaw_bearing_lubrication_program	YwBrgLuLev	YawLubPrgArUs	M	
converter_reset_program (v4.1)	ConvRstLev	ResetConverterPrgArUs	M	
hydraulic_pump_program (v4.1)	HyPmpLev	HydraulicPrgArUs	M	
gearbox_oil_pump_program (v4.1)	GbxOilPmpLev	GearOilPumpPrgArUs	M	
gearbox_oil_fan_program (v4.1)	GbxOilFanLev	GearOilCoolingPrgArUs	M	
gearbox_oil_heating_program (v4.1)	GbxOilHtLev	GearHeatingPrgArUs	M	
generator_fan_program (v4.1)	GnFanLev	GenFanPrgArUs	M	
generator_heating_program (v4.1)	GnHtLev	GenHeatingPrgArUs	M	
generator_bearing_lubrication_program (v4.1)	GnBrgLuLev	GenoLubPrgArUs	M	
rotor_bearing_lubrication_program (v4.1)	RotBrgLuLev	RotBearLubPrgArUs	M	
anemometer_heating_program (v4.1)	AneHtLev	VaneAneHeatPrgArUs	M	
medium_voltage_cb_program (v4.1)	MedCBLev	MeanVoltageCBPrgArUs	M	the generated name says "Mean"; read as a mistranslation of Mittelspannung (medium voltage), to be confirmed
tower_fan_program (v4.1)	TowFanLev	TowerFanPrgArUs	M	
transformer_oil_pump_program (v4.1)	TrfOilPmpLev	TrafoPumpPrgArUs	M	
pitch_error_program (v4.1)	Pitch Error Program	PtErrOpPrgArUs	H	0, 20 in the Hub workbook; 0 in all three programs read
pitch_warning_program (v4.1)	Pitch Warning Program	PtWarnOpPrgArUs	H	0 in all three programs read
rotor_brake_program (v4.1)	ROTOR_BREAK (Hub service export)	RotorBrakePrgArUs	H	20 for rotor overspeed (14000) in the generated program, while the export shows 0 for all 265 Hub rows: the export loses the value

The 4XM sheet names a 20th entry `AvlLev` with the old name "Not Available"; no generated array corresponds to it and its meaning is unknown, so it is not modelled. The 13 further Main columns are optional in the sense that 0 is a valid value; no allowed-value list is known for them yet, so, like the existing program fields, they are stored and soft-validated only.

2d. Audience / Access Fields (replaces the single access_rights_id enum)

Corrected 2026-09-17 per explicit decision: each of the eight audiences in the real export is an **independent** field with its own lookup-backed vocabulary, not one shared "Access Rights" level. A status code can simultaneously carry different values for different audiences (e.g. `service_access = "Commissioning"` while `customer_access = "See only"`). Each Engineer-proposed value is confirmed or overridden at the Administrator's Review sign-off, following the same restricted-field pattern as before (04-domain-model.md §2.9).

Field	Datatype	Required	Default	Lookup	Example
development_access	INTEGER (FK -> lookup_development_access)	No	NULL	§9d.1	SCADA User
sales_access	INTEGER (FK -> lookup_sales_access)	No	NULL	§9d.2	See only

Field	Datatype	Required	Default	Lookup	Example
tcc_access	INTEGER (FK -> lookup_tcc_access)	No	NULL	§9d.3	Telenotdienst
service_access	INTEGER (FK -> lookup_service_access)	No	NULL	§9d.4	Service and Maintenance
turbine_operator_package_access	INTEGER (FK -> lookup_top_access)	No	NULL	§9d.5	Preventive Maintenance
grid_operator_access	INTEGER (FK -> lookup_grid_operator_access)	No	NULL	§9d.6	See only
service_partner_access	INTEGER (FK -> lookup_service_partner_access)	No	NULL	§9d.7	Ext. Service Provider
customer_access	INTEGER (FK -> lookup_customer_access)	No	NULL	§9d.8	Professional

Visibility (v4, decision D-02): every row of the eight access vocabularies (§9d) carries `audience_visible` (INTEGER 0/1, NOT NULL). An audience export contains a code only when the Administrator-confirmed value for that audience has `audience_visible = 1`; a code with no value for the audience is hidden; the seeded value `Covert` has `audience_visible = 0` (BR-016).

2d.1 Numeric user-group levels (v4.1)

The generated programs do not store the named audience values; they store a read level and a write level (BYTE) for each of 16 user-group slots (`UserGroupReadArBy[Idx, slot]`, `UserGroupWriteArBy[Idx, slot]`). **Slots 1–8 are the eight audiences in the order of the export columns** (`lookup_user_group_slot`): 1 Development, 2 Sales, 3 TCC, 4 Service, 5 Turbine Operator Package, 6 Grid Operator, 7 Service Partner, 8 Customer. This held wherever it could be checked: on 6 programs for all eight audiences (a `Covert` value is read level 50 in exactly the slot of its audience) and on all 11 Main programs for the Turbine Operator Package and Service Partner slots, where the write level follows the named value. Slot 13 is never written; slots 0 and 9–15 were 50/50 in every sample (slot 9 reads 1 on Main and 50 on Hub codes, so it may mean something) and stay unassigned until Controls Engineering names them.

Observed levels: read is 1 (readable) or 50 (no access); write is 50 (no write access) or one of 5, 10, 12, 15, 20. A named value does not always fix the number: `See only` is 1/50 and `Covert` 50/50 for every audience seen, `Preventive Maintenance` 1/10, `Troubleshooting` 1/15, `Ext. Service Provider` 1/15, `Customer after warranty` `Advanced` 1/10 and Development `Operation Control` 1/20 each held on every code that carried them, but `Service and Maintenance` was 20 on one Main code and 10 on another, and `Telenotdienst` 10 on one and 20 on another. So the levels are stored per revision in `status_code_revision_user_group` (one row per slot, `read_level` and `write_level` 0–255) and the eight audience fields stay the human-readable layer. `default_read_level` and `default_write_level` on the access lookups (§9d.9) pre-fill the numbers when a named value is chosen; the Engineer may override them, and a value without a default must be entered. The existing revisions get no rows: their numbers cannot be derived from the named values for every code, so they arrive with the import from the generated programs.

Value normalization: the raw export contains inconsistent casing (`See Only` vs `See only`) and one apparent typo (`ProfessionaProfessional`). The lookups in §9d.1–§9d.8 are seeded with the corrected, de-duplicated values; the typo is normalized to `Professional`.

2e. Technical Addressing Fields (derived, read-only)

Field	Description	Datatype	Required	Default	Notes
logical_node	Controller addressing identifier	TEXT	Yes	derived	Always equals the parent Status Code's Functional System Group prefix (e.g. WCNV) — auto-populated, never independently entered; confirmed by the real export where this column exactly mirrors the sheet's group.
node_value	Controller addressing index	INTEGER	Yes	derived	Always equals the parent Functional System Group's sort_order (1–9) — auto-populated.

Both fields exist in the real export as their own columns (presumably for direct firmware/SCADA tag generation) but carry no information beyond what the Functional System Group already encodes; CSCM Tool derives and displays them rather than accepting manual entry, to guarantee they can never drift out of sync with the group assignment.

2f. Main Controller and Hub Controller fields (v4.1)

The controller of a code is the `controller_type` (`MAIN` | `HUB`) of its Functional System Group. Compared field by field, the Main workbooks, the Hub workbooks and the generated programs of both controllers give this split.

Fields	Main	Hub	Notes
Identity and classification (title, description, status_category, available_group, group and subgroup), trigger_snapshot, set_delay, reset_delay, up_down_counter, the repeat fields, manual_reset_program, auto_reset_program, brake_program, the eight audience fields, the user-group levels	yes	yes	brake_program is the Hub "Stop Program" (§2c.1). The repeat-escalation code is 9084–9090 on Main and 10050 on Hub
alarm_flag	yes	yes	the Hub generated programs carry it (FALSE in all 3) although no Hub workbook has the column; a Hub import therefore leaves it at 0
operating_mode_mask, task_call_frequency, availability_calc	yes	yes	the mask is 20 bits on Main and 6 bits on Hub; availability_calc was TRUE on all 11 Main codes and FALSE on the 3 Hub codes (all Available Group 7); none of the three is in any workbook
yaw_program, converter_reactive_power_program, umrichterprogramm_gsc, umrichterprogramm_msc, yaw_bearing_lubrication_program and the 13 further subsystem programs (§2c.1)	yes	no	the Hub programs have no such arrays
loadless_spinning_permitted, zero_load_operation	yes	no	the Hub programs have no zero-load handling. loadless_spinning_permitted is ZeroLoadPrg_ArBo: the workbook value matched it on all 11 Main codes (FALSE only for 1002, 1009, 1010). zero_load_operation is ZeroLoadOperation_ArBo (may the code be set while zero-load operation is active; TRUE for all but 1008) and appears in no workbook
pitch_error_program, pitch_warning_program, rotor_brake_program	no	yes	
stop_program_40_flag, stop_program_80_flag, manual_operation_flag, automatic_operation_flag, vane_position_flag	no	yes	see below
logical_node, node_value	derived	derived	the Hub export has the group id (11, 12, 13) in its group sheet instead of a node value

Rules: a field that does not apply to the controller of the code is stored as 0 (programs) or `NULL` (flags and parameters) and is not offered in the form; the service layer rejects any other value for it. The database stays permissive here, so a later controller generation can reuse a column without a schema change. Exports use the controller's own header (Hub: "Stop Program" for `brake_program`).

The five Hub flags and the mask: all three Hub programs read have `OpModeArDw := 2#111011` (five bits set, one clear) and all five flags True, and 244 of the 265 Hub codes in the workbook have all five True, which suggests the flags are decoded from five of the six bits. The bit order is not verified, so until it is the mask and the flags are captured independently and neither is derived from the other. The programs of the Hub codes with a different flag pattern would settle it: workbook numbers 7027, 10070, 12511, 13511, 13901 and 10250.

Descriptive text: the workbooks carry `Cause` on both controllers and `Directive` on the Hub; the tool has no column of either. Proposed mapping, to be confirmed: `Directive` -> `remedy_text`, `Cause` -> `description`. The generated programs carry the German short name in their header while the workbooks are English; `revision_translation` (§13) holds both languages.

3. status_code_revision_platform (Turbine Platform, multi-select)

At least one Turbine Platform must be selected before a revision can be submitted for Review.

Field	Description	Datatype	Required	Default	Example
id	Surrogate PK	INTEGER	Yes	auto	77
status_code_revision_id	Parent revision	INTEGER (FK)	Yes	—	5031
turbine_platform_id	Applicable platform	INTEGER (FK -> lookup_turbine_platform)	Yes	—	3XM

4. change_request, review_comment, domain_signoff

Shape as in the prior revision (05-governance-handbook.md §7–§8 for the dual-review-then-Chief-Engineer-approval sequence these tables support), with these additions (2026-10-21, v4 2026-10-23):

- `change_request.change_set_id` (v4) (INTEGER, nullable, FK -> `change_set`) — FR-124.
- `change_request.assigned_reviewer_id` (INTEGER, nullable, FK -> `user`) — FR-039. An Administrator may assign, reassign, or unassign the Reviewer while the CR is in Draft/Review and the Reviewer has not yet signed; once set, only that Reviewer may record the Reviewer sign-off. The author can never be assigned (BR-005).
- `domain_signoff_request` (new table: `id`, `change_request_id`, `engineering_domain_id`, `requested_by`, `target_user_id` nullable, `created_at`; unique on CR + domain) — FR-028. The originating Engineer's explicit request for a pending domain's sign-off, optionally addressed to one Engineer holding that domain. It surfaces on the target's Dashboard and My Change Requests until a `domain_signoff` exists for that domain; an unaddressed request is visible to every Engineer holding the domain.

5. import_job (new — library import)

Field	Description	Datatype	Required	Default	Validation	Example
id	Surrogate PK	INTEGER	Yes	auto	—	12
requested_by	Engineer or Administrator who initiated the import	INTEGER (FK -> user.id)	Yes	current user	—	14
source_format	Format of the uploaded file; ST (v4.2) = generated StCd_.st programs	TEXT (enum)	Yes	—	CHECK IN ('CSV','XLSX','ST')	CSV, XLSX, ST
source_file_reference	Path/pointer to the uploaded source file, retained for audit	TEXT	Yes	—	—	/imports/12/source.csv
row_count	Total rows in the source file	INTEGER	Yes	—	—	134
status	Job state	TEXT (enum)	Yes	PENDING	—	PENDING,VALIDATING,COMPLETE,FAILED
created_count	Number of rows successfully turned into Draft Change Requests	INTEGER	No	0	—	128
error_count	Number of rows that failed validation and were not imported	INTEGER	No	0	—	6
error_report_reference	Path/pointer to a downloadable per-row error report	TEXT	No	NULL	—	/imports/12/errors.csv
created_at	Job start timestamp	TEXT (ISO-8601)	Yes	now()	—	—
completed_at	Job completion timestamp	TEXT (ISO-8601)	No	NULL	—	—
processed_count (v4)	Rows processed so far (progress of a background job, FR-048n)	INTEGER	Yes	0	<= row_count	60
warning_count (v4)	Rows that imported with a warning (similar title, ...)	INTEGER	Yes	0	—	4
import_mode (v4)	CREATE or UPDATE (FR-048k)	TEXT (enum)	Yes	CREATE	CHECK IN ('CREATE', 'UPDATE')	UPDATE
error_report_xlsx_reference (v4)	Path of the XLSX error report (FR-048m)	TEXT	No	NULL	—	/imports/12/errors.xlsx
change_set_id (v4.2)	The change set that holds the Drafts of a status code import from programs (FR-048t)	INTEGER (FK -> change_set.id)	No	NULL	—	7

Governance note: every row that successfully validates becomes a normal Draft `StatusCode`/`StatusCodeRevision`/`ChangeRequest` triple, exactly as if an Engineer had typed it in by hand — it still must pass Cross-Domain Sign-Off (if applicable), both Review sign-offs, and Chief Engineer approval before it can be Released. Import is a faster data-entry path, not a governance bypass (confirmed 2026-09-17; see 05-governance-handbook.md §6b).

5b. The mapping and its approvals (v4.17)

Docs: 03-srs.md FR-191 … FR-198, BR-027. The tables are in `docs/artifacts/schema.sql` (v4.17) and `app/models/mapping.py`; the logic is `app/services/mapping_service.py`.

Table	Content
status_code_mapping	One row per new number (new_number unique): family (MC/HC), legacy_number (unique per family, empty for a code without a predecessor), description (the English title, 1...120), functional_group, logical_node, status_type (ERROR/WARNING/INFO), up_down_counter, parameter_set_delay / parameter_reset_delay (a time with unit, * marks a parameter value: *5s), the eight ..._access columns (the label of the named level), cause, directive, note, status, source, version (+1 at every applied change), created_at, updated_at
mapping_change	A request: title, reason, origin (FORM, WORKBOOK, SELECTION), requested_by, requested_at, state (OPEN, APPLIED, REJECTED, WITHDRAWN), closed_at
mapping_change_item	One row of a request: action (ADD, CHANGE, CONFIRM, REJECT), mapping_id (none for ADD), base_version (the version the request was made on), before_json, after_json
mapping_approval	The decisions: role_code (STCD_OWNER, CHIEF_ENGINEER, ADMINISTRATOR), user_id, decision, comment, decided_at; unique per request and role and per request and person

Status. `PROPOSED` - loaded or in the plan, not approved; `FROM_PROGRAM` - the pair is carried by a generated program; `APPROVED` - the three roles approved it; `ISSUED` - a code of the catalogue carries the number (set by the import; the numbers of the row are frozen); `REJECTED` - never to be used (the row keeps its numbers, so they are not given twice).

What is not in the mapping. Everything the programs carry: trigger snapshot, alarm, delays that are times, repeats, the subsystem programs, the numbers of the user groups, `PrgIdx_GUi`. The workbook columns for planning (priority, estimation, comment) are not carried over. The row `source` says where it came from: a workbook name and row, or `mapping change .`

The workbook (`seed_data/legacy_mapping.xlsx`, and the export): the sheets `MC` and `HC` with the columns `legacy_number`, `new_number`, `description`, `functional_group`, `logical_node`, `type`, `up_down_counter`, `parameter_set_delay`, `parameter_reset_delay`, `development`, `sales`, `tcc`, `service`, `turbine_operator_package`, `grid_operator`, `service_partner`, `customer`, `cause`, `directive`, `note`, `status`, `source`; further sheets (README, HC_groups, Glossary, Approvals, Lists) are for people.

5a. Status code import from generated programs (v4.2)

Docs: 03-srs.md §1.16, 05-governance-handbook.md §6b. The reader (`app/importers/stcd.py`) scans the text of each `PROGRAM StCd_`; it never runs or compiles anything.

What a program becomes. The number the program carries becomes `status_code.reserved_identifier` (`StCd-` + five digits); the subgroup is the one whose band holds the number; the Drafts of one confirmation are collected in one `change_set` named on `import_job.change_set_id`. Hub Controller programs carry the legacy Hub number: it is stored as `legacy_reference_number`, and the new number comes from the numbering table (proposal 1, five digits, or proposal 2, inside the Main scheme).

Program (file)	Catalogue field	Note
header StCd: , ''	legacy_reference_number; the name goes into the comment	Main Controller only; a Hub program's legacy number is its file number. The title is the English description of the mapping row (§5b); the German name is compared with it through the glossary (FR-193)
StCdArUi[Idx]	reserved_identifier	Main: the current number
AvailabilityGroupArUs	available_group	required
AlarmArBo	alarm_flag	
SetTimer(PT := ...), ResetTimer(PT := ...)	set_delay, reset_delay	T#10ms -> 10ms, T#120000ms -> 2min; a variable (StCd14000SetDelayGpTi) -> the name of the variable is entered and listed in the comment of the Draft, with a warning; no timer -> not in the program (FR-189, no value is assumed)
MaxRepeatNbArBy, RepeatTimeArDw, RepeatStCdArUi	number_of_times_repeated, repeated_over_the_course (86400 s -> 1d), status_code_for_repeated_error	
StCdWithSnap_Bo, ZeroLoadOperation_ArBo	trigger_snapshot, loadless_spinning_permitted	

Program (file)	Catalogue field	Note
AzimutPrgArUs, ManResetPrgArUs, AutoResetPrgArUs, QConverterPrgArUs, BrakePrgArUs, ConverterGSCPrgArUs, ConverterMSCPrgArUs, YawLubPrgArUs	yaw_program, manual_reset_program, auto_reset_program, converter_reactive_power_program, brake_program, umrichterprogramm_gsc, umrichterprogramm_msc, yaw_bearing_lubrication_program	
other StCd programs named in the set condition	related_status_codes	Hub: legacy numbers translated to new ones
PrgIdx_GUi	prg_idx	stored only when free and 1–65535, otherwise a warning
everything else (user-group levels, remaining subsystem programs, operating-mode mask, ...)	import_source_item.parsed_json	kept for the catalogue columns that wire them later (21-improvement-ideas.md)
not in the file	title, status_category (the type), up_down_counter, the eight audience-access values, remedy_text (cause and directive) and the value of a delay that is set by a parameter: from the mapping row (§5b), nowhere else (FR-192); a program without a row, or with a row without a type, is blocked. description (generated: file, legacy number, name in the program, where the mapping and the program differ, the list of what neither gives, set condition), turbine platforms (the uploader's choice)	

Provenance. `import_source_item` holds one row per imported program: the file name, its SHA-256, program number and line, the controller and every parsed attribute as JSON. The files themselves are kept under `instance/imports//`.

Legacy -> new mapping. Since v4.17 the mapping has tables of its own (§5b) and is the source of the pair; the catalogue view Legacy mapping is still the pair as the codes of the catalogue hold it: `status_code.legacy_reference_number` read with the Functional System Group's `controller_type` (`MAIN` -> Main Controller, `HUB` -> Hub Controller) and the identifier the code holds (`status_code_identifier`, or `reserved_identifier` while it is a Draft). A legacy number is unique per controller (the import blocks a second code with the same one). The direction is legacy -> new only: the new library holds more codes than the old one, so a code without a legacy number is normal ("n/a") and nothing maps back (FR-048u).

Guards. The service checks the reservation first; the database backs it with the unique index `ux_status_code_reserved_identifier` and the triggers `trg_identifier_matches_reservation` (a reserved code can only be issued its own number), `trg_reservation_not_issued_elsewhere` (a reservation cannot be made for an identifier already issued) and `trg_identifier_not_reserved_elsewhere` (no code can be issued a number another code reserves).

6. release / release_item

`release` has the columns of `docs/artifacts/schema.sql`; (v4) `release_target_version` (`id`, `release_id`, `software_version_id`; unique per pair) records the controller software versions a Release is published for. It can be changed only while the Release is BUILDING (BR-007, BR-021).

7. user / user_engineering_domain

`user.email` (v4.7) is the stable identity of the person: it is entered when the account is created, stored in lower case, unique, and **cannot be changed** — the trigger `trg_user_email_immutable` aborts any update that changes it, and the service and the API refuse it first (FR-162). The Windows sign-in matches the identity of the gateway with this address (FR-163).

`user.ui_language` (v4) (TEXT, a two-letter lower-case language code, default `en`; the CHECK no longer lists the codes, the application decides which exist: `en`, `de`, `fr`, `hi`, `ka`, `ar`, `bn`, `mr`, `te`, `ta`, `gu`, `ur`, `kn`, `or`, `ml`, `pa`, `es`, `it`, `da`) is the interface language preference (FR-155, FR-157).

`user.deleted_at` (v4.9) (TEXT, UTC timestamp, NULL by default): set when an Administrator deletes an account that has history (BR-025). The user name, e-mail address and name are replaced (`deleted-`, `deleted-@deleted.invalid`, Deleted user), the password is destroyed and `is_active` is 0; the row stays so that every record and audit entry keeps its author. The list and every lookup of users leave such an account out. The

trigger `trg_user_email_immutable` lets exactly this change of the address through.

`user_password_history` (added 2026-10-21): `id`, `user_id` (FK -> `user`), `password_hash` (Argon2id hash of a retired password, never plaintext), `retired_at`. Written whenever a user's password is changed or reset; a new password is rejected if it matches the current one or any of the 4 most recently retired ones (the last 5 passwords, `11-security-architecture.md` §4).

8. audit_log_entry / export_job

`audit_log_entry` gains (v4) `prev_hash` and `entry_hash` (TEXT, 64 hex characters; the first row's `prev_hash` is 64 zeros, the literal `GENESIS`); `entry_hash` is SHA-256 — HMAC-SHA-256 when `AUDIT_HMAC_KEY` is set — over the canonical JSON of the row's other fields and `prev_hash` (AUD-074). Rows written before v4 are hashed by the migration.

`export_job.export_type` accepts `CSV`, `JSON`, `XLSX`, `PDF`, `SQLITE_BACKUP` and, since v4, `HTML`, `MD`, `ZIP`; `export_scope_type` is `CATALOGUE`, `FULL_DATABASE`, `RELEASE_NOTES` or `OFFLINE_BUNDLE`. v4 adds `sha256`, `size_bytes`, `row_count`, `profile_code`, `audience`, `language_code`, `software_version_code` and `manifest` (JSON text).

9. lookup_functional_system_group

Important (2026-09-17): `code` **is not a unique key.** The Hub Controller status code library (`HC Status Code Number.xlsx`, 264 rows) confirmed `WTUR` and `WROT` are each reused for a second, numerically distinct group — same thematic prefix, different status code range, confirmed explicitly by the business. Uniqueness is on `(code, range_start)`; always resolve a group by its numeric range, never by prefix alone (`artifacts/schema.sql`).

code	label	range_start	range_end	Rows
WCNV	Converter & Grid Interface	1000	1999	134 (2026-09-17 export)
WGEN	Generator System	2000	2999	55
WNAC	Meteorology & Nacelle Environment	3000	3999	103
WROT	Rotor & Pitch System	4000	4999	165
WTOW	Tower & Structure	5000	5999	43
WTRF	Transformer & MV System	6000	6999	35
WTRM	Drivetrain & Gearbox	7000	7999	127
WYAW	Yaw System	8000	8999	33
WTUR	Turbine Control & Operation	9000	9999	202
WTUR	Hub Controller Control System	11000	11999	51 (Hub export)
WROT	Hub Controller Rotor & Pitch	12000	12999	212
WPPD	Wind Farm / Plant Dispatch	13000	13999	1

Confirmed reserved capacity (2026-09-17): `10000?10999`, `14000?14999`, `15000?15999`, and `16000?16999` are explicitly free — deliberately available for a whole new Functional System Group, not an open question needing resolution (00-INDEX.md).

9.1 Administrator-Created Functional System Groups

New capability (2026-09-17), mirroring §9a.1 one level up: an Administrator can create a brand-new Functional System Group — code, label, and numeric range — into any free range above, without a schema change. The service layer validates the proposed range does not overlap any existing group's range and returns 409 naming the conflict if it does; a database trigger (`trg_prevent_group_overlap`, `artifacts/schema.sql`) backs this up. A new group can immediately have subgroups added underneath it via §9a.1.

9a. lookup_functional_subgroup — real names, from the live export

Rule (2026-10-22): the **first** subgroup of every Main Controller group covers the group's **first 200 numbers** — e.g. `050xx` Tower 5000–5199, `020xx` Generator 2000–2199 — so the later subgroups start at the next hundred pair (x200, ...). The code prefixes (`010xx` ...) are kept as stable identifiers for imports and legacy references even though each now spans two hundred-blocks. The Hub Controller groups keep their contiguous bands.

Rule (2026-10-23) — Hub Controller bands and subgroup codes: like the Main groups, every Hub subgroup is aligned to the hundreds digit of its prefix: `123xx` owns 12300–12399, and a band shorter than a hundred is named by its first ten with a digit in place of the second `x` (`1110x` is 11100–11149, `1115x` is 11150–11199, `1205x` starts at

12050). The same rule names the Main sub-bands of the 099xx hundred (`0990x` … `0999x`). A code has **no dash and no letter**: it is the digits of the first number of the band with `x` for the free positions, so a Hub code reads like a Main code (`110xx` next to `010xx`); migration `f6c8e0a2b459` renamed the earlier `11-1xxA`, `12-0xxB`, `099xx-a` and the like. The first seed copied the legacy export, which counts from x001 (so `12-3xx` was 12301–12400 and `11-0xx` was 11000–11100, 101 numbers); migration `a1d4f6b8c203` shifted each band down by one number. The Observed usage column below is the **legacy** numbering of the export — those numbers are kept in `legacy_reference_number`, they are not identifiers — and every observed row lies inside its aligned band. `test_hub_sub_bands_are_aligned_to_the_hundreds_of_their_prefix` and `test_subgroup_codes_have_no_dash_and_no_letter_after_the_x` keep this true.

Correction (2026-09-17): the subgroup names in the prior revision (sourced from a hand-typed summary table) did not match the authoritative row-level export, confirming the "Sub Group not named properly" issue raised by the business. The table below is built directly from the real `Functional Group Name` column of `New Statuscode for sharing.xlsx` and **supersedes** the earlier version. WTUR in particular has **nine** subgroups, not four, with six of them packed into narrow sub-bands within 9900–9999 rather than one hundred-block each.

Subgroup Prefix	Group	Subgroup Name (real)	Declared sub-range	Observed usage
010xx	WCNV	Converter	01000–01199	1000–1075 (76 rows)
012xx	WCNV	Grid	01200–01299	1200–1257 (58 rows)
020xx	WGEN	Generator	02000–02199	2000–2051 (52 rows)
022xx	WGEN	Protection	02200–02299	2200–2202 (3 rows)
030xx	WNAC	Meteorology	03000–03199	3000–3052 (53 rows)
032xx	WNAC	Nacelle	03200–03299	3200–3249 (50 rows)
040xx	WROT	Rotor	04000–04199	4000–4096 (97 rows)
042xx	WROT	Hot Air De-icing	04200–04299	4200–4246 (47 rows)
043xx	WROT	Lighting	04300–04399	4300–4304 (5 rows)
044xx	WROT	Blade Sensors	04400–04499	4400–4415 (16 rows)
050xx	WTOW	Tower	05000–05199	5000–5042 (43 rows)
060xx	WTRF	Switches	06000–06199	6000–6007 (8 rows)
062xx	WTRF	Transformer	06200–06299	6200–6226 (27 rows)
070xx	WTRM	Gearbox	07000–07199	7000–7058 (59 rows)
072xx	WTRM	Drive train	07200–07299	7200–7230 (31 rows)
073xx	WTRM	Hydraulics	07300–07399	7300–7316 (17 rows)
074xx	WTRM	Brake	07400–07499	7400–7419 (20 rows)
080xx	WYAW	Yaw	08000–08199	8000–8032 (33 rows)
090xx	WTUR	System	09000–09199	9001–9091 (91 rows)
092xx	WTUR	Control System	09200–09299	9200–9258 (59 rows)
093xx	WTUR	Safety	09300–09399	9300–9319 (20 rows)
0990x	WTUR	External	09900–09929	9900–9908 (9 rows)
0993x	WTUR	Farm	09930–09959	9930–9942 (13 rows)
0996x	WTUR	Feedback Control	09960–09969	9960–9961 (2 rows)
0997x	WTUR	Interface	09970–09979	9970–9970 (1 row)
0998x	WTUR	Power Management	09980–09989	9980–9981 (2 rows)
0999x	WTUR	UPS	09990–09999	9990–9994 (5 rows)
110xx	WTUR (11000–11999, Hub)	Control System	11000–11099	11001–11036 (36 rows)
1110x	WTUR (11000–11999, Hub)	Field Bus	11100–11149	11101–11108 (8 rows)
1115x	WTUR (11000–11999, Hub)	System	11150–11199	11151–11157 (7 rows)
1200x	WROT (12000–12999, Hub)	Blade	12000–12049	12001–12025 (25 rows)
1205x	WROT (12000–12999, Hub)	Batteries	12050–12199	12051–12101 (51 rows)
1220x	WROT (12000–12999, Hub)	Pitch	12200–12249	12201–12235 (35 rows)
1225x	WROT (12000–12999, Hub)	Pitch Control	12250–12299	12251–12259 (9 rows)
123xx	WROT (12000–12999, Hub)	Pitch Drivers	12300–12399	12301–12333 (33 rows)
124xx	WROT (12000–12999, Hub)	Pitch Converter	12400–12499	12401–12445 (45 rows)
125xx	WROT (12000–12999, Hub)	Rotor	12500–12599	12501–12514 (14 rows)

Subgroup Prefix	Group	Subgroup Name (real)	Declared sub-range	Observed usage
130xx	WPPD (13000–13999)	Wind Farm	13000–13099	13001 (1 row)

38 subgroups total (27 Main Controller + 11 Hub Controller, added 2026-09-17 from `HC Status Code Number.xlsx`). Note the subgroup prefix column now includes a disambiguating group tag where the group code alone (`WTUR`, `WROT`) is not unique (§9). **Reserved capacity, by design:** every hundred-block not listed above (e.g. `011xx` between Converter and Grid, `045xx–049xx` within Rotor & Pitch System, the large `9400xx–9899xx` span within Turbine Control & Operation, and everything past `12600` within the Hub Rotor & Pitch group) is deliberately unassigned, reserved for new subgroups added later (§9a.1). A code that already exists inside a currently-unassigned block (e.g. legacy `StCd-04960`, found earlier in the live 4XM codebase, sits in the unassigned `049xx` block) predates this taxonomy and needs formal reclassification before the mandatory-subgroup rule can apply to it retroactively.

9a.1 Administrator-Created Subgroups

New requirement (2026-09-17): an Administrator can create a new Functional Subgroup within any Functional System Group. The system validates the proposed `sub_range_start`/`sub_range_end` does not overlap any existing subgroup (active or inactive) in that group — overlap is rejected with a 409 (09-api-specification.md), and the Administrator must choose a genuinely free sub-range (typically one of the reserved blocks above). No alignment to hundreds-digit boundaries is enforced by the schema — the WTUR `0990x` … `0999x` sub-blocks above prove ten-number sub-ranges are legitimate — but the UI should visually suggest the next hundred-aligned free block as a sane default (10-ui-ux-specification.md).

Inline-help text (NFR-007b, 2026-10-21): every `lookup_*` table carries a nullable `description` column. The UI shows the selected option's description under each lookup-backed dropdown, or an explicit "no description available yet" note while it is NULL. Administrators maintain descriptions in Lookup Administration (audited as `LOOKUP_CHANGE`); the real wording for the 23 Available Group codes and the audience-access values still has to come from Controls Engineering.

9b. lookup_turbine_platform

`2XM`, `3XM`, `4XM`, extensible by an Administrator.

9c. lookup_available_group

Seeded with the 23 numeric codes currently observed in the export (1,2,4,5,6,9,11,12,13,14,15,16,18,20,21,22,23,24,25,27,28,30,32); **labels are TBD** — the source export carries only the numeric code, no description of what each availability group means. One row in the raw export contained a stray `\` value, treated as a data-quality artifact from the source spreadsheet, not a valid code, and excluded from the seed.

9d. Audience / Access Lookups (8 tables, real observed values)

Each shares the shape `id`, `code`, `label`, `is_active`, `sort_order`, `audience_visible` and, (v4.1), `default_read_level` and `default_write_level` (§9d.9).

9d.1 `lookup_development_access`: `SCADA User`, `Department`, `Operation Control`, `See only`

9d.2 `lookup_sales_access`: `Sales`, `Covert`, `See only`

9d.3 `lookup_tcc_access`: `Telenotdienst`, `Telenotdienst-read`, `PMS Dispatcher`, `See only`

9d.4 `lookup_service_access`: `Commissioning`, `Service and Maintenance`, `See only`

9d.5 `lookup_top_access` (Turbine Operator Package): `Preventive Maintenance`, `Troubleshooting`, `Troubleshooting Advanced`, `Covert`, `See only`

9d.6 `lookup_grid_operator_access`: `Covert`, `See only`

9d.7 `lookup_service_partner_access`: `Service Partner`, `Ext. Service Provider`, `Customer after warranty Professional`, `Customer after warranty Advanced`, `Covert`, `See only`

9d.8 `lookup_customer_access`: `Standard`, `Advanced`, `Professional`, `Premium`, `Covert`, `See only`

Open item (handled by decision D-02): the exact meaning of `Covert` (appearing across several of these lookups) is not confirmed — it may be a mistranslation of "Covered" (i.e. included/in-scope) rather than its literal English sense; flagged for Controls Engineering confirmation before go-live, not guessed at here.

9d.9 Default user-group levels (v4.1): seeded only where every code in the 14 generated programs that carried the

named value showed the same numbers (§2d.1); all other values stay `NULL` and the level is entered per revision.

Named value	Read	Write	Lookups
See only	1	50	all eight
Covert	50	50	sales, turbine operator package, grid operator, service partner, customer
Operation Control	1	20	development
Preventive Maintenance	1	10	turbine operator package
Troubleshooting	1	15	turbine operator package
Ext. Service Provider	1	15	service partner
Customer after warranty Advanced	1	10	service partner

No default: `Service and Maintenance` and `Telenotdienst` (10 on one code, 20 on another), and the values seen on a single code only (`SCADA User` 5, `Department` 10, `Sales` 10, `PMS Dispatcher` 10, `Commissioning` 20, `Customer after warranty Professional` 12).

10. lookup_engineering_domain

`CONTROLS`, `ELECTRICAL`, `MECHANICAL`, `GRID`, `SAFETY`; extensible by an Administrator.

11. Export Formats

Since v4 an export may be shaped by an export profile, an audience, a language and a software version, can ship as a ZIP bundle with `MANIFEST.json` (and release notes), and the formats `HTML` (offline bundle) and `MD` (release notes) exist besides the four below (FR-100 … FR-111). Corrected 2026-09-17: **CSV, JSON, XLSX, and PDF** (adds XLSX; the prior revision listed only CSV/JSON/PDF). Every format is table/flat-row structured — one row per Status Code Revision with all fields as columns — including JSON, which is emitted as a flat array of row objects rather than a nested structure, mirroring the legacy tool's grid layout and the "Export for: Customer / Service" pattern observed in the legacy UI (which maps onto filtering by the relevant audience-access field, §2d).

12. Legacy Migration Traceability

The `legacy_reference_number` field (§1) exists specifically to carry the old "Status Code Number" from a source system (as seen in both the legacy UI screenshot and the `New Statuscode for sharing.xlsx` export's `Status Code Number` column) through the CSV/XLSX import process (§5), so every migrated code remains traceable back to where it came from without that old number ever being mistaken for a governed StCd identifier.

13. v4 Tables (improvement pack, schema v4)

All timestamps are ISO-8601 UTC text. FK = foreign key; "unique" lists the unique constraint. `docs/artifacts/schema.sql` has the exact DDL, CHECKs and indexes.

Table	Fields	Rules
lookup_controller_software_version	id, code (unique), label, description, is_active, sort_order	sort_order is the release order used by ranges (FR-106)
lookup_language	id, code (unique), label, description, is_active, sort_order, is_source	exactly one source language (en); de seeded (D-03); fr, hi, ka added in v4.3, ar in v4.4 and bn, mr, te, ta, gu, ur, kn, or, ml, pa in v4.5 and es, it, da in v4.6, with Georgian moved to the end (FR-157); every label reads country letters, name, flag emoji (GB English GB)
revision_translation	id, status_code_revision_id, language_id, title, description, remedy_text, translation_status (DRAFT|APPROVED), translated_by, translated_at, approved_by, approved_at	unique per revision + language; approved_by <> translated_by (BR-018)
export_profile	id, code (unique), name, description, columns (JSON list of {field, label}), language_code, audience, title_page, pdf_orientation (L|P), is_active, sort_order, created_by, created_at	seeded: service manual, SCADA tag list, customer list (FR-100)
change_set	id, name (unique), description, target_software_version_id, status (OPEN|CLOSED), created_by, created_at, closed_at	FR-124

Table	Fields	Rules
delegation	id, delegator_id, delegate_id, role_code (REVIEWER|ADMINISTRATOR|CHIEF_ENGINEER), valid_from, valid_to, created_by, created_at, revoked_at	delegator_id <> delegate_id, valid_to >= valid_from (FR-125)
notification	id, user_id, kind, message, link, change_request_id, dedupe_key, created_at, read_at, emailed_at	dedupe_key is unique; it is built from user, CR, stage, level and day, so a reminder is created at most once per combination (FR-121)
cr_attachment	id, change_request_id, filename, content_type, size_bytes, sha256, storage_path, description, uploaded_by, uploaded_at	size_bytes >= 0; limits in FR-130
field_comment	id, change_request_id, field_name, author_id, comment_text (1–2000), created_at, resolved_by, resolved_at	FR-127
review_checklist_item	id, functional_system_group_id (NULL = all groups), text (3–300), is_active, sort_order, created_at	FR-128
cr_checklist_confirmation	id, change_request_id, checklist_item_id, item_text, confirmed_by, confirmed_at	unique per CR + item; the confirmed wording is a copy, so later edits never rewrite history
saved_search	id, user_id, name (1–80), definition (JSON filters + columns), created_at	unique per user + name (FR-142)
login_attempt	id, scope (IP|USER), key, attempted_at (epoch seconds, REAL)	rows older than 15 minutes are pruned (NFR-009)
app_setting	key (PK), value, updated_by, updated_at	keys and defaults: bulk.sample_percent 10, bulk.sample_minimum 8, bulk.sample_above 5 (the sample of a bulk decision, FR-180), sla.review.amber_days 3, sla.review.red_days 5, sla.approval.amber_days 2, sla.approval.red_days 3, reviewer.auto_assign 0, health.capacity_alert_percent 15, health.capacity_alert_numbers 20; amber must not exceed red
occurrence_import	id, source_label, period_start, period_end, imported_by, imported_at, row_count, unmatched_count	period_end >= period_start (FR-143)
code_occurrence	id, occurrence_import_id, status_code_id, occurrences	occurrences >= 0; one row per code per import
api_token	id, user_id, name (1–80), token_prefix, token_hash (SHA-256), scopes (space separated), expires_at, last_used_at, revoked_at, created_at	the secret is never stored; at most 365 days (FR-144)
webhook	id, name, url, secret, events, is_active, created_by, created_at, last_delivery_at, last_status	target must be private or loopback (FR-146)
webhook_delivery	id, webhook_id, event, payload, http_status, ok, error, delivered_at	the outcome of every delivery
hub_mc_mapping	id, hub_status_code_id, mc_status_code_id, created_by, created_at	hub_status_code_id <> mc_status_code_id; one mapping per Hub code (FR-147)
lookup_user_group_slot (v4.1)	id, slot (unique, 0–15), audience_code (unique, NULL for an unassigned slot), label, description, is_active, sort_order	16 rows; slots 1–8 are the eight audiences, the others are inactive (§2d.1)
import_source_item (v4.2)	id, import_job_id, status_code_id, source_file, source_sha256, program_number, controller (MC|HC), source_line, parsed_json	unique per job, file and program; source_sha256 is 64 characters (FR-048v)
status_code_revision_user_group (v4.1)	id, status_code_revision_id, user_group_slot_id, read_level, write_level	unique per revision and slot; both levels 0–255 (§2d.1)
language_reviewer (v4.7)	id, user_id, language_code, assigned_by, assigned_at	unique per user and language; the native speakers assigned to review one language of the interface (FR-164)
translation_review (v4.7)	id, language_code, reviewer_id, status (OPEN | COMPLETED), started_at, completed_at, fingerprint, string_count	one review of one language by one reviewer; at most one OPEN review per language and reviewer (partial unique index ux_translation_review_open); fingerprint is that of the catalogue (shipped text plus overrides) the review started from
translation_review_item (v4.7)	id, review_id, string_id, english, current_text, verdict (OK | CHANGE | UNCLEAR), proposal, comment, decision (PENDING | ACCEPTED | REJECTED | NOTED), decided_by, decided_at, updated_at	unique per review and string; proposal and comment at most 1,000 characters; a CHANGE needs a proposal (CHECK); a proposal without current_text translates a string that was missing
translation_override (v4.7)	id, language_code, string_id, english, text, review_item_id, created_by, created_at	unique per language and string; the text the application shows instead of the shipped one (1–1,000 characters); deleting the row restores the shipped text

Table	Fields	Rules
terminology_term (v4.9)	id, code, category (PRODUCT | DATA | STATE | PROCESS | ROLE | AUDIENCE | KEEP), english, definition, keep_english, is_active, created_by, created_at, updated_at	unique code (lower-case letters, digits, hyphens); the technical nomenclature, one row per term (FR-166); keep_english marks a term that is never translated
terminology_translation (v4.9)	id, term_id, language_code, text (1–200 characters), status (DRAFT | APPROVED), note, proposed_by, proposed_at, approved_by, approved_at	unique per term and language; a draft until a second person approves it (BR-018 applied to the terminology); the note is the reason of a send-back; approved exactly when approved_at is set
bulk_sample (v4.13)	id, change_set_id, step (domain | reviewer | admin | chief-engineer), domain_code, drawn_by, drawn_at, seed, population, population_hash, rate_percent, status (OPEN | PASSED | FAILED | USED | VOID), used_at	the sample drawn before a bulk decision; domain_code exactly for the domain step, used_at exactly when used; the population hash ties it to the batch as it was
bulk_sample_item (v4.13)	id, sample_id, change_request_id, verdict (OK | DEFECT), note, checked_at	one sampled code; unique per sample; the verdict and the time are set together

Also (v4): `lookup_functional_system_group.controller_type` (`MAIN` \| `HUB`, default `MAIN`; Hub groups are those with `range_start >= 11000`, decision D-06), the `audience_visible` column on the eight access lookups, `change_request.change_set_id`, and the optional search index `status_code_fts` (+ `status_code_fts_vocab`, created by migration `f8b2d4e6a012` when the SQLite build has FTS5; `docs/artifacts/schema-fts5.sql`).

Also (v4.1): `status_code.prg_idx` (§1), the 25 further columns of `status_code_revision` (§2c.1, §2f), `default_read_level` and `default_write_level` on the eight access lookups (§9d.9), the triggers `trg_prevent_prg_idx_change_after_release` and `trg_require_prg_idx_for_review`, and the setting `status_code.require_prg_idx` (default 0), which only the second trigger reads and which is not yet editable in the Administration screen.

Also (v4.2): `status_code.reserved_identifier` (§1), `import_job.change_set_id` and the `ST` source format (§5), `import_source_item` (above, §5a) and the triggers `trg_identifier_matches_reservation`, `trg_reservation_not_issued_elsewhere`, `trg_identifier_not_reserved_elsewhere`.

Also (v4.7): `user.email` is fixed once the account exists, enforced by the trigger `trg_user_email_immutable` (BEFORE UPDATE OF `email` ... aborts when the value changes; FR-162), and the four translation tables above; the Windows sign-in (FR-163) adds no column, it matches the identity with `user.email`.

07. Database Design Document — CSCM Tool

1. Overview

MVP persistence is **SQLite** (single file, per C-001 in 03-srs.md), accessed exclusively through a repository layer (18-claude-code-implementation-guide.md) so a future migration to PostgreSQL requires no changes above the repository interface. The database runs on a local, on-premises server only — see 14-deployment-architecture.md §1 — and is itself the Master Database referenced throughout this package; no external upstream system exists. The full DDL is maintained at artifacts/schema.sql (schema v4.9: 65 tables, 14 triggers; the optional search index is in artifacts/schema-fts5.sql) and is the executable source of truth; this document explains the design decisions behind it.

2. Logical Schema

The logical model mirrors 04-domain-model.md. Key v2 decisions (delta from the original Controller-Type-based design):

- **Functional System Group replaces Controller Type entirely.** `status_code.functional_system_group_id` is now the sole classification and numbering axis; the Controller Type of v1 is gone as a numbering axis. (v4 re-introduces a `controller_type` flag `MAIN | HUB` on the group — not on the code — purely so Hub groups can be listed and consolidated into Main groups; it never influences numbering, decision D-06, §12.) `lookup_functional_system_group` carries the numeric range (`range_start`/`range_end`) that drives allocation (§7a).
- **Functional Subgroup is a child lookup, not a separate table per group.** One `lookup_functional_subgroup` table, FK'd to its parent group, holding a sub-range — this lets subgroups (the real taxonomy is seeded per 06-data-dictionary.md §9a) be added or amended later with a pure data change, never a schema migration. Unassigned hundred-blocks within a group's range are intentional reserved capacity, not gaps to be closed.
- **The identifier is nullable until Review.** `status_code.status_code_identifier` is `NULL` throughout Draft and is set exactly once, atomically, at the Draft->Review transition (07-database-design.md §7a) — this is why it is a plain `UNIQUE` column rather than `NOT NULL UNIQUE`: SQLite's `UNIQUE` constraint allows multiple `NULL`s by design, which is exactly the behavior needed (many concurrent Drafts, all with `NULL` identifiers, one real identifier per code from Review onward).
- **Turbine Platform is genuinely M:N.** `status_code_revision_platform` is a true join table (unlike the single-valued lookups), because a revision can legitimately apply to more than one platform.
- **Dual review is modeled as two nullable sign-off columns plus a role-tagged comment log**, not a generic "approvals" table — `reviewer_signoff_by/at` and `admin_signoff_by/at` are both on `status_code_revision` directly so the "has this revision cleared Review" check is a simple two-column NULL check, while `review_comment.decision` carries the full role-tagged history for audit purposes.
- **Chief Engineer approval is a third, independent set of columns** (`chief_engineer_approved_by/date`) plus `change_request.chief_engineer_decided_by/at`, kept structurally distinct from the Reviewer/Administrator sign-offs so segregation-of-duties checks (no overlap between author, reviewer, admin, chief engineer) are simple column comparisons, enforced redundantly at the `CHECK` constraint level (§4).
- **Sandbox status codes reuse the existing** `status_code` **table** (`is_sandbox` flag) rather than a parallel table, so the same lifecycle/validation code paths apply to sandbox testing as to real records — the only behavioral differences are numbering source, Release-eligibility (blocked by trigger, §5), and delete permission (also trigger-enforced).

3. Physical Schema

See artifacts/schema.sql for full DDL. Summary of tables:

Table	Purpose
role	6-row role lookup (Administrator, Engineer, Reviewer, Chief Engineer, Viewer, StCd Owner)
lookup_engineering_domain	Engineer specialty tags, drives Cross-Domain Sign-Off
lookup_functional_system_group	12-row numbering band lookup (9 Main Controller + 3 Hub Controller groups, §5a), with controller_type (v4)
lookup_functional_subgroup	Real sub-band lookup (38 seeded subgroups: 27 Main + 11 Hub), child of the above, with reserved capacity for more

Table	Purpose
lookup_turbine_platform	Multi-select platform lookup (2XM/3XM/4XM, extensible)
lookup_available_group	Soft-validated numeric availability codes, labels TBD (corrected 2026-09-17)
lookup_development_access, lookup_sales_access, lookup_tcc_access, lookup_service_access, lookup_top_access, lookup_grid_operator_access, lookup_service_partner_access, lookup_customer_access	Eight independent audience-access vocabularies, real observed values (corrected 2026-09-17, replaces the single lookup_access_rights)
user, user_engineering_domain	Accounts and their domain tags; the e-mail address cannot be changed (trigger, v4.7)
language_reviewer, translation_review, translation_review_item, translation_override (v4.7)	Native-speaker review of the interface translations: assignments, reviews, answers and proposals, accepted overrides (FR-164)
bulk_sample, bulk_sample_item (v4.13)	The sample inspection of a bulk decision: which codes of a batch were drawn, by whom, with which seed, and the verdict per code (FR-180)
terminology_term, terminology_translation (v4.9)	The technical nomenclature: the terms of the controller and the workflow and their translation per language, a draft until a second person approves it (FR-166)
status_code	Governed identity, nullable identifier, sandbox flag
status_code_revision	Versioned governed content, dual-review + Chief Engineer approval columns
status_code_revision_platform	Revision <-> Turbine Platform join
change_request	Workflow carrier for a revision
review_comment	Role-tagged comments/decisions on a CR
domain_signoff	Cross-Domain Sign-Off records
user_password_history	Hashes of retired passwords for the last-5 reuse rule
domain_signoff_request	Originating Engineer's request for a domain sign-off, optionally addressed to a named Engineer (FR-028)
release, release_item	Named, versioned export bundle and membership
audit_log_entry	Immutable action log
export_job	Export generation record: scope (CATALOGUE, FULL_DATABASE, RELEASE_NOTES, OFFLINE_BUNDLE), SHA-256, size, rows, profile, audience, language, software version, manifest (v4)
lookup_controller_software_version, release_target_version (v4)	Ordered software versions; the versions a Release targets (FR-106)
lookup_language, revision_translation (v4)	Content languages and per-revision translations (FR-110)
export_profile (v4)	Administrator-managed export column sets (FR-100)
change_set (v4)	Grouping of change requests for one software release (FR-124)
delegation, notification, app_setting, login_attempt (v4)	Out-of-office cover; in-app notices; Administrator settings; shared rate-limit counters
cr_attachment, field_comment, review_checklist_item, cr_checklist_confirmation (v4)	Evidence, field-level comments, review checklist and its confirmations
saved_search, occurrence_import, code_occurrence (v4)	Saved views; fleet-feedback counts
api_token, webhook, webhook_delivery, hub_mc_mapping (v4)	Machine access and Hub-to-Main mapping
status_code_fts, status_code_fts_vocab (v4, optional)	FTS5 search index and its vocabulary; absent where SQLite has no FTS5 (the search falls back to substring matching)

Field-level detail of every table: 06-data-dictionary.md (§13 for the v4 tables).

4. Primary Keys, Foreign Keys, Constraints

Surrogate `INTEGER PRIMARY KEY` throughout; foreign keys enforced (`PRAGMA foreign_keys = ON` on every connection). Natural-key uniqueness: `status_code.status_code_identifier` (nullable-safe unique, §2). `status_code_revision (status_code_id, revision_number)` unique. Segregation-of-duties is enforced at three independent `CHECK` constraints on `status_code_revision`: `reviewer_signoff_by <> created_by`, `admin_signoff_by <> created_by`, and a compound check that `chief_engineer_approved_by` differs from all three of `created_by`, `reviewer_signoff_by`, and `admin_signoff_by`. `change_request.chief_engineer_decided_by <> requested_by` mirrors this at the workflow-object level.

5. Indexes & Enforced Invariants

Index / Trigger	Enforces
ux_revision_current_per_code	Exactly one current revision per code
ux_one_open_cr_per_code (now spans DRAFT,REVIEW,PENDING_APPROVAL)	Only one open change per code at a time
trg_prevent_edit_locked_revision	No in-place edit of Approved/Released/Archived revisions
trg_prevent_release_item_delete	Published Release contents are immutable
trg_prevent_sandbox_release	A sandbox-flagged revision can never be inserted into release_item — hard-blocks the one dangerous path a sandbox code could otherwise take
trg_audit_no_update / trg_audit_no_delete (new 2026-10-21)	audit_log_entry is append-only at the database layer — any UPDATE or DELETE aborts, so not even a defect or direct SQL access through the application connection can rewrite history (FR-072, FR-073)
trg_prevent_nonsandbox_delete	DELETE FROM status_code is rejected unless is_sandbox = 1 — the single, narrow exception to the no-hard-delete rule (05-governance-handbook.md §12) is enforced at the database layer, not only in application code
ux_status_code_reserved_identifier, trg_identifier_matches_reservation, trg_reservation_not_issued_elsewhere, trg_identifier_not_reserved_elsewhere (v4.2)	A number reserved by an imported program belongs to one code only and is issued to that code unchanged (BR-023)
trg_prevent_subgroup_overlap	A new lookup_functional_subgroup row is rejected if its sub-range overlaps an existing subgroup in the same group (06-data-dictionary.md §9a.1)
Unique constraints on notification.dedupe_key, api_token.token_hash, hub_mc_mapping.hub_status_code_id, revision_translation (revision, language), release_target_version (release, version) (v4)	One reminder per user/CR/stage/level/day; no duplicate tokens; one MC replacement per Hub code; one translation per language; no duplicate targets
CHECK (approved_by <> translated_by) on revision_translation, CHECK (delegator_id <> delegate_id) on delegation, CHECK (hub_status_code_id <> mc_status_code_id) on hub_mc_mapping (v4)	BR-018, BR-017 and the mapping sanity rule at the database layer
trg_prevent_group_overlap (new 2026-09-17)	A new lookup_functional_system_group row is rejected if its range overlaps an existing group's range — lets an Administrator assign a whole new group into 10000–10999/14000–14999/15000–15999/16000–16999 safely (06-data-dictionary.md §9.1)

5a. Hub Controller Groups & Non-Unique Prefixes (new 2026-09-17)

A real export of the Hub Controller status code library (`HC Status Code Number.xlsx`, 264 rows, range 11000–13999) revealed that Hub Controller content is **not** folded into the existing nine Main Controller groups — it gets three entirely new, numerically distinct groups (`WTUR` 11000–11999, `WROT` 12000–12999, `WPPD` 13000–13999), confirmed by the business as intentional and structurally identical to the existing model (each subdivides into real subgroups the same way, 06-data-dictionary.md §9a). The one design consequence: `WTUR` and `WROT` are each now deliberately reused across two different groups (one Main Controller, one Hub Controller), so `lookup_functional_system_group.code` **is no longer a unique key on its own** — uniqueness moved to `(code, range_start)`. Every seed-data join and every application-layer lookup of a group must resolve by `(code, range_start)` or by `id`, never by `code` alone; `artifacts/schema.sql`'s subgroup seed INSERT was rewritten with an explicit `group_range_start` disambiguator for exactly this reason.

6. Soft-Delete Strategy

No hard delete of governance-relevant data, **except** sandbox Status Codes, which may be permanently deleted by an Administrator (`trg_prevent_nonsandbox_delete` is the only thing that makes this possible at all — every other record is structurally undeletable). The service layer must write a `SANDBOX_DELETE` audit entry, capturing a full JSON snapshot of the record in `before_value`, in the same transaction as the delete, so the fact of creation-and-deletion survives even though the row does not (12-audit-compliance.md §2).

7. Versioning Strategy

Revisions are stored row-per-revision and a Release is published atomically. Optimistic concurrency on Draft edits is, since v4, real: `status_code_revision.row_version` (INTEGER, starts at 1, +1 on every save) and `updated_at` back the `expected_row_version` / `expected_updated_at` check of FR-090, and a stale save changes nothing. Two further points:

7a. Atomic Identifier Allocation

Allocating the `status_code_identifier` at Draft->Review is a single transaction that must be race-safe under SQLite's single-writer model: `BEGIN IMMEDIATE` acquires the write lock before reading the current max-allocated number in the target range/sub-range, guaranteeing no two concurrent submissions can be handed the same number. The allocation query is:

```
-- within functional_system_group (no subgroup selected)
SELECT COALESCE(MAX(CAST(SUBSTR(status_code_identifier, 6) AS INTEGER)), range_start - 1) + 1
FROM status_code sc
JOIN lookup_functional_system_group g ON g.id = sc.functional_system_group_id
WHERE sc.functional_system_group_id = :group_id
  AND sc.status_code_identifier IS NOT NULL
  AND CAST(SUBSTR(sc.status_code_identifier, 6) AS INTEGER) BETWEEN g.range_start AND g.range_end;

-- within a selected subgroup, bounded to its sub_range_start/sub_range_end instead
```

If the computed next number exceeds the (sub)range's upper bound, the service layer raises a capacity-exhausted domain error rather than allocating out of range — this is treated as an operational event requiring Administrator/Controls-Engineering attention (range exhaustion should be rare given 1000-number bands, but the system does not silently overflow into a neighboring group).

7b. Dual Sign-Off & Approval Atomicity

`Review -> PendingApproval` is system-triggered, not user-triggered: the service method that records a Reviewer or Administrator sign-off checks, within the same transaction, whether the other required sign-off column is already non-null; if so, it also sets `lifecycle_status = 'PENDING_APPROVAL'` in that same write. This means the second sign-off action and the state transition are always one atomic operation — there is no window where both sign-offs exist but the state has not yet advanced.

8. Audit Strategy

One `audit_log_entry` per state-changing service call, in the same transaction. Action codes are listed in 12-audit-compliance.md §2. Since v4 a `before_flush` hook gives every new row `prev_hash` and `entry_hash` (the hash of the previous row's fields chained forward, §12); the append-only triggers stop in-application edits and the chain makes edits by someone with database access detectable.

9. Migration Path Beyond SQLite

The schema avoids SQLite-specific typing quirks and uses portable ISO-8601 text timestamps; what is SQLite-specific is inventoried in §12 and counted by CI. The PL/pgSQL equivalents of the triggers are to be produced at the time of a Phase 3 PostgreSQL migration (16-mvp-roadmap.md).

10. Locking and Transactions (v4)

SQLite runs in WAL mode (readers never block the writer) with a 15 s busy timeout, so a writer queues behind another instead of failing. Every write transaction starts with `BEGIN IMMEDIATE` (taking the write lock up front is what makes number allocation §7a, the audit chain §12 and the login rate limit race-safe across worker processes); read-only requests (GET/HEAD/OPTIONS not flagged as writing) use a plain deferred `BEGIN`. The time spent waiting for the write lock is measured and exported as the histogram `cscm_db_lock_wait_seconds` (FR-153). `PRAGMA foreign_keys = ON` is set on every connection. Both settings are applied only to SQLite connections (dialect guard in `app/__init__.py`).

11. Migrations (v4)

Alembic, one linear chain. The baseline executes `docs/artifacts/schema.sql` verbatim, so a fresh database is by construction the documented schema; later revisions are guarded (each step checks whether its change is already present) and idempotent, which is what lets the same chain upgrade both a v3 database and a fresh one.

Revision	Purpose
9dc07a07a48b	initial schema from schema.sql
b4d2f6a8c0e1	reviewer assignment, domain sign-off requests, audit append-only triggers, lookup descriptions
c5e7a9b1d3f2	password history (last-5 reuse block)
d6f8b0c2e4a3	first subgroup of each Main Controller group covers the first 200 numbers
e7a1c3d5f901	v4 schema: improvement pack I-01 ... I-36 (new tables and columns, widened export_job, audience visibility, language and profile seeds)
f8b2d4e6a012	audit hash chain (back-fills prev_hash/entry_hash with the append-only trigger temporarily dropped) and the FTS5 index (populated from existing revisions)
a1d4f6b8c203	Hub sub-bands aligned to the hundreds (each Hub subgroup shifted down by one number, e.g. Pitch Drivers 12301–12400 -> 12300–12399; guarded; refuses to run if a code is already allocated outside its new band)
c9e2a4b6d815	schema v4.1: status_code.prg_idx (unique index, frozen-after-release trigger, optional required-from-Review trigger), the further Main and Hub revision columns, the 16 user-group slots with per-revision levels, default levels on the eight audience lookups; guarded and idempotent, existing rows keep NULL / 0
b3f5a7c9d124	schema v4.2: status_code.reserved_identifier with its unique index and three guard triggers, import_job rebuilt for the ST source format and change_set_id (run with legacy_alter_table so change_request keeps pointing at it), the import_source_item provenance table; every step guarded
d4a6c8e0f236	schema v4.3: the role STCD_OWNER, the languages French, Hindi and Georgian with flags in their names, user.ui_language widened (the user table is rebuilt; foreign keys are switched off for the migration and checked at its end, migrations/env.py)
e5b7d9f1a347	schema v4.4: the language Arabic, language names as country letters, name, flag emoji; user.ui_language accepts six codes (second rebuild of user, same technique)
f6c8e0a2b459	subgroup codes without letters or dashes: 099xx-a ... -f become 0990x ... 0999x, 11-0xx becomes 110xx, 12-0xxB becomes 1205x and so on (matched on group, range and old code)
a7d9b1c3e560	schema v4.5: ten more languages (Bengali ... Punjabi) in the language lookup; user.ui_language becomes a two-letter code without a list (third and last rebuild of user for a language)
b8e0c2d4f671	schema v4.6: Spanish, Italian and Danish in the language lookup; Georgian moved to the end of the list (sort order 20)
c9f1d3e5a782	schema v4.7: the trigger trg_user_email_immutable — the e-mail address of a user can no longer be updated (FR-162); guarded, so it is created once
d0a2e4f6b893	schema v4.7: the translation review — language_reviewer, translation_review (with the partial unique index of the open review), translation_review_item, translation_override (FR-164)
e1b3f5a7c904	schema v4.9: the technical nomenclature — terminology_term, terminology_translation (FR-166)
f2c4a6b8d015	schema v4.9: user.deleted_at (an account with history is erased, BR-025) and status_code.withdrawn_at (an empty shell can be deleted, BR-026); the triggers trg_user_email_immutable and trg_prevent_nonsandbox_delete are replaced; shells that already exist are marked
h4e6c8d0f237	schema v4.13: the sample inspection of a bulk decision — bulk_sample, bulk_sample_item (FR-180)
g3d5b7c9e126	schema v4.9.2: status_code.deleted_at (a withdrawn code is erased in place, never removed, so no id is reused); the delete trigger is strict again

Tests keep this honest (`tests/integration/test_schema_upgrade.py`): a frozen copy of the v3 schema (`tests/fixtures/schema_d6f8b0c2e4a3.sql`) is filled with data and upgraded — data survives, defaults are sensible, `Covert` is hidden, Hub groups are `HUB`, old audit rows are chained and verify, the index is filled, an upgraded database has the same tables and columns as a fresh one, and the downgrade to v3 keeps the data. Before upgrading a live database the installer takes a verified backup (14-deployment-architecture.md §4).

12. Schema v4 and Portability (v4)

What v4 adds (details: 06-data-dictionary.md §13): the controller software version model (D-01), translations (D-03), export profiles, change sets, delegation, notifications, evidence, field comments and review checklist, saved searches, occurrence feedback, API tokens and webhooks, the Hub-to-Main mapping, Administrator settings, shared rate-limit counters, `audience_visible` on the access lookups (D-02), `controller_type` on the group (D-06), concurrency columns on the revision, and the audit hash chain.

Audit hash chain (AUD-074). `entry_hash = SHA-256(canonical JSON [entity_type, entity_id, action, actor_id, occurred_at, before_value, after_value, ip_address, request_id, prev_hash])`, HMAC-SHA-256 with `AUDIT_HMAC_KEY` when configured; the first row's `prev_hash` is 64 zeros. Verification walks the

table in `id` order and reports the first row whose link or hash does not fit; a stored anchor (row count and head hash, written by each successful verification to `instance/audit_chain_anchor.json`) additionally exposes truncation at the end of the table. Migration and application share one canonical form (`app/audit_hash.py`), and a test proves the migrated hash equals the one the application would have written.

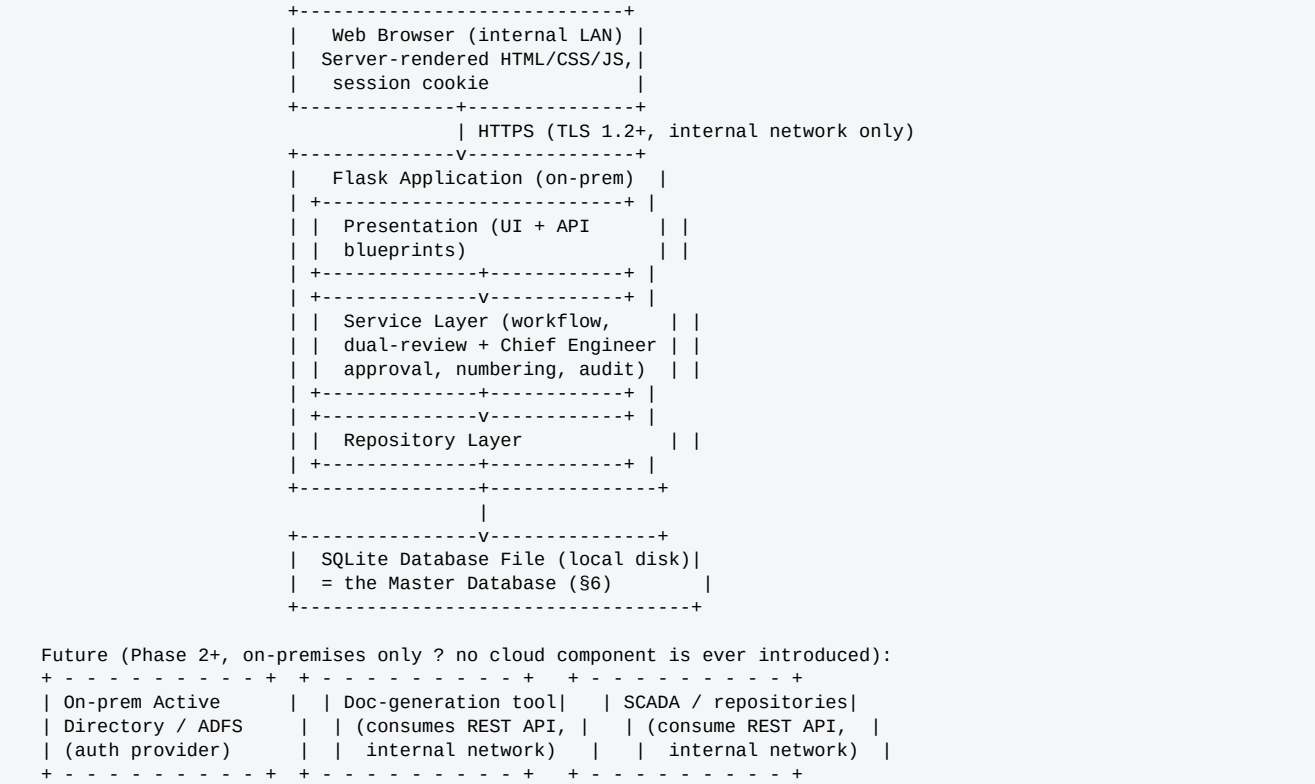
Full-text search (FR-148). `status_code_fts` (FTS5; title, description, remedy text, identifier) is maintained by four triggers (`trg_fts_revision_insert/update/delete`, `trg_fts_code_identifier`); ranking is `bm25` over the four columns, so a word in a short title outranks the same word buried in a long description; typo tolerance expands each word with close spellings from `status_code_fts_vocab`. It is optional derived data: without it the repository uses substring search.

PostgreSQL readiness (NFR-014). `python scripts/pg_readiness.py` inventories the SQLite-specific constructs in the schema, the migrations and the application code, with a porting note each, and `--check` compares the counts with `scripts/pg_baseline.json`: triggers (17 incl. FTS) and `RAISE(ABORT, ?)` -> PL/pgSQL; `strftime` defaults and `INTEGER PRIMARY KEY` -> `TIMESTAMPTZ`/identity columns; FTS5 -> `tsvector` + GIN; `BEGIN IMMEDIATE` -> `SELECT ? FOR UPDATE` on a counter row or `pg_advisory_xact_lock`; `GLOB`, `sqlite_master` and `PRAGMA` -> regex / `information_schema`; the `sqlite3` module and the online-backup API (backups, the audit check on a backup file, the full-database export) -> `pg_dump`-based equivalents. CI fails when SQLite-specific code grows, and when it shrinks until the baseline is lowered. The engine set-up is already guarded by dialect. **The application has not been run on PostgreSQL**; this is preparation, not a port.

08. System Architecture Document — CSCM Tool

1. Architecture Overview

CSCM Tool (MVP) is a **monolithic, server-rendered web application**: Python/Flask backend, SQLite database, HTML/CSS/vanilla JavaScript frontend rendered via Jinja2 templates, with a REST API layer exposed alongside the server-rendered UI for future internal integrations. The entire system runs on a single local, on-premises server (or an on-prem virtualization cluster) with **no public-cloud (SaaS/PaaS/IaaS) component of any kind** — this is an explicit, non-negotiable deployment constraint (C-004, 14-deployment-architecture.md §1), not merely a default.



1a. Presentation Layer & OneTool Shell (new 2026-09-17)

CSCM Tool's UI renders inside the shared OneTool platform shell (top bar, sidebar, navigation), per the OneTool design system supplied by the central tools team (10-ui-ux-specification.md §0). This affects only the `presentation/ui` component below — the shared shell is markup/CSS/JS that CSCM Tool includes and themes to, not a service CSCM Tool calls. **Authentication remains entirely CSCM Tool's own** (confirmed decision): the `auth` component is unaffected by this integration; only the visual chrome is shared, never identity, session, or credentials.

2. Component Diagram

Component	Responsibility	Depends on
presentation (Flask Blueprint: ui)	Server-rendered pages, forms, navigation	service (via internal calls)
api (Flask Blueprint: api_v1)	JSON REST endpoints	service
service	Business rules, lifecycle transitions, numbering allocation, dual-review/Chief-Engineer-approval orchestration, Cross-Domain Sign-Off gating, validation, audit writes	repository
repository	Data access abstraction (one repository class per aggregate)	models (SQLAlchemy), DB connection
auth	Session management, password hashing, RBAC decorators (6 roles), pluggable on-premises auth provider interface	service.user
export	CSV/JSON/XLSX/PDF catalogue export (all roles per matrix) + full-database export/backup (Administrator only)	service.release, service.status_code

Component	Responsibility	Depends on
audit	Central audit-log writer used by all services; since v4 also the hash chain hook, verification and archiving (app/audit_chain.py)	repository.audit
publication (v4)	Export pipeline with profiles, audiences, languages, software versions, manifests, release notes, offline bundle (export_service, manifest_service, audience_service, release_notes_service, translation_service, software_version_service)	service.release, service.status_code
governance extensions (v4)	Step-up, delegation, SLA ageing and notices, change sets, field comments and checklists, attachments, KPIs (auth_service, delegation_service, sla_service, notification_service, change_set_service, field_review_service, attachment_service, kpi_service)	service.change_request, repository
import (v4)	Library import toolkit: template, dry run, update mode, reconciliation, error reports, background worker (import_service)	service.status_code
integration (v4)	Scoped API tokens, change feed, webhooks, fleet-feedback import, Hub-to-Main consolidation (integration_service, occurrence_service, hub_consolidation_service)	service, repository
operations (v4)	Backups with verified restore, readiness and metrics, health report, settings, logging, CLI jobs (backup_service, observability, health_report_service, settings_service, cli)	repository, OS scheduler
i18n (v4)	English/German interface catalogue, help content (app/i18n.py, app/i18n_de.py, app/help_content.py)	presentation

Full detail and code layout: 18-claude-code-implementation-guide.md.

3. Deployment Diagram

See 14-deployment-architecture.md for complete environment detail — a single on-premises server (or on-prem cluster), reverse proxy terminating TLS, WSGI-hosted Flask app, local-disk SQLite file with nightly backup, all inside the corporate network with no external connectivity by default.

4. Data Flow Diagram (Change Request Lifecycle, v2)

```
[Engineer] --create Draft (select Functional Group [+Subgroup])--> [StatusCodeService.create_draft()]
      |                                    (identifier = NULL at this point)
      |
      +--[if cross-domain fields unset]--> flagged "awaiting domain input"
      |         |
      |         +--[second Engineer, correct domain]--> [DomainSignoffService.sign()] --> domain_signoff row
      |
      +--submit for Review--> [ChangeRequestService.submit()]
                  |
                  +--validates completeness + all domain sign-offs present
                  +--ALLOCATES status_code_identifier atomically (07-database-design.md §7a)
                  +--transitions Draft -> Review
                  |
      [Reviewer] --sign off--> [ChangeRequestService.reviewer_signoff()] --\
      [Administrator] --sign off--> [ChangeRequestService.admin_signoff()] --+--> on 2nd sign-off,
                                                                               system auto-transitions
                                                                               Review -> PendingApproval
                  |
      [Chief Engineer] --approve/reject--> [ChangeRequestService.chief_engineer_decide()]
                  |
                  +--approve--> PendingApproval -> Approved
                  +--reject (mandatory comment)--> PendingApproval -> Draft
                  |
      [Administrator] --publish Release (bundles Approved revisions)--> [ReleaseService.publish()]
                  |
                  +--atomic: Approved -> Released, effective_date stamped, release locked
```

Every arrow above writes exactly one (or, for the Release publish fan-out, one-per-item) `audit_log_entry` in the same transaction (12-audit-compliance.md).

5. Sequence Diagrams

5.1 Submit for Review (identifier allocation)

```
Engineer -> UI: click "Submit for Review"
UI -> API: POST /api/v1/change-requests/{id}/submit
API -> ChangeRequestService: submit(cr_id, actor=Engineer)
ChangeRequestService -> ValidationService: validate_revision(revision)
ChangeRequestService -> DomainSignoffService: all_required_domains_signed(revision)
alt validation fails OR domain sign-offs incomplete
```

```
    ChangeRequestService --> API: 422 (field errors / missing domain sign-offs)
else all clear
    ChangeRequestService -> Repository: BEGIN IMMEDIATE
    ChangeRequestService -> Repository: allocate_next_identifier(group, subgroup)  [07-database-design.md §7a]
    ChangeRequestService -> Repository: update status_code.status_code_identifier
    ChangeRequestService -> Repository: update revision.lifecycle_status = REVIEW
    ChangeRequestService -> Repository: update change_request.state = REVIEW, submitted_at
    ChangeRequestService -> AuditService: log(SUBMIT_FOR_REVIEW, IDENTIFIER_ALLOCATED)
    Repository -> SQLite: COMMIT
    ChangeRequestService --> API: 200 OK (StCd-01232 now visible)
end
```

5.2 Dual Review Sign-Off

```
Reviewer -> UI: click "Sign Off" (with optional comment)
UI -> API: POST /api/v1/change-requests/{id}/reviewer-signoff
API -> AuthZ: assert role == REVIEWER AND actor != cr.requested_by
API -> ChangeRequestService: reviewer_signoff(cr_id, actor, decision, comment)
ChangeRequestService -> Repository: BEGIN IMMEDIATE
ChangeRequestService -> Repository: insert review_comment (decision=REVIEWER_APPROVE|REJECT)
alt decision == REJECT
    ChangeRequestService -> Repository: revision.lifecycle_status = DRAFT; clear reviewer_signoff_by/at
else decision == APPROVE
    ChangeRequestService -> Repository: revision.reviewer_signoff_by/at = actor/now
    ChangeRequestService -> Repository: if revision.admin_signoff_by IS NOT NULL:
                                            revision.lifecycle_status = PENDING_APPROVAL
end
ChangeRequestService -> AuditService: log(REVIEWER_SIGNOFF)
Repository -> SQLite: COMMIT
```

(The Administrator sign-off endpoint is symmetric, with `admin_signoff_by/at` and the same second-signoff check against `reviewer_signoff_by`.)

5.3 Chief Engineer Approval

```
Chief Engineer -> UI: click "Approve" / "Reject"
UI -> API: POST /api/v1/change-requests/{id}/chief-engineer-decide
API -> AuthZ: assert role == CHIEF_ENGINEER AND actor NOT IN {requested_by, reviewer_signoff_by, admin_signoff_by}
API -> ChangeRequestService: chief_engineer_decide(cr_id, actor, decision, comment)
ChangeRequestService -> Repository: BEGIN IMMEDIATE
ChangeRequestService -> Repository: insert review_comment (CHIEF_ENGINEER_APPROVE|REJECT)
alt approve
    ChangeRequestService -> Repository: revision.chief_engineer_approved_by/date = actor/now; lifecycle_status = APP
  ROVED
else reject
    ChangeRequestService -> Repository: revision.lifecycle_status = DRAFT; clear both sign-off pairs
end
ChangeRequestService -> AuditService: log(CHIEF_ENGINEER_APPROVE | CHIEF_ENGINEER_REJECT)
Repository -> SQLite: COMMIT
```

5.4 Sandbox Status Code Create/Delete

```
Administrator -> UI: "New Sandbox Status Code"
UI -> API: POST /api/v1/status-codes {is_sandbox: true, ...}
API -> StatusCodeService: create_sandbox(actor=Administrator)
StatusCodeService -> Repository: allocate from isolated StCd-T##### counter (never touches real ranges)
StatusCodeService -> AuditService: log(SANDBOX_CREATE)
...
Administrator -> UI: "Delete Sandbox Code"
UI -> API: DELETE /api/v1/status-codes/{id}
API -> AuthZ: assert role == ADMINISTRATOR AND status_code.is_sandbox == true
API -> StatusCodeService: delete_sandbox(id, actor)
StatusCodeService -> AuditService: log(SANDBOX_DELETE, before_value=full snapshot)
StatusCodeService -> Repository: DELETE FROM status_code WHERE id=:id   [DB trigger permits only is_sandbox=1]
Repository -> SQLite: COMMIT
```

5a. Added Request Flows (v4)

- **Step-up and delegation (FR-125, FR-126).** A decision route is guarded by `require_role_or_delegate`: the actor's role, or an active `delegation` for that role, admits the call; `AuthService.step_up` then verifies the password in the request body (once per request — a change-set decision reuses `g.step_up_verified`) and audits `STEP_UP_SUCCESS` / `STEP_UP_FAILURE`; only then does the service run its usual segregation-of-duties checks and write the audit entry naming actor and `on_behalf_of`.
- **Export (FR-100 … FR-111).** `ExportService._prepare` selects revisions (release, released set or working set),

applies the audience filter and software-version filter, resolves translations, and builds the rows with the profile's columns; `_render` produces CSV/JSON/XLSX/PDF; `manifest_service` hashes the file(s) and signs the manifest; `_record` stores the `export_job` (hash, size, rows, options, manifest) and an audit entry; with `bundle` the data file, `MANIFEST.json` and the release notes are zipped.

- **Publish (FR-043, FR-106, FR-146).** Password check -> state checks (Release still BUILDING, at least one item) -> atomic transition of all items to Released with one audit entry each -> commit -> webhook dispatch after the commit (the candidate list, limited to the target versions, is built earlier by the release builder) (a delivery problem is recorded and never undoes or fails the publish).
- **Import (FR-048i ... FR-048n).** Upload -> header check -> per-row evaluation (rules of manual entry plus duplicate and legacy-number checks) -> dry run stops here, otherwise rows are applied one transaction each (create, update in place, new revision, skip) -> error report CSV/XLSX -> job COMPLETE; above the row threshold the apply phase runs in a worker thread and the request returns 202.
- **Audit write (FR-151).** Every `audit_log_entry` insert passes the `before_flush` hook, which reads the last stored `entry_hash` inside the same write transaction and chains the new rows in insertion order.

6. Master Database & Integration Architecture

CSCM Tool's own on-premises SQLite database **is** the Master Database — there is no separate, external "master" system this application synchronizes with. The only ways data leaves the system:

Consumer	Mechanism	Scope	Role Required
Technical Publications / manuals	Administrator- or role-scoped catalogue export (CSV/JSON/XLSX/PDF) of Approved/Released status codes	Approved/Released only, per role matrix	Per 11-security-architecture.md §3 (all roles except sandbox-only data)
Disaster recovery / migration	Full database export/backup	Entire database	Administrator only (FR-047, 04-domain-model.md §2.17)
Documentation-generation tools, SCADA, software repositories	REST API pull, internal corporate network only, never internet-exposed, never a cloud endpoint	Released catalogues only (Viewer-level visibility)	Scoped, expiring API token, Administrator-issued (FR-144, SEC-027)
The same consumers, event-driven	GET /changes change feed (polling, ETag) and signed webhooks on release.published to private/loopback addresses	Release and lifecycle events	Token with scope changes:read; webhook secret (FR-145, FR-146, SEC-028)
Monitoring	GET /health, GET /health/ready (unauthenticated, no business data), GET /metrics (allow-listed networks, optional bearer token)	Operational state only	Network allow-list (SEC-029)

No system pushes data **into** CSCM Tool — status codes are authored exclusively inside the application (this is the entire premise of the tool per 01-executive-summary.md §1), so there is no inbound "connect to Master DB" integration to build; the relevant integration surface is outbound/export, as summarized above. The only data that enters other than by typing is file uploads started by a person: the legacy library import (FR-048a, which creates Drafts only, BR-015) and the anonymised occurrence counts of the fleet-feedback import (FR-143, which change no status code).

7. Authentication Provider Abstraction

The `auth` component defines an `AuthProvider` interface with one MVP implementation (`SessionPasswordAuthProvider`) and a reserved seam for a future **on-premises** identity provider:

```
class AuthProvider(Protocol):
    def authenticate(self, credentials: dict) -> AuthenticatedUser | None: ...
    def get_login_url(self) -> str | None: ...   # None for session/password
```

Correction from the original stack assumption: the future SSO path is **on-premises Active Directory (LDAP/Kerberos) or on-prem ADFS/SAML federation**, not Azure AD/OIDC — Azure AD is a Microsoft-hosted cloud identity service and is incompatible with the no-cloud-infrastructure requirement (00-INDEX.md "Deployment Principle", 11-security-architecture.md §2, 14-deployment-architecture.md §1). Flask route handlers and the service layer depend only on `AuthenticatedUser`, never on `AuthProvider` implementation details, so this remains an additive change (16-mvp-roadmap.md Phase 2), just against a different, on-prem-compatible target.

8. Scalability, Caching, Backup Strategy

MVP sizing targets, SQLite WAL-mode read concurrency, in-process lookup caching and nightly SQLite Online Backup API snapshots (now verified, see below). All infrastructure referenced there is local/on-premises; no external or cloud storage target is introduced by this update.

Multi-process operation (v4, NFR-009). Waitress may run as several processes against the one SQLite file: anything that must be consistent between them lives in the database (the login rate limit in `login_attempt`, the audit chain read from the last stored row inside the write transaction, number allocation under `BEGIN IMMEDIATE`); the in-process metrics registry is per process and each process is scraped separately. Large imports run in a daemon thread of the process that received the upload (FR-048n); the job row, not the thread, is the source of truth for progress, so another process or a restart can report it. Behind a reverse proxy the real client address reaches the rate limit through `CSCM_TRUSTED_PROXY` (`scripts/serve.py`).

Backups, readiness and metrics (v4). `flask backup-db` copies the live database with SQLite's online-backup API, proves the copy (integrity check, restore into a scratch database, row-count comparison, audit-chain check) and writes `backup_status.json`; `GET /health/ready` reads it together with the schema's Alembic head and the newest audit rows; `GET /metrics` exposes request, latency, lock-wait, queue and backup gauges (FR-150, FR-152, FR-153; procedures in 14-deployment-architecture.md §6–§8).

Transaction modes (added 2026-10-21, NFR-004): only writers take the SQLite write lock up front (`BEGIN IMMEDIATE`, required for race-safe identifier allocation, 07-database-design.md §7a). Read-only requests — `GET`/`HEAD`/`OPTIONS` not marked with `@writes_on_get` (`app/db_modes.py`; the export endpoints are, because they record an `ExportJob` and an audit entry) — use a deferred `BEGIN`, so in WAL mode readers neither block nor are blocked by a writer. Taking the write lock for every transaction serialised the whole application and produced `database is locked` errors with 50 concurrent users; `busy_timeout` is 15 s so writers queue instead of failing fast. Covered by `tests/performance/test_performance.py`.

09. API Specification — CSCM Tool

Full machine-readable contract: artifacts/openapi.yaml (OpenAPI 3.0.3, **v4.0.0**, 171 operations — field set corrected 2026-09-17 against a real export of the live status code library, extended 2026-10-23 by the improvement pack). A test (`test_openapi_conformance`) fails when the code and this contract differ, and the endpoint table in §7 is generated from the code (`scripts/gen_api_summary.py`). This document explains the conventions that apply across every endpoint.

1. Versioning

All endpoints are namespaced under `/api/v1`. The v2 documentation-package revision (this update) changes the contract, not the version prefix, since the application has not yet shipped — `/api/v1` remains correct for the first production release, which will already reflect this model. A genuinely breaking change post-launch requires `/api/v2`.

2. Authentication

MVP: session-cookie authentication (`cscm_session`, HttpOnly, Secure, SameSite=Strict), served entirely over the internal on-premises network. Future non-browser API consumers (documentation-generation tools, SCADA, software repositories — 08-system-architecture.md §6) will use a scoped, Administrator-issued API token (`Authorization: Bearer`), still internal-network-only, never internet-exposed and never involving a cloud identity provider. Every endpoint requires authentication except `POST /auth/login`.

3. Authorization

Role-based across six roles (Administrator, StCd Owner, Engineer, Reviewer, Chief Engineer, Viewer), enforced identically whether the request originates from the server-rendered UI or a direct API call. The canonical permission matrix is 11-security-architecture.md §3. Segregation-of-duties is enforced server-side on all three decision endpoints:

- `POST /change-requests/{id}/reviewer-signoff` — 403 if `actor.id == change_request.requested_by`.
- `POST /change-requests/{id}/admin-signoff` — 403 if `actor.id == change_request.requested_by`.
- `POST /change-requests/{id}/chief-engineer-decide` — 403 if `actor.id` matches the requester, the Reviewer sign-off, or the Administrator sign-off on this CR.

`DELETE /status-codes/{id}` (sandbox deletion) additionally requires `status_code.is_sandbox == true`; the same endpoint against a non-sandbox record returns 403 regardless of role.

`POST /exports/full-database` (a `GET` before v4) requires the Administrator role and the Administrator's `current_password`; no other role can reach it (04-domain-model.md §2.17). Since v4 a delegate is admitted to the **decision** routes (`reviewer-signoff`, `admin-signoff`, `chief-engineer-decide`) when an active delegation covers the role (FR-125); no other route honours delegation.

4. Request/Response Conventions

JSON bodies, ISO-8601 UTC timestamps, lookup-backed fields returned/accepted by `code` string (`functional_system_group: "WCNV"`, `turbine_platforms: ["2XM","3XM"]`, etc.) rather than internal surrogate IDs. `status_code_identifier` is `null` in every response until the revision reaches `REVIEW` or later.

5. Error Handling

HTTP Status	Meaning	Body
400	Malformed request	Error
401	Not authenticated	Error
403	Not authorized (role, segregation-of-duties, sandbox-only-delete, domain-signoff-wrong-domain); error: step_up_required / step_up_failed when a sensitive action lacks or mistypes the actor's password (FR-126); account_locked	Error
404	Resource does not exist	Error

HTTP Status	Meaning	Body
409	Conflict — duplicate identifier (should not occur given atomic allocation, but guarded), optimistic-concurrency mismatch (error: conflict, a stale expected_row_version / expected_updated_at, FR-090), invalid state transition, open-CR-already-exists (BR-003), sandbox code added to a Release	Error
422	Semantically invalid input, OR required Cross-Domain Sign-Offs incomplete at Submit-for-Review	ValidationErrorBody
429	Rate limit exceeded (login attempts; shared by all worker processes, NFR-009)	Error
503	Readiness probe: database unreachable, schema not at the expected head, or audit chain broken (GET /health/ready only)	Readiness
202	Success, not an error: an import is accepted — finished for small files, processed in the background above the row threshold (FR-048n)	ImportJob
304	Success, not an error: GET /changes with a matching If-None-Match	—
500	Unexpected server error	Error

6. Pagination, Filtering, Sorting

`page`/`page_size` envelope (default 50, maximum 200), `sort` with a `-` prefix for descending. Filters on `GET /status-codes`: `functional_system_group`, `functional_subgroup`, `turbine_platform`, `is_sandbox` (defaults to `false`, so sandbox records never appear in default search results — matching their exclusion from catalogues). Full-text search (`q`) covers `status_code_identifier`, `title`, `description`.

6a. Mandatory Fields on Creation (corrected 2026-09-17)

`POST /status-codes` requires both `functional_system_group` **and** `functional_subgroup` — the prior contract treated the subgroup as optional; real production data confirmed every status code carries one, so `StatusCodeCreate` now lists it in `required` (`artifacts/openapi.yaml`). A request omitting it returns 422, the same as any other missing mandatory field.

6b. Library Import

`POST /imports` accepts a CSV or XLSX file (`multipart/form-data`; since v4 also `mode=CREATE|UPDATE`, `dry_run=true` — a report that creates nothing, not even a job — and `async=true`). The answer is **202 Accepted** with the `ImportJob`: files up to the row threshold (default 300) are already finished when the response is sent (`status COMPLETE`), larger files or `async=true` are processed in a background worker and the job is polled; a dry run answers 200 with the per-row report. `GET /imports/template` returns the template (CSV or XLSX); `GET/POST /imports/reconciliation` return the migration progress (FR-048i … FR-048n). `GET /imports/{id}` polls status; `created_count/error_count` report progress, and `GET /imports/{id}/errors` downloads a per-row error report once the job completes. Every successfully validated row produces a normal Draft `ChangeRequest` (`cr_type=IMPORT`) that must still pass Cross-Domain Sign-Off, both Review sign-offs, and Chief Engineer approval — import never creates an Approved or Released record directly (05-governance-handbook.md §6b).

6c. Subgroup & Group Administration

`POST /lookups/functional-subgroups` (Administrator only) creates a new Functional Subgroup within an existing Functional System Group. The service layer validates the proposed `sub_range_start`/`sub_range_end` against every existing subgroup in that group and returns 409 with the conflicting subgroup's name on overlap; a non-overlapping request succeeds even without hundreds-digit alignment (06-data-dictionary.md §9a.1).

`POST /lookups/functional-system-groups` (Administrator only, new 2026-09-17) creates a brand-new Functional System Group into any currently free numeric range (`10000?10999`, `14000?14999`, `15000?15999`, `16000?16999`), validated for non-overlap against every existing group the same way (06-data-dictionary.md §9.1). No schema change is ever required for either operation.

6d. Status Code Import from Generated Programs (v4.2)

Two steps, so nothing is created before a person has looked (FR-048o … FR-048w). `POST /imports/st/preview` takes `multipart/form-data` with one or more `files` (`.st`, or one `.zip`) and the options `controller` (`auto|MC|HC`), `hc_proposal` (`1|2`), `default_category` (optional - no category is assumed; a program without one is refused at confirmation with 422), `alarm_type_categories` (JSON, alarm type -> category) and `platforms`; it answers **200** with

a `token`, a summary and one item per program (`key`, `identifier`, `legacy_number`, `controller`, `alarm_type`, `category`, `group`, `subgroup`, `state` = `ready|exists|blocked`, `issues`, `warnings`) and creates nothing but the caller's staged copy. `POST /imports/st` with `{token, selected?, categories?, change_set_name?, async?}` (`categories` sets the status category per item key, FR-181) creates the Drafts for the chosen items (default: all that are ready) and answers **202** with the `ImportJob` (`source_format` `ST`, `change_set_id`); an unknown, expired or foreign token is 404 or 403, nothing selected or a program that is no longer ready is 422. `GET /status-codes/{id}/import-source` returns the provenance of an imported code. `GET /legacy-mapping` (filters `controller`, `q`, `state`) and `GET /legacy-mapping/export` (`format=csv|xlsx`) give the legacy -> new mapping for the internal roles; there is no reverse lookup (FR-048u).

7. Endpoint Summary

Generated from `docs/artifacts/openapi.yaml` and the access declared on each route (`python scripts/gen_api_summary.py --write`; a test fails if this table is out of date). "Who may call it" is the route-level gate; row-level rules — own change request, segregation of duties, sandbox-only delete, Released-only visibility for tokens and Viewers — are applied by the service layer (§3). Audit-relevant detail of each operation is in the OpenAPI description.

<!-- api-summary:start -->

Method	Path	Purpose	Who may call it
POST	/auth/windows	Sign in with the Windows identity the authenticating gateway supplies (v4.7, FR-163): the header WINDOWS_AUTH_HEADER (default X-Remote-Us...	Public
POST	/auth/login	Authenticate and establish a session	Public
POST	/auth/logout	Invalidate current session	Any signed-in user
GET	/auth/csrf-token	Fetch the CSRF token to send as X-CSRFToken on every state-changing request	Public
GET	/health	Liveness probe (no authentication)	Public
GET	/auth/me	Current authenticated user	Any signed-in user
GET	/users	List users (Administrator only)	Administrator, StCd Owner
POST	/users	Create a user, including role and Engineering Domain tags (Administrator only)	Administrator
GET	/users/{userId}	Get a user	Administrator, StCd Owner
PATCH	/users/{userId}	Update a user, role, or Engineering Domain tags (Administrator only)	Administrator, StCd Owner
POST	/users/me/change-password	Change own password (policy enforced; other sessions are invalidated; FR-013)	Any signed-in user
POST	/users/{userId}/force-reset-password	Force a password reset for a user (Administrator only; invalidates their sessions; FR-014)	Administrator
GET	/status-codes	List/search status codes (current revision per code)	Any signed-in user; API token catalogue:read
POST	/status-codes	Create a new Status Code (Draft; no identifier assigned yet — functional_subgroup is mandatory, never optional) or a Sandbox Status Code...	Administrator, Engineer
GET	/status-codes/{statusCodeId}	Get current revision of a status code	Any signed-in user; API token catalogue:read
DELETE	/status-codes/{statusCodeId}	Permanently delete a Sandbox Status Code (Administrator only; blocked at the DB layer for any non-sandbox record)	Administrator
POST	/status-codes/{statusCodeId}/deprecation-requests	Open a Deprecation change request against a Released code (Administrator or Engineer)	Administrator, Engineer
POST	/status-codes/{statusCodeId}/archive	Archive a Deprecated code once the minimum retention period has elapsed (Administrator only; FR-051, BR-006)	Administrator
POST	/status-codes/{statusCodeId}/reinstate	Reinstate a Deprecated code to Released (Administrator only; exception path, reason required)	Administrator
GET	/status-codes/{statusCodeId}/revisions	Full revision history for a status code	Any signed-in user
POST	/status-codes/{statusCodeId}/revisions	Open a new Draft revision (change request) against an existing code	Administrator, Engineer
GET	/status-codes/{statusCodeId}/revisions/{revisionA}/diff/{revisionB}	Field-level diff between two revisions	Any signed-in user

Method	Path	Purpose	Who may call it
PATCH	/revisions/{revisionId}	Edit a Draft revision's fields (Engineer, own CR only)	Administrator, Engineer
GET	/change-requests	List change requests (review/approval queues)	Working roles (not Viewer) and StCd Owner (read only)
GET	/change-requests/{crId}	Get change request detail (incl. comments and domain sign-offs)	Working roles (not Viewer) and StCd Owner (read only)
POST	/change-requests/{crId}/submit	Submit Draft for Review (Draft -> Review)	Administrator, Engineer
POST	/change-requests/{crId}/reviewer-signoff	Reviewer (Sub-PO for StCd) sign-off	Reviewer
POST	/change-requests/{crId}/admin-signoff	Administrator (Product Owner) sign-off	Administrator
POST	/change-requests/{crId}/chief-engineer-decide	Chief Engineer final Approval decision (PendingApproval -> Approved / Draft)	Chief Engineer
POST	/change-requests/{crId}/withdraw	Withdraw own Draft/Review/PendingApproval CR	Administrator, Engineer
POST	/change-requests/{crId}/comments	Add a comment without a decision, at any stage (Reviewer, Administrator, Chief Engineer; FR-038)	Administrator, Reviewer, Chief Engineer
PUT	/change-requests/{crId}/reviewer-assignment	Assign, reassign or (reviewer_id null) unassign the Reviewer (Administrator only; FR-039)	Administrator
POST	/change-requests/{crId}/domain-signoff-requests	The originating Engineer requests a pending domain's sign-off, optionally from one named Engineer holding that domain (FR-028)	Administrator, Engineer
GET	/domain-signoff-requests	Open domain sign-off requests awaiting the current Engineer (addressed to them, or open to their domains)	Engineer
GET	/change-requests/{crId}/domain-signoffs	List required and completed Cross-Domain Sign-Offs for a CR	Working roles (not Viewer) and StCd Owner (read only)
POST	/change-requests/{crId}/domain-signoffs	Record a Cross-Domain Sign-Off (actor must hold the named Engineering Domain)	Engineer
GET	/users/{userId}/usage	What the account has done in the tool (the rows of every other table that point at it) and whether a delete removes the account or erases...	Administrator, StCd Owner
POST	/users/{userId}/delete	Delete an account (Administrator, with the Administrator's own password)	Administrator
POST	/change-requests/withdraw	Withdraw several of your own open change requests in one call (FR-159)	Administrator, Engineer
GET	/imports	List import jobs	Administrator, Engineer
POST	/imports	Upload a CSV or XLSX status code library file	Administrator, Engineer
GET	/imports/{importJobId}	Get import job status, including created/error counts	Administrator, Engineer
GET	/imports/{importJobId}/errors	Download the per-row error report (row, column, value, problem, how to fix) for a completed import job (CSV, or XLSX with format=xlsx)	Administrator, Engineer
GET	/releases	List releases	Any signed-in user; API token catalogue:read
POST	/releases	Create a Release (Administrator only)	Administrator
GET	/releases/{releaseId}	Get release detail (incl. items)	Any signed-in user; API token catalogue:read
DELETE	/releases/{releaseId}	Delete an unpublished Release (Administrator only; BR-008)	Administrator
GET	/releases/{releaseId}/candidates	List Approved (non-sandbox) revisions matching the release's scope filter	Administrator
PUT	/releases/{releaseId}/items	Set the included revision set (pre-publish only; sandbox revisions are rejected)	Administrator
POST	/releases/{releaseId}/publish	Publish the release (atomic; Administrator only)	Administrator
GET	/releases/{releaseId}/export	Export a release catalogue as a flat table (role-scoped, per 11-security-architecture.md §3)	Any signed-in user; API token catalogue:read
GET	/exports/released	Export the current set of all Released (non-sandbox) status codes as a flat table (any authenticated role; FR-045)	Any signed-in user; API token catalogue:read
GET	/exports/working-set	Ad-hoc filtered working set in any lifecycle state (never sandbox), for the StCd Owner and the Administrator only (FR-046, FR-175)	Administrator, StCd Owner
POST	/exports/full-database	Export a full raw database backup (Administrator only)	Administrator

Method	Path	Purpose	Who may call it
GET	/lookups/{vocabulary}	List active values of a controlled vocabulary	Any signed-in user
POST	/lookups/{vocabulary}	Add a value to a simple controlled vocabulary (Administrator only; functional groups/subgroups have their own endpoints)	Administrator
PATCH	/lookups/{vocabulary}/{valueId}	Relabel, describe (inline-help text), (de)activate or reorder a vocabulary value (Administrator only; values are never deleted)	Administrator
POST	/lookups/functional-system-groups	Create a brand-new Functional System Group (Administrator only), into any currently free range (10000-10999, 14000-14999, 15000-15999, 16...	Administrator
POST	/lookups/functional-subgroups	Create a new Functional Subgroup (Administrator only)	Administrator
GET	/status-codes/next-available	Preview (non-binding) the next available identifier for a Functional Group + Subgroup, for UI display only — not allocated until Submit f...	Administrator, Engineer
GET	/audit-log	Search audit log (Administrator; Reviewer/Chief Engineer read-only subset)	Administrator, Reviewer, Chief Engineer
GET	/health/ready	Readiness probe (no authentication, I-21): database reachable, schema at the expected Alembic head, newest audit rows chain correctly, la...	Public
GET	/status-codes/similar	Existing codes of a subgroup with a similar title ("did you mean to revise ...?", I-12)	Administrator, Engineer
GET	/status-codes/{statusCodeId}/where-used	Every other code whose current revision refers to this one (repeated-error pointer, related codes, supersession, platform variant) - I-04	Working roles (not Viewer) and StCd Owner (read only)
GET	/change-sets/{changeSetId}/readiness	What still stops each Draft of the batch from being submitted, with the rule of the sample inspection (FR-180)	Working roles (not Viewer) and StCd Owner (read only)
POST	/change-sets/{changeSetId}/bulk-edit	Set the same status category, platforms, available group or access levels on the Drafts of the batch (FR-180); each Draft is saved on its...	Administrator, Engineer
POST	/change-sets/{changeSetId}/bulk-submit	Submit for review the Drafts of the batch that the caller wrote (FR-180); one that is not ready is reported with the reason	Administrator, Engineer
POST	/change-sets/{changeSetId}/domain-signoffs	Sign off an engineering domain on the Drafts of the batch (FR-180); more than a handful of codes need a passed sample_id	Administrator, Engineer
POST	/change-sets/{changeSetId}/samples	Draw the sample a bulk decision at step needs (FR-180) - stratified by subgroup, drawn for the caller, valid 24 hours	Working roles (not Viewer)
GET	/change-sets/{changeSetId}/samples	The samples drawn for the batch, newest first (FR-180)	Working roles (not Viewer) and StCd Owner (read only)
GET	/change-sets/{changeSetId}/dossier.xlsx	The change set as an XLSX dossier for an audit - codes with their decisions, samples with their verdicts, the audit entries (FR-180)	Working roles (not Viewer) and StCd Owner (read only)
GET	/bulk-samples/{sampleId}	One sample with its sampled codes and their verdicts (FR-180)	Working roles (not Viewer) and StCd Owner (read only)
PUT	/bulk-samples/{sampleId}/items/{changeRequestId}	Mark a sampled code OK or DEFECT (a defect needs a note); only by the person who drew the sample (FR-180)	Working roles (not Viewer)
POST	/bulk-samples/{sampleId}/reject-defects	Send back the codes of a failed sample that were found defective, with the note as the reason, and close the sample (FR-180)	Working roles (not Viewer)
GET	/status-codes/{statusCodeId}/connections	What suppresses this code, what it suppresses, what it requires and where it escalates, read from the imported programs, with the chain o...	Working roles (not Viewer) and StCd Owner (read only)
GET	/status-codes/connections	The causes with the codes they suppress (a tree), the escalation targets and what looks wrong in the connections between codes (FR-179)	Working roles (not Viewer) and StCd Owner (read only)
GET	/status-codes/{statusCodeId}/family	The code and its platform variants (I-25)	Any signed-in user
POST	/status-codes/{statusCodeId}/platform-variants	Copy the latest decided revision into a new Draft for another turbine platform, linked to the source (I-25)	Administrator, Engineer
PUT	/status-codes/{statusCodeId}/planned-dates	Set planned effective / deprecation dates on the current revision (I-27, Administrator)	Administrator
POST	/status-codes/{statusCodeId}/mc-drafts	Create a Main Controller Draft from a Hub Controller code (I-36)	Administrator, Engineer

Method	Path	Purpose	Who may call it
GET	/hub-consolidation/mapping	Every Hub Controller code with its Main Controller replacement and how far that is (I-36)	Working roles (not Viewer) and StCd Owner (read only)
POST	/hub-consolidation/deprecation-requests	Open a deprecation request for every Released Hub code of a group whose MC replacement is Released (I-36, Administrator)	Administrator
GET	/change-requests/stalled	Requests waiting past their amber/red SLA threshold (I-08, Administrator)	Administrator
GET	/change-requests/{crId}/sla	How long the request has waited at its stage, in business days, and its SLA level (I-08)	Working roles (not Viewer)
POST	/change-requests/{crId}/reviewer-assignment/auto	Assign the least-loaded available Reviewer (I-30, Administrator)	Administrator
GET	/reviewers/workload	Open reviews per active, available Reviewer (I-30, Administrator)	Administrator
GET	/change-requests/{crId}/field-comments	Comments attached to single fields of the revision (I-23)	Working roles (not Viewer)
POST	/change-requests/{crId}/field-comments	Comment on one field while the CR is in Review or Pending Approval (I-23)	Administrator, Reviewer, Chief Engineer
POST	/field-comments/{commentId}/resolve	Mark a field comment resolved (the commenter, the CR author or an Administrator)	Working roles (not Viewer)
GET	/review-checklist-items	Active review-checklist items (I-23); Administrators may add include_inactive=true	Working roles (not Viewer)
POST	/review-checklist-items	Add a checklist item for one Functional System Group or all (Administrator)	Administrator
PATCH	/review-checklist-items/{itemId}	Edit or (de)activate a checklist item (Administrator)	Administrator
GET	/change-requests/{crId}/review-checklist	The checklist items that apply to the CR and the confirmations already recorded (I-23)	Working roles (not Viewer)
GET	/change-requests/{crId}/attachments	Evidence attached to the CR (I-13)	Working roles (not Viewer)
POST	/change-requests/{crId}/attachments	Attach evidence (author or Administrator, CR open)	Administrator, Engineer
GET	/attachments/{attachmentId}	Download evidence as an attachment (never rendered in the browser)	Working roles (not Viewer)
DELETE	/attachments/{attachmentId}	Remove evidence while the CR is still a Draft (uploader or Administrator)	Administrator, Engineer
GET	/change-sets	Change sets - groups of CRs shipping with one controller software release (I-11)	Working roles (not Viewer) and StCd Owner (read only)
POST	/change-sets	Create a change set (Administrator, Engineer, Chief Engineer)	Administrator, Engineer, Chief Engineer
GET	/change-sets/{changeSetId}	A change set with its CRs, their states and SLA levels	Working roles (not Viewer) and StCd Owner (read only)
POST	/change-sets/{changeSetId}/change-requests	Add an open CR to the set (its author or an Administrator)	Administrator, Engineer
DELETE	/change-sets/{changeSetId}/change-requests/{crId}	Take a CR out of the set	Administrator, Engineer
POST	/change-sets/{changeSetId}/close	Close the set (Administrator)	Administrator
POST	/change-sets/{changeSetId}/decisions	Decide several CRs of the set in one sitting	Administrator, Reviewer, Chief Engineer
GET	/delegations	Delegations you gave or received (Administrators see all). include_inactive=true adds ended and revoked ones	Working roles (not Viewer)
POST	/delegations	Name a delegate for a date range (I-16)	Administrator, Reviewer, Chief Engineer
DELETE	/delegations/{delegationId}	Revoke a delegation (the delegator or an Administrator)	Administrator, Reviewer, Chief Engineer
GET	/notifications	Your in-app notices (I-08), newest first	Any signed-in user
POST	/notifications/read-all	Mark all your notices as read	Any signed-in user
POST	/notifications/{notificationId}/read	Mark one notice as read	Any signed-in user
GET	/settings	Administrator-managed settings (SLA thresholds, reviewer auto-assignment, capacity alert level) with defaults	Administrator

Method	Path	Purpose	Who may call it
PUT	/settings	Change settings (Administrator)	Administrator
GET	/reports/catalogue-health	Catalogue health and number-range capacity forecast (I-19)	Administrator, Reviewer, Chief Engineer
GET	/reports/kpis	The business-case KPIs - approval time, cross-domain turnaround, audit-trail completeness, queue ages, ownerless codes (I-10)	Administrator, Chief Engineer
GET	/reports/occurrences	Codes that never fire and the most frequent ones, from imported occurrence counts (I-28)	Administrator, Chief Engineer
GET	/occurrence-imports	Imported occurrence batches (I-28)	Administrator, Chief Engineer
POST	/occurrence-imports	Import occurrence counts per status code for a period (I-28): CSV/XLSX with the columns status_code_identifier and occurrences, aggregate...	Administrator, Chief Engineer
GET	/operations/backup-status	State and age of the last verified backup (I-09, Administrator)	Administrator
POST	/operations/audit-verify	Walk the audit-log hash chain now (I-18, Administrator). ok:false names the first bad row	Administrator
GET	/imports/template	Download the import template - exact headers (CSV) or headers, instructions and the lookup codes (XLSX) (I-05)	Administrator, Engineer
GET	/imports/reconciliation	Migration progress - legacy rows imported, in review, approved, released, retired and (with expected_total) still missing (I-05)	Administrator, Engineer
POST	/imports/reconciliation	Same report with the legacy master list uploaded so the missing legacy numbers are named	Administrator, Engineer
POST	/imports/st/preview	Status code import from generated StCd_.st programs, step 1 (v4.2): upload several files or a ZIP; nothing is created	Administrator, Engineer
POST	/imports/st	Status code import from generated StCd_.st programs, step 2 (v4.2): create the Drafts for the chosen programs of a preview	Administrator, Engineer
GET	/status-codes/{statusCodeId}/import-source	Where an imported status code came from - file, SHA-256, program and every attribute parsed from it, including those without a catalogue...	Working roles (not Viewer)
GET	/legacy-mapping	Legacy number -> new identifier for every code that carries a legacy number (v4.2)	Working roles (not Viewer)
GET	/legacy-mapping/export	The legacy -> new mapping as CSV (default) or XLSX, with the filters of the list (v4.2)	Working roles (not Viewer)
GET	/mapping	The legacy -> new mapping (v4.17, FR-191): the rows with what the .st programs do not carry	Working roles (not Viewer) and StCd Owner (read only)
GET	/mapping/export	The mapping as a workbook with the sheets MC and HC - the layout the upload reads (FR-197)	Working roles (not Viewer) and StCd Owner (read only)
POST	/mapping/seed	Load the initial mapping from the shipped workbook into an empty table (Administrator)	Administrator
GET	/mapping/{mappingId}	One row of the mapping	Working roles (not Viewer) and StCd Owner (read only)
GET	/mapping-changes	The requests to change the mapping, newest first; filter state	Working roles (not Viewer) and StCd Owner (read only)
POST	/mapping-changes	Request a change of the mapping (FR-194, FR-195)	Working roles (not Viewer) and StCd Owner (read only)
POST	/mapping-changes/workbook	Compare an uploaded workbook (sheets MC and HC) with the mapping and request one change with every new and every differing row (FR-197)	Working roles (not Viewer) and StCd Owner (read only)
GET	/mapping-changes/{changeId}	One request with its rows and the three approvals	Working roles (not Viewer) and StCd Owner (read only)
POST	/mapping-changes/{changeId}/decisions	Approve or reject (FR-194, BR-027)	Administrator, StCd Owner, Chief Engineer
POST	/mapping-changes/{changeId}/withdraw	The requester withdraws an open request	Working roles (not Viewer) and StCd Owner (read only)
GET	/export-profiles	Export profiles - Administrator-managed column sets (I-02). include_inactive=true is honoured for Administrators	Any signed-in user
POST	/export-profiles	Create an export profile (Administrator)	Administrator
PATCH	/export-profiles/{profileId}	Change or deactivate an export profile (Administrator)	Administrator
GET	/audiences	The eight audiences an export can be scoped to (I-03)	Any signed-in user

Method	Path	Purpose	Who may call it
GET	/releases/{releaseId}/notes	Release notes - added / changed (field-level diff) / deprecated against the previous published Release (I-06)	Any signed-in user
GET	/releases/{releaseId}/offline-bundle	One self-contained HTML lookup page for the Release, for use at the turbine without connectivity (I-24)	Any signed-in user
GET	/releases/{releaseId}/exports	Every export recorded for the Release with its SHA-256 (I-06)	Working roles (not Viewer)
PUT	/releases/{releaseId}/target-versions	Set the controller software versions a Release is published for (I-07, Administrator, before publish only)	Administrator
GET	/releases/{releaseId}/baseline	Counts of the complete baseline of a Release - every code valid as of it - by controller and platform (FR-173)	Any signed-in user
GET	/releases/{releaseId}/readiness	What is still missing before the documents of a Release are complete - translated codes per language and codes without confirmed access l…	Working roles (not Viewer) and StCd Owner (read only)
POST	/releases/{releaseId}/packages	Build the package of a published Release - the complete baseline in every language for every access level, with release notes and a manif…	Any signed-in user
GET	/releases/{releaseId}/packages	The packages built for a Release, newest first (FR-174)	Any signed-in user
GET	/packages/{jobId}	State and progress of a package (FR-174)	Any signed-in user
GET	/packages/{jobId}/download	Download the finished package as a ZIP (FR-174)	Any signed-in user
GET	/exports/{jobId}/manifest	The manifest recorded for an export (I-06)	Working roles (not Viewer)
POST	/exports/verify	Is this file one the system produced (I-06)?	Any signed-in user
GET	/software-versions/matrix	Which Released codes apply to a controller software version (I-07)	Any signed-in user
GET	/software-versions/changes	What was added, removed and changed between two software versions (I-07)	Any signed-in user
GET	/revisions/{revisionId}/translations	Translation status per language - MISSING, DRAFT or APPROVED (I-15)	Any signed-in user
PUT	/revisions/{revisionId}/translations/{language}	Save a translation as a DRAFT (Engineer, Administrator; source text must be Approved)	Administrator, Engineer
POST	/revisions/{revisionId}/translations/{language}/approve	Approve a translation - a light review by a Reviewer or Administrator other than the translator (I-15)	Administrator, Reviewer
POST	/revisions/{revisionId}/translations/carry-over	Copy the previous revision's translations as DRAFTs for languages this revision lacks	Administrator, Engineer
GET	/translations/worklist	Current Approved/Released codes still lacking an approved translation in a language (I-15)	Working roles (not Viewer)
GET	/saved-searches	Your saved status-code searches and the columns the list can show (I-26)	Any signed-in user
POST	/saved-searches	Save the filters and visible columns under a name (saving under an existing name replaces it)	Any signed-in user
DELETE	/saved-searches/{searchId}	Remove one of your saved searches	Any signed-in user
GET	/api-tokens	Machine API tokens (I-29, Administrator)	Administrator
POST	/api-tokens	Create a scoped, expiring token (max 365 days)	Administrator
DELETE	/api-tokens/{tokenId}	Revoke a token (Administrator)	Administrator
GET	/webhooks	Webhooks fired on release.published (I-29, Administrator)	Administrator
POST	/webhooks	Register a webhook	Administrator
PATCH	/webhooks/{webhookId}	Pause or resume a webhook	Administrator
GET	/webhooks/{webhookId}/deliveries	Recent deliveries of a webhook with their outcome	Administrator
GET	/changes	Change feed - releases, deprecations, reinstatements and archivals in audit order (I-29)	Any signed-in user; API token changes:read

<!-- api-summary:end -->

Export file names follow `cscm_[_]_YYYY-MM-DD_HH-MM.` (generation time, UTC): `released`, `release_`, `working-set_NOT_FOR_DISTRIBUTION`, `full-database-backup`, `release-notes_`, `offline-lookup_`. Non-API endpoints: `GET /metrics` (Prometheus text, §11) and the UI pages under `/`.

8. Idempotency & Concurrency

Optimistic concurrency on Draft edits is real since v4 (§10, FR-090) and publishing is idempotent (a second publish of a published Release is refused with 409). Identifier allocation on `POST /change-requests/{id}/submit` is race-safe by construction (07-database-design.md §7a) — two concurrent submissions in the same Functional Group/Subgroup can never receive the same number; the loser of the race simply receives the next number after, not an error.

9. Rate Limiting

`POST /auth/login` is rate-limited per IP (10 attempts in 5 minutes) and per username (5 in 15 minutes) independently; the counters are stored in the database, so they are shared by all worker processes and survive a restart (NFR-009).

10. v4 Conventions

Step-up (FR-126). The request body of `reviewer-signoff`, `admin-signoff`, `chief-engineer-decide`, `POST /releases/{id}/publish`, `POST /exports/full-database` and `POST /change-sets/{id}/decisions` carries `current_password`. Absent -> 403 `step_up_required`; wrong -> 403 `step_up_failed`. The check is audited and never logs the password.

Concurrency (FR-090). `PATCH /revisions/{id}` may carry `expected_row_version` and/or `expected_updated_at`; a stale value -> 409 `conflict` and nothing is changed. Responses return `row_version` and `updated_at` so a client can send them back. Saves also return `warnings` — soft findings (unknown or retired references, similar titles) that never block (FR-091, FR-093).

Export shaping (FR-100 ... FR-111). `GET /releases/{id}/export` and `GET /exports/released` accept `format` (`csv|json|xlsx|pdf`) and optionally `profile` (an export-profile code), `audience`, `language`, `strict_language=true`, `software_version` and `bundle=true` (a ZIP holding the data file, `MANIFEST.json` and release notes). `audience` is rejected on the unapproved working set. Every export response carries `X-Content-SHA256` (of the data file) and is recorded as an export job whose manifest is available at `GET /exports/{jobId}/manifest`; `POST /exports/verify` checks an uploaded file against recorded exports and an uploaded manifest. Release notes, the offline bundle and the export history are `GET /releases/{id}/notes`, `/offline-bundle` and `/exports`.

API tokens (FR-144, SEC-027). An Administrator creates a token (`POST /api-tokens`; the secret is returned once). It is sent as `Authorization: Bearer cscm_?` and is honoured only on `GET` operations whose row above names its scope (`catalogue:read`, `changes:read`), with Viewer-level visibility. It can never write; a revoked, expired or unknown token gets 401.

Change feed (FR-145). `GET /changes?since_id=&limit=<1..500>` returns `{items, next_since_id, has_more}`; poll with the previous `next_since_id`. The response has a weak `ETag`; `If-None-Match` with the current value answers 304. Events: `code.released`, `code.deprecated`, `code.reinstated`, `code.archived`.

Webhooks (FR-146, SEC-028). A webhook registered for `release.published` receives `POST {"event": "release.published", "data": {release_id, name, version_label, published_at, item_count}}` with the headers `X-CSCM-Event` and `X-CSCM-Signature: sha256=`. Targets must resolve to private or loopback addresses; redirects are not followed; the outcome of each delivery is listed at `GET /webhooks/{id}/deliveries`.

Idempotent reads, paging. List endpoints that return `items` follow §6; the change feed uses a cursor instead of pages.

11. Operational Endpoints (v4)

Endpoint	Purpose	Access
GET /api/v1/health	Liveness; answers {status, version} - the version of the running application (FR-184)	Public
GET /api/v1/health/ready	Readiness: database, schema head, newest audit rows, last verified backup. 200 with status ok or degraded, 503 with unavailable; carries the version too. Reveals no business data.	Public (internal network)
GET /metrics	Prometheus text: cscm_http_requests_total, cscm_http_request_duration_seconds, cscm_db_lock_wait_seconds, cscm_change_requests, cscm_revisions, cscm_cr_waiting, cscm_backup_age_hours, cscm_backup_ok	Allow-listed networks (METRICS_ALLOWED_CIDRS) or Authorization: Bearer ; otherwise 403 (SEC-029)

10. UI/UX Specification — CSCM Tool

0. OneTool Design System Integration

CSCM Tool is a **OneTool-integrated application** (2026-09-17 decision): it renders inside the shared OneTool shell (top bar, sidebar, navigation) rather than as a standalone page shell, following the canonical `ONETOOL_UI_DESIGN.md` design system supplied by the central tools team. Per the explicit decision on authentication scope (03-srs.md, 11-security-architecture.md §2), CSCM Tool **keeps its own login and its own user/credential store** — only the visual shell, tokens, and component treatments are shared, not identity or session.

What is shared (adopt as-is, do not reinvent):

- Shared top bar (64px, fixed) and sidebar (256px, fixed) — CSCM Tool's own pages occupy only the main workspace region (`margin-left: 256px`, `padding: 32px`, `padding-top: 96px`), using `max-width: 1280px` (CSCM Tool qualifies as a "dense data tool" per the design system's own guidance, given its tables and multi-section forms).
- Design tokens: `--ot-primary #002e6d`, `--ot-background #f7f9fb`, `--ot-surface-raised #fff`, `--ot-error #ba1a1a`, and the full token set in the design doc — CSCM Tool introduces no new primary color, font, or button style.
- Typography: Manrope (headings), Inter (body/labels/controls), Material Symbols Outlined icons.
- Component treatments: cards (12px radius, soft shadow), buttons (primary/secondary/outline/danger per the design doc's table), form inputs (40px height, 8px radius, blue focus ring), tabs, alerts, loading/empty states — all per the design doc verbatim.
- Responsive/accessibility rules: sidebar collapse at tablet width, 16–20px mobile padding, 44×44px minimum pointer targets, WCAG AA contrast.

What CSCM Tool still owns:

- Its own login screen (S1) and Account Settings (S18) — not delegated to OneTool, per the auth-scope decision.
- Its own navigation item within the OneTool sidebar (marked active when a CSCM Tool page is open), registered once, not duplicated.
- All domain-specific screens, forms, and workflow (S2–S17 below).

Integration checklist (from the design doc, applied to CSCM Tool specifically): render the shared shell with the CSCM Tool nav item active; use the standard main workspace offsets; use the documented tokens/components exclusively; test desktop, collapsed-sidebar, tablet, keyboard-only, loading, error, and empty states before release (15-development-standards.md).

1. Screen Inventory

#	Screen	Primary Role(s)	Purpose
S1	Login	All	Authenticate (CSCM Tool's own — not delegated to OneTool)
S2	Dashboard	All	Role-relevant summary + quick links
S3	Status Code List / Search	All	Browse, filter, search all codes (sandbox excluded by default)
S4	Status Code Detail	All	View current revision, history, related CR
S5	New Status Code	Engineer, Administrator	Create a new code (Draft, no fixed identifier yet)
S5a	New Sandbox Status Code	Administrator	Create a test code from the isolated StCd-T##### band
S6	Edit Status Code Revision	Engineer	Edit a Draft
S7	Revision History / Diff	All	Compare any two revisions
S8	My Change Requests	Engineer	Track own submissions, incl. pending domain sign-off requests; tick several and Withdraw selected (v4.6)
S9	Review Queue	Reviewer, Administrator	List CRs awaiting their sign-off
S9a	Approval Queue	Chief Engineer	List CRs in PendingApproval awaiting final decision
S10	Change Request Detail / Decision	Reviewer, Administrator, Chief Engineer	Sign off / approve / reject with diff + comments + domain sign-off status
S10a	Domain Sign-Off Request	Engineer (second domain)	Complete domain-owned fields, record sign-off

#	Screen	Primary Role(s)	Purpose
S11	Release List	All	Browse releases
S12	Release Builder	Administrator	Build candidate set, publish
S13	Release Detail	All	View published contents, export the complete baseline (by platform, controller, language, access level), build the package of all languages and access levels (FR-173, FR-174)
S14	User Management	Administrator, StCd Owner (roles only)	CRUD users, roles, Engineering Domain tags; every role is a coloured badge, red = highest level; the StCd Owner sees the list and moves users between Viewer, Engineer and Reviewer (v4.3). Per user: View (the user's page, /admin/users/{id}: profile and what the account has done), Edit, Deactivate / Activate and Delete (v4.6; delete only while the account has no history and with the Administrator's password)
S15	Lookup/Vocabulary Management	Administrator	Manage Functional Groups/Subgroups (incl. create new — §4 S15), Turbine Platforms, Engineering Domains, Available Group, the 8 access vocabularies
S16	Audit Log Search	Administrator, Reviewer/Chief Engineer (scoped)	Investigate history
S17	Export Center	All (scoped by role); Full-Database Export	Ad-hoc catalogue exports (CSV/JSON/XLSX/PDF); full DB export (Administrator only)
S18	Account Settings	All	Change own password (CSCM Tool's own)
S19	Library Import	Engineer, Administrator	Upload CSV/XLSX, review progress and errors; since v4 template download, Validate only dry run, mode (create / update), background progress
S20	Import Reconciliation ("Migration progress") (v4)	Engineer, Administrator	Legacy rows imported / in review / released / still missing, with the master list uploaded to name the missing numbers (/imports/reconciliation)
S21	Notices (v4)	All	The user's in-app notices, mark as read (/notifications)
S22	Delegation (v4)	Reviewer, Chief Engineer, Administrator	Name a delegate for a date range, see and revoke delegations (/delegation)
S23	Change Sets (v4)	Administrator, Engineer, Reviewer, Chief Engineer (create: not Reviewer)	List, create, open a set, add or remove requests, decide several in one sitting (/change-sets)
S24	Stalled Requests (v4)	Administrator	Requests past their amber/red SLA with reassignment and Reviewer workload (/admin/stalled)
S25	Translations (v4)	All (Viewers see approved text only)	Status per language, edit, approve, carry over (/status-codes/{id}/translations)
S26	Release Notes, Shaped Export, Offline Bundle (v4)	All	Added / changed / deprecated against the previous Release; export with profile, audience, language, version and bundle; offline lookup page (/releases/{id}/notes, /shaped-export, /offline-bundle)
S27	Software Versions (v4)	All	The version matrix and the changes between two versions (/software-versions)
S28	Verify a File (v4)	All	Upload an export and optionally its manifest; result of the hash and signature check (part of S17, posts to /exports/verify)
S29	Catalogue Health (v4)	Administrator, Reviewer, Chief Engineer	Capacity forecast (Main Controller rows blue, Hub Controller rows light blue, each with an MC / HC chip), missing data, inconsistencies, unused numbers, upcoming dates (/reports/catalogue-health)
S30	Success Metrics (v4)	Administrator, Chief Engineer	The business-case KPIs (/reports/success-metrics)
S31	Fleet Feedback (v4)	Administrator, Chief Engineer	Import occurrence counts; never-fired and most-frequent codes (/reports/fleet-feedback)
S32	Hub Consolidation (v4)	Working roles (actions: Engineer, Administrator)	Mapping of Hub codes to Main replacements; create MC draft; bulk deprecation (/hub-consolidation)
S33	Export Profiles (v4)	Administrator	Column sets, default audience and language (/admin/export-profiles)
S34	Review Checklist (v4)	Administrator	Checklist items per Functional System Group (/admin/review-checklist)
S35	Settings (v4)	Administrator	SLA thresholds, reviewer auto-assignment, capacity alert (/admin/settings)
S36	Integrations (v4)	Administrator	API tokens and webhooks with delivery log (/admin/integrations)
S37	Operations (v4)	Administrator	Backup status, audit-chain verification (/admin/operations)
S38	Help (v4)	All	Per-role quick start, glossary of states, roles and terms (/help)

#	Screen	Primary Role(s)	Purpose
S39	Import from .st files (v4.2)	Engineer, Administrator	Upload several generated StCd_.st files, a folder or a ZIP (/imports/st), read the preview, tick the programs to import, confirm into Drafts in one change set; programs that differ from a code already in the catalogue offer update the Draft, new revision or restore (not ticked by default, v4.6)
S40	Legacy mapping (v4.2)	Engineer, Administrator, Reviewer, Chief Engineer	Legacy number -> new identifier with filters and CSV/XLSX export (/legacy-mapping); legacy -> new only
S41	Windows sign-in (v4.7)	All	The page behind Sign in with your Windows account (/login/windows): in a deployment the gateway has already told the application who the user is and the page signs them in; on a developer machine it asks for the identity the gateway would have supplied (/login/windows/simulate)
S42	Translations (v4.7)	Reviewers of a language, Administrator, StCd Owner	The languages the user may review with the state of each (not reviewed, in progress, reviewed, changed since the review); managers see all languages (/translations)
S43	Translation review (v4.7)	Assigned reviewers	One row per string with screens, English, current translation, verdict, proposal and comment; the missing translations; XLSX sheet download and upload (/translations/{language})
S44	Translation decisions (v4.7)	Administrator, StCd Owner	Reviewers of the language, proposals waiting for a decision, accepted overrides, recent decisions (/translations/{language}/decisions)
S45	Terminology (v4.9)	Everybody signed in; managers add and change terms	The vocabulary: coverage per language, the terms with category, definition and the state of every language, search and filter, add a term (/terminology)
S46	Terminology of a language (v4.9)	Everybody signed in; the technical roles propose, the approving roles approve	One row per term with the English text, the proposal and its state; save translations, send back with a note, approve one or all drafts of others; XLSX and CSV export (/terminology/{language})
S47	Terminology check (v4.9)	Everybody signed in	The interface strings that contain a term whose translation is missing from the translation of the string (/terminology/{language}/check)
S50	Change set workbench (v4.13)	Engineer, Reviewer, Administrator, Chief Engineer (the StCd Owner reads)	Readiness of the Drafts of a batch, bulk edit, bulk submit, bulk domain sign-off, the sample inspection (draw, verdicts, send the defective codes back) and the decision with its sample number; link to the dossier (/change-sets/{id}/workbench)
S49	Connections (v4.12)	Engineer, Reviewer, Administrator, Chief Engineer, StCd Owner	How codes are connected, read from the imported programs: the causes with the codes they suppress as a tree, the escalation targets and what looks wrong; filter Main or Hub controller; the code page (S7) has a Connections card (/status-codes/connections)
S48	Withdrawn Codes (v4.9)	Administrator	The codes whose last revision was withdrawn and that still hold a number, a legacy number or a PrgIdx_GUi; tick, enter the password, delete (/admin/withdrawn-codes)

2. Navigation Map

```
[OneTool shared top bar + sidebar, CSCM Tool nav item active]
  Login (S1, CSCM Tool's own)
  +-> Dashboard (S2)
        +-> Status Codes (S3) --> Status Code Detail (S4) --> Revision History/Diff (S7)
        |                              |
        |                              +-> Edit Revision (S6) [Engineer, own open CR]
        |
        +-> New Status Code (S5) [Engineer/Admin] --> Status Code Detail (S4)
        +-> New Sandbox Status Code (S5a) [Admin only] --> Status Code Detail (S4, sandbox-flagged)
        +-> Library Import (S19) [Engineer/Admin] --> My Change Requests (S8) [imported Drafts appear here]
        +-> Import from .st files (S39) [Engineer/Admin] --> Import job --> Change Sets (S23) [the batch change set]
        +-> Legacy mapping (S40) [every role except Viewer]
        +-> Translations (S42) [reviewers of a language, Administrator, StCd Owner] --> Review (S43) / Decisions (S4
  4)
        |
        +-> My Change Requests (S8) [Engineer] --> Domain Sign-Off Request (S10a) [if flagged]
        |
        +-> Review Queue (S9) [Reviewer/Admin] --> Change Request Detail/Decision (S10)
        +-> Approval Queue (S9a) [Chief Engineer] --> Change Request Detail/Decision (S10)
        |
        +-> Notices (S21) · Help (S38) · Language switch [every role, top bar]
        +-> Change Sets (S23) · Delegation (S22) [decision roles] · Stalled Requests (S24) [Admin]
        |
        +-> Insight: Catalogue Health (S29) · Success Metrics (S30) · Fleet Feedback (S31) · Hub Consolidation (S32)
        |
        +-> Releases (S11) --> Release Detail (S13) --> Notes / Shaped Export / Offline Bundle (S26) --> Export Cent
  er (S17)
        |         +-> New Release / Release Builder (S12) [Admin]
```

```
        |
        +-> Administration [Admin only, top-level menu]
        |         +-> User Management (S14)
        |         +-> Lookup Management (S15) --> New Subgroup form (non-overlap validated)
        |         +-> Export Profiles (S33) · Review Checklist (S34) · Settings (S35)
        |         +-> Integrations (S36) · Operations (S37)
        |         +-> Audit Log (S16)
        |         +-> Full-Database Export (S17, Admin-only section)
        |
        +-> Account Settings (S18) [every role, via user menu]
```

3. Role-Based Navigation Visibility

(v4.9, FR-169) The menu is grouped by the work of the user: **Catalogue** (Status Codes, Releases, Export Center, New Sandbox Code), **Workflow** (My Change Requests, Review Queue, Approval Queue, Change Sets, Delegation), **Import and Migration** (Library Import, Import from .st files, Legacy mapping, Hub Consolidation, Withdrawn Codes), **Insight** (Catalogue Health, Stalled Requests, Success Metrics, Fleet Feedback), **Languages** (Terminology, Translations) and **Administration** with the blocks People (Users), Configuration (Lookups, Export Profiles, Review Checklist, Settings, Integrations) and Operations (Operations, Audit Log); Dashboard stays on top, Help at the bottom, and a + New Status Code button sits above the groups for the roles that create codes. A group opens and closes and remembers it; a badge shows what waits (Review Queue, Approval Queue, Stalled Requests, Withdrawn Codes, and one's own open requests at My Change Requests), and a closed group still shows the total. Below 768 px a Menu button in the top bar opens the menu as a panel. The table below lists the same links by role as they were before the grouping. In the language menu a language that no native speaker has reviewed carries a beta mark (FR-170); Engineers, Reviewers, Chief Engineers, Administrators and the StCd Owner see all languages, a Viewer and the login page only the reviewed ones (FR-172).

Nav Item	Administrator	Engineer	Reviewer	Chief Enginee r	Viewer
Status Codes	yes	yes	yes	yes	yes
New Status Code	yes	yes	—	—	—
New Sandbox Status Code	yes	—	—	—	—
Library Import	yes	yes	—	—	—
My Change Requests	yes	yes	yes (as any sub mitter)	yes	—
Review Queue	yes	—	yes	—	—
Approval Queue	—	—	—	yes	—
Releases	yes	yes (read)	yes (read)	yes (read)	yes (read)
Release Builder	yes	—	—	—	—
Administration menu	yes	—	—	—	—
Full-Database Export	yes	—	—	—	—
Catalogue Export Center	yes	yes	yes	yes	yes (Released/Approved only)
Notices, Help, Software Versions	yes	yes	yes	yes	yes
Change Sets	yes	yes	yes	yes	—
Delegation	yes	—	yes	yes	—
Stalled Requests	yes	—	—	—	—
Catalogue Health	yes	—	yes	yes	—
Success Metrics, Fleet Feedback	yes	—	—	yes	—
Hub Consolidation	yes	yes	yes	yes	—
Export Profiles, Review Checklist, Settings, I ntegrations, Operations	yes	—	—	—	—

4. Wireframe Descriptions

S5 — New Status Code

(v4) The form additionally offers the optional remedy text, affected component, related codes, standard references and the software-version range; shows a **capacity hint** for the chosen subgroup ("158 numbers left in this band"); warns, without blocking, about unknown or retired references and similar titles after saving; and on a stale save (409) re-shows the user's entries beside the current values and the current `row_version`.

A single-page form on the standard `ot-card` surfaces, laid out in labeled sections per the design system's two-column grid guidance:

1. **Classification** — Functional System Group (dropdown, required) -> Functional Subgroup (dependent dropdown, populated from the selected group, **required — never optional**, disabled with a "select a Functional System Group first" hint until a group is chosen) -> Turbine Platform (multi-select chip input, at least one required). The identifier field shows **"Not yet assigned — fixed on Submit for Review"**, never a real or placeholder number; a small, explicitly-labeled non-binding grey preview may appear ("likely next: `StCd-01233` — not reserved").
2. **Description** — Title, Description.
3. **Status & Timing** — Status Category (Error/Warning/Info), Available Group (numeric picker, sourced from the soft-validated lookup), Alarm (checkbox), Set Delay / Reset Delay (each a value + unit-select pair — ms/s/min/h/d — composed into the stored free-text form, so Engineers never have to type the unit suffix by hand), Up/Down Counter, Trigger Snapshot, Loadless Spinning Permitted (checkboxes).
4. **Repeated-Error Escalation** (collapsible, optional) — Number of Times Repeated, Repeated Over the Course (value + unit), Status Code for Repeated Error (plain numeric field, labeled "reference number in the separate escalation code pool — not a CSCM Tool status code").
5. **Program References** (collapsible, labeled "Advanced — controller firmware programs", CONTROLS-domain-owned) — Yaw Program, Manual Reset Program, Auto Reset Program, Converter Reactive Power Program, Brake Program, Umrichterprogramm GSC, Umrichterprogramm MSC, Yaw Bearing Lubrication Program — each a plain numeric field with the currently-observed valid values shown as a soft-suggestion list (autocomplete), not a hard-enforced dropdown, since the firmware-side table can grow independently of CSCM Tool (06-data-dictionary.md §2c).

5a. (v4.9, FR-168) **Further Program References** (collapsible: the 13 Main Controller programs, then the 3 Hub Controller programs), **Operation Parameters** (operating-mode mask as a bit string, task call frequency, the flags as not captured / Yes / No) and **User-Group Levels** (a read and a write level for each of the 16 slots). The detail page shows the sections of the controller of the code and the levels that exist.

6. **Audience Access** (Governance-restricted — badge "Proposed, confirmed by Product Owner at review" on every field, per §5 below) — eight independent dropdowns: Development, Sales, TCC, Service, Turbine Operator Package, Grid Operator, Service Partner, Customer.
7. **Ownership** — Owner (defaults to current user).

If the current Engineer's tagged Engineering Domain(s) do not cover a required domain (e.g. Program References is CONTROLS-owned and the originating Engineer is Electrical), that section is marked **"Needs Controls input"** with a "Request Domain Sign-Off" button; Submit for Review stays disabled until every required `DomainSignoff` exists.

No data loss on failure (explicit fix, 2026-09-17): a failed Submit-for-Review or Save never clears the form. The legacy tool resets all fields on any creation failure — flagged directly by the business as "cumbersome." CSCM Tool's client keeps every entered value in place after a 422 response and highlights only the specific invalid fields inline (per §9's inline-validation pattern); nothing is ever discarded because of a validation error, a network retry, or a session-timeout-triggered re-login.

Sticky footer: "Save as Draft" (secondary) and "Submit for Review" (primary, disabled until validation and all domain sign-offs pass).

S5a — New Sandbox Status Code (Administrator only)

Same form as S5, minus the Functional Group/Subgroup/numbering section — a persistent banner reads **"Sandbox status code — never numbered from a real range, never released, deletable."** The resulting record's identifier is shown as `StCd-T#####`, with a red "SANDBOX" badge visible everywhere the code appears, and a "Delete Permanently" button on S4 for this record only, gated behind a typed-confirmation modal.

S19 — Library Import

1. **Upload** — drag-and-drop or file picker for a `.csv` or `.xlsx` file, with a short format note ("one row per status code; column headers must match the exported template — download a template from here").
2. **Progress** — once uploaded, the job runs (in the background above the row threshold); the screen polls the progress endpoint and shows a bar with `processed_count` of `row_count` plus `created_count`, `error_count` and `warning_count`.

(v4) Before importing, **Template** (CSV or XLSX) downloads the exact headers and lookup codes; **Validate only** runs the dry run and shows every problem per row and column with the value and a fix hint, and a banner states that nothing has been created; the **mode** selector chooses create or update (matching on the legacy number). Warnings (similar titles) are listed but do not block.

3. **Result** — on completion, a summary ("128 of 134 rows imported as Drafts, 6 failed validation") with "Download error report" links (CSV and XLSX; row, column, value, problem, how to fix) and a "View imported Drafts" link to My Change Requests (S8), filtered to this import job. Imported rows are ordinary Drafts — the screen never implies they are already Approved.

S39 — Import from .st files (v4.2)

1. **Upload** — the generated `StCd_.st` files (several at once), a whole folder or a ZIP; options: controller (detect / Main / Hub), the new number of Hub programs (proposal 1 five-digit, the default, or proposal 2 inside the Main scheme), the status category of the new codes (the files carry none) and the turbine platforms. A note states that every imported program becomes an ordinary Draft with its number reserved.
2. **Preview** — nothing is created. Chips count files, programs, ready, already in the catalogue, blocked, warnings and programs without a legacy number. One row per program: a checkbox (disabled unless the program is ready), new identifier, legacy number (n/a when there is none), controller, name with file and line, group and subgroup, state, and every problem and warning. A header checkbox selects every ready program, a filter box narrows the list by text. Optional name of the change set.
3. **Create Drafts** — creates the ticked programs (a large batch runs in the background with the progress bar of S19) and opens the import job page, which links to the batch change set (S23); **Cancel** discards the staged upload.

S40 — Legacy mapping (v4.2)

Chips: codes with a legacy number (Main / Hub), the number of new codes without one (n/a), and how many identifiers are issued, reserved or not yet assigned. Filter by text, controller and number state. Table: controller, legacy number, new identifier, number state, name and lifecycle state; a row opens the status code. CSV and XLSX export honour the filter. The page says that the mapping runs from legacy to new only.

S41 — Windows sign-in (v4.7)

The login page shows the password form, the link Sign in with your Windows account, or both, as `AUTH_MODE` says (FR-163). The Windows page signs in the account that matches the identity of the gateway and says clearly when there is no account ("No active account matches this Windows identity") or when no identity arrived (No Windows identity reached the application — the gateway is not configured or the request did not come from a trusted proxy). On a developer machine (`WINDOWS_AUTH_SIMULATE`) it shows a field for the identity and a note Simulation. that says a deployment signs in without this page; the production configuration never shows it. In `windows` mode the account page offers no password change. The e-mail address of a user is shown as fixed in the user form and the user view (FR-162).

S42 — Translations (v4.7)

A table of the languages the user reviews (all languages for an Administrator or StCd Owner): flag and name, strings translated and missing, the state of the language, the date of the last completed review, Review and, for managers, Decisions and reviewers. A user who reviews nothing sees a note instead of the table; the sidebar entry appears only for reviewers and managers.

S43 — Translation review (v4.7)

The header shows the answered strings of the total as a bar, a What to check checklist and the two views Check translations and Missing translations; a filter by text, screen and answer; 40 rows per page. A row has the screens and the string id, the English, the current translation (right to left for Arabic and Urdu), three radio buttons OK / CHANGE / UNCLEAR, a proposal box and a comment. Mark the rest of this page OK answers the rows without a verdict; Save and Save and next page store the page; a row that is decided is locked. Download the sheet and Upload are the offline route. Complete the review is possible when every translated string has a verdict, otherwise the message says how many are missing.

S44 — Translation decisions (v4.7)

Three blocks: the reviewers of the language (remove, and assign any active user), the proposals waiting for a decision — string, current text, what the reviewer says; **Accept** with the text pre-filled and editable, **Reject**, or for an UNCLEAR item **Noted** and **Dismiss**; a proposal of the signed-in manager shows a note instead of buttons — and the accepted overrides with Remove, followed by the recent decisions.

S15 — Lookup/Vocabulary Management, incl. New Subgroup and New Functional System Group

In addition to the original vocabularies, an Administrator can select a Functional System Group and click "Add Subgroup," entering a name and a proposed numeric sub-range. The form shows the group's existing subgroups and their ranges visually (a simple horizontal bar from `range_start` to `range_end` with existing subgroups shaded and reserved blocks highlighted) and pre-fills a suggested next hundred-aligned free block. Submitting a range that overlaps an existing subgroup returns the specific conflict inline ("overlaps 'Hot Air De-icing' (4200–4299)") without losing the entered name — the same no-data-loss principle as S5.

A second, top-level "Add Functional System Group" action (new 2026-09-17, FR-048g/h) works the same way one level up: an overview strip shows the full 1000–16999 span with occupied groups (including both the Main and Hub Controller ranges) shaded and the confirmed free bands (`10000?10999`, `14000?14999`, `15000?15999`, `16000?16999`) highlighted as available. The Administrator enters a code, label, and range within a free band; an overlapping range is rejected inline the same way, naming the conflicting group, without losing the entered name.

S4 — Status Code Detail

(v4) Additional panels: **Where-used**, **Family** (platform variants) with Create platform variant, **reference and similar-title warnings**, **planned dates** (Administrator), **remedy, component, related codes and standards**, the software-version range, and a link to **Translations** (S25). The list (S3) offers ranked, typo-tolerant search, sort links, paging, a **column chooser**, **Save search** and a share link. Header band: Status Code Identifier (or "Not yet assigned" for Draft) + Title + Lifecycle State badge (§6) + Functional System Group **and Functional Subgroup** tags (both always present) + Turbine Platform chips. A red "SANDBOX" badge replaces the identifier styling for sandbox records. Tabs: **Current**, **History**, **Change Requests**. Action buttons (role- and state-gated): "Open New Revision" (Engineer, if no open CR), "Deprecate" (Administrator, if Released), "Delete Permanently" (Administrator, sandbox only).

S9a — Approval Queue (Chief Engineer)

Table: Identifier, Title, Functional Group/Subgroup, Reviewer (who signed off), Administrator (who signed off), Age since both sign-offs completed. Row click -> S10. (v4) An **SLA chip** (days waited, amber/red) replaces the plain age column on this queue and on the Review Queue; the Review Queue also shows the assigned Reviewer.

S10 — Change Request Detail / Decision

Split layout: left panel field-level diff (now covering every field category in §4 S5); right panel CR metadata, a **Sign-Off Status strip** (Reviewer / Administrator / Chief Engineer, each pending or signed-with-name-and-date), and a **Domain Sign-Off strip**. Action buttons render per the viewer's role and the CR's current state, hidden entirely (not merely disabled) for the CR's own author and, at the Chief Engineer stage, for whichever individuals already signed off as Reviewer or Administrator on this same CR.

S10a — Domain Sign-Off Request

Shows only the fields tagged to the requested domain (06-data-dictionary.md §2a), pre-filled with the originating Engineer's best-effort values where present, editable, with a "Sign Off This Domain" button.

S10 additions (v4)

The decision panel asks for the actor's **password** (step-up); a banner names the person being covered when the actor decides as a delegate; **field comments** can be added per field in the diff and the author sees the open ones inline; the configured **review checklist** appears as checkboxes the approving Reviewer must tick; **evidence** (attachments) is listed and downloadable; the **SLA chip** shows days waited and amber/red; **Where-used**, **reference warnings** and **similar-title warnings** appear above the diff; the change set the request belongs to is linked.

S12 — Release Builder

Candidate table shows Functional System Group **and Subgroup**; sandbox-flagged revisions are structurally absent from the candidate list. (v4) A Target software versions form (before publishing) limits the candidates to revisions whose version range includes a target; revisions with a future planned effective date are flagged; **Publish** asks for the Administrator's password and keeps the typed-confirmation pattern.

S17 — Export Center

Two sections: **Catalogue Export** — format selector now offers **CSV, JSON, XLSX, PDF** (adds XLSX, 2026-09-17), all producing the same flat, table-shaped output (one row per status code, every field as a column — including JSON, emitted as a flat array of row objects, mirroring the legacy tool's grid and its "Export for: Customer / Service" pattern, which maps onto filtering by the relevant audience-access field). **Full-Database Export** — visible only to Administrator, with the same data-sensitivity warning as before.

5. Field Definitions, Validation, and Inline Help (corrected 2026-09-17)

Field set and validation rules mirror 06-data-dictionary.md exactly (superseding the earlier, less accurate field list). Two explicit UX requirements, both direct responses to feedback on the legacy tool:

- **"No explanations given on options"** (legacy tool complaint) — every dropdown/lookup-backed field in CSCM Tool shows the selected lookup value's `description` (06-data-dictionary.md §9, `LookupValue.description` in the API) as inline help text: a small `(?)` icon next to the field label opens a tooltip with the description, and the dropdown's own option list shows the description as secondary text under each option's label, not just the raw code. Lookups currently without a populated `description` (e.g. Available Group codes, whose meanings are still TBD per 06-data-dictionary.md §9c) show "Description not yet available — confirm with Controls Engineering" rather than leaving the tooltip empty or absent.
- **No data loss on validation failure** — see §4 S5.

The eight audience-access fields carry a "Proposed — confirmed by Product Owner at review" badge (§2.9 restricted-field pattern, 04-domain-model.md) until `admin_signoff_at` is non-null, after which they render as confirmed/locked on the read-only Current tab. The identifier field is never directly editable by any role in the UI.

6. Lifecycle State Visual Language

State	Badge Color	Icon
Draft	Grey	pencil
Review	Amber	clock
PendingApproval	Deep amber / gold	hourglass
Approved	Blue	check
Released	Green	check-circle (filled)
Deprecated	Orange	alert-triangle
Archived	Dark grey / muted	archive box
(Sandbox, any state)	Red outline overlay on the badge above	flask/test-tube

Uses the OneTool `--ot-error`/`--ot-success` tokens where applicable (§0); color is never the sole signal (icon + text label always accompany the badge), satisfying WCAG 1.4.1 (§8).

7. Responsive Behaviour

Per the OneTool design system (§0): sidebar collapses at tablet width with main content taking full width; mobile padding reduces to 16–20px with grids stacking and toolbars scrolling horizontally; desktop remains the primary target given CSCM Tool's dense tables and multi-section forms (`max-width: 1280px`).

8. Accessibility Requirements

(v4, NFR-010) These are now checked by automated tests on every page in both languages: contrast of text and badge colours >= 4.5 : 1, one `main` landmark, skip link, language attribute, heading order, labelled controls, table header scope, and a style sheet that honours `prefers-reduced-motion`; a manual screen-reader pass stays on the pre-go-live checklist (13-test-strategy.md §8a). Per the OneTool design system (§0): WCAG 2.1 AA target, full keyboard operability, associated labels, `aria-live` validation announcements, WCAG AA contrast pairs (dark blue on white, white on primary blue), 44×44px minimum pointer targets, semantic headings and native controls, text alternatives for icons, proper table semantics (`, aria-sort`).

9. User Interaction Patterns

- **No data loss on failure** (§4 S5) — the single highest-priority UX fix from legacy-tool feedback; applies to every form in the application, not just S5.
- **Inline explanations on all lookup-backed options** (§5) — the second highest-priority fix.
- **Sandbox deletion** requires a typed-confirmation pattern identical in rigor to Release publish.
- **Domain sign-off requests** produce a notification-style entry on the target Engineer's Dashboard (S2) and My Change Requests (S8).
- **Import progress** (S19) never blocks the UI synchronously for large files — always asynchronous with pollable status, so an Engineer can navigate away and come back.
- Destructive/irreversible actions always require explicit confirmation; optimistic UI is avoided for state-changing actions; empty states include a next action; toasts confirm non-navigating actions while navigating actions land on the resulting screen with a success banner.

10. v4 Interaction Patterns

- **Password confirmation.** Every sensitive action (sign-off, approval, publish, full-database export, change-set decision) has a Your password field in the same form; a wrong or missing password re-shows the form with the user's entries intact and an inline error (NFR-007a).
- **Language.** The translations, their review by native speakers and the way the manuals reach every language are described in 26-translations.md. A language menu in the top bar (country letters, name, drawn national flag; nineteen languages from GB English to GE Soft war, Georgian last, the list scrolls) stores the preference with the user; before login it follows the browser. Untranslated strings fall back to English, never to a blank. Flash messages from the service layer and the administration screens stay English (further improvement in 21-improvement-ideas.md).
- **Notices.** The bell in the top bar shows the number of unread notices; the Notices page marks one or all as read.
- **Tooltips and help.** State badges and role names carry their glossary text as a tooltip; each change-request state shows a "What happens next" line; `/help` gives the per-role quick start (FR-154).
- **Shareable views.** The status code list keeps filters, sort, page and chosen columns in the URL, so a link reproduces the view; Save search stores it under a name.
- **Background work.** Imports above the row threshold show progress and can be left; the job page reports `COMPLETE` or `FAILED`, never an endless spinner.
- **Tables.** Every table uses ``; sortable headers are links; empty states name the next action.

10a. Translation scope (v4.4)

A critical review of what the interface translates and what it must not. The rule: **translate what a person reads to operate the tool; never translate what is data, evidence, identity or contract.**

Item	Translated?	Reason
Interface text: navigation, buttons, labels, hints, table headings, help	Yes, through the catalogues (German complete; French, Hindi, Georgian, Arabic the main strings; the ten Indian-subcontinent languages after Arabic about sixty strings; the rest falls back to English)	A person needs to operate the tool in their language.
Role names	Yes, as display words; the role code stays in data, API and audit	Same code, readable name.
Identifiers (StCd-01000), numbers, versions, dates, file names	No; Latin letters and digits in every language, shown left-to-right inside right-to-left text	An identifier must be identical wherever it is quoted, searched or typed on a controller.
Stored and exchanged values: lifecycle states (DRAFT), categories (ERROR, WARNING, INFO), engineering-domain and platform codes, CSV and JSON field names, API error codes	No	They are the contract of the API, the exports and the database. A translated label may be shown beside the code later; it never replaces it.
Title, description and remedy text of a status code	Not by the interface catalogue. Only through the translation workflow (FR-110): one translation per language, approved by a second person	This text carries the safety-relevant meaning; a machine or unreviewed translation must never stand in for the source.
Audit log entries, action codes, the hash chain and audit exports	No	Compliance evidence must read the same for every reviewer (BR-020).

Item	Translated?	Reason
Technical vocabulary: codes and abbreviations (MC, HC, StCd, PrgIdx_GUi, 2XM / 3XM / 4XM), IEC terms, product names	No; explained in help where needed	They are terms of the controller software and its documentation. The descriptive names Main Controller and Hub Controller are translated, one fixed word per language (26-translations.md §4.3), the abbreviations are not.
Messages from the service layer and the API (message fields), log lines	Not yet; English	Clients and operators search and parse them. Messages shown to people in the UI could be translated later; API messages never.
PDF, XLSX, DOCX and CSV exports	Content language is chosen at export (FR-111); column headers and codes stay English	Audience files must stay machine-readable.
Names of people, free text typed by users, comments	Never	They are the user's words.
The documentation set and the manual	English source; the manual may be translated later	One authoritative text.

Findings of the review: (1) the lifecycle and category badges show the English code by design, a translated word beside the code is an open improvement; (2) flash messages from the administration screens and the service layer are still English; (3) French, Hindi, Georgian and Arabic cover about a hundred strings each, the ten languages after them about sixty each, Spanish, Italian and Danish about a hundred, and none has been checked by native speakers, so they should not be switched on for people who rely on them before that happens; (4) a flag stands for a country, not a language: Saudi Arabia for Arabic (confirmed by the business), Pakistan for Urdu and India for the other Indian languages are conventions, and Bengali and Punjabi are also spoken in Bangladesh and Pakistan; (5) in right-to-left pages (Arabic, Urdu) the layout is mirrored by `app.css` (`html[dir="rtl"]`), mixed text such as `.st` in a heading can show its punctuation on the wrong side; (6) Urdu is normally set in the Nastaliq style, which many computers do not offer; the page then shows the Arabic-script font of the system, usually Naskh.

13. Colour themes (v4.14, FR-183)

Choice. A selector in the top bar (and at the top right of the login page) offers Automatic (system), Light and Dark. Automatic is the default: the page follows the operating system's light/dark setting and follows it when it changes. The choice is stored in the browser of the device (`localStorage`, key `cscm-theme`); it is not part of the user's account, so a shared workstation shows each person's own browser setting and nothing is written to the server. If the browser blocks storage the page follows the system for that visit.

Mechanics. `static/js/theme.js` is loaded in the head without defer, before the stylesheet, and sets data-theme="light|dark" on the html element before the first paint. The stylesheet defines every colour as a token on :root; the dark values override the tokens under :root[data-theme="dark"], inside @media screen, so a printout keeps the light colours. The dark theme also sets color-scheme: dark, so the browser draws its own controls (check boxes, scroll bars, date pickers) dark.

Palette. Background `#0f1419`, surface `#1a2029`, border `#2f3a47`, text `#e6eaef`, muted text `#9aa7b6`, links and primary fills `#8ab4f8` (with dark text `#0b1b33` on them), error `#ff8a80`, success `#6fcf97`. Lifecycle badges, the SLA pills, flash messages, diffs and the review cells are dark tints with light text. Fills that carry their own text colour - the role badges, the Main/Hub controller chips, the translation confidence colours and the flags - are the same in both themes: they are recognised by their colour and must not change meaning.

Rules for new screens. Never write a colour into a rule or an inline style: use a token (`var(--color-surface)`, `var(--tint-warn-bg)`, ...) or add one with its dark twin. `tests/system/test_dark_mode.py` fails when a hex colour appears outside the token blocks (except the fills named above), when a light token has no dark twin, and when any text/fill pair of the dark palette falls under 4.5 : 1.

Not themed. Exports and printouts (light), the PDF of the status code catalogue, the language flags.

11. Security Architecture — CSCM Tool

1. Overview

CSCM Tool holds engineering-governed data with compliance/audit obligations (05-governance-handbook.md, 12-audit-compliance.md) and is deployed **exclusively on-premises, on a local server, with no public-cloud component of any kind** (00-INDEX.md "Deployment Principle", 14-deployment-architecture.md §1). Controls below are sized to that threat model while following OWASP baseline practice.

2. Authentication

- **SEC-001** Session-based authentication; credentials verified against `user.password_hash` using Argon2id (preferred) or bcrypt (cost >= 12). Plaintext/reversible storage is prohibited absolutely.
- **SEC-002** Session cookie attributes: `HttpOnly`, `Secure`, `SameSite=Strict`; identifier is a cryptographically random 128-bit+ value.
- **SEC-003** Idle session timeout 30 minutes; absolute session lifetime 12 hours.
- **SEC-004** Account lockout: 5 consecutive failed attempts locks the account for 15 minutes.
- **SEC-005** The authentication subsystem sits behind the `AuthProvider` interface (08-system-architecture.md §7) so a future SSO integration is additive. **Since v4.7 the Windows sign-in is built behind a gateway (SEC-033) and can be simulated on a developer machine; the long-term SSO target remains on-premises Active Directory (LDAP/Kerberos) or on-prem ADFS/SAML federation — not a cloud identity provider such as Azure AD.** This corrects the original stack assumption and is a hard requirement flowing from the no-cloud-infrastructure constraint; no OIDC/cloud-IdP code path may be introduced at any phase without an explicit, separate decision to relax that constraint.
- **SEC-006** (new 2026-09-17) CSCM Tool's integration with the shared OneTool platform (10-ui-ux-specification.md §0) is **visual only** — shared top bar, sidebar, design tokens, and component treatments. It shares no authentication, session, or credential mechanism with OneTool: `SessionPasswordAuthProvider` (and, later, the on-prem AD/ADFS provider above) remains CSCM Tool's sole identity source. No code path may accept a OneTool-issued session or token as proof of CSCM Tool identity without an explicit, separate decision to relax this boundary — confirmed explicitly by the business, not a default worth quietly reconsidering during implementation.

3. Authorization — Role-Based Access Control

Six fixed roles: Administrator, StCd Owner, Engineer, Reviewer, Chief Engineer, Viewer (00-INDEX.md "Roles"); the StCd Owner was added in v4.3 (FR-156, BR-024). Permission matrix (yes = allowed; blank = denied; enforced identically at UI and API):

Action	Administrator	Engineer	Reviewer	Chief Engineer	Viewer	StCd Owner
View Approved/Released/Deprecated/Archived codes	yes	yes	yes	yes	yes	yes
View Draft/Review/PendingApproval-state codes	yes	yes	yes	yes	—	yes
Create Status Code / Change Request	yes	yes	—	—	—	—
Create Sandbox Status Code	yes	—	—	—	—	—
Delete Sandbox Status Code	yes	—	—	—	—	—
Edit own Draft revision	yes	yes	—	—	—	—
Provide Cross-Domain Sign-Off	—	yes (matching domain)	—	—	—	—
Submit CR for Review (allocates identifier)	yes	yes (own CR)	—	—	—	—
Reviewer sign-off (Review)	—	—	yes (not own CR)	—	—	—
Administrator sign-off (Review)	yes (not own CR)	—	—	—	—	—
Chief Engineer approval (PendingApproval)	—	—	—	yes (not requester/reviewer/admin on this CR)	—	—
Build/Publish Release	yes	—	—	—	—	—

Action	Administrator	Engineer	Reviewer	Chief Engineer	Viewer	StCd Owner
Deprecate Released code	yes	yes (via full CR cycle)	—	—	—	—
Archive Deprecated code	yes	—	—	—	—	—
Manage Users/Roles/Engineering Domain tags	yes	—	—	—	—	role only, among Viewer / Engineer / Reviewer (BR-024)
Manage all Lookup Vocabularies (incl. Functional Group/Subgroup/Platform)	yes	— (propose only)	—	—	—	—
View Audit Log	yes	—	yes (scoped)	yes (scoped)	—	—
Export catalogue (CSV/JSON/XLSX/PDF) — released data and packages (v4.10: everybody; working set and unpublished Releases: Administrator and StCd Owner only, FR-175)	yes	yes	yes	yes	yes (Released only)	yes
Export full database (password confirmation)	yes	—	—	—	—	—
Read the mapping; request a change of it (FR-191, FR-195)	yes	yes	yes	yes	—	yes
Approve or reject a change of the mapping (password confirmation, one role each, never the requester, BR-027)	yes (one Administrator)	—	—	yes	—	yes
Publish Release (password confirmation)	yes	—	—	—	—	—
Delegate own decision role / act as delegate (decision actions only)	yes	—	yes	yes	—	—
Attach / download evidence on a CR	yes	yes (own CR)	yes (download)	yes (download)	—	—
Comment on a field; confirm review checklist	yes	—	yes	yes (comment)	—	—
Create change sets / decide several requests in one sitting	yes	yes (create)	yes (decide)	yes	—	—
Save / approve a translation	yes	yes (save)	yes (approve, not own)	—	read approved only	—
Manage export profiles, review checklist, settings	yes	—	—	—	—	—
Issue / revoke API tokens; manage webhooks	yes	—	—	—	—	—
View catalogue health report	yes	—	yes	yes	—	—
View success metrics; import and view fleet feedback	yes	—	—	yes	—	—
Backup status, audit-chain verification	yes	—	—	—	—	—
Create users; force a password reset	yes	—	—	—	—	—
Change a user's role	yes (any role)	—	—	—	—	Viewer / Engineer / Reviewer only
Read change requests, change sets and the Hub mapping without acting	yes	yes	yes	yes	—	yes (read only)

- **SEC-010** Authorization checks live in the service layer, not only route decorators/UI conditionals. Implemented as `require_roles()` (`app/domain/authz.py`) at the top of every role-restricted service entry point, raising `ForbiddenError` (HTTP 403); `tests/integration/test_governance_extras.py` calls each service directly with a wrong-role actor.
- **SEC-011** Segregation-of-duties is a hard server-side check at all three decision points (§3 of 09-api-specification.md): Reviewer sign-off, Administrator sign-off, and Chief Engineer approval each independently reject if the actor matches the CR's author or, for the Chief Engineer step, either of the two prior sign-offs. Since v4 it holds for **delegates** as well (they are checked as themselves, the two Review sign-offs must come from two different people) and for **change-set decisions** (each request is decided by its own service call with its own check).
- **SEC-012** Principle of least privilege: default role on new user creation is Viewer.
- **SEC-013** Restricted-field enforcement: each of the eight audience-access fields (corrected 2026-09-17 — replaces the single `access_rights_id`: `development_access`, `sales_access`, `tcc_access`, `service_access`, `turbine_operator_package_access`, `grid_operator_access`, `service_partner_access`, `customer_access`) on a revision may be written by an Engineer only while the revision is in `DRAFT`; none becomes authoritative until stamped by the Administrator's sign-off action, which is the sole path by which `admin_signoff_at` becomes non-null (06-data-dictionary.md §2d, "restricted field"). This is enforced in the service layer, not merely a UI

convention.

- **SEC-014** `DELETE /status-codes/{id}` is authorized only when both `actor.role == ADMINISTRATOR` **and** `status_code.is_sandbox == true`; the second condition is additionally enforced by a database trigger (`trg_prevent_nonsandbox_delete`, 07-database-design.md §5) as defense in depth against a service-layer bug.
- **SEC-015** `POST /exports/full-database` (a `GET` before v4) is authorized only for `actor.role == ADMINISTRATOR` and needs the actor's password (SEC-026); no other role, including Chief Engineer, may reach it. A delegate never inherits it (delegation covers decision actions only, SEC-011, BR-017).

3a. Review of the role rights (2026-10-23)

The matrix above was reviewed role by role when the StCd Owner was added. What the review found, and what was done:

Role	Finding	Outcome
Administrator	Holds data entry, the Administrator sign-off, publishing, lookups and user administration at once. Segregation of duties per change request (SEC-011) keeps the sign-offs apart, but the account is powerful.	Kept as the highest role (red). Handing out the three lower roles moved to the StCd Owner as well, so an Administrator is no longer needed for everyday role changes; creating users and the three upper roles stay here.
StCd Owner	New. Needs to see what is going on in the library and to place people, not to decide.	Read-only oversight of the workflow and the full catalogue; role changes among Viewer, Engineer, Reviewer; no decisions, no data entry, no import, no lookups, no audit log.
Chief Engineer	Final approval only; cannot see the user list or the lookups, which is right.	Unchanged.
Reviewer	Sees the audit log (scoped), the health report and decides the first review step.	Unchanged.
Engineer	Creates and edits Drafts, can import. Domain tags decide which cross-domain sign-offs the Engineer may give.	Unchanged.
Viewer	Sees decided data only.	Unchanged.

Open points for the business (not built): (1) an StCd Owner who promotes somebody to Engineer cannot tag their engineering domains, so the new Engineer cannot give cross-domain sign-offs until an Administrator tags them — decide whether the owner may tag domains; (2) delegation covers Reviewer, Administrator and Chief Engineer only, the StCd Owner cannot be covered by a colleague; (3) whether the StCd Owner should read the audit log; (4) a second person's confirmation when an Administrator role is handed out.

4. Password Policy

12-char minimum, no mandatory character-class rules, breach-corpus blocklist, no mandatory expiry, last-5 reuse block, Argon2id/bcrypt storage, TLS-only transmission.

5. Session Management

Server-side session store, immediate revocation on logout/forced reset/deactivation, login rate limiting (10/IP/5min, 5/username/15min independently), CSRF synchronizer-token protection on all state-changing requests. Since v4 the rate-limit counters are rows in `login_attempt`, written inside the login request's write transaction, so they are shared by every worker process and survive a restart (NFR-009); behind a reverse proxy the real client address is taken from `X-Forwarded-For` only when the proxy is named in `CSCM_TRUSTED_PROXY`.

6. Input Validation

- **SEC-020** All input is validated server-side against the rules in 06-data-dictionary.md regardless of client-side validation state (client-side validation is a UX convenience only, never a trust boundary).
- **SEC-021** Parameterized queries / ORM bound parameters exclusively — no string-concatenated SQL anywhere in the codebase, enforced via code review checklist (15-development-standards.md).
- **SEC-022** All user-supplied text rendered in HTML (Title, Description, comments, justifications, the eight audience-access field labels) is output-encoded by the templating engine's auto-escaping (Jinja2 default autoescape must remain enabled; raw/`|safe` filters are prohibited on user-supplied content).
- **SEC-023** File export generation (CSV/XLSX/PDF) sanitizes any field that could trigger CSV formula injection (leading `=`, `+`, `-`, `@` characters are prefixed with a single quote or tab on export).

- **SEC-024** Any file-upload feature must go through file-type allow-listing, size limits, and storage outside the web root before it ships. The library import (05-governance-handbook.md §6b) was the first: uploaded CSV/XLSX files are validated for type and size before parsing, and the raw upload is never executed or served back to a browser. **Evidence attachments (v4, FR-130)** add: an allow-list (PDF, PNG, JPG, TXT, LOG, MD, CSV, XLSX, DOCX), content checks against the claimed type (magic bytes), at most 10 MB and 20 files per CR, a SHA-256 per file, storage under `instance/attachments` outside the web root, download only as an attachment (`Content-Disposition`, `nosniff`) to the working roles, removal only while the CR is a Draft. Import error reports neutralise values that start with a formula character. The `.st` program import (FR-048o, SEC-031) follows the same rules.
- **SEC-025** Every response carries standard security headers (`Content-Security-Policy` with self-only sources and no inline script, `X-Content-Type-Options: nosniff`, `X-Frame-Options: DENY`, `Referrer-Policy: same-origin`, `Cache-Control: no-store` outside `/static/`, and HSTS on HTTPS requests). Implemented in `app/security_headers.py`; templates therefore carry no inline `` ` or on*= handlers ( tests/system/test_security_headers.py` ``).

The Cross-Domain Sign-Off text fields and comment fields (`review_comment.comment_text`) follow the same output-encoding rule (SEC-022) as Title/Description.

7. Transport & Data Protection

TLS 1.2+ (internal network only — never internet-facing, per the no-cloud constraint), HSTS, standard security headers, OS-level ACL on the SQLite file. **Data-at-rest note specific to full-database exports (SEC-015):** the exported backup file carries the same sensitivity as the live database and must be stored with equivalent access control from the moment it is generated — the export feature itself does not weaken protection, but it does create a second copy of the Master Database that the Administrator is responsible for handling per this section (14-deployment-architecture.md §5).

8. Role-Based Access Control — Enforcement Points

API route decorator -> service-layer check -> database constraint/trigger -> UI conditional rendering), now covering five roles and the additional decision/sandbox/export checks in §3.

9. Audit Trail (Security-Relevant Events)

Plus, since v4: `STEP_UP_SUCCESS`, `STEP_UP_FAILURE`, `DELEGATION_CREATE`, `DELEGATION_REVOKE`, `API_TOKEN_CREATE`, `API_TOKEN_REVOKE`, `WEBHOOK_CREATE`, `WEBHOOK_UPDATE`, `SETTINGS_CHANGE`, `ATTACHMENT_ADD`, `ATTACHMENT_DELETE`, `AUDIT_ARCHIVE`. Earlier: `REVIEWER_SIGNOFF`, `ADMIN_SIGNOFF`, `CHIEF_ENGINEER_APPROVE`, `CHIEF_ENGINEER_REJECT`, `DOMAIN_SIGNOFF`, `SANDBOX_CREATE`, `SANDBOX_DELETE`, `FULL_DATABASE_EXPORT`. Full catalogue: 12-audit-compliance.md §2.

10. OWASP Top 10 (2021) Control Mapping

A01 (Broken Access Control) now explicitly covering the five-role matrix and the three-decision-point segregation-of-duties model, and A04 (Insecure Design) explicitly covering the sandbox/real-record separation as a design-level control against accidental production pollution, not just a workflow nicety. v4 adds: A02 (Cryptographic Failures) — API tokens and webhook payloads use SHA-256 / HMAC-SHA-256, keys come from the environment and are never logged (§13); A07 (Identification and Authentication Failures) — step-up re-authentication, shared rate limiting; A08 (Software and Data Integrity Failures) — signed manifests, hash-chained audit log, verified backups; A09 (Logging and Monitoring Failures) — JSON request log file, readiness and metrics; A10 (SSRF) — webhook targets are restricted to private/loopback addresses and redirects are not followed (SEC-028).

11. Logging Standards

Structured JSON application logs (stderr, and since v4 optionally a rotating file `LOG_FILE`), never containing credentials, session tokens, API tokens, webhook secrets or step-up passwords, distinct from and complementary to the `audit_log_entry` compliance record of truth (12-audit-compliance.md §1).

12. No-Cloud Verification Checklist

Added in v2 as an explicit, standing check (to be re-run at every architecture review): no dependency on a public-cloud SaaS/PaaS/IaaS service anywhere in the stack; no cloud identity provider (§2); no cloud object storage for exports or backups (14-deployment-architecture.md §5); no CDN or external script/style source reachable only via the public internet in the UI (10-ui-ux-specification.md renders with local static assets only); no telemetry/analytics SDK phoning home to a third-party cloud endpoint. Any future proposal that would introduce one of these requires an explicit, separate decision — it is never a default.

v4 re-check (2026-10-23): nothing in the improvement pack contacts a public service. The only outbound connections are the optional SMTP relay named in `SMTP_HOST` (the organisation's own) and webhook POSTs to private/loopback addresses (SEC-028); with neither configured the application makes no outbound connection, and `test_whole_lifecycle_works_with_all_outbound_network_blocked` still passes. The offline lookup bundle and every page load nothing from the network.

13. v4 Security Controls (improvement pack)

- **SEC-026** Step-up re-authentication (I-17, decision D-07). Reviewer and Administrator sign-off, Chief Engineer decision, publishing a Release and the full-database export require the actor's own current password in the request. The check re-verifies the Argon2id hash, counts a failure toward the account lock-out of SEC-004, audits `STEP_UP_SUCCESS` / `STEP_UP_FAILURE`, never logs or stores the value, and answers 403 `step_up_required` / `step_up_failed`. One verification per request covers a whole change-set decision. `STEP_UP_REQUIRED=false` is a documented, deliberate deviation for test or lab systems only.
- **SEC-027** Machine API tokens (I-29). Random (`cscm_` + 256 bits), stored only as a SHA-256 hash, shown once, expiring (1–365 days), revocable, scoped (`catalogue:read`, `changes:read`). A token is honoured only as `Authorization: Bearer ?` on `GET` operations that declare its scope, with **Viewer-level visibility** whoever owns the token, and stops working when revoked, expired or when its owner is deactivated. It can never write; it never bypasses the role matrix.
- **SEC-028** Webhooks (I-29). Targets must resolve to private, loopback or link-local addresses (`WEBHOOK_ALLOW_PUBLIC_TARGETS` is false and stays so), are re-validated at delivery, redirects are not followed, the timeout is 3 s, the body is signed `X-CSCM-Signature: sha256=` with a per-webhook secret shown once, and a failure is recorded and never affects the publish.
- **SEC-029** Operational endpoints (I-21). `/metrics` answers only clients inside `METRICS_ALLOWED_CIDRS` (loopback and RFC 1918 by default) or presenting `METRICS_TOKEN`, otherwise 403; it carries counts and timings, never business data. `/health` and `/health/ready` are unauthenticated and reveal only component states.
- **SEC-030** Integrity keys and verification (I-06, I-18, I-09). `MANIFEST_SIGNING_KEY` and `AUDIT_HMAC_KEY` are random values (>= 32 characters) read from the environment or the protected configuration file (mode 640 / ACL), generated by the installer, kept in the organisation's password safe, and **not recoverable** — losing the audit key makes existing chain entries unverifiable. Backups are verified (integrity check, scratch restore, row counts, audit chain) and carry the same sensitivity as the database (§7).
- **SEC-031** Program import (v4.2, FR-048o ... FR-048w). The `.st` upload accepts only `.st`, `.scl`, `.exp`, `.txt` and `.zip` ; at most 2 MB per program file, 2,000 files and 50 MB unpacked per upload, at most 5,000 programs per file; ZIP members are used by base name only (no path ever reaches the file system) and nothing in a program is executed, compiled or served back. The preview keeps the upload under `instance/imports/staging/` (a random 128-bit token) outside the web root; it belongs to the uploading user (403 for anybody else, including a replayed token), expires after 24 hours and is deleted on confirm or cancel; the confirmed source files are stored under `instance/imports//`. Upload and confirm are for Administrator and Engineer; the legacy mapping and a code's import source are for the four internal roles, never for Viewers, because they show titles and states of Drafts.
- **SEC-032** Role administration (v4.3, FR-156, BR-024). The user list (`GET /users`, `/admin/users`) is open to the Administrator and the StCd Owner; creating a user and the password reset are Administrator-only at the route and again in the service. `UserService.update_user` refuses an StCd Owner who changes anything but the role, and any role change that starts or ends outside Viewer, Engineer and Reviewer, with 403; the last active Administrator stays protected (FR-012). The StCd Owner's read access to the workflow is limited to the read-only routes `GET /change-requests`, `/change-requests/{id}`, its domain sign-offs, `/change-sets`, `/change-sets/{id}`, the Hub mapping and `/status-codes/{id}/where-used`; it can neither write nor decide anything. The migration that widens `user.ui_language` rebuilds the `user` table with foreign keys switched off and ends with a foreign-key check (migrations/env.py), so no table is left pointing at a discarded copy. Deleting a user (`POST /users/{id}/delete`) is Administrator-only, needs the Administrator's own password again (SEC-026), is refused for the own account and for the last active Administrator; an account with history is erased

instead of removed (BR-025) and the audit log is not touched; the StCd Owner cannot deactivate or delete. Deleting a withdrawn code (S48, `POST /admin/withdrawn-codes/delete`) is Administrator-only, needs the password again and is refused for any code that has a revision.

- **SEC-033** Windows sign-in and its simulation (v4.7, FR-163). The gateway's identity header is a credential, so it is trusted only when `request.remote_addr` is in `WINDOWS_AUTH_TRUSTED_PROXIES` (empty by default: nobody); from any other address the header is ignored and the attempt is logged as a warning. The proxy must strip the header from incoming requests and set it itself from the authenticated Windows session (IIS/ARR: server variable; Apache: `RequestHeader set X-Remote-User %{REMOTE_USER}s`; nginx: `proxy_set_header X-Remote-User $remote_user`), and the application must not be reachable except through the proxy. The identity is matched by e-mail address, which cannot change (FR-162); no account is created on the fly, a deactivated account is refused, and the session that results is the ordinary one (SEC-002, SEC-003, same revocation on deactivation). `WINDOWS_AUTH_SIMULATE` lets a developer type an identity; it exists only in the development configuration and `create_app` refuses to start in production with it set, or with an unknown `AUTH_MODE`. In `windows` mode the password routes answer 403 so that a stolen password cannot bypass the gateway.

13.1 Threats and mitigations specific to v4

Threat	Mitigation
An unattended browser approves or publishes	Step-up password per sensitive action (SEC-026)
A delegate becomes an escalation path	Decision actions only; SoD applies personally; audit names both (BR-017)
Audience export discloses a code to the wrong audience	Visibility is data; no confirmed value or Covert means hidden; Draft values never count (BR-016); a test checks that a hidden code is absent from every format
Someone edits the exported file after release	Manifest SHA-256, optional HMAC signature, offline verification (FR-103, FR-104)
Someone edits or removes audit rows with database access	Hash chain, anchor, weekly verification, append-only triggers (AUD-074)
Token leakage	Hash-only storage, short expiry, scopes, revocation, Viewer-level reach, read-only
Webhook or mail used to reach external systems	Private-address allow-list, no redirects; mail only through the organisation's relay
Malicious upload (attachment or import file)	Allow-list, magic-byte check, size and count limits, storage outside the web root, download-only, formula neutralisation (SEC-024)
Lost update overwrites engineering work	row_version check, 409 with the user's entries kept (FR-090)
Brute force across several worker processes	Rate limit in the database (NFR-009)
A bad backup is trusted	Every backup is restored into a scratch database and compared before it counts (FR-150)

12. Audit & Compliance Specification — CSCM Tool

1. Purpose and Relationship to Other Logging

This document governs the **business audit trail** (`audit_log_entry`), the compliance record of truth for every governed change in CSCM Tool. Distinct from operational application/security logging (11-security-architecture.md §11).

2. Audit Events Catalogue

Action Code	Entity	Trigger	Captures
CREATE	StatusCode, StatusCodeRevision, ChangeRequest	New Status Code created (05-governance-handbook.md §6)	full initial values, including Functional System Group/Subgroup and Turbine Platform(s)
SANDBOX_CREATE	StatusCode	Administrator creates a test/sandbox code	actor, isolated identifier allocated
EDIT	StatusCodeRevision	Draft field edit	before/after per changed field
DOMAIN_SIGNOFF	ChangeRequest, DomainSignoff	Second Engineer completes a required domain's fields	engineering domain, signer identity, timestamp
SUBMIT_FOR_REVIEW	ChangeRequest, StatusCode	Draft -> Review; identifier allocated	actor, timestamp, allocated status_code_identifier; (v4.2) identifier_source = reserved when the number came from an imported program (BR-023)
REVIEWER_SIGNOFF	ChangeRequest, StatusCodeRevision	Reviewer (Sub-PO for StCd) sign-off, approve or reject	reviewer identity, decision, comment id and text
ADMIN_SIGNOFF	ChangeRequest, StatusCodeRevision	Administrator (Product Owner) sign-off, approve or reject; may confirm/override restricted fields	administrator identity, decision, comment reference, any restricted-field override
(system) Review -> PendingApproval	StatusCodeRevision	Both required sign-offs now present	logged as part of whichever sign-off action was second
CHIEF_ENGINEER_APPROVE	ChangeRequest, StatusCodeRevision	PendingApproval -> Approved	Chief Engineer identity, timestamp
CHIEF_ENGINEER_REJECT	ChangeRequest, StatusCodeRevision	PendingApproval -> Draft	Chief Engineer identity, mandatory comment reference
DOMAIN_SIGNOFF_REQUEST	ChangeRequest	Originating Engineer requests a domain sign-off (FR-028)	domain, optional target user
REVIEWER_ASSIGN	ChangeRequest	Administrator assigns/reassigns/unassigns the Reviewer (FR-039)	before/after assigned reviewer
IDENTIFIER_OVERRIDE	StatusCode	Administrator submits with a manual identifier (FR-021a)	identifier and mandatory documented reason
WITHDRAW	ChangeRequest	Author cancels open CR	actor, timestamp, and a snapshot in before_value of the discussion that would otherwise be deleted with the CR (every comment/decision with author and time, domain sign-offs, title, identifier)
COMMENT	ReviewComment	Comment added without a decision	comment id and full text
RELEASE_CREATE	Release	New Release container created	scope filter (Functional Group/Subgroup), actor
RELEASE_ITEMS_SET	Release	Candidate set confirmed/changed pre-publish	before/after item list
RELEASE_DELETE	Release	Administrator deletes an unpublished Release (BR-008)	before: name, version, candidate revision ids
RELEASE_PUBLISH	Release, StatusCodeRevision (per item)	Publish action	publisher identity, effective date, item count
DEPRECATE	StatusCodeRevision	Released -> Deprecated	reason, superseding code (if any), actor
REINSTATE	StatusCodeRevision	Deprecated -> Released (exception path)	reason, actor
ARCHIVE	StatusCodeRevision	Deprecated -> Archived	actor, retention-period check result

Action Code	Entity	Trigger	Captures
SANDBOX_DELETE	StatusCode	Administrator permanently deletes a sandbox code	full pre-delete snapshot of the StatusCode and its revision(s) in before_value — the one case where entity_id no longer resolves to a live row after this entry is written
CATALOGUE_EXPORT	ExportJob	Any catalogue export generated	requester, scope, format
FULL_DATABASE_EXPORT	ExportJob	Administrator generates a full raw database export/backup	requester (always Administrator), timestamp, file reference
LOOKUP_CHANGE	Lookup* (incl. Functional Group/Subgroup/Platform/Domain)	Controlled vocabulary value added/deactivated	before/after, actor
SUBGROUP_CREATE	FunctionalSubgroup	Administrator creates a new Functional Subgroup (new 2026-09-17)	group, code, label, sub-range, actor
IMPORT_STARTED / IMPORT_COMPLETED / IMPORT_FAILED	ImportJob	Library import lifecycle (new 2026-09-17)	requester, row_count, created_count, error_count; (v4.2) for an import from programs also source_format ST, the file names and the change_set_id
USER_CREATE / USER_UPDATE / USER_DEACTIVATE / ROLE_CHANGE / DOMAIN_TAG_CHANGE	User	Administration action	before/after, actor
USER_DELETE (v4.6)	User	An account is deleted: removed when nothing points at it, erased when it has history (BR-025, AUD-076)	before_value: username, e-mail, full name, role, domain tags; after_value: mode (removed or erased), references; actor
WITHDRAWN_CODE_DELETE (v4.9)	StatusCode	An Administrator deletes a code whose change request was withdrawn; the row stays and gives up its numbers (BR-026, AUD-076)	before_value: identifier, reserved identifier, legacy number, PrgIdx_GUi, group, subgroup, withdrawal time; actor
LOGIN_SUCCESS / LOGIN_FAILURE / LOGOUT / ACCOUNT_LOCKED	User (session)	Authentication event; (v4.7) a Windows sign-in (FR-163) writes the same codes, with method = windows, the identity the gateway supplied and whether it came from the gateway or the developer simulation	outcome, reason
LANGUAGE_REVIEWER_ASSIGN / LANGUAGE_REVIEWER_REMOVE (v4.7)	LanguageReviewer	An Administrator or StCd Owner assigns a native speaker to a language or takes the assignment away (FR-164)	language, reviewer, actor
TRANSLATION_REVIEW_COMPLETE (v4.7)	TranslationReview	A reviewer completes the review of a language	language, fingerprint at start and at completion, strings, number of OK / CHANGE / UNCLEAR, actor
TRANSLATION_PROPOSAL_DECIDE (v4.7)	TranslationReviewItem	An Administrator or StCd Owner accepts, rejects or notes a proposal; an accepted one creates the override	before: language, string id, verdict; after: language, English text, decision, the text shown (accept); actor
TRANSLATION_OVERRIDE_REMOVE (v4.7)	TranslationOverride	An override is removed and the shipped text shows again	before_value: language, English text, removed text; actor
TERM_CREATE / TERM_UPDATE (v4.9)	TerminologyTerm	A manager adds a term or changes its English text, category, definition, never translated flag or state (FR-166)	after: code, English text, category; on an update also before and approvals_reopened; actor
TERM_TRANSLATE / TERM_APPROVE / TERM_REJECT (v4.9)	TerminologyTranslation	A translation of a term is proposed, approved by a second person or sent back with a note (FR-166, BR-018)	term code, language, text (the note on a send-back); before on a change; actor
PASSWORD_CHANGE / PASSWORD_RESET	User	Credential change	actor (self or Administrator)
STEP_UP_SUCCESS / STEP_UP_FAILURE (v4)	User	Password confirmation for a sensitive action (SEC-026)	actor, purpose (e.g. RELEASE_PUBLISH) — never the password
PLANNED_DATES_SET (v4)	StatusCodeRevision	Administrator stamps planned effective / deprecation dates (FR-109)	before/after dates
RELEASE_TARGETS_SET (v4)	Release	Target software versions set before publish (FR-106)	before/after version codes

Action Code	Entity	Trigger	Captures
BULK_SAMPLE_DRAW / BULK_SAMPLE_VERDICT / BULK_SAMPLE_VOID (v4.13)	ChangeSet / BulkSample	A sample was drawn (population, size, seed, the codes), a sampled code was marked OK or DEFECT, a sample was closed after its defects were sent back (FR-180)	person, step, domain, sample, verdict, note
BULK_DECISION / BULK_EDIT / BULK_SUBMIT / BULK_DOMAIN_SIGNOFF (v4.13)	ChangeSet	A bulk action finished: how many codes were done and how many failed, the sample that covered it, the step and the decision or the fields set (FR-180)	person, counts, sample
PACKAGE_REQUEST / PACKAGE_EXPORT (v4.10)	ExportJob	A package of all languages and access levels was requested and finished (FR-174)	requester, release, scope, file count, SHA-256
OFFLINE_BUNDLE_EXPORT / RELEASE_NOTES_EXPORT (v4)	ExportJob	Offline lookup bundle or release notes generated (FR-105, FR-108)	requester, release, SHA-256, options
EXPORT_PROFILE_CREATE / EXPORT_PROFILE_UPDATE (v4)	ExportProfile	Administrator changes an export profile (FR-100)	before/after columns, audience, language
TRANSLATION_SAVE / TRANSLATION_APPROVE / TRANSLATION_CARRY_OVER (v4)	StatusCodeRevision	Translation workflow (FR-110)	language, status, translator / approver
DELEGATION_CREATE / DELEGATION_REVOKE (v4)	Delegation	Out-of-office cover (FR-125)	delegator, delegate, role, dates
MAPPING_SEEDED (v4.17)	StatusCodeMapping	The initial mapping was loaded from the workbook (FR-191)	rows, file
MAPPING_CHANGE_REQUESTED / MAPPING_CHANGE_WITHDRAWN (v4.17)	MappingChange	A request to change the mapping was made, or withdrawn by its requester (FR-194, FR-195)	title, rows, origin
MAPPING_APPROVAL / MAPPING_REJECTION (v4.17)	MappingChange	The StCd Owner, the Chief Engineer or an Administrator approved or rejected a request (BR-027)	role, decision, comment
MAPPING_CHANGE_APPLIED / MAPPING_CHANGE_REJECTED (v4.17)	MappingChange	The third approval applied the request, or a rejection closed it (FR-194)	rows, or the rejecting role
MAPPING_ROW_ISSUED (v4.17)	StatusCodeMapping	An import created a code with an approved number; the row is frozen (FR-192)	new number
CHANGE_SET_CREATE / CHANGE_SET_ADD_CR / CHANGE_SET_REMOVE_CR / CHANGE_SET_CLOSE (v4)	ChangeSet	Change-set management (FR-124)	set, request, actor
FIELD_COMMENT / FIELD_COMMENT_RESOLVE (v4)	ChangeRequest	Field-level comment and its resolution (FR-127)	field, text, author / resolver
CHECKLIST_ITEM_CREATE / CHECKLIST_ITEM_UPDATE (v4)	ReviewChecklistItem	Administrator edits the review checklist (FR-128)	before/after
ATTACHMENT_ADD / ATTACHMENT_DELETE (v4)	ChangeRequest	Evidence attached or removed (FR-130)	file name, size, SHA-256, actor
SETTINGS_CHANGE (v4)	Settings	Administrator changes SLA, auto-assignment or capacity settings	before/after per key
OCCURRENCE_IMPORT (v4)	OccurrenceImport	Fleet-feedback counts imported (FR-143)	source, period, matched codes, unmatched rows
API_TOKEN_CREATE / API_TOKEN_REVOKE (v4)	ApiToken	Token issued or revoked (FR-144)	name, scopes, expiry — never the secret

Action Code	Entity	Trigger	Captures
WEBHOOK_CREATE / WEBHOOK_UPDATE (v4)	Webhook	Webhook registered, paused or resumed (FR-146)	name, URL, events, active flag
HUB_MC_DRAFT_CREATE / HUB_BULK_DEPRECATION (v4)	StatusCode, LookupFunctionalSystemGroup	Hub-to-Main consolidation (FR-147)	replaced Hub code; counts of requests created and skipped
AUDIT_ARCHIVE (v4)	AuditArchive	Old audit entries archived (AUD-073)	file name, count, first and last entry id, SHA-256, cutoff

Notes on the catalogue. `REVIEWER_SIGNOFF`, `ADMIN_SIGNOFF` and the Chief Engineer codes are written per decision; `ACCOUNT_LOCKED` is written instead of `LOGIN_FAILURE` on the attempt that locks an account; `USER_DEACTIVATE` instead of `USER_UPDATE` when an account is switched off. `CATALOGUE_EXPORT` also covers profile / audience / language / bundle exports; its `after_value` records those options and the file's SHA-256. A delegate's decision entries carry `on_behalf_of` (BR-017). `tests/system/test_docs_consistency.py` fails if the application can write an action code that this table does not list.

3. Retention Rules

Data	Minimum Retention	Rationale
audit_log_entry rows	7 years, never purged by the application, including every SANDBOX_CREATE/SANDBOX_DELETE pair even after the underlying sandbox record is gone	05-governance-handbook.md §12 makes the audit trail the only surviving evidence of a deleted sandbox record
Archived Status Codes and all revisions/CRs/comments/domain sign-offs	7 years from archival, never hard-deleted	Standing rule
Published Releases	Indefinite, immutable	Standing rule
Sandbox Status Codes themselves	No minimum retention — this is the one entity category explicitly exempted from retention, since it is designed to be created and deleted freely by an Administrator for testing (05-governance-handbook.md §12)	The audit trail, not the record, is what is retained
Application/security logs	1 year	Standing rule
Audit archive files (v4) (audit_archive_*.jsonl.gz + .sha256)	Same as the audit rows they copy: 7 years; the rows themselves also stay in the database	Archiving writes a copy and an AUDIT_ARCHIVE entry; nothing is deleted (AUD-073, BR-020)
Backups (v4)	The shipped schedule keeps 35 generations of verified nightly backups (14-deployment-architecture.md §6)	Recoverability (NFR-012); a backup carries the same sensitivity as the database

No feature provides deletion of `audit_log_entry` rows, non-sandbox Status Codes, or published Releases.

4. Traceability Requirements

- **AUD-001** For any non-sandbox Status Code, the system shall produce a complete, chronologically ordered trace from `CREATE` through every `EDIT`, `DOMAIN_SIGNOFF`, `SUBMIT_FOR_REVIEW`, `REVIEWER_SIGNOFF`, `ADMIN_SIGNOFF`, `CHIEF_ENGINEER_APPROVE/REJECT`, `RELEASE_PUBLISH`, and any `DEPRECATE/ARCHIVE` event, with actor and timestamp at each step.
- **AUD-002** For any Release, the system shall list every included revision and, transitively, its full change history.
- **AUD-003** For any User, the system shall list every action they performed as `actor_id`.
- **AUD-004** For any deleted sandbox Status Code, the system shall be able to retrieve the `SANDBOX_CREATE` and `SANDBOX_DELETE` audit pair, including the full pre-delete snapshot, even though the live record no longer exists.
- **AUD-072** (v4) No application path updates or deletes an `audit_log_entry`: there is no endpoint for it, and the triggers `trg_audit_no_update` / `trg_audit_no_delete` abort any attempt through the application connection (FR-072).
- **AUD-073** (v4) Entries older than the retention period (default 7 years) are archived — never deleted — by `flask audit-archive`: a gzip JSON-lines file with a `.sha256` file, plus an `AUDIT_ARCHIVE` entry naming the file, count, id range and hash. The next run continues after the last archived entry, so running it twice is harmless.
- **AUD-074** (v4) Each entry stores `prev_hash` and `entry_hash` (SHA-256, or HMAC-SHA-256 when `AUDIT_HMAC_KEY` is set) over its own fields and its predecessor's hash, written inside the same write transaction as the entry. Rows that existed before v4 were chained by the migration; a hash-less row after hashed rows is itself a failure.

- **AUD-075** (v4) The chain is verified on demand (`flask audit-verify`, Administrator UI and `POST /operations/audit-verify`) and weekly by the scheduler; a failure names the first bad entry and the reason (broken link, changed content, rows removed since the last verification — detected through the stored anchor), exits non-zero and shows as `broken` in the readiness probe; every backup is checked for chain integrity before it counts as verified (FR-150, FR-151).
- **AUD-076** (v4.6, changed in v4.9) Deleting a user writes `USER_DELETE` with the account's username, e-mail, name, role and domain tags before the account is removed or erased, the mode and the actor. The audit log itself is never modified: an account with history is erased in place (BR-025), so no earlier audit entry is left without its actor, and the hash chain still verifies. Deleting a withdrawn code writes `WITHDRAWN_CODE_DELETE` with what the code held (BR-026). The `WITHDRAW` snapshot of a change request also records what the code gave back to the catalogue (`released_to_the_catalogue`: reserved identifier, legacy number, PrgIdx_GUi, BR-026).
- **AUD-077** (v4.7) Reviewing translations is audited: assigning or removing a reviewer, completing a review, deciding on a proposal and removing an override each write one entry (`LANGUAGE_REVIEWER_ASSIGN`, `LANGUAGE_REVIEWER_REMOVE`, `TRANSLATION_REVIEW_COMPLETE`, `TRANSLATION_PROPOSAL_DECIDE`, `TRANSLATION_OVERRIDE_REMOVE`); the text of a decided proposal stays in its row and the text of a removed override is kept in the entry, so what the application showed in a language can always be reconstructed. The terminology (FR-166) is audited the same way (v4.9): adding or changing a term writes `TERM_CREATE` or `TERM_UPDATE`; proposing, approving and sending back a translation write `TERM_TRANSLATE`, `TERM_APPROVE` and `TERM_REJECT` with the term, the language and the text.

5. Change History Requirements

Field-level `EDIT` entries batched per explicit Save (v4: each save also increments `row_version`, so a lost update can be told from the entries), full historical content independently queryable via the row-per-revision model, audit log complements rather than replaces it.

6. Electronic Approval Rules

- `REVIEWER_SIGNOFF`, `ADMIN_SIGNOFF`, and `CHIEF_ENGINEER_APPROVE/REJECT` each constitute a distinct electronic approval record, bound to the deciding user's authenticated session, and each independently enforces segregation of duties (05-governance-handbook.md §7–§8, 11-security-architecture.md §3 SEC-011).
- The pairing of `audit_log_entry` and `review_comment` together constitute durable evidence of each decision; both are retained under the same 7-year minimum.
- (v4) Each of those approvals is preceded by a password confirmation (`STEP_UP_SUCCESS` with the purpose) by the same actor in the same request, so an unattended session cannot approve (SEC-026); a delegate's approval is recorded with `on_behalf_of`, and every approval remains bound to the delegate's own identity.
- No approval action can be back-dated or edited after the fact; all sign-off/approval timestamps are system-generated at the moment of the transaction.
- The automatic `Review -> PendingApproval` transition (triggered by the second sign-off, 07-database-design.md §7b) is itself traceable: the audit entry for whichever sign-off action was second also records the resulting state change, so there is no silent/unaudited system-triggered transition anywhere in the lifecycle.

7. Compliance Controls Summary

Control	Implementation
Segregation of duties (3 decision points)	05-governance-handbook.md §7–§8, SEC-011, DB CHECK constraints (07-database-design.md §4)
Immutable audit trail	No delete/update endpoint exists; survives even sandbox record deletion (§3); append-only triggers (AUD-072)
Tamper-evident audit trail (v4)	Hash chain, anchor, weekly verification, archive that never deletes (AUD-073 … AUD-075)
Deliberate approvals (v4)	Step-up password, delegation audit with on_behalf_of, change-set decisions one audit entry per request
Immutable released data	trg_prevent_edit_locked_revision, atomic publish transaction
Sandbox isolation	trg_prevent_sandbox_release blocks any sandbox revision from ever reaching a Release; is_sandbox excluded from default search/export (09-api-specification.md §6)
Complete traceability	AUD-001–AUD-004, AUD-072–AUD-075
Defined retention, with one explicit, narrow exception	§3

Control	Implementation
Access control over audit data	Administrator (full), Reviewer/Chief Engineer (scoped); no role can write to it directly
Full-database export control	Administrator-only (SEC-015), itself audited (FULL_DATABASE_EXPORT)

13. Test Strategy — CSCM Tool

1. Test Levels

Unit, Integration, System (API), Performance, Security, UAT — tooling: `pytest`, in-memory or file SQLite, the Flask test client, `schemathesis`/`hypothesis` for contract and property tests, timing tests without coverage tracing for performance, `black`/`ruff`/`mypy`/`bandit`/`pip-audit` as gates. Since v4 the suite is verified three ways (§11) and every requirement id must be accounted for (§12).

2. Unit Tests

Minimum coverage target: 80% on `service/` and `validation/`. Required areas, updated for v2:

- Lifecycle transition validator: every legal transition in the 7-state model (05-governance-handbook.md §5) succeeds; every illegal transition (exhaustive pairwise) raises the correct domain exception.
- Identifier allocation: sequential within a Functional System Group's range; sequential within a Functional Subgroup's sub-band when selected; independence between groups; capacity-exhaustion behavior (07-database-design.md §7a); sandbox codes never draw from a real range.
- Dual sign-off logic: recording the first sign-off leaves state at `REVIEW`; recording the second (either order) atomically advances to `PENDING_APPROVAL`; a reject from either sign-off returns to `DRAFT` and clears both sign-off columns.
- Chief Engineer approval: segregation-of-duties check against all three of author, Reviewer sign-off, Administrator sign-off.
- Cross-Domain Sign-Off gating: Submit-for-Review blocked while any required domain lacks a `DomainSignoff`; unblocked once all present.
- Restricted-field handling: each Engineer-proposed audience-access field (development/sales/tcc/service/turbine-operator-package/grid-operator/service-partner/customer, corrected 2026-09-17) is not authoritative until `admin_signoff_at` is set.
- Field-format rules (corrected 2026-09-17, replacing the retired Safety-category/Alarm-Behaviour and Auto-reset/delay cross-field rules, neither of which the real data model supports): `set_delay`/`reset_delay` match the `^\*?\d+(\.\d+)?\s?(ms|s|min|MIN|h|d)$` pattern; `status_category` is restricted to `ERROR`/`WARNING`/`INFO`; Functional Subgroup must belong to the selected Functional System Group; at least one Turbine Platform is selected.
- Sandbox delete: only permitted when `is_sandbox = true`; blocked for every other record, with the DB trigger as a second line of defense.

3. Integration Tests

- Repository layer against real (temp-file) SQLite: uniqueness of `status_code_identifier`, partial unique indexes (`is_current`, one-open-CR now spanning three states), all eight triggers (`trg_prevent_edit_locked_revision`, `trg_prevent_release_item_delete`, `trg_prevent_sandbox_release`, `trg_prevent_nonsandbox_delete`, `trg_prevent_subgroup_overlap`, `trg_prevent_group_overlap`, `trg_audit_no_update`, `trg_audit_no_delete`) and the four FTS triggers.
- Full Change Request lifecycle: Draft -> (domain sign-offs) -> Review (identifier allocated) -> (dual sign-off) -> PendingApproval -> Chief Engineer Approve -> Approved -> Release publish -> Released.
- Atomic Release publish, including the sandbox-exclusion guarantee: attempt to add a sandbox revision to a Release candidate set and assert it is never offered as a candidate and, if forced via direct repository call in a test, is rejected by the trigger.
- Concurrent identifier allocation: two simulated concurrent submissions within the same Functional Subgroup never receive the same number (07-database-design.md §7a).
- Sandbox delete: full lifecycle create -> delete, assert the row is gone, assert the `SANDBOX_CREATE` and `SANDBOX_DELETE` audit entries both exist and the latter's `before_value` contains a complete pre-delete snapshot.
- Audit log completeness: every new v2 action code (12-audit-compliance.md §2) produces exactly one audit entry per triggering call.

4. System (API) Tests

- Full request/response contract conformance against `artifacts/openapi.yaml` v4.0.0 (161 operations; every operation of the code is in the contract and every unauthenticated call is 401, except the documented public routes).
- Authorization matrix: six roles × every endpoint, including the three decision endpoints' segregation-of-duties checks and the sandbox-only-delete / Administrator-only-full-export restrictions.
- Identifier lifecycle via API: `status_code_identifier` is `null` in every response while Draft; becomes non-null only after `POST .../submit`; `GET /status-codes/next-available` never reserves a number (two sequential calls without an intervening submit return the same preview).
- Cross-Domain Sign-Off: a Submit attempt with an outstanding required domain returns 422 with a field error naming the missing domain; a matching-domain Engineer's `POST .../domain-signoffs` call from a non-matching domain returns 403.
- Export correctness: catalogue export never includes sandbox-flagged records even if explicitly requested; `POST /exports/full-database` returns 403 for every non-Administrator role.

5. Performance Tests

Targets: p95 500ms search, p95 200ms single-record read, 5s CSV/JSON export, 15s PDF export, 50 concurrent users — re-validated against a synthetic dataset now distributed across all nine Functional System Groups to confirm identifier allocation contention does not create a hot-row bottleneck under concurrent submission load within a single popular group.

6. Security Tests

`bandit` and `pip-audit` in CI (§11 gates); ZAP baseline pre-release; manual OWASP checklist. Since v4 also: the route inventory (every route declares its access and the declared role list is what each of the five roles actually gets), step-up, token and webhook tests, audience leakage tests, attachment hardening and the no-egress lifecycle test. Added: a specific manual check that no outbound network call in the codebase targets a public-cloud endpoint (11-security-architecture.md §12 "No-Cloud Verification Checklist") — grep-based CI check for known cloud SDK imports (e.g. `azure.*`, `boto3`, `google.cloud`) failing the build if found, since this is now a hard architectural constraint, not a preference.

7. User Acceptance Test Scenarios

- UAT-01: Engineer creates a new Status Code under Converter & Grid Interface (WCNV), sees no fixed identifier while Draft, submits for Review, and confirms an `StCd-01xxx` identifier appears immediately after submission.
- UAT-02: Reviewer and Administrator each sign off independently (in either order); the CR reaches PendingApproval automatically after the second sign-off, with no separate action required.
- UAT-03: Chief Engineer approves; the revision reaches Approved; Engineer sees the updated status on My Change Requests.
- UAT-04: Reviewer rejects with a comment; Engineer sees the comment, edits, resubmits; confirms the Administrator's prior sign-off (if any) was cleared and must be re-given.
- UAT-05: An Electrical-domain Engineer creates a Generator-group (WGEN) code, is blocked from submitting due to outstanding Controls-domain fields, requests sign-off, a Controls-domain Engineer completes and signs off, and submission then succeeds.
- UAT-06: Administrator builds and publishes a Release from Approved codes, confirms an Approved-but-excluded candidate remains Approved, and confirms no sandbox code ever appears as a candidate.
- UAT-07: Viewer searches and exports the current Released catalogue filtered to a single Functional System Group.
- UAT-08: Administrator creates a sandbox Status Code, exercises it through Draft/Review/PendingApproval for training purposes, confirms it can never be added to a real Release, and permanently deletes it, then confirms via Audit Log Search that both the creation and deletion are retained with a full snapshot.
- UAT-09: Any user attempts (via UI and via direct API call) to sign off or approve their own submission at each of the three decision points and is blocked in every case.
- UAT-10: Administrator performs a full-database export; confirms no other role can reach that action.
- UAT-11 (new 2026-09-17): Engineer uploads a CSV library file; confirms every valid row becomes a Draft

requiring the full sign-off/approval cycle (none land as Approved/Released), and the error report correctly lists rows that failed validation.

- UAT-12 (new 2026-09-17): Engineer attempts to create a Status Code without selecting a Functional Subgroup; confirms Submit is blocked. Administrator creates a new Functional Subgroup in a reserved block, then successfully overlaps-tests a conflicting range and receives a clear, non-destructive error (the form retains the entered name).
- UAT-13 (new 2026-09-17): Engineer deliberately submits an incomplete New Status Code form; confirms the resulting validation error leaves every already-entered field intact (no reset), and confirms every classification dropdown shows an inline description for its selected option.

8. Full Test Matrix

ID	Level	Area	Scenario	Priority
TC-001	Unit	Lifecycle	All valid 7-state transitions succeed	P1
TC-002	Unit	Lifecycle	All invalid transitions rejected (exhaustive pairwise)	P1
TC-003	Unit	Validation	Each mandatory field rejects null/empty	P1
TC-004	Unit	Numbering	Identifier format ^StCd-\d{5}$ / ^StCd-T\d{5}$ enforced	P1
TC-005	Unit	Validation	set_delay/reset_delay reject a value not matching the unit pattern (e.g. 10minutes, abc)	P1
TC-006	Unit	Validation	status_category rejects any value outside ERROR/WARNING/INFO	P1
TC-007	Unit	Numbering	Sequential allocation per Functional System Group range	P1
TC-008	Unit	Numbering	Sequential allocation per Functional Subgroup sub-band, independent of siblings	P1
TC-009	Unit	SoD	Reviewer sign-off by requester raises domain error	P1
TC-009a	Unit	SoD	Administrator sign-off by requester raises domain error	P1
TC-009b	Unit	SoD	Chief Engineer approval by requester/reviewer/admin raises domain error	P1
TC-010	Integration	Uniqueness	Duplicate status_code_identifier rejected at DB layer	P1
TC-011	Integration	Concurrency	One current revision per code enforced under concurrent writes	P1
TC-012	Integration	Concurrency	Only one open (Draft/Review/PendingApproval) CR per code	P1
TC-013	Integration	Immutability	Trigger blocks edit of Approved/Released/Archived revision	P1
TC-014	Integration	Release	Atomic publish: partial failure rolls back fully	P1
TC-014a	Integration	Release	Sandbox revision can never be inserted into release_item	P1
TC-015	Integration	Audit	Every v2 state-changing action produces exactly one audit entry	P1
TC-015a	Integration	Audit	Sandbox delete produces SANDBOX_DELETE with full before_value snapshot	P1
TC-016	System	Auth	Login lockout after 5 failed attempts, clears after 15 min	P1
TC-017	System	Auth	Session expires after 30 min idle	P2
TC-018	System	AuthZ	Full 5-role × endpoint permission grid	P1
TC-018a	System	AuthZ	Sandbox delete: 403 for non-Administrator and for non-sandbox records	P1
TC-018b	System	AuthZ	Full-database export: 403 for every non-Administrator role	P1
TC-019	System	AuthZ	Direct API call cannot bypass SoD at any of the three decision points	P1
TC-020	System	Domain Sign-Off	Submit blocked with 422 while a required domain is unsigned	P1
TC-021	System	Domain Sign-Off	Sign-off attempt from a non-matching domain returns 403	P1
TC-022	System	Search	Full-text search matches status_code_identifier, title, description	P2
TC-023	System	Export	Catalogue export never includes sandbox-flagged records	P1
TC-024	Performance	Search	p95 <= 500ms at 20,000 revisions across all 9 groups	P2
TC-025	Performance	Numbering	Concurrent submission within one popular group does not breach identifier-allocation latency target	P2
TC-026	Security	Injection	XSS payload in Description/comment fields is escaped on render	P1
TC-027	Security	Injection	SQL injection probe on search filter has no effect	P1
TC-028	Security	CSRF	State-changing request without CSRF token rejected	P1
TC-029	Security	No-Cloud	CI grep check fails the build on any cloud-SDK import	P1
TC-030	UAT	End-to-end	UAT-01 through UAT-13 (§7)	P1
TC-031	Unit	Validation	Status Code create rejected (422) with no functional_subgroup	P1
TC-032	Integration	Lookups	New subgroup insert rejected on sub-range overlap (trigger + service layer)	P1

ID	Level	Area	Scenario	Priority
TC-033	System	Import	Import job creates Drafts only; none reach Approved/Released without normal sign-offs	P1
TC-034	System	Import	Invalid rows produce a downloadable error report and create nothing	P1
TC-035	System	Export	XLSX export produces the same flat row set as CSV/JSON for the same scope	P2
TC-036	System	UI/UX	A rejected (422) form submission never clears previously entered field values	P1
TC-037	Integration	Lookups	New Functional System Group insert rejected on range overlap (trigger + service layer)	P1
TC-038	Integration	Security / SoD	Step-up password required and audited on every sensitive action; delegation grants decision actions only and keeps segregation of duties (test_step_up_and_delegation)	P1
TC-039	Integration	Data integrity	Stale Draft save refused with 409; reference, duplicate and where-used warnings; platform variants; planned dates; remedy fields; capacity hint (test_content_integrity)	P1
TC-040	Integration	Review aids	Field comments, review checklist, evidence attachments (hardening), change sets, reviewer balancing (test_review_aids)	P1
TC-041	Integration	Workflow	SLA ageing in business days, stalled requests, notices and reminders (idempotent), KPIs (test_sla_and_notifications)	P2
TC-042	Integration	Publication	Profiles and wrapped PDF, audience scoping, manifests / bundles / verification, release notes, software versions, translations, offline bundle (test_publication)	P1
TC-043	Integration	Import	Template, dry run, update mode, reconciliation, error reports, background job on a real file database (test_import_toolkit)	P1
TC-044	Integration	Operations	Audit hash chain (edit / remove / truncate detected, key, archive), verified backups and restore, CLI jobs, readiness, metrics gating, log file (test_audit_chain_and_ops)	P1
TC-045	Integration	Migration	Upgrade of a v3 database keeps data, hashes old audit rows, fills the index, matches a fresh schema, downgrades (test_schema_upgrade)	P1
TC-046	Integration	Integration / reports	API tokens, change feed with ETag, signed webhooks, saved searches, fleet feedback, catalogue health, KPIs, Hub-to-Main consolidation (test_integrations_and_reports)	P2
TC-047	Integration	Search / help	Ranked, typo-tolerant search and its substring fallback; help content complete and consistent with the glossary (test_search_and_help)	P2
TC-048	System	AuthZ	Route inventory: every route declares access, anonymous callers are refused, declared roles equal enforced roles for all six roles (test_route_inventory)	P1
TC-049	System	i18n / a11y	German catalogue complete and every other language complete (FR-167); contrast >= 4.5 : 1 in the light and in the dark theme, no colour outside the tokens, the theme choice on every page; landmarks, labels, headings, table scope on every page in both languages (test_i18n, test_accessibility, test_dark_mode); the version is shown and agrees with pyproject.toml (test_version)	P2
TC-050	System	Multi-process	Two server processes share the login rate limit and survive 16 concurrent writes (test_multiprocess, unit/test_rate_limit)	P1
TC-051	System	Deployment	Installer syntax and dry runs, systemd units, settings written are settings read, production entry point starts and refuses a weak key, CI definitions, PostgreSQL ratchet (test_deploy_artifacts)	P2
TC-052	System	Documentation	The documents agree with the code: endpoint table, schema counts, audit codes, glossary, requirement ids, links (test_docs_consistency)	P1
TC-053	Integration	Status code import	The six real .st programs (three Main, three Hub) are read, previewed with multi-select, imported as Drafts with reserved numbers in one change set, issued exactly the reserved number at Submit, the legacy -> new mapping is listed and exported, staged uploads are private and expire (test_st_import); the status category follows the alarm type, the default or the choice per program (test_st_category_and_draft_delete); no value and no category is assumed - a variable stays its name, what the program lacks is listed in the comment; the user-group numbers become named access levels only where exactly one level fits (test_st_no_assumptions)	P0
TC-054	Integration	Roles and languages	The StCd Owner role and its limits (only an Administrator creates users; the owner moves users between Viewer, Engineer and Reviewer and nothing else), role colours, the language names (letters left, flag right), the drawn flags, French / Hindi / Georgian / Arabic pages with English fallback and right-to-left Arabic, catalogue integrity (test_roles_and_languages)	P1

ID	Level	Area	Scenario	Priority
TC-055	Integration	User administration	View, activate / deactivate and delete users: only an unused account is deleted, the password is asked again, an account with history is refused and deactivated, the StCd Owner has no such buttons (test_user_administration)	P1
TC-056	Integration	Withdraw and re-import	Several change requests withdrawn at once with a result each; a withdrawn import Draft gives its number back, a withdrawn code with an identifier keeps it and is restored; changed programs update the Draft or open a revision (test_withdraw_and_reimport)	P1
TC-057	System	Translations	The translation document equals the generator output; ids and fingerprints are stable; every language's review sheet is complete (review, missing, checklist), right to left where needed (test_translations_doc)	P1
TC-058	Integration	E-mail and Windows sign-in	The e-mail address cannot be changed (service, API, database trigger); Windows identity matched by e-mail or DOMAIN\name, accepted only from a trusted proxy, no auto-creation, deactivated accounts refused, password routes off in windows mode, the developer simulation uses the same path and is refused in production (test_email_and_windows_signin)	P1
TC-059	Integration	Translation review	Assigned reviewers only, managers assign and decide; save validations (a change needs a proposal, placeholders, length), completion rule, XLSX download and upload, no self-decision and final decisions, an accepted proposal shown at once and removable, missing translations proposed, right to left, audit entries (test_translation_review)	P1
TC-060	Integration	PDF in every script	Each of the twelve script families gets its own font, Latin needs none, Arabic is told from Urdu, right-to-left cells are right-aligned, shaping is switched on only for the cells that need it, every character of every translation is covered by a bundled font, the fonts and their licence are bundled, a 1,500-row Latin export stays fast (test_pdf_scripts); the rendered pages were looked at	P1
TC-061	Integration	Technical nomenclature	The vocabulary ships with every term in every language as a draft and agrees with the interface catalogue, a never-translated term has no translation, everybody reads and only the technical roles propose, an interface reviewer has no say, a proposal is a draft until another person approves it, send-back needs a note and reopens an approval, validation, only managers add and change terms and a new English text reopens approvals, the consistency check, export per language and for all, a formula is neutralised in the CSV (test_terminology)	P1
TC-062	System	Windows sign-in against a real Windows identity	SSPI (Negotiate / NTLM) names the Windows user who runs the tests; the whole chain client, gateway, application signs that user in; a request without credentials gets a challenge and never reaches the application; a header the client sends is dropped by the gateway and ignored when it does not come from the trusted gateway; an unknown identity and a deactivated account are refused (test_windows_gateway, Windows only)	P1
TC-063	Integration	Fields of the generated programs	A Main and a Hub program import keeps every field of the file (further programs, mask, task call frequency, flags, 15 user-group levels), the two zero-load attributes are not mixed up, a field of the other controller is refused (API and form, which keeps what was typed), values out of range are refused, a new revision and a Draft update keep what the caller does not send, the diff lists changes, the exports carry the columns and an audience export never does, withdrawing removes the levels (test_generated_fields)	P1
TC-064	Integration	The menu and the beta mark	What each of the six roles sees and does not see, the groups, no link twice, only the current page marked, the counts at the links and on the group, groups that open and close, the Menu button and its script and style, the menu in German, a language is beta until a review of its current text is completed, a Viewer and a guest see only reviewed languages (and an assigned reviewer his own), a stored preference reads as English, the mark and the hiding can be switched off (test_menu)	P1
TC-070	Integration	Bulk workflow with sample inspection	The size of the sample follows the settings, the sample is spread over the subgroups and reproducible, readiness names what stops each Draft, bulk edit sets a value on every Draft with its own audit entry and refuses unknown fields and other people's Drafts, a bulk domain sign-off needs a sample and gives each code its own sign-off, only a holder of the domain signs, a small batch needs no sample, a defect blocks the decision until the defective codes are sent back and a new sample is passed, a sample cannot be borrowed, reused or kept after the batch changed, a defect needs a note, bulk submit sends what is ready and reports the rest, the dossier holds codes, samples and audit, Viewers see nothing, the workbench works end to end and is translated, and every list with a checkbox per row has a header box that selects them all (test_bulk_workflow, test_select_all)	P1
TC-071	System	First start on a new machine	scripts/first_start.py lists its six steps from any folder, changes nothing with --dry-run, honours --port and --no-admin; a real run on a clean copy (environment, packages, database, first Administrator, second run) was done by hand (test_first_start)	P2

ID	Level	Area	Scenario	Priority
TC-072	System	Release documents as PDF	The Markdown reader finds every kind of block, inline markup becomes styles, every character of the documents can be drawn (a new symbol needs a word), the four files carry the version in their names, the full documentation has a contents list and appendices, the user manual is built for real (test_docs_pdf)	P2
TC-073	System	Design showcase	The slide model refuses a text that overflows its box, both decks build from placeholder pictures with every picture described, the demo database and its two demo-only routes work and the application itself has none of them (test_showcase)	P2
TC-074	System	Identity	The long name, the owner line and the logo are on every title page, the guide names the same owner, the logo files exist and are drawn, the menu bar and the login page show the logo (test_branding)	P2
TC-075	System	Hand-over to the team repository	The audit finds a planted account name, key, .env, data file, personal path, stray address, wrong version and missing release file and passes a clean tree; the export is one commit on main by the named author with the annotated tag; a push without the address and the confirmation is refused (test_handover)	P2
TC-076	Integration	The mapping and its three approvals	The seed workbook loads once into an empty table, is consistent (unique numbers, the Hub rows in the Hub range, the four Rotor gaps numbered) and a bad one is refused as a whole; a new row takes effect only after the StCd Owner, the Chief Engineer and an Administrator have approved, two are not enough, one rejection (with a reason) closes the request, only the three roles decide and each once, the requester cannot decide, a decision asks for the password, the requester can withdraw, a Viewer neither reads nor proposes (test_mapping)	P1
TC-077	Integration	What the import takes from the mapping	Cause and directive become the remedy text, the mapping gives the value of a delay the program sets with a parameter and a time in the program wins, the header legacy number must be the row's, a row that is only proposed is used with a warning, a rejected row or one of the other family is as good as none, a title that does not say what the German name says is flagged and never blocks, a code whose row is gone keeps its values (test_st_mapping_rules, test_st_no_assumptions, test_st_category_and_draft_delete)	P1
TC-078	Integration	Checks of a request	Every wrong value is named and makes no request (type, counter, numbers, delays, access levels, completeness, the controller of the number range), numbers stay unique, a change, a confirmation and a rejection work on stored rows, a row changed in between makes the older request unusable, the numbers of an issued row cannot change (test_mapping)	P1
TC-079	Integration	The workbook and the pages	The export is the layout the upload reads and compares as unchanged, a workbook makes one request with the new and the different rows, a wrong row names its place and requests nothing, a formula is never written, the pages show the mapping, the requests and the three places, the form posts make and decide requests, the menu counts the requests that wait, Show all gives one page with every row, an old database gets the page that names the two commands, the sign-off sits below the changes (test_mapping)	P2
TC-069	Integration	Connections between status codes	A negated active flag in a set condition is a cause and a plain one a precondition, the real programs name their cause and repeat code, one code shows what it depends on and what depends on it with the chain of causes, the hierarchy puts the cause on top with the codes it suppresses below, escalation targets say which are outside the catalogue, what looks wrong is listed (unknown and deprecated causes, loops), Main and Hub are separated, the code page and the overview show the connections, a Viewer does not, the API returns both views and refuses a bad controller, the menu offers it to the working roles only, and an imported program brings its cause and repeat code (test_connections)	P1
TC-068	Integration	Language impact of a change of the vocabulary	A new term tells the StCd Owner that all 18 languages need attention, an assigned reviewer gets the notice and nobody else, a term kept in English touches no language, a changed term and an approved translation are reported again, a failing notice never undoes the change, the notice shows on the Notices page, nothing is sent while the confidence check is current, changed shipped text is announced once per change and a further change again, a changed nomenclature is named, the deployment command reports and notifies, and the server start and both installers run the check (test_translation_impact)	P1

ID	Level	Area	Scenario	Priority
TC-067	Integration	Approval of new and changed translations; confidence check	A reviewed language has nothing pending until a string changes, a Viewer reads a changed string in English while the approved ones stay in the language and the working roles see the new text, a language never reviewed stays hidden, a follow-up review lists and requires only the new and changed strings, an accepted or rejected proposal is approved and an undecided one is not, the colour intervals run from red (<= 50 %) to green (> 95 %), the overview shows the percentage of every language in its colour, the record holds a confidence and the findings of every language, a changed string or a changed nomenclature must get a verdict before the record is current, and a test fails while the text or the nomenclature changed without the check (test_translation_approval)	P1
TC-066	Integration	Baseline and package	A Release exports every code valid at its time (not only its own items), an earlier Release keeps its smaller baseline, changes only gives the delta, the baseline narrows to a platform and a controller, the summary counts by controller and platform, an unpublished Release previews its baseline, the package holds every language for every access level with one file per format, its manifest lists every file with a matching SHA-256 and the coverage per language, it narrows to platform and controller, everybody downloads what is released and only the StCd Owner and the Administrator what is not (working set, unpublished Release), invalid choices and unpublished Releases are refused, readiness shows the missing translations, the Release page offers it and the job is audited (test_baseline_and_packages)	P1
TC-065	Integration	The words around an export	A PDF in French prints its title page, digest note, column headings, footer and banner in French and leaves the data alone, English and no language keep the English text, every word exists in every language, a placeholder survives (test_export_language)	P2

8a. v4 Test Suites

Module (tests/...)	Cases	Requirements
integration/test_step_up_and_delegation.py	16	FR-125, FR-126, SEC-026, BR-017, BR-022
integration/test_content_integrity.py	16	FR-090 ... FR-094, FR-109, FR-131, FR-141
integration/test_review_aids.py	17	FR-124, FR-127 ... FR-130
integration/test_sla_and_notifications.py	16	FR-120 ... FR-123
integration/test_publication.py	25	FR-100 ... FR-111, BR-016, BR-018, BR-021
integration/test_import_toolkit.py	20	FR-048i ... FR-048n
integration/test_st_import.py	29	FR-048o ... FR-048w, BR-023, SEC-031
integration/test_roles_and_languages.py	19	FR-156, FR-157, BR-024, SEC-032
integration/test_user_administration.py	10	FR-158, BR-025, SEC-032, AUD-076
integration/test_withdraw_and_reimport.py	11	FR-159, FR-160, FR-048x, BR-026
integration/test_audit_chain_and_ops.py	27	FR-150 ... FR-153, AUD-072 ... AUD-075, SEC-029
integration/test_schema_upgrade.py	16	upgrade, AUD-074 back-fill, BR-016, D-06
integration/test_prg_idx_and_user_groups.py	9	schema v4.1: PrgIdx_GUi, Main/Hub fields, user-group levels (no FR yet)
integration/test_integrations_and_reports.py	32	FR-140 ... FR-147, SEC-027, SEC-028
integration/test_search_and_help.py	16	FR-148, FR-154
system/test_route_inventory.py	8	SEC-010, FR-081
system/test_i18n.py, system/test_accessibility.py	12 + 8	FR-155, FR-167, NFR-010
system/test_multiprocess.py	4	NFR-009
system/test_deploy_artifacts.py	22	NFR-013, NFR-014
system/test_docs_consistency.py	17	TC-052
system/test_translations_doc.py	8	FR-161
integration/test_email_and_windows_signin.py	16	FR-162, FR-163, SEC-033
integration/test_translation_review.py	14	FR-164, AUD-077
integration/test_pdf_scripts.py	9	FR-165, FR-157
integration/test_terminology.py	13	FR-166, BR-018, AUD-077, SEC-010
system/test_windows_gateway.py	6	FR-163, SEC-033
integration/test_generated_fields.py	12	FR-168, FR-048o, FR-025, FR-032

Module (tests/...)	Cases	Requirements
integration/test_menu.py	14	FR-169, FR-170, SEC-010, NFR-010
integration/test_export_language.py	4	FR-171, FR-165, FR-110, FR-045

The authoritative, always-current counts are in `24-test-report.md` (generated).

Shared fixtures (v4). `StepUpClient` answers the password prompt for the acting user so the many tests about other behaviour do not each repeat it (`raw_client` / `no_step_up=True` for the tests about the prompt itself); `file_app` is an application on a real SQLite file (background import, backups, upgrade, multi-process); every `app` gets its own instance directory (attachments, import files, the audit anchor); `CSCM_TEST_ORDER` reorders the run and `CSCM_TEST_DB=file` puts the whole suite on a file database. A frozen copy of the v3 schema (`fixtures/schema_d6f8b0c2e4a3.sql`) is the upgrade source.

9. Test Environments

Local, CI, staging, production, all on-premises per the deployment constraint. CI is defined for the internal Git server (`.gitea/workflows/ci.yml`, `.gitlab-ci.yml`, both calling `scripts/ci.py`).

10. Definition of "Test Complete" for a Release

All P1 cases pass in all three verification runs (§11), coverage threshold met, the five static gates clean, every defined requirement accounted for (§12), the documentation consistency test green, security checklist clean, at least one full UAT pass signed off.

11. Verification Runs: Three Methodologies (v4)

A change is verified three times, each in a way that can expose a different class of defect. `python scripts/generate_test_report.py [options]` runs one methodology, appends it to `artifacts/test-run-history.json` and regenerates `24-test-report.md`.

#	Methodology	Command	What it can catch that the others cannot
1	Sequential functional run + static gates. The complete suite in file order with line coverage, then black, ruff, mypy, bandit, pip-audit; the load tests run afterwards without coverage tracing.	scripts/generate_test_report .py	Functional regressions, type errors, insecure constructs, vulnerable runtime packages, coverage loss
2	Order-independence and property testing. The same suite in a reversed or shuffled order (seeded, reproducible), including the schemathesis / hypothesis tests that generate requests from the OpenAPI contract.	scripts/generate_test_report .py --order reverse and --order shuffle:	Hidden coupling between tests, leaked global state (rate limiter, caches, instance files), contract violations on inputs nobody wrote by hand
3	Real database file and real processes. Every test on a SQLite file in WAL mode instead of memory; inside the suite, two Waitress processes against one file, a background import thread, the upgrade of a frozen v3 database, the installer dry runs and the production entry point.	scripts/generate_test_report .py --db file	Locking and visibility problems that only appear with several connections or processes, migration and upgrade defects, packaging defects, file-handling and path errors

Additionally, `scripts/ci.py` (CI) runs methodology 1 plus the PostgreSQL ratchet, and `--fast` stops at the first failure for a developer's inner loop.

12. Requirement Accounting and Documentation Checks (v4)

`24-test-report.md` §6a states, for every requirement id defined in the specification (FR, NFR, BR in `03-srs.md`, SEC in `11-security-architecture.md`, AUD in `12-audit-compliance.md`), whether it is cited by a test, covered per the traceability matrix, or marked there as out of scope or needing a person. **No id may be left unaccounted for.** `tests/system/test_docs_consistency.py` additionally fails when: the endpoint table of `09-api-specification.md` differs from the code; table, trigger or operation counts quoted in the documents differ from the database or the contract; the application can write an audit action code that `12-audit-compliance.md` does not list; a term the help content defines is missing from the glossary; an id cited in a document is not defined anywhere; a link between documents points to a file or heading that does not exist; or a document set version differs between the index and the documents.

14. Deployment Architecture — CSCM Tool

1. Deployment Principle: Fully On-Premises, No Cloud

The system runs **entirely on local, on-premises servers**. No public cloud (SaaS/PaaS/IaaS) component is used anywhere — not for compute, not for the database, not for backups, not for authentication, not for logging/monitoring shipment. This is a hard constraint (C-004a, new in v2), not a default that may quietly drift: every environment below is a physical or on-prem-virtualized server inside the corporate network, and every future integration point in this package (08-system-architecture.md §6–§7) is explicitly scoped to stay on-premises. See 11-security-architecture.md §12 for the standing verification checklist.

2. Environment Overview

Environment	Purpose	Data	Access	Location
Development	Individual developer workstations	Local seeded fixtures, disposable SQLite file	Developers only	On-prem workstation
Test / CI	Automated test execution	Ephemeral, recreated per run	CI runner only	On-prem CI server
Staging	Pre-production validation, UAT, performance/security testing	Synthetic production-scale dataset	Project team, UAT participants	On-prem server
Production	Live system of record (the Master Database, 08-system-architecture.md §6)	Real governed status code data	All licensed end users, per RBAC	On-prem server

3. Infrastructure

- **Compute:** one application server (physical or on-prem VM) per environment; minimum 2 vCPU / 4 GB RAM for MVP load (NFR-004).
- **OS:** Windows Server (IIS + Waitress WSGI) or Linux (nginx + gunicorn), whichever matches the organization's standard on-prem server platform.
- **Storage:** local attached disk for the SQLite database file, export/backup working directory, and full-database export output (11-security-architecture.md §7), on a volume covered by the organization's standard on-prem encryption-at-rest and antivirus/EDR policy.
- **Network:** internal corporate network only; no public internet ingress or egress is required or permitted for any core function. Reverse proxy terminates TLS on the internal network and forwards to the Flask WSGI process on a private interface.
- **DNS/TLS certificates:** issued by the organization's internal CA, renewed per its standard rotation process — never a public cloud-issued certificate service if that would imply an external dependency; an internet-facing public CA (e.g. for TLS only) is acceptable only if the organization's policy already treats that as compatible with an otherwise fully on-prem system — flagged here for the platform team to confirm, since it is the one edge case where "no cloud" and "valid TLS certificate" can appear to be in tension.
- **Identity:** on-premises Active Directory / ADFS only for any future SSO integration (08-system-architecture.md §7, 11-security-architecture.md §2) — never a cloud identity provider.

4. Application Deployment Process

CI builds and tests on every push (`scripts/ci.py`, defined for the internal Git server in `.gitea/workflows/ci.yml` and `.gitlab-ci.yml`; NFR-013); tagged release produces a versioned artifact published to an **on-premises** artifact store (not a cloud package registry, unless the organization's existing on-prem mirror/proxy of a public registry is used for dependency resolution only — the application's own artifacts are never published externally); deployment to Staging automatic on tag, Production manual-approval gated and done with the installer below; standard stop -> backup -> deploy -> migrate -> start -> smoke-test -> confirm-or-rollback sequence; brief planned outage acceptable per NFR-002.

5. Master Database Export Process

CSCM Tool's production database is the Master Database (08-system-architecture.md §6) — there is no external master system it synchronizes with. Two, and only two, ways data leaves the system, both Administrator-gated:

1. **Catalogue export** (CSV/JSON/XLSX/PDF) — role-scoped per 11-security-architecture.md §3, generated on demand, never automatic/scheduled in MVP, used by Technical Publications and other downstream consumers.
2. **Full-database export/backup** — Administrator-only (SEC-015), a raw SQLite backup file, used for disaster recovery, environment migration, or handing a complete dataset to an authorized downstream consumer under separate organizational process. This is distinct from the nightly backup procedure (§6) in that it is a manually-triggered, audited (`FULL_DATABASE_EXPORT`, 12-audit-compliance.md §2), on-demand action rather than a scheduled operational job — though it uses the same underlying SQLite Online Backup mechanism.

No system pushes data into CSCM Tool; all status codes are authored inside the application by design (01-executive-summary.md §1). (v4) Catalogue exports can be shaped by profile, audience, language and software version and carry a manifest (FR-100 … FR-111); machine consumers may pull with scoped API tokens, poll the change feed or receive signed webhooks — all inside the network (08-system-architecture.md §6).

4a. Installation and Upgrade (v4, NFR-013)

`cscm_tool/deploy/` holds one installer per platform; both are idempotent, take **nothing** from the internet except the Python packages of `requirements.txt` (point `PIP_INDEX_URL` at the internal mirror), never overwrite configuration, data or backups, and offer a dry run that prints every step and changes nothing.

	Linux (systemd)	Windows Server
Run	sudo deploy/linux/install.sh [--dry-run]	powershell -ExecutionPolicy Bypass -File deploy\windows\install.ps1 [-DryRun] [-InstallDir C:\CSCM] [-RunAs]
Code, environment	/opt/cscm/app, /opt/cscm/venv	C:\CSCM\app, C:\CSCM\venv
Configuration (created once, generated keys)	/etc/cscm/cscm.env, mode 640	C:\CSCM\config\cscm.env, ACL: Administrators and the service account
Data (database, attachments, imports, audit anchor)	/var/lib/cscm (CSCM_INSTANCE_PATH)	C:\CSCM\data
Backups, log	/var/backups/cscm, /var/log/cscm/cscm.log	C:\CSCM\backups, C:\CSCM\logs\cscm.log
Web server	cscm.service -> scripts/serve.py (Waitress, unprivileged user, systemd hardening)	scheduled task CSCM Tool – web server at start-up
Jobs	cscm-backup.timer 02:15 daily, cscm-reminders.timer weekdays 07:30, cscm-audit-verify.timer Sundays 03:30	tasks nightly backup, reminders, audit chain check with the same times
Reverse proxy	deploy/linux/nginx-cscm.conf (TLS from the internal CA, X-Forwarded-*, /metrics limited to the monitoring network)	IIS/ARR or nginx, same headers

First start on a developer or trial machine (v4.14.1). `python cscm_tool/scripts/first_start.py` does what a person would type by hand, in the same way in PowerShell, Command Prompt, Git Bash and Linux and from any folder: it creates `.venv` (never recreating an existing one), installs `requirements.txt`, creates `.env` from `.env.example` once, runs `flask db upgrade`, creates the first Administrator when the database has none, and prints the start command. It is safe to run again, `--dry-run` changes nothing. Step-by-step instructions and the usual error messages are in `DEPLOYMENT-QUICKSTART.md` at the repository root. The production installers above remain the way to install for a team.

Release documents (v4.15, FR-185). `python scripts/build_docs_pdf.py` writes the full documentation, the software description, the user manual and the test report as PDF into `releases/v/`, which is committed with the release, each with the version in its file name; `python scripts/ci.py --pdf` does so after a green pipeline. They are made from the Markdown sources with the application's own fonts, so a build machine needs nothing besides the packages of `requirements.txt`.

Design showcase (v4.16, FR-186). `python scripts/build_showcase.py` builds the two decks (management, team) from screenshots of a throw-away copy of the application taken with the headless Edge or Chrome of the build machine (`CSCM_BROWSER` names a specific executable); `python scripts/ci.py --showcase` does so after a green pipeline. The decks need `python-pptx` (`requirements.txt`) and are written to `releases/v/`.

Hand-over to the team repository (v4.16, FR-188). `scripts/handover.py` audits the tree of a release tag and exports it as a clean repository (one commit, the annotated tag); the GitHub Actions workflow `.github/workflows/ci.yml` is the same file the internal Gitea server runs.

PDF fonts (v4.7, FR-165). The Noto fonts for the PDF export are part of the code (`app/static/fonts/`, SIL Open Font

License, licence file included), so nothing is installed on the server; the only addition to `requirements.txt` is the binary wheel `uharfbuzz`, which an on-premises package mirror must carry like the others.

What the installer does, in order: create the service account and directories; copy the code (and `docs/artifacts/schema.sql`, which the baseline migration executes); build the virtual environment from the runtime part of `requirements.txt`; create the configuration **once** with freshly generated `SECRET_KEY`, `AUDIT_HMAC_KEY` and `MANIFEST_SIGNING_KEY` (**keep a copy in the password safe — they cannot be recovered**); take a verified backup of an existing database, then `flask db upgrade`; ask for the first Administrator if the database has no users (`flask create-admin`, which also reads `CSCM_ADMIN_USERNAME`, `?_EMAIL`, `?_FULL_NAME`, `?_PASSWORD` for unattended installs); install and start the units or tasks. `scripts/serve.py` refuses to start in production with a missing, default or short `SECRET_KEY` (SEC-002) and honours `CSCM_HOST`, `CSCM_PORT`, `CSCM_THREADS` and `CSCM_TRUSTED_PROXY` (so the login rate limit sees the real client address behind the proxy). `deploy/windows/uninstall.ps1` removes the tasks and never deletes data. The settings are listed in `cscm_tool/.env.example`; a test checks that every setting the installers write is read by the application.

Windows sign-in (v4.7, FR-163, SEC-033). In a deployment people sign in with their Windows account, not a password. The proxy authenticates the user against Active Directory (IIS or ARR with Windows authentication; Apache with Kerberos; nginx with spnego), **removes** any `X-Remote-User` header the client sent and sets its own from the authenticated session; the application is reachable only through the proxy. In `cscm.env`: `AUTH_MODE=windows` (`both` keeps the password form for the transition; `password` is the default), `WINDOWS_AUTH_TRUSTED_PROXIES=` (required: without it the header is ignored), optionally `WINDOWS_AUTH_HEADER` and `WINDOWS_AUTH_EMAIL_DOMAIN` (for `CORP

The Windows sign-in tested with a real Windows identity (v4.9, TC-062). `scripts/windows_gateway.py` is a small stand-in for IIS: it answers a request without credentials with a `401` that offers `Negotiate` and `NTLM`, completes the handshake with the Windows SSPI of the operating system (`secur32` through `ctypes`, no package to install), takes the user name from the security context, drops any identity header the client sent and passes the user on in `X-Remote-User`. `tests/system/test_windows_gateway.py` runs the real handshake — a client with the credentials of the Windows user who runs the tests against the SSPI server side of the gateway — and then the whole chain client, gateway, application; these tests run on Windows only. By hand: `python scripts/windows_gateway.py --listen 127.0.0.1:8080 --upstream http://127.0.0.1:5000` with `AUTH_MODE=windows` and `WINDOWS_AUTH_TRUSTED_PROXIES` set to the address the gateway connects from (`--source`, default `127.0.0.2`), then `curl.exe --ntlm -u : http://127.0.0.1:8080/login/windows`. What it proves: the application signs in the account that the operating system names, refuses an unknown or deactivated account and ignores the header when it does not come from the gateway. What it does not prove: Kerberos, a domain controller, IIS or ARR. On a computer that is not in a domain the identity is the local account `COMPUTER\user`, which the application matches through the user name (FR-163); a browser sends integrated credentials only to hosts of the intranet zone, so a development browser does not sign in through `127.0.0.1`. ame `identities). The account is found by e-mail address, so every user needs an account with the address of their user principal name, created by an Administrator (nobody is created on the fly).` WINDOWS_AUTH_SIMULATE` exists for developer machines only; the application refuses to start in production with it. The pre-go-live checklist gains: a real sign-in through the proxy, a request that carries the header but comes from another address (must be ignored), and a deactivated account (must be refused).

What cannot be proven on a development machine and is therefore on the pre-go-live checklist (20-traceability-matrix.md §6): a real install on a clean Windows Server and a clean Linux host, systemd's own verification of the units, TLS termination and the proxy headers, the Windows sign-in through the real gateway (the simulation and the test gateway of the paragraph above prove the application's side and the operating system's handshake, not Kerberos, a domain controller or IIS), and the egress block on the production server.

6. Backup Procedure

Nightly SQLite Online Backup API snapshot (never a raw file copy) by `flask backup-db --keep 35` (the shipped timer; the command's own default is 14 generations), copied within 1 hour to an on-premises storage location separate from the application server (`BACKUP_DIR` should be that location or be replicated to it), 12 monthly copies kept by the organisation's own tooling, 7-day WAL archiving for point-in-time recovery. No backup target is ever a cloud storage service.

Every backup is proven (v4, FR-150, NFR-012). After copying, the job runs `PRAGMA integrity_check` on the copy, restores it into a scratch database and compares the row counts of the key tables with the live database, checks the audit hash chain of the copy, and writes `backup_status.json` (file, size, SHA-256, `ok`, the time of the last verified success). A backup that fails any check is kept for diagnosis but never counts as the last good one, the command exits non-zero (the scheduler shows the unit as failed) and the readiness probe reports `degraded` (

`last_attempt_failed`, or `stale` when the newest verified backup is older than `BACKUP_MAX_AGE_HOURS`, default 36). `flask verify-backup` repeats the proof for any file.

Audit archive (AUD-073). `flask audit-archive --dest [--older-than-years 7]` writes entries past the retention period to a compressed file with a `.sha256`, records an `AUDIT_ARCHIVE` entry and deletes nothing; run it yearly.

7. Recovery Procedure

RPO <= 24 h (<= 1 h with WAL), RTO <= 4 h.

Restore runbook.

1. Stop the web service (`systemctl stop cscm` / stop the scheduled task).
2. Choose the newest generation named `last_good_file` in `backup_status.json`; run `flask verify-backup` — it must print `"ok": true` (integrity, restore test, audit chain).
3. Move the damaged `cscm.db` (and its `-wal` / `-shm` files) aside; copy the backup to the database path.
4. Run `flask db upgrade` (a no-op when the backup is at the current schema, an upgrade when it is older).
5. Start the service; `GET /api/v1/health/ready` must answer `ok` or `degraded` (degraded only because no backup is younger than 36 h yet).
6. Run `flask audit-verify` (the first run after a restore may report that the log is shorter than at the last verification — delete `audit_chain_anchor.json` in the data directory to accept the restored state, and note this in the incident record) and spot-check the newest status codes in the UI.
7. Run `flask backup-db` once so the readiness probe is green again.

Quarterly drill. Perform steps 2–6 on a scratch host with the latest backup, record the time taken and the result; the test suite proves the mechanics (`test_a_restored_backup_runs_the_application`, `test_a_tampered_audit_log_inside_a_backup_is_flagged`) but not your hardware.

8. Monitoring

`GET /api/v1/health` (liveness) and `GET /api/v1/health/ready` (readiness: database, schema at the Alembic head, newest audit rows, last verified backup; 200 `ok`/`degraded`, 503 `unavailable`) for the load balancer and monitoring; `GET /metrics` (Prometheus text) from the monitoring network only — `METRICS_ALLOWED_CIDRS` (default loopback and RFC 1918) or `METRICS_TOKEN`; each worker process is scraped separately. Suggested alerts: readiness not 200; `cscm_backup_ok == 0` or `cscm_backup_age_hours > 36`; `cscm_cr_waiting{level="red"}` rising; `cscm_db_lock_wait_seconds` p95 above 1 s; 5xx rate above 0.1 % (NFR-001); disk space; failed-login spikes. All shipped to an **on-premises** log/monitoring store, never a cloud observability SaaS product, unless the organization already runs such tooling entirely within its own infrastructure (self-hosted, not a hosted cloud offering).

9. Logging

Structured JSON request logs (one line per request with the request id, never credentials or tokens) on stderr and, with `LOG_FILE`, in a rotating file (10 MB × 5) for the log shipper, shipped to an on-prem central log store; reverse-proxy access logs retained per organizational standard; the `audit_log_entry` compliance record lives in the application database itself, never in log files.

10. Environment Parity

Staging kept at application-version and OS/runtime parity with Production, synthetic data volume scaled to meet or exceed NFR-001's sizing assumption, now distributed realistically across all nine Functional System Groups rather than a single Controller Type split.

15. Development Standards — CSCM Tool

1. Folder Structure

See 18-claude-code-implementation-guide.md §2 for the complete, authoritative folder layout used during implementation (package root `cscm_tool/`). Summary of top-level conventions:

```
cscm_tool/
+-- app/
|   +-- __init__.py           # application factory (create_app(config_name, overrides)), engine hooks
|   +-- api/v1/               # REST blueprints (one module per area; _errors.py maps service errors)
|   +-- ui/                   # server-rendered UI blueprints
|   +-- models/               # SQLAlchemy ORM models (operations.py holds the v4 tables)
|   +-- repositories/         # repository pattern (status_code_repository, fts)
|   +-- services/             # business logic, workflow, validation, publication, operations
|   +-- auth/                 # providers, RBAC decorators (declare `view.access`), rate limit
|   +-- export/               # CSV/JSON/XLSX/PDF/offline-bundle + full-database export
|   +-- audit/ audit_chain.py audit_hash.py   # audit writer; hash chain, verification, archive
|   +-- observability.py logging_config.py    # metrics registry, JSON request log
|   +-- i18n.py i18n_de.py help_content.py    # English/German catalogue and help text
|   +-- cli.py                # `flask` jobs: backup-db, verify-backup, audit-verify, send-reminders, ?
|   +-- templates/            # Jinja2 templates
|   +-- static/               # CSS/JS (no external/CDN sources ? 11-security-architecture.md §12)
+-- migrations/               # Alembic revision chain (07-database-design.md §11)
+-- scripts/                  # generate_test_report.py, ci.py, serve.py, pg_readiness.py, gen_api_summary.py, gen_
  translations_doc.py
+-- deploy/linux/ deploy/windows/   # installers, systemd units, scheduled-task scripts, nginx sample
+-- tests/
|   +-- unit/ integration/ system/ performance/ fixtures/
|   +-- conftest.py helpers.py
+-- pyproject.toml requirements.txt .env.example
docs/                         # this documentation package (repository root, next to cscm_tool/)
.gitea/workflows/ci.yml  .gitlab-ci.yml  .gitattributes   # at the repository root
```

1a. OneTool Design System Compliance (new 2026-09-17)

All UI code (templates, CSS, JS) must use the shared OneTool design tokens, typography (Manrope/Inter), and component treatments (10-ui-ux-specification.md §0) rather than introducing new colors, fonts, or button/card styles. Before any UI PR is merged: render inside the shared shell with the CSCM Tool nav item active, and verify desktop, collapsed-sidebar, tablet, keyboard-only, loading, error, and empty states, per the design system's own integration checklist. Authentication (`app/auth/`) is explicitly out of scope for this integration — it stays CSCM Tool's own.

2. Coding Standards

PEP 8 via `black`/`ruff` (line length 100; long SQL strings and banners in `tests/` are exempt from E501), type hints required on service/repository signatures (`mypy`), no business logic in route handlers, no raw SQL string concatenation, configuration values as named constants, vanilla JS with no framework and no CDN-hosted dependency (11-security-architecture.md §12 — every static asset is vendored locally, never fetched from an external host at runtime or build time).

3. Naming Conventions

Element	Convention	Example
Python modules/files	snake_case	change_request_service.py
Python classes	PascalCase	ChangeRequestService
Python functions/variables	snake_case	reviewer_signoff()
SQLAlchemy model classes	Singular PascalCase	StatusCodeRevision
Database tables/columns	snake_case, singular	status_code_revision
API URL paths	plural, kebab/lowercase	/api/v1/change-requests
API JSON fields	snake_case	status_code_identifier
Constants	UPPER_SNAKE_CASE	MAX_PAGE_SIZE
Git branches	type/short-description	feature/dual-review-workflow

Element	Convention	Example
Requirement IDs in code comments/commits	as defined in 00-INDEX.md	FR-033, BR-005

4. API Standards

Contract-first against `artifacts/openapi.yaml`: every new endpoint ships with schema entries, a system test, and a role declaration on the route (`require_role?` / `login_required` set `view.access`; `allow_token(scope)` for token-readable GETs) — the route inventory test then checks it for all five roles, and `python scripts/gen_api_summary.py --write` regenerates the endpoint table of 09-api-specification.md. Breaking changes require a version bump.

5. Database Standards

All schema changes via reversible, guarded and idempotent Alembic migrations (07-database-design.md §11; a frozen copy of the previous schema in `tests/fixtures/` proves the upgrade), no ad hoc schema edits, new tables/columns reflected in the same PR across 06-data-dictionary.md, 07-database-design.md, and `artifacts/schema.sql`, every foreign key indexed unless proven unnecessary.

6. Documentation Standards

- This `docs/` package is the living source of truth; a PR that changes behavior described by a specific FR/NFR/BR/SEC/AUD identifier must update that document in the same PR.
- **Glossary discipline (new in v2):** any PR that introduces, renames, or retires a term, role, lifecycle state, or domain concept must update 19-glossary.md in the same PR — the glossary is treated as a required artifact, not optional polish, per the explicit instruction that it "be kept updated with every change."
- **Paragraph formatting:** prose paragraphs in every document under `docs/` are written as a single continuous source line (no manual mid-paragraph line breaks); only tables, code blocks, and list items may span multiple source lines. This applies to new and edited content alike — an edit that reintroduces hard-wrapped paragraphs should be corrected before merge.
- Code comments explain **why**, not **what**; no commented-out code is committed.
- Every service-layer public method has a docstring stating its pre/postconditions and which FR/BR it implements, e.g. `"""Records the Administrator sign-off. Enforces BR-005 and may confirm a restricted field. See FR-034."""`.

7. Testing Standards

Tests required in the same PR as new logic/endpoints, deterministic (no wall-clock or ordering dependence), a fresh in-memory SQLite database and a fresh instance directory per test (a SQLite file with `CSCM_TEST_DB=file`). A test module's docstring cites the TC and requirement ids it covers (`TC-0nn. Requirements: FR-?`) so the report can account for every requirement (13-test-strategy.md §12). The suite is verified three ways before a release (13-test-strategy.md §11).

8. Dependency Management

Pinned requirements (`requirements.txt`: runtime first, tooling after the `# --- dev` marker, which the installers cut at), `pip-audit` in CI blocking on new High/Critical findings, zero external JS dependencies by default, any exception vendored/pinned and never loaded from an external CDN at runtime — consistent with the no-cloud, on-premises-only constraint (11-security-architecture.md §12), which is a stricter reason for this rule than offline-deployment convenience alone.

9. Git Workflow

Trunk-based, short-lived feature branches, CI gate (lint/format/type/tests/`bandit`/`pip-audit`) required green before merge, no direct pushes to `main`.

The team repository (v4.16, FR-188). The team repository starts with release 4.16.0 as a clean start: `python scripts/handover.py --tag v4.16.0 --author "Name " --out` audits the tree of the tag, exports it as one commit on `main` with the annotated tag, and pushes only when `--remote --push --yes` are all given. The development history stays on the development machine, which becomes an ordinary clone of the team repository.

Settings of the team repository: `main` protected (pull request required, the checks `verify` and `deploy-scripts` required, a review by the code owner required, no force push, linear history), tags `v*` protected, `.github/CODEOWNERS` names the Software Owner. How to work in it: `CONTRIBUTING.md`.

10. Pull Request Process

Standard checklist, with the glossary-update requirement (§6) added to the standard PR checklist alongside the existing documentation-update requirement.

11. Definition of Done

Approved PR merged, acceptance criteria demonstrated, tests passing at coverage target, relevant documentation **and glossary** updated to match actual behavior, no known High/Critical security findings, deployed and smoke-tested in Staging at least once before counting toward a release milestone.

Living artefacts (added 2026-10-21). Any change to the software also updates, in the same PR: the test report — regenerate it with `python cscm_tool/scripts/generate_test_report.py` (runs the full suite, coverage and the static gates, and appends to the run history; never edit `24-test-report.md` by hand); the traceability matrix `20-traceability-matrix.md` when a requirement's code or evidence moves (the report flags matrix citations that no longer resolve to a test); and, when behaviour visible to users or integrators changes, the software description `22-software-description.md`, the user manual `23-user-manual.md` and `artifacts/openapi.yaml`. Generated blocks (the endpoint table of 09-api-specification.md, the test report) are regenerated, never edited by hand, and `tests/system/test_docs_consistency.py` must be green. A PR whose report verdict is not PASS is not done.

12. v4 Conventions

- **Quality gates** (all must be clean before merge; `scripts/ci.py` runs them): `black --check app tests migrations/versions`, `ruff check app tests migrations/versions`, `mypy app wsgi.py`, `bandit -q -r app`, `pip-audit -r requirements.txt` (advisories in development-only tools are accepted and listed in the report; a runtime advisory fails the gate).
- **No** `assert` **for control flow in** `app/` (bandit B101) — raise a domain error; no silent `except: pass`.
- **SQL built from names** is allowed only from constants and carries `# nosec B608` with the reason; user input only ever reaches the database through bound parameters (SEC-021).
- **Secrets** (keys, tokens, passwords, webhook secrets) come from the environment or the protected configuration file, are never logged, never stored in clear (tokens are hashed), and are shown once when generated (SEC-030).
- **New SQLite-specific code** (a trigger, `BEGIN IMMEDIATE`, `GLOB`, a `sqlite3` call, FTS) must be justified: `python scripts/pg_readiness.py --check` fails when the count grows, and the baseline is lowered with `--update-baseline` when it shrinks (NFR-014).
- **User-visible strings in templates** are wrapped in `_()` and get a German entry in `app/i18n_de.py`; new help text lives in `app/help_content.py` and new terms in 19-glossary.md (`test_i18n`, `test_search_and_help`).
- **Accessibility**: every new control has a label or accessible name, tables use `<th scope="col">`, colours come from the tokens (`test_accessibility`).
- **Releases**: raise `VERSION` in `app/version.py` and `version` in `pyproject.toml` (`test_version` keeps them equal), then run `python scripts/ci.py --pdf --showcase` (the pipeline, then the four PDFs and the two showcase decks of the release, each deck as `.pptx` and `.pdf`; the decks need Edge or Chrome on the machine) and tag the commit `v`. The version is shown in the menu, on the login page and in `/api/v1/health`; the PDFs are `releases/v/CSCM-Tool_v_{Full-Documentation,Software-Description,User-Manual,Test-Report}.pdf` (small, so they are committed with the release).
- **Line endings**: Linux-executed files (`*.sh`, `deploy/linux/*`) keep LF whatever `core.autocrlf` says (`.gitattributes`).
- **Documents**: numbers quoted in prose (operations, tables, triggers, action codes) are checked against the code by `test_docs_consistency`; regenerate generated blocks (endpoint table, test report) instead of editing them.

16. MVP Roadmap — CSCM Tool

1. Phasing Overview

Phase	Theme	Outcome
Phase 1 (MVP)	Core governance: create, dual-review, Chief Engineer approval, release, audit, sandbox testing	Production-ready system of record replacing document-first status code authoring, fully on-premises
Phase 2	Enterprise integration (on-premises only)	On-prem Active Directory/ADFS SSO, internal-network API tokens for external consumers, first live integration (documentation generation tool or SCADA), real Functional Subgroup taxonomy in place
Phase 3	Scale & automation	PostgreSQL migration (if warranted), webhook-based push integrations (internal network only), configurable/multi-stage workflows if governance needs grow further
v4 improvement pack (delivered 2026-10-23, between Phase 1 and Phase 2)	Governance extensions, publication, operations — the 36 ideas of 21-improvement-ideas.md	Several Phase 2/3 items arrive early: internal API tokens, change feed and webhooks, PostgreSQL readiness (not the migration), multi-language content (see §2a)

2. Phase 1 — MVP

Scope: all FR/NFR/BR items in 03-srs.md, the full 7-state lifecycle with dual-review-then-Chief-Engineer-approval (05-governance-handbook.md), Functional System Group/Subgroup numbering with `StCd-XXXXX` identifiers, Turbine Platform multi-select, Cross-Domain Sign-Off, sandbox status codes, REST API (contract-complete, internal-consumer-only), UI, security controls, audit — all running entirely on-premises with no cloud component (14-deployment-architecture.md §1).

Explicit exclusions: on-prem AD/ADFS SSO itself (the hook is built, the integration is not — FR-005), live external integrations, configurable approval chains beyond the fixed Reviewer+Administrator+Chief-Engineer model, mobile-optimized layout. (Bulk legacy import was excluded in v2 and is in scope since 2026-09-17; localization is partly in scope since v4 — see §2a.)

Resolved since v2.0: the real Functional Subgroup taxonomy has been supplied and is seeded directly (06-data-dictionary.md §9a) — no longer a placeholder. The `15000?15999` range is likewise resolved: confirmed as deliberate free capacity (alongside `10000?10999`, `14000?14999`, `16000?16999`), not an unclassified gap — no further action needed on it.

Exit criteria:

- All P1 test cases in 13-test-strategy.md §8 pass.
- At least one full UAT pass signed off (13-test-strategy.md §7, §10), including the dual-review, Chief Engineer approval, Cross-Domain Sign-Off, and sandbox-delete scenarios specifically.
- Security checklist executed with no unresolved High/Critical findings, including the no-cloud verification checklist (11-security-architecture.md §12).
- ~~The `15000?15999` range classified into (or added as) a Functional System Group~~ — resolved: confirmed free/reserved capacity, no classification needed.
- Technical Publications team has successfully produced one manual section sourced from a CSCM Tool export end-to-end.

2a. v4 Improvement Pack (delivered)

All 36 ideas of 21-improvement-ideas.md are implemented, specified (03-srs.md §1.10–§1.15) and tested (13-test-strategy.md §8a). Mapping to the epics proposed there: **EPIC-17** Concurrency & Data Integrity (I-01, I-04, I-12, I-25); **EPIC-18** Release Integrity & Software-Version Traceability (I-06, I-07, I-24); **EPIC-19** Workflow SLAs, Notifications & Metrics (I-08, I-10, I-16); **EPIC-20** Audience-Scoped & Profile-Based Publication (I-02, I-03); **EPIC-22** Multilingual Content (I-15); **EPIC-23** Knowledge Enrichment & Evidence (I-13, I-14); **EPIC-24** Fleet Feedback Loop (I-28); **EPIC-25** Operations & Resilience (I-09, I-21, I-33, I-20) — EPIC-21 stays unused; plus extensions of EPIC-02, 04, 05, 06/08, 10, 11, 12, 13, 14, 15 and 16 (17-implementation-backlog.md).

What this changes in the phasing below: "scoped API token issuance" (Phase 2) and "webhook-based push on

Release publish" (Phase 3) already exist for the internal network (FR-144 … FR-146); bulk legacy import is no longer excluded (FR-048a, since 2026-09-17) and has a migration toolkit; localization is partially in scope (English/German interface and content, FR-110, FR-155); a verified backup, installers and CI exist (FR-150, NFR-013). **Still open** in Phase 2/3: the AD/ADFS provider, the PostgreSQL migration itself (only the inventory and ratchet exist, NFR-014), mobile layout beyond the responsive shell, and the business decisions listed in 21-improvement-ideas.md §B.6 (several are answered by the safest default of 03-srs.md §6).

3. Phase 2 — Enterprise Integration (On-Premises Only)

- On-premises Active Directory (LDAP/Kerberos) or ADFS/SAML authentication provider (behind the existing `AuthProvider` interface, 08-system-architecture.md §7; FR-005), including group-to-role mapping policy — explicitly **not** a cloud identity provider.
- ~~Scoped API token issuance for machine consumers on the internal network only~~ — delivered in v4 (FR-144, SEC-027); what remains is connecting the first real consumer.
- Multi-role support for users if operational need is confirmed.
- Revisit mobile/responsive scope if Viewer-role field usage on tablets/phones proves common.
- Confirm the final Functional Subgroup taxonomy has been in stable production use and re-validate numbering-capacity headroom per group.

Trigger for starting Phase 2: Phase 1 in stable production use for at least one full quarterly Release cycle with success metrics trending toward target.

4. Phase 3 — Scale & Automation

Evaluate PostgreSQL migration against observed volume/contention (the SQLite-specific code is inventoried and capped by `scripts/pg_readiness.py`, 07-database-design.md §12; **no PostgreSQL run exists yet**); webhook-based push notification on Release publish is delivered (FR-146, internal network only — no cloud message broker); reassess whether the fixed Reviewer+Administrator+Chief-Engineer model remains sufficient only if governance complexity has genuinely grown.

5. Priority Matrix (Phase 1 Internal Sequencing)

Capability	Business Value	Technical Risk/Complexity	Priority
Data model + 7-state lifecycle engine	Critical — everything depends on this	Medium-High (more states than v1)	P0
Auth + 5-role RBAC	Critical	Low-Medium	P0
Functional Group/Subgroup numbering + atomic identifier allocation	Critical — replaces the entire original numbering scheme	Medium-High (concurrency-sensitive)	P0
Status Code create/edit (Draft) + validation	Critical	Medium	P0
Cross-Domain Sign-Off gating	High — explicit business requirement, blocks Submit without it	Medium	P0
Dual Review sign-off (Reviewer + Administrator)	Critical — the core governance value proposition	Medium	P0
Chief Engineer approval	Critical — explicit new governance requirement	Low-Medium	P0
Audit trail (incl. sandbox-delete snapshot)	Critical, compliance requirement	Low-Medium	P0
Search/browse/filter	High	Low	P1
Release management + publish (sandbox-exclusion enforced)	Critical	Medium-High	P0
Catalogue export (CSV/JSON/XLSX/PDF)	High	Medium	P1
Full-database export (Administrator only)	Medium — explicit new requirement, narrow scope	Low	P1
Sandbox status codes (create/delete)	Medium — explicit new requirement	Low-Medium	P1
Deprecation/Archival	Medium	Low	P1
User/lookup administration (incl. Engineering Domain tags, Functional Group/Subgroup/Platform vocab)	High	Medium	P1
REST API (beyond internal UI needs)	Medium for MVP, High for future value	Medium	P1
Revision diff view	Medium	Low-Medium	P2

Capability	Business Value	Technical Risk/Complexity	Priority
(v4) Backups with verified restore, step-up, concurrency control, audit hash chain	High — close real risks before go-live	Low-Medium	P0 for go-live
(v4) Import toolkit, profiles / audiences / manifests, software versions	High — needed while the legacy library is migrated and for manuals	Medium	P1
(v4) SLAs, delegation, change sets, notices, KPIs	High — protects the throughput the business case promises	Medium	P1
(v4) Translations, offline bundle, fleet feedback, Hub consolidation, saved searches	Medium	Medium	P2

6. Risk Analysis

Risk	Likelihood	Impact	Mitigation
(retired 2026-09-17) The 15000–15999 range remains unclassified	—	—	Resolved: confirmed free/reserved capacity alongside 10000–10999, 14000–14999, 16000–16999; an Administrator can assign a new Functional System Group into any of them later via FR-048g without a schema change
Legacy status code data creates pressure to bulk-import, undermining governance-first principle	High	High	Import exists but only creates Drafts that pass the whole cycle (BR-015); the dry run, update mode and reconciliation report (FR-048i … FR-048n) make the migration iterative without a governance bypass
Identifier-allocation contention under concurrent submission within one popular Functional Group	Low-Medium	Low (SQLite BEGIN IMMEDIATE serializes correctly; worst case is added latency, not incorrect numbers)	Performance-tested explicitly (TC-025); repository pattern keeps PostgreSQL migration available if contention proves material
Dual-review-plus-approval sequence slows throughput versus the original single-Reviewer model	Medium	Medium — undermines the "faster manual production" value proposition if turnaround degrades materially	Track sign-off and approval turnaround KPIs separately from week one; Administrator can still reassign Reviewer load
Cross-Domain Sign-Off requests stall in practice (second Engineer unavailable)	Medium	Medium	Dashboard/queue visibility (S8/S10a in 10-ui-ux-specification.md) makes outstanding requests hard to miss; revisit if turnaround data shows a bottleneck
Sandbox codes accidentally treated as production data by a downstream consumer	Low (system enforces exclusion structurally)	High if it occurred	is_sandbox excluded from default search, catalogue export, and Release candidates at the query/trigger level, not just UI convention (07-database-design.md §5)
No-cloud constraint conflicts with a convenient off-the-shelf cloud service later in the project	Medium	Medium	Standing verification checklist (11-security-architecture.md §12) reviewed at every architecture decision point, not just at go-live
Scope creep toward configurable workflows or SSO before Phase 1 ships	Medium	Medium	Explicit out-of-scope statement reinforced in sprint planning
(v4) An integrity key (AUDIT_HMAC_KEY, MANIFEST_SIGNING_KEY) is lost, making the audit chain or old manifests unverifiable	Low	High	Keys generated by the installer, copied to the password safe at install time (14-deployment-architecture.md §4a); restore drill checks the chain
(v4) Step-up and SLA reminders irritate users or are routed around	Medium	Low-Medium	Reminders are one notice per user and request per day; delegation exists for absences; thresholds are Administrator settings
(v4) PostgreSQL migration is harder than assumed	Medium	Medium	Ratchet keeps SQLite-specific code from growing; the port is estimated from the inventory before it is started (NFR-014)

Risk	Likelihood	Impact	Mitigation
(v4) A single SQLite file is a single point of failure	Low-Medium	High	Verified nightly backups, readiness shows backup age, quarterly restore drill (NFR-012)

7. Effort Estimation (T-Shirt Sizing)

Epic (see 17-implementation-backlog.md)	Size
EPIC-01 Foundation (scaffold, auth, 5-role RBAC, DB schema/migrations)	L
EPIC-02 Status Code Data Model, Numbering & Validation	L (up from M — Functional Group/Subgroup allocation adds complexity)
EPIC-03 Cross-Domain Sign-Off	M (new epic)
EPIC-04 Dual Review & Chief Engineer Approval Workflow	L (up from M — three decision points instead of one)
EPIC-05 Search, Browse, Filter, Revision History/Diff	M
EPIC-06 Release Management (build/publish/export, sandbox-exclusion)	L
EPIC-07 Sandbox Status Codes	S (new epic)
EPIC-08 Deprecation & Archival	S
EPIC-09 User & Lookup Administration (incl. new vocabularies)	M
EPIC-10 Audit Trail & Reporting	M
EPIC-11 REST API Hardening & OpenAPI Conformance	M
EPIC-12 Security Hardening, Compliance & No-Cloud Verification	M
EPIC-13 Non-functional: Performance, Deployment, Monitoring	M
EPIC-14 Library Import (new 2026-09-17)	M
EPIC-15 OneTool Shell Integration & UX Fixes (new 2026-09-17: no-data-loss-on-failure, inline field help, design system adoption)	M
EPIC-16 Subgroup & Group Administration (new 2026-09-17)	S
EPIC-17 Concurrency & Data Integrity (v4, delivered)	M
EPIC-18 Release Integrity & Software-Version Traceability (v4, delivered)	L
EPIC-19 Workflow SLAs, Notifications & Metrics (v4, delivered)	M
EPIC-20 Audience-Scoped & Profile-Based Publication (v4, delivered)	M
EPIC-22 Multilingual Content (v4, delivered)	L
EPIC-23 Knowledge Enrichment & Evidence (v4, delivered)	M
EPIC-24 Fleet Feedback Loop (v4, delivered as a file import)	M
EPIC-25 Operations & Resilience (v4, delivered)	M

Sizing scale: S ~ 1 sprint, M ~ 2 sprints, L ~ 3–4 sprints, small (2–4 engineer) team; see 18-claude-code-implementation-guide.md §7 for sequencing.

8. Corrections Log (2026-09-17)

Grounded against a real 897-row export of the live status code library and the legacy tool's screenshot, this update corrects several assumptions made without real data: Functional Subgroup is mandatory, not optional; subgroup names and the WTUR nine-way split replace an earlier, incorrect hand-typed summary; `status_category`/`available_group`/`alarm`/delay fields/reset-and-brake programs all use different real types than originally assumed; the single Access Rights enum is replaced by eight independent audience fields; CSV/XLSX library import is now in scope (reversing the original "no bulk import" position) but still requires full governance per row; export formats gain XLSX; and the UI adopts the OneTool shared design system for its visual shell while keeping its own authentication. See 06-data-dictionary.md's provenance note and 19-glossary.md for the full corrected vocabulary.

17. Implementation Backlog — CSCM Tool

Conventions: `EPIC-##` -> `FEAT-##.#` -> `US-###`. Priority: P0 (must-have for MVP go-live), P1 (should-have), P2 (nice-to-have, may slip to Phase 2). Dependencies reference other story IDs or epics. Sizing: 16-mvp-roadmap.md §7.

EPIC-01 — Foundation

FEAT-01.1 Project Scaffold & CI

- **US-001** Flask application factory, blueprint skeleton, folder structure (15-development-standards.md §1). AC: app boots, health endpoint returns 200. Dependencies: none. Priority: P0.
- **US-002** CI configured (lint, format, type-check, test, security scan, no-cloud grep check). AC: CI runs on every push/PR; a deliberately failing test blocks merge; a deliberate cloud-SDK import blocks merge (TC-029). Dependencies: US-001. Priority: P0.

FEAT-01.2 Database Schema & Migrations

- **US-003** Initial Alembic migration implementing `artifacts/schema.sql` (v2 at the time; v4.7 now — 61 tables, 14 triggers; all lookups, 7-state lifecycle, dual sign-off + Chief Engineer columns, sandbox support). AC: fresh DB matches schema.sql exactly; migration reversible. Dependencies: US-001. Priority: P0.
- **US-004** Seed all lookup tables, including the 12 Functional System Groups (9 Main + 3 Hub Controller), the 38 real Functional Subgroups (06-data-dictionary.md §9a), Turbine Platforms (2XM/3XM/4XM), and Engineering Domains. AC: seed migration inserts all documented rows. Dependencies: US-003. Priority: P0.

FEAT-01.3 Authentication & Session Management

- **US-005–US-008** Login/session/lockout/logout/timeouts (FR-001–FR-004, SEC-001–SEC-004). Dependencies: US-003. Priority: P0/P1.

FEAT-01.4 Role-Based Access Control (6 roles)

- **US-009** Administrator can create/edit/deactivate users, assign exactly one of the 6 roles (v4.3: an StCd Owner may move users between Viewer, Engineer and Reviewer), and tag Engineer-role users with one or more Engineering Domains (FR-010–FR-012 + new domain-tagging). AC: last active Administrator cannot be deactivated. Dependencies: US-005. Priority: P0.
- **US-010** Full 6-role × endpoint permission matrix enforced identically on UI and API (11-security-architecture.md §3). AC: TC-018 parameterized grid passes. Dependencies: US-009, incrementally with every endpoint. Priority: P0.
- **US-011/US-012** Self password change / Administrator-forced reset. Priority: P1/P2.

EPIC-02 — Status Code Data Model, Numbering & Validation

FEAT-02.1 Functional Group / Subgroup Classification

- **US-020** Engineer selects Functional System Group and Functional Subgroup (both required, subgroup as a dependent dropdown) when starting a new Status Code; no identifier is assigned at this point (05-governance-handbook.md §2, §6). AC: Draft shows "Not yet assigned"; Submit is blocked without a subgroup. Dependencies: EPIC-01. Priority: P0.
- **US-021** Turbine Platform multi-select (minimum one required) on the same form. AC: Submit blocked with zero platforms selected. Dependencies: US-020. Priority: P0.

FEAT-02.2 Identifier Allocation

- **US-022** Atomic identifier allocation at Draft->Review: `StCd-XXXXX` drawn from the selected group's (or subgroup's) numeric range (07-database-design.md §7a). AC: TC-007, TC-008 pass; two concurrent submissions never collide (integration test with simulated concurrency). Dependencies: US-020. Priority: P0.
- **US-023** `GET /status-codes/next-available` non-binding preview endpoint and UI display. AC: preview never

reserves a number. Dependencies: US-022. Priority: P1.

- **US-024** Administrator manual-override allocation for a documented migration/reservation case, with reason logged. Dependencies: US-022. Priority: P2.
- **US-025** Capacity-exhaustion handling: allocation beyond a (sub)range's upper bound raises a domain error rather than overflowing. Dependencies: US-022. Priority: P2.

FEAT-02.3 Field Validation

- **US-026** Mandatory field validation, client- and server-side (06-data-dictionary.md). Dependencies: US-020. Priority: P0.
- **US-027** Field-format and cross-field rules. (The v2 rules Safety × Alarm Behaviour, Auto reset × delay and Emergency Brake × Fault/Safety were retired on 2026-09-17: the real data model supports none of them; the current rules are the delay pattern, the category enum, the repeat-field coupling and the subgroup-belongs-to-group check, 06-data-dictionary.md.) AC: TC-005, TC-006. Dependencies: US-026. Priority: P0.
- **US-028** Restricted-field handling: each of the eight audience-access fields (corrected 2026-09-17) is Engineer-proposable but only authoritative once the Administrator's sign-off sets it (06-data-dictionary.md §2d, SEC-013). Dependencies: US-026. Priority: P0.
- **US-029** Save-as-Draft any number of times without triggering submission validation. Dependencies: US-020. Priority: P0.

FEAT-02.4 Revision Management

- **US-030** No in-place editing of Approved/Released/Archived revisions; new Draft revision required. AC: TC-013. Dependencies: US-026. Priority: P0.
- **US-031** Only one open (Draft/Review/PendingApproval) revision per code at a time. AC: TC-012. Dependencies: US-030. Priority: P0.
- **US-032** Engineer can withdraw own open CR. Dependencies: US-030. Priority: P1.

EPIC-03 — Cross-Domain Sign-Off

- **US-040** System derives, from the field-to-domain ownership table (06-data-dictionary.md §2a), which Engineering Domains are "required" on a given revision, and flags any not covered by the originating Engineer's own domain tags. Dependencies: EPIC-02, US-009 (domain tags must exist). Priority: P0.
- **US-041** Originating Engineer can request sign-off from an Engineer holding a required domain; request surfaces on that Engineer's Dashboard and My Change Requests (S10a). Dependencies: US-040. Priority: P0.
- **US-042** Second Engineer completes domain-owned fields and records a `DomainSignoff`; system verifies they actually hold that domain (403 otherwise). AC: TC-020, TC-021. Dependencies: US-041. Priority: P0.
- **US-043** Submit-for-Review is blocked (422) while any required domain lacks a sign-off. Dependencies: US-042. Priority: P0.

EPIC-04 — Dual Review & Chief Engineer Approval Workflow

FEAT-04.1 Submission

- **US-050** Engineer submits Draft for Review; identifier allocated (EPIC-02 FEAT-02.2); requires all domain sign-offs complete (EPIC-03). Dependencies: EPIC-02, EPIC-03. Priority: P0.

FEAT-04.2 Dual Sign-Off

- **US-051** Reviewer (Sub-PO for StCd) records sign-off (approve/reject); cannot act on own CR. Dependencies: US-050. Priority: P0.
- **US-052** Administrator (Product Owner) records sign-off (approve/reject); cannot act on own CR; approving also confirms/overrides restricted fields (US-028). Dependencies: US-050. Priority: P0.
- **US-053** System auto-transitions Review->PendingApproval atomically on the second sign-off, in either order. AC: integration test covers both orderings. Dependencies: US-051, US-052. Priority: P0.
- **US-054** Either sign-off rejecting (mandatory comment) returns the CR to Draft and clears both sign-off columns.

Dependencies: US-051, US-052. Priority: P0.

FEAT-04.3 Chief Engineer Approval

- **US-055** Chief Engineer views Approval Queue (S9a) listing PendingApproval CRs. Dependencies: US-053. Priority: P0.
- **US-056** Chief Engineer approves (PendingApproval->Approved) or rejects (->Draft, mandatory comment); cannot act if they are the requester, Reviewer sign-off, or Administrator sign-off on this CR. AC: TC-009b, TC-019. Dependencies: US-055. Priority: P0.

FEAT-04.4 Review/Approval Queues & Tracking

- **US-057** Reviewer/Administrator see items awaiting their sign-off, filterable. Dependencies: US-050. Priority: P0.
- **US-058** Engineer's "My Change Requests" shows sign-off/approval progress (three-slot status strip). Dependencies: US-050. Priority: P1.
- **US-059** Reviewer/Administrator/Chief Engineer can add plain comments without a decision. Dependencies: US-050. Priority: P1.

EPIC-05 — Search, Browse, History

Search, browse and history with the filters Functional System Group, Functional Subgroup, Turbine Platform, `is_sandbox` (default excluded) (FR-060–FR-064; they replaced Controller Type in v2). Dependencies: EPIC-02. Priority: P0/P1.

EPIC-06 — Release Management

Release management, with candidate resolution explicitly excluding `is_sandbox=true` revisions (enforced by `trg_prevent_sandbox_release`, TC-014a) and `scope_filter` expressed via Functional System Group/Subgroup. Dependencies: EPIC-04. Priority: P0.

EPIC-07 — Sandbox Status Codes

- **US-070** Administrator creates a sandbox Status Code (`is_sandbox=true`), identifier drawn from the isolated `StCd-T#####` band. Dependencies: EPIC-02. Priority: P1.
- **US-071** Sandbox codes are structurally excluded from default search, catalogue export, and Release candidates. AC: TC-014a, TC-023. Dependencies: US-070. Priority: P0 (verification story).
- **US-072** Administrator permanently deletes a sandbox code via typed-confirmation UI; blocked at both service and DB-trigger layers for any non-sandbox record. AC: TC-018a. Dependencies: US-070. Priority: P1.
- **US-073** Deletion writes a `SANDBOX_DELETE` audit entry with a full pre-delete snapshot; `SANDBOX_CREATE`/`SANDBOX_DELETE` pair remains queryable after the record is gone. AC: TC-015a. Dependencies: US-072. Priority: P0 (compliance-critical despite feature being P1).

EPIC-08 — Deprecation & Archival

Deprecation and archival (FR-050–FR-052); deprecating a Released code via an Engineer CR requires the **full** dual-review-plus-Chief-Engineer-approval cycle (it no longer needs only a single Reviewer). Dependencies: EPIC-06. Priority: P1.

EPIC-09 — User & Lookup Administration

- **US-090** User list/create/edit with role + Engineering Domain tag assignment (S14). Dependencies: US-009. Priority: P1.
- **US-091** Manage Functional System Group / Functional Subgroup / Turbine Platform / Engineering Domain vocabularies, alongside the original 7 (S15). Dependencies: US-004. Priority: P1.
- **US-092** Engineer can propose a new lookup value (any vocabulary) for Administrator review. Dependencies:

US-091. Priority: P2.

- **US-093** Administrator (or the baseline seed migration) confirms the real Functional Subgroup taxonomy (already seeded per 06-data-dictionary.md §9a) is complete before final seed data is locked. (2026-09-17: the `15000–15999` range is resolved — confirmed free/reserved capacity, not a classification gap — no longer part of this story's scope.) Dependencies: US-091. Priority: P1 (downgraded from P0 now that the blocking sub-item is resolved).

EPIC-10 — Audit Trail & Reporting

Audit trail and reporting, with the action catalogue in 12-audit-compliance.md §2 (`REVIEWER_SIGNOFF`, `ADMIN_SIGNOFF`, `CHIEF_ENGINEER_APPROVE/REJECT`, `DOMAIN_SIGNOFF`, `SANDBOX_CREATE/DELETE`, `FULL_DATABASE_EXPORT`), and audit search access extended to the Chief Engineer role (scoped). Dependencies: threads through EPIC-02–EPIC-07. Priority: P0.

EPIC-11 — REST API Hardening & Conformance

User and lookup administration API covering the decision endpoints (`reviewer-signoff`, `admin-signoff`, `chief-engineer-decide`), domain sign-off endpoints, sandbox delete, and full-database export. Dependencies: built alongside each corresponding UI story. Priority: P0/P1.

EPIC-12 — Security Hardening, Compliance & No-Cloud Verification

Security hardening and compliance (security headers, CSRF, XSS/SQLi prevention, `bandit/pip-audit`, manual OWASP checklist), plus:

- **US-120** CI grep-based check blocking any known cloud-SDK import. AC: TC-029. Dependencies: US-002. Priority: P0.
- **US-121** Manual no-cloud verification checklist (11-security-architecture.md §12) executed before each production deployment. Dependencies: all functional epics substantially complete. Priority: P0.

EPIC-13 — Non-Functional: Performance, Deployment, Monitoring

REST API hardening and conformance; all infrastructure is on-premises (14-deployment-architecture.md), performance validation covers identifier-allocation contention within a popular Functional Group (TC-025). Priority: P0/P1.

EPIC-14 — Library Import (new 2026-09-17)

- **US-140** As an Engineer or Administrator, I can upload a CSV or XLSX file and have it validated and converted into Draft Change Requests, one per valid row (FR-048a–b). AC: TC-033, TC-034. Dependencies: EPIC-02. Priority: P1.
- **US-141** As the system, I require every imported row to pass the full dual-review-plus-Chief-Engineer-approval cycle before Release, identical to a manually authored code (FR-048c). AC: TC-033. Dependencies: US-140, EPIC-04. Priority: P0 (governance-critical despite the feature being P1).
- **US-142** As an Engineer, I can download a per-row error report for a completed import job and see exactly why each failed row was skipped (FR-048b). Dependencies: US-140. Priority: P1.
- **US-143** As the system, I preserve the source file's legacy numbering as `legacy_reference_number`, never confusing it with the governed StCd identifier (FR-048d). Dependencies: US-140. Priority: P2.

EPIC-15 — OneTool Shell Integration & UX Fixes (new 2026-09-17)

- **US-150** As any user, I see CSCM Tool rendered inside the shared OneTool top bar and sidebar, using the OneTool design tokens and component treatments exclusively (10-ui-ux-specification.md §0). Dependencies: EPIC-01. Priority: P1.
- **US-151** As the system, I never discard already-entered form data on a validation failure anywhere in the application (NFR-007a). AC: TC-036. Dependencies: EPIC-02. Priority: P0 (direct, explicit legacy-tool complaint).

- **US-152** As a user, every lookup-backed dropdown shows its selected option's description as inline help text (NFR-007b). Dependencies: EPIC-02, EPIC-09 (lookup administration). Priority: P1.
- **US-153** As a user, CSCM Tool still uses its own login screen and credential store, not a shared OneTool identity (confirmed decision, 11-security-architecture.md §2). Dependencies: EPIC-01. Priority: P0 (verification story — confirms scope boundary is respected, not crossed).

EPIC-16 — Subgroup & Group Administration (new 2026-09-17)

- **US-160** As an Administrator, I can create a new Functional Subgroup within any Functional System Group, specifying a name and numeric sub-range (FR-048e). Dependencies: EPIC-02. Priority: P1.
- **US-161** As the system, I reject a proposed subgroup sub-range that overlaps any existing subgroup, naming the conflict (FR-048f). AC: TC-032. Dependencies: US-160. Priority: P0 (data-integrity critical despite the feature being P1).
- **US-162** As an Administrator, I can create a brand-new Functional System Group (code, label, numeric range) into any of the confirmed free ranges — `10000?10999`, `14000?14999`, `15000?15999`, `16000?16999` — without any deploy beyond the data change (FR-048g). Dependencies: EPIC-02. Priority: P2 (no near-term business need identified yet, but explicitly requested as future-proofing).
- **US-163** As the system, I reject a proposed Functional System Group range that overlaps any existing group, naming the conflict (FR-048h). AC: TC-037. Dependencies: US-162. Priority: P0 (data-integrity critical despite the feature being P2).

EPIC-17 — Concurrency & Data Integrity (v4, delivered)

- **US-170** As an Engineer, I am told — and lose nothing — when someone saved the Draft I am editing (FR-090, I-01). AC: TC-039. Priority: P0.
- **US-171** As an Engineer, I am warned about unknown, retired or self-referencing codes in the reference fields, and about similar titles (FR-091, FR-093; I-04, I-12). AC: TC-039. Priority: P1.
- **US-172** As an Administrator or requester, I see where a code is used and must acknowledge live dependants before deprecating it (FR-092). AC: TC-039. Priority: P1.
- **US-173** As an Engineer, I can create a platform variant of a code (FR-094, I-25). AC: TC-039. Priority: P2.

EPIC-18 — Release Integrity & Software-Version Traceability (v4, delivered)

- **US-180** As Technical Publications, I receive every Release export with a manifest of SHA-256 hashes, optionally signed, as a ZIP bundle, and can verify a file later without the database (FR-103, FR-104; I-06). AC: TC-042. Priority: P0.
- **US-181** As Technical Publications, I get release notes — added, changed (field diff), deprecated — against the previous Release (FR-105). AC: TC-042. Priority: P1.
- **US-182** As an Engineer, I state the software-version range a code applies to; as an Administrator I set the versions a Release targets; anyone can ask which codes apply to a version and what changed between two (FR-106, FR-107, D-01; I-07). AC: TC-042. Priority: P1.
- **US-183** As a service technician, I use an offline lookup page generated from a Release (FR-108, I-24). AC: TC-042. Priority: P2.

EPIC-19 — Workflow SLAs, Notifications & Metrics (v4, delivered)

- **US-190** As a Reviewer or Chief Engineer, I see how long each request has waited, amber and red (FR-120, I-08). AC: TC-041. Priority: P0.
- **US-191** As a decision holder, I receive one in-app notice per stalled request per day, and mail if the relay is configured; as an Administrator I see stalled requests and reassign (FR-121, FR-122, D-04). AC: TC-041. Priority: P1.
- **US-192** As an Administrator or Chief Engineer, I see the business-case KPIs (FR-123, I-10). AC: TC-041. Priority: P2.
- **US-193** As a decision maker, I name a delegate for my absence; the delegate acts as themselves, for decisions only, under segregation of duties (FR-125, D-08; I-16). AC: TC-038. Priority: P1.
- **US-194** As the organisation, I require the actor's password for sign-offs, approvals, publishing and the full export (FR-126, SEC-026, D-07; I-17). AC: TC-038. Priority: P0.

EPIC-20 — Audience-Scoped & Profile-Based Publication (v4, delivered)

- **US-200** As an Administrator, I define export profiles (columns, labels, default audience and language) (FR-100, I-02). AC: TC-042. Priority: P1.
- **US-201** As Technical Publications, the PDF never truncates text, repeats the header and carries a title page (FR-101). AC: TC-042. Priority: P0.
- **US-202** As the organisation, an export for an audience contains only codes that audience may see (FR-102, BR-016, D-02; I-03). AC: TC-042. Priority: P0.

EPIC-22 — Multilingual Content (v4, delivered)

- **US-220** As an Engineer or Administrator, I translate title, description and remedy text; as a Reviewer or Administrator other than the translator I approve it; exports in a language use approved text (FR-110, FR-111, BR-018, D-03; I-15). AC: TC-042. Priority: P2.

EPIC-23 — Knowledge Enrichment & Evidence (v4, delivered)

- **US-230** As an Engineer, I attach evidence to a change request (FR-130, SEC-024; I-13). AC: TC-040. Priority: P1.
- **US-231** As an Engineer, I record remedy text, affected component, related codes and standards (FR-131; I-14). AC: TC-039. Priority: P1.
- **US-232** As a Reviewer, I comment on single fields and confirm the configured checklist before approving (FR-127, FR-128; I-23). AC: TC-040. Priority: P1.
- **US-233** As an Engineer or Chief Engineer, I group change requests into a change set and decide several in one sitting without weakening any decision (FR-124; I-11). AC: TC-040. Priority: P2.
- **US-234** As an Administrator, I assign the least-loaded Reviewer, manually or automatically (FR-129; I-30). AC: TC-040. Priority: P2.

EPIC-24 — Fleet Feedback Loop (v4, delivered as a file import)

- **US-240** As an Administrator or Chief Engineer, I import anonymised occurrence counts and see codes that never fire and those that fire most (FR-143; I-28). AC: TC-046. Priority: P2.

EPIC-25 — Operations & Resilience (v4, delivered)

- **US-250** As an operator, every nightly backup is proven restorable and its age is visible (FR-150, NFR-012; I-09). AC: TC-044. Priority: P0.
- **US-251** As an auditor, an edited, removed or truncated audit entry is detected, and old entries are archived, never deleted (FR-151, AUD-072 … AUD-075; I-18). AC: TC-044. Priority: P0.
- **US-252** As an operator, readiness and metrics endpoints and a log file feed the on-premises monitoring (FR-152, FR-153, NFR-011; I-21). AC: TC-044. Priority: P1.
- **US-253** As an operator, the login rate limit is shared by all worker processes and survives a restart (NFR-009; I-20). AC: TC-050. Priority: P1.
- **US-254** As an operator, I install and upgrade with one script per platform; CI runs the whole verification (NFR-013; I-33). AC: TC-051. Priority: P1.
- **US-255** As the architect, SQLite-specific code is inventoried and capped (NFR-014; I-34). AC: TC-051. Priority: P2.

Extensions of earlier epics (v4, delivered)

- **EPIC-14: US-144** template download; **US-145** dry run; **US-146** update mode keyed on `legacy_reference_number`; **US-147** reconciliation report; **US-148** XLSX error report; **US-149** background import with progress (FR-048i … FR-048n; I-05, I-32). AC: TC-043. Priority: P0 for the migration period.
- **EPIC-14 (v4.2): US-256** read several generated `StCd_.st` programs (files, folder, ZIP) and preview them with a per-program state and multi-select (FR-048o, FR-048p); **US-257** import the ticked programs as Drafts with their numbers reserved, in one change set (FR-048q … FR-048t, BR-023); **US-258** the legacy -> new mapping with export (FR-048u); **US-259** provenance of every imported code and the re-check at confirm (FR-048v, FR-048w). AC: TC-053. Priority: P0.
- **EPIC-05: US-075** ranked, typo-tolerant search; **US-076** saved searches, column chooser, shareable filters (FR-148, FR-142; I-31, I-26). AC: TC-047, TC-046.

- **EPIC-06 / EPIC-08: US-095** planned effective and deprecation dates with release warnings (FR-109; I-27).
- **EPIC-02 / EPIC-08: US-164** Hub-to-Main consolidation: MC drafts from Hub codes, mapping report, bulk deprecation (FR-147, A-008; I-36). AC: TC-046.
- **EPIC-16: US-165** catalogue health report and capacity forecast (FR-140, FR-141; I-19). AC: TC-046, TC-039.
- **EPIC-11: US-125** scoped API tokens, change feed, signed webhooks (FR-144 … FR-146, SEC-027, SEC-028; I-29). AC: TC-046.
- **EPIC-15: US-153** German interface and accessibility (FR-155, NFR-010; I-22); **US-154** in-app help (FR-154; I-35). AC: TC-049, TC-047.

Dependency Graph Summary (Epic Level)

```
EPIC-01 (Foundation)
   |
   +--> EPIC-02 (Data Model, Numbering & Validation)
   |        |
   |        +--> EPIC-03 (Cross-Domain Sign-Off)
   |        |        |
   |        |        +--> EPIC-04 (Dual Review & Chief Engineer Approval)
   |        |                 |
   |        |                 +--> EPIC-06 (Release Management)
   |        |                 |        |
   |        |                 |        +--> EPIC-08 (Deprecation & Archival)
   |        |                 |
   |        |                 +--> EPIC-10 (Audit Trail) [threads through 02-08]
   |        |
   |        +--> EPIC-05 (Search, Browse, History)
   |        +--> EPIC-07 (Sandbox Status Codes)
   |
   +--> EPIC-09 (User & Lookup Administration)
   |        +--> EPIC-16 (Subgroup & Group Administration)
   +--> EPIC-11 (API Hardening) [parallel, per-endpoint]
   +--> EPIC-12 (Security & No-Cloud Verification) [parallel, cross-cutting]
   +--> EPIC-13 (Non-Functional / Deployment) [parallel, cross-cutting]
   +--> EPIC-14 (Library Import) [depends on EPIC-02 for validation reuse, EPIC-04 for governance]
   +--> EPIC-15 (OneTool Shell Integration & UX Fixes) [parallel, cross-cutting UI pass]
   +--> v4: EPIC-17 (Integrity) -> EPIC-18 (Release Integrity) -> EPIC-20 (Publication) -> EPIC-22 (Languages)
            EPIC-19 (SLAs, delegation, step-up) · EPIC-23 (Evidence & review aids) · EPIC-24 (Fleet feedback)
            EPIC-25 (Operations: backups, audit chain, deploy, CI) [cross-cutting, P0 items before go-live]
```

18. Claude Code Implementation Guide — CSCM Tool

This document is the implementation-facing companion to the rest of this package. It assumes the reader (human or Claude Code) has the full documentation set available and translates it into concrete build guidance. It does not restate business rules — it references them by ID.

1. Recommended Architecture Patterns

Layered monolith (Presentation -> Service -> Repository -> Data), repository pattern per aggregate root, service layer owning all business rules, Application Factory pattern, DTOs at the service boundary. The v2 model adds no new architectural pattern — it adds more service-layer state machine complexity (7 lifecycle states, three decision points instead of one) and one new cross-cutting concern (Cross-Domain Sign-Off gating), both of which fit cleanly inside the existing service layer.

2. Recommended Folder Structure

(v4) The tree below is the original recommendation; the layout **as built** — including the v4 services (`publication`, `governance`, `operations`, `integration`), `scripts/`, `deploy/` and `tests/` — is in 15-development-standards.md §1, and 08-system-architecture.md §2 maps each component to its modules. Notable deviations from the recommendation: the subgroup/group administration lives in `lookup_admin_service.py` (there is no `subgroup_admin_service.py`), the routes are one module per area under `app/api/v1/`, and the UI blueprints are split by area under `app/ui/`.

```
cscm_tool/
+-- app/
|   +-- __init__.py                    # create_app() application factory
|   +-- config.py
|   +-- extensions.py
|   |
|   +-- models/
|   |   +-- user.py                    # User, Role, EngineeringDomain, UserEngineeringDomain
|   |   +-- status_code.py             # StatusCode, StatusCodeRevision, StatusCodeRevisionPlatform
|   |   +-- change_request.py          # ChangeRequest, ReviewComment, DomainSignoff
|   |   +-- release.py                 # Release, ReleaseItem
|   |   +-- lookup.py                  # FunctionalSystemGroup, FunctionalSubgroup, TurbinePlatform,
|   |   |                               #   LookupStatusCategory, ... (7 original vocabularies)
|   |   +-- audit.py                   # AuditLogEntry
|   |   +-- export_job.py              # ExportJob
|   |
|   +-- repositories/
|   |   +-- base.py
|   |   +-- user_repository.py
|   |   +-- status_code_repository.py  # incl. allocate_next_identifier() ? see §6
|   |   +-- change_request_repository.py
|   |   +-- domain_signoff_repository.py
|   |   +-- release_repository.py
|   |   +-- lookup_repository.py
|   |   +-- audit_repository.py
|   |
|   +-- services/
|   |   +-- auth_service.py
|   |   +-- user_service.py
|   |   +-- status_code_service.py     # create Draft, create sandbox, delete sandbox, edit
|   |   +-- numbering_service.py       # NEW ? Functional Group/Subgroup identifier allocation
|   |   +-- domain_signoff_service.py  # NEW ? Cross-Domain Sign-Off gating and recording
|   |   +-- change_request_service.py  # submit, reviewer_signoff, admin_signoff,
|   |   |                               #   chief_engineer_decide, withdraw
|   |   +-- release_service.py
|   |   +-- lookup_service.py
|   |   +-- export_service.py          # catalogue + full-database export
|   |   +-- import_service.py          # CSV/XLSX library import -> Draft CRs (new)
|   |   +-- subgroup_admin_service.py  # new-subgroup creation, overlap validation (new)
|   |   +-- audit_service.py
|   |   +-- validation.py
|   |
|   +-- auth/
|   |   +-- providers.py               # AuthProvider protocol, SessionPasswordAuthProvider
|   |   |                               #   (future: OnPremActiveDirectoryAuthProvider ? never a cloud IdP)
|   |   +-- decorators.py              # @login_required, @require_role(...), @not_author(...)
|   |   +-- password.py
|   |
|   +-- api/v1/
|   |   +-- auth_routes.py
|   |   +-- user_routes.py
|   |   +-- status_code_routes.py      # incl. next-available preview, sandbox delete
```

```
|   |   +-- revision_routes.py
|   |   +-- change_request_routes.py   # incl. reviewer-signoff, admin-signoff, chief-engineer-decide
|   |   +-- domain_signoff_routes.py
|   |   +-- release_routes.py
|   |   +-- lookup_routes.py
|   |   +-- export_routes.py           # incl. /exports/full-database (Administrator only)
|   |   +-- audit_routes.py
|   |
|   +-- ui/
|   |   +-- auth_views.py
|   |   +-- dashboard_views.py
|   |   +-- status_code_views.py       # S3-S7, S5a (sandbox)
|   |   +-- change_request_views.py    # S8-S10, S9a (approval queue), S10a (domain sign-off)
|   |   +-- release_views.py           # S11-S13
|   |   +-- admin_views.py             # S14-S16
|   |   +-- export_views.py            # S17
|   |
|   +-- export/
|   |   +-- csv_exporter.py
|   |   +-- json_exporter.py
|   |   +-- xlsx_exporter.py
|   |   +-- pdf_exporter.py
|   |   +-- full_database_exporter.py  # SQLite Online Backup API wrapper, Administrator only
|   |
|   +-- templates/
|   +-- static/                        # local-only assets, no CDN references (11-security-architecture.md §12)
|   |   +-- css/onetool-tokens.css     # OneTool design tokens (10-ui-ux-specification.md §0), vendored locally
|   |   +-- vendor/manrope, inter, material-symbols/  # self-hosted fonts/icons, per the no-CDN rule
|   +-- errors.py
|
+-- migrations/
+-- tests/{unit,integration,system}/
+-- docs/
+-- seed_data/lookup_seed.sql          # matches artifacts/schema.sql's seed inserts, incl. the real Functional Subg
  roup taxonomy
+-- requirements.txt
+-- pyproject.toml
+-- wsgi.py
+-- README.md
```

3. Flask Blueprint Structure

Route handlers stay thin; the new decision endpoints follow the same pattern as the original `decide` example, now split into three distinct, narrowly-scoped handlers rather than one generic one, because each has a different authorization check:

```
# app/api/v1/change_request_routes.py
from flask import Blueprint, request, jsonify
from app.auth.decorators import login_required, require_role
from app.services.change_request_service import ChangeRequestService

def register(bp: Blueprint):
    @bp.post("/change-requests/<int:cr_id>/reviewer-signoff")
    @login_required
    @require_role("REVIEWER")
    def reviewer_signoff(cr_id: int):
        body = request.get_json()
        result = ChangeRequestService.reviewer_signoff(
            cr_id=cr_id, actor=request.current_user,
            decision=body["decision"], comment=body.get("comment"),
        )
        return jsonify(result.to_dict()), 200

    @bp.post("/change-requests/<int:cr_id>/admin-signoff")
    @login_required
    @require_role("ADMINISTRATOR")
    def admin_signoff(cr_id: int):
        body = request.get_json()
        result = ChangeRequestService.admin_signoff(
            cr_id=cr_id, actor=request.current_user,
            decision=body["decision"], comment=body.get("comment"),
        )
        return jsonify(result.to_dict()), 200

    @bp.post("/change-requests/<int:cr_id>/chief-engineer-decide")
    @login_required
    @require_role("CHIEF_ENGINEER")
    def chief_engineer_decide(cr_id: int):
        body = request.get_json()
        result = ChangeRequestService.chief_engineer_decide(
            cr_id=cr_id, actor=request.current_user,
```

```
            decision=body["decision"], comment=body.get("comment"),
        )
        return jsonify(result.to_dict()), 200
```

The segregation-of-duties check (actor != author, and for the Chief Engineer step, actor != either prior sign-off) lives inside each service method, not in the decorator — the decorator only checks role membership, which is static; the author/sign-off comparison needs the specific `ChangeRequest` row (SEC-011).

4. Repository Pattern

Repositories hold queries and generic CRUD helpers, no business rules. A repository method of note:

```
# app/repositories/status_code_repository.py
class StatusCodeRepository(BaseRepository[StatusCode]):
    model = StatusCode

    def allocate_next_identifier(self, functional_system_group_id: int,
                                 functional_subgroup_id: int | None) -> str:
        """Must be called inside a BEGIN IMMEDIATE transaction ? see
        docs/07-database-design.md §7a. Returns e.g. 'StCd-01233'."""
        ...

    def allocate_next_sandbox_identifier(self) -> str:
        """Draws from the isolated StCd-T##### counter; never touches the
        real Functional System Group ranges."""
        ...
```

5. Service Layer Pattern

The dual-sign-off + Chief Engineer approval logic is the most state-sensitive part of the system; it is written so the "did both sign-offs land" check and the resulting transition are always one atomic operation (07-database-design.md §7b):

```
# app/services/change_request_service.py
from app.extensions import db
from app.repositories.change_request_repository import ChangeRequestRepository
from app.services.audit_service import AuditService
from app.domain.errors import SegregationOfDutiesViolation

class ChangeRequestService:
    """Implements the dual-review-then-Chief-Engineer-approval workflow.
    See 05-governance-handbook.md §7-8, BR-004/BR-005."""

    @staticmethod
    def admin_signoff(cr_id: int, actor, decision: str, comment: str | None):
        repo = ChangeRequestRepository()
        cr = repo.get(cr_id)
        revision = cr.revision

        if actor.id == cr.requested_by:
            raise SegregationOfDutiesViolation("BR-005: cannot sign off own change request")
        if revision.lifecycle_status != "REVIEW":
            raise InvalidTransition(f"Cannot sign off a revision in state {revision.lifecycle_status}")
        if decision == "REJECT" and not comment:
            raise ValueError("Rejection requires a comment")

        with db.session.begin():
            repo.add_comment(cr, author=actor, text=comment,
                             decision="ADMIN_APPROVE" if decision == "APPROVE" else "ADMIN_REJECT")
            if decision == "REJECT":
                repo.clear_review_signoffs(revision)          # both columns, whichever was set
                revision.lifecycle_status = "DRAFT"
            else:
                revision.admin_signoff_by, revision.admin_signoff_at = actor.id, now()
                # confirm/override any restricted field the Engineer proposed (SEC-013)
                repo.confirm_restricted_fields(revision, actor)
                if revision.reviewer_signoff_by is not None:
                    revision.lifecycle_status = "PENDING_APPROVAL"   # atomic 2nd-signoff transition
            AuditService.log(entity_type="StatusCodeRevision", entity_id=revision.id,
                             action="ADMIN_SIGNOFF", actor=actor,
                             before={"lifecycle_status": "REVIEW"},
                             after={"lifecycle_status": revision.lifecycle_status})
        return cr

    @staticmethod
    def chief_engineer_decide(cr_id: int, actor, decision: str, comment: str | None):
        repo = ChangeRequestRepository()
        cr = repo.get(cr_id)
        revision = cr.revision
```

```
        blocked = {cr.requested_by, revision.reviewer_signoff_by, revision.admin_signoff_by}
        if actor.id in blocked:
            raise SegregationOfDutiesViolation("BR-005: segregation of duties across all three decision points")
        if revision.lifecycle_status != "PENDING_APPROVAL":
            raise InvalidTransition(f"Cannot decide a revision in state {revision.lifecycle_status}")
        if decision == "REJECT" and not comment:
            raise ValueError("Rejection requires a comment")

        with db.session.begin():
            repo.add_comment(cr, author=actor, text=comment,
                             decision="CHIEF_ENGINEER_APPROVE" if decision == "APPROVE" else "CHIEF_ENGINEER_REJECT
 ")
            if decision == "APPROVE":
                revision.chief_engineer_approved_by, revision.chief_engineer_approval_date = actor.id, now()
                revision.lifecycle_status = "APPROVED"
            else:
                repo.clear_review_signoffs(revision)
                revision.lifecycle_status = "DRAFT"
            cr.chief_engineer_decided_by, cr.chief_engineer_decided_at = actor.id, now()
            AuditService.log(entity_type="StatusCodeRevision", entity_id=revision.id,
                             action="CHIEF_ENGINEER_APPROVE" if decision == "APPROVE" else "CHIEF_ENGINEER_REJECT",
                             actor=actor, before={"lifecycle_status": "PENDING_APPROVAL"},
                             after={"lifecycle_status": revision.lifecycle_status})
        return cr
```

`reviewer_signoff` is the mirror image of `admin_signoff` (no restricted-field confirmation step). `submit` (Draft->Review) additionally calls `NumberingService.allocate_next_identifier(...)` and `DomainSignoffService.assert_all_required_domains_signed(revision)` before performing its transaction.

6. Database Migration Strategy

Alembic baseline from `artifacts/schema.sql` (executed verbatim, so a fresh database is by construction the documented schema; later revisions are guarded and idempotent — 07-database-design.md §11; hand-reviewed autogenerate for `CHECK`/triggers/partial indexes, `PRAGMA foreign_keys = ON` on every SQLite connection). (v4) Every new migration is tested against a frozen copy of the previous schema (`tests/fixtures/`), including the downgrade; a back-fill that touches append-only data (the audit chain) drops and restores the protecting trigger inside the same migration, and the application and the migration share one canonical form (`app/audit_hash.py`). Note: the real Functional Subgroup data (06-data-dictionary.md §9a) is seeded by the same baseline migration as every other lookup; any future addition (a new subgroup into a currently-reserved hundred-block, or a whole new Functional System Group into one of the confirmed free ranges — `10000?10999`, `14000?14999`, `15000?15999`, `16000?16999`) is a **data migration** (an `INSERT`/re-seed script), never a schema migration — call this out explicitly in that future migration's docstring so nobody mistakes it for a breaking change.

7. Testing Framework

`pytest` and `pytest-cov`; role users are made with the `make_user(role, engineering_domains=?)` factory or `make_cast` (one user per role, plus a Controls-tagged Engineer for Cross-Domain Sign-Off tests) and logged in with `login_as`. `schemathesis` against `artifacts/openapi.yaml` v4.0.0. (v4) Fixtures: `StepUpClient` (answers the password prompt), `raw_client`, `file_app` (a real SQLite file for background threads, backups, upgrades and processes), `make_user`, `wcnv_ids`; helpers in `tests/helpers.py` (`drive_to_approved`, `release_revisions`, …). Three verification methodologies — sequential with gates, reversed/shuffled with property tests, whole suite on a file database — are defined in 13-test-strategy.md §11. `bandit -r app/` and `pip-audit`, plus the new no-cloud-import grep check (13-test-strategy.md §6, TC-029) wired into the same CI stage.

8. Suggested Coding Order

Updated sequencing reflecting the new epics (17-implementation-backlog.md dependency graph):

1. `app/config.py`, `app/__init__.py`, `wsgi.py`, health endpoint (US-001, US-002 incl. no-cloud CI check).
2. `app/models/` in full (all v2 tables from `artifacts/schema.sql`), then the baseline Alembic migration incl. seed data (US-003, US-004).
3. Auth: password hashing, `SessionPasswordAuthProvider`, `auth_service`, decorators (US-005–US-008).
4. `user_repository`, `user_service`, minimal user list/create UI+API, **including Engineering Domain tagging** (US-009–US-012) — needed early since Cross-Domain Sign-Off tests require tagged users.
5. `audit_service` as shared infrastructure before any consuming service (every service writes audit entries, so it must exist first).

6. `numbering_service` (Functional Group/Subgroup allocation, §5, §6) and `validation.py`, then `status_code_repository`/`status_code_service` for Draft creation (EPIC-02).
7. `domain_signoff_service` and its API/UI (EPIC-03) — build this **before** the submit/review workflow, since Submit-for-Review depends on it, not after.
8. `change_request_repository`, `change_request_service` covering `submit`, `reviewer_signoff`, `admin_signoff`, `chief_engineer_decide`, `withdraw` (EPIC-04) — the most transaction-sensitive service in the system; write the atomic-second-signoff and segregation-of-duties integration tests alongside this code, not after.
9. Search/filter/list on `StatusCodeRepository` plus corresponding UI/API, with the new Functional Group/Subgroup/Platform/sandbox filters (EPIC-05).
10. `release_repository`/`release_service` (candidate resolution now excluding sandbox, atomic publish) plus `export/` (catalogue formats, then `full_database_exporter.py`) (EPIC-06).
11. Sandbox create/delete (EPIC-07) — deliberately sequenced after the main workflow is solid, since it reuses the same state machine and its only new surface area is the isolated numbering band, the delete trigger, and the `SANDBOX_DELETE` snapshot-audit path.
12. Deprecation/archival methods, now routed through the full dual-review-plus-approval cycle when initiated by an Engineer CR (EPIC-08).
13. Remaining lookup administration UI/API for the new vocabularies (EPIC-09).
14. Audit search/reporting UI/API, extended to the Chief Engineer role (EPIC-10's consumer-facing half).
15. Fill in remaining OpenAPI-contract gaps, `schemathesis` conformance (EPIC-11).
16. Security headers, CSRF, `bandit`/`pip-audit`, manual OWASP + no-cloud checklist sign-off (EPIC-12).
17. Performance testing (incl. identifier-allocation contention, TC-025), deployment scripts, backup/restore drill, monitoring (EPIC-13), immediately preceding go-live.
18. `import_service.py` and the Library Import UI/API (EPIC-14), reusing the validation logic already built in step 6 — an import row and a manually typed Draft go through the exact same validators.
19. `subgroup_admin_service.py` and its group-level counterpart (EPIC-16) — straightforward once the lookup administration screens from step 13 exist; the group-level "create a new Functional System Group into a free range" capability (FR-048g/h) is the lowest-priority piece of this epic and can slip past MVP if time is tight.
20. OneTool shell integration and the two explicit UX fixes — no-data-loss-on-failure and inline lookup descriptions (EPIC-15) — applied as a pass across every existing form, not a new module; do this once the forms it touches are stable, to avoid rework.
21. (v4) Foundations that other v4 features rely on, in this order: the schema v4 migration and `app/schema_sql.py`; `settings_service`; `audit_hash` / `audit_chain` (the hash chain must exist before more audit-writing code does); `auth_service.step_up` and the `StepUpClient` test fixture; `delegation_service`.
22. (v4) Content integrity (EPIC-17): `row_version` checks in `status_code_service`, `reference_service`, `duplicate_service`, platform variants, planned dates — each with its tests in `test_content_integrity`.
23. (v4) Workflow aids (EPIC-19, EPIC-23): `sla_service`, `notification_service`/`mail_service`, `kpi_service`, `change_set_service`, `field_review_service`, `attachment_service`, reviewer balancing in `change_request_service`.
24. (v4) Publication (EPIC-18, EPIC-20, EPIC-22): `software_version_service`, `translation_service`, `audience_service`, `export_profile_service`, the rewritten `export_service` pipeline with `manifest_service`, release notes, offline bundle, wrapped PDF.
25. (v4) Import toolkit (EPIC-14 extension): template, dry run, update mode, reconciliation, error reports, background worker — the dry run and the real run share the same `_evaluate` path so they cannot disagree.
26. (v4) Insight and integration: `health_report_service`, `saved_search_service`, `occurrence_service`, `integration_service` (tokens, feed, webhooks), `hub_consolidation_service`, FTS5 search.
27. (v4) Operations (EPIC-25): `backup_service`, `observability`, readiness, the CLI jobs, `scripts/serve.py`, `deploy/`, CI definitions, `pg_readiness`.
28. (v4) UI pass: new screens (10-ui-ux-specification.md S20–S38), `app/i18n_de.py`, `app/help_content.py`, accessibility fixes; then regenerate the documents' generated blocks and the test report.

19. Glossary — CSCM Tool

This glossary is a living document. Any change to terminology, roles, lifecycle states, or domain concepts elsewhere in this package must be reflected here in the same pull request (see 15-development-standards.md §6). Terms are alphabetical.

Administrator — System role. Organizationally equivalent to the Product Owner Controls Software / Component Owner Controls Software. Owns user administration, lookup/vocabulary governance (incl. creating new Functional Subgroups), Release publishing, database export, sandbox status codes, and is one of the two mandatory Review sign-offs (05-governance-handbook.md §7).

API Token — A scoped, expiring, revocable secret an Administrator issues to a machine consumer on the internal network (scopes `catalogue:read`, `changes:read`; at most 365 days). Sent as `Authorization: Bearer cscm_?`, honoured only on read-only `GET` operations that declare its scope, with Viewer-level visibility; stored only as a SHA-256 hash and shown once (FR-144, SEC-027).

Applicability Matrix — The answer to "which Released codes apply to controller software version X" and "what was added, removed and changed between versions X and Y", computed from the revisions' version ranges (FR-107, decision D-01).

Approved — Lifecycle state reached once a Chief Engineer has approved a status code revision that has already completed dual Review sign-off. Awaiting inclusion in a Release.

Archived — Terminal lifecycle state for long-retained, inactive status codes, reached from Deprecated after the minimum retention period.

Audience — One of the eight groups an export can be scoped to: Development, Sales, TCC, Service, Turbine Operator Package, Grid Operator, Service Partner, Customer. An audience export contains only the codes the Administrator-confirmed access value makes visible to that audience (FR-102, BR-016).

Audience Access Fields — Eight independent governed fields (Development, Sales, TCC, Service, Turbine Operator Package, Grid Operator, Service Partner, Customer), each with its own real vocabulary, controlling visibility per audience (corrected 2026-09-17 from a real library export; replaces the earlier single "Access Rights" enum). Proposed by the authoring Engineer but only authoritative once confirmed by the Administrator's Review sign-off (06-data-dictionary.md §2d, §9d).

Audience Visibility — The flag `audience_visible` on each value of the eight access vocabularies. A code is visible to an audience only if its confirmed value for that audience has the flag set; no value, or `Covert`, means not visible (decision D-02).

Audit Anchor — The row count and head hash written to `instance/audit_chain_anchor.json` by every successful chain verification; the next verification uses it to detect entries removed from the end of the log.

Audit Hash Chain — The tamper-evidence of the audit log: each entry stores `prev_hash` and `entry_hash` (SHA-256, or HMAC-SHA-256 with `AUDIT_HMAC_KEY`) over its own fields and its predecessor's hash. Verification finds an edited, removed, reordered or truncated entry (AUD-074, AUD-075).

Audit Log Entry — Immutable system record of every state-changing action (12-audit-compliance.md). Never edited or deleted, with one narrow exception: the deletion of a Sandbox Status Code, which is itself logged (the deletion event persists even though the underlying record does not).

Available Group — A numeric availability-accounting code (corrected 2026-09-17: real data uses a plain integer, not a lettered A–D group). Soft-validated against the currently-observed set; descriptive labels not yet available from source data (06-data-dictionary.md §9c).

Backup, Verified — A copy of the database made with SQLite's online-backup API and proven usable: integrity check, restore into a scratch database, row-count comparison and audit-chain check. Only a verified backup counts as the last good one (FR-150, NFR-012).

Business Day — Monday to Friday; there is no holiday calendar. Service levels are counted in business days (FR-120).

Change Feed — `GET /api/v1/changes`: releases, deprecations, reinstatements and archivals in audit order, paged with a `since_id` cursor and an ETag, for consumers that poll (FR-145).

Change Set — A group of change requests that ship with one controller software release, shown and queued together. Grouping changes presentation only: each request keeps its own decisions (FR-124).

Chief Engineer — System role. Performs the final Approval decision after both required Review sign-offs (Reviewer and Administrator) are complete. Cannot approve a change request they authored, or one they already signed off on as Reviewer or Administrator (segregation of duties).

Code Number — See Status Code Identifier (StCd).

Controller Software Version — An entry of the ordered vocabulary of controller software versions. A revision declares an inclusive from/to range of versions it applies to; a Release targets one or more versions (FR-106, decision D-01).

Covert — A value that appears in several audience-access vocabularies. Its business meaning is unconfirmed (it may be a mistranslation of "Covered"); the system treats it as not visible to that audience until the business defines it (decision D-02, assumption A-010).

Cross-Domain Sign-Off — The workflow step where a second Engineer, holding an Engineering Domain the originating Engineer does not, completes the fields tagged to that domain before a Change Request can be submitted for Review (05-governance-handbook.md §6a).

Delegation — Out-of-office cover: a Reviewer, Administrator or Chief Engineer names a delegate for a date range. The delegate acts as themselves, for the decision actions of that role only, under full segregation of duties; the audit entry names both (FR-125, BR-017, decision D-08).

Deprecated — Lifecycle state for a previously Released status code that is no longer active but is retained for historical/service reference.

Draft — Initial, freely editable lifecycle state. A status code in Draft has no fixed Status Code Identifier — the identifier is allocated only when the Change Request is submitted for Review (05-governance-handbook.md §2).

Dry Run — An import validation that reports every problem per row and column and creates nothing, not even an import job (FR-048j).

Engineer — System role. A controls engineer or cross-functional engineer (e.g. electrical, mechanical) who authors and edits status codes. Tagged with one or more Engineering Domains.

Engineering Domain — Classification of an Engineer's specialty (e.g. Controls, Electrical, Mechanical, Grid, Safety). Used to route Cross-Domain Sign-Off requirements.

Evidence — A file attached to a change request — test report, simulation, design note, safety assessment. Allow-listed types, content-checked, at most 10 MB and 20 per request, stored outside the web root, removable only while the request is a Draft (FR-130).

Export Profile — An Administrator-managed, named column set with header labels and an optional default audience and language, used to shape an export (FR-100).

Field Comment — A comment a Reviewer, Administrator or Chief Engineer attaches to one specific field of a change request in Review or Pending Approval; the author sees open ones inline and resolves them (FR-127).

Fleet Feedback — Anonymised occurrence counts per status code and period, imported from the monitoring side, used to find codes that never fire and those that fire most (FR-143).

Functional Subgroup — A sub-division of a Functional System Group, **mandatory on every Status Code — never optional** (corrected 2026-09-17). Most subgroups own a 100-number sub-band aligned to the hundreds digit of the identifier (e.g. `020xx` = 02000–02199 for the first subgroup of a group, which owns 200 numbers; later ones such as `022xx` = 02200–02299 own 100); within Turbine Control & Operation, several subgroups share the `099xx` block in finer ten-number sub-bands, named `0990x` … `0999x`. A subgroup code is the first digits of its band with `x` for the free positions — never a dash or a letter (`110xx`, `1110x`). The real taxonomy, corrected 2026-09-17 against an authoritative row-level export (the earlier hand-typed summary used incorrect subgroup names), is seeded in full (06-data-dictionary.md §9a). Unassigned blocks within a group are deliberate reserved capacity — an Administrator can create a new subgroup there at any time, provided it does not overlap an existing one.

Functional System Group — The primary classification and numbering axis for status codes. Twelve groups as of 2026-09-17: nine Main Controller groups (WCNV, WGEN, WNAC, WROT, WTOW, WTRF, WTRM, WYAW, WTUR, ranges 1000–9999) plus three Hub Controller groups added from a real export (WTUR 11000–11999, WROT 12000–12999, WPPD 13000–13999) — see 05-governance-handbook.md §2. The four-letter prefix is **not** a unique key on its own: `WTUR` and `WROT` are each intentionally reused across a Main Controller group and a Hub Controller group with a different numeric range — always resolve by range, not prefix alone. `10000?10999`, `14000?14999`, `15000?15999`, and `16000?16999` are confirmed free/reserved capacity for future groups, not currently assigned to anything.

Hub Consolidation — Moving the status codes of the Hub Controller to the Main Controller side without renumbering: Main Controller drafts created from Hub codes, a mapping report, and bulk deprecation of the Hub codes once replaced (FR-147, assumption A-008).

Import Job — A record of a CSV or XLSX status code library upload (new 2026-09-17). Every row that passes validation becomes an ordinary Draft Change Request subject to the full governance cycle — import is a faster data-entry path, never a bypass (05-governance-handbook.md §6b). Row failures are reported per-row and create nothing. Since v4 it also records its mode (`CREATE` or `UPDATE`), progress and warning counts, and runs in the

background above a row threshold.

Legacy Mapping — Which new status code replaces which legacy number, for the Main Controller and the Hub Controller separately: the code's `legacy_reference_number` read with its controller type and its identifier (or the number it has reserved). It runs from legacy to new only — the new library holds more codes than the old one, so a code without a legacy number is normal (n/a) and nothing maps back (FR-048u).

Legacy Reference Number — Free-text field on a Status Code preserving a pre-migration/source-system number (e.g. the old "Status Code Number" seen in the legacy tool) carried through library import, for traceability only — never mistaken for the governed StCd identifier (06-data-dictionary.md §1, §12).

Logical Node / **Node Value** — Derived, read-only technical addressing fields on a Status Code Revision, always equal to the parent Functional System Group's prefix and sort order respectively. Exist in the real export as their own columns but carry no information CSCM Tool doesn't already have from the group assignment (06-data-dictionary.md §2e).

Manifest — The `MANIFEST.json` that accompanies an export: release, scope, row count, SHA-256 of every file, content digest, generator, timestamp and applied options, signed with HMAC-SHA-256 when `MANIFEST_SIGNING_KEY` is configured. A file can be verified against it without the database (FR-103, FR-104).

Master Database — CSCM Tool's own on-premises server database. It is the single system of record; no external "master" system exists upstream of it. Data leaves only via Administrator-triggered export (08-system-architecture.md §6).

Notice — An in-app message for whoever holds the next decision on a stalled change request (and anyone covering for them), created at most once per user, request, stage, level and day; optionally also mailed through the organisation's relay (FR-121, decision D-04).

Offline Bundle — A single self-contained HTML page generated from a Release, with the codes embedded and a search box, for use at the turbine without connectivity (FR-108).

OneTool — The shared internal tools platform CSCM Tool integrates with visually (shared top bar, sidebar, design tokens, and component treatments — 10-ui-ux-specification.md §0). CSCM Tool keeps its own login and credential store; only the visual shell is shared, not authentication (confirmed decision, 2026-09-17).

PendingApproval — Lifecycle state entered automatically once both required Review sign-offs (Reviewer and Administrator) are recorded. Awaiting Chief Engineer decision.

Planned Date — A planned effective date or planned deprecation date stamped on the current revision by an Administrator. It warns at release time and appears in the health report but never changes a lifecycle state (FR-109).

Platform Variant Link — An optional cross-reference between two independent Status Code records that represent the same underlying condition on different Turbine Platforms. Used when a code was renamed or regenerated between platforms rather than being modeled as a single multi-platform record (05-governance-handbook.md §2a). Since v4 it is created by the Create platform variant action (FR-094).

Product Owner — Organizational title equivalent to the Administrator role in CSCM Tool.

Program Reference Fields — Eight numeric fields (Yaw Program, Manual Reset Program, Auto Reset Program, Converter Reactive Power Program, Brake Program, Umrichterprogramm GSC, Umrichterprogramm MSC, Yaw Bearing Lubrication Program) referencing controller-firmware subroutine numbers CSCM Tool does not itself govern — soft-validated against observed values, not hard-enforced lookups, since the descriptive meaning of each code is not available from source data (06-data-dictionary.md §2c).

Readiness Probe — `GET /api/v1/health/ready`: database reachable, schema at the expected Alembic head, newest audit entries chain correctly, last verified backup recent. 200 with status `ok` or `degraded`, 503 when not ready (FR-152).

Reconciliation Report — The migration progress of the legacy library: rows imported, in review, approved, released, retired and — given the legacy master list — the legacy numbers still missing (FR-048l).

Release — A named, versioned, immutable bundle of Approved status code revisions, published by the Administrator. Since v4 it may target one or more controller software versions and has generated release notes and a signed manifest on every export.

Release Notes — A generated document listing the codes a Release added, changed (with the field-level diff) and deprecated compared with the previous published Release (FR-105).

Released — Lifecycle state reached when a revision is included in a published Release; authoritative and citable.

Remedy Text — An optional field with the recommended action for service staff, on a revision and per translation (FR-131).

Reserved Identifier — The number an imported `StCd_.st` program already carries, held for the Draft created from it (`status_code.reserved_identifier`). It is issued unchanged at Submit for Review; the allocator skips it, no override can take it and no other code can be given it (BR-023).

Review Checklist — Administrator-defined items, per Functional System Group or for all, that an approving Reviewer must confirm; the confirmed wording is stored with the sign-off (FR-128).

Reviewer — System role. Organizationally the Sub-PO for Status Codes. One of the two mandatory Review sign-offs, alongside the Administrator.

Review — Lifecycle state entered on submission. Requires two independent sign-offs (Reviewer and Administrator) before automatically advancing to PendingApproval. Either sign-off may reject back to Draft with a mandatory comment.

Row Version — A counter on each revision that increases on every save. A save based on an older value is refused with 409 and nothing is overwritten (FR-090, BR-019).

Sandbox Status Code — A test-only status code created by an Administrator, drawn from an isolated `StCd-T#####` identifier band outside the real Functional System Group ranges. Never included in a Release, catalogue export, or manual. The only category of record that may be permanently deleted (05-governance-handbook.md §12).

Segregation of Duties — The rule that no single person may both author and decide on the same Change Request at any of the three decision points (Reviewer sign-off, Administrator sign-off, Chief Engineer approval).

SLA (Service Level) — How long a change request may wait at its stage, in business days, before it is shown amber and then red; thresholds are Administrator settings (defaults review 3 / 5, approval 2 / 3) (FR-120).

Status Code — The governed identity record for a wind turbine controller status/fault/warning condition, identified by its Status Code Identifier (StCd).

Status Code Identifier (StCd) — The five-digit, zero-padded human identifier, e.g. `StCd-01232`. The numeric value is the code's position within its Functional System Group's range (e.g. 1000–1999 for Converter & Grid Interface), so the identifier itself encodes classification. Fixed only at the Draft -> Review transition, never before.

StCd Owner — (v4.3) System role. The owner of the status code library: reads the whole library and the workflow without acting, and may change a user's role between Viewer, Engineer and Reviewer — nothing else about a person, and never a higher role. Creating users and all other role changes stay with the Administrator (BR-024). Shown purple; the colour of a role badge is its level, red is the highest (the Administrator).

Status Code Revision — A specific versioned set of attribute values for a Status Code, carrying its own lifecycle state. Field set corrected 2026-09-17 against a real 897-row library export — see 06-data-dictionary.md's provenance note for the full list of corrections (Status Category is Error/Warning/Info; Available Group is numeric; Alarm is boolean; Set/Reset Delay are free text with unit; Manual and Automatic Reset are separate fields; eight Audience Access fields replace the single Access Rights enum; eight numeric Program Reference fields; repeated-error escalation fields; Logical Node/Node Value derived fields).

StCd Program — One `PROGRAM StCd_` of the generated controller code, written in IEC 61131-3 Structured Text: a status code with its number, name, delays, repeat rule, alarm and availability settings, subsystem programs and user-group levels. The status code import reads these programs as text, without running them (FR-048o).

Step-up — Re-entering the actor's own password for a sensitive action — Reviewer or Administrator sign-off, Chief Engineer decision, publishing a Release, the full-database export. A missing password gives `step_up_required`, a wrong one `step_up_failed`; both are audited (FR-126, SEC-026, decision D-07).

Translation — The title, description and remedy text of a revision in a language other than the English source, with a status of MISSING, DRAFT or APPROVED; approved by someone other than the translator (FR-110, BR-018, decision D-03).

Turbine Platform — A multi-select classification (minimum seed values `2XM`, `3XM`, `4XM`, extensible) indicating which turbine platform(s) a status code revision applies to. See Platform Variant Link for cross-platform divergence handling.

Update Mode — An import mode keyed on `legacy_reference_number`: a changed row revises the open Draft in place or opens a new revision of a Released code, an unchanged row is skipped, an unmatched row is created (FR-048k).

Viewer — System role. Read-only consumer of Approved/Released/Deprecated/Archived data.

WCNV / WGEN / WNAC / WROT / WTOW / WTRF / WTRM / WYAW / WTUR / WPPD — Four-letter prefixes identifying each Functional System Group (Converter & Grid Interface, Generator System, Meteorology & Nacelle Environment, Rotor & Pitch System, Tower & Structure, Transformer & MV System, Drivetrain & Gearbox, Yaw System, Turbine Control & Operation, and Wind Farm / Plant Dispatch respectively). `WROT` and `WTUR` each also identify a second, Hub Controller group with a different numeric range (12000–12999 and 11000–11999 respectively, added 2026-09-17) —

see Functional System Group. Displayed alongside the StCd identifier for cross-reference, not part of the identifier itself.

Webhook — A registered internal URL that receives a signed POST when a Release is published. Targets must be private or loopback addresses, redirects are not followed, a failure never affects the publish (FR-146, SEC-028).

Where-used — The list, on a code's detail page, of other codes whose current revision refers to it (repeated-error pointer, related codes, supersession, platform variant); deprecating a code with live dependants requires acknowledging them (FR-092).

20. Traceability Matrix — Spec vs. Code vs. Tests

Reviewed: 2026-10-23 against branch `dev` (code in `cscm_tool/`), baseline v4.0. **607 automated tests pass** (598 functional + 9 load tests) in every final verification run (§7), in three different ways: file order, reversed and shuffled order, and the whole suite on a real SQLite file; line coverage is 92% overall; `black`, `ruff`, `mypy` and `bandit` are clean; `pip-audit` reports no vulnerability in any runtime dependency. The full per-test results, measurements, requirement accounting (§6a) and run history are in the generated 24-test-report.md.

Legend: OK implemented **and** covered by an automated test · - out of MVP scope by design · (manual) needs a human or the production environment (cannot be proven by the test suite — listed in §6).

A OK means "an automated test exercises it", not "a person signed it off". Test paths are under `cscm_tool/tests/`.

1. Summary

Area	OK	-	(manual)
Functional requirements (FR) — 130	130	0	0
Business rules (BR) — 23	23	0	0
Security (SEC) — 24	24	0	0 (TLS termination and key custody are deployment checks, §6)
Audit (AUD) — 8	8	0	0
Non-functional (NFR) — 14	12	1 (NFR-002 availability)	1 (NFR-007 usability needs UAT)
Test matrix (every TC in 13-test-strategy.md §8 and §8a, TC-001 … TC-070)	all	0	TC-030: scripted OK, human sign-off (manual)

1a. The v4 pass (2026-10-23)

The improvement pack of 21-improvement-ideas.md was implemented, specified (03-srs.md §1.10–§1.15), documented and verified three times (13-test-strategy.md §11). Rows for every new requirement are in §3a and §4; the new test cases are TC-038 … TC-052. **Defects found by the verification itself, all fixed and covered:**

1. **Fuzzy search crashed for queries with several words** (`fts5: syntax error near "("`): each word that expands to a group of close spellings must be joined with `AND`; found by the 20,000-revision load test, now `test_search_and_help::test_several_misspelt_words_are_all_tolerated_together`.
2. **A test assumed an in-memory suite database** and failed when the whole suite ran on a file (`test_backup_needs_a_file_database`); found by methodology 3.
3. **Test isolation:** a shared instance directory let one test's audit-chain anchor make another test's audit log look truncated; every test now has its own instance directory.
4. **Admin pages missed accessibility basics** (table header scope, unlabelled inputs) — found by `test_accessibility`.
5. **Hub sub-bands did not follow the handbook** (they counted from x001, e.g. `12-3xx` = 12301–12400 and `11-0xx` = 11000–11100; later renamed `123xx` and `110xx`, and `099xx-a` … `-f` became `0990x` … `0999x`, migration `f6c8e0a2b459`): aligned to the hundreds by migration `a1d4f6b8c203`, `test_schema_migration::test_hub_sub_bands_are_aligned_to_the_hundreds_of_their_prefix`, `test_schema_upgrade::test_hub_sub_bands_are_shifted_to_the_hundreds_by_the_upgrade`.
6. **Documentation errors** found by `test_docs_consistency` and two reading passes: dangling section references (a handbook §2a that did not exist), two wrong UI labels in the manual, a stale per-module test count, and statements about the old behaviour that the code had outgrown (for example the claim that no `controller_type` column exists).

2. Changes made in the v3 pass (2026-10-21)

The review found 10 missing features and 8 defects; all are fixed and tested.

Features added — restricted audience-access confirm/override at Administrator sign-off (FR-025a/034, BR-012, SEC-013); revision diff API + CR/history UI (FR-032/064); Administrator identifier override with reason (FR-021a); Reviewer assignment (FR-039); "all Released" and labelled working-set exports + XLSX in the UI (FR-045/046); domain sign-off requests surfaced on Dashboard / My Change Requests (FR-028); change-request comments endpoint + UI (FR-038); pre-publish Release delete (BR-008); `IMPORT_*`, `SUBGROUP_CREATE`, `DOMAIN_SIGNOFF_REQUEST`, `REVIEWER_ASSIGN`, `IDENTIFIER_OVERRIDE`, `RELEASE_DELETE` audit events; lookup `description` inline help

(NFR-007b); default role Viewer (SEC-012); append-only audit triggers (FR-072).

Defects found by the new tests and fixed

1. A CR rejected back to Draft and resubmitted — and every new revision of an already-numbered code — **allocated a fresh identifier**, violating BR-001/FR-021. Now allocated once.
2. Every read took the SQLite write lock (`BEGIN IMMEDIATE` on all transactions), so 50 concurrent users produced `database is locked`. Reads now use a deferred transaction (WAL read concurrency); only writers take the lock (NFR-004).
3. `lifecycle_status` could be set to any value by any code path; the 7-state machine is now enforced centrally (`app/domain/lifecycle.py`, BR-004).
4. Role checks lived only in route decorators; every role-restricted service entry point now enforces them too (`app/domain/authz.py`, SEC-010).
5. Comments (FR-038) had a service method but no endpoint or screen.
6. Nine implemented API operations were missing from `openapi.yaml`, and four response schemas under-declared fields; the contract and code now match and a test keeps them in sync.
7. Response security headers / CSP absent (SEC-025; earlier in this branch).
8. Three runtime dependencies had published vulnerabilities (Flask 3.0.3, waitress 3.0.0, python-dotenv 1.0.1) — upgraded to 3.1.3 / 3.0.2 / 1.2.2.

Second verification pass (same day) — re-testing in detail, with the permission grid corrected to the API specification, found six more defects, all fixed and covered:

1. **Viewers could read change requests** (`GET /change-requests`, `/change-requests/{id}` and the matching UI pages), exposing Draft and in-review content that BR-009 reserves for the four working roles. The first grid had encoded the wrong expectation.
2. **A Released code vanished for Viewers** (detail view and search) as soon as anyone opened a new Draft revision of it, because the "current" revision was the invisible Draft. Viewers now always see the latest Approved/Released/Deprecated/Archived revision.
3. `GET /status-codes/next-available` was open to every role; the specification limits it to Engineers and Administrators.
4. A production instance would start with the built-in development `SECRET_KEY`, making session cookies forgeable. It now refuses to start with the default, the `.env.example` placeholder, or any key shorter than 32 characters.
5. The password policy of `11-security-architecture.md` §4 lacked the **last-5 reuse block** (new `user_password_history` table, migration `c5e7a9b1d3f2`) and had a 7-entry block-list; the list is now a data file (`seed_data/common_passwords.txt`) that can be replaced with an internal breach corpus.
6. The **structured JSON logging** of §11 did not exist; request and error logs now carry the request id and never credentials or query strings. (Running a migration also used to silence all existing loggers; fixed.)
7. **Exports issued ~13 SQL queries per row** (lazy-loaded relationships): a 5,000-row CSV took ~4 s. Eager loading brings it to ~1 s with a constant number of queries, and a test now guards it.
8. **Review comments were not traceable**: audit entries for decisions and comments carried no text, withdrawing a change request deleted its discussion, and the status-code page showed none of it. Audit entries now carry the comment id and text, `WITHDRAW` keeps a snapshot, a Decisions & Comments tab lists everything per code (working roles only), and the Audit Log can be filtered by action.
9. Export file names now carry the generation time (`cscm__YYYY-MM-DD_HH-MM.`, UTC) — `test_exports_and_diff::test_export_file_names_carry_a_timestamp`.
10. **Subgroup bands:** the first subgroup of each Main Controller group now covers the group's first 200 numbers (e.g. Tower 5000–5199) via schema seed and migration `d6f8b0c2e4a3` — `test_schema_migration::test_first_subgroup_of_every_main_group_covers_the_first_200_numbers`; the unseeded placeholder group `WNEW` was removed from the dev database (`test_no_placeholder_group_is_seeded`).

Also: `record_property` measurements are now carried into the report (`junit_family = xunit1`), and the report generator itself cross-checks that every test cited in this matrix still exists.

Schema — baseline `schema.sql` plus follow-on migration `b4d2f6a8c0e1` (idempotent, reversible, tested): `change_request.assigned_reviewer_id`, `domain_signoff_request`, `trg_audit_no_update/delete`, `description` on all 13 lookup tables. **Existing databases must run** `flask db upgrade`.

3. Functional requirements

ID	Status	Evidence	
FR-001	OK	test_auth::test_login_success..., test_wrong_password...; test_injection_and_session::test_session_cookie_attributes	
FR-002	OK	test_injection_and_session::test_idle_session_expires_after_thirty_minutes, ...recent_activity_keeps...	
FR-003	OK	test_auth::test_logout_invalidates_session	
FR-004	OK	test_auth::test_five_failed_attempts_locks_the_account; unit/test_rate_limit.py; test_governance_extras::test_account_lock_and_password_events_are_audited	
FR-005	OK	test_governance_extras::test_auth_provider_seam_is_substitutable (SSO itself -, Phase 2)	
FR-010	OK	test_users::test_admin_can_list_and_create_users; test_governance_extras::test_user_administration_events_are_audited	
FR-011	OK	test_governance_extras::test_a_user_holds_exactly_one_role, ...user_administration_events... (ROLE_CHANGE)	
FR-011a	OK	...user_administration_events... (DOMAIN_TAG_CHANGE); domain sign-off tests	
FR-012	OK	test_users::test_cannot_deactivate_last_active_administrator	
FR-013	OK	test_users::test_self_password_change; unit/test_password.py	
FR-014	OK	test_users::test_admin_can_force_reset...; test_auth::test_password_reset_invalidates...	
FR-020	OK	test_status_codes::test_engineer_can_create_draft..., ...rejects_missing_functional_subgroup	
FR-021	OK	test_numbering.py; test_governance_extras::test_next_available_preview_does_not_reserve_a_number; test_invariants::test_identifier_never_changes..., ...new_revision_of_a_released_code_keeps_its_identifier	
FR-021a	OK	test_workflow_gaps::test_admin_can_override_identifier..., ...requires_reason, ...forbidden_for_engineer, ...rejects_out_of_range_malformed_and_duplicate	
FR-022	OK	unit/test_validation.py; test_status_codes::test_create_rejects_invalid_fields; test_injection_and_session::test_server_side_validation...	
FR-023	OK	test_invariants::test_duplicate_identifier_rejected_by_database	
FR-023a	OK	test_invariants::test_submit_requires_a_turbine_platform	
FR-024	OK	test_invariants::test_draft_can_be_saved_repeatedly...	
FR-025	OK	test_invariants::test_in_flight_revision_cannot_be_edited_in_place; copy-on-write in test_deprecation	
FR-025a	OK	test_workflow_gaps::test_admin_signoff_can_override_proposed_audience_access; test_ui_screens::test_ui_comment_reviewer_assignment_and_override_flow	
FR-026	OK	test_change_requests::test_withdraw_*	
FR-027	OK	test_change_requests::test_submit_blocked_until_domain_signoff_complete	
FR-028	OK	test_workflow_gaps::test_domain_signoff_request_* (4 tests); test_ui_screens::test_sign_off_request_listed_on_my_change_requests; test_uat::test_uat_05...	
FR-029	OK	test_change_requests::test_domain_signoff_rejects_user_without_that_domain	
FR-030	OK	CR created for every draft/revision/import/deprecation (test_import, test_deprecation)	
FR-031	OK	test_uat::test_uat_01... (review queue is the notification; see §5 note)	
FR-032	OK	test_exports_and_diff::test_revision_diff_endpoint_and_cr_baseline_diff, ...new_code_diff_is_marked_new, ...diff_card_renders...	
FR-033	OK	test_change_requests::test_reviewer_rejection_without_comment..., ...rejection_returns_to_draft...	
FR-034	OK	test_change_requests admin paths; test_workflow_gaps override tests	
FR-035	OK	test_change_requests::test_dual_signoff_in_either_order...; test_uat::test_uat_02...	
FR-036	OK	test_change_requests::test_full_happy_path_to_approved; test_uat::test_uat_03...	
FR-037	OK	test_change_requests::test_author_cannot_sign_off_own_cr, ...chief_engineer_cannot_be_a_prior_signer; test_uat::test_uat_09*	
FR-038	OK	test_decision_trail::test_audit_entries_carry_the_comment_text_and_id, ...status_code_page_lists_all_decisions_and_comments, ...viewers_cannot_see_the_review_discussion, ...withdrawing_keeps_the_discussion_in_the_audit_log;	test_ui_screens::test_comments_api_validates_and_is_role_restricted; test_governance_extras::test_comment_leaves_state_and_signoffs_untouched

ID	Status	Evidence
FR-039	OK	test_workflow_gaps::test_assigned_reviewer_only_may_sign_off, ...reassigned_and_unassigned, ...validation_and_authorization, ...no_reassignment_after...
FR-040	OK	test_releases::test_full_release_lifecycle, ...non_admin_cannot_create_release
FR-041	OK	lifecycle test; test_uat::test_uat_06...
FR-042	OK	test_uat::test_uat_06... (excluded candidate stays Approved)
FR-043	OK	test_releases::test_full_release_lifecycle
FR-044	OK	test_releases::test_cannot_change_items_after_publish
FR-045	OK	test_exports_and_diff::test_released_export_*, test_xlsx_export_has_same_rows_as_csv, test_json_export_is_flat_rows; test_releases::test_csv_export_of_a_release
FR-046	OK	test_exports_and_diff::test_working_set_* (4 tests)
FR-047	OK	test_releases::test_full_database_export_requires_administrator; test_uat::test_uat_10...
FR-048	OK	test_releases::test_sandbox_revision_rejected_from_release_items + trg_prevent_sandbox_release
FR-048a	OK	test_import::test_non_admin_engineer_can_upload_import, ...viewer_cannot..., ...unsupported_file_extension...
FR-048b	OK	test_import::test_invalid_row_is_reported_and_creates_nothing; test_uat::test_uat_11...
FR-048c	OK	test_import::test_imported_row_becomes_import_type_draft_change_request
FR-048d	OK	test_governance_extras::test_import_preserves_legacy_reference_number
FR-048e	OK	test_lookup_admin::test_create_functional_subgroup_with_range
FR-048f	OK	test_lookup_admin::test_functional_subgroup_overlap_rejected
FR-048g	OK	test_lookup_admin::test_admin_can_create_new_functional_system_group_in_free_rang e; test_uat::test_uat_12...
FR-048h	OK	test_lookup_admin::test_new_functional_system_group_range_overlap_rejected
FR-050	OK	test_deprecation::test_full_deprecation_cycle
FR-051	OK	test_deprecation::test_archive_*
FR-052	OK	test_schema_migration::test_only_sandbox_status_codes_can_be_deleted; test_status_codes::test_sandbox_delete_requires_admin_and_sandbox_flag
FR-053	OK	test_status_codes::test_non_admin_cannot_create_sandbox, ...admin_sandbox_gets_identifier_immediately
FR-054	OK	test_status_codes::test_search_defaults_exclude_sandbox; test_exports_and_diff::test_released_export_contains_only..., test_working_set_is_labelled_and_excludes_sandbox
FR-055	OK	test_status_codes::test_sandbox_delete_requires_admin_and_sandbox_flag
FR-056	OK	test_audit_coverage::test_sandbox_delete_keeps_full_snapshot_and_creation_entry
FR-060	OK	test_invariants::test_full_text_search_matches_identifier_title_and_description
FR-061	OK	test_invariants::test_filters_sort_and_pagination
FR-062	OK	same (ascending and descending sort)
FR-063	OK	same (page, page_size, cap at 200)
FR-064	OK	test_invariants::test_revision_history_visible_to_viewer_only_once_decided; test_exports_and_diff::test_diff_hidden_from_viewers...
FR-070	OK	test_audit_coverage::test_each_lifecycle_action_is_audited_exactly_once, ...import_lifecycle_is_audited, ...subgroup_creation..., ...user_and_auth_events...; all 40 catalogue actions are emitted
FR-071	OK	test_audit_log.py; test_decision_trail::test_audit_log_can_be_filtered_by_action_and_actor
FR-072	OK	test_audit_coverage::test_audit_log_entries_cannot_be_updated_or_deleted (DB triggers)
FR-073	OK	test_audit_coverage::test_sandbox_delete_keeps_full_snapshot...
FR-080	OK	test_openapi_conformance::test_every_documented_operation_is_implemented_and_vi ce_versa
FR-081	OK	test_permission_grid.py (5 roles × 46 API and 20 UI routes, plus anonymous access; expectations taken from the role column of 09-api-specification.md §7)
FR-082	OK	same pagination/filter/sort params as UI (test_filters_sort_and_pagination)
FR-083	OK	test_governance_extras::test_whole_lifecycle_works_with_all_outbound_network_blocke d; test_no_cloud.py

3a. Functional requirements added in v4 (03-srs.md §1.10–§1.15)

ID	Status	Evidence
FR-048i	OK	test_import_toolkit::test_template_csv_and_xlsx_carry_the_exact_headers_and_lookup_codes, …the_downloaded_template_imports_cleanly_as_is
FR-048j	OK	test_import_toolkit::test_dry_run_reports_every_problem_and_creates_nothing, …dry_run_warns_about_a_possible_duplicate…, …missing_required_columns_are_named_up_front
FR-048k	OK	test_import_toolkit::test_update_mode_revises_the_open_draft_in_place, …skips_unchanged_rows_and_creates_missing_ones, …opens_a_new_revision_of_a_released_code, …update_mode_errors_are_specific, …update_of_someone_elses_draft…
FR-048l	OK	test_import_toolkit::test_reconciliation_counts_states_and_names_the_missing_numbers
FR-048m	OK	test_import_toolkit::test_error_report_has_row_column_value_and_a_fix_hint_as_csv_and_xlsx, …formula_like_values_in_the_error_report_are_neutralised
FR-048n	OK	test_import_toolkit::test_a_big_file_is_processed_in_the_background_with_progress, …files_above_the_threshold_go_to_the_background_automatically
FR-048o	OK	test_st_import::test_the_reader_takes_the_attributes_of_a_main_controller_program, test_st_import::test_a_zip_and_loose_files_are_read_alike_and_other_files_are_noted, test_st_import::test_a_file_without_programs_or_too_large_is_refused_cleanly, test_st_import::test_nothing_usable_is_a_clear_error
FR-048p	OK	test_st_import::test_the_preview_lists_every_program_and_creates_nothing, test_st_import::test_confirming_without_a_selection_takes_every_ready_program, test_st_import::test_the_controller_can_be_forced_for_a_file
FR-048q	OK	test_st_import::test_submitting_issues_exactly_the_reserved_number, test_st_import::test_an_administrator_cannot_override_a_reserved_identifier, test_st_import::test_allocation_skips_a_reserved_number_and_a_manual_override_cannot_take_it, test_st_import::test_the_database_refuses_a_second_reservation_of_a_number
FR-048r	OK	test_st_import::test_the_preview_lists_every_program_and_creates_nothing, test_st_import::test_the_controller_can_be_forced_for_a_file
FR-048s	OK	test_st_import::test_a_hub_program_keeps_its_legacy_number_and_gets_the_new_one_from_the_mapping, test_st_import::test_a_legacy_hub_number_missing_from_the_mapping_leaves_no_identifier, test_st_import::test_the_command_line_reads_the_hub_numbers_of_the_seed_workbook, test_st_import::test_the_new_number_of_a_hub_program_is_the_one_of_its_mapping_row
FR-048t	OK	test_st_import::test_confirm_creates_drafts_that_reserve_their_numbers_in_one_change_set, test_st_import::test_the_draft_carries_the_attributes_of_the_program
FR-048u	OK	test_st_import::test_the_mapping_lists_legacy_to_new_and_counts_codes_without_a_legacy_number, test_st_import::test_the_mapping_exports_as_csv_and_xlsx, test_st_import::test_the_mapping_is_not_for_viewers
FR-048v	OK	test_st_import::test_the_source_of_an_imported_code_can_be_read, test_st_import::test_confirm_creates_drafts_that_reserve_their_numbers_in_one_change_set, test_st_import::test_a_used_prg_idx_is_a_warning_and_is_not_stored_twice
FR-048w	OK	test_st_import::test_a_second_preview_shows_what_is_already_there_and_confirm_refuses_it, test_st_import::test_a_legacy_number_that_belongs_to_another_code_blocks_the_program, test_st_import::test_a_number_defined_twice_in_one_upload_blocks_the_second
FR-048x	OK	test_withdraw_and_reimport::test_a_changed_program_updates_the_open_draft_when_it_is_ticked, …a_changed_program_opens_a_new_revision_of_an_approved_code, …the_preview_screen_lists_the_choices_and_leaves_changed_programs_unticked, …a_withdrawn_code_that_has_its_identifier_keeps_it_and_can_be_restored
FR-090	OK	test_content_integrity::test_stale_save_is_refused_with_the_current_values, …expected_updated_at_is_enforced_too, …save_without_a_version_still_works…, …ui_stale_edit_keeps_the_users_entries…
FR-091	OK	test_content_integrity::test_unknown_escalation_number_warns_but_does_not_block, …reference_to_a_deprecated_code_warns, …related_codes_are_validated_and_checked
FR-092	OK	test_content_integrity::test_where_used_lists_dependants_and_deprecation_needs_acknowledgement
FR-093	OK	test_content_integrity::test_similar_title_in_the_subgroup_is_flagged_not_blocked, …similar_endpoint_supports_the_did_you_mean_hint; test_import_toolkit::test_similar_titles_are_reported_as_warnings
FR-094	OK	test_content_integrity::test_platform_variant_copies_the_code_and_links_the_family, …variant_needs_a_decided_source_and_an_engineer
FR-100	OK	test_publication::test_three_profiles_are_seeded_and_shape_the_columns, …administrator_manages_profiles_as_data, …export_profile_fields_match_the_export_row
FR-101	OK	test_publication::test_pdf_cells_are_wrapped_not_truncated, …pdf_survives_characters_outside_latin_1, …pdf_column_widths_fit_the_page
FR-102	OK	test_publication::test_audience_export_only_contains_codes_the_audience_may_see, …covert_visibility_is_data_an_administrator_can_flip, …customer_profile_applies_its_default_audience…

ID	Status	Evidence
FR-103	OK	test_publication::test_bundle_contains_the_file_and_a_manifest_that_matches, …signed_manifest_detects_edits_to_the_manifest_itself, …export_history_lists_hashes_per_release
FR-104	OK	test_publication::test_verify_recognises_an_untouched_file_and_a_tampered_one, …verify_export_cli_works_without_the_database
FR-105	OK	test_publication::test_release_notes_list_added_changed_and_deprecated, …unpublished_release_notes_are_an_administrator_preview
FR-106	OK	test_publication::test_version_range_must_be_ordered_and_known, …release_targets_limit_the_candidates_and_freeze_at_publish
FR-107	OK	test_publication::test_applicability_matrix_and_changes_between_versions
FR-108	OK	test_publication::test_offline_bundle_is_one_self_contained_page_with_the_data, …offline_bundle_respects_audiences_and_needs_a_published_release
FR-109	OK	test_content_integrity::test_planned_dates_are_administrator_set_audited_and_warn_at_publish
FR-110	OK	test_publication::test_translation_workflow_with_light_review_and_language_exports, …translation_validation_worklist_and_carry_over, …draft_source_text_cannot_be_translated_yet
FR-111	OK	same (language exports use approved text; strict mode)
FR-120	OK	test_sla_and_notifications::test_business_days_skip_weekends, …levels_follow_the_thresholds, …queues_show_the_waiting_time, …thresholds_are_configurable_and_validated
FR-121	OK	test_sla_and_notifications::test_reminders_go_to_whoever_holds_the_next_decision_once_a_day, …reminders_also_reach_a_covering_delegate_and_can_be_mailed, …mail_failure_never_breaks_the_reminder_job, …send_reminders_cli
FR-122	OK	test_sla_and_notifications::test_stalled_view_and_reassignment
FR-123	OK	test_sla_and_notifications::test_kpis_are_computed_from_stored_timestamps, …kpis_flag_an_incomplete_audit_trail…; test_integrations_and_reports::test_kpis_have_the_documented_shape
FR-124	OK	test_review_aids::test_change_set_groups_requests_and_bulk_decisions_stay_individual, …bulk_decision_skips_what_the_actor_may_not_decide, …bulk_decision_needs_the_password_once…, …change_set_rules
FR-125	OK	test_step_up_and_delegation::test_delegate_can_sign_off_as_the_absent_reviewer_and_audit_shows_both, …delegation_is_limited_to_its_period_and_can_be_revoked, …delegation_validation_and_who_may_create_it, …delegating_does_not_grant_other_privileges, …deactivated_delegator_ends_the_delegation
FR-126	OK	test_step_up_and_delegation::test_reviewer_signoff_without_password_is_refused, …wrong_password_is_refused_and_audited, …admin_signoff_chief_decision_publish_and_full_export_all_need_the_password, …ui_decision_forms_carry_a_password_field
FR-127	OK	test_review_aids::test_reviewer_comments_on_a_field_and_the_author_sees_and_resolves_it, …field_comment_rules, …only_the_commenter_author_or_admin_can_resolve
FR-128	OK	test_review_aids::test_checklist_must_be_confirmed_to_approve_and_is_recorded, …checklist_is_administrator_managed…, …rejection_does_not_need_the_checklist
FR-129	OK	test_review_aids::test_least_loaded_reviewer_is_suggested_and_assigned, …out_of_office_reviewers_are_not_auto_assigned, …auto_assignment_on_submit_is_a_setting
FR-130	OK	test_review_aids::test_author_attaches_evidence_and_reviewers_download_it, …attachment_type_size_and_content_checks, …attachment_permissions_and_removal_rules, …removing_a_draft_attachment_deletes_the_file…
FR-131	OK	test_content_integrity::test_troubleshooting_fields_round_trip_and_reach_the_export, …remedy_text_length_is_validated
FR-140	OK	test_integrations_and_reports::test_health_report_covers_capacity_missing_data_and_inconsistencies, …health_report_finds_an_info_code_with_an_alarm, …reports_are_limited_to_the_right_roles
FR-141	OK	test_integrations_and_reports::test_health_report_flags_a_nearly_full_band, …the_capacity_table_colours_main_blue_and_hub_light_blue; test_content_integrity::test_next_available_reports_how_many_numbers_are_left
FR-142	OK	test_integrations_and_reports::test_saved_searches_round_trip_through_the_url, …saving_under_an_existing_name…, …saved_searches_are_private_to_their_owner, …saved_search_ui_save_and_delete
FR-143	OK	test_integrations_and_reports::test_occurrence_report_names_codes_that_never_fire_and_those_that_fire_most, …occurrence_import_validation_and_permissions
FR-144	OK	test_integrations_and_reports::test_a_token_is_shown_once_and_stored_only_as_a_hash, …a_token_reads_released_data_only_and_never_writes, …a_token_only_works_for_its_scope, …revoked_expired_unknown_and_orphaned_tokens_are_refused
FR-145	OK	test_integrations_and_reports::test_the_feed_lists_releases_in_order_with_a_cursor_and_an_etag, …the_feed_needs_authentication, …sandbox_activity_never_appears_in_the_feed
FR-146	OK	test_integrations_and_reports::test_a_publish_calls_the_webhook_with_a_verifiable_signature, …failing_webhook_is_recorded_and_never_blocks_the_publish, …redirects_are_not_followed, …webhook_targets_must_be_internal…

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FR-147	OK	test_integrations_and_reports::test_a_hub_code_is_carried_over_as_a_main_controller_draft, …the_whole_consolidation_loop_ends_with_a_deprecation_request, …bulk_deprecation_is_for_administrators_and_hub_groups
FR-148	OK	test_search_and_help::test_prefix_and_multi_word_search, …a_hit_in_the_title_outranks_a_hit_in_the_description, …typos_still_find_the_code, …without_the_index_search_falls_back_to_substrings
FR-150	OK	test_audit_chain_and_ops::test_a_backup_is_written_verified_and_recorded, …only_the_newest_generations_are_kept, …a_corrupt_backup_fails_verification, …a_restored_backup_runs_the_application, …cli_backup_verify_and_audit_commands
FR-151	OK	test_audit_chain_and_ops::test_an_edited_row_is_found_and_named, …a_removed_row_breaks_the_link, …truncating_the_tail_is_caught_by_the_anchor, …archive_exports_old_rows_with_a_hash_and_deletes_nothing; test_schema_upgrade::test_old_audit_rows_are_chained_and_the_chain_verifies
FR-152	OK	test_audit_chain_and_ops::test_ready_reports_ok_and_degraded_by_backup_state, …ready_is_503_when_the_chain_is_broken_or_the_schema_outdated, …ready_needs_no_login…
FR-153	OK	test_audit_chain_and_ops::test_metrics_are_refused_outside_the_allowed_networks, …metrics_count_requests_and_hold_no_business_data, …requests_are_logged_to_the_configured_file
FR-154	OK	test_search_and_help::test_every_lifecycle_state_role_and_cr_state_has_help, …the_help_page_shows_the_quick_start_of_the_signed_in_role, …state_badges_carry_their_glossary_text_as_a_tooltip, …the_change_request_screen_says_what_happens_next
FR-155	OK	test_i18n::test_every_template_string_has_a_german_entry, …the_language_choice_is_remembered_for_the_user, …main_pages_render_in_each_language…; test_accessibility::test_every_page_is_structurally_accessible
FR-156	OK	test_roles_and_languages::test_the_stcd_owner_moves_users_between_viewer_engineer_and_reviewer, …the_stcd_owner_cannot_reach_the_other_roles, …the_stcd_owner_changes_nothing_but_the_role, …the_user_page_colours_every_role_and_red_is_the_administrator, …the_stcd_owner_sees_the_list_without_the_administrator_forms; test_permission_grid::test_api_permission_grid, …test_ui_permission_grid
FR-157	OK	test_roles_and_languages::test_the_language_names_have_the_country_letters_left_and_the_flag_right, …the_menu_shows_letters_left_and_a_drawn_flag_right_for_every_language, …arabic_is_right_to_left_but_identifiers_codes_and_numbers_stay_as_they_are, …each_added_language_translates_the_login_page_and_falls_back_to_english, …the_choice_of_an_added_language_is_remembered_for_the_user, …a_partly_translated_page_still_shows_every_string, …the_catalogues_agree_and_only_translate_real_strings; test_schema_migration::test_nineteen_languages_seeded_with_country_letters_left_and_the_flag_right; test_schema_upgrade::test_the_table_rebuilds_leave_every_foreign_key_intact
FR-158	OK	test_user_administration::test_an_account_that_was_never_used_can_be_deleted_and_the_deletion_is_audited, …an_account_with_history_is_erased_and_the_audit_log_is_untouched, …a_deactivated_account_can_be_activated_again, …the_user_list_offers_view_edit_deactivate_and_delete_to_an_administrator, …the_detail_page_shows_history_and_the_buttons_follow_it, …the_buttons_deactivate_activate_and_delete_through_the_screens, …nobody_deletes_their_own_account
FR-159	OK	test_withdraw_and_reimport::test_several_change_requests_are_withdrawn_in_one_go_and_each_is_reported, …a_change_request_of_somebody_else_is_refused_and_the_others_go_on, …bulk_withdraw_needs_a_list_and_a_role_that_can_withdraw, …the_list_has_tick_boxes_and_a_withdraw_selected_button
FR-160	OK	test_withdraw_and_reimport::test_a_withdrawn_import_draft_gives_its_number_and_legacy_number_back, …a_withdrawn_code_that_has_its_identifier_keeps_it_and_can_be_restored
FR-161	OK	test_translations_doc::test_the_document_is_exactly_what_the_generator_produces_from_the_code, …every_language_has_its_section_and_its_coverage_row, …string_ids_are_stable_and_never_collide, …the_fingerprint_changes_when_one_translation_changes, …the_review_sheet_of_every_language_is_complete, …right_to_left_languages_get_a_right_to_left_translation_column, …the_command_line_writes_one_sheet_per_language_and_refuses_unknown_ones, …the_concept_names_the_layers_the_checks_and_the_open_items
FR-162	OK	test_email_and_windows_signin::test_an_address_is_stored_trimmed_and_lower_case_and_must_be_valid_and_new, …the_address_of_a_user_cannot_be_changed, …the_database_refuses_a_changed_address_too, …the_screens_show_the_address_as_fixed
FR-163	OK	test_email_and_windows_signin::test_the_gateways_identity_signs_the_matching_account_in_and_is_audited, …no_identity_an_unknown_one_and_a_deactivated_account_are_all_refused, …a_domain_name_is_matched_through_the_mail_domain_or_the_user_name, …with_windows_only_passwords_are_refused_and_the_login_page_says_so, …with_both_modes_the_login_page_offers_the_button_and_the_form, …the_simulation_page_stands_in_for_the_gateway_and_signs_in_by_address, …the_simulation_refuses_an_unknown_identity_and_is_off_by_default, …the_development_defaults_are_the_simulation_next_to_the_password, …the_shipped_example_configuration_documents_the_settings

ID	Status	Evidence
FR-164	OK	test_translation_review::test_the_overview_shows_a_reviewer_only_their_languages_and_a_manager_all, …only_an_administrator_or_an_stcd_owner_assigns_reviewers_and_it_is_audited, …an_unassigned_user_cannot_open_a_review_and_an_unknown_language_is_404, …a_reviewer_answers_strings_and_is_told_what_is_wrong, …a_review_is_completed_only_when_every_string_has_a_verdict_and_is_recorded, …an_accepted_proposal_is_shown_at_once_and_can_be_taken_back, …nobody_decides_on_their_own_proposal_and_a_decision_is_final, …a_proposal_must_keep_the_placeholders_and_an_unclear_string_is_only_noted, …a_missing_translation_can_be_proposed_and_accepted, …a_completed_review_shows_as_changed_after_a_later_acceptance, …the_sheet_can_be_downloaded_filled_in_offline_and_uploaded, …a_sheet_that_is_not_one_is_refused_clearly, …right_to_left_languages_are_shown_right_to_left_in_the_review
FR-165	OK	test_pdf_scripts::test_a_script_gets_its_own_font, …latin_text_needs_no_script_font_and_arabic_is_told_from_urdu, …every_script_builds_a_pdf_with_its_font, …the_title_page_and_notice_may_carry_any_script, …shaping_is_switched_on_per_cell_only_where_a_script_needs_it, …every_character_of_every_translation_is_drawn_by_a_bundled_font, …the_fonts_and_their_licence_are_bundled_and_shaping_is_available, …a_large_export_stays_fast_for_latin_text; test_publication::test_pdf_survives_characters_no_font_has
FR-166	OK	test_terminology::test_the_vocabulary_ships_with_every_term_in_every_language_as_a_draft, …the_shipped_terms_agree_with_the_interface_catalogue_where_both_have_the_word, …terms_that_are_never_translated_have_no_translation, …everybody_reads_but_only_the_technical_roles_propose, …an_interface_reviewer_has_no_say_in_the_terminology, …a_proposal_is_a_draft_until_another_person_approves_it, …a_shipped_draft_can_be_approved_by_anybody_who_may_approve_and_all_at_once, …send_back_needs_a_note_and_reopens_an_approved_translation, …validation_of_a_translation, …only_the_managers_add_and_change_terms_and_a_new_english_reopens_approvals, …the_check_finds_interface_strings_that_do_not_use_the_term, …the_vocabulary_is_exported_per_language_and_for_all, …a_formula_in_a_cell_is_neutralised_in_the_csv
FR-167	OK	test_i18n::test_every_language_translates_the_whole_interface_inventory, …every_template_string_has_a_german_entry, …placeholders_survive_translation; test_translations_doc::test_every_language_has_its_section_and_its_coverage_row
FR-168	OK	test_generated_fields::test_a_main_program_import_keeps_every_field_of_the_program, …the_two_zero_load_attributes_are_not_mixed_up, …a_hub_program_import_keeps_its_hub_fields_and_nothing_of_the_main_ones, …a_field_of_the_other_controller_is_refused_and_the_rest_is_stored, …values_out_of_range_are_refused, …a_new_revision_carries_over_what_the_caller_does_not_send_and_the_diff_shows_changes, …the_form_stores_the_fields_and_the_pages_show_them, …a_rejected_form_keeps_what_was_typed, …the_levels_and_programs_are_internal_and_never_leave_in_an_audience_export, …withdrawing_and_deleting_a_code_removes_its_levels
FR-169	OK	test_menu::test_a_viewer_sees_the_catalogue_and_nothing_to_work_with, …each_role_sees_the_links_of_its_work_and_no_other, …no_link_appears_twice_and_the_groups_are_the_planned_ones, …only_the_link_of_the_current_page_is_marked, …what_waits_for_the_user_is_counted_in_the_menu, …groups_can_be_closed_and_a_narrow_window_has_a_menu_button, …the_menu_is_translated_and_a_group_title_too
FR-170	OK	test_menu::test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta, …a_review_removes_the_mark_and_a_changed_string_brings_it_back, …the_marker_can_be_switched_off_in_the_settings
FR-171	OK	test_export_language::test_a_pdf_in_french_prints_its_own_words_in_french, …english_and_no_language_leave_the_text_as_it_is, …every_word_around_an_export_exists_in_every_language, …a_placeholder_survives_in_every_language
FR-172	OK	test_menu::test_a_viewer_sees_only_the_reviewed_languages_and_nothing_marked, …a_viewer_cannot_choose_or_keep_a_language_nobody_has_reviewed, …a_viewer_who_reviews_a_language_sees_that_one, …the_login_page_offers_only_the_reviewed_languages, …the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta
FR-173	OK	test_baseline_and_packages::test_the_baseline_of_a_release_holds_every_code_valid_at_that_release, …a_release_can_still_be_exported_as_only_its_own_changes, …the_baseline_narrows_to_a_platform, …the_baseline_narrows_to_a_controller, …the_summary_counts_the_baseline_by_controller_and_platform, …an_unpublished_release_previews_what_its_manual_will_contain, …the_title_page_names_the_scope, …the_readiness_check_shows_what_is_still_missing
FR-174	OK	test_baseline_and_packages::test_a_package_holds_every_language_for_every_access_level, …the_manifest_vouches_for_every_file_and_reports_the_coverage, …a_package_can_be_narrowed_to_a_platform_and_a_controller, …everybody_may_download_what_is_released_and_only_owner_and_admin_what_is_not, …a_package_needs_a_published_release_and_valid_choices, …the_release_page_offers_the_baseline_and_the_package, …the_package_is_audited
FR-175	OK	test_baseline_and_packages::test_everybody_may_download_what_is_released_and_only_owner_and_admin_what_is_not, test_exports_and_diff::test_working_set_not_available_to_viewers_or_anonymous, test_permission_grid

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FR-176	OK	test_translation_approval::test_a_reviewed_language_has_nothing_pending_until_a_string_changes, …a_viewer_reads_a_changed_string_in_english_until_it_is_approved, …a_language_never_reviewed_stays_hidden_from_a_viewer, …a_follow_up_review_covers_only_what_is_new_or_changed, …an_accepted_proposal_is_approved_and_an_undecided_one_is_not, test_translation_review::test_a_proposal_awaits_approval_until_it_is_decided
FR-177	OK	test_translation_approval::test_the_colour_intervals_run_from_red_to_green, …the_overview_shows_the_confidence_of_every_language_in_colour, …the_record_holds_a_confidence_for_every_language_with_its_findings, …every_change_of_the_text_or_the_nomenclature_has_been_through_the_check, …a_changed_string_must_get_a_verdict_before_the_record_is_current, …a_change_of_the_nomenclature_needs_the_flag_as_well, …a_delta_read_one_by_one_can_lower_or_keep_the_confidence
FR-178	OK	test_translation_impact::test_a_new_term_tells_the_stcd_owner_which_languages_need_attention, …the_reviewer_of_an_affected_language_is_told_too, …a_term_that_stays_in_english_touches_no_language, …a_changed_term_and_an_approved_translation_are_reported_again, …a_failing_notice_never_undoes_the_change, …the_notice_appears_on_the_notices_page_of_the_stcd_owner, …nothing_is_reported_while_the_confidence_check_is_current, …changed_shipped_text_is_reported_once_per_change, …a_changed_nomenclature_is_named_in_the_notice, …the_command_for_the_deployment_reports_and_notifies, …the_start_of_the_server_and_the_installers_run_the_check, test_translation_approval::test_every_change_of_the_text_or_the_nomenclature_has_been_through_the_check
FR-179	OK	test_connections::test_a_negated_active_flag_is_a_cause_and_a_plain_one_a_precondition, …the_real_programs_name_their_cause_and_their_repeat_code, …one_code_shows_what_it_depends_on_and_what_depends_on_it, …the_hierarchy_puts_the_cause_on_top_with_what_it_suppresses_below, …escalation_targets_say_which_are_outside_the_catalogue, …what_looks_wrong_is_listed, …the_hub_and_main_controller_can_be_looked_at_separately, …the_code_page_and_the_overview_show_the_connections, …a_viewer_does_not_see_the_connections, …the_api_returns_the_connections_of_a_code_and_of_the_catalogue, …the_menu_offers_the_connections_to_the_working_roles_only, …an_imported_program_brings_its_cause_and_its_repeat_code
FR-180	OK	test_bulk_workflow::test_the_size_of_the_sample_follows_the_settings, …the_sample_is_spread_over_the_subgroups_and_reproducible, …the_readiness_names_what_stops_each_draft, …bulk_edit_sets_the_same_value_on_every_draft_with_its_own_audit_entry, …bulk_edit_refuses_unknown_fields_and_other_peoples_drafts, …a_bulk_domain_signoff_needs_a_sample_and_every_code_gets_its_own_signoff, …a_domain_signoff_is_only_for_a_holder_of_the_domain, …a_small_batch_needs_no_sample, …a_defect_in_the_sample_blocks_the_decision_until_it_is_sent_back, …a_sample_cannot_be_borrowed_reused_or_kept_after_the_batch_changed, …a_defect_needs_a_note_and_a_verdict_must_be_ok_or_defect, …bulk_submit_sends_the_drafts_that_are_ready_and_reports_the_rest, …the_dossier_puts_the_codes_the_samples_and_the_audit_in_one_workbook, …viewers_see_nothing_of_the_bulk_workbench, …the_workbench_shows_the_batch_the_readiness_and_the_rule, …a_whole_domain_signoff_can_be_done_in_the_workbench, …a_decision_in_the_workbench_names_its_sample, …the_bulk_screens_are_translated, test_select_all::test_every_list_with_a_checkbox_per_row_can_be_selected_as_a_whole
FR-181	OK	superseded by FR-192: test_st_category_and_draft_delete::test_a_category_chosen_in_the_request_changes_nothing, …the_pages_have_no_category_controls_any_more
FR-182	OK	test_st_category_and_draft_delete::test_the_list_offers_tick_boxes_for_the_own_drafts_and_deletes_the_ticked, …all_own_drafts_of_a_filter_are_deleted_on_every_page, …somebody_elses_drafts_cannot_be_ticked_or_deleted
FR-183	OK	test_dark_mode::test_the_dark_palette_meets_aa_on_its_backgrounds, …every_badge_and_sla_pill_meets_aa_in_the_dark_theme, …the_dark_theme_gives_every_light_tint_a_dark_value, …colours_are_only_written_out_where_the_dark_theme_cannot_matter, …the_dark_theme_is_for_the_screen_only, …theme_js_is_a_plain_file_that_follows_the_system_and_remembers_the_choice, …the_login_page_and_every_signed_in_page_offer_the_theme_choice, …the_theme_choice_is_translated
FR-184	OK	test_version::test_the_version_is_a_release_number_and_pyproject_agrees, …the_health_endpoints_name_the_version, …the_login_page_and_the_menu_show_the_version, …the_api_contract_documents_the_version_in_both_health_answers
FR-185	OK	test_docs_pdf::test_the_reader_finds_every_kind_of_block, …inline_markup_is_removed_and_kept_as_styles, …symbols_a_pdf_font_cannot_draw_become_words, …every_character_of_the_documents_can_be_drawn, …a_document_is_built_with_the_version_in_its_name, …the_full_documentation_has_a_contents_list_and_the_artefacts, …the_four_documents_have_their_file_names, …the_user_manual_is_built_from_the_real_document, …an_unknown_document_is_refused
FR-186	OK	test_showcase::test_a_text_that_does_not_fit_stops_the_build, …both_decks_are_built_from_the_screenshots_and_every_picture_is_described, …the_decks_are_written_as_powerpoint_files, …the_demo_has_its_data_and_the_application_has_no_demo_routes, …the_browser_is_found_or_named

ID	Status	Evidence
FR-187	OK	test_branding::test_the_owner_the_long_name_and_the_logo_are_written_once, …the_guide_names_the_same_owner, …the_logo_files_exist_and_are_drawn, …the_menu_bar_and_the_login_page_show_the_logo, …every_title_page_carries_the_identity
FR-188	OK	test_handover::test_a_clean_tree_passes_the_audit, …every_kind_of_finding_stops_the_handover, …the_export_is_one_commit_by_the_named_author_with_the_tag, …a_tag_that_does_not_exist_and_a_push_without_confirmation_are_refused, …the_repository_carries_its_team_files
FR-189	OK	test_st_no_assumptions::test_a_delay_that_is_a_variable_stays_the_name_of_the_variable, …a_delay_the_program_does_not_have_says_so, …a_program_that_gives_everything_lists_only_what_no_program_carries, …the_preview_warns_with_the_name_of_the_parameter, …a_program_the_mapping_does_not_know_is_blocked_and_never_imported, …a_delay_may_be_the_name_of_a_parameter_but_not_a_malformed_time
FR-190	OK	test_st_no_assumptions::test_the_named_access_levels_come_from_the_mapping, …a_level_the_mapping_leaves_empty_stays_empty_and_is_listed, …a_level_whose_numbers_differ_from_the_program_is_named_and_warned_about, …a_level_that_is_no_longer_active_is_not_guessed
FR-191	OK	test_mapping::test_the_seed_workbook_loads_into_an_empty_table_once, …the_seed_workbook_is_consistent, …a_bad_seed_workbook_is_refused_as_a_whole
FR-192	OK	test_st_mapping_rules::test_cause_and_directive_become_the_remedy_text, …the_mapping_gives_the_value_of_a_delay_the_program_sets_with_a_parameter, …a_time_in_the_program_wins_over_the_mapping, …a_rejected_row_is_as_good_as_none, test_st_category_and_draft_delete::test_the_type_of_a_program_is_the_type_of_its_mapping_row, …a_row_without_a_type_blocks_the_program, test_mapping::test_a_number_a_code_of_the_catalogue_carries_cannot_change
FR-193	OK	test_st_mapping_rules::test_the_legacy_number_of_the_header_must_be_the_one_of_the_row, …a_title_that_does_not_say_what_the_name_says_is_flagged, …a_hub_row_for_a_main_program_is_not_taken, …the_glossary_is_a_check_not_a_translation
FR-194	OK	test_mapping::test_a_new_row_takes_effect_only_after_all_three_approvals, …one_rejection_closes_the_request_and_needs_a_reason, …only_the_three_roles_decide_and_each_only_once, …a_decision_asks_for_the_password_again, …the_requester_can_withdraw_and_nobody_else, …a_row_changed_in_between_makes_the_older_request_unusable
FR-195	OK	test_mapping::test_a_row_that_does_not_pass_the_checks_makes_no_request, …numbers_stay_unique, …a_change_confirm_and_reject_work_on_stored_rows
FR-196	OK	test_mapping::test_the_pages_show_the_mapping_the_requests_and_the_three_slots, …the_pages_make_requests_and_decide_with_the_form_posts, …the_menu_counts_the_requests_that_wait_for_a_decision, …a_viewer_neither_reads_nor_proposes, …show_all_puts_every_row_on_one_page_so_that_select_all_covers_them, …a_database_without_the_mapping_tables_says_what_to_run, …the_sign_off_of_a_domain_is_below_the_changes_and_the_status_above
FR-197	OK	test_mapping::test_the_export_is_the_layout_the_upload_reads, …a_workbook_makes_one_request_with_the_new_and_the_different_rows, …a_workbook_with_a_wrong_row_names_it_and_requests_nothing, …the_export_never_starts_a_formula
FR-198	OK	test_mapping::test_a_new_row_takes_effect_only_after_all_three_approvals (audit), test_docs_consistency::test_the_audit_catalogue_lists_every_code_the_application_can_write, test_openapi_conformance

4. Business rules, security, non-functional

ID	Status	Evidence
BR-001	OK	test_invariants::test_identifier_never_changes_after_allocation, …new_revision_of_a_released_code_keeps…
BR-002	OK	test_invariants::test_exactly_one_current_revision_per_code
BR-003	OK	test_invariants::test_only_one_open_revision_per_code (service 409 and DB index)
BR-004	OK	unit/test_lifecycle.py (the 9 defined edges accepted, all 40 other state pairs rejected, ORM guard)
BR-005	OK	SoD tests; test_uat::test_uat_09*
BR-006	OK	test_deprecation
BR-007	OK	test_releases::test_cannot_change_items_after_publish
BR-008	OK	test_exports_and_diff::test_release_can_be_deleted_before_publish_only; test_releases::test_non_admin_cannot_create_release
BR-009	OK	test_status_codes::test_viewer_cannot_see_draft_status_codes; test_invariants::test_released_code_stays_visible_to_viewers_while_a_new_revision_is_open; test_permission_grid (Viewers gated from change-request endpoints, API and UI); history/diff/working-set tests
BR-010	OK	sandbox release + delete tests

ID	Status	Evidence
BR-011	OK	domain sign-off tests
BR-012	OK	test_workflow_gaps override tests
BR-013	OK	test_status_codes::test_create_rejects_missing_functional_subgroup
BR-014	OK	test_lookup_admin::test_functional_subgroup_overlap_rejected
BR-015	OK	import tests; test_uat::test_uat_11…
SEC-001	OK	unit/test_password.py (Argon2id)
SEC-002	OK	test_injection_and_session::test_session_cookie_attributes; test_production_config::test_production_refuses_a_weak_secret_key, …starts_with_a_strong_secret_key
SEC-003	OK	idle and 12 h absolute expiry tests
SEC-004	OK	lockout tests
SEC-005	OK	test_auth_provider_seam_is_substitutable
SEC-006	OK	own login/credential store (test_auth.py); visual-only integration
SEC-010	OK	test_governance_extras::test_services_enforce_roles_independently_of_route_decorators (16 service entry points)
SEC-011	OK	SoD tests at all three decision points
SEC-012	OK	test_ui_screens::test_default_role_for_new_user_is_viewer
SEC-013	OK	override tests; access_confirmed flag in the API
SEC-014	OK	sandbox delete tests + trigger
SEC-015	OK	test_uat::test_uat_10…; permission grid
SEC-020	OK	test_injection_and_session::test_server_side_validation_ignores_client_side_constraints
SEC-021	OK	test_injection_and_session::test_sql_injection_probes_have_no_effect, …audit_log_filters_resist_injection; bandit clean
SEC-022	OK	test_injection_and_session::test_user_supplied_text_is_escaped_everywhere_it_renders
SEC-023	OK	test_exports_and_diff::test_exports_neutralise_formula_injection (CSV and XLSX)
SEC-024	OK	test_import::test_unsupported_file_extension_rejected; 20 MB cap
SEC-025	OK	test_security_headers.py
CSRF	OK	test_csrf.py
Password policy (11 §4)	OK	test_logging_and_passwords::test_policy_length_and_blocklist, …last_five_passwords_cannot_be_reused, …administrator_reset_also_honours_history, …only_hashes_are_stored_in_the_history; unit/test_password.py
Logging (11 §11)	OK	test_logging_and_passwords::test_requests_are_logged_as_json_without_credentials, …authenticated_requests_record_the_user_id, …unhandled_errors_are_logged_with_the_request_id
NFR-001	OK	performance/: search p95 ~ 110 ms and read p95 ~ 4 ms at 20,000 revisions; export of 5,000 rows: CSV ~ 0.7 s, JSON ~ 0.5 s, XLSX ~ 1.7 s, PDF ~ 3.1 s (current figures: 24-test-report.md §4); test_export_issues_a_constant_number_of_queries guards the eager loading that made exports ~4× faster
NFR-002	-	Operational availability target (deployment)
NFR-003 / 003a	OK	headers, CSRF, injection, hashing, test_no_cloud.py, pip-audit
NFR-004	OK	performance::test_fifty_concurrent_users_are_served_without_errors (p95 ~ 72 ms), …reads_are_not_blocked_by_an_open_write_transaction, …concurrent_identifier_allocation_in_one_group
NFR-005	OK	every service module >= 82% line coverage; ruff, mypy, bandit clean
NFR-006	OK	test_each_lifecycle_action_is_audited_exactly_once
NFR-007	(manual)	Responsive server-rendered UI, all screens render for every role (test_ui_screens); trained-user walkthrough is part of UAT
NFR-007a	OK	test_ui_screens::test_failed_form_submission_keeps_every_entered_value; test_uat::test_uat_13…
NFR-007b	OK	test_ui_screens::test_lookup_selects_carry_inline_help_hooks, …lookup_description_is_editable_and_flows_to_form_and_api (mechanism; wording is §6)
NFR-008	OK	egress-blocked lifecycle test
NFR-009	OK	test_multiprocess::test_the_username_limit_is_shared_by_both_processes, …the_ip_limit_is_shared…, …the_limit_survives_a_restart, …concurrent_writers_in_two_processes_all_succeed; unit/test_rate_limit.py
NFR-010	OK	test_accessibility::test_text_tokens_meet_aa_on_their_backgrounds, …every_state_badge_meets_aa, …every_page_is_structurally_accessible, …the_stylesheet_honours_user_preferences (a manual screen-reader pass is §6)
NFR-011	OK	test_audit_chain_and_ops::test_requests_are_logged_to_the_configured_file, …metrics_count_requests…, …ready_reports_ok_and_degraded_by_backup_state

ID	Status	Evidence
NFR-012	OK	test_audit_chain_and_ops::test_a_backup_is_written_verified_and_recorded, …a_restored_backup_runs_the_application (the quarterly restore drill on real hardware is §6)
NFR-013	OK	test_deploy_artifacts::test_linux_installer_parses_and_its_dry_run_changes_nothing, …windows_installer_dry_run_lists_every_step…, …units_parse_and_have_the_required_sections, …the_production_entry_point_serves_and_reports_its_state, …ci_files_parse_and_run_the_ci_script (a real install is §6)
NFR-014	OK	test_deploy_artifacts::test_the_postgres_ratchet_holds, …the_ratchet_notices_new_sqlite_specific_code, …non_sqlite_connections_are_left_alone (the port itself has not been done)
BR-015	OK	test_import::test_imported_row_becomes_import_type_draft_change_request; update mode and Hub carry-over also create Drafts only: test_import_toolkit::test_update_mode_opens_a_new_revision_of_a_released_code, test_integrations_and_reports::test_a_hub_code_is_carried_over_as_a_main_controller_draft
BR-016	OK	test_publication::test_audience_export_only_contains_codes_the_audience_may_see, …covert_visibility_is_data_an_administrator_can_flip; test_schema_upgrade::test_covert_is_not_visible_and_the_other_audiences_still_are
BR-017	OK	test_step_up_and_delegation::test_delegation_does_not_lift_segregation_of_duties, …one_person_cannot_give_both_review_signoffs_through_a_delegation, …delegating_does_not_grant_other_privileges
BR-018	OK	test_publication::test_translation_workflow_with_light_review_and_language_exports (approver differs from translator)
BR-019	OK	test_content_integrity::test_stale_save_is_refused_with_the_current_values
BR-020	OK	test_audit_chain_and_ops::test_archive_exports_old_rows_with_a_hash_and_deletes_nothing, …the_append_only_triggers_still_stop_edits_inside_the_application
BR-021	OK	test_publication::test_release_targets_limit_the_candidates_and_freeze_at_publish
BR-022	OK	test_step_up_and_delegation::test_admin_signoff_chief_decision_publish_and_full_export_all_need_the_password, …step_up_can_be_switched_off_by_configuration
BR-023	OK	test_st_import::test_submitting_issues_exactly_the_reserved_number, test_st_import::test_allocation_skips_a_reserved_number_and_a_manual_override_cannot_take_it, test_st_import::test_the_database_refuses_a_second_reservation_of_a_number
BR-024	OK	test_roles_and_languages::test_who_may_change_which_role_to_which, …only_an_administrator_creates_users, …the_stcd_owner_cannot_reach_the_other_roles, …the_administrator_still_gives_every_role, …the_last_administrator_is_still_protected
BR-025	OK	test_user_administration::test_an_account_with_history_is_erased_and_the_audit_log_is_untouched, …an_account_that_was_never_used_can_be_deleted_and_the_deletion_is_audited, …the_detail_page_shows_history_and_the_buttons_follow_it
BR-026	OK	test_withdraw_and_reimport::test_a_withdrawn_import_draft_gives_its_number_and_legacy_number_back, …a_withdrawn_code_that_has_its_identifier_keeps_it_and_can_be_restored, …a_withdrawn_code_is_listed_and_an_administrator_deletes_it_and_imports_again, …a_restored_code_is_no_shell_and_the_database_refuses_a_delete_around_the_service
BR-027	OK	test_mapping::test_a_second_administrator_cannot_replace_the_first_and_the_requester_cannot_decide, …only_the_three_roles_decide_and_each_only_once, …a_new_row_takes_effect_only_after_all_three_approvals
SEC-026	OK	test_step_up_and_delegation::test_wrong_password_is_refused_and_audited, …correct_password_lets_the_signoff_through_and_is_audited, …repeated_wrong_passwords_lock_the_account
SEC-027	OK	test_integrations_and_reports::test_a_token_is_shown_once_and_stored_only_as_a_hash, …a_token_reads_released_data_only_and_never_writes, …token_management_is_for_administrators_and_validated; test_route_inventory::test_a_tokenised_endpoint_is_always_a_get
SEC-028	OK	test_integrations_and_reports::test_webhook_targets_must_be_internal_and_inputs_valid, …redirects_are_not_followed, …a_publish_calls_the_webhook_with_a_verifiable_signature
SEC-029	OK	test_audit_chain_and_ops::test_metrics_are_refused_outside_the_allowed_networks, …a_metrics_token_admits_an_address_outside_the_networks, …ready_needs_no_login_and_reveals_no_business_data
SEC-030	OK	test_audit_chain_and_ops::test_a_configured_key_makes_the_chain_keyed, …a_forged_row_cannot_be_repaired_without_the_key; test_publication::test_signed_manifest_detects_edits_to_the_manifest_itself; installers generate the keys: test_deploy_artifacts::test_every_setting_an_installer_writes_is_read_by_the_application
SEC-031	OK	test_st_import::test_a_preview_belongs_to_its_uploader_and_expires, test_st_import::test_an_expired_preview_is_refused_and_removed, test_st_import::test_a_zip_that_unpacks_too_large_is_refused, test_st_import::test_only_engineers_and_administrators_import, test_st_import::test_the_mapping_is_not_for_viewers

ID	Status	Evidence
SEC-032	OK	test_roles_and_languages::test_only_an_administrator_creates_users, ...other_roles_cannot_open_the_user_list, ...the_stcd_owner_reads_the_workflow_but_changes_nothing_in_it, ...the_stcd_owner_changes_nothing_but_the_role; test_route_inventory::test_the_declared_role_list_is_what_each_role_gets; test_schema_upgrade::test_the_table_rebuilds_leave_every_foreign_key_intact
SEC-033	OK	test_email_and_windows_signin::test_a_header_from_an_address_that_is_not_a_trusted_proxy_is_not_believed, ...production_never_starts_with_the_simulation_or_an_unknown_mode, ...with_passwords_only_the_windows_sign_in_is_not_there, ...with_windows_only_passwords_are_refused_and_the_login_page_says_so
AUD-001	OK	test_decision_trail (complete trace per code), test_audit_coverage::test_each_lifecycle_action_is_audited_exactly_once
AUD-002	OK	test_releases::test_full_release_lifecycle (items of a Release); test_publication::test_export_history_lists_hashes_per_release
AUD-003	OK	test_audit_log::test_admin_sees_all_entries, ...reviewer_only_sees_own_actions; test_decision_trail::test_audit_log_can_be_filtered_by_action_and_actor
AUD-004	OK	test_audit_coverage::test_sandbox_delete_keeps_full_snapshot...
AUD-072	OK	test_audit_coverage::test_audit_log_entries_cannot_be_updated_or_deleted; test_audit_chain_and_ops::test_the_append_only_triggers_still_stop_edits_inside_the_application
AUD-073	OK	test_audit_chain_and_ops::test_archive_exports_old_rows_with_a_hash_and_deletes_nothing, ...archive_with_nothing_old_enough_does_nothing, ...cli_backup_verify_and_audit_commands
AUD-074	OK	test_audit_chain_and_ops::test_every_audit_row_is_chained_to_its_predecessor, ...the_hash_is_reproducible_from_the_stored_fields; test_schema_upgrade::test_the_migrated_hash_equals_what_the_application_would_have_written
AUD-075	OK	test_audit_chain_and_ops::test_an_edited_row_is_found_and_named, ...verify_tail_detects_recent_tampering_cheaply, ...a_tampered_audit_log_inside_a_backup_is_flagged, ...cli_audit_verify_fails_with_a_non_zero_exit_when_broken
AUD-076	OK	test_user_administration::test_an_account_that_was_never_used_can_be_deleted_and_the_deletion_is_audited, test_withdraw_and_reimport::test_a_withdrawn_import_draft_gives_its_number_and_legacy_number_back
AUD-077	OK	test_translation_review::test_only_an_administrator_or_an_stcd_owner_assigns_reviewers_and_it_is_audited, ...a_review_is_completed_only_when_every_string_has_a_verdict_and_is_recorded, ...an_accepted_proposal_is_shown_at_once_and_can_be_taken_back

5. Test matrix (13-test-strategy.md §8)

TC	Where
001, 002	unit/test_lifecycle.py
003, 005, 006, 031	unit/test_validation.py; test_status_codes
004, 007, 008	test_invariants::test_identifier_formats, test_sequential_submits_in_two_subgroups...; test_numbering.py
009, 009a, 009b, 019	test_change_requests; test_uat::test_uat_09*; permission grid
010, 011, 012, 013	test_invariants; test_schema_migration
014, 014a	test_releases
015, 015a	test_audit_coverage
016, 017	test_auth; test_injection_and_session
018, 018a, 018b	test_permission_grid.py
020, 021	test_change_requests
022	test_invariants::test_full_text_search_...
023	test_exports_and_diff (released, release and working-set exports exclude sandbox)
024, 025	performance/test_performance.py
026, 027, 028	test_injection_and_session; test_csrf.py
029	system/test_no_cloud.py
030	system/test_uat.py — UAT-01 ... UAT-13 scripted (human sign-off (manual))
032, 037	test_lookup_admin (+ DB triggers)
033, 034	test_import; test_uat::test_uat_11...
035	test_exports_and_diff::test_xlsx_export_has_same_rows_as_csv
036	test_ui_screens::test_failed_form_submission_keeps_every_entered_value
038	test_step_up_and_delegation
039	test_content_integrity

TC	Where
040	test_review_aids
041	test_sla_and_notifications
042	test_publication
043	test_import_toolkit
044	test_audit_chain_and_ops
045	test_schema_upgrade
046	test_integrations_and_reports
047	test_search_and_help
048	test_route_inventory
049	test_i18n; test_accessibility
050	test_multiprocess; unit/test_rate_limit.py
051	test_deploy_artifacts
052	test_docs_consistency

Note on "notification" (FR-031): the spec's notification is the Review Queue / Dashboard counters ("notification-style entries", 10-ui-ux-specification.md §6), not e-mail — an e-mail relay would be an external dependency the no-cloud principle avoids. That is what is built.

6. What still needs a person or the production environment

Item	Why it cannot be closed by the test suite
UAT sign-off (UAT-01 … 13, 13-test-strategy.md §7, §10)	The scenarios run automatically, but the roadmap requires a business sign-off pass
Technical Publications end-to-end (16-mvp-roadmap.md exit criterion)	Needs the real consumer team
Lookup wording — descriptions for the 23 Available Group codes, the eight program-reference fields, and the meaning of Covert	Content owned by Controls Engineering; Administrators enter it in Lookup Administration (the inline-help mechanism is done)
Deployment checks — TLS 1.2+ termination, OS ACL on the SQLite file, egress actually blocked on the server, nightly backup job	Properties of the target server, not of the code
Manual OWASP checklist (US-121 / EPIC-12)	Requires a reviewer; automated parts below
Open data question — software_version on revision vs. release	Resolved for now by decision D-01 (range per revision plus Release targets, 03-srs.md §6); the business may still revise it
Real installation (v4) — the installers on a clean Windows Server and a clean Linux host, systemd's own unit verification, TLS and proxy headers	Needs the target servers; the suite proves syntax, dry runs and the production entry point only
Restore drill and backup storage (v4) — quarterly restore on real hardware, the second volume, off-host copy	Properties of the target environment (14-deployment-architecture.md §7)
Screen-reader pass (v4) — WCAG 2.1 AA beyond the automated checks	Needs a person with assistive technology
Meaning of Covert, SMTP relay, languages of the manuals (v4)	Business answers; safe defaults are in 03-srs.md §6 (D-02, D-03, D-04)
PostgreSQL (v4)	Only the inventory and ratchet exist (NFR-014); the port and its test run are Phase 3
Key custody (v4) — keeping AUDIT_HMAC_KEY and MANIFEST_SIGNING_KEY in the password safe	Organisational procedure
Dev-tooling advisories — pytest 8.3.3, black 24.8.0, and starlette (via schemathesis) have advisories	Development-only packages, never deployed; upgrading pytest/black is a major-version change that would reformat the code base

No-cloud verification record (11-security-architecture.md §12, run 2026-10-21)

Checklist item	Result	Evidence
No public-cloud SaaS/PaaS/IaaS dependency	Pass	test_no_cloud_sdk_in_requirements, …imports_in_source (AWS, Azure, GCP, OCI, IBM, Firebase, telemetry SDKs)
No cloud identity provider	Pass	only SessionPasswordAuthProvider exists; provider seam test
No cloud storage for exports/backups	Pass	exporters return bytes to the caller; full-database export is a local SQLite backup

Checklist item	Result	Evidence
No CDN / external script or style source	Pass	test_templates_and_static_reference_no_external_assets; CSP default-src 'self' (test_security_headers.py)
No telemetry/analytics SDK	Pass	forbidden-module list includes sentry/datadog/newrelic/segment/mixpanel
Runs with egress blocked	Pass	test_whole_lifecycle_works_with_all_outbound_network_blocked
Dependency vulnerabilities	Pass (runtime)	pip-audit: none in runtime packages after the upgrades above

7. Verification runs

v4 (2026-10-23). The complete suite (607 tests: 598 functional, 9 load) with coverage and the five static gates was executed with three methodologies (13-test-strategy.md §11): (1) file order with the gates; (2) reversed order and a seeded shuffle (seed 7), including the `schemathesis` contract tests; (3) every test on a real SQLite file. Runs 8–16 of the history table of 24-test-report.md §9 are the discovery runs that exposed the defects of §1a (each shows one failed test); the final runs, 17–20, are clean:

Run	Methodology	Result
17	reversed order	607 passed, 0 failed, 5/5 gates
18	shuffled order (seed 7)	607 passed, 0 failed, 5/5 gates
19	whole suite on a SQLite file	607 passed, 0 failed, 5/5 gates
20 (final)	file order	607 passed, 0 failed, 0 retries, 5/5 gates, 92% line coverage, all 428 test references in this matrix resolve

v3 (2026-10-21). The complete suite (319 tests, plus coverage and the five static gates) was executed repeatedly, with the order changed on purpose to prove that no test depends on another. Every run of `python cscm_tool/scripts/generate_test_report.py` appends a line to the run history in 24-test-report.md §9.

Run	Order	Result
Baseline (before this pass)	file	302 passed — it did not catch the defects in §2 because the permission grid then encoded the wrong expectation
Report run 1	reverse	318 passed, 5/5 gates; one timing test needed a retry
Report run 2	shuffled (seed 11)	318 passed, 5/5 gates
Report run 3	file	318 passed, 5/5 gates; one timing test needed a retry
Report run 4	file	318 passed, 5/5 gates, no retry
Report run 5	shuffled (seed 23)	319 passed, 5/5 gates, no retry — after the export speed-up and its query-count test were added
Report run 6 (final)	file	319 passed, 0 failed, 0 retries, 5/5 gates, 88% line coverage, all 249 test references in this matrix resolve

What the repeated runs revealed about the tests themselves (all fixed): the load tests failed intermittently because they ran **inside the coverage-instrumented process**, which slows Python 2–3×; they now run in a separate process without coverage, scale their time limits to a measured host-speed factor (never below 1, capped at 3×), and the 50-user test was changed from a 4 s to an 8 s think time so it models named users rather than a saturating burst. A performance test that fails is re-run once alone and flagged "retry" in the report; any other failure is never retried.

Because the report is regenerated after every software change, **its history table, not this paragraph, is the authoritative record of which runs happened and what they returned.**

21. Improvement Ideas — CSCM Tool

This document has two halves. **Part A** records the 36 improvement ideas collected on 2026-10-21 (I-01 … I-36) and what became of each: all 36 are **implemented** in v4 (2026-10-23), specified in `03-srs.md` §1.10–§1.15 and traced to automated tests in 20-traceability-matrix.md. **Part B** is the new list of improvements (I-37 onward) collected while building, testing and documenting v4 — each grounded in something observed in the code, the tests or the deployment, or in a decision still open. Nothing in Part B is committed work; it is an input to backlog grooming.

How to read Part B

- **Priority** reflects value to the wind-turbine controls organisation weighed against risk and cost: **HIGH** closes a real risk or gap that will hurt soon after go-live; **MEDIUM** has clear value once the system is in production use; **LOW** is nice to have, speculative or depends on decisions not yet made.
- **Effort**: S ~ days, M ~ 1–2 sprints, L ~ 3+ sprints (the scale of `16-mvp-roadmap.md` §7).
- Constraint carried over from the project: **everything stays on-premises** — no cloud service, no external e-mail/SMS provider, no telemetry leaving the network (`11-security-architecture.md` §12). Ideas that need an outside connection say so and are opt-in by design.
- Items marked **(observed)** come from something actually seen in the code, the tests or the deployment, not from speculation.

Part A — The 36 ideas of 2026-10-21 and their outcome

A.1 Status at a glance

ID	Idea	Status	Requirements	Tests	Deviations from the idea as written
I-01	Optimistic concurrency on Draft edits	OK implemented	FR-090, BR-019	TC-039	—
I-02	Manual-ready PDF and export profiles	OK implemented	FR-100, FR-101	TC-042	PDF in every script since v4.7 (I-37, FR-165)
I-03	Audience-scoped publication	OK implemented	FR-102, BR-016	TC-042	Covert = not visible until defined (D-02)
I-04	Referential integrity and where-used	OK implemented	FR-091, FR-092	TC-039	soft warnings, never blocks (as proposed)
I-05	Migration toolkit	OK implemented	FR-048i … FR-048m	TC-043	—
I-06	Release integrity, notes, diff	OK implemented	FR-103 … FR-105, SEC-030	TC-042	HMAC signature, not asymmetric (D-05, see I-39)
I-07	Software-version binding and matrix	OK implemented	FR-106, FR-107, BR-021	TC-042	model decided: range per revision + Release targets (D-01)
I-08	SLA tracking, aging queues, reminders	OK implemented	FR-120 … FR-122	TC-041	business days without holiday calendar (see I-40)
I-09	Automated backups with verified restore	OK implemented	FR-150, NFR-012	TC-044	second volume / off-host copy is an operator task (see I-45)
I-10	Success-metrics dashboard	OK implemented	FR-123	TC-041, TC-046	—
I-11	Change sets	OK implemented	FR-124	TC-040	presentation and queueing only (as proposed)
I-12	Duplicate and near-duplicate detection	OK implemented	FR-093	TC-039, TC-043	—
I-13	Evidence and attachments	OK implemented	FR-130, SEC-024	TC-040	—
I-14	Remedy text, component, standards	OK implemented	FR-131	TC-039	"Any" owner: never gates cross-domain sign-off
I-15	Multilingual content	OK implemented	FR-110, FR-111, BR-018	TC-042	English + German only (D-03)
I-16	Delegation and out-of-office cover	OK implemented	FR-125, BR-017	TC-038	decision actions only (D-08)
I-17	Step-up re-authentication	OK implemented	FR-126, SEC-026, BR-022	TC-038	—
I-18	Tamper-evident audit log, retention job	OK implemented	FR-151, AUD-072 … AUD-075	TC-044	—

ID	Idea	Status	Requirements	Tests	Deviations from the idea as written
I-19	Catalogue health and capacity forecast	OK implemented	FR-140, FR-141	TC-046, TC-039	—
I-20	Persistent rate-limit state	OK implemented	NFR-009	TC-050	proven with two real processes
I-21	Observability	OK implemented	FR-152, FR-153, NFR-011	TC-044	—
I-22	Accessibility and German UI	OK implemented	FR-155, NFR-010	TC-049	German and every other language cover the whole interface since v4.9 (I-38, FR-167)
I-23	Field comments and review checklists	OK implemented	FR-127, FR-128	TC-040	—
I-24	Offline field-lookup bundle	OK implemented	FR-108	TC-042	single HTML file (no SQLite variant)
I-25	Platform variants	OK implemented	FR-094	TC-039	—
I-26	Saved searches, columns, shareable links	OK implemented	FR-142	TC-046	—
I-27	Planned effective / deprecation dates	OK implemented	FR-109	TC-039	warnings only
I-28	Fleet feedback	OK implemented	FR-143	TC-046	file import (the agreed shape); no scheduled feed
I-29	Tokens, change feed, webhooks	OK implemented	FR-144 … FR-146, SEC-027, SEC-028	TC-046	webhooks make a single attempt (see I-46)
I-30	Reviewer balancing	OK implemented	FR-129	TC-040	—
I-31	FTS5 search and typo tolerance	OK implemented	FR-148	TC-047	substring fallback without FTS5
I-32	Background imports with progress	OK implemented	FR-048n	TC-043	a thread of the web process (see I-44)
I-33	On-prem packaging, installer, CI	OK implemented	NFR-013	TC-051	not yet run on real servers (docs/20 §6)
I-34	PostgreSQL readiness	(o) partly	NFR-014	TC-051	inventory + ratchet + dialect guards; no PostgreSQL run (see I-41)
I-35	In-app help	OK implemented	FR-154	TC-047	—
I-36	Hub -> Main Controller consolidation	OK implemented	FR-147, D-06	TC-046	see 25-hc-integration-concept.md for the open numbering proposal

I-34 is the only idea delivered in part, and the table says so; everything else is complete as specified in 03-srs.md. The 21 new tables, 78 new API operations (one more changed: the full-database export became a POST) and 19 new screens are listed in 06 §13, 09 §7 and 10 §1.

A.2 The business questions of the original list and how v4 answered them

Question	Answer used (03-srs.md §6)	Still needs the business?
What does Covert mean; which values let an audience see a code?	Not visible; visibility is data on each value (D-02)	Yes — please define it; a data change, no release needed
Does software_version belong to the revision or the Release?	Both: a range per revision and the versions a Release targets (D-01)	Confirm
In which languages are manuals published?	English source, German first (D-03)	Confirm; more languages are data plus translation work
Is an internal SMTP relay available?	In-app notices are the mechanism, SMTP optional (D-04)	Yes — if mail is wanted, set SMTP_*
Which evidence does the SAFETY sign-off require?	Any allow-listed file type; no per-domain rule	Yes, if a rule is wanted
Is there an agreed feed of occurrence counts?	File import with an agreed shape (status_code_identifier, occurrences)	Yes, for a scheduled feed
Is an on-prem Git/CI server available?	CI definitions for Gitea and GitLab are provided	Yes — which one

A.3 Quick wins and sequencing of the original list — now moot

All seven quick wins and the four-step sequencing of the original document are delivered together in v4; the risk-reducing items (backups, concurrency, persistent rate limit, PDF wrapping) are covered by TC-044, TC-039, TC-050 and TC-042.

Part B — New improvement ideas (I-37 ...)

B.1 Priority at a glance

ID	Idea	Priority	Effort	Epic
I-37	Unicode font for the PDF export (observed) — OK delivered in v4.7 (FR-165)	HIGH	S	EPIC-20
I-38	Complete interface coverage in every language: flash messages, admin pages, service-layer sentences (observed) — OK delivered in v4.9 (FR-167); the text of the exports outside the catalogue is still English (26-translations.md §5)	HIGH	M	EPIC-15
I-39	Asymmetric (Ed25519) manifest signatures and key rotation (observed)	MEDIUM	M	EPIC-18, EPIC-12
I-40	Holiday calendar for business days (observed) — x not needed (decision of 2026-10-24)	—	—	EPIC-19
I-41	Run and test on PostgreSQL (execute the port prepared by I-34)	MEDIUM	L	EPIC-13
I-42	Git-linked status code library versions and tag proposals	MEDIUM	L	new EPIC-26
I-43	Template-driven DOCX export (customer / brand templates) — (o) the data side (complete baseline, platform and controller filter, package) is delivered in v4.10 (FR-173, FR-174); the Word output waits for the templates	MEDIUM	L	EPIC-20
I-44	Import worker that survives a restart and can resume (observed)	MEDIUM	M	EPIC-14
I-45	Backup replication and encryption: second volume, off-host copy	MEDIUM	S	EPIC-25
I-46	Webhook retry with back-off and a dead-letter view (observed)	MEDIUM	S	EPIC-11
I-47	Active Directory / ADFS authentication provider	MEDIUM	L	EPIC-12 (Phase 2)
I-48	Key custody helper: rotate, escrow and check the integrity keys (observed)	MEDIUM	M	EPIC-25
I-49	Bulk revision as a change set (e.g. adjust one field across many codes)	MEDIUM	M	EPIC-04
I-50	Pause the SLA clock while the request waits for its author	LOW	S	EPIC-19
I-51	Alert rules and dashboards shipped with the metrics (Prometheus / Grafana files)	LOW	S	EPIC-25
I-52	Audit-log export (CSV) and saved audit filters	LOW	S	EPIC-10
I-53	Command palette and keyboard shortcuts	LOW	S	EPIC-15
I-54	Translation memory and term consistency check	LOW	M	EPIC-22
I-55	Description linter: style rules, title != description, banned words	LOW	S	EPIC-02
I-56	Customer read-only view with permalinks (needs a decision on FR-083)	LOW	M	new EPIC-27
I-57	Mobile / tablet layout for field use	LOW	M	EPIC-15
I-58	Token usage report and per-token rate limit	LOW	S	EPIC-11
I-59	Stronger test techniques: mutation testing and state-machine property tests	LOW	M	EPIC-13
I-60	Document-to-code links: show which exports and releases used a code's wording	LOW	S	EPIC-18
I-61	Batch edit of imported Drafts: status category, platforms and audience access for the ticked codes of a change set — OK delivered in v4.13 (FR-180)	MEDIUM	M	EPIC-14
I-62	Bulk submit for review and bulk cross-domain sign-off for a change set — OK delivered in v4.13 (FR-180)	MEDIUM	M	EPIC-19
I-63	Re-import from the programs: show what changed against the catalogue (drift) and revise the codes	MEDIUM	M	EPIC-14
I-64	Wire the parsed user-group levels, the remaining subsystem programs and the operating-mode mask from import_source_item into catalogue columns — OK delivered in v4.9.1 (FR-168)	MEDIUM	M	EPIC-14
I-65	Per-program status category in the preview (derived from the alarm type, or set per program) — OK delivered in v4.13.2 (FR-181)	LOW	S	EPIC-14

B.2 HIGH priority

I-37 · Unicode font for the PDF export — HIGH · S

Problem (observed). The PDF exporter uses the built-in Latin-1 font; any character outside it (for example some typographic dashes and quotes, or text in a language beyond German) is replaced (`test_pdf_survives_characters_outside_latin_1` proves it does not crash, not that it is right). A German manual with an ? or a non-Latin-1 symbol, or a future language, would be wrong. **Idea.** Ship one on-premises, freely licensed Unicode font (for example DejaVu Sans) in `app/static/fonts/` and register it with `fpdf2`; keep the Latin-1 fallback if the file is missing. A test renders a sample with Greek, Cyrillic and typographic characters and extracts the text back. **Value.** The PDF becomes safe for every language the catalogue may get. **Epic:** EPIC-20. **Delivered (v4.7, FR-165).** Noto Sans plus the Noto font of each script (Devanagari, Bengali, Gurmukhi, Gujarati, Odia, Tamil, Telugu, Kannada, Malayalam, Georgian, Naskh Arabic, Nastaliq Urdu) are bundled in `app/static/fonts/` with their OFL licence, the text is shaped with `uharfbuzz` and Arabic and Urdu are laid out right to left. Checked by `tests/integration/test_pdf_scripts.py` (TC-060) and by looking at the rendered pages of a sample in all twelve script families. Remaining: bold type (only regular weights ship), a taller line for Nastaliq so its tall letters do not touch the cell border, and a translated title page and headings (layer C of 26-translations.md).

I-38 · Complete German coverage — HIGH · M

Problem (observed). The catalogue (`app/i18n_de.py`, ~690 strings) covers every template string, but messages built in the service layer (flash messages, validation errors), the administration screens and the column headers of exports stay English (documented limitation in 22 §13). **Idea.** Route service-layer messages through message codes with parameters, translate them in one catalogue, translate the administration templates, and make export headers part of the profile (a `label_de` next to `label`). Extend `test_i18n` to fail on any user-visible English literal in `app/` that has no catalogue entry. **Value.** German-speaking users get one language throughout; removes a source of confusing mixed-language screens. **Epic:** EPIC-15.

B.3 MEDIUM priority

ID	Idea	Detail	Epic
I-39	Asymmetric manifest signatures and key rotation (M) (observed)	Decision D-05 chose HMAC because it needs no PKI, but a verifier then needs the secret and could forge a manifest. An Ed25519 key pair (private key on the server, public key distributed with the manuals) lets Technical Publications verify without being able to sign; the manifest already carries a key_id, so old and new keys can coexist. Add a documented rotation procedure for MANIFEST_SIGNING_KEY and for the audit key (a signed checkpoint entry before and after a key change).	EPIC-18, EPIC-12
I-40	Holiday calendar for business days (S) (observed)	business_days_between skips only weekends, so a public holiday week looks like a stalled queue. An Administrator-maintained table of non-working days (per site or global) feeds SLA ages, reminders and the KPIs.	EPIC-19
I-41	Run and test on PostgreSQL (L)	I-34 delivered the inventory (47 blockers, 125 ports at the baseline of 2026-10-24, schema v4.7) and the dialect guards; the port itself is open: a PostgreSQL schema file, PL/pgSQL triggers, FOR UPDATE/advisory locks instead of BEGIN IMMEDIATE, tsvector search, pg_dump-based backup and a CI job that runs the suite on PostgreSQL. Do this when volume or availability needs justify it (Phase 3).	EPIC-13
I-42	Git-linked status code library versions and tag proposals (L)	Every controller software release tag gets a status code library version (SemVer-like: new code = minor, changed meaning/severity/reaction = major, text only = patch). Tags are pulled as proposals from a local or mirrored clone (optionally enriched from GitHub), a human accepts, edits or rejects them; an unchanged software release attaches to the existing library version; a drift scan compares the controller source with the library. Designed (state machine, data model, options, open questions) but not built: it needs a sample of the controller source files, the Git host and the tag conventions. Strictly opt-in and read-only toward the code repository, with an explicit exception recorded in 11 §12.	new EPIC-26

ID	Idea	Detail	Epic
I-43	Template-driven DOCX export (L)	Generate the status-code documentation straight into the customer or brand Word template (registry of versioned templates, placeholders validated on upload, audience-redacted content, headings and tables that feed the table of contents, document properties with library version and hash, optional PDF where LibreOffice exists). The Word output is designed but not built; it needs the three real templates (put in the project folder or uploaded, see I-66), the meaning of "2xm/3xm/4xm" (assumed: the turbine platforms) and the audience tiers per customer.	EPIC-20
I-44	Import worker that survives a restart (M) (observed)	A large import runs in a thread of the web process; a restart leaves the job at its last recorded state and the file must be re-uploaded. Persist the evaluated rows, process them from a small job table by a separate process (or flask import-worker), resume after a restart.	EPIC-14
I-45	Backup replication and encryption (S)	flask backup-db --mirror to copy each verified generation to a second location and record it in the status file; optional encryption of the copy with a key from the password safe; readiness reports an unreplicated backup.	EPIC-25
I-46	Webhook retry with back-off and a dead-letter view (S) (observed)	A delivery is attempted once; a consumer that is down for ten minutes misses the event (the change feed is the safety net). Retry a few times with back-off from the reminder job and show undelivered events on the Integrations page with a Resend button.	EPIC-11
I-47	Active Directory / ADFS authentication (L)	The AuthProvider seam exists (FR-005); Partly built (v4.7, FR-163, SEC-033): the Windows sign-in behind an authenticating gateway matches the identity with the (fixed) e-mail address and can be simulated on a developer machine; still open are a direct LDAP/Kerberos or ADFS/SAML provider, a group-to-role mapping policy and just-in-time user creation. Must not introduce any cloud identity provider.	EPIC-12
I-48	Key custody helper (M) (observed)	The three generated keys cannot be recovered; losing the audit key makes old chain entries unverifiable. A flask keys-check that confirms the keys are present and have the expected strength and a printable escrow sheet for the password safe at install time.	EPIC-25
I-49	Bulk revision as a change set (M)	Controls engineers sometimes change one field across many codes (a delay unit, a reaction). A bulk action creates one revision Draft per selected code, groups them in a change set and shows the diff of all of them, so review and approval stay per code but the work is one operation.	EPIC-04
I-61	Batch edit of imported Drafts (M)	The status code import gives every Draft the same category, platforms and empty audience access. A bulk action on a change set would set them for the ticked codes in one step, each still an ordinary edit of an open Draft by its author (audited per code).	EPIC-14
I-62	Bulk submit and bulk domain sign-off (M)	An imported batch of hundreds of Drafts must be submitted one by one and each needs the cross-domain sign-off. A change-set action would submit every ready Draft and let a domain owner sign off the batch with one password confirmation, as the bulk decisions already do (I-11, I-17).	EPIC-19
I-63	Re-import and drift report (M)	import_source_item keeps the SHA-256 and the parsed attributes of every program. Uploading the same program again could show which attributes changed against the catalogue and open a revision (or update the Draft) for the codes the user ticks.	EPIC-14
I-64	Wire the remaining program attributes (M) — delivered in v4.9.1 (FR-168)	User-group levels, the remaining subsystem programs, the operating-mode mask and the task frequency are parsed and kept as JSON only. Once the v4.1 columns are editable in the application the import should fill them; the mapping table of 06-data-dictionary.md §5a lists which.	EPIC-14

B.4 LOW priority

ID	Idea	Detail	Epic
I-50	Pause the SLA clock while waiting for the author (S)	A request rejected back to Draft is already outside the clock; add an explicit waiting for the author state for questions in Review so reviewers are not shown as slow.	EPIC-19
I-51	Alert rules and dashboards shipped with the metrics (S)	The suggested alerts of 14 §8 as a Prometheus rule file and a Grafana dashboard JSON that the platform team can import.	EPIC-25

ID	Idea	Detail	Epic
I-52	Audit-log export and saved filters (S)	A CSV export of the filtered audit log (itself audited) and saved filters for the usual auditor questions.	EPIC-10
I-53	Command palette and keyboard shortcuts (S)	Jump to StCd-..., search or a screen with the keyboard; g s style shortcuts documented in Help.	EPIC-15
I-54	Translation memory and term check (M)	Reuse translations of identical sentences across codes and flag a translated term that differs from the project glossary.	EPIC-22
I-55	Description linter (S)	Rules beyond the current "title must not equal description": minimum information, units spelled consistently, banned words — shown as warnings.	EPIC-02
I-56	Customer read-only view with permalinks (M)	A stable per-code URL for a read-only, audience-scoped view (and a QR code on printed documents). Conflicts with FR-083 (internal network only) unless the view is served inside the customer-reachable segment; needs a business and security decision first.	new EPIC-27
I-57	Mobile / tablet layout (M)	The shell collapses on narrow screens but dense tables are desktop-first; a card layout for the list and detail pages.	EPIC-15
I-58	Token usage report and per-token rate limit (S)	last_used_at exists; add request counts, the endpoints used and a limit so a runaway consumer cannot slow the application.	EPIC-11
I-59	Stronger test techniques (M)	Mutation testing on the lifecycle, numbering and segregation-of-duties code; a hypothesis state-machine test over random sequences of workflow actions; trend of the performance measurements across report runs.	EPIC-13
I-60	Document-to-code links (S)	From a code's page, show which export jobs and releases contained which wording (the export history already stores hashes per release); answers "which manuals need to change?".	EPIC-18

B.5 Proposed new epics

Epic	Name	Contains	Size	Suggested phase
EPIC-26	Git-Linked Library Versions	I-42	L	After the Git host, tag convention and a controller-source sample are known
EPIC-27	External Read-Only Access	I-56	M	Only after the FR-083 decision

Existing epics to extend: EPIC-02 (I-55), EPIC-04 (I-49), EPIC-10 (I-52), EPIC-11 (I-46, I-58), EPIC-12 (I-39, I-47), EPIC-13 (I-41, I-59), EPIC-14 (I-44), EPIC-15 (I-38, I-53, I-57), EPIC-18 (I-39, I-60), EPIC-19 (I-40, I-50), EPIC-20 (I-37, I-43), EPIC-22 (I-54), EPIC-25 (I-45, I-48, I-51).

B.6 Decisions needed from the business before work starts

Question	Unblocks
Which Git host is used, how are release tags named (platform, pre-release and hotfix markers), are GitHub Releases used, are tags signed, and may the tool ever write back? Please also supply a sample of the controller source files that carry status codes.	I-42
Where are the customer and brand Word templates, who owns their placeholder conventions, which audience tiers does each customer need, and does "2xm/3xm/4xm" mean the turbine platforms or the 2xxx/3xxx/4xxx number ranges? Is LibreOffice available for PDF output?	I-43
Which languages beyond English and German will the manuals need, and in which fonts?	I-37, I-38, I-54
What is the meaning of Covert (and of the other access values)?	I-56, and removes the safe default of D-02
Which public holidays apply to which site?	I-40
Is a PKI or a key-management system available on the network?	I-39, I-48
Is an off-host or second-volume location available for backups, and is encryption at rest mandatory for them?	I-45
Is the volume or availability requirement expected to outgrow one SQLite file, and when?	I-41
Which directory (Active Directory / ADFS) and which group-to-role policy?	I-47
May any part of the catalogue be reachable from a customer-facing segment?	I-56

B.8 Follow-ups of the roles and languages change (v4.3)

Open points, not built; none has an I-number yet because the business has not ranked them. (1) Let the StCd Owner tag engineering domains, so an Engineer they promote can give cross-domain sign-offs at once. (2) Let a colleague cover for the StCd Owner (delegation covers only Reviewer, Administrator and Chief Engineer). (3) Decide whether the StCd Owner may read the audit log. (4) A second person's confirmation when an Administrator role is handed out. (5) Have the fourteen added languages reviewed by native speakers and extend them to the full catalogue (German is complete); until then keep them off for people who depend on them. (6) Show a translated word beside the lifecycle and category codes (translation scope, `10-ui-ux-specification.md` §10a). (7) Translate the flash messages of the service layer for the interface, never the API messages. (8) Decide the flag for Bengali and Punjabi (India today; Bangladesh and Pakistan are the other homes of the languages).

B.9 Follow-ups of user administration, withdrawal and re-import (v4.6)

Open points, not built. (1) Deleting an account is possible only while it has no history; a person who must disappear for data-protection reasons needs an anonymise action that replaces name and e-mail but keeps the account row and its audit trail. (2) Bulk deactivate and bulk role change in the user list. (3) Bulk withdraw from the change-set page, not only from My Change Requests. (4) A Draft that is withdrawn after other people commented loses those comments except in the audit snapshot; a return to author state would keep them. (5) The preview of a re-import compares only the fields the catalogue has columns for; the user-group levels and the subsystem programs of the file are kept in the provenance JSON but not compared. (6) Update from a file for a code in Review has to wait for the decision; a replace the Draft under review path is open.

B.10 Translations (v4.7)

The concept and the eleven open items for checking the translations by native speakers and for exporting the status code manuals in every designated language are in 26-translations.md §8. Built since: the review itself runs in the application, in its own section Translations (FR-164, 26-translations.md §3.4). The ones that matter most: find one native-speaker reviewer per language and assign them there; mark unreviewed languages as beta; a terminology list per language; language competence on the user for approving status code translations; content sheets (export and import) for translating the status codes; a coverage matrix and a readiness rule per Release; Unicode fonts for the PDF (I-37). Delivered in v4.9: the whole interface in all nineteen languages (FR-167, I-38) and the technical nomenclature as a section of its own with a two-person approval (FR-166, 26-translations.md §8a); the critical review of the implementation is in §8b of that file.

B.11 Delivered since v4.9, and what the review of the release process added (v4.10)

Delivered after the list was written (each is specified in 03-srs.md §1.17 and traced in 20-traceability-matrix.md):

Delivered	Version	Requirements
The fields of the generated programs (further programs, flags, task call frequency, operating-mode mask, user-group levels) stored, edited, imported and exported — this is I-64	v4.9.1	FR-168
Users with history and withdrawn codes can be deleted (erased in place, the audit log is never changed)	v4.9.0 – v4.9.2	BR-025, BR-026
The menu grouped by the work of the user, with counts and a Menu button	v4.9.2	FR-169
The beta mark for unreviewed languages, hidden from Viewers and guests	v4.9.2 – v4.9.3	FR-170, FR-172
The words around an export in the language of the export	v4.9.2	FR-171
The complete baseline of a Release (every code valid as of it, ~900 MC + ~350 HC), narrowed by platform (2XM / 3XM / 4XM) and controller (MC / HC)	v4.10.0	FR-173
The package: every language x every access level x chosen formats, release notes and a manifest with a SHA-256 per file, built in the background	v4.10.0	FR-174
The download rule: everybody downloads what is released; the working set and unpublished Releases only the StCd Owner and the Administrator	v4.10.0	FR-175
Confidence check of the translations, separate approval of new and changed strings, language impact notices to the StCd Owner	v4.11.0	FR-176 … FR-178
Connections between status codes (suppression, preconditions, escalation) read from the imported programs	v4.12.0	FR-179

Delivered	Version	Requirements
Bulk workflow for the migration — I-70 with I-61 and I-62: the workbench of a change set (readiness, bulk edit, bulk submit, bulk domain sign-off, bulk decisions), the sample inspection (stratified, 10 % / at least 8, one defect blocks), and the dossier (XLSX) for an auditor	v4.13.0	FR-180
Select-all on every list with tick boxes; a status category per program of a .st import (by alarm type, per row or for the ticked rows); delete several own Drafts at once from the Status Codes list — I-65	v4.13.1 / v4.13.2	FR-181, FR-182
Dark mode: Automatic (system) / Light / Dark in the top bar, every colour a token, the dark palette checked for contrast	v4.14.0	FR-183
Visible version: in the menu, on the login page and in the health endpoints, from one place in the code	v4.14.1	FR-184
Release documents as PDF: full documentation, software description, user manual and test report, each with the version in its file name	v4.15.0	FR-185
Design showcase decks (management, team) from real screenshots, and the identity of every title page: long name, owner line, logo	v4.16.0	FR-186, FR-187
Hand-over to the team repository: audited clean export, code owners, pipeline for GitHub, contribution guide	v4.16.0	FR-188
No assumed values in the .st import: a variable stays its name, what the program lacks is listed in the comment, no status category is preselected; the user-group numbers become named access levels where exactly one level fits	v4.16.1	FR-189, FR-190

New ideas from the review of the release process (not built):

ID	Idea	Priority	Effort	Epic
I-66	Upload of the DOCX templates in the application (one per platform, replaceable by an Administrator, placeholders and styles checked on upload) — the way into I-43	MEDIUM	M	EPIC-20
I-67	Publish gate on translation coverage: warn or block the publication of a Release while a language is below a threshold (today the Release page only shows the numbers, FR-173)	MEDIUM	S	EPIC-22
I-68	Packages: retention and cleanup of the files under instance/packages, one running build per Release, a build that survives a restart (as I-44), a notice when it is ready	MEDIUM	M	EPIC-20
I-69	A manual edition number and document properties (library version, Release, hash) printed on every file of a package	LOW	S	EPIC-20

I-43 now. The baseline, the platform and controller filter and the package are the data side of it and are built. What is missing is the Word output itself: the three templates (2XM, 3XM, 4XM; assumed to be the turbine platforms, which the templates will confirm) are to be put in the project folder or uploaded in the application (I-66), together with their style names, the cover fields and an example entry page.

B.12 What would benefit the controls team most (2026-10-24)

Weighed from the work of a wind turbine controls team - getting ~1,250 codes (about 900 Main, 350 Hub) into governed use, keeping the manual equal to what the turbine does, and helping the technician at the turbine - not from the list of what is easy to build. The holiday calendar (I-40) is dropped: not needed.

Rank	ID	Idea	Why it matters to the team	Size
1 OK	I-70	Bulk workflow for the migration (delivered in v4.13): bulk edit (I-61), bulk submit and bulk cross-domain sign-off (I-62) and a bulk decision with a sample check for a change set of imported codes	The three decisions per code are right for a new code and impossible for 1,250 imported ones; without a bulk path go-live stalls in the queues of four roles	M
2	I-71	Build-against-catalogue check: upload the generated programs of a controller software build and get a drift report against the released baseline for that software version (extends I-63 and the first step of I-42)	Closes the loop manual = turbine: the one question the whole tool exists to answer, asked at every software release instead of once at import	M
3	I-72	Definition linter with turbine rules (extends I-55 and I-67): alarm without remedy text, set or reset delay inconsistent with the category, a repeat counter without a replacement code, a Hub code without a Main counterpart, a missing program field, duplicate titles; shown per code and as a gate on the Release	Catches the defects that cost service technicians time, before they are published; gives the Release a measurable completeness	S - M
4	I-73	Change propagation across platform variants: when a code of 3XM is revised, list its 2XM and 4XM variants with the same change prefilled for review	The same fault is maintained three times today; this is the main source of drift between platforms	M
5	I-74	Technician lookup at the turbine: the offline page (I-24) as an installable page with search by symptom and free text, the remedy steps first, related codes, and the language of the technician	The people who read the codes most are at the turbine without a network, not at a desk	M

Rank	ID	Idea	Why it matters to the team	Size
6	I-67 / I-68 / I-69	Release hardening: publish gate on coverage, package retention and resumable builds, edition number	Keeps what v4.10 delivered dependable for Technical Publications	S - M
7	I-75	Move the Hub codes into the Main range, systematically: the Hub rows of the mapping keep the Hub range (11000 and up); if the Hub codes are to join the Main scheme later, the new numbers are proposed from the bands, shown as one request (workbook or generated) and approved by the StCd Owner, the Chief Engineer and an Administrator in one step, never one by one	The mapping and its approvals (v4.17) make it one request instead of a migration script; the former "proposal 2" of the numbering workbook is the starting point	M

Items that wait for input from outside the project (templates for I-43 and I-66, the directory for I-47, the Git host for I-42, a PostgreSQL environment for I-41) stay where they are.

B.7 Suggested sequencing

1. **With go-live of v4:** I-45 (backup copy), I-48 (key custody) (I-37, the PDF font, is delivered) — small and they close observed gaps.
2. **First quarter in production:** I-40 (holidays), I-44 (resumable import) and, once its inputs exist, the design-ready I-42 and I-43.
3. **Second quarter:** I-39 (signatures and rotation), I-46 (webhook retry), I-49 (bulk revision), I-47 (SSO) as needed.
4. **Later / on demand:** the LOW items and I-41 when the volume justifies the port.

22. Software Description — CSCM Tool

Document type: software description (what the delivered software is and how it behaves). It is a separate artefact from the requirements specification (`03-srs.md`), the user manual (23-user-manual.md) and the test report (24-test-report.md). **Describes:** the implementation in `cscm_tool/` as of 2026-10-24 (v4.10), branch `dev`, against requirements baseline v4.0 (the v3 field model plus the improvement pack of 21-improvement-ideas.md; sections and rows added by v4 are marked (v4)). **Traceability:** every statement about behaviour cites the requirement it implements (FR-/BR-/SEC-/NFR- IDs from `03-srs.md` and `11-security-architecture.md`); 20-traceability-matrix.md maps each of those to code and automated tests.

---

1. Purpose and scope

1.1 Purpose

The **Controller Status Code Management Tool (CSCM Tool)** is the governed system of record for the status codes of Senvion 4XM/3XM/2XM wind-turbine **Main Controllers** and **Hub Controllers**: the faults, warnings and informational messages the controller software raises. It reverses the former document-first practice — codes are created, completed, reviewed, approved, versioned and released inside this tool, and technical manuals, specifications and SCADA tag lists are derived from it.

1.2 What the software does

1. Holds each status code with a stable identifier `StCd-XXXXX`, a rich field set (category, delays, counters, program references, the further programs, parameters and user-group levels of the generated programs, eight audience-access values) and complete revision history.
2. Enforces a **seven-state lifecycle** — Draft -> Review -> PendingApproval -> Approved -> Released -> Deprecated -> Archived — with **three independent decisions** before release: a Reviewer sign-off, an Administrator sign-off, and a Chief Engineer approval (FR-030–FR-039, BR-004, BR-005).
3. Routes work to the right engineers with **Cross-Domain Sign-Off** (FR-027–FR-029).
4. Allocates identifiers atomically per Functional System Group and Subgroup (FR-021).
5. Bundles approved codes into immutable **Releases** and exports them as CSV, JSON, XLSX or PDF (FR-040–FR-048).
6. Imports a legacy library from CSV/XLSX — always as governed Drafts, never as a shortcut (FR-048a–d).
7. Records every governed action in an append-only **audit log** (FR-070–FR-073).
8. Provides **sandbox** codes for training and testing that can never reach a Release (FR-053–FR-056).
9. Offers a server-rendered web UI (50 screens, nineteen languages) and a versioned JSON REST API (`/api/v1`, 171 operations).
10. (v4) Protects engineering work and approvals: lost-update detection on Drafts (FR-090), reference and duplicate warnings with a Where-used list (FR-091 … FR-093), password confirmation for sensitive actions (FR-126), delegation for absences (FR-125), SLA ageing with reminders (FR-120, FR-121), change sets, field comments, review checklists and evidence (FR-124, FR-127 … FR-130).
11. (v4) Publishes trustworthily: export profiles, audience-, language- and software-version-scoped exports, manifests with SHA-256 hashes (optionally signed) and ZIP bundles, release notes, an offline lookup page, translations, planned dates (FR-100 … FR-111).
12. (v4) Helps with the migration and with insight: import template, dry run, update mode and reconciliation (FR-048i … FR-048n); catalogue health and capacity forecast, KPIs, saved searches, fleet feedback, Hub-to-Main consolidation (FR-123, FR-140 … FR-147).
13. (v4) Operates safely on-premises: verified backups, a hash-chained audit log, readiness and metrics endpoints, scoped API tokens, a change feed and signed webhooks, installers, systemd units and CI definitions (FR-144 … FR-153, NFR-009 … NFR-014).

1.3 What it deliberately does not do

Not in scope	Reason / reference
Single sign-on with Active Directory/ADFS	Phase 2; the AuthProvider seam is built (FR-005, SEC-005)
Any cloud service — hosting, identity, storage, telemetry, CDN	Hard constraint C-004a, NFR-008
Configurable approval chains	The three-decision model is fixed (16-mvp-roadmap.md §2)

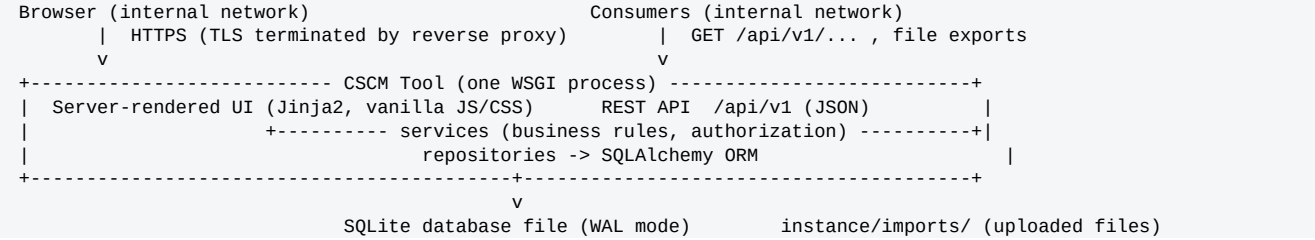
Not in scope	Reason / reference
Push integration into SCADA or documentation tools	Consumers pull via API, change feed or signed webhooks on the internal network (A-004, FR-144 ... FR-146)
Mobile-optimised layout; languages beyond the nineteen of FR-157	Excluded; see 21-improvement-ideas.md
Running on PostgreSQL	Only prepared (inventory and ratchet, NFR-014); not done
Bulk import that bypasses review	Reversed C-005: import is allowed but fully governed (FR-048c)

1.4 Intended users

Six roles (SEC-012 default is Viewer): **Administrator** (Product Owner / Component Owner Controls Software, the highest level, shown red), **StCd Owner** (owner of the library: read-only oversight and role changes among Viewer, Engineer and Reviewer), **Engineer**, **Reviewer** (Sub-PO for Status Codes), **Chief Engineer**, **Viewer**. Engineers may additionally hold one or more **Engineering Domain tags** (Controls, Electrical, Mechanical, Grid, Safety) that determine which cross-domain sign-offs they can give. Detailed role guidance is in the user manual.

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2. System context



- Everything runs on the organisation's own servers; there is **no outbound dependency** (NFR-008). A test runs the whole lifecycle with all sockets blocked.
- The UI shares the **OneTool visual shell** (top bar, sidebar, tokens) but keeps its **own login and credential store** (SEC-006).

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3. Technology

Layer	Choice	Notes
Language / framework	Python 3.12, Flask 3.1	Constraint C-003
Persistence	SQLite (WAL), SQLAlchemy 2.0, Alembic migrations	C-001; repository pattern keeps PostgreSQL possible (NFR-004)
UI	Jinja2 templates, hand-written CSS and vanilla JavaScript	No framework, no CDN; enforced by a Content-Security-Policy
Passwords	Argon2id (argon2-cffi)	SEC-001
CSRF	Flask-WTF synchroniser tokens	NFR-003
Exports	built-in csv/json/zipfile/hashlib/hmac, openpyxl (XLSX), fpdf2 (PDF)	Pure Python, no system packages
Search	SQLite FTS5 (optional, with substring fallback)	FR-148
Serving	Waitress (Windows) or any WSGI server behind a TLS-terminating proxy	14-deployment-architecture.md
Quality	pytest, coverage, schemathesis/hypothesis, black, ruff, mypy, bandit, pip-audit	15-development-standards.md, 13-test-strategy.md §11
Packaging (v4)	deploy/ installers (PowerShell, bash), systemd units, scripts/serve.py, scripts/ci.py, CI definitions for Gitea / GitLab	14-deployment-architecture.md §4a

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4. Architecture

4.1 Layering

```
app/
  api/v1/      JSON endpoints (routes only: parse request, call a service, serialise); one module per area
  ui/          Server-rendered views (same services as the API); one module per area
  services/    All business rules and authorization ? the only place rules live (v4: publication,
               governance, import, integration and operations services, see 08-system-architecture.md §2)
  repositories/ Query encapsulation over the ORM (status codes, releases, change requests, FTS)
  models/      SQLAlchemy tables (mirror docs/artifacts/schema.sql; operations.py = v4 tables)
  domain/      Cross-cutting domain code: errors, lifecycle state machine, role checks
  export/      CSV / JSON / XLSX / PDF / offline bundle / full-database writers
  auth/        Password hashing, session loading, rate limiting (database-backed), delegation-aware decorators
  audit_chain.py audit_hash.py   hash chain, verification, archive (v4)
  observability.py logging_config.py   metrics and JSON request log (v4)
  i18n.py i18n_de.py i18n_extra.py help_content.py   catalogues (German complete; French, Hindi, Georgian, Arabic an
  d ten more for the main strings) and help text
  cli.py       flask jobs (create-admin, backup-db, verify-backup, send-reminders, audit-verify, audit-archive, veri
  fy-export)
  templates/, static/   UI
migrations/    Alembic (baseline applies schema.sql; follow-on migrations are guarded and idempotent)
tests/         unit, integration, system, performance, fixtures
scripts/       generate_test_report.py, ci.py, serve.py, pg_readiness.py, gen_api_summary.py, gen_translations_doc.p
  y
deploy/        linux/ (installer, systemd units, nginx sample), windows/ (installer, task scripts)
```

Rules that follow from the layering (SEC-010, `15-development-standards.md`): routes never contain business logic; role checks exist **both** in route decorators and in the service layer (`require_roles`), so a future caller cannot reach a privileged operation by omission; the UI and API call the same service methods, so behaviour cannot diverge.

4.2 Request handling

1. `before_request` assigns a request id (written to every audit entry).
2. The auth decorator loads the user from the signed session cookie and re-validates, on **every request**: idle timeout (30 min), absolute lifetime (12 h), account still active, password unchanged since login (so deactivation and forced resets take effect immediately).
3. CSRF is checked for state-changing requests (UI via form token, API via `X-CSRFToken`).
4. The service performs the action in **one database transaction** and writes the audit entry in the same transaction.
5. `after_request` adds the security headers (§9.3).

4.3 Logging

Structured JSON, one object per line on standard error and, with `LOG_FILE`, in a rotating file (`app/logging_config.py`, `11-security-architecture.md` §11): timestamp, level, request id, method, path, status, duration, authenticated user id, client address; unhandled exceptions with the stack trace and request id. Credentials, session tokens, request bodies and query strings are never logged. Logs are diagnostics; the audit log is the business record.

4.4 Transaction model

(v4) Several worker processes may serve one database file: the rate limit, the audit chain and number allocation are all coordinated through the database write lock, and background imports report through their job row (NFR-009, 08-system-architecture.md §8). SQLite is single-writer. Writers start with `BEGIN IMMEDIATE` (race-safe identifier allocation, FR-021). Read-only requests (`GET/HEAD/OPTIONS`, except endpoints marked `@writes_on_get` such as the exports, which record an export job) use a deferred `BEGIN`, so in WAL mode readers do not block and are not blocked by a writer (NFR-004). `busy_timeout` is 15 s so writers queue rather than fail.

---

5. Domain model

5.1 Core entities (tables in docs/artifacts/schema.sql)

Entity	Purpose
status_code	The identity of a code: identifier (StCd-XXXXX, or StCd-T##### for sandbox), Functional System Group + mandatory Subgroup, owner, legacy reference, sandbox flag. Never deleted unless sandbox (trigger-enforced).

Entity	Purpose
status_code_revision	A versioned set of descriptive/technical fields plus the three decisions' columns, lifecycle state, and the single is_current flag. Locked by trigger once Approved/Released/Archived.
status_code_revision_platform	Revision <-> Turbine Platform (2XM/3XM/4XM), multi-select.
change_request	The workflow carrier for one revision: type (NEW / REVISION / DEPRECATION / IMPORT), requester, state, optional assigned Reviewer.
review_comment	Role-tagged decisions and plain comments.
domain_signoff, domain_signoff_request	Cross-domain sign-offs and the requests for them.
release, release_item	A named, versioned bundle; immutable once published.
import_job	One library upload with counts and error report; (v4.2) ST for an import from generated programs, with the batch change_set_id.
(v4.2) import_source_item	One imported program: file, SHA-256, program number and every parsed attribute as JSON.
audit_log_entry	Append-only action log.
export_job	Record of each export.
user, user_engineering_domain, role	Accounts, domain tags, the six roles.
user_password_history	Hashes of retired passwords for the last-5 reuse block.
(v4) tables	21 more: software versions and Release targets, languages and translations, export profiles, change sets, delegation, notifications, settings, rate-limit counters, evidence, field comments, review checklist and confirmations, saved searches, occurrence feedback, API tokens, webhooks and deliveries, Hub-to-Main mapping — field detail in 06-data-dictionary.md §13; the technical nomenclature (terminology_term, terminology_translation, v4.9); 65 tables in total (v4.2 adds import_source_item, status_code.reserved_identifier and the ST import source).
lookup_* (13 tables)	Controlled vocabularies: 12 Functional System Groups, 38 Subgroups, Turbine Platforms, Engineering Domains, Available Groups (23 codes), and eight audience-access vocabularies. Each has an optional description used as inline help; the eight audience-access vocabularies also carry audience_visible, and each group carries controller_type (MAIN/HUB) (v4).

5.2 Identifier scheme

`StCd-XXXXX` — five digits. The number falls in the range of the code's **Functional System Group** and, within it, the **Subgroup's sub-band** (for example Converter & Grid Interface WCNV 1000–1999, subgroup Converter 1000–1199: the first subgroup of every Main Controller group owns the group's first 200 numbers, later subgroups 100 each). Twelve groups exist: nine Main Controller groups (1000–9999) and three Hub Controller groups (11000–13999; `WTUR` and `WROT` are intentionally reused with different ranges, so a group is always resolved by id or range, never by prefix alone). The ranges 10000–10999, 14000–14999, 15000–15999 and 16000–16999 are reserved free capacity that an Administrator can assign to a new group without a deployment (FR-048g/h).

Rules: an identifier is **assigned once, at Submit for Review**, never while Draft (FR-021); a Draft shows "Not yet assigned". It never changes (BR-001) — a rejected-and-resubmitted CR and every later revision keep it. Numbers are not reused. An Administrator may submit with a manual identifier for a documented migration or reservation case, which is validated (format, sub-band, unused), requires a reason, and is audited (FR-021a). Sandbox codes draw from a separate `StCd-T#####` counter that never overlaps a real range (FR-053).

5.3 Field set of a revision

Grouped as in the data dictionary (`06-data-dictionary.md`): Classification (group, subgroup, turbine platforms, legacy reference); Description (title <= 120 characters, description 10–4000, software version); Behaviour (status category ERROR/WARNING/INFO, available group, alarm flag, set delay and reset delay such as `10ms` or `2min`, up/down counter, trigger snapshot, loadless spinning permitted); Repeated-error escalation (times repeated, window, escalation code); Program references (eight numeric references: yaw, manual reset, auto reset, converter reactive power, brake, converter programs GSC/MSC, yaw-bearing lubrication); Audience access (Development, Sales, TCC, Service, Turbine Operator Package, Grid Operator, Service Partner, Customer — eight independent values).

5.4 Ownership of fields and cross-domain sign-off

Title, description, category and available group belong to no domain. The timing, repeated-error and program-reference fields belong to **Controls**. If the originating Engineer does not hold the Controls tag, a Controls Engineer must record a sign-off before the CR can be submitted (FR-027, FR-029, BR-011). The originating Engineer

can send an explicit **sign-off request**, optionally addressed to a named Engineer; it appears on that person's Dashboard and My Change Requests until signed (FR-028). The audience-access fields are governed separately (below).

(v4) A revision also carries the optional remedy text, affected component, related codes and standard references, an inclusive software-version range, planned effective/deprecation dates, and `row_version` / `updated_at` for lost-update detection (06-data-dictionary.md §2).

5.5 Restricted audience-access fields

The Engineer may propose the eight audience-access values while drafting. They are **not authoritative** until the Administrator's Review sign-off confirms them; at that moment the Administrator can change any of them in the same action, and the change is audited (FR-025a, FR-034, BR-012, SEC-013). Until then the UI marks them "Proposed".

---

6. Lifecycle and workflow

6.1 State machine

```
        submit                  both sign-offs               Chief Engineer              publish
 DRAFT --------> REVIEW -------------------> PENDING_APPROVAL -----------> APPROVED ---------> RELEASED
   ^                |  reject (comment)              |  reject (comment)       |
   +----------------+--------------------------------+          (deprecation CR approved)
                                                    PENDING_APPROVAL -----> DEPRECATED ---> ARCHIVED
                                                                                | reinstate (Administrator, reaso
  n)
                                                                                +------------> RELEASED
```

The nine permitted transitions are defined once in `app/domain/lifecycle.py` and enforced for every assignment to a revision's state; any other transition raises an error (BR-004). Tests check all 49 state pairs.

6.2 The three decisions

Step	Who	Rules
Reviewer sign-off	Reviewer	Not the author (BR-005); if a Reviewer is assigned, only that Reviewer (FR-039). Rejection needs a comment and returns the revision to Draft, clearing both sign-offs.
Administrator sign-off	Administrator	Not the author; may confirm/override audience-access values; rejection as above.
Chief Engineer approval	Chief Engineer	Not the author, and not whoever gave either earlier sign-off on that CR (BR-005). Rejection needs a comment.

(v4) Each decision asks for the actor's password (step-up, SEC-026); a delegate with an active delegation may take a decision as themselves, with the same checks and the audit entry naming both people (FR-125, BR-017). A Chief Engineer may decide several requests of a change set in one sitting — each by its own check, comment and audit entry (FR-124).

The first two may happen in either order; the revision moves to PendingApproval **automatically, in the same transaction as the second sign-off** (FR-035). Segregation of duties is enforced in the service layer, by CHECK constraints on the table, and tested at all three points through the API and the UI.

6.3 Change request types

- **NEW** — a brand-new code. **REVISION** — copy-on-write of an existing code (editing in place is impossible after Draft, FR-025); only one open revision per code at a time (BR-003). **DEPRECATION** — opened against a Released code with a mandatory reason; runs the full three-decision cycle and lands the revision in Deprecated. **IMPORT** — a Draft created by a library import.
- An author can **withdraw** a CR in Draft, Review or PendingApproval (FR-026). Reviewers, Administrators and Chief Engineers can add plain **comments** that change nothing (FR-038). Administrators can **assign** the Reviewer (FR-039).

6.4 Releases

An Administrator creates a Release (name, version label, scope filter by group/subgroup), reviews the **candidates** (Approved, non-sandbox revisions matching the scope), chooses the included set, and **publishes**. Publishing moves every included revision to Released, stamps the effective date and freezes the Release — atomically (FR-040–FR-044). A Release can be deleted only before publishing (BR-008). (v4) Before publishing, target software versions may be set, which limit the candidates to revisions whose version range includes a target (BR-021); publishing needs the Administrator's password and warns about future planned dates; the published Release has release notes, a manifest and bundle exports, an offline lookup page, and fires the `release.published` webhooks after the commit. Sandbox revisions are excluded by the service and by a database trigger (FR-048).

6.5 Deprecation and archival

Deprecating a Released code is a CR with the full cycle (FR-050). An Administrator can archive a Deprecated code once the retention period (default 180 days, counted from the DEPRECATE audit entry) has passed (FR-051, BR-006), or reinstate it with a reason. Archived codes are read-only.

6.6 Sandbox codes

Administrator-only test codes with a `StCd-T#####` identifier, excluded from default search, exports and releases. They can be permanently deleted — the only deletable record type (database trigger) — and the deletion writes an audit entry containing a full snapshot of what was removed (FR-055, FR-056).

---

7. Functional description by area

Area	Behaviour	Requirements
Authentication	Username or e-mail plus password; session cookie HttpOnly, Secure, SameSite=Strict; 30-minute idle and 12-hour absolute expiry; five failed logins lock the account 15 minutes; per-IP and per-username throttling; logout invalidates immediately.	FR-001–FR-004, SEC-002–SEC-004
Users and roles	Administrators create/edit/deactivate users, assign exactly one role and domain tags, force password resets; the last active Administrator cannot be removed; passwords need 12+ characters, must not be on the block-list (seed_data/common_passwords.txt, replaceable with an internal corpus) and must differ from the user's last 5 passwords; default role is Viewer.	FR-010–FR-014, SEC-012, 11 §4
Search and browse	Full-text over identifier, title, description; filters for group, subgroup, platform, state, owner, sandbox (excluded by default); sorting; pagination (default 50, max 200). Viewers see only Approved/Released/Deprecated/Archived data and keep seeing the last Released revision while a newer Draft is open.	FR-060–FR-064, BR-009
Revision history and diff	All visible revisions; field-level comparison of any two; the CR page shows the differences from the last decided revision, or marks a new code.	FR-032, FR-064
Exports	Per Release; all current Released codes (optionally filtered); an unapproved working set for the StCd Owner and the Administrator only (FR-175), labelled "Not for distribution – unapproved data" on every row, in the file name and on the PDF; all as flat tables in CSV, JSON, XLSX, PDF; cells that could be read as formulas are neutralised. Full-database backup is Administrator-only.	FR-045–FR-047, SEC-015, SEC-023
Library import	CSV/XLSX upload (extension allow-list, 20 MB cap); each row validated like manual entry; valid rows become Drafts, invalid rows create nothing and appear in a downloadable error report; legacy numbers are kept in a separate field.	FR-048a–d, SEC-024
Lookup administration	Administrators add/relabel/describe/deactivate vocabulary values; create Subgroups and whole Functional System Groups in free ranges; overlaps are refused (service and database trigger).	FR-048e–h
Audit log	Search by entity, actor, action, date. Administrators see everything; Reviewers and Chief Engineers a scoped subset. Entries cannot be changed or removed (database triggers).	FR-070–FR-073
Dashboard	Role-specific counters and shortcuts (open CRs, requested domain sign-offs, awaiting sign-off/approval); since v4 also unread notices, stalled requests and backup state for Administrators.	S2
(v4) Concurrency and soft checks	Stale Draft saves refused (409) with the user's entries kept; warnings for unknown/retired references and similar titles; Where-used; platform variants; planned dates.	FR-090 … FR-094, FR-109
(v4) Workflow aids	SLA chips and stalled list; in-app notices and optional mail; delegation; change sets; field comments; review checklist; evidence; reviewer balancing; KPIs.	FR-120 … FR-131

Area	Behaviour	Requirements
(v4) Publication	Export profiles; audience, language and version scoping; wrapped PDF; manifests, bundles and verification; release notes; offline bundle; translations; software-version matrix.	FR-100 … FR-111
(v4.9 – v4.10) Menu, program fields, deletion	A menu grouped by the work of the user with counts (FR-169); the fields of the generated programs stored, edited, imported and exported (FR-168); users with history and withdrawn codes erased in place with the audit log untouched (BR-025, BR-026).	FR-168, FR-169
(v4.13) Bulk workflow	The workbench of a change set: readiness of the Drafts, bulk edit, bulk submit, bulk domain sign-off and the bulk decisions of the three decision roles - each code through its own checks, a decision on more than a handful of codes only after a sample inspection by the person who decides - and the dossier (XLSX) for an auditor.	FR-180
(v4.12) Connections	What suppresses a code, what it requires and where it escalates, read from the set condition and the repeat code of the imported programs; the code page shows them, the page Connections shows the causes as a tree, the escalation targets and what looks wrong.	FR-179
(v4.10) Baseline and package	A Release export is the complete baseline of every code valid as of it, narrowed by platform and controller; one package holds every language for every access level with release notes and a manifest; everybody downloads released data, only the StCd Owner and the Administrator what is not released.	FR-173 … FR-175
(v4) Import toolkit	Template, dry run, update mode, reconciliation, XLSX error report, background job with progress.	FR-048i … FR-048n
(v4.2) Status code import from programs	The generated StCd_.st programs (several files, a folder or a ZIP) are read as text, previewed per program with multi-select, and imported as ordinary Drafts that reserve the number the program carries and join one change set; Hub numbers are translated through the numbering table; every code keeps its source file, SHA-256 and parsed attributes; the legacy -> new mapping is listed and exported (one direction only). Uploads are size-limited, private to the uploader and expire after 24 hours.	FR-048o … FR-048w, BR-023, SEC-031
(v4) Insight and integration	Catalogue health and capacity; saved searches and shareable views; fleet feedback; API tokens, change feed, webhooks; Hub-to-Main consolidation; ranked, typo-tolerant search.	FR-140 … FR-148
(v4) Operations	Verified backups; audit hash chain, verification and archive; readiness and metrics; settings; German UI and help.	FR-150 … FR-155
(v4.7 – v4.9) Languages	Nineteen interface languages, complete in all of them (every template text, flash message and service-layer sentence, 1,093 strings, FR-167); a language per user; the review of the translations by native speakers in the application; the PDF in every script; the technical nomenclature as a vocabulary of its own with a two-person approval and a consistency check against the interface; the Windows sign-in with a test gateway for a real Windows identity; the beta mark of unreviewed languages and the export text in the language of the export.	FR-155, FR-157, FR-162 … FR-171

7.1 Audit events

One hundred and one action codes can be written (`12-audit-compliance.md` §2; a test keeps the catalogue complete) — among them `CREATE`, `EDIT`, `SUBMIT_FOR_REVIEW`, `REVIEWER_SIGNOFF`, `ADMIN_SIGNOFF`, `CHIEF_ENGINEER_APPROVE/REJECT`, `DOMAIN_SIGNOFF(_REQUEST)`, `REVIEWER_ASSIGN`, `IDENTIFIER_OVERRIDE`, `RELEASE_CREATE/ITEMS_SET/PUBLISH/DELETE`, `DEPRECATE`, `REINSTATE`, `ARCHIVE`, `SANDBOX_CREATE/DELETE`, `CATALOGUE_EXPORT`, `FULL_DATABASE_EXPORT`, `IMPORT_STARTED/COMPLETED/FAILED`, `LOOKUP_CHANGE`, `SUBGROUP_CREATE`, the user and authentication events. Each entry records actor, UTC time, entity, before/after values, client address and request id, and is written in the same transaction as the change. Decision and comment entries include the comment id and full text; a withdrawal keeps a snapshot of the deleted discussion, so review comments are never lost.

8. User interface

Fifty screens (S1–S50, `10-ui-ux-specification.md`; the nineteen of v3 are listed first, the v4 screens — notices, delegation, change sets, stalled requests, translations, release notes and shaped export, software versions, catalogue health, success metrics, fleet feedback, Hub consolidation, export profiles, review checklist, settings, integrations, operations and help, and in v4.2 the status code import from `.st` files and the legacy mapping — are added in the sidebar groups Workflow, Releases, Insight and Administration): Login; Dashboard; Status Code list; Status Code detail (Current / History / Deprecation tabs); New Status Code; New Sandbox Code; Edit Draft; My Change Requests; Review Queue; Approval Queue; Change Request detail/decision (diff, sign-off strip,

cross-domain strip, comments, role-specific actions); Release list/new/builder/detail; User administration; Lookup administration (including Functional System Group and Subgroup creation); Audit log; Export Center; Account settings; Library Import.

Design rules implemented: sidebar navigation that adapts to the role; every validation failure re-displays the form with **all entered values intact** (NFR-007a); each lookup-backed dropdown shows the selected option's description as inline help, or an explicit "no description yet" note (NFR-007b); destructive actions use a confirmation dialog or a typed confirmation; no inline scripts, so the Content-Security-Policy can forbid them.

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9. Interfaces

9.1 REST API

Versioned under `/api/v1`, JSON, session-cookie authentication plus `X-CSRFToken` on state-changing calls (`GET /auth/csrf-token`). The contract is `docs/artifacts/openapi.yaml` (v4.0.0, 171 operations; since v4 read-only machine consumers may also use `Authorization: Bearer cscm_?` tokens on the endpoints that declare a scope) and is checked against the running code by an automated test in both directions. Groups: auth and users; status codes and revisions (including diff); change requests (submit, the three decisions, withdraw, comments, reviewer assignment, domain sign-offs and requests); imports; releases; exports; lookups; audit log; health. Errors use `{"error": , "message": }` with 401 (not signed in), 403 (role or segregation of duties), 404, 409 (state conflict), 422 (validation, with a `field_errors` list), 429 (throttled).

9.2 Files

Imports: CSV or XLSX with a header row; columns include `functional_system_group`, `functional_subgroup`, `turbine_platforms` (semicolon-separated), `title`, `description`, `status_category`, `available_group`, `set_delay`, `reset_delay`, the optional behaviour/program/audience columns, and `legacy_reference_number`. Exports: one row per status code, every field a column. Uploaded files are stored under `instance/imports//` (outside any served path) and never served back to a browser. (v4.2) Status code programs: `.st`, `.scl`, `.exp`, `.txt` or one `.zip` (2 MB per file, 2,000 files and 50 MB unpacked per upload) through `POST /imports/st/preview` and `POST /imports/st`; the preview is kept under `instance/imports/staging/` and the confirmed files under `instance/imports//`.

9.3 Security headers

Every response carries a self-only Content-Security-Policy (no inline script, no framing), `X-Content-Type-Options: nosniff`, `X-Frame-Options: DENY`, `Referrer-Policy: same-origin`, `Cache-Control: no-store` (except static files), and HSTS on HTTPS requests (SEC-025).

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10. Security summary

Control	Implementation
Authentication	Argon2id hashes; lockout; throttling; per-request session re-validation
Authorization	Six roles; checked at the route and again in the service; domain-tag checks for cross-domain sign-off
Segregation of duties	Service checks + table CHECK constraints + tests at all three decision points
Data integrity	Foreign keys on; unique/partial indexes (one current and one open revision per code); triggers for locked revisions, sandbox-only deletion, release immutability, range overlap, append-only audit
Injection / XSS / CSRF	ORM-bound parameters only (bandit clean); Jinja auto-escaping; CSRF tokens; formula neutralisation in exports
Transport / headers	TLS at the reverse proxy; CSP and companion headers
Supply chain	Pinned requirements; pip-audit on every report run; static check that no cloud SDK or external asset is referenced
Confidentiality of exports	Role-scoped; working sets are watermarked; full-database export Administrator-only, password-confirmed and audited; audience exports disclose only what the Administrator confirmed as visible (v4)
(v4) Deliberate approvals	Step-up password; delegation limited to decisions, under segregation of duties
(v4) Tamper evidence	Hash-chained audit log with anchor and weekly verification; signed export manifests; verified backups
(v4) Machine access	Hash-only, scoped, expiring, read-only API tokens; webhooks to internal addresses only, signed, no redirects

Control	Implementation
(v4) Uploads	Allow-listed, content-checked, size-limited, stored outside the web root, download-only (evidence, imports)

11. Configuration and operation

Setting	Source	Default
FLASK_ENV	environment	development (production for live use)
SECRET_KEY	environment	required in production: at least 32 characters, not the development default and not the .env.example placeholder. A production instance refuses to start otherwise (session cookies are signed with it)
DATABASE_URL	environment	sqlite:///instance/cscm.db
Session idle timeout / absolute lifetime	app/config.py	30 minutes / 12 hours
Lockout	app/config.py	5 failures, 15 minutes
Upload size	app/config.py	20 MB
Archive retention	application constant	180 days
(v4) STEP_UP_REQUIRED	app/config.py	true
(v4) MANIFEST_SIGNING_KEY, AUDIT_HMAC_KEY	environment	unset (generated by the installers; recommended in production)
(v4) BACKUP_DIR, BACKUP_MAX_AGE_HOURS	environment	unset / 36
(v4) CSCM_INSTANCE_PATH, LOG_FILE	environment	./instance / unset
(v4) CSCM_HOST, CSCM_PORT, CSCM_THREADS, CSCM_TRUSTED_PROXY	environment (scripts/serve.py)	127.0.0.1, 8000, 8, unset
(v4) METRICS_ALLOWED_CIDRS, METRICS_TOKEN	environment	loopback and RFC 1918 / unset
(v4) SMTP_HOST, SMTP_PORT, SMTP_SENDER, SMTP_STARTTLS, SMTP_USERNAME, SMTP_PASSWORD	environment	mail off
(v4) IMPORT_ASYNC_ROW_THRESHOLD	environment	300 rows
(v4.13) the sample of a bulk decision (bulk.sample_percent, bulk.sample_minimum, bulk.sample_above)	Administration -> Settings (database)	10 % of the batch, at least 8 codes, none for 5 or fewer
(v4) SLA thresholds, reviewer.auto_assign, capacity alert	Administration -> Settings (database)	review 3/5, approval 2/3 days; off; 15 % / 20 numbers

(v4) Preferred bring-up and upgrade: the installer (`deploy/linux/install.sh` or `deploy/windows/install.ps1`, 14-deployment-architecture.md §4a), which does everything below, generates the keys, takes a verified backup before migrating and registers the web server, the nightly backup, the reminders and the weekly audit check. Manual bring-up on a new server:

```
python -m venv .venv && .venv\Scripts\activate
pip install -r requirements.txt
flask db upgrade            # creates the schema and seed vocabularies
flask create-admin          # the one place a first Administrator can be created
python scripts/serve.py                          # Waitress; behind the TLS reverse proxy
flask backup-db                                  # first verified backup (also scheduled)
```

Upgrading an existing installation: take a verified backup (`flask backup-db`), deploy the new code, run `flask db upgrade` (migrations are idempotent and reversible), run the smoke checks in the test report. Backup, TLS, file permissions and egress blocking are deployment responsibilities described in `14-deployment-architecture.md`.

12. Quality attributes (measured)

Attribute	Target	Measured (see 24-test-report.md §4)
Search latency, 20,000 revisions	p95 <= 500 ms	~ 110 ms (ranked FTS search)
Single-record read	p95 <= 200 ms	~ 4 ms
Export of 5,000 records	CSV/JSON <= 5 s; XLSX/PDF <= 15 s	~ 0.5–0.7 s; ~ 1.7 s / 3.1 s
50 concurrent users	keep the search target	~ 72 ms p95
Identifier allocation under contention	unique, gap-free	verified with 8 parallel writers

Attribute	Target	Measured (see 24-test-report.md §4)
Maintainability	>= 80% coverage of service code; clean static analysis	92% line coverage overall; black, ruff, mypy, bandit clean

13. Known limitations and open items

1. Descriptions for lookup values (23 Available Group codes, eight program-reference fields, audience-access values) must be supplied by Controls Engineering; until then the inline help says so. The meaning of the value `Covert` is unconfirmed; it is treated as not visible (D-02).
2. The PDF prints every script of the offered languages (v4.7, FR-165) with bundled Noto fonts, but only in the regular weight (headings are not bold) and Urdu in Nastaliq needs a taller line than the table uses, so its tall letters can touch the cell border; a character that no bundled font draws is replaced by `?`.
3. The interface is translated completely into all nineteen languages (v4.9, FR-167), but the 18 translations besides English are proposals written by the maintainer's tooling that no native speaker has checked yet (an AI pre-check estimates that 78 % to 95 % of the strings would be accepted unchanged, FR-177, `26-translations.md` §8c; it is not a review); the messages of the API and the user manual stay English; the words around an export follow the language of the export (FR-171).
4. Business days have no holiday calendar.
5. Manifests are signed with a shared secret (HMAC); there is no asymmetric signature, so a verifier needs the same key.
6. Single-node SQLite bounds write throughput; the repository layer and the ratchet keep a PostgreSQL migration possible, but the application has never been run on PostgreSQL (Phase 3).
7. Background imports run in a thread of the web process; a restart during a large import leaves the job in its last recorded state and the file must be re-uploaded.
8. Business UAT sign-off and the production deployment checks (real installation, TLS, file permissions, blocked egress, restore drill, key custody, screen-reader pass) are outside what automated tests can prove (20-traceability-matrix.md §6).
9. Packages are built by a thread of the web process (as the imports, item 7) and their files stay under `instance/packages` until someone removes them; there is no Word output yet (I-43, I-66), and anybody signed in may start a build (see I-68).

23. User Manual — CSCM Tool

Controller Status Code Management Tool — for the people who create, review, approve, release and look up wind-turbine controller status codes. This is a separate artefact from the requirements, the software description and the test report. It describes the application as delivered on 2026-10-23 (version 4; sections and rows added in v4 are marked (v4)); where behaviour is not yet available it says so.

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2. Getting started
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25. Software versions
26. Insight: health, metrics, fleet feedback, saved searches
27. Hub Controller to Main Controller consolidation
28. Administrator: settings, integrations and operations
29. Help inside the tool

1. What the tool is for

Every status code of the Main and Hub Controllers (faults, warnings, information) lives in this tool and nowhere else as the master. A code is written here, completed by the right specialists, checked by two independent people, approved by the Chief Engineer, and only then **released**. Manuals, specifications and SCADA lists are produced from released data.

A code travels through seven states:

State	Meaning
Draft	Being written. No identifier yet. Freely editable by its author.
Review	Submitted. Has its StCd-XXXXX number. Waiting for the Reviewer and the Administrator.
Pending Approval	Both reviews done. Waiting for the Chief Engineer.
Approved	Chief Engineer agreed. Waiting to be included in a Release.
Released	Published in a Release. Locked. This is the version the world sees.

State	Meaning
Deprecated	Obsolete; kept for history.
Archived	Deprecated long enough; read-only.

2. Getting started

1. Open the address your administrator gave you (an internal web address).
2. Sign in with your **username (or e-mail) and password**. This tool has its own login; it does not use your OneTool sign-in. (v4.7) Where your organisation has switched on the **Windows sign-in**, you are signed in with your Windows account: open the address from your company computer, or use Sign in with your Windows account on the login page; there is no password to type. The account is found by your e-mail address, so an Administrator must have created your account with it; if the page says that no active account matches, ask an Administrator.
3. After **5 wrong passwords** the account is locked for **15 minutes**. After **30 minutes without activity**, or **12 hours** after signing in, you must sign in again.
4. Passwords need at least 12 characters, must not be a common password, and must differ from your last 5 passwords. Change yours under **Account** (§17).
5. A new account starts as a **Viewer**. Ask an Administrator to give you the role you need.

3. Finding your way around

- **Top bar:** your name and role (and, (v4), a chip when you are covering for a colleague), the **Notices** bell with the number of unread notices, the **language** switch (nineteen languages from GB English to GE ქართული, each with its country letters and flag; the list scrolls; Georgian is last), **Account**, **Log out**.
- **Menu on the left** (it shows only what your role may use). A **+ New Status Code** button (Engineer, Administrator) sits on top, then Dashboard, then groups you can open and close (the groups you closed stay closed): **Catalogue** (Status Codes, Releases, Export Center, New Sandbox Code for an Administrator), **Workflow** (My Change Requests, Review Queue, Approval Queue, Change Sets, Delegation), **Import and Migration** (Library Import, Import from .st files, Legacy mapping, Mapping, Mapping changes, Hub Consolidation, Withdrawn Codes), **Insight** (Catalogue Health, Stalled Requests, Success Metrics, Fleet Feedback), **Languages** (Terminology, Translations) and **Administration** (People: Users; Configuration: Lookups, Export Profiles, Review Checklist, Settings, Integrations; Operations: Operations, Audit Log); **Help** is at the bottom. A number next to a link says how many items wait for you (reviews, approvals, stalled requests, withdrawn codes; at My Change Requests your own open requests); a closed group still shows the total.
- **Dashboard:** counters for the work waiting for you, with links.
- On narrow windows (below 768 px) the menu is hidden until you press **Menu** in the top bar; **Esc** closes it.

4. Who can do what

	Viewer	Engineer	Reviewer	Chief Engineer	Administrator	StCd Owner
Search and read Approved / Released / Deprecated / Archived codes	yes	yes	yes	yes	yes	yes
See Drafts and codes under review	—	yes	yes	yes	yes	yes
Create a code, edit own Draft, submit, withdraw	—	yes	—	—	yes	—
Give a cross-domain sign-off	—	yes (own domain)	—	—	—	—
Reviewer sign-off	—	—	yes	—	—	—
Administrator sign-off (and confirm audience access)	—	—	—	—	yes	—
Chief Engineer approval	—	—	—	yes	—	—
Comment on a change request	—	—	yes	yes	yes	—
Assign the Reviewer	—	—	—	—	yes	—
Build, publish, delete (unpublished) releases	—	—	—	—	yes	—
Archive / reinstate deprecated codes; sandbox codes	—	—	—	—	yes	—
Library import	—	yes	—	—	yes	—
Export released data, build and download the package of a published Release (v4.10)	yes	yes	yes	yes	yes	yes
Export the unapproved working set (and the files of an unpublished Release)	—	—	—	—	yes	yes

	Viewer	Engineer	Reviewer	Chief Engineer	Administrator	StCd Owner
Full-database backup (password required)	—	—	—	—	yes	—
Confirm a decision with your password (v4)	—	—	yes	yes	yes	—
Name a delegate / act as a delegate (v4)	—	—	yes	yes	yes	—
Attach evidence to own change request (v4)	—	yes	—	—	yes	—
Comment on a field, confirm the review checklist (v4)	—	—	yes	yes (comment)	yes	—
Change sets: create / decide (v4)	—	yes / —	— / yes	yes / yes	yes / yes	—
Translate / approve a translation (v4)	read approved	yes / —	— / yes	—	yes / yes	—
Publish with audience, language, profile; release notes; offline page (v4)	read	read	read	read	yes	—
Catalogue health	—	—	yes	yes	yes	—
Success metrics, fleet feedback	—	—	—	yes	yes	—
Settings, profiles, checklist, API tokens, webhooks, backups, audit verification (v4)	—	—	—	—	yes	—
Users, vocabularies	—	—	—	—	yes	—
Audit log	—	—	scoped	scoped	yes	—
Create users; change any role	—	—	—	—	yes	—
Change a role between Viewer, Engineer and Reviewer	—	—	—	—	yes	yes

Nobody decides their own work. The author of a change request can never give the Reviewer sign-off, the Administrator sign-off or the Chief Engineer approval on it, and the Chief Engineer also cannot be the person who gave either earlier sign-off (§18).

5. Looking up status codes (everyone)

1. Choose **Status Codes**.
2. **Search** by identifier (e.g. `StCd-01042`), words in the title, or words in the description.
3. Narrow with **Functional System Group**, **Lifecycle** and the other filters, then **Filter**. (v4) Search is ranked (a word in the title ranks above one in a long description) and forgives small spelling mistakes ("convertor"); click a column heading to sort; use **Columns** to choose what the table shows; **Save search** keeps the filters and columns under a name, and copying the page address shares exactly this view.
4. Open a row to see the detail page with three tabs:
   - **Current** — all fields, grouped (classification, status and timing, repeated-error escalation, program references, audience access). (v4.9) Below them: **Further Program References** (the subsystem programs of the controller of the code), **Operation Parameters** (operating-mode mask as a bit string such as `111011`, task call frequency, flags) and **User-Group Levels** (read and write level per slot, as the generated program holds them).
   - **History** — every revision you may see. To compare two, use **Compare Revisions**: pick From and To and choose **Compare**; changed fields are listed with the old and new value.
   - **Actions** — what you may do with the code (revise, deprecate, create a platform variant (**Create variant**), …). (v4) The detail page also shows **Where-used** (other codes that refer to this one), the **family** of platform variants, remedy text, affected component, related codes, standards, the software versions the code applies to, planned dates and a link to its **Translations**.
5. Sandbox test codes are hidden unless filtered for explicitly.

Viewers: you see only decided data. If someone is working on a new revision of a released code, you keep seeing the released one until the new one is released.

6. Engineer: creating and submitting a status code

6.1 Create the Draft

1. **New Status Code**.
2. **Classification** — choose the **Functional System Group**, then the **Functional Subgroup** (the list fills in after the group; the subgroup is **mandatory**). Tick at least one **Turbine Platform** (2XM, 3XM, 4XM). Optionally enter the **Legacy Reference Number** from the old tool; it is for traceability only and is never the new identifier.

3. **Description** — **Title** (up to 120 characters) and **Description** (10–4000 characters); they must not be identical.
4. **Status & Timing** — **Status Category** (Error, Warning, Info), **Available Group**, optional **Software Version**, **Set Delay** and **Reset Delay** written as a number and a unit: `10ms`, `2min`, `1.5s`, `*500ms` (units: ms, s, min, h, d), and the checkboxes Alarm, Up/Down counter, Trigger snapshot, Loadless spinning permitted.
5. **Repeated-error escalation** (optional) — times repeated (at least 1), the time window (e.g. `1d`) and the escalation code.
6. **Program references** — the eight numeric program numbers.
7. **Audience Access** — one choice for each of Development, Sales, TCC, Service, Turbine Operator Package, Grid Operator, Service Partner, Customer. These are **proposals**: they are marked Proposed until the Administrator confirms them at review.
8. Under every dropdown a line explains the option you selected. If it says No description available yet, the vocabulary has no text yet — ask Controls Engineering or an Administrator.
9. **Save as Draft**.

If something is wrong the page comes back with the error messages **and everything you typed still in place**. Fix the fields and save again.

The Draft shows **Not yet assigned** as its identifier. That is normal: the number is only given at submission.

6.2 Keep working

Open the Draft (from **My Change Requests** -> View -> View Status Code -> **Edit Draft**) and save as often as you like; nothing else happens until you submit. Another person (the Administrator) may also edit your Draft. (v4) If someone saved it after you opened it, your save is **refused** with the message that the Draft changed; nothing is overwritten and your entries stay in the form beside the current values — compare, adjust and save again. After saving you may see **warnings** (a reference to an unknown or retired code, a similar title in the same subgroup): they never block, but check them. New optional fields: remedy text, affected component, related codes, standard references and the software-version range.

6.3 Submit for review

1. Open the change request (**My Change Requests** -> View).
2. If cross-domain sign-offs are missing (§7), submission is refused with a message naming them.
3. Choose **Submit for Review**. The code receives its `StCd-` number and moves to Review. The number never changes afterwards, even if the request is rejected and resubmitted.
4. Changed your mind before a decision? Use **Withdraw** (a Draft, a code in Review, or one waiting for approval can be withdrawn by its author). To withdraw several at once, tick them in **My Change Requests** and press **Withdraw selected** — or, for Drafts, on the **Status Codes** list: tick your own Drafts (the header box selects all on the page) and press **Delete selected Drafts**; when the list is filtered to Draft, **Delete all my Drafts of this filter** takes every page (v4.13.2); each is reported, and one that cannot be withdrawn does not stop the others. **What happens to a withdrawn change request:** it is deleted, its content stays in the audit log, and the code stays hidden. If the code never received its identifier, the number an import reserved for it and its legacy number are free again, so the same program can be imported again. If it was already in review, it has its identifier and keeps it (numbers are never reused); import the program file again and choose **restore** to open a new Draft on the same number, or — if the number is to be given up — an Administrator deletes the empty code under **Administration -> Withdrawn Codes** (tick it, enter the password, Delete selected; the deletion is final and audited), after which the program or row can be imported as a new code.

6.4 If it is rejected

A reviewer, the Administrator or the Chief Engineer rejected it with a comment. The code returns to **Draft**, and **all earlier sign-offs are cleared**. Read the comment on the change request page, edit, and submit again.

7. Engineer: cross-domain sign-off

Some fields belong to a specialist domain. The timing, repeated-error and program fields belong to **Controls**. If you are not tagged for that domain, a Controls Engineer must confirm them before you can submit.

Asking for a sign-off (author). On the change request page, under Request a Sign-Off, choose a named Engineer who holds the domain (or Any ... Engineer) and press **Request**. The request appears on that person's Dashboard and under **My Change Requests -> Sign-Off Requests For You**.

Giving a sign-off (domain Engineer). Open the request, check the fields, choose your domain and press **Sign Off**

This Domain. You can only sign for a domain you are tagged with.

8. Changing or retiring an existing code

- **Revise a Released code:** open it -> **Actions** -> **Open New Revision**. This creates a new Draft copy; the released version stays valid until the new one is released. Only one open revision per code is allowed. It goes through the same review and approval, keeps the same identifier, and needs a new Release.
- **Deprecate a Released code: Actions** -> Deprecation reason (10–2000 characters, mandatory), optionally the replacement code -> **Initiate Deprecation**. It opens a change request that needs the full review and approval before it takes effect.
- **Archive / reinstate:** Administrators only (§10).

9. Reviewer: reviewing a change request

1. **Review Queue** lists requests in Review (with who is assigned, and whether the Reviewer / Administrator signed already).
2. Open one. Check:
   - **Changes** — what differs from the last decided version, or (new code).
   - **Revision Fields**, the **Sign-Off Status** strip, and the **Cross-Domain Sign-Off** strip.
   - **Comments** from others.
3. Decide: **Approve**, or **Reject** with a mandatory comment explaining what to fix.
4. You cannot decide a request you authored, and if an Administrator assigned the request to another Reviewer, only that Reviewer can decide it.
5. You may add a plain **comment** at any time without deciding. (v4) You can also comment on **one field** (§22), you must tick every item of the configured **review checklist** before you approve, and **Approve** / **Reject** asks for **your password** (§21). The queue shows how long each request has waited (amber/red).

Where comments and decisions are kept. (1) On the change request page, card **Comments** — author (name and role), decision in words, text and time. (2) On the status code page, tab **Decisions & Comments** — the same for every change request of that code, oldest first (not visible to Viewers). (3) In the **Audit Log** (filter by Action, e.g. `REVIEWER_SIGNOFF` or `COMMENT`, and by actor) — each entry includes the comment text. If a change request is withdrawn its comments disappear from the pages above, but the `WITHDRAW` audit entry keeps a full copy.

10. Administrator: sign-off, assignment and special actions

10.1 Administrator sign-off

In the review queue, open the request. Alongside the normal checks you confirm the **Audience Access**:

- The eight dropdowns under Confirm or override Audience Access show what the Engineer proposed. **Approve** confirms them as shown; change any dropdown first to override. Overrides are recorded in the audit log. If you **Reject**, the dropdowns are ignored.
- The Reviewer sign-off and yours can come in either order; when the second one is recorded the request moves to Pending Approval by itself.

10.2 Assigning the Reviewer

On the request page choose a Reviewer under Assigned Reviewer and **Assign Reviewer** (or unassigned). Possible until the Reviewer has signed. The author cannot be assigned.

10.3 Submitting with a manual identifier

For a documented migration or reservation case, an Administrator who authored the Draft can open Override identifier on the submit panel, enter an unused `StCd-NNNNN` inside the subgroup's number range and a reason, then **Submit for Review**. The tool refuses malformed numbers, numbers outside the subgroup, numbers already in use, and any override without a reason; every override is audited. It is not available once the code already has a number.

10.4 Archiving and reinstating

On a Deprecated code -> **Actions**: **Archive** (only after the 180-day retention period; refused before) or **Reinstate to Released** (with a reason, exception path).

10.5 Comments

Add a comment to any request with Add a comment. It changes nothing but is recorded.

11. Chief Engineer: final approval

1. **Approval Queue** lists requests in Pending Approval.
2. Open one, review the changes, comments and sign-offs.
3. **Approve** (for a deprecation request the button reads Approve (Deprecate)) or **Reject** with a comment; (v4) enter your password to confirm. From **Change Sets** you can work through the requests of one software release in one sitting (§21).
4. You are not offered the buttons if you authored the request or gave either earlier sign-off.

12. Administrator: releases

1. **Releases -> New Release**: Name, Version Label, optional Description, optional scope (Functional System Group). **Create Release**.
2. (v4) In the Release Builder, **Target software versions** limits the candidate list to codes that apply to the chosen versions (§25). **Edit Candidates**: the list shows Approved codes within the scope (never sandbox codes). Tick the ones to include -> **Save Candidate Set**. Approved codes you leave out stay Approved for a later Release.
3. **Publish** (confirm the dialog; (v4) you also enter your password, and codes whose planned effective date is still in the future are listed as warnings — they do not block). All included codes become **Released**, receive the effective date, and the Release is frozen. This cannot be undone and a published Release cannot be changed or deleted. Corrections need a new revision and a new Release.
4. A Release not yet published can be removed with **Delete Release**. The Release page shows the **complete baseline** - the count of every code valid as of the Release, split by controller (Main / Hub) and platform (2XM, 3XM, 4XM): a manual of a Release always contains all of them, not only the codes the Release changed. In **Shaped export** choose Complete baseline or Changes of this release only, narrow it to a **platform** and a **controller**, a language and an access level. Everybody who may see the Release (it is published) also finds **Package: all languages and access levels**: choose the formats (PDF, XLSX, CSV, JSON), optionally a platform and a controller, and Build package - one ZIP with the baseline in every language for every access level, the release notes and a `MANIFEST.json` with a SHA-256 for each file. Building takes a few minutes; reload the page to follow it and download the ZIP when it says COMPLETE. The line above the form shows how many codes are translated per language, so you know what will still be in English. (v4) A published Release has **Release notes**, **exports with a manifest**, an **offline lookup page** and — if webhooks are configured — notifies them (§24).

13. Exporting data

Bulk work on a change set (v4.13). A change set groups a batch of codes - an import puts its Drafts in one - and its **Workbench** (link on the change set page) lets the people who hold each step work on the whole batch, without making the step weaker. Readiness shows what still stops each Draft (no platform, a domain sign-off pending). Set values on the Drafts (Engineer, Administrator) sets the same status category, platforms, available group or access level on the ticked Drafts - the ones you wrote; with none ticked, on all of them. Submit for review submits your Drafts that are ready and reports the rest. A holder of an engineering domain signs it off for the batch, a Reviewer, an Administrator and the Chief Engineer decide it - each code still gets its own checks, its own audit entry and, for a decision, your password. **More than 5 codes (an Administrator can change this in Settings) need a sample inspection first:** choose the step and **Draw a sample** - 10 percent of the batch, at least 8 codes, spread over the subgroups - open every sampled code, mark it **OK** or **Defect** (a defect needs a note) and **Save the verdicts**; then enter the sample number with the decision. One defect blocks the batch: **Send the defective codes back** (Reviewer, Administrator, Chief Engineer; a defective Draft is for its author), then draw a new sample of what is left. A sample is yours, valid for 24 hours and for the batch as it is when you draw it, and serves one decision. The box in the header of a list ticks or clears every row (withdrawn codes, the batch, the candidates of a Release). **Dossier (XLSX)** puts the codes with their decisions, the samples with their verdicts and the audit entries of the change set in one workbook for an audit.

Connections between codes (v4.12; not for Viewers). The program of a code says what it depends on, and the tool shows it. On a code's page the card **Connections** lists the codes it is suppressed by (the cause: the code is not raised while the cause is active), the codes it is only set while active for, where it escalates to (its repeat code; one outside the catalogue is marked external) and the same the other way round (Suppresses, Required by, Escalated from), with the chain of causes above it. In the menu, **Catalogue -> Connections** shows the causes with the codes they suppress as a tree, the repeat codes with the codes that escalate to them, and **What looks wrong**: a cause that is not in the catalogue, a deprecated or archived cause, a loop. Choose Main controller or Hub controller to look at one. The connections are read from the text of the imported programs, not from the controller logic; they change when a program is imported again, and an imported Draft appears under the number it has reserved.

Export Center offers, in **CSV, JSON, XLSX and PDF**:

Export	Who	Content
All Released Status Codes	everyone	the latest Released version of every code (sandbox excluded)
Release Export	everyone	the contents of one published Release
Working-Set Export	StCd Owner and Administrator	every current code in any state, including unapproved Drafts. Every row carries Not for distribution - unapproved data, and the file name says so too. Never publish it.
Full-Database Backup	Administrator	complete raw database file. Store it with the same protection as the database itself.

File names end with the time of generation, e.g. `cscm_released_2026-10-22_11-05.csv`, `cscm_release_v1.0_2026-10-22_11-05.xlsx`, `cscm_working-set_NOT_FOR_DISTRIBUTION_2026-10-22_11-05.csv` (format `YYYY-MM-DD_HH-MM`, UTC).

(v4.7) The **PDF prints every language**: Hindi, Marathi, Bengali, Punjabi, Gujarati, Odia, Tamil, Telugu, Kannada, Malayalam, Georgian, Arabic (right to left) and Urdu as well as the Latin-script languages; the fonts come with the tool. Headings are not set in bold, and a character that none of the fonts has shows as ?.

(v4) Exports can be shaped: choose a **profile** (a fixed column set), an **audience** (only codes that audience may see), a **language** (approved translations), a **software version**, and **bundle** (a ZIP with the data file, `MANIFEST.json` and the release notes); see §24. Full-Database Backup asks for your password. Files are flat tables: one row per status code, every field a column. Values beginning with =, +, - or @ get a leading apostrophe so that spreadsheet programs do not treat them as formulas. (v4) The PDF now wraps long text over several lines (nothing is cut), repeats the header on every page and starts with a title page showing release, version, effective date, filters, row count and a content digest.

14. Importing the legacy library

Engineers and Administrators can load many codes at once, but **import is only a faster way to enter Drafts**. Every imported row still goes through cross-domain sign-off, both reviews and the Chief Engineer's approval.

1. Prepare a **.csv** or **.xlsx** (max 20 MB), first row = column names, one row per status code.
2. Required columns: `functional_system_group` (code, e.g. `WCNV`), `functional_subgroup` (code, e.g. `010xx`), `turbine_platforms` (e.g. `3XM;4XM`), `title`, `description`, `status_category`, `available_group`, `set_delay`, `reset_delay`. Optional: `functional_system_group_range_start` (needed when a code such as `WTUR` exists twice), `software_version`, `alarm_flag`, `up_down_counter`, `trigger_snapshot`, `loadless_spinning_permitted`, `number_of_times_repeated`, `repeated_over_the_course`, `status_code_for_repeated_error`, the eight program columns (`yaw_program`, `manual_reset_program`, `auto_reset_program`, `converter_reactive_power_program`, `brake_program`, `umrichterprogramm_gsc`, `umrichterprogramm_msc`, `yaw_bearing_lubrication_program`), the eight audience columns (`development_access`, `sales_access`, `tcc_access`, `service_access`, `turbine_operator_package_access`, `grid_operator_access`, `service_partner_access`, `customer_access`) and `legacy_reference_number`.
3. (v4) use **Start from the template** (XLSX with instructions and lookup codes, or CSV with headers only) if you start from scratch; the untouched template imports cleanly. Choose the **mode**: create (default) or update (match rows to codes already imported through `legacy_reference_number`: an open Draft is revised in place, a Released code gets a new revision, an unchanged row is skipped, a missing one is created; a request in review, a Draft of another owner or a change of group or subgroup is reported on that row). Press **Validate only** first: every problem is listed per row and column with the value and how to fix it, and **nothing is created**. Then **Upload and Import**.
4. The result page shows how many rows were read, created, warned and rejected; (v4) a large file (over 300 rows) runs in the background and the page shows a progress bar — you may leave and come back. Warnings (for example a similar title already in the subgroup) do not block. **Download Error Report** (CSV or XLSX) lists each rejected problem — row, column, value, problem, how to fix — for each rejected row (row 1 is the header) with

the reason; fix those rows and upload them again. A rejected row creates nothing. **View Imported Drafts** opens the new Drafts in My Change Requests.

(v4) **Migration progress:** Library Import -> Migration progress shows how many legacy rows are imported, in review, approved, released, retired and — when you give the expected total, or upload the legacy master list — which legacy numbers are still missing.

14.1 Importing from the generated .st programs (v4.2)

The generated controller programs (`StCd_1000.st`, `StCd_14000.st`, ...) already hold the status codes, so you can import them directly. As with the spreadsheet import, every program only becomes a **Draft**: it still needs cross-domain sign-off, both reviews and the Chief Engineer's approval.

1. Open **Import from .st files** (sidebar, or the button on Library Import). Choose the files — several at once, a whole folder, or one ZIP. Leave Controller on Detect (a file number from 10000 up is a Hub Controller program) unless a Hub file has a smaller number. The new number of a Hub program, and for every program the **title, the type, the up/down counter and the access levels**, come from the **mapping** (section 14.2): nothing is chosen here, and a program whose number is not in the mapping is blocked. Choose the turbine platforms, then **Read files and preview**.
2. The **preview** creates nothing. Each program is one row: its new identifier, the legacy number (n/a when there is none), the controller, the name with file and line, the group and subgroup its number lies in, and the state — ready, already in the catalogue or blocked with the reason (for example a number no subgroup covers, a Hub number missing from the numbering table, a legacy number already used by another code, or the same number twice in the upload). **Access levels:** the program gives a read and a write number for each audience. Where exactly one named access level carries exactly those numbers, it is proposed in the Draft (the comment lists each proposal, for example Development = See only (read 1, write 50)), and the Administrator confirms it at the sign-off. Where none or more than one fits, the level stays empty and the comment names the audience with its numbers; the numbers are kept either way. **Nothing is assumed:** a delay that the program gives as a variable keeps the **name of the variable** (for example `StCd14000SetDelayGpTi`); you look the time up in the controller's parameters and enter it in the Draft. A program without a timer shows not in the program. The comment (description) of every Draft lists what the program did not give - the parameters, and the flags and program references that were entered as No or 0 because the file had none - so nothing looks like a real value that is not. Warnings never block: a delay that is a parameter and not a time (`0ms` is entered), a `PrgIdx_GUi` that is already used (it is not stored), no `PrgIdx_GUi`. Each row shows the alarm type and a **status category** list (choose while none is set; such a row is not ticked) you can change per program; to set many at once, tick the programs, choose a category above the table and press **Apply to the ticked programs** (v4.13.2). Tick the programs you want (the header box selects all ready ones; the filter box narrows the list) and optionally name the change set.
3. **Create Drafts for the ticked programs.** The tool checks them again (somebody may have imported one meanwhile), creates the Drafts and puts all of them into **one change set**, so the batch can be reviewed and decided together (chapter 21). The result page links to the change set; large batches run in the background with a progress bar. Each Draft **reserves the number of its program**: when you submit it for review it receives exactly that identifier, no other number and no override. Open the Draft to correct the category, the description (generated from the file and the set condition) and the audience access, which the file does not supply.

Everything else the program says — the user-group levels and the remaining subsystem programs — is kept with the Draft as its import source and is not lost (`GET /status-codes//import-source`).

Legacy mapping. Legacy mapping (sidebar) lists which new status code replaces which legacy number, separately for the Main Controller and the Hub Controller, with the state of the number (issued, reserved by a Draft, no number yet), filters and a CSV or XLSX export. It runs from legacy to new only: the new library has more codes than the old one, so a code without a legacy number is normal (n/a) and there is no way back from a new number to a legacy one. Viewers do not see it.

14.2 The mapping and how to change it (v4.17)

The **mapping** (Import and Migration -> Mapping) says which legacy status code becomes which new one and holds what the program files do not carry: the English **title**, the **type** (Error, Warning, Info), the up/down counter, the **access level of each audience**, cause and directive. The import reads it; you cannot choose these values at the import any more.

- **Read it.** Filter by controller, status and text. The status says how far a number has come: `PROPOSED` (loaded, nobody has approved it), `APPROVED` (the StCd Owner, the Chief Engineer and an Administrator approved it), `ISSUED` (a code of the catalogue carries it; its numbers are frozen), `REJECTED` (never to be used). The import uses

a PROPOSED row with a warning.

- **Change one row.** Open it, change the fields, write the **reason** and press Request this change. Request approval of the number asks the three to approve the row as it is; Request rejection of the number asks that the number is never used. **New row** adds a number.
- **Several rows.** The list shows 100 rows a page; Show all puts every row on one page, and the box at the head of the table then ticks all of them. Tick them and Make the request (it asks for approval of the rows as they are), or use Compare a workbook: download the current mapping, correct it in Excel, upload it - every new and every different row becomes one request; nothing else changes.
- **Decide.** Mapping changes lists the requests; the menu shows how many wait for you. On a request the three places are at the top and the decision form is at the bottom, below the rows. A request takes effect when the **StCd Owner, the Chief Engineer and an Administrator** have each approved it - three people, none of them the one who asked, each confirming with the password. One rejection (with a reason) closes it. You can withdraw your own request while it is open. If a row changed after the request was made, the request cannot be applied: withdraw it and ask again.
- **Loading.** An empty mapping is loaded from `seed_data/legacy_mapping.xlsx` (`flask mapping-seed`, or the button an Administrator sees on the empty page).

15. Sandbox (test) codes

Administrators only. **New Sandbox Code** creates a code with a `StCd-T#####` number for training or tests. It is excluded from searches by default, from every export and from every Release. When no longer needed, open it -> **Actions** -> type the identifier to confirm -> **Delete Permanently**. This is the only thing in the tool that can be permanently deleted; the audit log keeps a full copy of what was removed.

16. Administration: users, vocabularies, audit log

16.1 Users

Roles and colours. Every role is a coloured badge; the colour is the level and red is the highest: **Administrator** (red), **StCd Owner** (purple), **Chief Engineer** (amber), **Reviewer** (blue), **Engineer** (green), **Viewer** (grey). The StCd Owner owns the status code library: reads everything and may move people between Viewer, Engineer and Reviewer under Administration -> Users, but cannot create users, change any other role, edit data or decide. Everything else about a person needs an Administrator.

Administration -> Users. Every row has **View** (the user's page with profile and what the account has done), **Edit**, **Deactivate / Activate** and **Delete**. Deactivate stops the person signing in at once and loses nothing; Delete removes an account for good and works only for an account that was never used — as soon as it has a change request, a review, an audit entry or any other record it can only be deactivated, so everything stays attributable. Deleting asks for your password again. The StCd Owner sees the list and the user pages and can only change the role between Viewer, Engineer and Reviewer.

Administration -> Users. Create User (username, e-mail, full name, role — default Viewer, engineering domains, initial password). (v4.7) **The e-mail address cannot be changed afterwards** — it is the person's identity (the Windows sign-in finds the account by it), so enter it carefully; a person with a new address gets a new account and the old one is deactivated. For an existing user use Edit to change name, role, engineering domains or active state; **Reset Password** forces a new password and signs the person out. (v4) Each user also has a language preference (set by the language switch). **Delete** (on the user's page, with your own password) works for every account: one that was never used is removed; one with history is **erased** — name, address and password go, the person leaves every list and can no longer sign in, and what they did stays in the audit log and in the records as Deleted user. The audit log is never changed. Their user name and address can be used again. Deactivate keeps the name and is the gentler choice. The last active Administrator cannot be deactivated, demoted or deleted, and nobody deletes their own account. Give **Engineers** the domain tags they are qualified for (Controls, Electrical, Mechanical, Grid, Safety) — they decide who can give cross-domain sign-offs.

16.2 Lookups (vocabularies)

Administration -> Lookups: choose a vocabulary (turbine platforms, engineering domains, available groups, the eight audience-access lists, functional groups and subgroups).

- Edit a value's **label**, **description** (shown as inline help to everyone) and active state. Values are deactivated, never deleted.

- **Add** a value to a simple vocabulary.
- **Create a Functional Subgroup:** pick the group, a code, a label and a sub-range of numbers. Ranges may not overlap another subgroup of the same group; the tool names the conflict.
- **Create a Functional System Group:** a code, a label and a range inside free capacity (10000–10999, 14000–14999, 15000–15999, 16000–16999). It may not overlap any existing group.

16.3 Audit log

Audit Log (Administrators see everything; Reviewers and Chief Engineers a scoped part). Filter by entity type and id, **action** (e.g. `REVIEWER_SIGNOFF`, `COMMENT`, `WITHDRAW`), actor id, and date range. Each line shows who did what and when, with before/after values. Entries cannot be edited or removed by anyone.

17. Your account

Account -> Change Password (enter the current and the new password). (v4) The same page shows your language preference (English or Deutsch); the language switch in the top bar changes it at any time, and before you sign in it follows your browser. Choose a passphrase of at least 12 characters that you do not use elsewhere. (v4.7) Where the Windows sign-in is the only way in, your password is managed in Windows and the page has no password form; your e-mail address is shown but cannot be changed.

18. Rules the tool enforces — and why

Rule	Why
The identifier is allocated once, at submission, and never changes	stable references in manuals and software
Subgroup is mandatory	every code belongs to exactly one subgroup
Author cannot sign off or approve own request; Chief Engineer also cannot be an earlier signer	independent control
Rejection needs a comment and clears all sign-offs	the fix is reviewed again from scratch
No editing outside Draft; changes are new revisions	released data is reliable
One open revision per code	no conflicting parallel edits
Released codes are never deleted	history; only deprecation and archiving
Sandbox codes can never be released and are the only deletable records	safe testing
Audience-access values are authoritative only after Administrator sign-off	access decisions are a governance act
A published Release cannot change	what was published is what was published
(v4) Sign-offs, approvals, publishing and the full backup ask for your password	each is a deliberate act that an unattended screen cannot perform
(v4) A delegate acts as themselves and only for decisions	covering for a colleague never widens anyone's rights; the audit shows both names
(v4) A save based on an outdated Draft is refused	nobody silently overwrites someone else's work
(v4) An audience export shows only what the audience may see; no confirmed value, or Covert, means hidden	the safe default is to disclose nothing
(v4) Audit entries are chained and never deleted	tampering is detectable; history is complete

19. Troubleshooting

You see	What it means / what to do
"Password must differ from your last 5 passwords."	Choose a passphrase you have not used recently.
"Invalid username or password."	Check caps lock; after 5 failures wait 15 minutes or ask an Administrator to reset the password.
Signed out unexpectedly	30 minutes idle or 12 hours since sign-in. Sign in again; unsaved form data is not kept across a sign-out.
"Cross-Domain Sign-Off still pending for: CONTROLS"	Ask a Controls-tagged Engineer to sign off (§7).
"At least one Turbine Platform is required"	Tick a platform and save the Draft, then submit.
Set Delay / Reset Delay rejected	Use a number plus a unit, e.g. 10ms, 2min.
"Title must not equal Description"	Make the description longer and more specific.
"Cannot sign off own change request"	By design; another person must do it.

You see	What it means / what to do
"This change request is assigned to a different Reviewer"	The Administrator assigned someone else; ask them to reassign.
Cannot find a code that exists	You may be a Viewer and it is not yet Approved/Released; or it is a sandbox code (filter for it).
Import rejects a row	Open the error report: it names the row and the problem (unknown group or subgroup code, bad delay, bad category ...).
"Archive refused – retention period"	180 days must pass after deprecation.
Page says forbidden (403)	Your role may not do that (§4).
Too many login attempts (429)	Wait a few minutes; the limit protects accounts.
(v4) "This Draft was changed by someone else after you opened it"	Your entries are kept below the form; compare with the current values and save again.
(v4) "Your password is required" / "The password is wrong"	Sign-offs, approvals, publishing and the full backup need your own password; after 5 wrong ones the account locks (15 minutes).
(v4) A code is missing from an audience or language export	The audience's confirmed value does not make it visible, or no approved translation exists (strict mode); check the access fields and the translation worklist.
(v4) "Evidence can only be attached while the request is open" / type not allowed	Attach to an open request; allowed: PDF, PNG, JPG, TXT, LOG, MD, CSV, XLSX, DOCX, up to 10 MB, 20 files.
(v4) Backup shows "stale" or "last attempt failed"	See §28; ask the operator to check the nightly job.
(v4) Import says nothing was created after "Validate only"	That is what it does; fix the listed problems and import.
Anything unexpected	Note the time and what you did; an Administrator can find it in the Audit Log.

20. Glossary

Audience access — eight values stating how each audience (Development, Sales, TCC, Service, Turbine Operator Package, Grid Operator, Service Partner, Customer) may use a code. **Change request (CR)** — the workflow envelope around one revision. **Cross-domain sign-off** — confirmation by a specialist of fields owned by their domain. **Engineering domain** — Controls, Electrical, Mechanical, Grid, Safety. **Functional System Group / Subgroup** — the classification that also defines the number range, e.g. WCNV "Converter & Grid Interface" -> subgroup "Converter". **Hub Controller / Main Controller** — the two controller types whose codes are managed. **Identifier** — `StCd-XXXXX`; sandbox codes use `StCd-T#####`. **Legacy reference number** — the number in the old tool. **Release** — a named, versioned, immutable bundle of Approved codes. **Revision** — one version of a code. **Sandbox code** — a test code that never reaches a Release. **Working set** — export of current codes in any state; not for distribution.

21. Notices, delegation and change sets (v4)

Notices. The bell in the top bar counts your unread **notices**: when a request waiting for your decision has stayed longer than the amber or red limit, you get one notice per request and day (and an e-mail if your organisation configured one). Open **Notices** to read them; **Mark all as read** clears them. **Stalled Requests** (Administrators) lists everything past its limit, with Assign least-loaded Reviewer and a per-request reassignment; the page also shows each Reviewer's open reviews.

Entering your password. Reviewer and Administrator sign-off, Chief Engineer approval, publishing and the full backup have a Your password field. It is checked every time — a wrong one cancels the action, is audited and counts toward the account lock.

Delegation (when you are away). Delegation -> Create delegation: choose a colleague (not a Viewer, not yourself) and the dates. For those dates the colleague may take the decision actions of your role — sign off, approve — **as themselves**: they see the queue, a chip in their top bar says whom they cover, the audit trail names both of you, and the usual rules still apply to them personally (they cannot decide something they wrote or already signed; the two Review sign-offs must come from two different people). They cannot administer users or do anything else an Administrator can. You or an Administrator can **Revoke** it at any time; it ends by itself after the last date, or if you are deactivated. While you are away the automatic Reviewer assignment skips you.

Change sets. An Engineer, Chief Engineer or Administrator creates a **change set** (a name, optionally a target software version) and adds the open requests that ship together (author or Administrator adds; Remove takes one out). The set page shows each request's state and waiting time. A decision maker can tick several requests and press **Approve selected** / **Reject selected**: each request is still decided on its own — its own checks, comment and audit entry; a request you may not decide (for instance one you wrote) is skipped with the reason and the others go

through; the password is asked once. An Administrator **closes** the set when the release is done.

22. Evidence, field comments and the review checklist (v4)

Evidence. On your open change request, **Attach**: test reports, simulations, design notes, a safety assessment (PDF, PNG, JPG, TXT, LOG, MD, CSV, XLSX, DOCX; up to 10 MB each, 20 per request; the file must really be what its extension says). Reviewers and the Chief Engineer download it at decision time; it is always downloaded, never shown in the browser. You can remove it only while the request is a Draft.

Field comments. In Review or Pending Approval, a Reviewer, Administrator or Chief Engineer can press **Add field comment** next to a field and say what is wrong with that field. The author sees the open comments next to the field when the request is back in Draft and presses **Mark resolved** (the commenter, the author and Administrators can do so).

Review checklist. Administrators maintain Administration -> Review Checklist: sentences such as "delay values checked against the control logic", for one Functional System Group or for all. When you approve as Reviewer you must tick every item that applies; the wording you confirmed is stored with your sign-off. Rejecting does not need the checklist.

23. Translations and language (v4)

Which version is this? (v4.14.1) The version number is at the foot of the menu on the left (CSCM Tool v4.16.1) and at the foot of the login page; give it when you report a problem. An administrator can also read it in the answer of `/api/v1/health`.

Light or dark display (v4.14). The list Automatic (system) / Light / Dark in the top bar (and on the login page) sets the colours. Automatic follows your operating system, so the tool turns dark with your computer in the evening. The choice is remembered by the browser on this computer only; on another computer or in another browser you choose again. Printouts and exports are always light.

Interface language. The switch in the top bar (English, Deutsch, Français, हिन्दी, العربية, then বাংলা, मराठी, తెలుగు, தமிழ், ગુજરાતી, اردو, ಕನ್ನಡ, ଓଡ଼ିଆ, മലയാളം, ਪੰਜਾਬੀ, Español, Italiano, Dansk and last ქართული — the country letters on the left and the flag on the right; Arabic and Urdu switch the page to right-to-left) is remembered for your account. French, Hindi, Arabic, Spanish, Italian, Danish and Georgian cover only the most visible text (navigation, login, dashboard, status code lists and the role names), the ten languages after Arabic only the top bar, the navigation, the login page and the role names, and all show English for the rest; the translations are unreviewed. German covers navigation, dashboard, login, lists, forms, change-request screens and help; some messages produced deep inside the tool and the administration pages stay English. (v4.9) Every text of the interface — also the messages that appear after an action and the reasons for a refusal — exists in all languages. The translations are proposals until a native speaker has checked them (next paragraph); a language nobody has reviewed carries a small **beta** mark in the language menu for Engineers, Reviewers, Chief Engineers, Administrators and the StCd Owner; Viewers and the login page offer only the reviewed languages (a Viewer who was assigned a language to review sees that one). An Administrator can switch both off in Settings. (v4.9.2) The words around an export — the title page, the column headings of the default PDF, the footer and the banner of the working set — follow the language you choose for the export.

Translating a code. English is the source. Once a code is Approved or Released, open **Translations** on its page: for each language you see MISSING, DRAFT or APPROVED. An Engineer or Administrator enters title, description and remedy text and saves a **draft**; a Reviewer or Administrator **other than the translator** presses **Approve translation**. Editing an approved translation sends it back to draft. **Copy from previous revision** takes over the translations of the previous revision as drafts. The translation **worklist** lists the codes that still lack an approved translation in a language. Viewers see approved translations only.

Checking the interface translations (native speakers, v4.7). The sidebar entry **Translations** appears for people who were assigned a language and for Administrators and StCd Owners. Concept and background: 26-translations.md.

- Reviewer. Open **Translations**, choose your language, **Review**. Each row shows the English, the current translation and where it is used. Mark **OK** if it is right, **CHANGE** and write your version in the Proposal box, or **UNCLEAR** and say why in the comment. The seven things to look at are under What to check; keep markers such as `%(n)s` exactly as they are. **Mark the rest of this page OK** saves time on a clean page; **Save** or **Save and next page** stores your answers, and you can stop and continue later. **Missing translations** lists the strings that have none yet: write a proposal for those you can. If you prefer a spreadsheet, **Download the sheet (XLSX)**, fill the Verdict, Proposal and Comment columns and **Upload** it. When every translated string has a verdict, **Complete the review**.
- Administrator or StCd Owner. Open **Translations**, choose a language, **Decisions and reviewers**. Assign the native

speaker (an external person gets a Viewer account and sees only this section) or Remove them. Under Proposals waiting for a decision **Accept** (you may correct the text first), **Reject**, or for an unclear string mark it **Noted**. You cannot decide on your own proposal. An accepted proposal is shown in the application at once, for that language; under Accepted changes **Remove** brings the shipped text back.

Pre-check confidence and follow-up reviews (v4.10). The overview of **Translations** shows, for every language, how many strings await approval and the **pre-check confidence**: the estimated share of strings a native speaker would accept unchanged, on a colour from red (50 % or less) through yellow to green (above 95 %). It comes from a checker that reads every string mechanically and a sample of 160 by eye; it is **not a review** and does not remove the beta mark. Point the mouse at the number to see what it is made of. What a native speaker has approved stays approved: when strings are added or changed later, **Review** opens a follow-up review with only New and changed translations (All translations shows the rest), and until they are approved a Viewer reads exactly those strings in English. Whoever changes the text or the nomenclature records the confidence check for the change (`scripts/translation_precheck.py`, `docs/26-translations.md` §8c); a test fails otherwise. (v4.11) The **StCd Owner** gets a notice (the bell) whenever the vocabulary changes - a new version with other interface text, or a term added, changed or its translation approved in Terminology - saying which languages need attention and how many strings are concerned; the reviewers assigned to an affected language get it too. Open **Terminology** or **Translations** from the notice.

Technical nomenclature (v4.9). The sidebar entry **Terminology** holds the words of the controller and of the workflow (Status Code, Hub Controller, Set delay, Draft, the roles ...) with their translation in every language, apart from the interface translations. Everybody who is signed in can read it. Overview: the coverage per language, the terms with category, meaning and the state of each language, a search and a filter by category; an Administrator, StCd Owner or Chief Engineer adds a term (Add a term: a code of lower-case letters, digits and hyphens that never changes, the English text, the category and a meaning for a translator who does not know the product) and edits the English text, the meaning or never translated (a changed English text reopens the approved translations). One language: an Engineer, Reviewer, Administrator, StCd Owner or Chief Engineer writes a translation and saves it as a **draft**; a second person with the right to approve presses **Approve** (or Approve all drafts of others for the whole language) — you cannot approve your own proposal. Send back needs a note and turns an approved translation into a draft again. Check against the interface lists the interface strings that contain a term whose translation does not appear in the translated string: it is a hint, not a verdict, because languages that inflect words change the ending. Export all (XLSX) and (CSV) give the vocabulary to a native speaker or a translation office. Being the reviewer of an interface language gives no right here.

24. Publishing: profiles, audiences, manifests, notes, offline page (v4)

- **Profiles.** Administration -> Export Profiles defines named column sets — Service manual, SCADA tag list and Customer list are provided — with the header labels, an optional default audience and language. Choose one in the export form.
- **Audience.** The eight audiences are Development, Sales, TCC, Service, Turbine Operator Package, Grid Operator, Service Partner, Customer. An audience export contains only the codes whose confirmed value for that audience is visible, and none of the other audiences' columns; a code without a value, or with `Covert`, is hidden. Administrators decide visibility per value in Lookups. The unapproved working set can never be exported for an audience.
- **Manifest and verification.** Every Release export is recorded with its SHA-256 and has a manifest (release, scope, row count, hashes, generator, time, options; signed if the organisation configured a key). With **bundle** you get a ZIP containing the data file, `MANIFEST.json` and the release notes. **Export Center -> Verify a file**: upload a file and optionally its manifest to learn whether it is exactly what the tool produced, whether it is listed in the manifest and whether the signature is valid. The same check works without the tool: `flask verify-export  .`
- **Release notes.** On a published Release: **Release notes** lists the codes added, changed (field by field, old and new value) and deprecated compared with the previous published Release, as Markdown, CSV or PDF.
- **Offline page. Offline lookup (HTML)** (or **Offline lookup + manifest (ZIP)**) produces one HTML file with the Release's codes and a search box, to open at the turbine without a network; it respects audience, language and software version.

25. Software versions (v4)

A code applies to a range of controller software versions: on the form choose Applies to software versions **from** and **to** (either may stay open). An Administrator sets the versions a Release targets (§12). **Software Versions** answers two questions: which Released codes apply to version X, and what was added, removed and changed between versions X and Y.

26. Insight: health, metrics, fleet feedback, saved searches (v4)

- **Catalogue Health** (Administrator, Reviewer, Chief Engineer): how full every subgroup's number band is (alert when little is left; Main Controller rows are blue, Hub Controller rows light blue, each with an MC or HC chip), codes missing expected data (remedy text, component, version range, audience access), suspicious combinations (an INFO code with an alarm, a repeat count without an escalation code or the reverse, a planned deprecation date that has passed), gaps in a number band, and planned dates coming up. On New Status Code a hint shows how many numbers are left in the chosen subgroup.
- **Success Metrics** (Administrator, Chief Engineer): median days from submission to approval (target <= 7), median days for a cross-domain sign-off (<= 3), the share of Released codes with a complete audit trail (100 %), how long open requests have waited, and codes whose owner is no longer an active user.
- **Fleet Feedback** (Administrator, Chief Engineer): upload a CSV or XLSX with `status_code_identifier` and `occurrences` (anonymised, per period) with a source label and the period dates. The report names Released codes that never occurred (candidates for deprecation) and the most frequent ones, flagging those without remedy text.
- **Saved searches** — see §5.

27. Hub Controller to Main Controller consolidation (v4)

If the Hub Controller leaves the hardware setup, its codes are defined anew on the Main Controller side. **Hub Consolidation** lists every Hub code with the state of its replacement (open, draft, live, retired). On an Approved or Released Hub code, an Engineer or Administrator presses **Create MC draft**, chooses a Main Controller group and subgroup, and a new Draft is created with the same descriptive fields; the Hub number is kept as its legacy reference, and the new number is allocated at submission like for any code. It then goes through the whole cycle. When replacements are Released, an Administrator uses **Deprecate replaced Hub codes** for a Hub group: a deprecation request is opened for each replaced code — each still needs review and approval.

28. Administrator: settings, integrations and operations (v4)

- **Settings:** the amber and red limits for the review and approval stages (business days; amber must not exceed red), automatic assignment of the least-loaded Reviewer at submission, and the number-band alert level. Every change is audited.
- **Integrations: API tokens** (name, scope catalogue:read and/or changes:read, validity up to 365 days): the secret is shown **once** — copy it; a token can only read, with the visibility of a Viewer, and can be **Revoked**. **Webhooks:** a URL on the internal network that receives a signed notice when a Release is published; the page shows each delivery's outcome; a failing webhook never stops a publish; **Pause** / resume as needed. Machine consumers may also poll the **change feed** (`/api/v1/changes`).
- **Operations:** shows the last **verified backup** (file, size, age, result of the integrity, restore and audit-chain checks) and has a **Verify now** button for the audit log (it also runs weekly by schedule). A red result names the first entry that does not fit. The backup and archive commands and the restore drill are in the deployment document (14).

29. Help inside the tool (v4)

Help (sidebar) shows the quick start for your role and the explanations of states, roles and terms. Hover over a state or role name for its explanation; every change request states **what happens next**.

24. Test Report — CSCM Tool

> **Generated file — do not edit by hand.** Regenerate with `python cscm_tool/scripts/generate_test_report.py` after every software change (Definition of Done, 15-development-standards.md). It runs the complete pytest suite, measures coverage, runs the static quality gates, and appends the run to the history below.

1. Run summary

Verdict	PASS	
Generated (UTC)	2026-10-28 19:51:46	
Branch / commit	main @ a696bc6 +uncommitted changes	
Test order	file	
Tests	1082 run — 1082 passed, 0 failed, 0 skipped	
Duration	1982.9 s	
Line coverage (app/)	93% (13625 of 14728 statements)	
Python / platform	3.12.10 / Windows 11	
pytest summary line	`functional: 1073 passed, 9 deselected, 5 warnings in 1945.38s (0:32:25)	performance: 9 passed, 1 warning in 65.51s (0:01:05)`

2. Results by suite

Suite	Tests	Passed	Failed	Skipped	Time
unit	74	74	0	0	5.6 s
integration	735	735	0	0	1490.1 s
system	264	264	0	0	422.1 s
performance	9	9	0	0	65.1 s

3. Static quality and security gates

Gate	Result	Detail
black --check	PASS	273 files would be left unchanged.
ruff check	PASS	All checks passed!
mypy	PASS	Success: no issues found in 162 source files
bandit -r app	PASS	[tester] WARNING nosec encountered (B324), but no failed test on line 74
pip-audit	PASS	runtime packages with advisories: none; dev-only advisories (accepted): black, pytest, starlette

4. Measured performance (NFR-001, NFR-004, TC-024, TC-025)

Test	Measured
test_search_and_read_latency_at_20000_revisions	read_p95_ms = 8.0, revisions = 20000, search_median_ms = 85.6, search_p95_ms = 125.5, speed_factor = 1.0
test_export_of_5000_records_within_target[csv-5]	export_seconds = 0.98, limit_seconds = 5, records = 5000, speed_factor = 1.0
test_export_of_5000_records_within_target[json-5]	export_seconds = 1.26, limit_seconds = 5, records = 5000, speed_factor = 1.0
test_export_of_5000_records_within_target[xlsx-15]	export_seconds = 3.03, limit_seconds = 15, records = 5000, speed_factor = 1.0
test_export_of_5000_records_within_target[pdf-15]	export_seconds = 8.33, limit_seconds = 15, records = 5000, speed_factor = 1.0
test_concurrent_identifier_allocation_in_one_group	allocation_p95_ms = 111.4, speed_factor = 1.0, submissions = 64
test_fifty_concurrent_users_are_served_without_errors	p95_ms_50_users = 114.7, speed_factor = 1.0
test_export_issues_a_constant_number_of_queries	statements_for_two_500_row_exports = 27

Targets: search p95 <= 500 ms and single read p95 <= 200 ms at 20,000 revisions; CSV/JSON export of 5,000 records <= 5 s, XLSX/PDF <= 15 s; 50 concurrent users keep the 500 ms search target; identifier allocation under contention correct and gap-free.

5. Coverage by module

Module	Statements	Missed	Coverage
app/__init__.py	83	0	100%
app/api/__init__.py	0	0	100%
app/api/v1/__init__.py	3	0	100%
app/api/v1/_errors.py	20	2	90%
app/api/v1/audit_routes.py	12	0	100%
app/api/v1/auth_routes.py	60	4	93%
app/api/v1/bulk_routes.py	92	0	100%
app/api/v1/change_request_routes.py	136	11	92%
app/api/v1/domain_signoff_routes.py	60	3	95%
app/api/v1/export_routes.py	77	3	96%
app/api/v1/governance_routes.py	244	3	99%
app/api/v1/health_routes.py	30	2	93%
app/api/v1/import_routes.py	93	11	88%
app/api/v1/lookup_routes.py	70	1	99%
app/api/v1/mapping_routes.py	130	12	91%
app/api/v1/operations_routes.py	165	0	100%
app/api/v1/package_routes.py	71	8	89%
app/api/v1/publication_routes.py	196	7	96%
app/api/v1/release_routes.py	79	8	90%
app/api/v1/revision_routes.py	26	3	88%
app/api/v1/serializers.py	58	0	100%
app/api/v1/st_import_routes.py	71	2	97%
app/api/v1/status_code_routes.py	211	18	91%
app/api/v1/user_routes.py	96	7	93%
app/audit/__init__.py	0	0	100%
app/audit_chain.py	134	7	95%
app/audit_hash.py	10	0	100%
app/auth/__init__.py	0	0	100%
app/auth/decorators.py	122	3	98%
app/auth/password.py	30	0	100%
app/auth/providers.py	32	1	97%
app/auth/rate_limit.py	34	0	100%
app/auth/roles.py	27	0	100%
app/auth/windows.py	43	2	95%
app/branding.py	6	0	100%
app/cli.py	96	13	86%
app/config.py	59	0	100%
app/db_modes.py	15	0	100%
app/domain/__init__.py	0	0	100%
app/domain/authz.py	5	0	100%
app/domain/errors.py	29	0	100%
app/domain/lifecycle.py	19	0	100%
app/errors.py	24	3	88%
app/export/__init__.py	0	0	100%
app/export/csv_exporter.py	17	0	100%
app/export/full_database_exporter.py	22	1	95%
app/export/json_exporter.py	3	0	100%
app/export/offline_bundle.py	13	0	100%
app/export/pdf_exporter.py	214	3	99%

Module	Statements	Missed	Coverage
app/export/xlsx_exporter.py	20	0	100%
app/extensions.py	9	0	100%
app/help_content.py	6	0	100%
app/i18n.py	100	9	91%
app/i18n_ar.py	1	0	100%
app/i18n_de.py	3	0	100%
app/i18n_eu.py	4	0	100%
app/i18n_extra.py	10	0	100%
app/i18n_generated.py	13	0	100%
app/i18n_in.py	4	0	100%
app/i18n_messages.py	26	0	100%
app/importers/__init__.py	0	0	100%
app/importers/glossary.py	28	0	100%
app/importers/stcd.py	227	5	98%
app/logging_config.py	45	0	100%
app/models/__init__.py	12	0	100%
app/models/audit.py	16	0	100%
app/models/change_request.py	136	0	100%
app/models/export_job.py	38	0	100%
app/models/import_job.py	30	0	100%
app/models/lookup.py	87	0	100%
app/models/mapping.py	69	0	100%
app/models/operations.py	121	0	100%
app/models/release.py	30	0	100%
app/models/status_code.py	145	0	100%
app/models/terminology.py	25	0	100%
app/models/translation.py	42	0	100%
app/models/user.py	33	0	100%
app/observability.py	102	11	89%
app/repositories/__init__.py	0	0	100%
app/repositories/audit_repository.py	22	1	95%
app/repositories/base.py	20	1	95%
app/repositories/change_request_repository.py	26	0	100%
app/repositories/fts.py	35	2	94%
app/repositories/release_repository.py	4	0	100%
app/repositories/status_code_repository.py	82	3	96%
app/repositories/user_repository.py	33	1	97%
app/schema_sql.py	45	1	98%
app/security_headers.py	15	0	100%
app/services/__init__.py	0	0	100%
app/services/attachment_service.py	106	11	90%
app/services/audience_service.py	31	1	97%
app/services/audit_service.py	23	0	100%
app/services/auth_service.py	108	9	92%
app/services/backup_service.py	131	7	95%
app/services/baseline_service.py	68	0	100%
app/services/bulk_service.py	354	36	90%
app/services/change_request_service.py	264	13	95%
app/services/change_set_service.py	135	15	89%
app/services/connection_service.py	131	5	96%
app/services/delegation_service.py	86	6	93%

Module	Statements	Missed	Coverage
app/services/domain_signoff_service.py	91	5	95%
app/services/duplicate_service.py	41	1	98%
app/services/export_profile_service.py	97	19	80%
app/services/export_service.py	269	8	97%
app/services/field_review_service.py	126	12	90%
app/services/health_report_service.py	81	4	95%
app/services/hub_consolidation_service.py	106	8	92%
app/services/import_service.py	483	43	91%
app/services/integration_service.py	192	7	96%
app/services/kpi_service.py	72	7	90%
app/services/legacy_mapping_service.py	70	6	91%
app/services/lookup_admin_service.py	96	5	95%
app/services/lookup_service.py	33	4	88%
app/services/mail_service.py	30	3	90%
app/services/manifest_service.py	46	1	98%
app/services/mapping_service.py	633	63	90%
app/services/notification_service.py	98	1	99%
app/services/numbering_service.py	78	3	96%
app/services/occurrence_service.py	88	7	92%
app/services/package_service.py	199	27	86%
app/services/reference_service.py	72	11	85%
app/services/release_notes_service.py	115	9	92%
app/services/release_service.py	136	6	96%
app/services/revision_diff_service.py	35	0	100%
app/services/saved_search_service.py	61	0	100%
app/services/settings_service.py	74	6	92%
app/services/sla_service.py	57	0	100%
app/services/software_version_service.py	48	1	98%
app/services/st_import_service.py	642	85	87%
app/services/status_code_service.py	456	19	96%
app/services/terminology_service.py	293	21	93%
app/services/translation_impact_service.py	127	8	94%
app/services/translation_review_service.py	362	33	91%
app/services/translation_service.py	123	10	92%
app/services/user_service.py	162	3	98%
app/services/validation.py	88	11	88%
app/status_code_fields.py	164	6	96%
app/terminology_seed.py	14	1	93%
app/terminology_seed_process.py	2	0	100%
app/terminology_seed_product.py	1	0	100%
app/translation_catalogue.py	123	0	100%
app/translation_precheck.py	141	21	85%
app/ui/__init__.py	73	8	89%
app/ui/admin_views.py	170	21	88%
app/ui/auth_views.py	95	26	73%
app/ui/bulk_views.py	114	17	85%
app/ui/change_request_views.py	192	22	89%
app/ui/dashboard_views.py	34	0	100%
app/ui/export_views.py	23	1	96%
app/ui/governance_views.py	255	40	84%
app/ui/import_views.py	133	15	89%

Module	Statements	Missed	Coverage
app/ui/insight_views.py	118	17	86%
app/ui/mapping_views.py	201	31	85%
app/ui/nav.py	72	2	97%
app/ui/publication_views.py	198	50	75%
app/ui/release_views.py	95	29	69%
app/ui/status_code_views.py	330	55	83%
app/ui/terminology_views.py	114	5	96%
app/ui/translation_views.py	138	4	97%
app/utils.py	9	0	100%
app/version.py	1	0	100%

6. Traceability evidence check

`docs/20-traceability-matrix.md` cites 618 test names / test files as evidence. **618 resolve** to a test that exists; **0 are stale**.

6a. Requirement coverage

The specification defines **251** requirement ids (FR, NFR, BR in `03-srs.md`, SEC in `11-security-architecture.md`, AUD in `12-audit-compliance.md`). **213** are cited by at least one test (module docstring, test docstring or test name); **37** more are marked covered, with the tests named, in `20-traceability-matrix.md`; **1** are marked there as out of scope or needing a person (`NFR-002`). **0 are not accounted for**.

7. Failures

None.

8. Complete test inventory

Every executed test, grouped by file, with its outcome, duration, and the requirement IDs it cites (module/test docstring or name). Parametrised cases appear individually.

tests/unit/test_lifecycle.py — 52 tests, 52 passed

TC-001 / TC-002 / BR-004: the 7-state lifecycle and its guard.

Result	Test	Purpose	Requirements	Time
OK	test_every_defined_transition_is_accepted[APPROVED-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[DEPRECATED-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[DEPRECATED-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[DRAFT-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[PENDING_APPROVAL-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[PENDING_APPROVAL-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[PENDING_APPROVAL-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[REVIEW-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_defined_transition_is_accepted[REVIEW-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[APPROVED-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[APPROVED-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[APPROVED-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[APPROVED-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[APPROVED-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[APPROVED-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s

Result	Test	Purpose	Requirements	Time
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[ARCHIVED-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DEPRECATED-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DEPRECATED-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DEPRECATED-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DEPRECATED-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DEPRECATED-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DRAFT-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DRAFT-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DRAFT-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DRAFT-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DRAFT-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[DRAFT-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[PENDING_APPROVAL-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[PENDING_APPROVAL-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[PENDING_APPROVAL-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[PENDING_APPROVAL-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-DRAFT]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-PENDING_APPROVAL]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[RELEASED-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[REVIEW-APPROVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[REVIEW-ARCHIVED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[REVIEW-DEPRECATED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[REVIEW-RELEASED]		BR-004, TC-001, TC-002	0.00 s
OK	test_every_other_pairwise_transition_is_rejected[REVIEW-REVIEW]		BR-004, TC-001, TC-002	0.00 s
OK	test_orm_guard_blocks_an_illegal_status_assignment		BR-004, TC-001, TC-002	1.57 s
OK	test_terminal_and_gated_states_cannot_be_skipped		BR-004, TC-001, TC-002	0.00 s
OK	test_there_are_exactly_seven_states		BR-004, TC-001, TC-002	0.02 s

tests/unit/test_password.py — 5 tests, 5 passed

Result	Test	Purpose	Requirements	Time
OK	test_hash_and_verify_round_trip			0.11 s
OK	test_policy_accepts_reasonable_password			0.00 s
OK	test_policy_rejects_blocklisted_password			0.00 s
OK	test_policy_rejects_short_password			0.00 s
OK	test_verify_rejects_wrong_password			0.10 s

tests/unit/test_rate_limit.py — 6 tests, 6 passed

SEC-004 / I-20: the login rate limiter, backed by the database.

Result	Test	Purpose	Requirements	Time
OK	test_a_refused_attempt_is_not_recorded		NFR-009, SEC-004, TC-050	0.45 s
OK	test_allows_attempts_under_the_limit		NFR-009, SEC-004, TC-050	0.53 s

Result	Test	Purpose	Requirements	Time
OK	test_attempts_age_out_of_the_window		NFR-009, SEC-004, TC-050	0.46 s
OK	test_blocks_after_five_username_attempts		NFR-009, SEC-004, TC-050	0.45 s
OK	test_blocks_after_ten_ip_attempts_across_usernames		NFR-009, SEC-004, TC-050	1.34 s
OK	test_username_limit_is_case_insensitive		NFR-009, SEC-004, TC-050	0.46 s

tests/unit/test_validation.py — 11 tests, 11 passed

FR-022, FR-027 field rules from docs/06-data-dictionary.md, v3

Result	Test	Purpose	Requirements	Time
OK	test_description_too_short		FR-022, FR-027	0.00 s
OK	test_number_of_times_repeated_must_be_at_least_one		FR-022, FR-027	0.00 s
OK	test_repeated_over_the_course_format		FR-022, FR-027	0.01 s
OK	test_reset_delay_rejects_missing_unit		FR-022, FR-027	0.00 s
OK	test_set_delay_accepts_leading_star_and_units		FR-022, FR-027	0.00 s
OK	test_set_delay_rejects_missing_unit		FR-022, FR-027	0.00 s
OK	test_status_category_must_be_error_warning_or_info		FR-022, FR-027	0.00 s
OK	test_title_must_not_equal_description		FR-022, FR-027	0.00 s
OK	test_title_too_short		FR-022, FR-027	0.00 s
OK	test_turbine_platforms_required_when_provided_empty		FR-022, FR-027	0.00 s
OK	test_valid_data_has_no_errors		FR-022, FR-027	0.00 s

tests/integration/test_audit_chain_and_ops.py — 27 tests, 27 passed

I-18 tamper-evident audit log, I-09 verified backups, I-21 readiness / metrics / log file.

Result	Test	Purpose	Requirements	Time
OK	test_a_backup_is_written_verified_and_recorded		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.24 s
OK	test_a_configured_key_makes_the_chain_keyed		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.17 s
OK	test_a_corrupt_backup_fails_verification		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.36 s
OK	test_a_forged_row_cannot_be_repaired_without_the_key		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.86 s
OK	test_a_metrics_token_admits_an_address_outside_the_networks		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.82 s
OK	test_a_removed_row_breaks_the_link		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.34 s
OK	test_a_restored_backup_runs_the_application	The restore drill from docs/14 §5 in miniature: back up, start an app on the copy, log in.	AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.63 s
OK	test_a_tampered_audit_log_inside_a_backup_is_flagged		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.28 s
OK	test_an_edited_row_is_found_and_named		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.18 s

Result	Test	Purpose	Requirements	Time
OK	test_archive_exports_old_rows_with_a_hash_and_deletes_nothing		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.19 s
OK	test_archive_with_nothing_old_enough_does_nothing		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.41 s
OK	test_backup_needs_a_file_database	An in-memory database cannot be backed up; built explicitly so the test also holds when	AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.17 s
OK	test_cli_audit_verify_fails_with_a_non_zero_exit_when_broken		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.11 s
OK	test_cli_backup_verify_and_audit_commands		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.25 s
OK	test_every_audit_row_is_chained_to_its_predecessor		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	2.62 s
OK	test_metrics_are_refused_outside_the_allowed_networks		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.58 s
OK	test_metrics_count_requests_and_hold_no_business_data		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.01 s
OK	test_only_the_newest_generations_are_kept		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.24 s
OK	test_ready_is_503_when_the_chain_is_broken_or_the_schema_outdated		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.54 s
OK	test_ready_needs_no_login_and_reveals_no_business_data		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.78 s
OK	test_ready_reports_ok_and_degraded_by_backup_state		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	2.02 s
OK	test_requests_are_logged_to_the_configured_file		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.17 s
OK	test_rows_without_hashes_before_the_chain_started_are_tolerated		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	0.96 s
OK	test_the_append_only_triggers_still_stop_edits_inside_the_application		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.33 s
OK	test_the_hash_is_reproducible_from_the_stored_fields		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.23 s
OK	test_truncating_the_tail_is_caught_by_the_anchor		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.05 s
OK	test_verify_tail_detects_recent_tampering_cheaply		AUD-072, AUD-073, AUD-074, AUD-075, BR-020, FR-150, FR-151, FR-152, FR-153, NFR-011, NFR-012, SEC-029, SEC-030, TC-044	1.14 s

tests/integration/test_audit_coverage.py — 7 tests, 7 passed

FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a: the audit trail

Result	Test	Purpose	Requirements	Time
OK	test_audit_log_entries_cannot_be_updated_or_deleted	FR-072: append-only at the database layer, not just by omission.	FR-072	0.66 s
OK	test_each_lifecycle_action_is_audited_exactly_once	TC-015: one entry per state-changing action across the whole lifecycle.	TC-015	2.24 s
OK	test_import_lifecycle_is_audited		FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a	1.04 s
OK	test_sandbox_delete_keeps_full_snapshot_and_creation_entry	TC-015a / FR-056 / FR-073.	FR-056, FR-073, TC-015a	1.15 s
OK	test_subgroup_creation_is_audited_as_subgroup_create		FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a	0.74 s
OK	test_unexpected_import_failure_marks_job_failed_and_is_audited		FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a	0.84 s
OK	test_user_and_auth_events_are_audited		FR-056, FR-070, FR-072, FR-073, NFR-006, TC-015, TC-015a	0.73 s

tests/integration/test_audit_log.py — 4 tests, 4 passed

FR-070-FR-071: audit log search, Administrator full access,

Result	Test	Purpose	Requirements	Time
OK	test_admin_sees_all_entries		FR-070, FR-071	0.95 s
OK	test_engineer_cannot_view_audit_log		FR-070, FR-071	0.59 s
OK	test_filter_by_entity_type		FR-070, FR-071	0.71 s
OK	test_reviewer_only_sees_own_actions		FR-070, FR-071	0.92 s

tests/integration/test_auth.py — 7 tests, 7 passed

FR-001-FR-004, SEC-001-SEC-004.

Result	Test	Purpose	Requirements	Time
OK	test_deactivated_user_is_immediately_rejected		FR-001, FR-004, SEC-001, SEC-004	0.67 s
OK	test_five_failed_attempts_locks_the_account	The 5th consecutive failure both locks the account (SEC-004) and	SEC-004	0.82 s
OK	test_login_success_returns_user_and_sets_session		FR-001, FR-004, SEC-001, SEC-004	0.58 s
OK	test_logout_invalidates_session		FR-001, FR-004, SEC-001, SEC-004	0.69 s
OK	test_me_requires_login		FR-001, FR-004, SEC-001, SEC-004	0.74 s
OK	test_password_reset_invalidates_existing_session		FR-001, FR-004, SEC-001, SEC-004	0.96 s
OK	test_wrong_password_is_rejected		FR-001, FR-004, SEC-001, SEC-004	0.58 s

tests/integration/test_baseline_and_packages.py — 16 tests, 16 passed

The baseline of a Release and the all-languages / all-access-levels package (docs/03-srs.md

Result	Test	Purpose	Requirements	Time
OK	test_a_hub_controller_group_exists_for_the_controller_filter		BR-007, FR-045, FR-173, FR-174, TC-066	0.58 s
OK	test_a_package_can_be_narrowed_to_a_platform_and_a_controller		BR-007, FR-045, FR-173, FR-174, TC-066	2.74 s
OK	test_a_package_holds_every_language_for_every_access_level		BR-007, FR-045, FR-173, FR-174, TC-066	3.50 s
OK	test_a_package_needs_a_published_release_and_valid_choices		BR-007, FR-045, FR-173, FR-174, TC-066	2.79 s
OK	test_a_release_can_still_be_exported_as_only_its_own_changes		BR-007, FR-045, FR-173, FR-174, TC-066	2.88 s

Result	Test	Purpose	Requirements	Time
OK	test_an_unpublished_release_previews_what_its_manual_will_contain		BR-007, FR-045, FR-173, FR-174, TC-066	2.79 s
OK	test_everybody_may_download_what_is_released_and_only_owner_and_admin_what_is_not		BR-007, FR-045, FR-173, FR-174, TC-066	4.33 s
OK	test_the_baseline_narrows_to_a_controller		BR-007, FR-045, FR-173, FR-174, TC-066	2.07 s
OK	test_the_baseline_narrows_to_a_platform		BR-007, FR-045, FR-173, FR-174, TC-066	3.12 s
OK	test_the_baseline_of_a_release_holds_every_code_valid_at_that_release		BR-007, FR-045, FR-173, FR-174, TC-066	2.91 s
OK	test_the_manifest_vouches_for_every_file_and_reports_the_coverage		BR-007, FR-045, FR-173, FR-174, TC-066	3.24 s
OK	test_the_package_is_audited		BR-007, FR-045, FR-173, FR-174, TC-066	3.23 s
OK	test_the_readiness_check_shows_what_is_still_missing		BR-007, FR-045, FR-173, FR-174, TC-066	3.17 s
OK	test_the_release_page_offers_the_baseline_and_the_package		BR-007, FR-045, FR-173, FR-174, TC-066	3.18 s
OK	test_the_summary_counts_the_baseline_by_controller_and_platform		BR-007, FR-045, FR-173, FR-174, TC-066	2.95 s
OK	test_the_title_page_names_the_scope		BR-007, FR-045, FR-173, FR-174, TC-066	3.62 s

tests/integration/test_bulk_workflow.py — 19 tests, 19 passed

Bulk work on a change set with a sample inspection (docs/03-srs.md FR-180): edit, submit, domain

Result	Test	Purpose	Requirements	Time
OK	test_a_bulk_domain_signoff_needs_a_sample_and_every_code_gets_its_own_signoff		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.60 s
OK	test_a_decision_in_the_workbench_names_its_sample		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	2.32 s
OK	test_a_defect_in_the_sample_blocks_the_decision_until_it_is_sent_back		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	2.93 s
OK	test_a_defect_needs_a_note_and_a_verdict_must_be_ok_or_defect		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.93 s
OK	test_a_domain_signoff_is_only_for_a_holder_of_the_domain		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.36 s
OK	test_a_sample_cannot_be_borrowed_reused_or_kept_after_the_batch_changed		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	2.53 s
OK	test_a_small_batch_needs_no_sample		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.46 s
OK	test_a_whole_domain_signoff_can_be_done_in_the_workbench		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	2.13 s
OK	test_bulk_edit_refuses_unknown_fields_and_other_peoples_drafts		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.67 s
OK	test_bulk_edit_sets_the_same_value_on_every_draft_with_its_own_audit_entry		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.43 s
OK	test_bulk_submit_sends_the_drafts_that_are_ready_and_reports_the_rest		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.69 s
OK	test_the_bulk_screens_are_translated		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.81 s
OK	test_the_dossier_puts_the_codes_the_samples_and_the_audit_in_one_workbook		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	2.25 s
OK	test_the_person_who_wrote_a_code_is_not_part_of_the_batch_they_decide		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	2.34 s
OK	test_the_readiness_names_what_stops_each_draft		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.16 s
OK	test_the_sample_is_spread_over_the_subgroups_and_reproducible		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	0.00 s
OK	test_the_size_of_the_sample_follows_the_settings		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	0.96 s
OK	test_the_workbench_shows_the_batch_the_readiness_and_the_rule		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	2.06 s
OK	test_viewers_see_nothing_of_the_bulk_workbench		BR-004, BR-005, FR-029, FR-031, FR-180, SEC-010, TC-070	1.27 s

tests/integration/test_change_requests.py — 11 tests, 11 passed

FR-030-FR-039, FR-027-FR-029, BR-004/BR-005. Full submit -> dual-review ->

Result	Test	Purpose	Requirements	Time
OK	test_author_cannot_sign_off_own_cr		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	1.08 s
OK	test_chief_engineer_cannot_be_a_prior_signer		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	1.64 s
OK	test_domain_signoff_rejects_user_without_that_domain		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	0.83 s
OK	test_dual_signoff_in_either_order_reaches_pending_approval		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	1.72 s
OK	test_full_happy_path_to_approved		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	1.98 s
OK	test_rejection_returns_to_draft_and_clears_signoffs		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	1.10 s
OK	test_reviewer_rejection_without_comment_is_rejected		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	1.16 s
OK	test_submit_blocked_until_domain_signoff_complete		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	1.00 s
OK	test_withdraw_forbidden_for_non_author		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	0.90 s
OK	test_withdraw_removes_abandoned_new_draft		BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	0.80 s
OK	test_withdraw_revision_type_restores_prior_current_revision	Regression test: withdrawing a REVISION-type CR must restore the	BR-004, BR-005, FR-027, FR-029, FR-030, FR-039	2.25 s

tests/integration/test_connections.py — 12 tests, 12 passed

How status codes are connected: suppression by a cause, preconditions and escalation, read from the

Result	Test	Purpose	Requirements	Time
OK	test_a_negated_active_flag_is_a_cause_and_a_plain_one_a_precondition		FR-048o, FR-092, FR-179, TC-069	0.52 s
OK	test_a_viewer_does_not_see_the_connections		FR-048o, FR-092, FR-179, TC-069	1.44 s
OK	test_an_imported_program_brings_its_cause_and_its_repeat_code		FR-048o, FR-092, FR-179, TC-069	1.32 s
OK	test_escalation_targets_say_which_are_outside_the_catalogue		FR-048o, FR-092, FR-179, TC-069	1.08 s
OK	test_one_code_shows_what_it_depends_on_and_what_depends_on_it		FR-048o, FR-092, FR-179, TC-069	1.09 s
OK	test_the_api_returns_the_connections_of_a_code_and_of_the_catalogue		FR-048o, FR-092, FR-179, TC-069	1.50 s
OK	test_the_code_page_and_the_overview_show_the_connections		FR-048o, FR-092, FR-179, TC-069	1.59 s
OK	test_the_hierarchy_puts_the_cause_on_top_with_what_it_suppresses_below		FR-048o, FR-092, FR-179, TC-069	1.08 s
OK	test_the_hub_and_main_controller_can_be_looked_at_separately		FR-048o, FR-092, FR-179, TC-069	1.09 s
OK	test_the_menu_offers_the_connections_to_the_working_roles_only		FR-048o, FR-092, FR-179, TC-069	1.29 s
OK	test_the_real_programs_name_their_cause_and_their_repeat_code		FR-048o, FR-092, FR-179, TC-069	0.52 s
OK	test_what_looks_wrong_is_listed		FR-048o, FR-092, FR-179, TC-069	1.41 s

tests/integration/test_content_integrity.py — 16 tests, 16 passed

I-01 optimistic concurrency, I-04 reference warnings and where-used, I-12 duplicate

Result	Test	Purpose	Requirements	Time
OK	test_expected_updated_at_is_enforced_too		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.19 s
OK	test_next_available_reports_how_many_numbers_are_left		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.94 s
OK	test_planned_dates_are_administrator_set_audited_and_warn_at_publish		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	2.35 s
OK	test_platform_variant_copies_the_code_and_links_the_family		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.98 s
OK	test_reference_to_a_deprecated_code_warns		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	2.77 s

Result	Test	Purpose	Requirements	Time
OK	test_related_codes_are_validated_and_checked		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.20 s
OK	test_remedy_text_length_is_validated		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.09 s
OK	test_save_without_a_version_still_works_for_old_clients		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.50 s
OK	test_similar_endpoint_supports_the_did_you_mean_hint		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.49 s
OK	test_similar_title_in_the_subgroup_is_flagged_not_blocked		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.19 s
OK	test_stale_save_is_refused_with_the_current_values		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.31 s
OK	test_troubleshooting_fields_round_trip_and_reach_the_export		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	2.11 s
OK	test_ui_stale_edit_keeps_the_users_entries_and_shows_the_conflict		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.60 s
OK	test_unknown_escalation_number_warns_but_does_not_block		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.15 s
OK	test_variant_needs_a_decided_source_and_an_engineer		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	1.30 s
OK	test_where_used_lists_dependants_and_deprecation_needs_acknowledgement		BR-019, FR-090, FR-091, FR-092, FR-093, FR-094, FR-109, FR-131, FR-141, TC-039	2.43 s

tests/integration/test_csrf.py — 2 tests, 2 passed

docs/11-security-architecture.md §5: CSRF synchronizer-token protection

Result	Test	Purpose	Requirements	Time
OK	test_state_changing_request_with_csrf_token_succeeds			0.72 s
OK	test_state_changing_request_without_csrf_token_is_rejected			0.73 s

tests/integration/test_decision_trail.py — 5 tests, 5 passed

Where review comments live: the change-request page, the status-code

Result	Test	Purpose	Requirements	Time
OK	test_audit_entries_carry_the_comment_text_and_id			2.26 s
OK	test_audit_log_can_be_filtered_by_action_and_actor			2.14 s
OK	test_status_code_page_lists_all_decisions_and_comments			1.88 s
OK	test_viewers_cannot_see_the_review_discussion			1.65 s
OK	test_withdrawing_keeps_the_discussion_in_the_audit_log			1.61 s

tests/integration/test_deprecation.py — 7 tests, 7 passed

FR-050-FR-052, BR-006, docs/05-governance-handbook.md §5, §11.

Result	Test	Purpose	Requirements	Time
OK	test_archive_blocked_before_retention_period_elapses		BR-006, FR-050, FR-052	2.43 s
OK	test_archive_succeeds_after_retention_period		BR-006, FR-050, FR-052	1.97 s
OK	test_deprecation_requires_released_state		BR-006, FR-050, FR-052	0.75 s
OK	test_full_deprecation_cycle		BR-006, FR-050, FR-052	2.82 s
OK	test_non_admin_cannot_archive_or_reinstate		BR-006, FR-050, FR-052	0.69 s
OK	test_reinstate_deprecated_code		BR-006, FR-050, FR-052	1.84 s
OK	test_second_revision_does_not_violate_current_uniqueness	Regression test: create_revision must flip the old current revision's	BR-006, FR-050, FR-052	2.00 s

tests/integration/test_email_and_windows_signin.py — 16 tests, 16 passed

The e-mail address of a user is fixed, and the Windows sign-in that will replace the password in a

Result	Test	Purpose	Requirements	Time
OK	test_a_domain_name_is_matched_through_the_mail_domain_or_the_user_name		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	1.51 s
OK	test_a_header_from_an_address_that_is_not_a_trusted_proxy_is_not_believed		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	1.00 s
OK	test_an_address_is_stored_trimmed_and_lower_case_and_must_be_valid_and_new		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	1.10 s
OK	test_no_identity_an_unknown_one_and_a_deactivated_account_are_all_refused		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	1.14 s
OK	test_production_never_starts_with_the_simulation_or_an_unknown_mode		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.16 s
OK	test_the_address_of_a_user_cannot_be_changed		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	1.49 s
OK	test_the_database_refuses_a_changed_address_too		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.94 s
OK	test_the_development_defaults_are_the_simulation_next_to_the_password		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.74 s
OK	test_the_gateways_identity_signs_the_matching_account_in_and_is_audited		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	1.08 s
OK	test_the_screens_show_the_address_as_fixed		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	1.24 s
OK	test_the_shipped_example_configuration_documents_the_settings		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.02 s
OK	test_the_simulation_page_stands_in_for_the_gateway_and_signs_in_by_address		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	1.19 s
OK	test_the_simulation_refuses_an_unknown_identity_and_is_off_by_default		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	1.15 s
OK	test_with_both_modes_the_login_page_offers_the_button_and_the_form		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	0.68 s
OK	test_with_passwords_only_the_windows_sign_in_is_not_there		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	1.17 s
OK	test_with_windows_only_passwords_are_refused_and_the_login_page_says_so		FR-001, FR-010, FR-162, FR-163, SEC-033, TC-058	1.13 s

tests/integration/test_export_language.py — 4 tests, 4 passed

The words around an export follow the language of the export (docs/03-srs.md FR-171): the title

Result	Test	Purpose	Requirements	Time
OK	test_a_pdf_in_french_prints_its_own_words_in_french		FR-045, FR-110, FR-165, FR-171, TC-065	0.11 s
OK	test_a_placeholder_survives_in_every_language		FR-045, FR-110, FR-165, FR-171, TC-065	0.00 s
OK	test_english_and_no_language_leave_the_text_as_it_is		FR-045, FR-110, FR-165, FR-171, TC-065	0.09 s
OK	test_every_word_around_an_export_exists_in_every_language		FR-045, FR-110, FR-165, FR-171, TC-065	0.00 s

tests/integration/test_exports_and_diff.py — 15 tests, 15 passed

FR-032, FR-045, FR-046, FR-054, FR-064, BR-008, SEC-023, TC-023, TC-035:

Result	Test	Purpose	Requirements	Time
OK	test_diff_card_renders_on_change_request_page		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.54 s
OK	test_diff_hidden_from_viewers_for_open_revisions_and_unknown_revisions		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.98 s

Result	Test	Purpose	Requirements	Time
OK	test_export_file_names_carry_a_timestamp	Every export file name ends with the generation time, YYYY-MM-DD_HH-MM (UTC).	BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	2.81 s
OK	test_exports_neutralise_formula_injection		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.26 s
OK	test_json_export_is_flat_rows		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	2.10 s
OK	test_new_code_diff_is_marked_new		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.59 s
OK	test_release_can_be_deleted_before_publish_only		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	2.27 s
OK	test_released_export_contains_only_released_non_sandbox_codes		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	2.28 s
OK	test_released_export_filters_and_formats		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	2.29 s
OK	test_revision_diff_endpoint_and_cr_baseline_diff		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	2.08 s
OK	test_working_set_filters_by_lifecycle_state		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.38 s
OK	test_working_set_is_labelled_and_excludes_sandbox		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.40 s
OK	test_working_set_not_available_to_viewers_or_anonymous		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.67 s
OK	test_working_set_pdf_and_xlsx_render		BR-008, FR-032, FR-045, FR-046, FR-054, FR-064, SEC-023, TC-023, TC-035	1.39 s
OK	test_xlsx_export_has_same_rows_as_csv	TC-035.	TC-035	2.41 s

tests/integration/test_generated_fields.py — 20 tests, 20 passed

The fields of the generated controller programs (schema v4.1, docs/06-data-dictionary.md sections

Result	Test	Purpose	Requirements	Time
OK	test_a_field_of_the_other_controller_is_refused_and_the_rest_is_stored		FR-025, FR-032, FR-048o, FR-168, TC-063	1.17 s
OK	test_a_hub_program_import_keeps_its_hub_fields_and_nothing_of_the_main_ones		FR-025, FR-032, FR-048o, FR-168, TC-063	1.25 s
OK	test_a_main_program_import_keeps_every_field_of_the_program[StCd_1000.st]		FR-025, FR-032, FR-048o, FR-168, TC-063	1.45 s
OK	test_a_main_program_import_keeps_every_field_of_the_program[StCd_1001.st]		FR-025, FR-032, FR-048o, FR-168, TC-063	1.24 s
OK	test_a_main_program_import_keeps_every_field_of_the_program[StCd_1002.st]		FR-025, FR-032, FR-048o, FR-168, TC-063	1.69 s
OK	test_a_named_audience_value_brings_its_default_levels_and_explicit_levels_win		FR-025, FR-032, FR-048o, FR-168, TC-063	1.50 s
OK	test_a_new_revision_carries_over_what_the_caller_does_not_send_and_the_diff_shows_changes		FR-025, FR-032, FR-048o, FR-168, TC-063	1.90 s
OK	test_a_rejected_form_keeps_what_was_typed		FR-025, FR-032, FR-048o, FR-168, TC-063	1.35 s
OK	test_the_csv_import_carries_the_new_columns_and_an_update_without_them_keeps_the_values		FR-025, FR-032, FR-048o, FR-168, TC-063	1.21 s
OK	test_the_form_stores_the_fields_and_the_pages_show_them		FR-025, FR-032, FR-048o, FR-168, TC-063	2.07 s
OK	test_the_levels_and_programs_are_internal_and_never_leave_in_an_audience_export		FR-025, FR-032, FR-048o, FR-168, TC-063	1.85 s
OK	test_the_two_zero_load_attributes_are_not_mixed_up		FR-025, FR-032, FR-048o, FR-168, TC-063	1.27 s
OK	test_values_out_of_range_are_refused[extra0]		FR-025, FR-032, FR-048o, FR-168, TC-063	1.13 s

Result	Test	Purpose	Requirements	Time
OK	test_values_out_of_range_are_refused[extra1]		FR-025, FR-032, FR-048o, FR-168, TC-063	1.15 s
OK	test_values_out_of_range_are_refused[extra2]		FR-025, FR-032, FR-048o, FR-168, TC-063	1.55 s
OK	test_values_out_of_range_are_refused[extra3]		FR-025, FR-032, FR-048o, FR-168, TC-063	1.11 s
OK	test_values_out_of_range_are_refused[extra4]		FR-025, FR-032, FR-048o, FR-168, TC-063	1.10 s
OK	test_values_out_of_range_are_refused[extra5]		FR-025, FR-032, FR-048o, FR-168, TC-063	1.13 s
OK	test_values_out_of_range_are_refused[extra6]		FR-025, FR-032, FR-048o, FR-168, TC-063	1.13 s
OK	test_withdrawing_and_deleting_a_code_removes_its_levels		FR-025, FR-032, FR-048o, FR-168, TC-063	1.19 s

tests/integration/test_governance_extras.py — 24 tests, 24 passed

Remaining spec checks: SEC-010 service-layer authorization, FR-010/011/011a/014

Result	Test	Purpose	Requirements	Time
OK	test_a_user_holds_exactly_one_role		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.73 s
OK	test_account_lock_and_password_events_are_audited		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	1.23 s
OK	test_auth_provider_seam_is_substitutable	Callers depend on AuthenticatedUser only: a different provider returning	FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.76 s
OK	test_comment_leaves_state_and_signoffs_untouched		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	1.75 s
OK	test_import_preserves_legacy_reference_number		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	1.32 s
OK	test_next_available_preview_does_not_reserve_a_number		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	1.84 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ADMINISTRATOR]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.54 s
OK	test_services_enforce_roles_independently_of_route_decorators[-CHIEF_ENGINEER0]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	1.01 s
OK	test_services_enforce_roles_independently_of_route_decorators[-CHIEF_ENGINEER1]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.55 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER0]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.59 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER1]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.57 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER2]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.57 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER3]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.56 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER4]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.52 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER5]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.45 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER6]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.73 s
OK	test_services_enforce_roles_independently_of_route_decorators[-ENGINEER7]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	1.22 s
OK	test_services_enforce_roles_independently_of_route_decorators[-REVIEWER0]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.60 s
OK	test_services_enforce_roles_independently_of_route_decorators[-REVIEWER1]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.51 s

Result	Test	Purpose	Requirements	Time
OK	test_services_enforce_roles_independently_of_route_decorators[-REVIEWER2]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.57 s
OK	test_services_enforce_roles_independently_of_route_decorators[-VIEWER0]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.48 s
OK	test_services_enforce_roles_independently_of_route_decorators[-VIEWER1]		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.63 s
OK	test_user_administration_events_are_audited		FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	0.98 s
OK	test_whole_lifecycle_works_with_all_outbound_network_blocked	The application must keep operating when it can reach nothing beyond	FR-005, FR-010, FR-021, FR-038, FR-048d, NFR-008, SEC-010	2.52 s

tests/integration/test_import.py — 7 tests, 7 passed

FR-048a-d, EPIC-14: CSV/XLSX library import, still fully governed

Result	Test	Purpose	Requirements	Time
OK	test_ambiguous_group_code_requires_range_hint	WTUR is reused for a second, numerically distinct Hub Controller	FR-048a	0.85 s
OK	test_imported_row_becomes_import_type_draft_change_request		FR-048a	0.82 s
OK	test_invalid_row_is_reported_and_creates_nothing		FR-048a	0.84 s
OK	test_non_admin_engineer_can_upload_import		FR-048a	0.79 s
OK	test_unknown_functional_system_group_code_reported_as_error		FR-048a	0.69 s
OK	test_unsupported_file_extension_rejected		FR-048a	0.73 s
OK	test_viewer_cannot_upload_import		FR-048a	1.03 s

tests/integration/test_import_toolkit.py — 20 tests, 20 passed

The migration toolkit around the library import (docs/21-improvement-ideas.md): I-05 template, dry

Result	Test	Purpose	Requirements	Time
OK	test_a_big_file_is_processed_in_the_background_with_progress		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	3.06 s
OK	test_a_legacy_number_already_imported_is_refused_in_create_mode		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	1.36 s
OK	test_dry_run_reports_every_problem_and_creates_nothing		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.76 s
OK	test_dry_run_warns_about_a_possible_duplicate_without_blocking		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.83 s
OK	test_error_report_has_row_column_value_and_a_fix_hint_as_csv_and_xlsx		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.74 s
OK	test_files_above_the_threshold_go_to_the_background_automatically		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.80 s
OK	test_formula_like_values_in_the_error_report_are_neutralised		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.73 s
OK	test_import_ui_pages_render_and_the_validate_button_creates_nothing		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	1.10 s
OK	test_missing_required_columns_are_named_up_front		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.64 s
OK	test_new_content_columns_are_imported		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.82 s
OK	test_reconciliation_counts_states_and_names_the_missing_numbers		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	2.05 s
OK	test_similar_titles_are_reported_as_warnings		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.83 s
OK	test_template_csv_and_xlsx_carry_the_exact_headers_and_lookup_codes		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.82 s

Result	Test	Purpose	Requirements	Time
OK	test_the_downloaded_template_imports_cleanly_as_is		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	1.26 s
OK	test_the_same_legacy_number_in_another_group_is_a_different_code		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.75 s
OK	test_update_mode_errors_are_specific		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	1.56 s
OK	test_update_mode_opens_a_new_revision_of_a_released_code		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	2.10 s
OK	test_update_mode_revises_the_open_draft_in_place		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.86 s
OK	test_update_mode_skips_unchanged_rows_and_creates_missing_ones		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	1.25 s
OK	test_update_of_someone_elses_draft_is_reported_on_the_row		BR-015, FR-048i, FR-048j, FR-048k, FR-048l, FR-048m, FR-048n, FR-093, TC-043	0.99 s

tests/integration/test_integrations_and_reports.py — 32 tests, 32 passed

I-29 API tokens, change feed and webhooks; I-26 saved searches; I-28 fleet occurrences;

Result	Test	Purpose	Requirements	Time
OK	test_a_disabled_webhook_is_not_called		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	2.42 s
OK	test_a_failing_webhook_is_recorded_and_never_blocks_the_publish		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	2.31 s
OK	test_a_hub_code_is_carried_over_as_a_main_controller_draft		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.86 s
OK	test_a_publish_calls_the_webhook_with_a_verifiable_signature		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	2.60 s
OK	test_a_token_is_shown_once_and_stored_only_as_a_hash		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.05 s
OK	test_a_token_only_works_for_its_scope		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.66 s
OK	test_a_token_reads_released_data_only_and_never_writes		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	2.14 s
OK	test_an_unreachable_webhook_target_does_not_block_the_publish		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	4.44 s
OK	test_bulk_deprecation_is_for_administrators_and_hub_groups		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.24 s
OK	test_carry_over_is_refused_for_the_wrong_kind_of_code_or_target		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	2.52 s
OK	test_health_report_covers_capacity_missing_data_and_inconsistencies		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	2.08 s
OK	test_health_report_finds_an_info_code_with_an_alarm		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.79 s
OK	test_health_report_flags_a_nearly_full_band		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.93 s
OK	test_hub_consolidation_ui_page_renders		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	2.02 s
OK	test_kpis_have_the_documented_shape		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	2.08 s
OK	test_occurrence_import_validation_and_permissions		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.23 s
OK	test_occurrence_report_names_codes_that_never_fire_and_those_that_fire_most		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	3.42 s
OK	test_redirects_are_not_followed		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	3.01 s
OK	test_reports_are_limited_to_the_right_roles		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	2.06 s
OK	test_revoked_expired_unknown_and_orphaned_tokens_are_refused		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.30 s

Result	Test	Purpose	Requirements	Time
OK	test_sandbox_activity_never_appears_in_the_feed		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.64 s
OK	test_saved_search_ui_save_and_delete		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.29 s
OK	test_saved_searches_are_private_to_their_owner		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.24 s
OK	test_saved_searches_round_trip_through_the_url		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.37 s
OK	test_saving_under_an_existing_name_replaces_it_and_validates_the_name		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.07 s
OK	test_the_capacity_table_colours_main_blue_and_hub_light_blue		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.23 s
OK	test_the_feed_lists_releases_in_order_with_a_cursor_and_an_etag		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	3.78 s
OK	test_the_feed_needs_authentication		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	0.59 s
OK	test_the_whole_consolidation_loop_ends_with_a_deprecation_request		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	3.50 s
OK	test_token_events_are_audited_without_the_secret		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.15 s
OK	test_token_management_is_for_administrators_and_validated		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.19 s
OK	test_webhook_targets_must_be_internal_and_inputs_valid		FR-123, FR-140, FR-141, FR-142, FR-143, FR-144, FR-145, FR-146, FR-147, SEC-027, SEC-028, TC-046	1.27 s

tests/integration/test_invariants.py — 15 tests, 15 passed

Business-rule and data-integrity checks: BR-001..BR-003, BR-009, FR-023a,

Result	Test	Purpose	Requirements	Time
OK	test_draft_can_be_saved_repeatedly_without_workflow_transitions		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	1.78 s
OK	test_duplicate_identifier_rejected_by_database		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	1.83 s
OK	test_each_subgroup_sub_band_numbers_independently		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	0.59 s
OK	test_exactly_one_current_revision_per_code	BR-002 / TC-011.	BR-002, TC-011	1.97 s
OK	test_filters_sort_and_pagination		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	1.53 s
OK	test_full_text_search_matches_identifier_title_and_description		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	1.45 s
OK	test_identifier_formats		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	1.44 s
OK	test_identifier_never_changes_after_allocation	BR-001: a rejected-and-resubmitted CR keeps its identifier.	BR-001	1.63 s
OK	test_in_flight_revision_cannot_be_edited_in_place	FR-025: Review and later states are locked.	FR-025	1.40 s
OK	test_new_revision_of_a_released_code_keeps_its_identifier	BR-001 / FR-021: the identifier is allocated once, not again per revision.	BR-001, FR-021	2.35 s
OK	test_only_one_open_revision_per_code	BR-003 / TC-012: a second revision cannot be opened while one is open.	BR-003, TC-012	2.04 s
OK	test_released_code_stays_visible_to_viewers_while_a_new_revision_is_open	BR-009: opening revision 2 (a Draft) must not make the Released revision 1	BR-009	2.22 s
OK	test_revision_history_visible_to_viewer_only_once_decided	FR-064 / BR-009.	BR-009, FR-064	2.45 s

Result	Test	Purpose	Requirements	Time
OK	test_sequential_submits_in_two_subgroups_do_not_interfere		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	2.04 s
OK	test_submit_requires_a_turbine_platform		BR-001, BR-003, BR-009, FR-023a, FR-024, FR-060, FR-064, TC-004, TC-008, TC-010, TC-013	1.07 s

tests/integration/test_lookup_admin.py — 9 tests, 9 passed

US-091: Administrator manages controlled vocabularies. docs/05-governance-

Result	Test	Purpose	Requirements	Time
OK	test_admin_can_add_turbine_platform		US-091	0.73 s
OK	test_admin_can_create_new_functional_system_group_in_free_range	FR-048g: into any currently free range (10000-10999, 14000-14999,	FR-048g	0.65 s
OK	test_admin_can_deactivate_and_relabel_a_value		US-091	0.78 s
OK	test_create_functional_subgroup_with_range		US-091	0.72 s
OK	test_duplicate_code_rejected		US-091	0.74 s
OK	test_functional_subgroup_overlap_rejected	FR-048f: rejected with 409, naming the conflicting subgroup.	FR-048f	0.64 s
OK	test_functional_system_group_not_creatable_via_generic_endpoint	v3 (EPIC-16): creation moved to a dedicated, non-overlap-validated	US-091	0.75 s
OK	test_new_functional_system_group_range_overlap_rejected	FR-048h: rejected with 409, naming the conflicting group.	FR-048h	1.20 s
OK	test_non_admin_cannot_create_lookup_value		US-091	0.71 s

tests/integration/test_mapping.py — 37 tests, 37 passed

The mapping of legacy to new status codes kept in the database and changed only with the three

Result	Test	Purpose	Requirements	Time
OK	test_a_bad_seed_workbook_is_refused_as_a_whole		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	0.53 s
OK	test_a_change_confirm_and_reject_work_on_stored_rows		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	2.00 s
OK	test_a_database_without_the_mapping_tables_says_what_to_run		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.80 s
OK	test_a_decision_asks_for_the_password_again		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.30 s
OK	test_a_new_row_takes_effect_only_after_all_three_approvals		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.33 s
OK	test_a_number_a_code_of_the_catalogue_carries_cannot_change		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.21 s
OK	test_a_number_held_by_the_catalogue_is_found_in_the_identifiers		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	0.67 s
OK	test_a_row_changed_in_between_makes_the_older_request_unusable		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	2.43 s
OK	test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides0-description]		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.03 s
OK	test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides1-type must be Error, Warning or Info]		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.05 s
OK	test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides2-up/down counter must be Yes or No]		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.13 s
OK	test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides3-new number must be a whole number]		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.08 s
OK	test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides4-between 1 and 99999]		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	0.94 s
OK	test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides5-neither a time]		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.58 s

Result	Test	Purpose	Requirements	Time
OK	test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides6-not a named level]		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.13 s
OK	test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides7-belongs to the]		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.00 s
OK	test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides8-cannot be approved without type]		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.11 s
OK	test_a_row_that_does_not_pass_the_checks_makes_no_request[overrides9-without logical node]		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.10 s
OK	test_a_second_administrator_cannot_replace_the_first_and_the_requester_cannot_decide		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.52 s
OK	test_a_viewer_neither_reads_nor_proposes		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.22 s
OK	test_a_withdrawn_import_is_blocked_when_its_mapping_row_is_gone		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.26 s
OK	test_a_workbook_makes_one_request_with_the_new_and_the_different_rows		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.57 s
OK	test_a_workbook_that_unpacks_to_too_much_or_has_too_many_rows_is_refused		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.15 s
OK	test_a_workbook_with_a_wrong_row_names_it_and_requests_nothing		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.10 s
OK	test_numbers_stay_unique		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.15 s
OK	test_one_rejection_closes_the_request_and_needs_a_reason		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.30 s
OK	test_only_the_three_roles_decide_and_each_only_once		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.51 s
OK	test_show_all_puts_every_row_on_one_page_so_that_select_all_covers_them		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.40 s
OK	test_the_export_is_the_layout_the_upload_reads		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.17 s
OK	test_the_export_never_starts_a_formula		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.17 s
OK	test_the_menu_counts_the_requests_that_wait_for_a_decision		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.33 s
OK	test_the_pages_make_requests_and_decide_with_the_form_posts		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	2.13 s
OK	test_the_pages_show_the_mapping_the_requests_and_the_three_slots		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.51 s
OK	test_the_requester_can_withdraw_and_nobody_else		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.79 s
OK	test_the_seed_workbook_is_consistent		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	2.32 s
OK	test_the_seed_workbook_loads_into_an_empty_table_once		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.89 s
OK	test_the_sign_off_of_a_domain_is_below_the_changes_and_the_status_above		BR-027, FR-191, FR-192, FR-193, FR-194, FR-195, FR-196, FR-197, FR-198, TC-076, TC-079	1.43 s

tests/integration/test_menu.py — 22 tests, 22 passed

The menu on the left (docs/10-ui-ux-specification.md section 3, docs/03-srs.md FR-169): what each role

Result	Test	Purpose	Requirements	Time
OK	test_a_review_removes_the_mark_and_a_changed_string_brings_it_back		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.60 s
OK	test_a_viewer_cannot_choose_or_keep_a_language_nobody_has_reviewed		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.45 s
OK	test_a_viewer_sees_only_the_reviewed_languages_and_nothing_marked		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	2.25 s
OK	test_a_viewer_sees_the_catalogue_and_nothing_to_work_with		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.19 s

Result	Test	Purpose	Requirements	Time
OK	test_a_viewer_who_reviews_a_language_sees_that_one		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.13 s
OK	test_each_role_sees_the_links_of_its_work_and_no_other[admin]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.21 s
OK	test_each_role_sees_the_links_of_its_work_and_no_other[chief_engineer]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.00 s
OK	test_each_role_sees_the_links_of_its_work_and_no_other[engineer]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.08 s
OK	test_each_role_sees_the_links_of_its_work_and_no_other[reviewer]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.19 s
OK	test_each_role_sees_the_links_of_its_work_and_no_other[stcd_owner]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.60 s
OK	test_groups_can_be_closed_and_a_narrow_window_has_a_menu_button		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.28 s
OK	test_no_link_appears_twice_and_the_groups_are_the_planned_ones		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	0.93 s
OK	test_only_the_link_of_the_current_page_is_marked		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.55 s
OK	test_the_login_page_offers_only_the_reviewed_languages		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.31 s
OK	test_the_marker_can_be_switched_off_in_the_settings		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.85 s
OK	test_the_menu_is_translated_and_a_group_title_too		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.09 s
OK	test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[admin]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.26 s
OK	test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[chief_engineer]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.25 s
OK	test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[engineer]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.74 s
OK	test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[reviewer]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.22 s
OK	test_the_working_roles_see_every_language_and_the_unreviewed_ones_are_marked_beta[stcd_owner]		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.21 s
OK	test_what_waits_for_the_user_is_counted_in_the_menu		FR-169, FR-170, FR-172, NFR-010, SEC-010, TC-064	1.52 s

tests/integration/test_numbering.py — 4 tests, 4 passed

docs/07-database-design.md §7a: atomic Functional Group/Subgroup allocation.

Result	Test	Purpose	Requirements	Time
OK	test_allocation_increments_and_persists			0.65 s
OK	test_capacity_exhaustion_raises			0.65 s
OK	test_first_allocation_in_a_group_starts_at_range_start			0.59 s
OK	test_sandbox_allocation_never_collides_with_real_range			0.60 s

tests/integration/test_pdf_scripts.py — 20 tests, 20 passed

The PDF export prints every script of the languages the tool offers (docs/03-srs.md FR-165,

Result	Test	Purpose	Requirements	Time
OK	test_a_large_export_stays_fast_for_latin_text		FR-045, FR-157, FR-165, NFR-008, TC-060	11.65 s
OK	test_a_plain_latin_export_loads_only_noto_sans		FR-045, FR-157, FR-165, NFR-008, TC-060	0.05 s
OK	test_a_script_gets_its_own_font[arabic-\u0631\u0645\u0632 \u0627\u0644\u062d\u0627\u0644\u0629 \u0648\u062d\u062f\u0629 \u0627\u0644\u062a\u062d\u0643\u0645 StCd-01000]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s

Result	Test	Purpose	Requirements	Time
OK	test_a_script_gets_its_own_font[bengali-\u09b8\u09cd\u09a5\u09bf\u09a4\u09bf \u0995\u09cb\u09a1 \u09a8\u09bf\u09af\u09bc\u09a8\u09cd\u09a4\u09cd\u09b0\u0995 \u09aa\u09b0\u09c0\u0995\u09cd\u09b7\u09be]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[devanagari-\u0938\u094d\u0925\u093f\u0924\u093f \u0915\u094b\u0921 \u0928\u093f\u092f\u0902\u0924\u094d\u0930\u0915 \u0915\u0940 \u091c\u093e\u0901\u091a]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[georgian-\u10e1\u10e2\u10d0\u10e2 \u10e3\u10e1\u10d8\u10e1 \u10d9\u10dd\u10d3\u10d8 \u10d9\u10dd\u10dc\u10e2\u10e0\u10dd\u10da\u10d4\u10e0\u10d8\u10e1 \u10e8\u10d4\u10db\u10dd\u10ec\u10db\u10d4\u10d1\u10d0]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[gujarati-\u0ab8\u0acd\u0aa5\u0abf\u0aa4\u0abf \u0a95\u0acb\u0aa1 \u0aa8\u0abf\u0aaf\u0a82\u0aa4\u0acd\u0ab0\u0a95 \u0aa4\u0aaa\u0abe\u0ab8]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[gurmukhi-\u0a38\u0a25\u0a3f\u0a24 \u0a40 \u0a15\u0a4b\u0a21 \u0a15\u0a70\u0a1f\u0a30\u0a4b\u0a32\u0a30 \u0a1c\u0a3e\u0a02\u0a1a]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[kannada-\u0cb8\u0ccd\u0ca5\u0cbf\u0ca4\u0cbf \u0c95\u0ccb\u0ca1\u0ccd \u0ca8\u0cbf\u0caf\u0c82\u0ca4\u0ccd\u0cb0\u0c95 \u0caa\u0cb0\u0cbf\u0cb6\u0cc0\u0cb2\u0ca8\u0cc6]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[malayalam-\u0d38\u0d4d\u0d25\u0d3f\u0d24\u0d3f \u0d15\u0d4b\u0d21\u0d4d \u0d28\u0d3f\u0d2f\u0d28\u0d4d\u0d24\u0d4d\u0d30\u0d15\u0d02 \u0d2a\u0d30\u0d3f\u0d36\u0d4b\u0d27\u0d28]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[odia-\u0b38\u0b4d\u0b25\u0b3f\u0b24\u0b3f \u0b15\u0b4b\u0b21\u0b4d \u0b28\u0b3f\u0b5f\u0b28\u0b4d\u0b24\u0b4d\u0b30\u0b15 \u0b2f\u0b3e\u0b1e\u0b4d\u0b1a]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[tamil-\u0ba8\u0bbf\u0bb2\u0bc8 \u0b95\u0bc1\u0bb1\u0bbf\u0baf\u0bc0\u0b9f\u0bc1 \u0b95\u0b9f\u0bcd\u0b9f\u0bc1\u0baa\u0bcd\u0baa\u0b9f\u0bc1\u0ba4\u0bcd\u0ba4\u0bbf \u0b9a\u0bcb\u0ba4\u0ba9\u0bc8]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[telugu-\u0c38\u0c4d\u0c25\u0c3f\u0c24\u0c3f \u0c15\u0c4b\u0c21\u0c4d \u0c28\u0c3f\u0c2f\u0c02\u0c24\u0c4d\u0c30\u0c3f\u0c15 \u0c24\u0c28\u0c3f\u0c16\u0c40]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_a_script_gets_its_own_font[urdu-\u0627\u0633\u0679\u06cc\u0679\u0633 \u06a9\u0648\u0688 \u06a9\u0646\u0679\u0631\u0648\u0644\u0631 \u06a9\u06cc \u062c\u0627\u0646\u0686]		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_every_character_of_every_translation_is_drawn_by_a_bundled_font		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_every_script_builds_a_pdf_with_its_font		FR-045, FR-157, FR-165, NFR-008, TC-060	0.50 s
OK	test_latin_text_needs_no_script_font_and_arabic_is_told_from_urdu		FR-045, FR-157, FR-165, NFR-008, TC-060	0.00 s
OK	test_shaping_is_switched_on_per_cell_only_where_a_script_needs_it		FR-045, FR-157, FR-165, NFR-008, TC-060	0.07 s
OK	test_the_fonts_and_their_licence_are_bundled_and_shaping_is_available		FR-045, FR-157, FR-165, NFR-008, TC-060	0.04 s
OK	test_the_title_page_and_notice_may_carry_any_script		FR-045, FR-157, FR-165, NFR-008, TC-060	0.16 s

tests/integration/test_prg_idx_and_user_groups.py — 9 tests, 9 passed

_Schema v4.1 (migration c9e2a4b6d815): PrgIdx_GUi on the status code, the controller-specific_

Result	Test	Purpose	Requirements	Time
OK	test_default_levels_are_seeded_only_where_the_programs_agree			0.55 s
OK	test_leaving_draft_without_prg_idx_is_blocked_only_once_the_switch_is_on			0.96 s

Result	Test	Purpose	Requirements	Time
OK	test_new_revision_columns_start_at_zero_or_not_captured			0.43 s
OK	test_prg_idx_is_an_optional_uint_that_is_unique_among_real_codes			0.57 s
OK	test_prg_idx_is_frozen_once_a_revision_was_released			0.47 s
OK	test_the_locked_revision_trigger_covers_the_new_content_columns			0.47 s
OK	test_the_migration_is_reversible_and_re_applies_cleanly			1.43 s
OK	test_the_sixteen_user_group_slots_map_slots_one_to_eight_to_the_audiences			0.49 s
OK	test_user_group_levels_are_bytes_and_unique_per_revision_and_slot			0.51 s

tests/integration/test_publication.py — 25 tests, 25 passed

Publishing the catalogue (docs/21-improvement-ideas.md): I-02 profiles and wrapped PDF, I-03

Result	Test	Purpose	Requirements	Time
OK	test_administrator_manages_profiles_as_data		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.20 s
OK	test_applicability_matrix_and_changes_between_versions		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	3.63 s
OK	test_audience_export_only_contains_codes_the_audience_may_see		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	4.06 s
OK	test_bundle_contains_the_file_and_a_manifest_that_matches		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.02 s
OK	test_covert_visibility_is_data_an_administrator_can_flip		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.00 s
OK	test_customer_profile_applies_its_default_audience_and_working_set_refuses_audiences		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	3.00 s
OK	test_draft_source_text_cannot_be_translated_yet		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.24 s
OK	test_export_history_lists_hashes_per_release		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.10 s
OK	test_export_profile_fields_match_the_export_row		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.79 s
OK	test_offline_bundle_is_one_self_contained_page_with_the_data		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.16 s
OK	test_offline_bundle_respects_audiences_and_needs_a_published_release		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	3.00 s
OK	test_pdf_cells_are_wrapped_not_truncated		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.20 s
OK	test_pdf_column_widths_fit_the_page		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	0.61 s
OK	test_pdf_survives_characters_no_font_has		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	0.64 s
OK	test_publication_ui_pages_render		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	3.38 s
OK	test_release_notes_list_added_changed_and_deprecated		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	5.95 s

Result	Test	Purpose	Requirements	Time
OK	test_release_targets_limit_the_candidates_and_freeze_at_publish		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	3.43 s
OK	test_signed_manifest_detects_edits_to_the_manifest_itself		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.58 s
OK	test_three_profiles_are_seeded_and_shape_the_columns		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.55 s
OK	test_translation_validation_worklist_and_carry_over		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	3.00 s
OK	test_translation_workflow_with_light_review_and_language_exports		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	3.48 s
OK	test_unpublished_release_notes_are_an_administrator_preview		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.65 s
OK	test_verify_export_cli_works_without_the_database		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.12 s
OK	test_verify_recognises_an_untouched_file_and_a_tampered_one		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	2.28 s
OK	test_version_range_must_be_ordered_and_known		BR-016, BR-018, BR-021, FR-100, FR-101, FR-102, FR-103, FR-104, FR-105, FR-106, FR-107, FR-108, FR-110, FR-111, TC-042	1.20 s

tests/integration/test_releases.py — 6 tests, 6 passed

FR-040-FR-048, BR-006-BR-008.

Result	Test	Purpose	Requirements	Time
OK	test_cannot_change_items_after_publish		BR-006, BR-008, FR-040, FR-048	1.77 s
OK	test_csv_export_of_a_release		BR-006, BR-008, FR-040, FR-048	1.87 s
OK	test_full_database_export_requires_administrator		BR-006, BR-008, FR-040, FR-048	1.61 s
OK	test_full_release_lifecycle		BR-006, BR-008, FR-040, FR-048	1.92 s
OK	test_non_admin_cannot_create_release		BR-006, BR-008, FR-040, FR-048	0.99 s
OK	test_sandbox_revision_rejected_from_release_items		BR-006, BR-008, FR-040, FR-048	0.79 s

tests/integration/test_review_aids.py — 17 tests, 17 passed

I-23 field comments and review checklists, I-13 evidence attachments, I-11 change sets,

Result	Test	Purpose	Requirements	Time
OK	test_attachment_permissions_and_removal_rules		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.88 s
OK	test_attachment_type_size_and_content_checks		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.25 s
OK	test_author_attaches_evidence_and_reviewers_download_it		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	2.38 s
OK	test_auto_assignment_on_submit_is_a_setting		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.60 s
OK	test_bulk_decision_needs_the_password_once_and_a_wrong_one_counts_once		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	2.53 s
OK	test_bulk_decision_skips_what_the_actor_may_not_decide		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.96 s
OK	test_change_set_groups_requests_and_bulk_decisions_stay_individual		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	2.16 s
OK	test_change_set_rules		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	2.04 s

Result	Test	Purpose	Requirements	Time
OK	test_checklist_is_administrator_managed_and_inactive_items_are_not_required		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.24 s
OK	test_checklist_must_be_confirmed_to_approve_and_is_recorded		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.73 s
OK	test_field_comment_rules		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.57 s
OK	test_least_loaded_reviewer_is_suggested_and_assigned		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.80 s
OK	test_only_the_commenter_author_or_admin_can_resolve		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.70 s
OK	test_out_of_office_reviewers_are_not_auto_assigned		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.65 s
OK	test_rejection_does_not_need_the_checklist		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.57 s
OK	test_removing_a_draft_attachment_deletes_the_file_and_withdrawal_purges_all		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.26 s
OK	test_reviewer_comments_on_a_field_and_the_author_sees_and_resolves_it		FR-124, FR-127, FR-128, FR-129, FR-130, SEC-024, TC-040	1.96 s

tests/integration/test_roles_and_languages.py — 40 tests, 40 passed

The StCd Owner role, the role colours, who may create users and hand out roles, and the three

Result	Test	Purpose	Requirements	Time
OK	test_a_string_missing_from_a_catalogue_shows_in_english	Every language is complete (FR-167), but a string that is ever missing must still show.	FR-167	1.28 s
OK	test_arabic_is_right_to_left_but_identifiers_codes_and_numbers_stay_as_they_are		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	2.18 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[ar-\u062a\u0633\u062c\u064a\u0644 \u0627\u0644\u062f\u062e\u0648\u0644]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.68 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[bn-\u09b2\u0997 \u0987\u09a8]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.67 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[da-Log ind]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.68 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[es-Iniciar sesi\xf3n]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.71 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[fr-Se connecter]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.64 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[gu-\u0ab2\u0ac9\u0a97 \u0a87\u0aa8]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.61 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[hi-\u0932\u0949\u0917 \u0907\u0928 \u0915\u0930\u0947\u0902]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.31 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[it-Accedi]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.69 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[ka-\u10e8\u10d4\u10e1\u10d5\u10da \u10d0]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.68 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[kn-\u0cb2\u0cbe\u0c97\u0ccd \u0c87\u0ca8\u0ccd]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.70 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[ml-\u0d32\u0d4b\u0d17\u0d3f\u0d7b]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.68 s

Result	Test	Purpose	Requirements	Time
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[mr-\u0932\u0949\u0917 \u0907\u0928]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.69 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[or-\u0b32\u0b17\u0b4d \u0b07\u0b28\u0b4d]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.31 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[pa-\u0a32\u0a3e\u0a17 \u0a07\u0a28]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.72 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[ta-\u0b89\u0bb3\u0bcd\u0ba8\u0bc1\u0bb4\u0bc8]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.66 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[te-\u0c32\u0c3e\u0c17\u0c3f\u0c28\u0c4d]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.69 s
OK	test_each_added_language_translates_the_login_page_and_falls_back_to_english[ur-\u0644\u0627\u06af \u0627\u0646]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.61 s
OK	test_only_an_administrator_creates_users		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.60 s
OK	test_other_roles_cannot_open_the_user_list		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.30 s
OK	test_the_administrator_still_gives_every_role		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.18 s
OK	test_the_catalogues_agree_and_only_translate_real_strings		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.79 s
OK	test_the_choice_of_an_added_language_is_remembered_for_the_user		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.42 s
OK	test_the_language_names_have_the_country_letters_left_and_the_flag_right		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.59 s
OK	test_the_last_administrator_is_still_protected		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.07 s
OK	test_the_levels_order_the_roles_and_red_is_the_highest		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	0.62 s
OK	test_the_menu_shows_letters_left_and_a_drawn_flag_right_for_every_language		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.26 s
OK	test_the_stcd_owner_cannot_reach_the_other_roles[admin-VIEWER]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.09 s
OK	test_the_stcd_owner_cannot_reach_the_other_roles[chief_engineer-REVIEWER]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.11 s
OK	test_the_stcd_owner_cannot_reach_the_other_roles[engineer-ADMINISTRATOR]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.14 s
OK	test_the_stcd_owner_cannot_reach_the_other_roles[reviewer-STCD_OWNER]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.16 s
OK	test_the_stcd_owner_cannot_reach_the_other_roles[stcd_owner-VIEWER]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.17 s
OK	test_the_stcd_owner_cannot_reach_the_other_roles[viewer-CHIEF_ENGINEER]		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.21 s
OK	test_the_stcd_owner_changes_nothing_but_the_role		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.78 s
OK	test_the_stcd_owner_moves_users_between_viewer_engineer_and_reviewer		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.23 s
OK	test_the_stcd_owner_reads_the_workflow_but_changes_nothing_in_it		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.44 s
OK	test_the_stcd_owner_sees_the_list_without_the_administrator_forms		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.36 s
OK	test_the_user_page_colours_every_role_and_red_is_the_administrator		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.32 s
OK	test_who_may_change_which_role_to_which		BR-024, FR-010, FR-012, FR-156, FR-157, SEC-032, TC-054	1.22 s

tests/integration/test_schema_migration.py — 14 tests, 14 passed

US-003/US-004 acceptance criteria: fresh DB matches docs/artifacts/schema.sql

Result	Test	Purpose	Requirements	Time
OK	test_all_documented_tables_exist		US-003, US-004	0.49 s
OK	test_first_subgroup_of_every_main_group_covers_the_first_200_numbers	Rule of 2026-10-22: e.g. Tower 5000-5199, Generator 2000-2199; the next	US-003, US-004	0.54 s
OK	test_foreign_keys_enforced	SEC/BR data-integrity depends on FK enforcement being on for every	US-003, US-004	0.61 s
OK	test_functional_system_group_code_is_reused_for_hub_controller	WTUR and WROT are each intentionally reused for a second,	US-003, US-004	0.60 s
OK	test_hub_sub_bands_are_aligned_to_the_hundreds_of_their_prefix	docs/05-governance-handbook.md section 2: 123xx owns 12300-12399; a band shorter than a	US-003, US-004	0.65 s
OK	test_nineteen_languages_seeded_with_country_letters_left_and_the_flag_right		US-003, US-004	0.61 s
OK	test_no_placeholder_group_is_seeded	Only the 12 real groups exist out of the box; free ranges stay free.	US-003, US-004	0.62 s
OK	test_only_sandbox_status_codes_can_be_deleted	Exercises trg_prevent_nonsandbox_delete from docs/artifacts/schema.sql.	US-003, US-004	0.67 s
OK	test_second_migration_is_reversible_and_idempotent	A database already at the earlier baseline is brought up to date by the	US-003, US-004	1.82 s
OK	test_six_roles_seeded		US-003, US-004	0.59 s
OK	test_subgroup_codes_have_no_dash_and_no_letter_after_the_x		US-003, US-004	0.62 s
OK	test_thirty_eight_functional_subgroups_seeded		US-003, US-004	0.56 s
OK	test_three_turbine_platforms_seeded		US-003, US-004	1.29 s
OK	test_twelve_functional_system_groups_seeded	9 Main Controller groups + 3 Hub Controller groups (WTUR/WROT/WPPD	US-003, US-004	0.67 s

tests/integration/test_schema_upgrade.py — 16 tests, 16 passed

Upgrading a database that was created by the previous release (Alembic d6f8b0c2e4a3, schema v3)

Result	Test	Purpose	Requirements	Time
OK	test_a_second_upgrade_is_a_no_op_and_the_downgrade_returns_to_v3_with_the_data		AUD-074, BR-016, FR-147, FR-148, TC-045	2.54 s
OK	test_an_upgraded_database_has_the_same_shape_as_a_fresh_one		AUD-074, BR-016, FR-147, FR-148, TC-045	1.87 s
OK	test_covert_is_not_visible_and_the_other_audiences_still_are		AUD-074, BR-016, FR-147, FR-148, TC-045	1.33 s
OK	test_every_trigger_of_a_fresh_database_exists_after_the_upgrade		AUD-074, BR-016, FR-147, FR-148, TC-045	1.72 s
OK	test_existing_data_survives_the_upgrade		AUD-074, BR-016, FR-147, FR-148, TC-045	1.36 s
OK	test_hub_groups_become_hub_controller_groups		AUD-074, BR-016, FR-147, FR-148, TC-045	1.27 s
OK	test_hub_sub_bands_are_shifted_to_the_hundreds_by_the_upgrade	The v3 database still has the legacy x001 bands; the upgrade aligns them.	AUD-074, BR-016, FR-147, FR-148, TC-045	1.31 s
OK	test_import_job_is_rebuilt_for_programs_without_losing_rows_or_references	v4.2: the CHECK of import_job.source_format has to widen to 'ST', which SQLite can only do by	AUD-074, BR-016, FR-147, FR-148, TC-045	17.68 s
OK	test_new_columns_default_sensibly_on_old_rows		AUD-074, BR-016, FR-147, FR-148, TC-045	1.92 s
OK	test_old_audit_rows_are_chained_and_the_chain_verifies		AUD-074, BR-016, FR-147, FR-148, TC-045	17.71 s
OK	test_the_languages_and_export_profiles_are_seeded		AUD-074, BR-016, FR-147, FR-148, TC-045	1.33 s
OK	test_the_migrated_hash_equals_what_the_application_would_have_written	Migration and application share one canonical form (app/audit_hash.py).	AUD-074, BR-016, FR-147, FR-148, TC-045	1.26 s

Result	Test	Purpose	Requirements	Time
OK	test_the_search_index_is_filled_with_the_old_codes		AUD-074, BR-016, FR-147, FR-148, TC-045	1.87 s
OK	test_the_subgroup_codes_lose_their_letters_and_dashes_in_the_upgrade		AUD-074, BR-016, FR-147, FR-148, TC-045	1.28 s
OK	test_the_table_rebuilds_leave_every_foreign_key_intact	import_job (revision b3f5a7c9d124) and user (d4a6c8e0f236) are rebuilt to widen a CHECK; the	AUD-074, BR-016, FR-147, FR-148, TC-045	17.72 s
OK	test_the_upgraded_database_works_for_a_whole_workflow		AUD-074, BR-016, FR-147, FR-148, TC-045	1.46 s

tests/integration/test_search_and_help.py — 16 tests, 16 passed

I-31 ranked, typo-tolerant search (with a substring fallback), I-35 in-app help content.

Result	Test	Purpose	Requirements	Time
OK	test_a_hit_in_the_title_outranks_a_hit_in_the_description		FR-148, FR-154, TC-047	1.24 s
OK	test_a_new_code_and_an_edit_are_reflected_without_a_rebuild		FR-148, FR-154, TC-047	1.89 s
OK	test_every_lifecycle_state_role_and_cr_state_has_help		FR-148, FR-154, TC-047	0.00 s
OK	test_help_texts_are_plain_sentences		FR-148, FR-154, TC-047	0.00 s
OK	test_new_terms_are_in_the_project_glossary		FR-148, FR-154, TC-047	0.00 s
OK	test_prefix_and_multi_word_search		FR-148, FR-154, TC-047	1.24 s
OK	test_search_ignores_fts_operators_and_punctuation		FR-148, FR-154, TC-047	1.22 s
OK	test_several_misspelt_words_are_all_tolerated_together	Each word may expand to a group of close spellings; the groups must be joined with AND	FR-148, FR-154, TC-047	1.32 s
OK	test_state_badges_carry_their_glossary_text_as_a_tooltip		FR-148, FR-154, TC-047	1.40 s
OK	test_the_change_request_screen_says_what_happens_next		FR-148, FR-154, TC-047	1.56 s
OK	test_the_help_page_shows_the_quick_start_of_the_signed_in_role		FR-148, FR-154, TC-047	1.42 s
OK	test_the_roles_and_states_are_named_in_the_project_glossary		FR-148, FR-154, TC-047	0.00 s
OK	test_the_search_box_works_on_the_ui_list		FR-148, FR-154, TC-047	1.37 s
OK	test_the_search_index_exists_on_a_fresh_database		FR-148, FR-154, TC-047	17.62 s
OK	test_typos_still_find_the_code		FR-148, FR-154, TC-047	1.29 s
OK	test_without_the_index_search_falls_back_to_substrings		FR-148, FR-154, TC-047	1.48 s

tests/integration/test_select_all.py — 2 tests, 2 passed

A list with a checkbox per row has a checkbox in its header that ticks them all (the helper is

Result	Test	Purpose	Requirements	Time
OK	test_every_list_with_a_checkbox_per_row_can_be_selected_as_a_whole		FR-159, FR-180, NFR-007, TC-036, TC-070	0.37 s
OK	test_the_header_box_and_the_row_boxes_share_their_class		FR-159, FR-180, NFR-007, TC-036, TC-070	0.02 s

tests/integration/test_sla_and_notifications.py — 16 tests, 16 passed

I-08 SLA ages, stalled requests, in-app notices and reminders (docs/21-improvement-ideas.md),

Result	Test	Purpose	Requirements	Time
OK	test_assigned_reviewer_only_and_pending_approval_goes_to_chief_engineers		FR-120, FR-121, FR-122, FR-123, TC-041	1.93 s
OK	test_business_days_skip_weekends		FR-120, FR-121, FR-122, FR-123, TC-041	0.00 s
OK	test_kpis_are_computed_from_stored_timestamps		FR-120, FR-121, FR-122, FR-123, TC-041	2.33 s
OK	test_kpis_flag_an_incomplete_audit_trail_and_an_ownerless_code		FR-120, FR-121, FR-122, FR-123, TC-041	2.10 s

Result	Test	Purpose	Requirements	Time
OK	test_kpis_without_data_do_not_divide_by_zero		FR-120, FR-121, FR-122, FR-123, TC-041	1.33 s
OK	test_levels_follow_the_thresholds		FR-120, FR-121, FR-122, FR-123, TC-041	2.19 s
OK	test_mail_failure_never_breaks_the_reminder_job		FR-120, FR-121, FR-122, FR-123, TC-041	1.44 s
OK	test_mark_all_read		FR-120, FR-121, FR-122, FR-123, TC-041	1.65 s
OK	test_queues_show_the_waiting_time		FR-120, FR-121, FR-122, FR-123, TC-041	1.58 s
OK	test_reminders_also_reach_a_covering_delegate_and_can_be_mailed		FR-120, FR-121, FR-122, FR-123, TC-041	2.32 s
OK	test_reminders_go_to_whoever_holds_the_next_decision_once_a_day		FR-120, FR-121, FR-122, FR-123, TC-041	1.85 s
OK	test_send_reminders_cli		FR-120, FR-121, FR-122, FR-123, TC-041	0.62 s
OK	test_stalled_view_and_reassignment		FR-120, FR-121, FR-122, FR-123, TC-041	1.79 s
OK	test_the_default_thresholds_are_the_documented_ones		FR-120, FR-121, FR-122, FR-123, TC-041	0.59 s
OK	test_thresholds_are_configurable_and_validated		FR-120, FR-121, FR-122, FR-123, TC-041	1.29 s
OK	test_withdrawn_cr_notices_are_removed_with_it		FR-120, FR-121, FR-122, FR-123, TC-041	1.42 s

tests/integration/test_st_category_and_draft_delete.py — 7 tests, 7 passed

The status category of a .st import is the type of the mapping row (FR-181 superseded by FR-192:

Result	Test	Purpose	Requirements	Time
OK	test_a_category_chosen_in_the_request_changes_nothing		FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070	1.39 s
OK	test_a_row_without_a_type_blocks_the_program		FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070	1.21 s
OK	test_all_own_drafts_of_a_filter_are_deleted_on_every_page		FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070	2.29 s
OK	test_somebody_elses_drafts_cannot_be_ticked_or_deleted		FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070	1.69 s
OK	test_the_list_offers_tick_boxes_for_the_own_drafts_and_deletes_the_ticked		FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070	1.58 s
OK	test_the_pages_have_no_category_controls_any_more		FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070	1.53 s
OK	test_the_type_of_a_program_is_the_type_of_its_mapping_row		FR-026, FR-159, FR-181, FR-182, FR-192, TC-053, TC-070	1.39 s

tests/integration/test_st_import.py — 29 tests, 29 passed

Status code import from the generated StCd.st programs and the legacy -> new mapping_

Result	Test	Purpose	Requirements	Time
OK	test_a_file_without_programs_or_too_large_is_refused_cleanly		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.60 s
OK	test_a_hub_program_keeps_its_legacy_number_and_gets_the_new_one_from_the_mapping		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.55 s
OK	test_a_legacy_hub_number_missing_from_the_mapping_leaves_no_identifier		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.62 s
OK	test_a_legacy_number_that_belongs_to_another_code_blocks_the_program		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.90 s
OK	test_a_number_defined_twice_in_one_upload_blocks_the_second		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.70 s
OK	test_a_preview_belongs_to_its_uploader_and_expires		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.90 s
OK	test_a_second_preview_shows_what_is_already_there_and_confirm_refuses_it		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	1.19 s
OK	test_a_used_prg_idx_is_a_warning_and_is_not_stored_twice		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.96 s
OK	test_a_zip_and_loose_files_are_read_alike_and_other_files_are_noted		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.72 s

Result	Test	Purpose	Requirements	Time
OK	test_a_zip_that_unpacks_too_large_is_refused		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	1.40 s
OK	test_allocation_skips_a_reserved_number_and_a_manual_override_cannot_take_it		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	1.48 s
OK	test_an_administrator_cannot_override_a_reserved_identifier		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	1.48 s
OK	test_an_expired_preview_is_refused_and_removed		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.84 s
OK	test_confirm_creates_drafts_that_reserve_their_numbers_in_one_change_set		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.97 s
OK	test_confirming_without_a_selection_takes_every_ready_program		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	1.60 s
OK	test_nothing_usable_is_a_clear_error		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.96 s
OK	test_only_engineers_and_administrators_import		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.88 s
OK	test_submitting_issues_exactly_the_reserved_number		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	1.56 s
OK	test_the_command_line_reads_the_hub_numbers_of_the_seed_workbook		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.78 s
OK	test_the_controller_can_be_forced_for_a_file		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.76 s
OK	test_the_database_refuses_a_second_reservation_of_a_number		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.72 s
OK	test_the_draft_carries_the_attributes_of_the_program_and_the_mapping		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.93 s
OK	test_the_mapping_exports_as_csv_and_xlsx		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	1.01 s
OK	test_the_mapping_is_not_for_viewers		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.76 s
OK	test_the_mapping_lists_legacy_to_new_and_counts_codes_without_a_legacy_number		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	1.02 s
OK	test_the_new_number_of_a_hub_program_is_the_one_of_its_mapping_row		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	1.42 s
OK	test_the_preview_lists_every_program_and_creates_nothing		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.79 s
OK	test_the_reader_takes_the_attributes_of_a_main_controller_program		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.56 s
OK	test_the_source_of_an_imported_code_can_be_read		BR-023, FR-048o, FR-048p, FR-048q, FR-048r, FR-048s, FR-048t, FR-048u, FR-048v, FR-048w, SEC-031, TC-053	0.94 s

tests/integration/test_st_mapping_rules.py — 11 tests, 11 passed

What the .st import takes from the mapping and how it checks the program against the row (FR-192,

Result	Test	Purpose	Requirements	Time
OK	test_a_hub_row_for_a_main_program_is_not_taken		BR-027, FR-189, FR-192, FR-193, TC-077	1.15 s
OK	test_a_program_already_in_the_catalogue_keeps_its_title_when_the_row_is_gone		BR-027, FR-189, FR-192, FR-193, TC-077	1.37 s
OK	test_a_rejected_row_is_as_good_as_none		BR-027, FR-189, FR-192, FR-193, TC-077	1.12 s
OK	test_a_row_that_is_not_approved_yet_is_used_with_a_warning		BR-027, FR-189, FR-192, FR-193, TC-077	1.21 s
OK	test_a_row_without_cause_and_directive_leaves_the_remedy_text_empty		BR-027, FR-189, FR-192, FR-193, TC-077	1.39 s
OK	test_a_time_in_the_program_wins_over_the_mapping		BR-027, FR-189, FR-192, FR-193, TC-077	1.19 s
OK	test_a_title_that_does_not_say_what_the_name_says_is_flagged		BR-027, FR-189, FR-192, FR-193, TC-077	1.94 s
OK	test_cause_and_directive_become_the_remedy_text		BR-027, FR-189, FR-192, FR-193, TC-077	1.24 s
OK	test_the_glossary_is_a_check_not_a_translation		BR-027, FR-189, FR-192, FR-193, TC-077	0.60 s
OK	test_the_legacy_number_of_the_header_must_be_the_one_of_the_row		BR-027, FR-189, FR-192, FR-193, TC-077	1.22 s

Result	Test	Purpose	Requirements	Time
OK	test_the_mapping_gives_the_value_of_a_delay_the_program_sets_with_a_parameter		BR-027, FR-189, FR-192, FR-193, TC-077	1.35 s

tests/integration/test_st_no_assumptions.py — 10 tests, 10 passed

The import assumes no value (FR-189): a value is read from the program, chosen by the user, or - when

Result	Test	Purpose	Requirements	Time
OK	test_a_delay_may_be_the_name_of_a_parameter_but_not_a_malformed_time		BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053	1.20 s
OK	test_a_delay_that_is_a_variable_stays_the_name_of_the_variable		BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053	1.11 s
OK	test_a_delay_the_program_does_not_have_says_so		BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053	1.32 s
OK	test_a_level_that_is_no_longer_active_is_not_guessed		BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053	1.29 s
OK	test_a_level_the_mapping_leaves_empty_stays_empty_and_is_listed		BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053	1.32 s
OK	test_a_level_whose_numbers_differ_from_the_program_is_named_and_warned_about		BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053	1.30 s
OK	test_a_program_that_gives_everything_lists_only_what_no_program_carries		BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053	1.32 s
OK	test_a_program_the_mapping_does_not_know_is_blocked_and_never_imported		BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053	1.38 s
OK	test_the_named_access_levels_come_from_the_mapping		BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053	1.36 s
OK	test_the_preview_warns_with_the_name_of_the_parameter		BR-023, FR-048o, FR-189, FR-190, FR-192, TC-053	1.86 s

tests/integration/test_status_codes.py — 9 tests, 9 passed

FR-020-FR-029, FR-053-FR-056, FR-060-FR-064.

Result	Test	Purpose	Requirements	Time
OK	test_admin_sandbox_gets_identifier_immediately		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	0.77 s
OK	test_create_rejects_invalid_fields		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	0.74 s
OK	test_create_rejects_missing_functional_subgroup	BR-013: mandatory, never optional (corrected 2026-09-17).	BR-013	0.73 s
OK	test_engineer_can_create_draft_with_no_identifier		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	1.63 s
OK	test_non_admin_cannot_create_sandbox		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	0.72 s
OK	test_sandbox_delete_requires_admin_and_sandbox_flag		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	1.12 s
OK	test_search_defaults_exclude_sandbox		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	0.86 s
OK	test_viewer_cannot_create_status_code		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	0.75 s
OK	test_viewer_cannot_see_draft_status_codes		FR-020, FR-029, FR-053, FR-056, FR-060, FR-064	1.72 s

tests/integration/test_step_up_and_delegation.py — 16 tests, 16 passed

I-17 step-up re-authentication (SEC-026) and I-16 delegation / out-of-office cover.

Result	Test	Purpose	Requirements	Time
OK	test_admin_signoff_chief_decision_publish_and_full_export_all_need_the_password		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	2.10 s
OK	test_correct_password_lets_the_signoff_through_and_is_audited		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.62 s
OK	test_deactivated_delegator_ends_the_delegation		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.53 s

Result	Test	Purpose	Requirements	Time
OK	test_delegate_can_sign_off_as_the_absent_reviewer_and_audit_shows_both		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.56 s
OK	test_delegate_sees_the_review_queue_and_a_delegated_chief_engineer_can_approve		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	2.38 s
OK	test_delegating_does_not_grant_other_privileges		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	0.92 s
OK	test_delegation_does_not_lift_segregation_of_duties	The delegate acts as themselves: the author cannot sign their own CR through a delegation.	BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.55 s
OK	test_delegation_is_limited_to_its_period_and_can_be_revoked		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.78 s
OK	test_delegation_validation_and_who_may_create_it		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.70 s
OK	test_one_person_cannot_give_both_review_signoffs_through_a_delegation		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.79 s
OK	test_repeated_wrong_passwords_lock_the_account	The prompt must not be usable to guess a password (same lockout as login).	BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.86 s
OK	test_reviewer_signoff_without_password_is_refused		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.49 s
OK	test_step_up_can_be_switched_off_by_configuration		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.48 s
OK	test_ui_decision_forms_carry_a_password_field		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	2.47 s
OK	test_ui_signoff_without_password_shows_an_error		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.72 s
OK	test_wrong_password_is_refused_and_audited		BR-017, BR-022, FR-125, FR-126, SEC-011, SEC-026, TC-038	1.59 s

tests/integration/test_terminology.py — 13 tests, 13 passed

The technical nomenclature: a vocabulary of its own, apart from the interface translations

Result	Test	Purpose	Requirements	Time
OK	test_a_formula_in_a_cell_is_neutralised_in_the_csv		AUD-077, BR-018, FR-166, SEC-010, TC-061	1.83 s
OK	test_a_proposal_is_a_draft_until_another_person_approves_it		AUD-077, BR-018, FR-166, SEC-010, TC-061	2.79 s
OK	test_a_shipped_draft_can_be_approved_by_anybody_who_may_approve_and_all_at_once		AUD-077, BR-018, FR-166, SEC-010, TC-061	13.19 s
OK	test_an_interface_reviewer_has_no_say_in_the_terminology		AUD-077, BR-018, FR-166, SEC-010, TC-061	1.23 s
OK	test_everybody_reads_but_only_the_technical_roles_propose		AUD-077, BR-018, FR-166, SEC-010, TC-061	2.62 s
OK	test_only_the_managers_add_and_change_terms_and_a_new_english_reopens_approvals		AUD-077, BR-018, FR-166, SEC-010, TC-061	3.14 s
OK	test_send_back_needs_a_note_and_reopens_an_approved_translation		AUD-077, BR-018, FR-166, SEC-010, TC-061	13.76 s
OK	test_terms_that_are_never_translated_have_no_translation		AUD-077, BR-018, FR-166, SEC-010, TC-061	0.55 s
OK	test_the_check_finds_interface_strings_that_do_not_use_the_term		AUD-077, BR-018, FR-166, SEC-010, TC-061	2.20 s

Result	Test	Purpose	Requirements	Time
OK	test_the_shipped_terms_agree_with_the_interface_catalogue_where_both_have_the_word	Role names are interface strings and terms: they must be the same word.	AUD-077, BR-018, FR-166, SEC-010, TC-061	0.01 s
OK	test_the_vocabulary_is_exported_per_language_and_for_all		AUD-077, BR-018, FR-166, SEC-010, TC-061	1.25 s
OK	test_the_vocabulary_ships_with_every_term_in_every_language_as_a_draft		AUD-077, BR-018, FR-166, SEC-010, TC-061	0.64 s
OK	test_validation_of_a_translation		AUD-077, BR-018, FR-166, SEC-010, TC-061	1.31 s

tests/integration/test_translation_approval.py — 12 tests, 12 passed

New and changed translations are approved separately, and every change of the text or of the

Result	Test	Purpose	Requirements	Time
OK	test_a_change_of_the_nomenclature_needs_the_flag_as_well		FR-164, FR-170, FR-172, FR-176, TC-067	4.83 s
OK	test_a_changed_string_must_get_a_verdict_before_the_record_is_current		FR-164, FR-170, FR-172, FR-176, TC-067	5.08 s
OK	test_a_delta_read_one_by_one_can_lower_or_keep_the_confidence		FR-164, FR-170, FR-172, FR-176, TC-067	3.34 s
OK	test_a_follow_up_review_covers_only_what_is_new_or_changed		FR-164, FR-170, FR-172, FR-176, TC-067	2.67 s
OK	test_a_language_never_reviewed_stays_hidden_from_a_viewer		FR-164, FR-170, FR-172, FR-176, TC-067	1.30 s
OK	test_a_reviewed_language_has_nothing_pending_until_a_string_changes		FR-164, FR-170, FR-172, FR-176, TC-067	1.36 s
OK	test_a_viewer_reads_a_changed_string_in_english_until_it_is_approved		FR-164, FR-170, FR-172, FR-176, TC-067	1.67 s
OK	test_an_accepted_proposal_is_approved_and_an_undecided_one_is_not		FR-164, FR-170, FR-172, FR-176, TC-067	1.38 s
OK	test_every_change_of_the_text_or_the_nomenclature_has_been_through_the_check	The gate: this fails while a string or a term changed and the check was not done for it.	FR-164, FR-170, FR-172, FR-176, TC-067	0.33 s
OK	test_the_colour_intervals_run_from_red_to_green		FR-164, FR-170, FR-172, FR-176, TC-067	0.00 s
OK	test_the_overview_shows_the_confidence_of_every_language_in_colour		FR-164, FR-170, FR-172, FR-176, TC-067	7.39 s
OK	test_the_record_holds_a_confidence_for_every_language_with_its_findings		FR-164, FR-170, FR-172, FR-176, TC-067	0.01 s

tests/integration/test_translation_impact.py — 11 tests, 11 passed

A change of the vocabulary is tested automatically and its language impact reaches the StCd Owner

Result	Test	Purpose	Requirements	Time
OK	test_a_changed_nomenclature_is_named_in_the_notice		FR-166, FR-176, FR-177, FR-178, TC-068	1.09 s
OK	test_a_changed_term_and_an_approved_translation_are_reported_again		FR-166, FR-176, FR-177, FR-178, TC-068	1.74 s
OK	test_a_failing_notice_never_undoes_the_change		FR-166, FR-176, FR-177, FR-178, TC-068	1.07 s
OK	test_a_new_term_tells_the_stcd_owner_which_languages_need_attention		FR-166, FR-176, FR-177, FR-178, TC-068	1.09 s
OK	test_a_term_that_stays_in_english_touches_no_language		FR-166, FR-176, FR-177, FR-178, TC-068	1.20 s
OK	test_changed_shipped_text_is_reported_once_per_change		FR-166, FR-176, FR-177, FR-178, TC-068	1.13 s

Result	Test	Purpose	Requirements	Time
OK	test_nothing_is_reported_while_the_confidence_check_is_current		FR-166, FR-176, FR-177, FR-178, TC-068	1.90 s
OK	test_the_command_for_the_deployment_reports_and_notifies		FR-166, FR-176, FR-177, FR-178, TC-068	1.11 s
OK	test_the_notice_appears_on_the_notices_page_of_the_stcd_owner		FR-166, FR-176, FR-177, FR-178, TC-068	1.40 s
OK	test_the_reviewer_of_an_affected_language_is_told_too		FR-166, FR-176, FR-177, FR-178, TC-068	1.80 s
OK	test_the_start_of_the_server_and_the_installers_run_the_check		FR-166, FR-176, FR-177, FR-178, TC-068	0.06 s

tests/integration/test_translation_review.py — 14 tests, 14 passed

Translation review in the application (docs/26-translations.md, docs/03-srs.md FR-164): native

Result	Test	Purpose	Requirements	Time
OK	test_a_missing_translation_can_be_proposed_and_accepted		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	15.97 s
OK	test_a_proposal_awaits_approval_until_it_is_decided		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	24.68 s
OK	test_a_proposal_must_keep_the_placeholders_and_an_unclear_string_is_only_noted		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	42.67 s
OK	test_a_review_is_completed_only_when_every_string_has_a_verdict_and_is_recorded		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	17.31 s
OK	test_a_reviewer_answers_strings_and_is_told_what_is_wrong		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	17.07 s
OK	test_a_sheet_that_is_not_one_is_refused_clearly		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	15.56 s
OK	test_an_accepted_proposal_is_shown_at_once_and_can_be_taken_back		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	48.23 s
OK	test_an_account_that_reviews_has_history_in_the_usage_report		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	15.02 s
OK	test_an_unassigned_user_cannot_open_a_review_and_an_unknown_language_is_404		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	1.11 s
OK	test_nobody_decides_on_their_own_proposal_and_a_decision_is_final		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	64.78 s
OK	test_only_an_administrator_or_an_stcd_owner_assigns_reviewers_and_it_is_audited		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	41.14 s
OK	test_right_to_left_languages_are_shown_right_to_left_in_the_review		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	30.04 s
OK	test_the_overview_shows_a_reviewer_only_their_languages_and_a_manager_all		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	37.12 s
OK	test_the_sheet_can_be_downloaded_filled_in_offline_and_uploaded		AUD-077, BR-018, BR-025, FR-157, FR-161, FR-164, TC-059	17.33 s

tests/integration/test_ui_screens.py — 18 tests, 18 passed

S1–S19 server-rendered UI: every screen renders for the roles that can

Result	Test	Purpose	Requirements	Time
OK	test_admin_signoff_form_offers_access_override_and_submit_override		NFR-007a, TC-036, UAT-13	2.12 s
OK	test_comments_api_validates_and_is_role_restricted		NFR-007a, TC-036, UAT-13	1.52 s
OK	test_cr_detail_shows_role_specific_controls		NFR-007a, TC-036, UAT-13	1.65 s
OK	test_default_role_for_new_user_is_viewer	SEC-012.	SEC-012	0.84 s
OK	test_draft_cr_page_says_it_must_still_be_submitted		NFR-007a, TC-036, UAT-13	1.93 s
OK	test_every_screen_renders_for_an_administrator		NFR-007a, TC-036, UAT-13	3.33 s
OK	test_failed_form_submission_keeps_every_entered_value	NFR-007a / TC-036 / UAT-13.	NFR-007a, TC-036, UAT-13	1.45 s
OK	test_lookup_description_is_editable_and_flows_to_form_and_api		NFR-007a, TC-036, UAT-13	1.71 s
OK	test_lookup_selects_carry_inline_help_hooks	NFR-007b: each lookup-backed select has a help target and per-option text.	NFR-007b	1.46 s

Result	Test	Purpose	Requirements	Time
OK	test_people_are_shown_by_name_and_lists_show_the_title	Authors, signers and commenters appear as "Full Name (Role)", never "User #n";	NFR-007a, TC-036, UAT-13	2.29 s
OK	test_role_specific_screens_render[chief_engineer-/change-requests/approval-queue]		NFR-007a, TC-036, UAT-13	1.91 s
OK	test_role_specific_screens_render[engineer-/change-requests/mine]		NFR-007a, TC-036, UAT-13	1.43 s
OK	test_role_specific_screens_render[engineer-/exports]		NFR-007a, TC-036, UAT-13	1.33 s
OK	test_role_specific_screens_render[reviewer-/change-requests/review-queue]		NFR-007a, TC-036, UAT-13	1.27 s
OK	test_role_specific_screens_render[viewer-/exports]		NFR-007a, TC-036, UAT-13	1.28 s
OK	test_role_specific_screens_render[viewer-/status-codes]		NFR-007a, TC-036, UAT-13	1.37 s
OK	test_sign_off_request_listed_on_my_change_requests		NFR-007a, TC-036, UAT-13	1.52 s
OK	test_ui_comment_reviewer_assignment_and_override_flow		NFR-007a, TC-036, UAT-13	2.94 s

tests/integration/test_user_administration.py — 10 tests, 10 passed

Viewing, deactivating and deleting users (docs/03-srs.md FR-158, BR-025, SEC-032). An account that

Result	Test	Purpose	Requirements	Time
OK	test_a_deactivated_account_can_be_activated_again		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	1.22 s
OK	test_an_account_that_was_never_used_can_be_deleted_and_the_deletion_is_audited		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	2.14 s
OK	test_an_account_with_history_is_erased_and_the_audit_log_is_untouched		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	1.48 s
OK	test_deleting_needs_the_administrators_password_and_an_administrator		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	1.97 s
OK	test_nobody_deletes_their_own_account		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	1.07 s
OK	test_the_buttons_deactivate_activate_and_delete_through_the_screens		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	2.54 s
OK	test_the_detail_page_shows_history_and_the_buttons_follow_it		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	1.65 s
OK	test_the_stcd_owner_sees_view_and_set_role_but_no_delete_or_deactivate		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	1.28 s
OK	test_the_user_list_offers_view_edit_deactivate_and_delete_to_an_administrator		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	1.31 s
OK	test_usage_is_open_to_the_user_list_roles_only		AUD-076, BR-025, FR-012, FR-158, SEC-032, TC-055	1.35 s

tests/integration/test_users.py — 7 tests, 7 passed

FR-010-FR-014, SEC-003 §3 permission matrix, US-009.

Result	Test	Purpose	Requirements	Time
OK	test_admin_can_force_reset_another_users_password		FR-010, FR-014, SEC-003, US-009	1.00 s
OK	test_admin_can_list_and_create_users		FR-010, FR-014, SEC-003, US-009	0.84 s
OK	test_can_deactivate_administrator_when_another_remains_active		FR-010, FR-014, SEC-003, US-009	0.81 s
OK	test_cannot_deactivate_last_active_administrator		FR-010, FR-014, SEC-003, US-009	0.74 s
OK	test_create_user_rejects_weak_password		FR-010, FR-014, SEC-003, US-009	0.73 s
OK	test_non_admin_cannot_list_users		FR-010, FR-014, SEC-003, US-009	0.72 s
OK	test_self_password_change		FR-010, FR-014, SEC-003, US-009	1.10 s

tests/integration/test_withdraw_and_reimport.py — 11 tests, 11 passed

Withdrawing change requests (several at once) and what a withdrawal does to a program that was

Result	Test	Purpose	Requirements	Time
OK	test_a_change_request_of_somebody_else_is_refused_and_the_others_go_on		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.41 s
OK	test_a_changed_program_opens_a_new_revision_of_an_approved_code		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	2.17 s
OK	test_a_changed_program_updates_the_open_draft_when_it_is_ticked		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	2.12 s
OK	test_a_restored_code_is_no_shell_and_the_database_refuses_a_delete_around_the_service		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.79 s
OK	test_a_withdrawn_code_is_listed_and_an_administrator_deletes_it_and_imports_again		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	2.14 s
OK	test_a_withdrawn_code_that_has_its_identifier_keeps_it_and_can_be_restored		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.64 s
OK	test_a_withdrawn_import_draft_gives_its_number_and_legacy_number_back		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.41 s
OK	test_bulk_withdraw_needs_a_list_and_a_role_that_can_withdraw		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	2.16 s
OK	test_several_change_requests_are_withdrawn_in_one_go_and_each_is_reported		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.33 s
OK	test_the_list_has_tick_boxes_and_a_withdraw_selected_button		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.57 s
OK	test_the_preview_screen_lists_the_choices_and_leaves_changed_programs_unticked		BR-001, BR-026, FR-026, FR-048x, FR-159, FR-160, TC-056	1.46 s

tests/integration/test_workflow_gaps.py — 18 tests, 18 passed

Closes the gaps found in the spec-vs-code review (docs/20-traceability-matrix.md):

Result	Test	Purpose	Requirements	Time
OK	test_admin_can_override_identifier_with_reason_and_it_is_audited		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.45 s
OK	test_admin_signoff_can_override_proposed_audience_access		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	2.61 s
OK	test_assigned_reviewer_only_may_sign_off		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.87 s
OK	test_cr_page_explains_when_no_engineer_holds_a_pending_domain		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.15 s
OK	test_domain_request_is_refused_when_nobody_holds_the_domain	A request nobody could ever act on must not be silently accepted.	BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	0.98 s
OK	test_domain_signoff_request_surfaces_for_target_and_clears_once_signed		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	2.54 s
OK	test_domain_signoff_request_validation		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.33 s
OK	test_identifier_override_forbidden_for_engineer		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.37 s
OK	test_identifier_override_rejects_out_of_range_malformed_and_duplicate		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	2.00 s
OK	test_identifier_override_requires_reason		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.39 s
OK	test_no_reassignment_after_reviewer_signed_off		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.57 s
OK	test_only_administrator_endpoint_accepts_overrides		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.47 s

Result	Test	Purpose	Requirements	Time
OK	test_rejection_ignores_access_overrides		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.77 s
OK	test_reviewer_assignment_validation_and_authorization		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.60 s
OK	test_reviewer_can_be_reassigned_and_unassigned		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.98 s
OK	test_unknown_access_override_field_is_rejected		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.55 s
OK	test_untargeted_domain_request_is_visible_to_every_domain_holder		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.40 s
OK	test_withdrawn_cr_removes_its_domain_requests		BR-012, FR-021a, FR-025a, FR-028, FR-034, FR-039, SEC-013	1.38 s

tests/system/test_accessibility.py — 9 tests, 9 passed

I-22 accessibility (WCAG 2.1 AA, docs/10-ui-ux-specification.md §12): colour contrast of the

Result	Test	Purpose	Requirements	Time
OK	test_every_page_is_structurally_accessible[de]		FR-155, NFR-010, TC-049	2.96 s
OK	test_every_page_is_structurally_accessible[en]		FR-155, NFR-010, TC-049	4.19 s
OK	test_every_state_badge_meets_aa		FR-155, NFR-010, TC-049	0.00 s
OK	test_status_is_never_conveyed_by_colour_alone	Every state badge carries the state as text, not only a colour.	FR-155, NFR-010, TC-049	1.23 s
OK	test_text_tokens_meet_aa_on_their_backgrounds		FR-155, NFR-010, TC-049	0.00 s
OK	test_the_chief_engineers_queue_is_accessible		FR-155, NFR-010, TC-049	1.14 s
OK	test_the_contrast_helper_matches_the_wcag_reference_values		FR-155, NFR-010, TC-049	0.01 s
OK	test_the_login_page_is_accessible		FR-155, NFR-010, TC-049	0.59 s
OK	test_the_stylesheet_honours_user_preferences		FR-155, NFR-010, TC-049	0.00 s

tests/system/test_branding.py — 5 tests, 5 passed

The identity of the product (FR-187): the full name, the owner on top of every title page and

Result	Test	Purpose	Requirements	Time
OK	test_every_title_page_carries_the_identity		FR-185, FR-186, FR-187, TC-074	0.31 s
OK	test_the_guide_names_the_same_owner		FR-185, FR-186, FR-187, TC-074	0.03 s
OK	test_the_logo_files_exist_and_are_drawn		FR-185, FR-186, FR-187, TC-074	0.21 s
OK	test_the_menu_bar_and_the_login_page_show_the_logo		FR-185, FR-186, FR-187, TC-074	1.33 s
OK	test_the_owner_the_long_name_and_the_logo_are_written_once		FR-185, FR-186, FR-187, TC-074	0.00 s

tests/system/test_cli_create_admin.py — 2 tests, 2 passed

Result	Test	Purpose	Requirements	Time
OK	test_create_admin_cli_command			0.67 s
OK	test_create_admin_cli_rejects_weak_password			0.59 s

tests/system/test_dark_mode.py — 8 tests, 8 passed

Dark mode (docs/10-ui-ux-specification.md section 13, FR-183): the dark palette meets the same

Result	Test	Purpose	Requirements	Time
OK	test_colours_are_only_written_out_where_the_dark_theme_cannot_matter	Outside the token blocks a hex colour is allowed only for fills that carry their own text	FR-155, FR-183, NFR-010, SEC-025, TC-049	0.00 s
OK	test_every_badge_and_sla_pill_meets_aa_in_the_dark_theme		FR-155, FR-183, NFR-010, SEC-025, TC-049	0.00 s
OK	test_the_dark_palette_meets_aa_on_its_backgrounds		FR-155, FR-183, NFR-010, SEC-025, TC-049	0.00 s
OK	test_the_dark_theme_gives_every_light_tint_a_dark_value	A token added to the light palette without a dark twin would stay light on a dark page.	FR-155, FR-183, NFR-010, SEC-025, TC-049	0.00 s
OK	test_the_dark_theme_is_for_the_screen_only		FR-155, FR-183, NFR-010, SEC-025, TC-049	0.87 s
OK	test_the_login_page_and_every_signed_in_page_offer_the_theme_choice		FR-155, FR-183, NFR-010, SEC-025, TC-049	1.63 s
OK	test_the_theme_choice_is_translated		FR-155, FR-183, NFR-010, SEC-025, TC-049	1.39 s
OK	test_theme_js_is_a_plain_file_that_follows_the_system_and_remembers_the_choice		FR-155, FR-183, NFR-010, SEC-025, TC-049	0.01 s

tests/system/test_deploy_artifacts.py — 22 tests, 22 passed

I-33 / I-34: the packaging, installer, scheduler units and CI definitions are consistent with

Result	Test	Purpose	Requirements	Time
OK	test_backup_runs_every_night_and_reminders_on_weekdays		NFR-013, NFR-014, TC-051	0.01 s
OK	test_ci_files_parse_and_run_the_ci_script		NFR-013, NFR-014, TC-051	0.02 s
OK	test_create_admin_can_take_its_answers_from_the_environment		NFR-013, NFR-014, TC-051	0.54 s
OK	test_each_timer_has_a_matching_oneshot_service		NFR-013, NFR-014, TC-051	0.01 s
OK	test_every_expected_artifact_is_shipped		NFR-013, NFR-014, TC-051	0.03 s
OK	test_every_pattern_in_the_scan_is_classified		NFR-013, NFR-014, TC-051	0.00 s
OK	test_every_setting_an_installer_writes_is_read_by_the_application		NFR-013, NFR-014, TC-051	0.02 s
OK	test_every_setting_in_the_example_file_is_read_by_the_application		NFR-013, NFR-014, TC-051	0.01 s
OK	test_linux_installer_parses_and_its_dry_run_changes_nothing		NFR-013, NFR-014, TC-051	2.56 s
OK	test_non_sqlite_connections_are_left_alone	The engine listeners must not touch another driver's connection (docs/07 section 12).	NFR-013, NFR-014, TC-051	1.09 s
OK	test_production_refuses_to_start_with_a_weak_key		NFR-013, NFR-014, TC-051	8.90 s
OK	test_serve_options_follow_the_environment		NFR-013, NFR-014, TC-051	0.05 s
OK	test_services_call_only_commands_that_exist		NFR-013, NFR-014, TC-051	0.56 s
OK	test_the_ci_script_documents_itself		NFR-013, NFR-014, TC-051	2.29 s
OK	test_the_postgres_ratchet_holds		NFR-013, NFR-014, TC-051	10.40 s
OK	test_the_production_entry_point_serves_and_reports_its_state		NFR-013, NFR-014, TC-051	20.67 s
OK	test_the_ratchet_notices_new_sqlite_specific_code		NFR-013, NFR-014, TC-051	8.17 s
OK	test_units_parse_and_have_the_required_sections		NFR-013, NFR-014, TC-051	0.06 s
OK	test_web_service_runs_the_production_entry_point_as_an_unprivileged_user		NFR-013, NFR-014, TC-051	0.00 s
OK	test_windows_installer_dry_run_lists_every_step_and_creates_nothing		NFR-013, NFR-014, TC-051	3.12 s
OK	test_windows_scripts_parse		NFR-013, NFR-014, TC-051	0.73 s
OK	test_write_paths_of_the_units_cover_what_the_jobs_write		NFR-013, NFR-014, TC-051	0.01 s

tests/system/test_docs_consistency.py — 17 tests, 17 passed

TC-052. The documents agree with the code and with each other: the generated endpoint table, the

Result	Test	Purpose	Requirements	Time
OK	test_every_requirement_id_cited_in_a_document_is_defined		TC-052	0.62 s
OK	test_every_requirement_is_accounted_for_and_the_matrix_evidence_resolves		TC-052	1.19 s
OK	test_every_screen_path_in_the_ui_inventory_is_a_real_route	docs/10 section 1 names the path of each v4 screen; each must exist as a UI route.	TC-052	0.53 s
OK	test_every_test_case_id_cited_in_the_documents_exists		TC-052	0.17 s
OK	test_links_between_documents_point_at_existing_files_and_headings		TC-052	0.38 s
OK	test_paragraphs_in_documents_are_single_source_lines	docs/15 §6: prose paragraphs are one continuous source line (tables, lists and code may span).	TC-052	0.10 s
OK	test_section_references_between_documents_exist		TC-052	0.32 s
OK	test_the_audit_catalogue_lists_every_code_the_application_can_write		TC-052	0.80 s
OK	test_the_contract_version_and_operation_count_quoted_in_the_documents		TC-052	0.73 s
OK	test_the_decisions_and_assumptions_are_complete		TC-052	0.05 s
OK	test_the_document_set_version_is_current_and_the_report_is_generated		TC-052	0.01 s
OK	test_the_endpoint_table_of_the_api_document_is_the_generated_one		TC-052	0.99 s
OK	test_the_group_and_subgroup_counts_quoted_in_the_documents		TC-052	0.01 s
OK	test_the_screen_count_matches_the_inventory		TC-052	0.02 s
OK	test_the_table_and_trigger_counts_quoted_in_the_documents		TC-052	0.04 s
OK	test_the_test_suite_table_of_the_strategy_matches_the_test_files	docs/13 section 8a quotes how many test functions each v4 module has.	TC-052	0.14 s
OK	test_the_v4_tables_of_the_data_dictionary_name_real_columns	docs/06 section 13 lists each v4 table with its fields; every named field must exist.	TC-052	0.02 s

tests/system/test_docs_pdf.py — 34 tests, 34 passed

The documents of a release as PDF with the version in the file name (FR-185,

Result	Test	Purpose	Requirements	Time
OK	test_a_document_is_built_with_the_version_in_its_name		FR-184, FR-185, NFR-013, TC-072	1.05 s
OK	test_an_unknown_document_is_refused		FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[00-INDEX.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[01-executive-summary.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[02-prd.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[03-srs.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.03 s
OK	test_every_character_of_the_documents_can_be_drawn[04-domain-model.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[05-governance-handbook.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[06-data-dictionary.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.02 s
OK	test_every_character_of_the_documents_can_be_drawn[07-database-design.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s

Result	Test	Purpose	Requirements	Time
OK	test_every_character_of_the_documents_can_be_drawn[08-system-architecture.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[09-api-specification.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[10-ui-ux-specification.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[11-security-architecture.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[12-audit-compliance.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[13-test-strategy.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[14-deployment-architecture.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[15-development-standards.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[16-mvp-roadmap.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[17-implementation-backlog.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[18-claude-code-implementation-guide.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[19-glossary.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_every_character_of_the_documents_can_be_drawn[20-traceability-matrix.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.03 s
OK	test_every_character_of_the_documents_can_be_drawn[21-improvement-ideas.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.02 s
OK	test_every_character_of_the_documents_can_be_drawn[22-software-description.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.02 s
OK	test_every_character_of_the_documents_can_be_drawn[23-user-manual.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.02 s
OK	test_every_character_of_the_documents_can_be_drawn[24-test-report.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.05 s
OK	test_every_character_of_the_documents_can_be_drawn[25-hc-integration-concept.md]	A new symbol in a document must be given a word in _WORDS, or it prints as a gap.	FR-184, FR-185, NFR-013, TC-072	0.01 s
OK	test_inline_markup_is_removed_and_kept_as_styles		FR-184, FR-185, NFR-013, TC-072	0.00 s
OK	test_symbols_a_pdf_font_cannot_draw_become_words		FR-184, FR-185, NFR-013, TC-072	0.00 s
OK	test_the_four_documents_have_their_file_names		FR-184, FR-185, NFR-013, TC-072	0.15 s
OK	test_the_full_documentation_has_a_contents_list_and_the_artefacts		FR-184, FR-185, NFR-013, TC-072	0.44 s
OK	test_the_reader_finds_every_kind_of_block		FR-184, FR-185, NFR-013, TC-072	0.03 s
OK	test_the_user_manual_is_built_from_the_real_document		FR-184, FR-185, NFR-013, TC-072	6.58 s

tests/system/test_first_start.py — 3 tests, 3 passed

The first start on a new machine is one command that works from any folder

Result	Test	Purpose	Requirements	Time
OK	test_every_step_is_listed_from_any_folder_and_nothing_is_changed		NFR-013, TC-071	1.39 s
OK	test_the_port_and_the_skipped_administrator_are_honoured		NFR-013, TC-071	1.09 s
OK	test_the_script_names_the_application_folder_it_works_in		NFR-013, TC-071	1.07 s

tests/system/test_handover.py — 5 tests, 5 passed

The hand-over to the team repository (FR-188, scripts/handover.py): the audit, the clean export and

Result	Test	Purpose	Requirements	Time
OK	test_a_clean_tree_passes_the_audit		FR-188, NFR-013, TC-075	0.41 s
OK	test_a_tag_that_does_not_exist_and_a_push_without_confirmation_are_refused		FR-188, NFR-013, TC-075	0.90 s
OK	test_every_kind_of_finding_stops_the_handover		FR-188, NFR-013, TC-075	0.76 s
OK	test_the_export_is_one_commit_by_the_named_author_with_the_tag		FR-188, NFR-013, TC-075	1.07 s
OK	test_the_repository_carries_its_team_files		FR-188, NFR-013, TC-075	0.04 s

tests/system/test_health.py — 1 tests, 1 passed

US-001 acceptance criterion: app boots, health endpoint returns 200.

Result	Test	Purpose	Requirements	Time
OK	test_health_returns_ok		US-001	0.69 s

tests/system/test_i18n.py — 30 tests, 30 passed

I-22: the German interface. The catalogue must cover every string a template wraps in `()`,_

Result	Test	Purpose	Requirements	Time
OK	test_every_language_translates_the_whole_interface_inventory		FR-155, FR-167, TC-049	0.79 s
OK	test_every_template_string_has_a_german_entry		FR-155, FR-167, TC-049	1.04 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[ar]		FR-155, FR-167, TC-049	1.63 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[bn]		FR-155, FR-167, TC-049	1.59 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[da]		FR-155, FR-167, TC-049	1.71 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[de]		FR-155, FR-167, TC-049	1.57 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[en]		FR-155, FR-167, TC-049	1.61 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[es]		FR-155, FR-167, TC-049	1.59 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[fr]		FR-155, FR-167, TC-049	2.36 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[gu]		FR-155, FR-167, TC-049	1.58 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[hi]		FR-155, FR-167, TC-049	1.63 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[it]		FR-155, FR-167, TC-049	1.57 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[ka]		FR-155, FR-167, TC-049	1.64 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[kn]		FR-155, FR-167, TC-049	1.57 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[ml]		FR-155, FR-167, TC-049	1.64 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[mr]		FR-155, FR-167, TC-049	2.42 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[or]		FR-155, FR-167, TC-049	1.66 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[pa]		FR-155, FR-167, TC-049	1.59 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[ta]		FR-155, FR-167, TC-049	1.60 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[te]		FR-155, FR-167, TC-049	1.54 s
OK	test_main_pages_render_in_each_language_without_leaking_format_markers[ur]		FR-155, FR-167, TC-049	1.60 s
OK	test_placeholders_survive_translation		FR-155, FR-167, TC-049	0.59 s
OK	test_the_browser_language_picks_the_login_page_language		FR-155, FR-167, TC-049	0.70 s
OK	test_the_catalogue_has_no_empty_or_identical_entries		FR-155, FR-167, TC-049	0.62 s
OK	test_the_choice_before_login_applies_to_the_login_page		FR-155, FR-167, TC-049	1.41 s
OK	test_the_german_login_page_keeps_english_credentials_hint_free_of_secrets		FR-155, FR-167, TC-049	0.67 s
OK	test_the_language_choice_is_remembered_for_the_user		FR-155, FR-167, TC-049	0.97 s
OK	test_the_user_preference_is_exposed_on_the_account_page		FR-155, FR-167, TC-049	0.79 s
OK	test_there_are_no_stale_german_entries_for_removed_template_strings		FR-155, FR-167, TC-049	2.13 s
OK	test_translate_falls_back_to_the_english_text		FR-155, FR-167, TC-049	0.56 s

tests/system/test_injection_and_session.py — 8 tests, 8 passed

TC-017, TC-026, TC-027, SEC-002, SEC-003, SEC-020-SEC-022: injection

Result	Test	Purpose	Requirements	Time
OK	test_audit_log_filters_resist_injection		SEC-002, SEC-003, SEC-020, SEC-022, TC-017, TC-026, TC-027	0.77 s
OK	test_idle_session_expires_after_thirty_minutes	TC-017 / FR-002 / SEC-003.	FR-002, SEC-003, TC-017	0.77 s
OK	test_recent_activity_keeps_the_session_alive		SEC-002, SEC-003, SEC-020, SEC-022, TC-017, TC-026, TC-027	0.77 s
OK	test_server_side_validation_ignores_client_side_constraints	SEC-020: every rule is enforced server-side, whatever the browser did.	SEC-020	1.20 s
OK	test_session_cookie_attributes	SEC-002 (the Secure flag is off only in the dev/test configs).	SEC-002	0.72 s
OK	test_session_expires_at_the_absolute_lifetime	SEC-003: 12 hours regardless of activity.	SEC-003	0.74 s
OK	test_sql_injection_probes_have_no_effect	TC-027 / SEC-021.	SEC-021, TC-027	1.61 s
OK	test_user_supplied_text_is_escaped_everywhere_it_renders	TC-026 / SEC-022.	SEC-022, TC-026	3.18 s

tests/system/test_logging_and_passwords.py — 8 tests, 8 passed

Security policy details from 11-security-architecture.md: §4 password policy

Result	Test	Purpose	Requirements	Time
OK	test_administrator_reset_also_honours_history			1.22 s
OK	test_authenticated_requests_record_the_user_id			0.78 s
OK	test_blocklist_ignores_comments_and_short_entries			0.00 s
OK	test_last_five_passwords_cannot_be_reused			4.25 s
OK	test_only_hashes_are_stored_in_the_history			1.80 s
OK	test_policy_length_and_blocklist			0.00 s
OK	test_requests_are_logged_as_json_without_credentials			0.71 s
OK	test_unhandled_errors_are_logged_with_the_request_id			0.80 s

tests/system/test_multiprocess.py — 4 tests, 4 passed

Two real server processes on one database file (docs/14-deployment-architecture.md §3):

Result	Test	Purpose	Requirements	Time
OK	test_concurrent_writers_in_two_processes_all_succeed		NFR-009, TC-050	26.75 s
OK	test_the_ip_limit_is_shared_and_other_users_are_unaffected_until_it_is_reached		NFR-009, TC-050	25.95 s
OK	test_the_limit_survives_a_restart		NFR-009, TC-050	25.35 s
OK	test_the_username_limit_is_shared_by_both_processes		NFR-009, TC-050	27.42 s

tests/system/test_no_account_names.py — 2 tests, 2 passed

The documents and the guide do not name the Git account or the repository (a decision of the

Result	Test	Purpose	Requirements	Time
OK	test_no_document_or_script_names_the_account_or_the_repository		NFR-013	0.92 s
OK	test_the_guide_says_who_to_ask_for_the_repository		NFR-013	0.01 s

tests/system/test_no_cloud.py — 3 tests, 3 passed

US-120 / TC-029: block cloud-SDK imports, cloud identity providers and

Result	Test	Purpose	Requirements	Time
OK	test_no_cloud_sdk_imports_in_source		TC-029, US-120	1.48 s

Result	Test	Purpose	Requirements	Time
OK	test_no_cloud_sdk_in_requirements		TC-029, US-120	0.02 s
OK	test_templates_and_static_reference_no_external_assets		TC-029, US-120	0.44 s

tests/system/test_openapi_conformance.py — 5 tests, 5 passed

EPIC-11 / FR-080: the code and docs/artifacts/openapi.yaml describe the

Result	Test	Purpose	Requirements	Time
OK	test_every_documented_operation_is_implemented_and_vice_versa		FR-080	0.64 s
OK	test_live_payloads_only_use_declared_properties		FR-080	1.43 s
OK	test_lookup_and_domain_request_payloads_match_contract		FR-080	2.17 s
OK	test_revision_diff_payload_matches_contract		FR-080	1.47 s
OK	test_unauthenticated_requests_are_rejected_for_every_documented_operation		FR-080	0.72 s

tests/system/test_permission_grid.py — 14 tests, 14 passed

TC-018 / TC-018a / TC-018b / TC-019 / FR-081 / SEC-010: the full

Result	Test	Purpose	Requirements	Time
OK	test_api_endpoints_require_authentication		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	0.95 s
OK	test_api_permission_grid[ADMINISTRATOR]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.75 s
OK	test_api_permission_grid[CHIEF_ENGINEER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	2.65 s
OK	test_api_permission_grid[ENGINEER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.46 s
OK	test_api_permission_grid[REVIEWER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.68 s
OK	test_api_permission_grid[STCD_OWNER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.36 s
OK	test_api_permission_grid[VIEWER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.54 s
OK	test_ui_permission_grid[ADMINISTRATOR]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	2.50 s
OK	test_ui_permission_grid[CHIEF_ENGINEER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	2.84 s
OK	test_ui_permission_grid[ENGINEER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.74 s
OK	test_ui_permission_grid[REVIEWER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.85 s
OK	test_ui_permission_grid[STCD_OWNER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.55 s
OK	test_ui_permission_grid[VIEWER]		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.41 s
OK	test_ui_redirects_anonymous_users_to_login		FR-081, SEC-010, TC-018, TC-018a, TC-018b, TC-019	1.18 s

tests/system/test_production_config.py — 7 tests, 7 passed

SEC-002 / deployment hardening: a production instance must not start with a

Result	Test	Purpose	Requirements	Time
OK	test_development_and_testing_still_start_with_the_default_key		SEC-002	0.33 s
OK	test_production_refuses_a_weak_secret_key[]		SEC-002	0.00 s
OK	test_production_refuses_a_weak_secret_key[change-me-in-every-real-environment]		SEC-002	0.00 s
OK	test_production_refuses_a_weak_secret_key[dev-secret-key-change-me]		SEC-002	0.00 s
OK	test_production_refuses_a_weak_secret_key[too-short]		SEC-002	0.00 s
OK	test_production_refuses_a_weak_secret_key[xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx]		SEC-002	0.00 s
OK	test_production_starts_with_a_strong_secret_key		SEC-002	0.16 s

tests/system/test_route_inventory.py — 13 tests, 13 passed

Every route is access-controlled, and the declared role lists are what is enforced.

Result	Test	Purpose	Requirements	Time
OK	test_a_tokenised_endpoint_is_always_a_get	API tokens may only ever read (docs/11 §12): allow_token must never sit on a write.	FR-081, SEC-010, TC-018, TC-048	0.53 s
OK	test_every_gated_route_refuses_anonymous_callers		FR-081, SEC-010, TC-018, TC-048	0.74 s
OK	test_every_route_declares_its_access_control		FR-081, SEC-010, TC-018, TC-048	0.56 s
OK	test_login_only_routes_are_reachable_by_every_role		FR-081, SEC-010, TC-018, TC-048	19.25 s
OK	test_login_page_credentials_still_work_for_the_ui_walk		FR-081, SEC-010, TC-018, TC-048	0.57 s
OK	test_role_lists_are_never_empty_and_only_name_real_roles		FR-081, SEC-010, TC-018, TC-048	0.53 s
OK	test_the_declared_role_list_is_what_each_role_gets[ADMINISTRATOR]		FR-081, SEC-010, TC-018, TC-048	7.01 s
OK	test_the_declared_role_list_is_what_each_role_gets[CHIEF_ENGINEER]		FR-081, SEC-010, TC-018, TC-048	2.74 s
OK	test_the_declared_role_list_is_what_each_role_gets[ENGINEER]		FR-081, SEC-010, TC-018, TC-048	3.07 s
OK	test_the_declared_role_list_is_what_each_role_gets[REVIEWER]		FR-081, SEC-010, TC-018, TC-048	2.98 s
OK	test_the_declared_role_list_is_what_each_role_gets[STCD_OWNER]		FR-081, SEC-010, TC-018, TC-048	2.54 s
OK	test_the_declared_role_list_is_what_each_role_gets[VIEWER]		FR-081, SEC-010, TC-018, TC-048	2.72 s
OK	test_the_public_list_is_exactly_the_routes_that_need_no_session		FR-081, SEC-010, TC-018, TC-048	0.55 s

tests/system/test_security_headers.py — 4 tests, 4 passed

SEC-025: standard security headers on every response.

Result	Test	Purpose	Requirements	Time
OK	test_api_response_has_security_headers		SEC-025	1.33 s
OK	test_hsts_sent_over_https		SEC-025	0.56 s
OK	test_templates_have_no_inline_script_or_handlers	The CSP forbids inline script; guard against regressions.	SEC-025	0.21 s
OK	test_ui_page_has_csp_and_no_hsts_over_plain_http		SEC-025	0.64 s

tests/system/test_showcase.py — 6 tests, 6 passed

_The design showcase decks (FR-186, scripts/build_showcase.py): the slide model and its layout_

Result	Test	Purpose	Requirements	Time
OK	test_a_text_that_does_not_fit_stops_the_build		FR-186, FR-187, TC-073	0.22 s
OK	test_both_decks_are_built_from_the_screenshots_and_every_picture_is_described		FR-186, FR-187, TC-073	0.42 s
OK	test_the_browser_is_found_or_named		FR-186, FR-187, TC-073	0.01 s
OK	test_the_decks_are_written_as_powerpoint_files		FR-186, FR-187, TC-073	2.52 s
OK	test_the_demo_has_its_data_and_the_application_has_no_demo_routes		FR-186, FR-187, TC-073	7.53 s
OK	test_the_script_refuses_an_unknown_deck		FR-186, FR-187, TC-073	0.03 s

tests/system/test_sqlite_path.py — 2 tests, 2 passed

A new machine: the database file is found, and its folder created, from the example .env

Result	Test	Purpose	Requirements	Time
OK	test_a_relative_sqlite_path_is_read_from_the_application_directory		NFR-013	0.23 s
OK	test_an_absolute_path_gets_its_folder_and_memory_is_left_alone		NFR-013	0.33 s

tests/system/test_translations_doc.py — 27 tests, 27 passed

The translation catalogue document and the review sheets for native speakers

Result	Test	Purpose	Requirements	Time
OK	test_every_language_has_its_section_and_its_coverage_row		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.25 s
OK	test_right_to_left_languages_get_a_right_to_left_translation_column[ar-right]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.59 s
OK	test_right_to_left_languages_get_a_right_to_left_translation_column[fr-left]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.59 s
OK	test_right_to_left_languages_get_a_right_to_left_translation_column[ur-right]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	1.28 s
OK	test_string_ids_are_stable_and_never_collide		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.15 s
OK	test_the_command_line_writes_one_sheet_per_language_and_refuses_unknown_ones		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	9.15 s
OK	test_the_concept_names_the_layers_the_checks_and_the_open_items		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.02 s
OK	test_the_document_is_exactly_what_the_generator_produces_from_the_code		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.56 s
OK	test_the_fingerprint_changes_when_one_translation_changes		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.01 s
OK	test_the_review_sheet_of_every_language_is_complete[ar]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.91 s
OK	test_the_review_sheet_of_every_language_is_complete[bn]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.94 s
OK	test_the_review_sheet_of_every_language_is_complete[da]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.70 s
OK	test_the_review_sheet_of_every_language_is_complete[de]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.72 s
OK	test_the_review_sheet_of_every_language_is_complete[es]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	1.40 s
OK	test_the_review_sheet_of_every_language_is_complete[fr]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	1.64 s
OK	test_the_review_sheet_of_every_language_is_complete[gu]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.69 s
OK	test_the_review_sheet_of_every_language_is_complete[hi]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.61 s
OK	test_the_review_sheet_of_every_language_is_complete[it]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.72 s
OK	test_the_review_sheet_of_every_language_is_complete[ka]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.80 s
OK	test_the_review_sheet_of_every_language_is_complete[kn]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	1.42 s
OK	test_the_review_sheet_of_every_language_is_complete[ml]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.61 s
OK	test_the_review_sheet_of_every_language_is_complete[mr]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.73 s
OK	test_the_review_sheet_of_every_language_is_complete[or]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.65 s
OK	test_the_review_sheet_of_every_language_is_complete[pa]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.66 s
OK	test_the_review_sheet_of_every_language_is_complete[ta]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.79 s
OK	test_the_review_sheet_of_every_language_is_complete[te]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	1.39 s
OK	test_the_review_sheet_of_every_language_is_complete[ur]		FR-110, FR-111, FR-155, FR-157, FR-161, TC-057	0.68 s

tests/system/test_uat.py — 14 tests, 14 passed

TC-030: UAT-01 .. UAT-13 (docs/13-test-strategy.md §7) as scripted

Result	Test	Purpose	Requirements	Time
OK	test_uat_01_new_code_has_no_identifier_until_submitted		UAT-01	1.88 s
OK	test_uat_02_either_signoff_order_reaches_pending_approval_automatically		UAT-02	1.69 s
OK	test_uat_03_chief_engineer_approval_visible_to_the_engineer		UAT-03	2.10 s
OK	test_uat_04_rejection_comment_visible_and_prior_signoff_cleared		UAT-04	2.04 s
OK	test_uat_05_cross_domain_signoff_unblocks_submit		UAT-05	1.42 s
OK	test_uat_06_release_build_excludes_unselected_and_sandbox		UAT-06	3.58 s
OK	test_uat_07_viewer_searches_and_exports_released_catalogue_for_one_group		UAT-07	2.09 s
OK	test_uat_08_sandbox_lifecycle_and_traceable_deletion		UAT-08	1.19 s
OK	test_uat_09_nobody_can_decide_their_own_submission		UAT-09	1.90 s
OK	test_uat_09b_chief_engineer_cannot_be_a_prior_signer_or_author		TC-030, UAT-01, UAT-13	2.10 s
OK	test_uat_10_full_database_export_is_administrator_only		UAT-10	1.57 s
OK	test_uat_11_import_creates_only_governed_drafts_and_an_error_report		UAT-11	1.34 s
OK	test_uat_12_subgroup_is_mandatory_and_overlaps_are_refused		UAT-12	1.23 s
OK	test_uat_13_failed_form_keeps_input_and_dropdowns_explain_themselves		UAT-13	2.33 s

tests/system/test_version.py — 4 tests, 4 passed

The version of the application is written once and shown where a person or a monitor looks for

Result	Test	Purpose	Requirements	Time
OK	test_the_api_contract_documents_the_version_in_both_health_answers		FR-184, NFR-013, TC-049	0.03 s
OK	test_the_health_endpoints_name_the_version		FR-184, NFR-013, TC-049	0.78 s
OK	test_the_login_page_and_the_menu_show_the_version		FR-184, NFR-013, TC-049	1.42 s
OK	test_the_version_is_a_release_number_and_pyproject_agrees		FR-184, NFR-013, TC-049	0.04 s

tests/system/test_windows_gateway.py — 7 tests, 7 passed

The Windows sign-in against a real Windows identity (docs/03-srs.md FR-163, docs/11 SEC-033).

Result	Test	Purpose	Requirements	Time
OK	test_a_header_the_client_sends_is_dropped_by_the_gateway		FR-163, SEC-033, TC-062	1.22 s
OK	test_a_request_without_credentials_gets_a_challenge_and_never_reaches_the_application		FR-163, SEC-033, TC-062	1.57 s
OK	test_an_unknown_windows_user_and_a_deactivated_account_are_refused		FR-163, SEC-033, TC-062	1.18 s
OK	test_sspi_names_the_windows_user_who_runs_the_process[NTLM]		FR-163, SEC-033, TC-062	0.01 s
OK	test_sspi_names_the_windows_user_who_runs_the_process[Negotiate]		FR-163, SEC-033, TC-062	0.00 s
OK	test_the_header_is_ignored_when_it_does_not_come_from_the_trusted_gateway		FR-163, SEC-033, TC-062	1.74 s
OK	test_the_whole_chain_signs_the_windows_user_in		FR-163, SEC-033, TC-062	1.17 s

tests/performance/test_performance.py — 9 tests, 9 passed

NFR-001, TC-024, TC-025, NFR-004: performance targets on a temp-file SQLite

Result	Test	Purpose	Requirements	Time
OK	test_concurrent_identifier_allocation_in_one_group	TC-025: parallel submissions in one popular group yield unique, gap-free	TC-025	1.08 s
OK	test_export_issues_a_constant_number_of_queries	Regression guard for the N+1 problem: exporting 500 revisions must not cost	NFR-001, NFR-004, TC-024, TC-025	1.31 s
OK	test_export_of_5000_records_within_target[csv-5]	NFR-001: CSV/JSON <= 5 s, PDF <= 15 s for 5,000 records (XLSX held to the PDF limit).	NFR-001	2.40 s

Result	Test	Purpose	Requirements	Time
OK	test_export_of_5000_records_within_target[json-5]	NFR-001: CSV/JSON <= 5 s, PDF <= 15 s for 5,000 records (XLSX held to the PDF limit).	NFR-001	2.54 s
OK	test_export_of_5000_records_within_target[pdf-15]	NFR-001: CSV/JSON <= 5 s, PDF <= 15 s for 5,000 records (XLSX held to the PDF limit).	NFR-001	9.70 s
OK	test_export_of_5000_records_within_target[xlsx-15]	NFR-001: CSV/JSON <= 5 s, PDF <= 15 s for 5,000 records (XLSX held to the PDF limit).	NFR-001	4.38 s
OK	test_fifty_concurrent_users_are_served_without_errors	NFR-004: 50 concurrent named users (each searching every ~8 s —	NFR-004	30.49 s
OK	test_reads_are_not_blocked_by_an_open_write_transaction	WAL read concurrency: a reader proceeds while another connection holds	NFR-001, NFR-004, TC-024, TC-025	0.79 s
OK	test_search_and_read_latency_at_20000_revisions	TC-024.	TC-024	12.38 s

9. Run history

#	Date (UTC)	Commit	Order	Tests	Passed	Failed	Retried	Coverage	Gates	Duration
1	2026-10-21 16:36	f23ef18*	reverse	318	318	0	1	88%	5/5 pass	232 s
2	2026-10-21 16:42	f23ef18*	shuffle:11	318	318	0	0	88%	5/5 pass	235 s
3	2026-10-21 16:49	f23ef18*	file	318	318	0	1	88%	5/5 pass	237 s
4	2026-10-21 16:57	f23ef18*	file	318	318	0	0	88%	5/5 pass	244 s
5	2026-10-21 17:09	f23ef18*	shuffle:23	319	319	0	0	88%	5/5 pass	242 s
6	2026-10-21 17:15	f23ef18*	file	319	319	0	0	88%	5/5 pass	237 s
7	2026-10-22 10:02	f23ef18*	file	329	329	0	0	89%	5/5 pass	150 s
8	2026-10-23 06:16	c0aff94*	file	601	600	1	0	92%	5/5 pass	589 s
9	2026-10-23 06:27	c0aff94*	reverse	601	600	1	0	92%	5/5 pass	526 s
10	2026-10-23 06:38	c0aff94*	shuffle:7	601	600	1	0	92%	5/5 pass	473 s
11	2026-10-23 06:50	c0aff94*	file	602	602	0	0	92%	5/5 pass	472 s
12	2026-10-23 06:59	c0aff94*	reverse	602	602	0	0	92%	5/5 pass	470 s
13	2026-10-23 07:08	c0aff94*	shuffle:7	602	602	0	0	92%	5/5 pass	469 s
14	2026-10-23 07:18	c0aff94*	file+filedb	602	601	1	0	92%	5/5 pass	542 s
15	2026-10-23 07:31	c0aff94*	file+filedb	602	602	0	0	92%	5/5 pass	590 s
16	2026-10-23 07:41	c0aff94*	file	602	602	0	0	92%	5/5 pass	505 s
17	2026-10-23 07:54	c0aff94*	reverse	607	607	0	0	92%	5/5 pass	459 s
18	2026-10-23 08:04	c0aff94*	shuffle:7	607	607	0	0	92%	5/5 pass	491 s
19	2026-10-23 08:16	c0aff94*	file+filedb	607	607	0	0	92%	5/5 pass	628 s
20	2026-10-23 08:25	c0aff94*	file	607	607	0	0	92%	5/5 pass	508 s
21	2026-10-23 15:33	c0aff94*	file	807	807	0	0	92%	5/5 pass	685 s
22	2026-10-23 19:53	3670346*	file	830	828	2	0	93%	2/5 pass	850 s
23	2026-10-23 20:10	3670346*	file	830	830	0	0	93%	5/5 pass	728 s
24	2026-10-23 23:30	9f038bc*	file	848	847	1	0	93%	5/5 pass	741 s
25	2026-10-23 23:47	9f038bc*	file	848	848	0	0	93%	5/5 pass	748 s
26	2026-10-24 10:42	e07a82d*	file	867	867	0	0	93%	4/5 pass	768 s
27	2026-10-24 10:57	88bf318*	file	867	867	0	0	93%	5/5 pass	772 s
28	2026-10-24 11:50	9828815*	file	876	876	0	0	93%	5/5 pass	773 s
29	2026-10-24 12:42	1f1f0c9*	file	892	891	1	0	93%	5/5 pass	807 s
30	2026-10-24 12:58	1f1f0c9*	file	892	892	0	0	93%	5/5 pass	806 s
31	2026-10-24 14:37	a81555a*	file	904	904	0	0	93%	4/5 pass	967 s
32	2026-10-24 14:56	a81555a*	file	915	915	0	0	93%	5/5 pass	979 s
33	2026-10-24 16:00	2cd8fe6*	file	927	927	0	0	93%	5/5 pass	993 s
34	2026-10-24 17:31	afc84a3*	file	946	945	1	0	93%	3/5 pass	1051 s
35	2026-10-24 17:53	afc84a3*	file	946	943	3	0	93%	5/5 pass	1050 s
36	2026-10-24 18:12	afc84a3*	file	946	946	0	0	93%	5/5 pass	1054 s
37	2026-10-25 11:41	016e74f*	file	956	956	0	0	93%	3/5 pass	1207 s
38	2026-10-25 12:06	016e74f*	file	956	956	0	0	93%	5/5 pass	1205 s

#	Date (UTC)	Commit	Order	Tests	Passed	Failed	Retried	Coverage	Gates	Duration
39	2026-10-26 09:59	b0029bd*	file	958	958	0	0	93%	5/5 pass	1163 s
40	2026-10-26 10:26	2a5a8ff	file	966	966	0	0	93%	5/5 pass	1113 s
41	2026-10-26 11:39	97af72c	file	973	973	0	0	93%	5/5 pass	1099 s
42	2026-10-26 12:29	a62d0c6*	file	1007	1007	0	0	93%	5/5 pass	1209 s
43	2026-10-26 13:11	613aa61*	file	1020	1019	0	0	93%	5/5 pass	1187 s
44	2026-10-26 13:44	3b18219	file	1025	1023	1	0	93%	5/5 pass	1194 s
45	2026-10-26 14:07	5094e00	file	1025	1025	0	0	93%	5/5 pass	1205 s
46	2026-10-27 16:23	45bd50e*	file	1035	1030	5	0	93%	4/5 pass	1292 s
47	2026-10-27 16:51	45bd50e*	file	1035	1035	0	0	93%	4/5 pass	1285 s
48	2026-10-27 17:16	45bd50e*	file	1035	1035	0	0	93%	5/5 pass	1292 s
49	2026-10-27 19:40	78dcd55*	file	1079	1077	2	0	93%	5/5 pass	1409 s
50	2026-10-27 20:09	78dcd55*	file	1079	1079	0	0	93%	5/5 pass	1421 s
51	2026-10-28 19:13	a696bc6*	file	1082	1080	2	0	93%	5/5 pass	2077 s
52	2026-10-28 19:51	a696bc6*	file	1082	1082	0	0	93%	5/5 pass	1983 s

25. Concept — Integrating Hub Controller Codes into the 4xxx / 9xxx Scheme

Status: proposal for decision (2026-10-22). Nothing in this document is implemented. Supporting data: artifacts/hc-numbering-proposal.xlsx — old number, proposal 1 (5-digit), proposal 2 (this concept) for every Hub Controller code, plus the proposed bands and the existing Main Controller bands.

Inputs: `HC Status Code Number.xlsx` (268 HC codes: 51 WTUR, 216 WROT, 1 WPPD), the current Main Controller (MC) taxonomy, assumption A-008 (`03-srs.md`: if the Hub Controller is removed from the hardware setup, its codes are defined anew on the MC side), and the decisions of 2026-10-22: MC identifiers are four digits; WPPD is obsolete (hidden).

Relationship to what v4 implemented (2026-10-23). v4 (decision D-06, FR-147) put a `controller_type` flag `MAIN` | `HUB` on the **Functional System Group** — the three existing Hub groups are `HUB` — and built Hub -> Main consolidation (Main Controller drafts created from Hub codes, a mapping report, bulk deprecation of replaced Hub codes) for assumption A-008. This concept proposes something different and **still unimplemented**: no separate Hub groups at all, with the flag on the **subgroup** inside the existing groups. If this concept is adopted, the group-level flag and the consolidation screens become obsolete for the migration they serve, and these items need a decision: (a) move `controller_type` to `lookup_functional_subgroup` and drop it from the group (a migration with a data copy: every subgroup of a former Hub group becomes `HUB`); (b) keep both and let the subgroup value override the group value; (c) keep v4 as it is and use this concept's numbering only as the target for the Main drafts that Create MC draft produces. Option (c) needs no schema change and is the least risky; options (a) and (b) change `schema.sql`, the data dictionary and the consolidation service.

1. Idea in one sentence

Hub Controller functions live inside the Main Controller groups they belong to — WROT-HC inside group WROT (4000–4999), WTUR-HC inside group WTUR (9000–9999) — each as its own HC subgroup in a reserved block, flagged `controller = HUB`, so every status code is a plain four-digit `StCd-NNNN` and the separate five-digit Hub groups disappear.

2. Proposed bands (revised 2026-10-22)

Both HC blocks start at the **500 range** of their group. Order and block sizes follow the HC export (50-number blocks, as in the source); HC Pitch Converter gets an extra 50.

Group	HC subgroup	Band	Codes now	Spare
WTUR	HC Control System	9500–9599	36	64
WTUR	HC Field Bus	9600–9649	8	42
WTUR	HC System	9650–9699	7	43
WROT	HC Blades	4500–4549	25	25
WROT	HC Batteries	4550–4649	51	49
WROT	HC Pitch	4650–4699	35	15
WROT	HC Pitch Control	4700–4749	9	41
WROT	HC Pitch Drives	4750–4799	33	17
WROT	HC Pitch Converter	4800–4899	45	55
WROT	HC Rotor	4900–4949	18 (14 + 4 added since the export)	32

- **WTUR:** 9500–9699 (200 slots) for 51 codes. MC headroom left in WTUR: 9400–9499 and 9700–9899.
- **WROT:** 4500–4949 (450 slots) for 216 codes. MC headroom left in WROT: only 4950–4999 (the existing MC subgroups end at 4499).
- The new number of each existing code is `band start + (proposal-1 number ? block start)`, i.e. the order of the HC export is kept. The four codes without a number in the file (old 14321–14324) are placed at the end of the Rotor block as exported (4914–4917) — they describe the pitch ultracapacitor and fit HC Pitch Converter better; see §6.
- **Tightest bands:** HC Pitch (15 spare) and HC Pitch Drives (17 spare).

3. What changes in the tool

Area	Change
Taxonomy	Remove the three Hub groups (WTUR 11000–11999, WROT 12000–12999, WPPD — hidden today). Add the ten HC subgroups above under the existing groups WTUR (id 9) and WROT (id 4). The "reused prefix, resolve by range" special case disappears.
New attribute	lookup_functional_subgroup.controller_type = MAIN | HUB (default MAIN). Enables "HC only" filters and exports, a visible HC badge, and bulk deprecation later.
Identifier	Always StCd-NNNN (1000–9999); sandbox stays StCd-T#####. Needs the database check constraint on status_code_identifier to allow four digits (table rebuild in a migration) and the numbering/validation code and documents to drop the leading zero. Existing development codes are renumbered by the migration.
Names	HC subgroups are labelled "HC ..." because three names collide with MC subgroups (Control System, Rotor, System).
Legacy numbers	The old HC numbers (35 of them are four-digit and overlap MC numbers; 7 are below 1000, one is 0) go into legacy_reference_number, which is not unique.
Load	Import the 267 codes through the existing library import as Drafts (full governance still applies), using the mapping sheet.
Importer (v4.2)	Hub Controller programs (StCd_.st) can be imported directly: the legacy number in the file is translated to the new one through the Hub rows of the mapping (v4.17, kept in the database and changed only with three approvals, 03-srs.md FR-191 ... FR-198; the new numbers are in the Hub range), the legacy number is kept as legacy_reference_number, and the Draft reserves the new number (03-srs.md FR-048s, 06-data-dictionary.md §5a).

4. When the Hub Controller is removed (assumption A-008)

1. Deactivate the ten HC subgroups (a lookup change; data is kept, numbers are never reused).
2. The functions that move to the MC side are **defined as new MC codes** in the ordinary MC subgroups (4000–4499 / 9000–9399, or the free MC headroom 4950–4999 / 9400–9499 / 9700–9899), using the "create MC draft from HC code" action planned as I-36, which copies the descriptive fields and records the HC number as the legacy reference.
3. The retired HC bands (4500–4949, 9500–9699) simply stay retired — no renumbering of anything.

5. Compared with the alternative (5-digit HC numbers, column B of the export)

	This concept (4xxx / 9xxx)	5-digit HC numbers (11001–13001)
Identifier	One format, four digits	Two formats: four digits MC, five digits HC
MC vs HC at a glance	Only via subgroup / HC badge	By the number's length
Taxonomy	Simpler: no Hub groups, no repeated prefixes	Today's 12 groups incl. repeated WTUR/WROT prefixes
MC growth room	Reduced: WROT keeps only 4950–4999 free	Untouched
HC removal	Deactivate subgroups; retired numbers stay inside MC space	Deactivate the Hub groups; MC space untouched
Change effort now	Higher: taxonomy + controller_type + identifier migration + renumber	Lower: only the identifier format (MC four digits)

Trade-off to decide: the concept buys a single identifier format and a simpler taxonomy at the price of consuming most of the free capacity in WROT/WTUR and losing the digit-count distinction. Because HC codes are expected to be redefined on the MC side anyway, consuming MC number space for codes with a limited lifetime is the main argument against it.

6. Open decisions

1. Approve the concept, or keep the five-digit Hub numbers (then only the MC four-digit change is built).
2. Band sizes — whether to widen HC Pitch and HC Pitch Drives (15 and 17 spare) as well; WROT would then have no MC headroom left at all.
3. Where the 4 post-export codes (UCAP capacity low axis 1–3, Pitch system hardware mismatch) go: the export lists them under Rotor, but they belong functionally to the pitch system — moving them to HC Pitch Converter would use 4 of its 55 spare numbers.
4. Whether the digit-length rule "MC = 4, HC = 5" is a business requirement in itself (e.g. for manuals or field staff). If it is, this concept cannot be used.

CSCM Tool — Quick deployment guide

For a new machine. Two paths: **A. try it** (about 10 minutes, one person, no server setup) and **B. install it for the team** (Windows Server or Linux, runs as a service). The full reference is docs/14-deployment-architecture.md §4a.

The tool runs entirely on-premises. The only thing it downloads is the Python packages of `cscm_tool/requirements.txt` (use `PIP_INDEX_URL` for an internal mirror).

Who to ask

- **Software Owner and Administrator:** Bastian Ritter, bastian.ritter@senvion.in

They give you the address of the repository, access to it, and the first Administrator account of a team installation.

What you need

- **Python 3.12 or newer** — check with `python --version` (on Linux often `python3 --version`).
- **Git**, and read access to the project repository (branch `dev`). The address of the repository is given by the Software Owner or the Administrator (see "Who to ask" below).
- Windows 10/11/Server or Linux. SQLite is built in; there is no database server to install.

A. Try it on a new machine (development mode)

1. Get the code and find your place

Use **any** terminal: PowerShell, Command Prompt, or Git Bash. The commands below are the same in all of them.

```
git clone -b dev <address of the repository> cscm
cd cscm
```

If you already cloned it, you are probably in the right place. Check: the folder you are in must contain a folder named `cscm_tool` (`dir` in PowerShell/cmd, `ls` in Git Bash). **Do not change into the folder a second time**: if the prompt already ends in the project folder, stay there.

2. Run the first start

```
python cscm_tool/scripts/first_start.py
```

Linux or macOS: `python3 cscm_tool/scripts/first_start.py`.

The script works from whatever folder you are in and does everything, in order, telling you each step:

1. checks the Python version,
2. creates the virtual environment `cscm_tool/.venv` (an existing one is left alone),
3. installs the packages (about a minute),
4. creates `cscm_tool/.env` from `.env.example` (an existing one is kept),
5. creates the database `cscm_tool/instance/cscm.db`,
6. asks for the **first Administrator** (name, e-mail, password) if there is none yet.

It then prints the command that starts the tool. It is safe to run again: finished steps are skipped. `--dry-run` shows the steps without changing anything; `--port 5002` changes the printed port; `--no-admin` skips the Administrator.

3. Start it

Run the command the script printed, inside `cscm_tool`:

Shell	Start command
PowerShell or Command Prompt	cd cscm_tool then .venv\Scripts\python.exe dev_server.py 5001
Git Bash	cd cscm_tool then .venv/Scripts/python.exe dev_server.py 5001
Linux / macOS	cd cscm_tool then .venv/bin/python dev_server.py 5001

Open http://localhost:5001 and log in with the Administrator you created. Create the other users (Engineer, Reviewer, Chief Engineer, StCd Owner, Viewer) on the user administration page (`/admin/users`).

You do not need to "activate" the virtual environment: these commands use its Python directly, which is why they

work the same in every shell.

4. Check that it works

1. `http://localhost:5001/api/v1/health` shows `{"status": "ok", "version": "4.17.1"}`; the same version is at the foot of the menu and of the login page.
2. `http://localhost:5001/api/v1/health/ready` reports database, schema, audit chain and backup state.
3. Import the sample programs: Import from .st files -> choose `cscm_tool/tests/fixtures/st/*.st` -> preview -> create Drafts.
4. Run the automated tests (about 20 minutes for everything; `-m "not performance"` skips the load tests). Inside `cscm_tool`: `.venv\Scripts\python.exe -m pytest -m "not performance"` (Git Bash: `.venv/Scripts/python.exe`, Linux: `.venv/bin/python`).

The same by hand (only if you cannot use the script)

Inside `cscm_tool`:

Step	PowerShell / cmd	Git Bash	Linux
Environment	python -m venv .venv	python -m venv .venv	python3 -m venv .venv
Activate	.venv\Scripts\activate	source .venv/Scripts/activate	source .venv/bin/activate
Packages	pip install -r requirements.txt	same	same
Settings	copy .env.example .env	cp .env.example .env	cp .env.example .env
Database	flask db upgrade, then flask mapping-seed (loads the initial legacy-to-new mapping once)	same	same
Administrator	flask create-admin	same	same

B. Install for the team (production)

Run from a checkout of the repository, as an administrator.

Windows Server

```
cd cscm\cscm_tool
powershell -ExecutionPolicy Bypass -File deploy\windows\install.ps1 -DryRun    # prints every step, changes nothing
powershell -ExecutionPolicy Bypass -File deploy\windows\install.ps1             # installs to C:\CSCM
```

Linux (systemd)

```
cd cscm/cscm_tool
sudo deploy/linux/install.sh --dry-run
sudo deploy/linux/install.sh                                  # installs to /opt/cscm
```

The installer creates the directories and service account, builds the virtual environment, writes the configuration **once** with freshly generated `SECRET_KEY`, `AUDIT_HMAC_KEY` and `MANIFEST_SIGNING_KEY`, runs the migrations, creates the first Administrator, and sets up the web server plus the nightly verified backup, weekday reminders and the weekly audit-chain check. It never overwrites configuration, data or backups, so **running it again is also the upgrade**.

Then, before people use it

1. **Keep a copy of the three keys** from `C:\CSCM\config\cscm.env` (Linux: `/etc/cscm/cscm.env`) in your password safe. They cannot be recovered, and a backup is useless without them.
2. **Put a reverse proxy with TLS in front** (IIS/ARR or nginx; example: `deploy/linux/nginx-cscm.conf`). The application listens on `127.0.0.1:8000` only; set `CSCM_TRUSTED_PROXY` to the proxy's address.
3. **Sign-in with the Windows account** (optional): set `AUTH_MODE=windows` (or `both` while testing) and `WINDOWS_AUTH_TRUSTED_PROXIES` to the proxy address. The proxy must authenticate against Active Directory and set the `X-Remote-User` header itself. Every user needs an account whose e-mail equals their user principal name.
4. **Backups:** point `BACKUP_DIR` at storage on a different machine. Test a restore once: `flask verify-backup .`
5. Check `/api/v1/health/ready` through the proxy.

If something goes wrong

Read the message, find it in the left column. **Most first-start problems are "wrong folder" or "wrong shell".** `pwd` (Git Bash, Linux), `cd` (cmd) or `Get-Location` (PowerShell) tells you where you are.

Message	Cause and fix
cd: cscm/cscm_tool: No such file or directory	You are already inside the project folder. Do not change folder again; run python cscm_tool/scripts/first_start.py from where you are.
Could not open requirements file / cp: cannot stat '.env.example' / can't open file '...dev_server.py'	You are not in cscm_tool. Use the script (it finds the right folder itself), or cd cscm_tool first.
flask: command not found / 'flask' is not recognized	The virtual environment is not active or was not created. Use the script and the start command it prints; .venv/.../python -m flask ... needs no activation.
Permission denied: '...\.venv\Scripts\python.exe' (while creating .venv)	You tried to recreate .venv while it is active or in use. Close other terminals using it, run deactivate, delete the .venv folder, run the script again. The script never recreates an existing .venv.
unable to open database file	Fixed in v4.13.3: git pull. On an older checkout open .env and delete or comment out the line DATABASE_URL=sqlite:///instance/cscm.db. To put the database elsewhere use an absolute path, e.g. DATABASE_URL=sqlite:///C:/CSCM/data/cscm.db.
no such table / empty pages	The database was not created: run the script again, or flask db upgrade in cscm_tool.
Refusing to start in production	SECRET_KEY is missing, short, or still change-me.... Set 32+ random characters in .env.
A new release of pip is available	Only a notice. Ignore it.
Python is older than 3.12	Install Python 3.12 or newer and run the script with it (on Windows: py -3.12 cscm_tool/scripts/first_start.py).

Moving existing data to a new machine

On the old machine: `flask backup-db --dest .` Copy the backup file, install the tool on the new machine as above, stop the web server, replace `cscm.db` in the data folder (`instance/` or `C:\CSCM\data`) with the backup, run `flask db upgrade`, start the server and run `flask audit-verify`. The new machine needs the **same** `AUDIT_HMAC_KEY` and `MANIFEST_SIGNING_KEY` as the old one.

Upgrading

`git pull`, then run the installer again (production) or the first-start script again (development: it installs new packages and applies new migrations). The production installer takes a backup before it migrates.

Known limits (be honest with yourself before go-live)

The installers are tested in dry-run and by the automated tests, but **not yet on a clean Windows Server or Linux host**, and the Windows sign-in is proven against the Windows handshake of a test gateway, not against your domain controller or IIS. Do a trial installation on a spare machine first and work through the pre-go-live checklist in docs/20-traceability-matrix.md §6. The development first-start script was run end to end on a clean copy (new environment, packages, database, first Administrator, second run) and by an automated dry-run test.

Appendix A - REST API contract (OpenAPI)

The contract of the REST API.

```
openapi: 3.0.3
info:
  title: CSCM Tool API
  description: >
    Controller Status Code Management Tool REST API. Versioned at /api/v1. All endpoints
    require an authenticated session (see /auth/login) except where noted; a few read-only endpoints also accept
    a scoped API token (`Authorization: Bearer cscm_...`, I-29). Sign-offs, approvals, publishing and the
    full-database export need the caller's password again (`current_password`, I-17). Runs on the
    internal on-premises network only. Field set grounded in a real export of the live
    status code library (2026-09-17). See docs/09-api-specification.md for narrative detail.
  version: "4.0.0"
servers:
  - url: /api/v1
security:
  - sessionCookie: []
tags:
  - name: Auth
  - name: Users
  - name: StatusCodes
  - name: Revisions
  - name: ChangeRequests
  - name: DomainSignoffs
  - name: Import
  - name: Mapping
  - name: Releases
  - name: Lookups
  - name: AuditLog
  - name: Exports
  - name: Review
  - name: Evidence
  - name: ChangeSets
  - name: Delegation
  - name: Notifications
  - name: Settings
  - name: Reports
  - name: Operations
  - name: SoftwareVersions
  - name: Translations
  - name: SavedSearches
  - name: Integrations
  - name: Hub

paths:
  /auth/windows:
    post:
      tags: [Auth]
      security: []
      summary: >
        Sign in with the Windows identity the authenticating gateway supplies (v4.7, FR-163): the header
        WINDOWS_AUTH_HEADER (default X-Remote-User) holding name@domain or DOMAIN\name, believed only from
        a trusted proxy (WINDOWS_AUTH_TRUSTED_PROXIES). The account is found by e-mail address (UPN) or user
        name; nobody is created. No body. 401 when there is no identity or no active account, 403 when
        AUTH_MODE is password only. With AUTH_MODE=windows the password endpoints answer 403.
      responses:
        '200':
          content:
            application/json:
              schema: { $ref: '#/components/schemas/User' }
        '401': { $ref: '#/components/responses/Unauthorized' }
        '403': { description: The Windows sign-in is not enabled }
        '429': { description: Too many attempts }

  /auth/login:
    post:
      tags: [Auth]
      security: []
      summary: Authenticate and establish a session
      requestBody:
        required: true
        content:
          application/json:
            schema:
              type: object
              required: [username, password]
              properties:
                username: { type: string }
                password: { type: string, format: password }
      responses:
        '200':
          content:
            application/json:
              schema: { $ref: '#/components/schemas/User' }
        '401': { $ref: '#/components/responses/Unauthorized' }
```

```
      '403': { description: Passwords are disabled (AUTH_MODE=windows) }
      '423': { description: Account locked }

/auth/logout:
  post:
    tags: [Auth]
    summary: Invalidate current session
    responses:
      '204': { description: Logged out }

/auth/csrf-token:
  get:
    tags: [Auth]
    summary: Fetch the CSRF token to send as X-CSRFToken on every state-changing request
    security: []
    responses:
      '200':
        content:
          application/json:
            schema:
              type: object
              properties:
                csrf_token: { type: string }

/health:
  get:
    tags: [Auth]
    summary: Liveness probe (no authentication)
    security: []
    responses:
      '200':
        content:
          application/json:
            schema:
              type: object
              properties:
                status: { type: string, example: ok }
                version: { type: string, example: "4.16.1", description: "Version of the running application (FR-184)" }

/auth/me:
  get:
    tags: [Auth]
    summary: Current authenticated user
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/User' }

/users:
  get:
    tags: [Users]
    summary: List users (Administrator only)
    parameters: [{ $ref: '#/components/parameters/Page' }, { $ref: '#/components/parameters/PageSize' }]
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/UserList' }
  post:
    tags: [Users]
    summary: Create a user, including role and Engineering Domain tags (Administrator only)
    requestBody:
      content:
        application/json:
          schema: { $ref: '#/components/schemas/UserCreate' }
    responses:
      '201':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/User' }

/users/{userId}:
  parameters: [{ $ref: '#/components/parameters/UserId' }]
  get:
    tags: [Users]
    summary: Get a user
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/User' }
  patch:
    tags: [Users]
    summary: Update a user, role, or Engineering Domain tags (Administrator only)
    requestBody:
      content:
        application/json:
          schema: { $ref: '#/components/schemas/UserUpdate' }
```

```
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/User' }

/users/me/change-password:
  post:
    tags: [Users]
    summary: Change own password (policy enforced; other sessions are invalidated; FR-013)
    requestBody:
      content:
        application/json:
          schema:
            type: object
            required: [current_password, new_password]
            properties:
              current_password: { type: string }
              new_password: { type: string }
    responses:
      '204': { description: Password changed }
      '422': { $ref: '#/components/responses/ValidationError' }

/users/{userId}/force-reset-password:
  parameters: [{ $ref: '#/components/parameters/UserId' }]
  post:
    tags: [Users]
    summary: Force a password reset for a user (Administrator only; invalidates their sessions; FR-014)
    requestBody:
      content:
        application/json:
          schema:
            type: object
            required: [new_password]
            properties:
              new_password: { type: string }
    responses:
      '204': { description: Password reset }
      '422': { $ref: '#/components/responses/ValidationError' }

/status-codes:
  get:
    tags: [StatusCodes]
    summary: List/search status codes (current revision per code). Sandbox records excluded unless is_sandbox=true is pas
sed explicitly.
    parameters:
      - { $ref: '#/components/parameters/Page' }
      - { $ref: '#/components/parameters/PageSize' }
      - { $ref: '#/components/parameters/Sort' }
      - name: q
        in: query
        description: >
          Search across status_code_identifier, title, description. Substring matches always count; with the
          SQLite FTS5 index (I-31) results are ranked by relevance and a misspelt word still finds its
          close neighbours ("convertor" finds "converter").
        schema: { type: string }
      - name: functional_system_group
        in: query
        schema: { type: string, example: WCNV }
      - name: functional_subgroup
        in: query
        schema: { type: string }
      - name: turbine_platform
        in: query
        schema: { type: string, example: 3XM }
      - name: lifecycle_status
        in: query
        schema: { $ref: '#/components/schemas/LifecycleStatus' }
      - name: is_sandbox
        in: query
        schema: { type: boolean, default: false }
      - name: owner_id
        in: query
        schema: { type: integer }
      - name: status_category
        in: query
        schema: { type: string, enum: [ERROR, WARNING, INFO] }
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/StatusCodeList' }
  post:
    tags: [StatusCodes]
    summary: >
      Create a new Status Code (Draft; no identifier assigned yet ? functional_subgroup is
      mandatory, never optional) or a Sandbox Status Code (is_sandbox=true, Administrator
      only, draws from the isolated StCd-T##### band)
    requestBody:
```

```
      required: true
      content:
        application/json:
          schema: { $ref: '#/components/schemas/StatusCodeCreate' }
    responses:
      '201':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/StatusCodeRevision' }
      '403': { $ref: '#/components/responses/Forbidden' }
      '422': { $ref: '#/components/responses/ValidationError' }

/status-codes/{statusCodeId}:
  parameters: [{ $ref: '#/components/parameters/StatusCodeId' }]
  get:
    tags: [StatusCodes]
    summary: Get current revision of a status code
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/StatusCodeRevision' }
      '404': { $ref: '#/components/responses/NotFound' }
  delete:
    tags: [StatusCodes]
    summary: Permanently delete a Sandbox Status Code (Administrator only; blocked at the DB layer for any non-sandbox re
cord)
    responses:
      '204': { description: Deleted; a SANDBOX_DELETE audit entry with a full before-state snapshot is written }
      '403': { description: Not a sandbox record, or actor is not Administrator }

/status-codes/{statusCodeId}/deprecation-requests:
  parameters: [{ $ref: '#/components/parameters/StatusCodeId' }]
  post:
    tags: [StatusCodes]
    summary: >
      Open a Deprecation change request against a Released code (Administrator or Engineer).
      Runs the full dual-review + Chief Engineer approval cycle (FR-050). Reason is mandatory.
    requestBody:
      content:
        application/json:
          schema:
            type: object
            required: [reason]
            properties:
              reason: { type: string }
              superseded_by_status_code_id: { type: integer, nullable: true }
              acknowledge_dependants: { type: boolean, default: false, description: "Required (422) when other live codes
 still refer to this one - see /status-codes/{id}/where-used (I-04)" }
    responses:
      '201':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/StatusCodeRevision' }
      '409': { description: Code is not Released, or an open revision already exists }
      '422': { $ref: '#/components/responses/ValidationError' }

/status-codes/{statusCodeId}/archive:
  parameters: [{ $ref: '#/components/parameters/StatusCodeId' }]
  post:
    tags: [StatusCodes]
    summary: Archive a Deprecated code once the minimum retention period has elapsed (Administrator only; FR-051, BR-006)
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/StatusCodeRevision' }
      '409': { description: Code is not Deprecated }
      '422': { description: Retention period has not yet elapsed }

/status-codes/{statusCodeId}/reinstate:
  parameters: [{ $ref: '#/components/parameters/StatusCodeId' }]
  post:
    tags: [StatusCodes]
    summary: Reinstate a Deprecated code to Released (Administrator only; exception path, reason required)
    requestBody:
      content:
        application/json:
          schema:
            type: object
            required: [reason]
            properties:
              reason: { type: string }
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/StatusCodeRevision' }
      '409': { description: Code is not Deprecated }
```

```
/status-codes/{statusCodeId}/revisions:
  parameters: [{ $ref: '#/components/parameters/StatusCodeId' }]
  get:
    tags: [Revisions]
    summary: Full revision history for a status code
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
              items: { $ref: '#/components/schemas/StatusCodeRevision' }
  post:
    tags: [Revisions]
    summary: Open a new Draft revision (change request) against an existing code
    requestBody:
      content:
        application/json:
          schema: { $ref: '#/components/schemas/StatusCodeRevisionCreate' }
    responses:
      '201':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/StatusCodeRevision' }
      '409': { description: An open Draft/Review/PendingApproval revision already exists (BR-003) }

/status-codes/{statusCodeId}/revisions/{revisionA}/diff/{revisionB}:
  parameters:
    - { $ref: '#/components/parameters/StatusCodeId' }
    - { name: revisionA, in: path, required: true, schema: { type: integer } }
    - { name: revisionB, in: path, required: true, schema: { type: integer } }
  get:
    tags: [Revisions]
    summary: Field-level diff between two revisions
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/RevisionDiff' }

/revisions/{revisionId}:
  parameters: [{ name: revisionId, in: path, required: true, schema: { type: integer } }]
  patch:
    tags: [Revisions]
    summary: Edit a Draft revision's fields (Engineer, own CR only). Never resets other fields on a validation failure ?
a rejected request changes nothing; the client keeps its unsaved edits.
    requestBody:
      content:
        application/json:
          schema: { $ref: '#/components/schemas/StatusCodeRevisionUpdate' }
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/StatusCodeRevision' }
      '409': { description: Optimistic concurrency conflict, or revision not in DRAFT }
      '422': { $ref: '#/components/responses/ValidationError' }

/change-requests:
  get:
    tags: [ChangeRequests]
    summary: List change requests (review/approval queues)
    parameters:
      - { $ref: '#/components/parameters/Page' }
      - { $ref: '#/components/parameters/PageSize' }
      - { name: state, in: query, schema: { type: string } }
      - { name: requested_by, in: query, schema: { type: integer } }
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/ChangeRequestList' }

/change-requests/{crId}:
  parameters: [{ name: crId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [ChangeRequests]
    summary: Get change request detail (incl. comments and domain sign-offs)
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/ChangeRequest' }

/change-requests/{crId}/submit:
  parameters: [{ name: crId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [ChangeRequests]
```

```
    summary: >
      Submit Draft for Review (Draft -> Review). Atomically allocates the StCd-XXXXX
      identifier (once: a CR rejected back to Draft and resubmitted keeps it). Blocked (422)
      if required Cross-Domain Sign-Offs are incomplete. An Administrator may supply
      `identifier_override` plus a documented `override_reason` for a migration/reservation
      case (FR-021a); the override is validated (format, sub-band, not already allocated)
      and audited as IDENTIFIER_OVERRIDE.
    requestBody:
      required: false
      content:
        application/json:
          schema:
            type: object
            properties:
              identifier_override: { type: string, example: StCd-01234 }
              override_reason: { type: string }
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/ChangeRequest' }
      '403': { description: Non-Administrator attempted an identifier override }
      '422': { $ref: '#/components/responses/ValidationError' }

/change-requests/{crId}/reviewer-signoff:
  parameters: [{ name: crId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [ChangeRequests]
    summary: Reviewer (Sub-PO for StCd) sign-off. Not the requester. One of two required Review sign-offs.
    requestBody:
      content:
        application/json:
          schema: { $ref: '#/components/schemas/SignoffDecision' }
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/ChangeRequest' }
      '403': { description: Segregation-of-duties violation }

/change-requests/{crId}/admin-signoff:
  parameters: [{ name: crId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [ChangeRequests]
    summary: >
      Administrator (Product Owner) sign-off. Not the requester. Second of two required
      Review sign-offs; also confirms/overrides the eight restricted audience-access fields.
      Recording the second sign-off (either order) atomically advances Review -> PendingApproval.
    requestBody:
      content:
        application/json:
          schema: { $ref: '#/components/schemas/AdminSignoffDecision' }
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/ChangeRequest' }
      '422': { $ref: '#/components/responses/ValidationError' }

/change-requests/{crId}/chief-engineer-decide:
  parameters: [{ name: crId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [ChangeRequests]
    summary: >
      Chief Engineer final Approval decision (PendingApproval -> Approved | Draft).
      Actor must not be the requester, Reviewer sign-off, or Administrator sign-off on this CR.
    requestBody:
      content:
        application/json:
          schema: { $ref: '#/components/schemas/SignoffDecision' }
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/ChangeRequest' }

/change-requests/{crId}/withdraw:
  parameters: [{ name: crId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [ChangeRequests]
    summary: Withdraw own Draft/Review/PendingApproval CR
    responses:
      '200': { description: Withdrawn }

/change-requests/{crId}/comments:
  parameters: [{ name: crId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [ChangeRequests]
    summary: Add a comment without a decision, at any stage (Reviewer, Administrator, Chief Engineer; FR-038)
```

```
    requestBody:
      content:
        application/json:
          schema:
            type: object
            required: [comment]
            properties:
              comment: { type: string, minLength: 1, maxLength: 4000 }
    responses:
      '201':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/ChangeRequest' }
      '422': { $ref: '#/components/responses/ValidationError' }

/change-requests/{crId}/reviewer-assignment:
  parameters: [{ name: crId, in: path, required: true, schema: { type: integer } }]
  put:
    tags: [ChangeRequests]
    summary: >
      Assign, reassign or (reviewer_id null) unassign the Reviewer (Administrator only; FR-039).
      Once assigned, only that Reviewer may sign off. Not allowed after the Reviewer has signed.
    requestBody:
      content:
        application/json:
          schema:
            type: object
            properties:
              reviewer_id: { type: integer, nullable: true }
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/ChangeRequest' }
      '409': { description: CR no longer awaiting Reviewer sign-off }
      '422': { $ref: '#/components/responses/ValidationError' }

/change-requests/{crId}/domain-signoff-requests:
  parameters: [{ name: crId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [DomainSignoffs]
    summary: >
      The originating Engineer requests a pending domain's sign-off, optionally from one named
      Engineer holding that domain (FR-028). Surfaces on the target's Dashboard / My Change Requests.
    requestBody:
      content:
        application/json:
          schema:
            type: object
            required: [engineering_domain]
            properties:
              engineering_domain: { type: string, example: CONTROLS }
              target_user_id: { type: integer, nullable: true }
    responses:
      '201':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/DomainSignoffRequest' }
      '403': { description: Actor is not the originating Engineer }
      '422': { $ref: '#/components/responses/ValidationError' }

/domain-signoff-requests:
  get:
    tags: [DomainSignoffs]
    summary: Open domain sign-off requests awaiting the current Engineer (addressed to them, or open to their domains)
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
              items: { $ref: '#/components/schemas/DomainSignoffRequest' }

/change-requests/{crId}/domain-signoffs:
  parameters: [{ name: crId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [DomainSignoffs]
    summary: List required and completed Cross-Domain Sign-Offs for a CR
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
              items: { $ref: '#/components/schemas/DomainSignoff' }
  post:
    tags: [DomainSignoffs]
    summary: Record a Cross-Domain Sign-Off (actor must hold the named Engineering Domain)
    requestBody:
```

```
      content:
        application/json:
          schema:
            type: object
            required: [engineering_domain]
            properties:
              engineering_domain: { type: string, example: CONTROLS }
    responses:
      '201':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/DomainSignoff' }
      '403': { description: Actor does not hold the named domain }

/users/{userId}/usage:
  parameters: [{ name: userId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Users]
    summary: What the account has done in the tool (the rows of every other table that point at it) and whether a delete
removes the account or erases it (Administrator, StCd Owner; BR-025)
    responses:
      '200': { description: Usage, content: { application/json: { schema: { $ref: '#/components/schemas/UserUsage' } } }
}
      '404': { description: User not found }

/users/{userId}/delete:
  parameters: [{ name: userId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [Users]
    summary: >
      Delete an account (Administrator, with the Administrator's own password). An account that
      nothing points at is removed; one with history is erased (name, address and password go, the
      row stays so that everything it did keeps its author) and the audit log is never changed
      (BR-025). Never the own account, never the last active Administrator.
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [current_password], properties: { current_password
: { type: string } } } } }
    responses:
      '204': { description: Deleted }
      '403': { description: Not an Administrator, the own account, or the password is missing or wrong }
      '404': { description: User not found }
      '422': { description: The last active Administrator cannot be deleted }

/change-requests/withdraw:
  post:
    tags: [ChangeRequests]
    summary: >
      Withdraw several of your own open change requests in one call (FR-159). Each goes through the
      single-request rules and has its own audit entry; one that cannot be withdrawn is reported in
      its result and the others go on. A withdrawn Draft is deleted; a code left without any revision
      that never received its identifier gives its reserved number and legacy number back (BR-026).
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [cr_ids], properties: { cr_ids: { type: array, ite
ms: { type: integer }, minItems: 1 } } } } }
    responses:
      '200':
        description: One result per request
        content:
          application/json:
            schema:
              type: object
              properties:
                results: { type: array, items: { type: object, properties: { cr_id: { type: integer }, ok: { type: boolea
n }, message: { type: string } } } }
                withdrawn: { type: integer }
                failed: { type: integer }
      '422': { description: cr_ids missing or not a list of ids }

/imports:
  get:
    tags: [Import]
    summary: List import jobs
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
              items: { $ref: '#/components/schemas/ImportJob' }
  post:
    tags: [Import]
    summary: >
      Upload a CSV or XLSX status code library file. Every row that passes validation
      becomes a normal Draft Change Request (cr_type=IMPORT) subject to the full
      dual-review-then-Chief-Engineer-approval cycle ? import is a faster data-entry
      path, never a governance bypass (05-governance-handbook.md §6b).
    requestBody:
```

```
      required: true
      content:
        multipart/form-data:
          schema:
            type: object
            required: [file]
            properties:
              file: { type: string, format: binary }
              mode: { type: string, enum: [CREATE, UPDATE], default: CREATE, description: "UPDATE matches rows on legacy_
reference_number within the Functional System Group and revises the code instead of creating a duplicate (I-05)" }
              dry_run: { type: boolean, default: false, description: "Validate only: 200 with the per-row report, nothing
 is created (I-05)" }
              async: { type: boolean, default: false, description: "Force background processing; files above IMPORT_ASYNC
_ROW_THRESHOLD rows get it automatically (I-32)" }
    responses:
      '200':
        description: Dry-run report (dry_run=true)
        content:
          application/json:
            schema: { $ref: '#/components/schemas/ImportDryRun' }
      '202':
        description: Import job accepted (small files are already COMPLETE; large ones are processed in the background -
poll GET /imports/{id})
        content:
          application/json:
            schema: { $ref: '#/components/schemas/ImportJob' }

/imports/{importJobId}:
  parameters: [{ name: importJobId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Import]
    summary: Get import job status, including created/error counts
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/ImportJob' }

/imports/{importJobId}/errors:
  parameters: [{ name: importJobId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Import]
    summary: Download the per-row error report (row, column, value, problem, how to fix) for a completed import job (CSV,
 or XLSX with format=xlsx)
    parameters: [{ name: format, in: query, schema: { type: string, enum: [csv, xlsx], default: csv } }]
    responses:
      '200':
        description: File stream (CSV)
        content:
          application/octet-stream: {}

/releases:
  get:
    tags: [Releases]
    summary: List releases
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
              items: { $ref: '#/components/schemas/Release' }
  post:
    tags: [Releases]
    summary: Create a Release (Administrator only)
    requestBody:
      content:
        application/json:
          schema: { $ref: '#/components/schemas/ReleaseCreate' }
    responses:
      '201':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/Release' }

/releases/{releaseId}:
  parameters: [{ name: releaseId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Releases]
    summary: Get release detail (incl. items)
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/Release' }
  delete:
    tags: [Releases]
    summary: Delete an unpublished Release (Administrator only; BR-008). A published Release can never be deleted.
    responses:
```

```
      '204': { description: Deleted }
      '409': { description: Release is already published }

/releases/{releaseId}/candidates:
  parameters: [{ name: releaseId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Releases]
    summary: List Approved (non-sandbox) revisions matching the release's scope filter
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
              items: { $ref: '#/components/schemas/StatusCodeRevision' }

/releases/{releaseId}/items:
  parameters: [{ name: releaseId, in: path, required: true, schema: { type: integer } }]
  put:
    tags: [Releases]
    summary: Set the included revision set (pre-publish only; sandbox revisions are rejected)
    requestBody:
      content:
        application/json:
          schema:
            type: object
            properties:
              status_code_revision_ids: { type: array, items: { type: integer } }
    responses:
      '200': { description: Updated }

/releases/{releaseId}/publish:
  parameters: [{ name: releaseId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [Releases]
    summary: >
      Publish the release (atomic; Administrator only). Needs `current_password` (I-17). The response carries
      `warnings` for items whose planned effective date is still in the future (I-27); they never block.
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [current_password], properties: { current_password
: { type: string, format: password } } } } }
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/Release' }

/releases/{releaseId}/export:
  parameters:
    - { name: releaseId, in: path, required: true, schema: { type: integer } }
    - { name: format, in: query, required: true, schema: { type: string, enum: [csv, json, xlsx, pdf] } }
    - { name: profile, in: query, schema: { type: string }, description: "Export profile code (I-02): fixes columns, orde
r, header labels, language and a default audience" }
    - { name: audience, in: query, schema: { type: string, enum: [development, sales, tcc, service, turbine_operator_pack
age, grid_operator, service_partner, customer] }, description: "Only codes whose confirmed access value lets this audience
see them; other audiences' and internal columns are dropped (I-03)" }
    - { name: language, in: query, schema: { type: string }, description: "Approved translations (I-15); untranslated row
s fall back to the source language and say so in content_language" }
    - { name: strict_language, in: query, schema: { type: boolean }, description: "With language: omit codes that have no
 approved translation" }
    - { name: software_version, in: query, schema: { type: string }, description: "Only codes that apply to this controll
er software version (I-07)" }
    - { name: bundle, in: query, schema: { type: boolean }, description: "Return a ZIP with the file and MANIFEST.json (S
HA-256 of every file, optional HMAC signature, I-06)" }
    - { name: content, in: query, schema: { type: string, enum: [baseline, delta], default: baseline }, description: "bas
eline (default): every code valid as of the Release; delta: only the revisions the Release carries (FR-173)" }
    - { name: platform, in: query, schema: { type: string, enum: ['2XM', '3XM', '4XM'] }, description: "Only codes for th
is turbine platform (FR-173)" }
    - { name: controller, in: query, schema: { type: string, enum: [MC, HC] }, description: "Only the Main or the Hub Con
troller codes (FR-173)" }
  get:
    tags: [Exports]
    summary: Export a release catalogue as a flat table (role-scoped, per 11-security-architecture.md §3). All formats, i
ncluding JSON, are flat row-per-status-code, not nested.
    responses:
      '200':
        description: File stream
        content:
          application/octet-stream: {}

/exports/released:
  parameters:
    - { name: format, in: query, required: true, schema: { type: string, enum: [csv, json, xlsx, pdf] } }
    - { name: functional_system_group, in: query, schema: { type: integer } }
    - { name: functional_subgroup, in: query, schema: { type: integer } }
    - { name: turbine_platform, in: query, schema: { type: string } }
    - { name: profile, in: query, schema: { type: string }, description: "Export profile code (I-02): fixes columns, orde
r, header labels, language and a default audience" }
```

```
    - { name: audience, in: query, schema: { type: string, enum: [development, sales, tcc, service, turbine_operator_pack
age, grid_operator, service_partner, customer] }, description: "Only codes whose confirmed access value lets this audience
see them; other audiences' and internal columns are dropped (I-03)" }
    - { name: language, in: query, schema: { type: string }, description: "Approved translations (I-15); untranslated row
s fall back to the source language and say so in content_language" }
    - { name: strict_language, in: query, schema: { type: boolean }, description: "With language: omit codes that have no
 approved translation" }
    - { name: software_version, in: query, schema: { type: string }, description: "Only codes that apply to this controll
er software version (I-07)" }
    - { name: bundle, in: query, schema: { type: boolean }, description: "Return a ZIP with the file and MANIFEST.json (S
HA-256 of every file, optional HMAC signature, I-06)" }
  get:
    tags: [Exports]
    security: [{ sessionCookie: [] }, { bearerToken: [] }]
    summary: Export the current set of all Released (non-sandbox) status codes as a flat table (any authenticated role; F
R-045)
    responses:
      '200':
        description: File stream
        content:
          application/octet-stream: {}

/exports/working-set:
  parameters:
    - { name: format, in: query, required: true, schema: { type: string, enum: [csv, json, xlsx, pdf] } }
    - { name: q, in: query, schema: { type: string } }
    - { name: functional_system_group, in: query, schema: { type: integer } }
    - { name: functional_subgroup, in: query, schema: { type: integer } }
    - { name: turbine_platform, in: query, schema: { type: string } }
    - { name: lifecycle_status, in: query, schema: { type: string } }
    - { name: profile, in: query, schema: { type: string } }
    - { name: language, in: query, schema: { type: string } }
    - { name: software_version, in: query, schema: { type: string } }
    - { name: bundle, in: query, schema: { type: boolean } }
  get:
    tags: [Exports]
    summary: >
      Ad-hoc filtered working set in any lifecycle state (never sandbox), for the StCd Owner and the
      Administrator only (FR-046, FR-175). Every row carries a `distribution_notice`
      column "Not for distribution - unapproved data"; the file name and PDF banner repeat it.
    responses:
      '200':
        description: File stream
        content:
          application/octet-stream: {}
      '403': { description: Viewer role }

/exports/full-database:
  post:
    tags: [Exports]
    summary: >
      Export a full raw database backup (Administrator only). A POST that carries `current_password` (I-17),
      so the download needs a fresh confirmation; it used to be a GET.
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [current_password], properties: { current_password
: { type: string, format: password } } } } }
    responses:
      '200':
        description: File stream (SQLite backup file)
        content:
          application/octet-stream: {}
      '403': { description: Actor is not Administrator, or the step-up password is missing or wrong }

/lookups/{vocabulary}:
  parameters:
    - name: vocabulary
      in: path
      required: true
      schema:
        type: string
        enum: [functional-system-groups, functional-subgroups, turbine-platforms,
               engineering-domains, available-groups, development-access, sales-access,
               tcc-access, service-access, turbine-operator-package-access,
               grid-operator-access, service-partner-access, customer-access]
    - name: functional_system_group
      in: query
      description: Required when vocabulary=functional-subgroups, to scope the returned subgroups
      schema: { type: string }
  get:
    tags: [Lookups]
    summary: List active values of a controlled vocabulary. Each value includes its description field, surfaced by the UI
 as inline help text next to every dropdown option (addresses "no explanations given on options").
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
```

```
              items: { $ref: '#/components/schemas/LookupValue' }
  post:
    tags: [Lookups]
    summary: Add a value to a simple controlled vocabulary (Administrator only; functional groups/subgroups have their ow
n endpoints)
    requestBody:
      content:
        application/json:
          schema:
            type: object
            required: [code, label]
            properties:
              code: { type: string }
              label: { type: string }
              description: { type: string, nullable: true }
    responses:
      '201':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/LookupValue' }
      '409': { description: Duplicate code }
      '422': { $ref: '#/components/responses/ValidationError' }

/lookups/{vocabulary}/{valueId}:
  parameters:
    - { name: vocabulary, in: path, required: true, schema: { type: string } }
    - { name: valueId, in: path, required: true, schema: { type: integer } }
  patch:
    tags: [Lookups]
    summary: Relabel, describe (inline-help text), (de)activate or reorder a vocabulary value (Administrator only; values
 are never deleted)
    requestBody:
      content:
        application/json:
          schema:
            type: object
            properties:
              label: { type: string }
              description: { type: string, nullable: true }
              is_active: { type: boolean }
              sort_order: { type: integer }
    responses:
      '200':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/LookupValue' }
      '404': { $ref: '#/components/responses/NotFound' }

/lookups/functional-system-groups:
  post:
    tags: [Lookups]
    summary: >
      Create a brand-new Functional System Group (Administrator only), into any currently
      free range (10000-10999, 14000-14999, 15000-15999, 16000-16999 as of 2026-09-17).
      The proposed range must not overlap any existing group; validated at the service
      layer with a helpful error naming the conflicting group, and as a backstop by a
      database trigger (07-database-design.md §5).
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            required: [code, label, range_start, range_end]
            properties:
              code: { type: string, example: WNEW }
              label: { type: string, example: New Subsystem }
              range_start: { type: integer, example: 15000 }
              range_end: { type: integer, example: 15999 }
    responses:
      '201':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/LookupValue' }
      '409': { description: Proposed range overlaps an existing Functional System Group }

/lookups/functional-subgroups:
  post:
    tags: [Lookups]
    summary: >
      Create a new Functional Subgroup (Administrator only). The proposed sub-range must
      not overlap any existing subgroup in the same Functional System Group; validated at
      the service layer with a helpful error naming the conflicting subgroup, and as a
      backstop by a database trigger (07-database-design.md §5).
    requestBody:
      required: true
      content:
        application/json:
          schema:
```

```
            type: object
            required: [functional_system_group, code, label, sub_range_start, sub_range_end]
            properties:
              functional_system_group: { type: string, example: WROT }
              code: { type: string, example: 045xx }
              label: { type: string, example: Pitch Actuator Diagnostics }
              sub_range_start: { type: integer, example: 4500 }
              sub_range_end: { type: integer, example: 4599 }
    responses:
      '201':
        content:
          application/json:
            schema: { $ref: '#/components/schemas/LookupValue' }
      '409': { description: Proposed sub-range overlaps an existing subgroup }

/status-codes/next-available:
  get:
    tags: [StatusCodes]
    summary: Preview (non-binding) the next available identifier for a Functional Group + Subgroup, for UI display only ?
 not allocated until Submit for Review
    parameters:
      - { name: functional_system_group, in: query, required: true, schema: { type: string } }
      - { name: functional_subgroup, in: query, required: true, schema: { type: string } }
    responses:
      '200':
        content:
          application/json:
            schema:
              type: object
              properties:
                preview_identifier: { type: string, example: StCd-01233 }
                capacity: { type: object, description: "I-19: range_start, range_end, total, allocated, highest_allocated
, remaining, percent_remaining of the subgroup's number band" }
                note: { type: string, example: "Not allocated until Submit for Review; another submission may claim this
number first." }

/audit-log:
  get:
    tags: [AuditLog]
    summary: Search audit log (Administrator; Reviewer/Chief Engineer read-only subset)
    parameters:
      - { $ref: '#/components/parameters/Page' }
      - { $ref: '#/components/parameters/PageSize' }
      - { name: entity_type, in: query, schema: { type: string } }
      - { name: entity_id, in: query, schema: { type: integer } }
      - { name: actor_id, in: query, schema: { type: integer } }
      - { name: action, in: query, schema: { type: string }, description: "Exact action code, e.g. REVIEWER_SIGNOFF, COMM
ENT, WITHDRAW" }
      - { name: date_from, in: query, schema: { type: string, format: date-time } }
      - { name: date_to, in: query, schema: { type: string, format: date-time } }
    responses:
      '200':
        content:
          application/json:
            schema:
              type: array
              items: { $ref: '#/components/schemas/AuditLogEntry' }

# ============================================================================================
# v4 (2026-10-22) - improvement pack I-01..I-36, docs/21-improvement-ideas.md
# ============================================================================================

/health/ready:
  get:
    tags: [Operations]
    summary: >
      Readiness probe (no authentication, I-21): database reachable, schema at the expected Alembic
      head, newest audit rows chain correctly, last verified backup recent (I-09). 200 when ready
      (status ok or degraded), 503 when not.
    security: []
    responses:
      '200': { description: Ready, content: { application/json: { schema: { $ref: '#/components/schemas/Readiness' } } }
}
      '503': { description: Not ready, content: { application/json: { schema: { $ref: '#/components/schemas/Readiness' }
} } }

/status-codes/similar:
  get:
    tags: [StatusCodes]
    summary: Existing codes of a subgroup with a similar title ("did you mean to revise ...?", I-12). Engineer, Administr
ator.
    parameters:
      - { name: functional_subgroup, in: query, required: true, schema: { type: integer } }
      - { name: title, in: query, required: true, schema: { type: string } }
      - { name: exclude, in: query, schema: { type: integer }, description: Status code id to leave out }
    responses:
      '200': { description: Similar codes, content: { application/json: { schema: { type: object, properties: { items: {
type: array, items: { $ref: '#/components/schemas/SimilarCode' } } } } } } }
      '422': { $ref: '#/components/responses/ValidationError' }
```

```
/status-codes/{statusCodeId}/where-used:
  parameters: [{ $ref: '#/components/parameters/StatusCodeId' }]
  get:
    tags: [StatusCodes]
    summary: Every other code whose current revision refers to this one (repeated-error pointer, related codes, supersess
ion, platform variant) - I-04. Not available to Viewers.
    responses:
      '200': { description: Dependants, content: { application/json: { schema: { type: object, properties: { items: { typ
e: array, items: { $ref: '#/components/schemas/WhereUsedEntry' } } } } } } }
      '404': { $ref: '#/components/responses/NotFound' }

/change-sets/{changeSetId}/readiness:
  parameters: [{ name: changeSetId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [ChangeSets]
    summary: What still stops each Draft of the batch from being submitted, with the rule of the sample inspection (FR-18
0). Not available to Viewers.
    responses:
      '200': { description: Readiness, content: { application/json: { schema: { type: object, properties: { tally: { type
: object }, pending_by_domain: { type: object }, rules: { type: object }, items: { type: array, items: { type: object } } }
 } } } } }

/change-sets/{changeSetId}/bulk-edit:
  parameters: [{ name: changeSetId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [ChangeSets]
    summary: Set the same status category, platforms, available group or access levels on the Drafts of the batch (FR-180
); each Draft is saved on its own, only by its author or an Administrator.
    requestBody: { required: true, content: { application/json: { schema: { type: object, properties: { change_request_id
s: { type: array, items: { type: integer } }, changes: { type: object } } } } } }
    responses:
      '200': { description: One result per Draft, content: { application/json: { schema: { type: object, properties: { re
sults: { type: array, items: { type: object } } } } } } }
      '422': { $ref: '#/components/responses/ValidationError' }

/change-sets/{changeSetId}/bulk-submit:
  parameters: [{ name: changeSetId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [ChangeSets]
    summary: Submit for review the Drafts of the batch that the caller wrote (FR-180); one that is not ready is reported
with the reason.
    responses:
      '200': { description: One result per Draft, content: { application/json: { schema: { type: object, properties: { re
sults: { type: array, items: { type: object } } } } } } }

/change-sets/{changeSetId}/domain-signoffs:
  parameters: [{ name: changeSetId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [ChangeSets]
    summary: Sign off an engineering domain on the Drafts of the batch (FR-180); more than a handful of codes need a pass
ed `sample_id`.
    requestBody: { required: true, content: { application/json: { schema: { type: object, required: [domain], properties:
 { domain: { type: string }, sample_id: { type: integer }, change_request_ids: { type: array, items: { type: integer } } }
} } } }
    responses:
      '200': { description: One result per Draft, content: { application/json: { schema: { type: object, properties: { re
sults: { type: array, items: { type: object } } } } } } }
      '422': { $ref: '#/components/responses/ValidationError' }

/change-sets/{changeSetId}/samples:
  parameters: [{ name: changeSetId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [ChangeSets]
    summary: Draw the sample a bulk decision at `step` needs (FR-180) - stratified by subgroup, drawn for the caller, val
id 24 hours.
    requestBody: { required: true, content: { application/json: { schema: { type: object, required: [step], properties: {
 step: { type: string, enum: [domain, reviewer, admin, chief-engineer] }, domain: { type: string }, change_request_ids: { t
ype: array, items: { type: integer } } } } } } }
    responses:
      '201': { description: The sample, content: { application/json: { schema: { $ref: '#/components/schemas/BulkSample'
} } } }
      '422': { $ref: '#/components/responses/ValidationError' }
  get:
    tags: [ChangeSets]
    summary: The samples drawn for the batch, newest first (FR-180).
    responses:
      '200': { description: Samples, content: { application/json: { schema: { type: object, properties: { items: { type:
array, items: { $ref: '#/components/schemas/BulkSample' } } } } } } }

/change-sets/{changeSetId}/dossier.xlsx:
  parameters: [{ name: changeSetId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [ChangeSets]
    summary: The change set as an XLSX dossier for an audit - codes with their decisions, samples with their verdicts, th
e audit entries (FR-180).
    responses:
      '200': { description: Workbook, content: { application/vnd.openxmlformats-officedocument.spreadsheetml.sheet: {} }
}
```

```
/bulk-samples/{sampleId}:
  parameters: [{ name: sampleId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [ChangeSets]
    summary: One sample with its sampled codes and their verdicts (FR-180).
    responses:
      '200': { description: The sample, content: { application/json: { schema: { $ref: '#/components/schemas/BulkSample'
} } } }
      '404': { $ref: '#/components/responses/NotFound' }

/bulk-samples/{sampleId}/items/{changeRequestId}:
  parameters: [{ name: sampleId, in: path, required: true, schema: { type: integer } }, { name: changeRequestId, in: path
, required: true, schema: { type: integer } }]
  put:
    tags: [ChangeSets]
    summary: Mark a sampled code OK or DEFECT (a defect needs a note); only by the person who drew the sample (FR-180).
    requestBody: { required: true, content: { application/json: { schema: { type: object, required: [verdict], properties
: { verdict: { type: string, enum: [OK, DEFECT] }, note: { type: string } } } } } }
    responses:
      '200': { description: The sample, content: { application/json: { schema: { $ref: '#/components/schemas/BulkSample'
} } } }
      '422': { $ref: '#/components/responses/ValidationError' }

/bulk-samples/{sampleId}/reject-defects:
  parameters: [{ name: sampleId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [ChangeSets]
    summary: Send back the codes of a failed sample that were found defective, with the note as the reason, and close the
 sample (FR-180). Needs the password of the caller.
    requestBody: { content: { application/json: { schema: { type: object, properties: { current_password: { type: string
}, comment: { type: string } } } } } }
    responses:
      '200': { description: One result per defective code, content: { application/json: { schema: { type: object, propert
ies: { results: { type: array, items: { type: object } } } } } } }
      '422': { $ref: '#/components/responses/ValidationError' }

/status-codes/{statusCodeId}/connections:
  parameters: [{ $ref: '#/components/parameters/StatusCodeId' }]
  get:
    tags: [StatusCodes]
    summary: What suppresses this code, what it suppresses, what it requires and where it escalates, read from the import
ed programs, with the chain of causes above it (FR-179). Not available to Viewers.
    responses:
      '200': { description: Connections, content: { application/json: { schema: { type: object, properties: { outgoing: {
 type: array, items: { type: object } }, incoming: { type: array, items: { type: object } }, causes: { type: array, items:
{ type: object } }, escalation: { type: array, items: { type: object } }, known: { type: boolean } } } } } }
      '404': { $ref: '#/components/responses/NotFound' }

/status-codes/connections:
  get:
    tags: [StatusCodes]
    summary: The causes with the codes they suppress (a tree), the escalation targets and what looks wrong in the connect
ions between codes (FR-179). Not available to Viewers.
    parameters:
      - { name: controller, in: query, schema: { type: string, enum: [MC, HC] } }
    responses:
      '200': { description: Hierarchy, content: { application/json: { schema: { type: object, properties: { trees: { type
: array, items: { type: object } }, escalation: { type: array, items: { type: object } }, findings: { type: array, items: {
 type: object } }, counts: { type: object, properties: { SUPPRESSED_BY: { type: integer }, REQUIRES: { type: integer }, ESC
ALATES_TO: { type: integer } } }, codes_with_connections: { type: integer } } } } } }
      '422': { $ref: '#/components/responses/ValidationError' }

/status-codes/{statusCodeId}/family:
  parameters: [{ $ref: '#/components/parameters/StatusCodeId' }]
  get:
    tags: [StatusCodes]
    summary: The code and its platform variants (I-25)
    responses:
      '200': { description: Family members, root first, content: { application/json: { schema: { type: object, properties
: { items: { type: array, items: { $ref: '#/components/schemas/FamilyMember' } } } } } } }

/status-codes/{statusCodeId}/platform-variants:
  parameters: [{ $ref: '#/components/parameters/StatusCodeId' }]
  post:
    tags: [StatusCodes]
    summary: Copy the latest decided revision into a new Draft for another turbine platform, linked to the source (I-25).
 Engineer, Administrator.
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [turbine_platform], properties: { turbine_platform
: { type: string, example: 4XM } } } } }
    responses:
      '201': { description: The new Draft revision, content: { application/json: { schema: { $ref: '#/components/schemas/
StatusCodeRevision' } } } }
      '409': { description: Source has no Approved/Released revision }
      '422': { $ref: '#/components/responses/ValidationError' }

/status-codes/{statusCodeId}/planned-dates:
```

```
  parameters: [{ $ref: '#/components/parameters/StatusCodeId' }]
  put:
    tags: [StatusCodes]
    summary: Set planned effective / deprecation dates on the current revision (I-27, Administrator). Audited as PLANNED_
DATES_SET.
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, properties: { planned_effective_date: { type: string, format
: date, nullable: true }, planned_deprecation_date: { type: string, format: date, nullable: true } } } } }
    responses:
      '200': { description: Updated revision, content: { application/json: { schema: { $ref: '#/components/schemas/Status
CodeRevision' } } } }
      '422': { $ref: '#/components/responses/ValidationError' }

/status-codes/{statusCodeId}/mc-drafts:
  parameters: [{ $ref: '#/components/parameters/StatusCodeId' }]
  post:
    tags: [Hub]
    summary: Create a Main Controller Draft from a Hub Controller code (I-36). The new MC number is allocated at submit;
the HC identifier becomes the legacy reference.
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [functional_system_group, functional_subgroup], pr
operties: { functional_system_group: { type: integer }, functional_subgroup: { type: integer } } } } }
    responses:
      '201': { description: The new Draft revision, content: { application/json: { schema: { $ref: '#/components/schemas/
StatusCodeRevision' } } } }
      '409': { description: Already mapped, or the Hub code is not Approved/Released }
      '422': { $ref: '#/components/responses/ValidationError' }

/hub-consolidation/mapping:
  get:
    tags: [Hub]
    summary: Every Hub Controller code with its Main Controller replacement and how far that is (I-36)
    parameters: [{ name: group, in: query, schema: { type: integer }, description: Hub Functional System Group id }]
    responses:
      '200': { description: Mapping report, content: { application/json: { schema: { $ref: '#/components/schemas/HubMappi
ngReport' } } } }

/hub-consolidation/deprecation-requests:
  post:
    tags: [Hub]
    summary: Open a deprecation request for every Released Hub code of a group whose MC replacement is Released (I-36, Ad
ministrator). Each is an ordinary DEPRECATION CR.
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [functional_system_group], properties: { functiona
l_system_group: { type: integer } } } } }
    responses:
      '200': { description: Created and skipped codes, content: { application/json: { schema: { type: object, properties:
 { created: { type: array, items: { type: string } }, skipped: { type: array, items: { type: object, properties: { hub_iden
tifier: { type: string }, reason: { type: string } } } } } } } } } }

/change-requests/stalled:
  get:
    tags: [ChangeRequests]
    summary: Requests waiting past their amber/red SLA threshold (I-08, Administrator)
    responses:
      '200': { description: Stalled requests and the thresholds, content: { application/json: { schema: { type: object, p
roperties: { items: { type: array, items: { $ref: '#/components/schemas/StalledRequest' } }, thresholds: { type: object } }
 } } } }

/change-requests/{crId}/sla:
  parameters: [{ $ref: '#/components/parameters/CrId' }]
  get:
    tags: [ChangeRequests]
    summary: How long the request has waited at its stage, in business days, and its SLA level (I-08)
    responses:
      '200': { description: SLA info or null when not waiting, content: { application/json: { schema: { type: object, pro
perties: { sla: { $ref: '#/components/schemas/SlaInfo' } } } } } }

/change-requests/{crId}/reviewer-assignment/auto:
  parameters: [{ $ref: '#/components/parameters/CrId' }]
  post:
    tags: [ChangeRequests]
    summary: Assign the least-loaded available Reviewer (I-30, Administrator). Reviewers who are out of office are skippe
d.
    responses:
      '200': { description: Updated change request, content: { application/json: { schema: { $ref: '#/components/schemas/
ChangeRequest' } } } }
      '422': { $ref: '#/components/responses/ValidationError' }

/reviewers/workload:
  get:
    tags: [ChangeRequests]
    summary: Open reviews per active, available Reviewer (I-30, Administrator)
    responses:
      '200': { description: Workload, content: { application/json: { schema: { type: object, properties: { items: { type:
 array, items: { type: object, properties: { user_id: { type: integer }, full_name: { type: string }, open_reviews: { type:
```

```
 integer } } } } } } } }

/change-requests/{crId}/field-comments:
  parameters: [{ $ref: '#/components/parameters/CrId' }]
  get:
    tags: [Review]
    summary: Comments attached to single fields of the revision (I-23). Not available to Viewers.
    responses:
      '200': { description: Field comments, content: { application/json: { schema: { type: object, properties: { items: {
 type: array, items: { $ref: '#/components/schemas/FieldComment' } } } } } } }
  post:
    tags: [Review]
    summary: Comment on one field while the CR is in Review or Pending Approval (I-23). Reviewer, Administrator, Chief En
gineer (or a delegate).
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [field, comment], properties: { field: { type: str
ing, example: set_delay }, comment: { type: string } } } } }
    responses:
      '201': { description: Created, content: { application/json: { schema: { $ref: '#/components/schemas/FieldComment' }
 } } }
      '409': { description: CR is not in Review / Pending Approval }
      '422': { $ref: '#/components/responses/ValidationError' }

/field-comments/{commentId}/resolve:
  parameters: [{ name: commentId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [Review]
    summary: Mark a field comment resolved (the commenter, the CR author or an Administrator)
    responses:
      '200': { description: Resolved comment, content: { application/json: { schema: { $ref: '#/components/schemas/FieldC
omment' } } } }

/review-checklist-items:
  get:
    tags: [Review]
    summary: Active review-checklist items (I-23); Administrators may add include_inactive=true
    parameters: [{ name: include_inactive, in: query, schema: { type: boolean } }]
    responses:
      '200': { description: Items, content: { application/json: { schema: { type: object, properties: { items: { type: ar
ray, items: { $ref: '#/components/schemas/ChecklistItem' } } } } } } }
  post:
    tags: [Review]
    summary: Add a checklist item for one Functional System Group or all (Administrator)
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [text], properties: { text: { type: string }, func
tional_system_group: { type: integer, nullable: true }, sort_order: { type: integer } } } } }
    responses:
      '201': { description: Created, content: { application/json: { schema: { $ref: '#/components/schemas/ChecklistItem'
} } } }

/review-checklist-items/{itemId}:
  parameters: [{ name: itemId, in: path, required: true, schema: { type: integer } }]
  patch:
    tags: [Review]
    summary: Edit or (de)activate a checklist item (Administrator)
    requestBody:
      content: { application/json: { schema: { type: object, properties: { text: { type: string }, is_active: { type: boo
lean }, sort_order: { type: integer } } } } }
    responses:
      '200': { description: Updated, content: { application/json: { schema: { $ref: '#/components/schemas/ChecklistItem'
} } } }

/change-requests/{crId}/review-checklist:
  parameters: [{ $ref: '#/components/parameters/CrId' }]
  get:
    tags: [Review]
    summary: The checklist items that apply to the CR and the confirmations already recorded (I-23)
    responses:
      '200': { description: Items and confirmations, content: { application/json: { schema: { type: object, properties: {
 items: { type: array, items: { $ref: '#/components/schemas/ChecklistItem' } }, confirmations: { type: array, items: { type
: object, properties: { checklist_item_id: { type: integer }, item_text: { type: string }, confirmed_by: { type: integer },
 confirmed_at: { type: string, format: date-time } } } } } } } } }

/change-requests/{crId}/attachments:
  parameters: [{ $ref: '#/components/parameters/CrId' }]
  get:
    tags: [Evidence]
    summary: Evidence attached to the CR (I-13). Working roles only.
    responses:
      '200': { description: Attachments, content: { application/json: { schema: { type: object, properties: { items: { ty
pe: array, items: { $ref: '#/components/schemas/Attachment' } } } } } } }
  post:
    tags: [Evidence]
    summary: >
      Attach evidence (author or Administrator, CR open). Allow-listed types (pdf, png, jpg, txt, log, md, csv, xlsx, doc
x),
      content-checked against the type, at most 10 MB and 20 files per CR (SEC-024). Stored outside the web root.
```

```
    requestBody:
      required: true
      content: { multipart/form-data: { schema: { type: object, required: [file], properties: { file: { type: string, for
mat: binary }, description: { type: string } } } } }
    responses:
      '201': { description: Attached, content: { application/json: { schema: { $ref: '#/components/schemas/Attachment' }
} } }
      '422': { $ref: '#/components/responses/ValidationError' }

/attachments/{attachmentId}:
  parameters: [{ name: attachmentId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Evidence]
    summary: Download evidence as an attachment (never rendered in the browser). Working roles only.
    responses:
      '200': { description: File stream, headers: { X-Content-SHA256: { schema: { type: string } } }, content: { applicat
ion/octet-stream: {} } }
  delete:
    tags: [Evidence]
    summary: Remove evidence while the CR is still a Draft (uploader or Administrator)
    responses:
      '204': { description: Removed }

/change-sets:
  get:
    tags: [ChangeSets]
    summary: Change sets - groups of CRs shipping with one controller software release (I-11)
    responses:
      '200': { description: Change sets, content: { application/json: { schema: { type: object, properties: { items: { ty
pe: array, items: { $ref: '#/components/schemas/ChangeSet' } } } } } } }
  post:
    tags: [ChangeSets]
    summary: Create a change set (Administrator, Engineer, Chief Engineer)
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [name], properties: { name: { type: string }, desc
ription: { type: string }, target_software_version: { type: string, nullable: true } } } } }
    responses:
      '201': { description: Created, content: { application/json: { schema: { $ref: '#/components/schemas/ChangeSet' } }
} }

/change-sets/{changeSetId}:
  parameters: [{ $ref: '#/components/parameters/ChangeSetId' }]
  get:
    tags: [ChangeSets]
    summary: A change set with its CRs, their states and SLA levels
    responses:
      '200': { description: Overview, content: { application/json: { schema: { $ref: '#/components/schemas/ChangeSetOverv
iew' } } } }

/change-sets/{changeSetId}/change-requests:
  parameters: [{ $ref: '#/components/parameters/ChangeSetId' }]
  post:
    tags: [ChangeSets]
    summary: Add an open CR to the set (its author or an Administrator)
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [change_request_id], properties: { change_request_
id: { type: integer } } } } }
    responses:
      '200': { description: Updated set, content: { application/json: { schema: { $ref: '#/components/schemas/ChangeSet'
} } } }

/change-sets/{changeSetId}/change-requests/{crId}:
  parameters: [{ $ref: '#/components/parameters/ChangeSetId' }, { $ref: '#/components/parameters/CrId' }]
  delete:
    tags: [ChangeSets]
    summary: Take a CR out of the set
    responses:
      '204': { description: Removed }

/change-sets/{changeSetId}/close:
  parameters: [{ $ref: '#/components/parameters/ChangeSetId' }]
  post:
    tags: [ChangeSets]
    summary: Close the set (Administrator)
    responses:
      '200': { description: Closed, content: { application/json: { schema: { $ref: '#/components/schemas/ChangeSet' } } }
 }

/change-sets/{changeSetId}/decisions:
  parameters: [{ $ref: '#/components/parameters/ChangeSetId' }]
  post:
    tags: [ChangeSets]
    summary: >
      Decide several CRs of the set in one sitting. Each CR is decided by its own single-CR service call:
      its own segregation-of-duties check, comment and audit entry. The password is verified once (I-17).
    requestBody:
      required: true
```

```
      content: { application/json: { schema: { $ref: '#/components/schemas/BulkDecision' } } }
    responses:
      '200': { description: One result per CR, content: { application/json: { schema: { type: object, properties: { resul
ts: { type: array, items: { type: object, properties: { cr_id: { type: integer }, ok: { type: boolean }, message: { type: s
tring } } } } } } } } }
      '403': { description: Step-up password missing or wrong }

/delegations:
  get:
    tags: [Delegation]
    summary: Delegations you gave or received (Administrators see all). include_inactive=true adds ended and revoked ones
.
    parameters: [{ name: include_inactive, in: query, schema: { type: boolean } }]
    responses:
      '200': { description: Delegations, content: { application/json: { schema: { type: object, properties: { items: { ty
pe: array, items: { $ref: '#/components/schemas/Delegation' } } } } } } }
  post:
    tags: [Delegation]
    summary: >
      Name a delegate for a date range (I-16). Reviewer, Administrator and Chief Engineer delegate their own
      decision role; an Administrator may create one for anybody. The delegate acts as themselves; the
      audit shows both and segregation of duties still applies to the delegate.
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [delegate_id, valid_from, valid_to], properties: {
 delegator_id: { type: integer, description: Defaults to the caller }, delegate_id: { type: integer }, valid_from: { type:
string, format: date }, valid_to: { type: string, format: date } } } } }
    responses:
      '201': { description: Created, content: { application/json: { schema: { $ref: '#/components/schemas/Delegation' } }
 } }
      '422': { $ref: '#/components/responses/ValidationError' }

/delegations/{delegationId}:
  parameters: [{ name: delegationId, in: path, required: true, schema: { type: integer } }]
  delete:
    tags: [Delegation]
    summary: Revoke a delegation (the delegator or an Administrator)
    responses:
      '204': { description: Revoked }

/notifications:
  get:
    tags: [Notifications]
    summary: Your in-app notices (I-08), newest first
    parameters: [{ name: unread, in: query, schema: { type: boolean } }]
    responses:
      '200': { description: Notices and the unread count, content: { application/json: { schema: { type: object, properti
es: { items: { type: array, items: { $ref: '#/components/schemas/Notification' } }, unread: { type: integer } } } } } }

/notifications/read-all:
  post:
    tags: [Notifications]
    summary: Mark all your notices as read
    responses:
      '200': { description: Count updated, content: { application/json: { schema: { type: object, properties: { updated:
{ type: integer } } } } } }

/notifications/{notificationId}/read:
  parameters: [{ name: notificationId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [Notifications]
    summary: Mark one notice as read
    responses:
      '200': { description: Updated, content: { application/json: { schema: { $ref: '#/components/schemas/Notification' }
 } } }

/settings:
  get:
    tags: [Settings]
    summary: Administrator-managed settings (SLA thresholds, reviewer auto-assignment, capacity alert level) with default
s
    responses:
      '200': { description: Settings, content: { application/json: { schema: { type: object, properties: { items: { type:
 array, items: { $ref: '#/components/schemas/Setting' } } } } } } }
  put:
    tags: [Settings]
    summary: Change settings (Administrator). Every value is validated; amber must not exceed red.
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, additionalProperties: { type: string }, example: { 'sla.revi
ew.amber_days': '2' } } } }
    responses:
      '200': { description: All settings, content: { application/json: { schema: { type: object, properties: { items: { t
ype: array, items: { $ref: '#/components/schemas/Setting' } } } } } } }
      '422': { $ref: '#/components/responses/ValidationError' }

/reports/catalogue-health:
  get:
    tags: [Reports]
```

```
    summary: Catalogue health and number-range capacity forecast (I-19). Administrator, Reviewer, Chief Engineer.
    responses:
      '200': { description: Report, content: { application/json: { schema: { $ref: '#/components/schemas/CatalogueHealth'
 } } } }

/reports/kpis:
  get:
    tags: [Reports]
    summary: The business-case KPIs - approval time, cross-domain turnaround, audit-trail completeness, queue ages, owner
less codes (I-10). Administrator, Chief Engineer.
    responses:
      '200': { description: KPIs, content: { application/json: { schema: { type: object } } } }

/reports/occurrences:
  get:
    tags: [Reports]
    summary: Codes that never fire and the most frequent ones, from imported occurrence counts (I-28). Administrator, Chi
ef Engineer.
    parameters:
      - { name: top, in: query, schema: { type: integer, default: 20 } }
      - { name: min_periods, in: query, schema: { type: integer, default: 1 } }
    responses:
      '200': { description: Report, content: { application/json: { schema: { type: object } } } }

/occurrence-imports:
  get:
    tags: [Reports]
    summary: Imported occurrence batches (I-28)
    responses:
      '200': { description: Batches, content: { application/json: { schema: { type: object, properties: { items: { type:
array, items: { $ref: '#/components/schemas/OccurrenceImport' } } } } } } }
  post:
    tags: [Reports]
    summary: >
      Import occurrence counts per status code for a period (I-28): CSV/XLSX with the columns
      status_code_identifier and occurrences, aggregated and anonymised. Administrator, Chief Engineer.
    requestBody:
      required: true
      content: { multipart/form-data: { schema: { type: object, required: [file, source_label, period_start, period_end],
 properties: { file: { type: string, format: binary }, source_label: { type: string }, period_start: { type: string, format
: date }, period_end: { type: string, format: date } } } } }
    responses:
      '201': { description: Imported, content: { application/json: { schema: { $ref: '#/components/schemas/OccurrenceImpo
rt' } } } }
      '422': { $ref: '#/components/responses/ValidationError' }

/operations/backup-status:
  get:
    tags: [Operations]
    summary: State and age of the last verified backup (I-09, Administrator)
    responses:
      '200': { description: Backup status, content: { application/json: { schema: { type: object } } } }

/operations/audit-verify:
  post:
    tags: [Operations]
    summary: Walk the audit-log hash chain now (I-18, Administrator). `ok:false` names the first bad row.
    responses:
      '200': { description: Verification result, content: { application/json: { schema: { $ref: '#/components/schemas/Aud
itChainResult' } } } }

/imports/template:
  get:
    tags: [Import]
    summary: Download the import template - exact headers (CSV) or headers, instructions and the lookup codes (XLSX) (I-0
5)
    parameters: [{ name: format, in: query, schema: { type: string, enum: [csv, xlsx], default: xlsx } }]
    responses:
      '200': { description: File stream, content: { application/octet-stream: {} } }

/imports/reconciliation:
  get:
    tags: [Import]
    summary: Migration progress - legacy rows imported, in review, approved, released, retired and (with expected_total)
still missing (I-05)
    parameters: [{ name: expected_total, in: query, schema: { type: integer } }]
    responses:
      '200': { description: Reconciliation report, content: { application/json: { schema: { type: object } } } }
  post:
    tags: [Import]
    summary: Same report with the legacy master list uploaded so the missing legacy numbers are named
    requestBody:
      required: true
      content: { multipart/form-data: { schema: { type: object, required: [file], properties: { file: { type: string, for
mat: binary, description: "CSV/XLSX with a legacy_reference_number column" } } } } }
    responses:
      '200': { description: Reconciliation report, content: { application/json: { schema: { type: object } } } }

/imports/st/preview:
```

```
  post:
    tags: [Import]
    summary: >
      Status code import from generated StCd_<n>.st programs, step 1 (v4.2): upload several
      files or a ZIP; nothing is created. Every program is listed with its state (ready,
      exists, blocked), its legacy -> new number, the subgroup its number lies in and its
      problems; the `token` is the staged upload for step 2. Hub Controller programs carry the
      legacy HC number; the new number comes from the numbering table (proposal 1 or 2).
    requestBody:
      required: true
      content:
        multipart/form-data:
          schema:
            type: object
            required: [files]
            properties:
              files: { type: array, items: { type: string, format: binary }, description: "StCd_<n>.st files, or one .zip
 holding them" }
              controller: { type: string, enum: [auto, MC, HC], default: auto, description: "auto: a file number from 100
00 up is a Hub Controller program" }
              platforms: { type: string, description: "Turbine platform codes, comma separated (default: all active platf
orms)" }
    responses:
      '200':
        description: Preview
        content:
          application/json:
            schema: { $ref: '#/components/schemas/StImportPreview' }
      '422': { description: No usable file, or an invalid option }

/imports/st:
  post:
    tags: [Import]
    summary: >
      Status code import from generated StCd_<n>.st programs, step 2 (v4.2): create the Drafts
      for the chosen programs of a preview. Each becomes an ordinary Draft status code with its
      change request (cr_type=IMPORT), keeps its number as a reservation (the identifier issued
      at Submit for Review is exactly that number), and all of them join one new change set.
      Import is a faster data-entry path, never a governance bypass (FR-048c).
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            required: [token]
            properties:
              token: { type: string, description: "From the preview" }
              selected: { type: array, nullable: true, items: { type: string }, description: "Item keys to import; omitte
d or null = every item that is ready" }
              change_set_name: { type: string, description: "Default: ST import <job> (<date>)" }
              async: { type: boolean, default: false, description: "Force background processing; more than IMPORT_ASYNC_R
OW_THRESHOLD programs get it automatically" }
    responses:
      '202':
        description: Import job accepted (`change_set_id` names the batch change set)
        content:
          application/json:
            schema: { $ref: '#/components/schemas/ImportJob' }
      '404': { description: The preview is unknown or has expired }
      '422': { description: Nothing selected, or a selected program is no longer ready }

/status-codes/{statusCodeId}/import-source:
  parameters: [{ name: statusCodeId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Import]
    summary: Where an imported status code came from - file, SHA-256, program and every attribute parsed from it, includi
ng those without a catalogue column yet (v4.2)
    responses:
      '200': { description: Provenance, content: { application/json: { schema: { type: object } } } }
      '404': { description: The code was not imported from a program file }

/legacy-mapping:
  get:
    tags: [Import]
    summary: >
      Legacy number -> new identifier for every code that carries a legacy number (v4.2).
      Legacy to new only: the new library is larger than the old one, so there is no reverse
      lookup; `summary.new_without_legacy` counts the codes with no legacy number.
    parameters:
      - { name: controller, in: query, schema: { type: string, enum: [MC, HC] } }
      - { name: q, in: query, schema: { type: string }, description: "Matches legacy number, identifier or title" }
      - { name: state, in: query, schema: { type: string, enum: [issued, reserved, pending] }, description: "issued = ide
ntifier assigned; reserved = number held by an imported Draft; pending = no number yet" }
    responses:
      '200': { description: Mapping, content: { application/json: { schema: { $ref: '#/components/schemas/LegacyMapping'
} } } }
      '422': { description: Invalid filter }
```

```
/legacy-mapping/export:
  get:
    tags: [Import]
    summary: The legacy -> new mapping as CSV (default) or XLSX, with the filters of the list (v4.2)
    parameters:
      - { name: format, in: query, schema: { type: string, enum: [csv, xlsx], default: csv } }
      - { name: controller, in: query, schema: { type: string, enum: [MC, HC] } }
      - { name: q, in: query, schema: { type: string } }
      - { name: state, in: query, schema: { type: string, enum: [issued, reserved, pending] } }
    responses:
      '200': { description: File stream, content: { application/octet-stream: {} } }
      '422': { description: Invalid format }

/mapping:
  get:
    tags: [Mapping]
    summary: >
      The legacy -> new mapping (v4.17, FR-191): the rows with what the .st programs do not carry.
      Filters `family`, `status`, `q`; 100 rows a page. Not for Viewers.
    parameters:
      - { name: family, in: query, schema: { type: string, enum: [MC, HC] } }
      - { name: status, in: query, schema: { type: string, enum: [PROPOSED, FROM_PROGRAM, APPROVED, ISSUED, REJECTED] } }
      - { name: q, in: query, schema: { type: string }, description: "Number, title, group or cause" }
      - { name: page, in: query, schema: { type: integer, minimum: 1 } }
    responses:
      '200': { description: Rows, content: { application/json: { schema: { $ref: '#/components/schemas/MappingPage' } } }
 }
      '422': { description: Invalid filter }

/mapping/export:
  get:
    tags: [Mapping]
    summary: The mapping as a workbook with the sheets MC and HC - the layout the upload reads (FR-197)
    parameters:
      - { name: family, in: query, schema: { type: string, enum: [MC, HC] } }
    responses:
      '200': { description: XLSX workbook, content: { application/vnd.openxmlformats-officedocument.spreadsheetml.sheet:
{} } }

/mapping/seed:
  post:
    tags: [Mapping]
    summary: Load the initial mapping from the shipped workbook into an empty table (Administrator)
    responses:
      '200': { description: "Number of rows loaded; 0 when the table already had rows", content: { application/json: { sc
hema: { type: object, properties: { loaded: { type: integer } } } } } }
      '422': { $ref: '#/components/responses/ValidationError' }

/mapping/{mappingId}:
  parameters: [{ name: mappingId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Mapping]
    summary: One row of the mapping
    responses:
      '200': { description: Row, content: { application/json: { schema: { $ref: '#/components/schemas/MappingRow' } } } }
      '404': { description: Unknown row }

/mapping-changes:
  get:
    tags: [Mapping]
    summary: The requests to change the mapping, newest first; filter `state`
    parameters:
      - { name: state, in: query, schema: { type: string, enum: [OPEN, APPLIED, REJECTED, WITHDRAWN] } }
    responses:
      '200': { description: Requests, content: { application/json: { schema: { type: object, properties: { changes: { typ
e: array, items: { $ref: '#/components/schemas/MappingChange' } } } } } } } }
  post:
    tags: [Mapping]
    summary: >
      Request a change of the mapping (FR-194, FR-195). Nothing changes before the StCd Owner, the Chief
      Engineer and an Administrator have approved it. Every row is checked; one error refuses the whole request.
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            required: [title, reason, items]
            properties:
              title: { type: string, maxLength: 120 }
              reason: { type: string, maxLength: 2000 }
              items:
                type: array
                items:
                  type: object
                  required: [action]
                  properties:
                    action: { type: string, enum: [ADD, CHANGE, CONFIRM, REJECT], description: "ADD a new row; CHANGE fie
lds of a row; CONFIRM = ask for the approval of the number as it is; REJECT = the number is never used" }
```

```
                    mapping_id: { type: integer, description: "The stored row; not for ADD" }
                    fields: { type: object, description: "The fields of the row (legacy_number, new_number, description,
functional_group, logical_node, status_type, up_down_counter, parameter_set_delay, parameter_reset_delay, the eight ..._acc
ess fields, cause, directive, note, and family for ADD)" }
    responses:
      '201': { description: Requested, content: { application/json: { schema: { $ref: '#/components/schemas/MappingChange
' } } } }
      '403': { $ref: '#/components/responses/Forbidden' }
      '422': { $ref: '#/components/responses/ValidationError' }

/mapping-changes/workbook:
  post:
    tags: [Mapping]
    summary: >
      Compare an uploaded workbook (sheets MC and HC) with the mapping and request one change with every
      new and every differing row (FR-197). A wrong row refuses the whole file and names it.
    requestBody:
      required: true
      content:
        multipart/form-data:
          schema:
            type: object
            required: [file, title, reason]
            properties:
              file: { type: string, format: binary }
              title: { type: string, maxLength: 120 }
              reason: { type: string, maxLength: 2000 }
              confirm_unchanged: { type: string, enum: ["1"], description: "Also ask for the approval of unchanged rows t
hat are only proposed" }
    responses:
      '201': { description: Requested, content: { application/json: { schema: { type: object, properties: { change: { $re
f: '#/components/schemas/MappingChange' }, counts: { type: object }, unchanged: { type: integer } } } } } }
      '422': { $ref: '#/components/responses/ValidationError' }

/mapping-changes/{changeId}:
  parameters: [{ name: changeId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Mapping]
    summary: One request with its rows and the three approvals
    responses:
      '200': { description: Request, content: { application/json: { schema: { $ref: '#/components/schemas/MappingChange'
} } } }
      '404': { description: Unknown request }

/mapping-changes/{changeId}/decisions:
  parameters: [{ name: changeId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [Mapping]
    summary: >
      Approve or reject (FR-194, BR-027). Only the StCd Owner, the Chief Engineer and an Administrator, each
      once, none of them the requester; the password is asked again (I-17). One rejection closes the
      request; the third approval applies it.
    requestBody:
      required: true
      content:
        application/json:
          schema:
            type: object
            required: [decision, current_password]
            properties:
              decision: { type: string, enum: [APPROVE, REJECT] }
              comment: { type: string, maxLength: 1000, description: "Required to reject" }
              current_password: { type: string }
    responses:
      '200': { description: The request after the decision, content: { application/json: { schema: { $ref: '#/components/
schemas/MappingChange' } } } }
      '403': { $ref: '#/components/responses/Forbidden' }
      '409': { description: Closed, or this role has decided already }
      '422': { $ref: '#/components/responses/ValidationError' }

/mapping-changes/{changeId}/withdraw:
  parameters: [{ name: changeId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [Mapping]
    summary: The requester withdraws an open request
    responses:
      '200': { description: Withdrawn, content: { application/json: { schema: { $ref: '#/components/schemas/MappingChange
' } } } }
      '403': { $ref: '#/components/responses/Forbidden' }
      '409': { description: Already closed }

/export-profiles:
  get:
    tags: [Exports]
    summary: Export profiles - Administrator-managed column sets (I-02). include_inactive=true is honoured for Administra
tors.
    parameters: [{ name: include_inactive, in: query, schema: { type: boolean } }]
    responses:
      '200': { description: Profiles, content: { application/json: { schema: { type: object, properties: { items: { type:
```

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 array, items: { $ref: '#/components/schemas/ExportProfile' } } } } } } }
  post:
    tags: [Exports]
    summary: Create an export profile (Administrator). Columns are export field names with a header label.
    requestBody:
      required: true
      content: { application/json: { schema: { $ref: '#/components/schemas/ExportProfileInput' } } }
    responses:
      '201': { description: Created, content: { application/json: { schema: { $ref: '#/components/schemas/ExportProfile'
} } } }
      '422': { $ref: '#/components/responses/ValidationError' }

/export-profiles/{profileId}:
  parameters: [{ name: profileId, in: path, required: true, schema: { type: integer } }]
  patch:
    tags: [Exports]
    summary: Change or deactivate an export profile (Administrator)
    requestBody:
      content: { application/json: { schema: { $ref: '#/components/schemas/ExportProfileInput' } } }
    responses:
      '200': { description: Updated, content: { application/json: { schema: { $ref: '#/components/schemas/ExportProfile'
} } } }

/audiences:
  get:
    tags: [Exports]
    summary: The eight audiences an export can be scoped to (I-03)
    responses:
      '200': { description: Audiences, content: { application/json: { schema: { type: object, properties: { items: { type
: array, items: { type: object, properties: { code: { type: string }, label: { type: string } } } } } } } } } }

/releases/{releaseId}/notes:
  parameters: [{ $ref: '#/components/parameters/ReleaseId' }]
  get:
    tags: [Exports]
    summary: Release notes - added / changed (field-level diff) / deprecated against the previous published Release (I-06
). A BUILDING release is previewable by Administrators only.
    parameters:
      - { name: format, in: query, schema: { type: string, enum: [md, csv, pdf, json], default: json } }
      - { name: bundle, in: query, schema: { type: boolean }, description: ZIP with MANIFEST.json }
    responses:
      '200': { description: File stream, headers: { X-Content-SHA256: { schema: { type: string } } }, content: { applicat
ion/octet-stream: {} } }

/releases/{releaseId}/offline-bundle:
  parameters: [{ $ref: '#/components/parameters/ReleaseId' }]
  get:
    tags: [Exports]
    summary: One self-contained HTML lookup page for the Release, for use at the turbine without connectivity (I-24). Sam
e audience/language/software_version/bundle parameters as the export.
    parameters:
      - { name: audience, in: query, schema: { type: string } }
      - { name: language, in: query, schema: { type: string } }
      - { name: software_version, in: query, schema: { type: string } }
      - { name: bundle, in: query, schema: { type: boolean } }
    responses:
      '200': { description: HTML file or ZIP, headers: { X-Content-SHA256: { schema: { type: string } } }, content: { app
lication/octet-stream: {} } }
      '409': { description: Release is not published }

/releases/{releaseId}/exports:
  parameters: [{ $ref: '#/components/parameters/ReleaseId' }]
  get:
    tags: [Exports]
    summary: Every export recorded for the Release with its SHA-256 (I-06). Working roles.
    responses:
      '200': { description: Export jobs, content: { application/json: { schema: { type: object, properties: { items: { ty
pe: array, items: { $ref: '#/components/schemas/ExportRecord' } } } } } } }

/releases/{releaseId}/target-versions:
  parameters: [{ $ref: '#/components/parameters/ReleaseId' }]
  put:
    tags: [Releases]
    summary: Set the controller software versions a Release is published for (I-07, Administrator, before publish only)
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [software_versions], properties: { software_versio
ns: { type: array, items: { type: string } } } } } }
    responses:
      '200': { description: Targets, content: { application/json: { schema: { type: object, properties: { release_id: { t
ype: integer }, software_versions: { type: array, items: { type: string } } } } } } }
      '409': { description: Release already published }

/releases/{releaseId}/baseline:
  parameters: [{ $ref: '#/components/parameters/ReleaseId' }]
  get:
    tags: [Releases]
    summary: Counts of the complete baseline of a Release - every code valid as of it - by controller and platform (FR-17
3). Any authenticated role.
```

```
    responses:
      '200': { description: Counts, content: { application/json: { schema: { type: object, properties: { baseline: { type
: integer }, in_release: { type: integer }, by_controller: { type: object, properties: { MC: { type: integer }, HC: { type:
 integer } } }, by_platform: { type: object, additionalProperties: { type: integer } } } } } } }
      '404': { $ref: '#/components/responses/NotFound' }

/releases/{releaseId}/readiness:
  parameters: [{ $ref: '#/components/parameters/ReleaseId' }]
  get:
    tags: [Releases]
    summary: What is still missing before the documents of a Release are complete - translated codes per language and cod
es without confirmed access levels (FR-173). Engineer, Reviewer, Chief Engineer, Administrator, StCd Owner.
    responses:
      '200': { description: Coverage, content: { application/json: { schema: { type: object, properties: { baseline: { ty
pe: integer }, translation_coverage: { type: object, additionalProperties: { type: object, properties: { translated: { type
: integer }, of: { type: integer } } } }, unconfirmed_access: { type: object, properties: { count: { type: integer }, examp
les: { type: array, items: { type: string } } } } } } } } }
      '404': { $ref: '#/components/responses/NotFound' }

/releases/{releaseId}/packages:
  parameters: [{ $ref: '#/components/parameters/ReleaseId' }]
  post:
    tags: [Exports]
    summary: Build the package of a published Release - the complete baseline in every language for every access level, w
ith release notes and a manifest - as a ZIP (FR-174). Any authenticated role (FR-175).
    requestBody:
      content:
        application/json:
          schema:
            type: object
            properties:
              formats: { type: array, items: { type: string, enum: [csv, json, xlsx, pdf] }, default: [csv, xlsx, pdf] }
              languages: { type: array, items: { type: string }, description: "Default: every language" }
              audiences: { type: array, items: { type: string }, description: "Default: all eight audiences" }
              include_unfiltered: { type: boolean, default: true }
              platform: { type: string, enum: ['2XM', '3XM', '4XM'] }
              controller: { type: string, enum: [MC, HC] }
              strict_language: { type: boolean }
    responses:
      '201': { description: Built already, content: { application/json: { schema: { $ref: '#/components/schemas/PackageJo
b' } } } }
      '202': { description: "Being built; poll /packages/{jobId}", content: { application/json: { schema: { $ref: '#/comp
onents/schemas/PackageJob' } } } }
      '422': { $ref: '#/components/responses/ValidationError' }
  get:
    tags: [Exports]
    summary: The packages built for a Release, newest first (FR-174). Any authenticated role (FR-175).
    responses:
      '200': { description: Packages, content: { application/json: { schema: { type: object, properties: { items: { type:
 array, items: { $ref: '#/components/schemas/PackageJob' } } } } } } }

/packages/{jobId}:
  parameters: [{ name: jobId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Exports]
    summary: State and progress of a package (FR-174). Any authenticated role (FR-175).
    responses:
      '200': { description: Package job, content: { application/json: { schema: { $ref: '#/components/schemas/PackageJob'
 } } } }
      '404': { $ref: '#/components/responses/NotFound' }

/packages/{jobId}/download:
  parameters: [{ name: jobId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Exports]
    summary: Download the finished package as a ZIP (FR-174). Any authenticated role (FR-175).
    responses:
      '200': { description: ZIP file, content: { application/zip: {} } }
      '404': { $ref: '#/components/responses/NotFound' }
      '422': { $ref: '#/components/responses/ValidationError' }

/exports/{jobId}/manifest:
  parameters: [{ name: jobId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Exports]
    summary: The manifest recorded for an export (I-06). Working roles.
    responses:
      '200': { description: Manifest JSON, content: { application/json: { schema: { $ref: '#/components/schemas/Manifest'
 } } } }
      '404': { $ref: '#/components/responses/NotFound' }

/exports/verify:
  post:
    tags: [Exports]
    summary: Is this file one the system produced (I-06)? Upload `file` and optionally its MANIFEST.json as `manifest`. R
eports the SHA-256, the matching recorded exports and the manifest check (file listed, signature valid/invalid/unsigned).
    requestBody:
      required: true
      content: { multipart/form-data: { schema: { type: object, required: [file], properties: { file: { type: string, for
```

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mat: binary }, manifest: { type: string, format: binary } } } } }
    responses:
      '200': { description: Verification, content: { application/json: { schema: { $ref: '#/components/schemas/VerifyResu
lt' } } } }

/software-versions/matrix:
  get:
    tags: [SoftwareVersions]
    summary: Which Released codes apply to a controller software version (I-07)
    parameters: [{ name: version, in: query, required: true, schema: { type: string } }]
    responses:
      '200': { description: Matrix, content: { application/json: { schema: { type: object } } } }
      '404': { $ref: '#/components/responses/NotFound' }

/software-versions/changes:
  get:
    tags: [SoftwareVersions]
    summary: What was added, removed and changed between two software versions (I-07)
    parameters:
      - { name: from, in: query, required: true, schema: { type: string } }
      - { name: to, in: query, required: true, schema: { type: string } }
    responses:
      '200': { description: Changes, content: { application/json: { schema: { type: object } } } }

/revisions/{revisionId}/translations:
  parameters: [{ name: revisionId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Translations]
    summary: Translation status per language - MISSING, DRAFT or APPROVED (I-15). Viewers see approved text only.
    responses:
      '200': { description: Status matrix, content: { application/json: { schema: { type: object, properties: { items: {
type: array, items: { $ref: '#/components/schemas/TranslationStatus' } } } } } } }

/revisions/{revisionId}/translations/{language}:
  parameters:
    - { name: revisionId, in: path, required: true, schema: { type: integer } }
    - { name: language, in: path, required: true, schema: { type: string, example: de } }
  put:
    tags: [Translations]
    summary: Save a translation as a DRAFT (Engineer, Administrator; source text must be Approved). Editing an approved t
ranslation sends it back to DRAFT.
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [title, description], properties: { title: { type:
 string }, description: { type: string }, remedy_text: { type: string, nullable: true } } } } }
    responses:
      '200': { description: Saved, content: { application/json: { schema: { $ref: '#/components/schemas/Translation' } }
} }
      '409': { description: Source revision not yet Approved }

/revisions/{revisionId}/translations/{language}/approve:
  parameters:
    - { name: revisionId, in: path, required: true, schema: { type: integer } }
    - { name: language, in: path, required: true, schema: { type: string } }
  post:
    tags: [Translations]
    summary: Approve a translation - a light review by a Reviewer or Administrator other than the translator (I-15)
    responses:
      '200': { description: Approved, content: { application/json: { schema: { $ref: '#/components/schemas/Translation' }
 } } }
      '403': { description: Translator cannot approve their own text }

/revisions/{revisionId}/translations/carry-over:
  parameters: [{ name: revisionId, in: path, required: true, schema: { type: integer } }]
  post:
    tags: [Translations]
    summary: Copy the previous revision's translations as DRAFTs for languages this revision lacks
    responses:
      '200': { description: Languages copied, content: { application/json: { schema: { type: object, properties: { copied
_languages: { type: array, items: { type: string } } } } } } } }

/translations/worklist:
  get:
    tags: [Translations]
    summary: Current Approved/Released codes still lacking an approved translation in a language (I-15)
    parameters: [{ name: language, in: query, required: true, schema: { type: string } }]
    responses:
      '200': { description: Worklist, content: { application/json: { schema: { type: object, properties: { items: { type:
 array, items: { type: object } } } } } } }

/saved-searches:
  get:
    tags: [SavedSearches]
    summary: Your saved status-code searches and the columns the list can show (I-26)
    responses:
      '200': { description: Saved searches, content: { application/json: { schema: { type: object, properties: { items: {
 type: array, items: { $ref: '#/components/schemas/SavedSearch' } }, available_columns: { type: array, items: { type: objec
t } }, default_columns: { type: array, items: { type: string } } } } } } } }
  post:
```

```
    tags: [SavedSearches]
    summary: Save the filters and visible columns under a name (saving under an existing name replaces it)
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [name], properties: { name: { type: string }, filt
ers: { type: object }, columns: { type: array, items: { type: string } } } } } }
    responses:
      '201': { description: Saved, content: { application/json: { schema: { $ref: '#/components/schemas/SavedSearch' } }
} }

/saved-searches/{searchId}:
  parameters: [{ name: searchId, in: path, required: true, schema: { type: integer } }]
  delete:
    tags: [SavedSearches]
    summary: Remove one of your saved searches
    responses:
      '204': { description: Removed }

/api-tokens:
  get:
    tags: [Integrations]
    summary: Machine API tokens (I-29, Administrator). The secret is never shown again.
    responses:
      '200': { description: Tokens, content: { application/json: { schema: { type: object, properties: { items: { type: a
rray, items: { $ref: '#/components/schemas/ApiToken' } }, available_scopes: { type: array, items: { type: object } } } } } }
} }
  post:
    tags: [Integrations]
    summary: >
      Create a scoped, expiring token (max 365 days). It authenticates as `Authorization: Bearer cscm_...` on the
      read-only endpoints that declare its scope, with Viewer-level visibility; it can never write.
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [name, scopes], properties: { name: { type: string
 }, scopes: { type: array, items: { type: string, enum: ['catalogue:read', 'changes:read'] } }, expires_in_days: { type: in
teger, default: 90 } } } } }
    responses:
      '201': { description: Created; `token` is shown exactly once, content: { application/json: { schema: { $ref: '#/com
ponents/schemas/ApiTokenCreated' } } } }

/api-tokens/{tokenId}:
  parameters: [{ name: tokenId, in: path, required: true, schema: { type: integer } }]
  delete:
    tags: [Integrations]
    summary: Revoke a token (Administrator)
    responses:
      '204': { description: Revoked }

/webhooks:
  get:
    tags: [Integrations]
    summary: Webhooks fired on release.published (I-29, Administrator)
    responses:
      '200': { description: Webhooks, content: { application/json: { schema: { type: object, properties: { items: { type:
 array, items: { $ref: '#/components/schemas/Webhook' } } } } } } } }
  post:
    tags: [Integrations]
    summary: >
      Register a webhook. The target must resolve to a private/loopback address. Deliveries carry
      `X-CSCM-Signature: sha256=<HMAC-SHA-256 of the body with the secret>`; redirects are not followed.
    requestBody:
      required: true
      content: { application/json: { schema: { type: object, required: [name, url], properties: { name: { type: string },
 url: { type: string }, events: { type: array, items: { type: string, enum: [release.published] } } } } } }
    responses:
      '201': { description: Created; `secret` is shown exactly once, content: { application/json: { schema: { $ref: '#/co
mponents/schemas/WebhookCreated' } } } }
      '422': { $ref: '#/components/responses/ValidationError' }

/webhooks/{webhookId}:
  parameters: [{ name: webhookId, in: path, required: true, schema: { type: integer } }]
  patch:
    tags: [Integrations]
    summary: Pause or resume a webhook
    requestBody:
      content: { application/json: { schema: { type: object, properties: { is_active: { type: boolean } } } } }
    responses:
      '200': { description: Updated, content: { application/json: { schema: { $ref: '#/components/schemas/Webhook' } } }
}

/webhooks/{webhookId}/deliveries:
  parameters: [{ name: webhookId, in: path, required: true, schema: { type: integer } }]
  get:
    tags: [Integrations]
    summary: Recent deliveries of a webhook with their outcome
    responses:
      '200': { description: Deliveries, content: { application/json: { schema: { type: object, properties: { items: { typ
e: array, items: { type: object } } } } } } }
```

```
  /changes:
    get:
      tags: [Integrations]
      summary: >
        Change feed - releases, deprecations, reinstatements and archivals in audit order (I-29). Poll with the previous
        `next_since_id`; ETag / If-None-Match supported (304). Session or an API token with scope changes:read.
      security: [{ sessionCookie: [] }, { bearerToken: [] }]
      parameters:
        - { name: since_id, in: query, schema: { type: integer, default: 0 } }
        - { name: limit, in: query, schema: { type: integer, default: 100, maximum: 500 } }
      responses:
        '200': { description: Events, headers: { ETag: { schema: { type: string } } }, content: { application/json: { schem
  a: { $ref: '#/components/schemas/ChangeFeed' } } } }
        '304': { description: Not modified }

components:
  securitySchemes:
    sessionCookie: { type: apiKey, in: cookie, name: cscm_session }
    bearerToken: { type: http, scheme: bearer, description: "Scoped, expiring API token (I-29); read-only endpoints that de
  clare its scope" }

  parameters:
    Page: { name: page, in: query, schema: { type: integer, minimum: 1, default: 1 } }
    PageSize: { name: page_size, in: query, schema: { type: integer, minimum: 1, maximum: 200, default: 50 } }
    Sort: { name: sort, in: query, schema: { type: string } }
    UserId: { name: userId, in: path, required: true, schema: { type: integer } }
    StatusCodeId: { name: statusCodeId, in: path, required: true, schema: { type: integer } }
    CrId: { name: crId, in: path, required: true, schema: { type: integer } }
    ChangeSetId: { name: changeSetId, in: path, required: true, schema: { type: integer } }
    ReleaseId: { name: releaseId, in: path, required: true, schema: { type: integer } }

  responses:
    Unauthorized: { description: Not authenticated, content: { application/json: { schema: { $ref: '#/components/schemas/Er
  ror' } } } }
    Forbidden: { description: Not authorized, content: { application/json: { schema: { $ref: '#/components/schemas/Error' }
   } } }
    NotFound: { description: Not found, content: { application/json: { schema: { $ref: '#/components/schemas/Error' } } } }
    ValidationError: { description: "Validation failed. Field-level errors only ? the client's already-entered data is neve
  r discarded by this response (addresses 'when creating fails, all information is resetted')", content: { application/json:
  { schema: { $ref: '#/components/schemas/ValidationErrorBody' } } } }

  schemas:
    LifecycleStatus:
      type: string
      enum: [DRAFT, REVIEW, PENDING_APPROVAL, APPROVED, RELEASED, DEPRECATED, ARCHIVED]
    Error:
      type: object
      properties: { error_code: { type: string }, message: { type: string }, request_id: { type: string } }
    ValidationErrorBody:
      type: object
      properties:
        error_code: { type: string, example: VALIDATION_FAILED }
        message: { type: string }
        field_errors:
          type: array
          items:
            type: object
            properties: { field: { type: string }, message: { type: string } }
    User:
      type: object
      properties:
        id: { type: integer }
        username: { type: string }
        email: { type: string }
        full_name: { type: string }
        role: { type: string, enum: [ADMINISTRATOR, ENGINEER, REVIEWER, CHIEF_ENGINEER, VIEWER] }
        engineering_domains: { type: array, items: { type: string }, example: [CONTROLS, ELECTRICAL] }
        is_active: { type: boolean }
        last_login_at: { type: string, format: date-time, nullable: true }
    UserCreate:
      type: object
      required: [username, email, full_name, password]
      properties:
        username: { type: string }
        email: { type: string }
        full_name: { type: string }
        role: { type: string, enum: [ADMINISTRATOR, ENGINEER, REVIEWER, CHIEF_ENGINEER, VIEWER], default: VIEWER, descripti
  on: Least privilege by default (SEC-012) }
        engineering_domains: { type: array, items: { type: string } }
        password: { type: string }
    UserUpdate:
      type: object
      properties:
        full_name: { type: string }
        role: { type: string, enum: [ADMINISTRATOR, ENGINEER, REVIEWER, CHIEF_ENGINEER, VIEWER] }
        engineering_domains: { type: array, items: { type: string } }
        is_active: { type: boolean }
    UserList:
      type: object
```

```
    properties: { items: { type: array, items: { $ref: '#/components/schemas/User' } }, total: { type: integer }, page: {
 type: integer }, page_size: { type: integer } }
  LookupValue:
    type: object
    properties:
      id: { type: integer }
      code: { type: string }
      label: { type: string }
      description: { type: string, nullable: true, description: "Surfaced by the UI as inline help text next to the optio
n" }
      is_active: { type: boolean }
      functional_system_group_id: { type: integer, nullable: true, description: "Present for functional-subgroups only" }
      range_start: { type: integer, nullable: true }
      range_end: { type: integer, nullable: true }
      functional_system_group: { type: string, nullable: true, description: "Present for functional-subgroups only" }
  StatusCodeRevision:
    type: object
    properties:
      id: { type: integer }
      status_code_id: { type: integer }
      status_code_identifier: { type: string, nullable: true, example: StCd-01232, description: "NULL while Draft" }
      functional_system_group: { type: string, example: WCNV }
      functional_subgroup: { type: string, description: "Mandatory, never null from Draft onward" }
      turbine_platforms: { type: array, items: { type: string }, example: [2XM, 3XM] }
      is_sandbox: { type: boolean }
      platform_variant_of_status_code_id: { type: integer, nullable: true }
      legacy_reference_number: { type: string, nullable: true }
      revision_number: { type: integer }
      is_current: { type: boolean }
      title: { type: string }
      description: { type: string }
      software_version: { type: string, nullable: true, example: "1.1.1.4.2MW" }
      status_category: { type: string, enum: [ERROR, WARNING, INFO] }
      available_group: { type: integer, example: 27 }
      alarm_flag: { type: boolean }
      set_delay: { type: string, example: 10ms }
      reset_delay: { type: string, example: 2min }
      up_down_counter: { type: boolean }
      trigger_snapshot: { type: boolean }
      loadless_spinning_permitted: { type: boolean }
      number_of_times_repeated: { type: integer, nullable: true }
      repeated_over_the_course: { type: string, nullable: true, example: 1d }
      status_code_for_repeated_error: { type: integer, nullable: true, description: "Opaque reference to a separate code
pool, not an StCd" }
      yaw_program: { type: integer }
      manual_reset_program: { type: integer }
      auto_reset_program: { type: integer }
      converter_reactive_power_program: { type: integer }
      brake_program: { type: integer }
      umrichterprogramm_gsc: { type: integer }
      umrichterprogramm_msc: { type: integer }
      yaw_bearing_lubrication_program: { type: integer }
      converter_reset_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Converter Reset Pro
gram (Main Controller only); 0 = the code does not act on it" }
      hydraulic_pump_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Hydraulic Pump Progr
am (Main Controller only); 0 = the code does not act on it" }
      gearbox_oil_pump_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Gearbox Oil Pump P
rogram (Main Controller only); 0 = the code does not act on it" }
      gearbox_oil_fan_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Gearbox Oil Fan Pro
gram (Main Controller only); 0 = the code does not act on it" }
      gearbox_oil_heating_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Gearbox Oil Hea
ting Program (Main Controller only); 0 = the code does not act on it" }
      generator_fan_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Generator Fan Program
 (Main Controller only); 0 = the code does not act on it" }
      generator_heating_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Generator Heating
 Program (Main Controller only); 0 = the code does not act on it" }
      generator_bearing_lubrication_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Gener
ator Bearing Lubrication Program (Main Controller only); 0 = the code does not act on it" }
      rotor_bearing_lubrication_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Rotor Bea
ring Lubrication Program (Main Controller only); 0 = the code does not act on it" }
      anemometer_heating_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Anemometer Heati
ng Program (Main Controller only); 0 = the code does not act on it" }
      medium_voltage_cb_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Medium Voltage Ci
rcuit Breaker Program (Main Controller only); 0 = the code does not act on it" }
      tower_fan_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Tower Fan Program (Main C
ontroller only); 0 = the code does not act on it" }
      transformer_oil_pump_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Transformer Oi
l Pump Program (Main Controller only); 0 = the code does not act on it" }
      pitch_error_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Pitch Error Program (Hu
b Controller only); 0 = the code does not act on it" }
      pitch_warning_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Pitch Warning Program
 (Hub Controller only); 0 = the code does not act on it" }
      rotor_brake_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Rotor Brake Program (Hu
b Controller only); 0 = the code does not act on it" }
      zero_load_operation: { type: boolean, nullable: true, description: "Zero Load Operation (Main Controller only); nul
l = not captured" }
      availability_calc: { type: boolean, nullable: true, description: "Availability Calculation (both controllers); null
 = not captured" }
      stop_program_40_flag: { type: boolean, nullable: true, description: "Stop Program 40 (Hub Controller only); null =
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not captured" }
      stop_program_80_flag: { type: boolean, nullable: true, description: "Stop Program 80 (Hub Controller only); null =
not captured" }
      manual_operation_flag: { type: boolean, nullable: true, description: "Manual Operation (Hub Controller only); null
= not captured" }
      automatic_operation_flag: { type: boolean, nullable: true, description: "Automatic Operation (Hub Controller only);
 null = not captured" }
      vane_position_flag: { type: boolean, nullable: true, description: "Vane Position (Hub Controller only); null = not
captured" }
      operating_mode_mask: { type: integer, nullable: true, minimum: 0, maximum: 1048575, description: "OpModeArDw as a n
umber: 20 bits on the Main Controller, 6 on the Hub Controller (the forms show it as a bit string); null = not captured" }
      task_call_frequency: { type: integer, nullable: true, minimum: 1, description: "TaskCallFrequencyArBy (1, 10 or 100
 so far); null = not captured" }
      user_groups: { type: array, description: "The numeric user-group levels (UserGroupReadArBy / WriteArBy), one entry
per slot 0-15; omitted = unchanged", items: { type: object, required: [slot, read, write], properties: { slot: { type: inte
ger, minimum: 0, maximum: 15 }, read: { type: integer, minimum: 0, maximum: 255 }, write: { type: integer, minimum: 0, maxi
mum: 255 } } } }
      operating_mode_bits: { type: string, description: "operating_mode_mask as the bit string of the generated program"
}
      controller_type: { type: string, enum: [MAIN, HUB] }
      access_confirmed: { type: boolean, description: "True once the Administrator's Review sign-off has confirmed (or ov
erridden) the eight audience-access fields" }
      functional_system_group_id: { type: integer }
      functional_subgroup_id: { type: integer }
      functional_system_group_label: { type: string }
      functional_system_group_range_start: { type: integer }
      functional_system_group_range_end: { type: integer }
      functional_subgroup_label: { type: string }
      development_access_id: { type: integer, nullable: true }
      sales_access_id: { type: integer, nullable: true }
      tcc_access_id: { type: integer, nullable: true }
      service_access_id: { type: integer, nullable: true }
      turbine_operator_package_access_id: { type: integer, nullable: true }
      grid_operator_access_id: { type: integer, nullable: true }
      service_partner_access_id: { type: integer, nullable: true }
      customer_access_id: { type: integer, nullable: true }
      development_access: { type: string, nullable: true }
      sales_access: { type: string, nullable: true }
      tcc_access: { type: string, nullable: true }
      service_access: { type: string, nullable: true }
      turbine_operator_package_access: { type: string, nullable: true }
      grid_operator_access: { type: string, nullable: true }
      service_partner_access: { type: string, nullable: true }
      customer_access: { type: string, nullable: true, description: "Engineer-proposed until admin-signoff confirms it, l
ike all eight access fields" }
      logical_node: { type: string, description: "Derived, read-only ? always the parent Functional System Group's prefix
" }
      node_value: { type: integer, description: "Derived, read-only ? always the parent Functional System Group's sort_or
der" }
      owner_id: { type: integer }
      lifecycle_status: { $ref: '#/components/schemas/LifecycleStatus' }
      created_by: { type: integer }
      created_at: { type: string, format: date-time }
      reviewer_signoff_by: { type: integer, nullable: true }
      reviewer_signoff_at: { type: string, format: date-time, nullable: true }
      admin_signoff_by: { type: integer, nullable: true }
      admin_signoff_at: { type: string, format: date-time, nullable: true }
      chief_engineer_approved_by: { type: integer, nullable: true }
      chief_engineer_approval_date: { type: string, format: date-time, nullable: true }
      effective_date: { type: string, format: date-time, nullable: true }
      deprecated_reason: { type: string, nullable: true }
      superseded_by_status_code_id: { type: integer, nullable: true }
      row_version: { type: integer, description: "I-01: increases on every Draft edit; send it back as expected_row_versi
on" }
      updated_at: { type: string, format: date-time, nullable: true }
      remedy_text: { type: string, nullable: true, description: "I-14: recommended action for service staff" }
      affected_component: { type: string, nullable: true }
      related_status_codes: { type: string, nullable: true, example: "StCd-01234, StCd-01250" }
      standard_references: { type: string, nullable: true }
      applies_from_version: { type: string, nullable: true, description: "I-07: first controller software version this re
vision applies to (code; null = open)" }
      applies_from_version_id: { type: integer, nullable: true }
      applies_to_version: { type: string, nullable: true }
      applies_to_version_id: { type: integer, nullable: true }
      planned_effective_date: { type: string, format: date, nullable: true, description: "I-27" }
      planned_deprecation_date: { type: string, format: date, nullable: true }
      warnings: { type: array, items: { $ref: '#/components/schemas/Warning' }, description: "Present on create/edit resp
onses: soft findings that never block (I-04, I-12)" }
  StatusCodeCreate:
    type: object
    required: [functional_system_group, functional_subgroup, title, description, status_category,
               available_group, set_delay, reset_delay, turbine_platforms]
    properties:
      functional_system_group: { type: string, example: WCNV }
      functional_subgroup: { type: string, description: "Mandatory ? never optional", example: Converter }
      is_sandbox: { type: boolean, default: false, description: "Administrator only" }
      turbine_platforms: { type: array, items: { type: string }, minItems: 1 }
      title: { type: string }
      description: { type: string }
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      software_version: { type: string, nullable: true }
      status_category: { type: string, enum: [ERROR, WARNING, INFO] }
      available_group: { type: integer }
      alarm_flag: { type: boolean, default: false }
      set_delay: { type: string, example: 10ms }
      reset_delay: { type: string, example: 2min }
      up_down_counter: { type: boolean, default: false }
      trigger_snapshot: { type: boolean, default: false }
      loadless_spinning_permitted: { type: boolean, default: false }
      number_of_times_repeated: { type: integer, nullable: true }
      repeated_over_the_course: { type: string, nullable: true }
      status_code_for_repeated_error: { type: integer, nullable: true }
      yaw_program: { type: integer, default: 0 }
      manual_reset_program: { type: integer, default: 0 }
      auto_reset_program: { type: integer, default: 0 }
      converter_reactive_power_program: { type: integer, default: 0 }
      brake_program: { type: integer, default: 0 }
      umrichterprogramm_gsc: { type: integer, default: 0 }
      umrichterprogramm_msc: { type: integer, default: 0 }
      yaw_bearing_lubrication_program: { type: integer, default: 0 }
      converter_reset_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Converter Reset Pro
gram (Main Controller only); 0 = the code does not act on it" }
      hydraulic_pump_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Hydraulic Pump Progr
am (Main Controller only); 0 = the code does not act on it" }
      gearbox_oil_pump_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Gearbox Oil Pump P
rogram (Main Controller only); 0 = the code does not act on it" }
      gearbox_oil_fan_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Gearbox Oil Fan Pro
gram (Main Controller only); 0 = the code does not act on it" }
      gearbox_oil_heating_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Gearbox Oil Hea
ting Program (Main Controller only); 0 = the code does not act on it" }
      generator_fan_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Generator Fan Program
 (Main Controller only); 0 = the code does not act on it" }
      generator_heating_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Generator Heating
 Program (Main Controller only); 0 = the code does not act on it" }
      generator_bearing_lubrication_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Gener
ator Bearing Lubrication Program (Main Controller only); 0 = the code does not act on it" }
      rotor_bearing_lubrication_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Rotor Bea
ring Lubrication Program (Main Controller only); 0 = the code does not act on it" }
      anemometer_heating_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Anemometer Heati
ng Program (Main Controller only); 0 = the code does not act on it" }
      medium_voltage_cb_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Medium Voltage Ci
rcuit Breaker Program (Main Controller only); 0 = the code does not act on it" }
      tower_fan_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Tower Fan Program (Main C
ontroller only); 0 = the code does not act on it" }
      transformer_oil_pump_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Transformer Oi
l Pump Program (Main Controller only); 0 = the code does not act on it" }
      pitch_error_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Pitch Error Program (Hu
b Controller only); 0 = the code does not act on it" }
      pitch_warning_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Pitch Warning Program
 (Hub Controller only); 0 = the code does not act on it" }
      rotor_brake_program: { type: integer, minimum: 0, maximum: 65535, default: 0, description: "Rotor Brake Program (Hu
b Controller only); 0 = the code does not act on it" }
      zero_load_operation: { type: boolean, nullable: true, description: "Zero Load Operation (Main Controller only); nul
l = not captured" }
      availability_calc: { type: boolean, nullable: true, description: "Availability Calculation (both controllers); null
 = not captured" }
      stop_program_40_flag: { type: boolean, nullable: true, description: "Stop Program 40 (Hub Controller only); null =
not captured" }
      stop_program_80_flag: { type: boolean, nullable: true, description: "Stop Program 80 (Hub Controller only); null =
not captured" }
      manual_operation_flag: { type: boolean, nullable: true, description: "Manual Operation (Hub Controller only); null
= not captured" }
      automatic_operation_flag: { type: boolean, nullable: true, description: "Automatic Operation (Hub Controller only);
 null = not captured" }
      vane_position_flag: { type: boolean, nullable: true, description: "Vane Position (Hub Controller only); null = not
captured" }
      operating_mode_mask: { type: integer, nullable: true, minimum: 0, maximum: 1048575, description: "OpModeArDw as a n
umber: 20 bits on the Main Controller, 6 on the Hub Controller (the forms show it as a bit string); null = not captured" }
      task_call_frequency: { type: integer, nullable: true, minimum: 1, description: "TaskCallFrequencyArBy (1, 10 or 100
 so far); null = not captured" }
      user_groups: { type: array, description: "The numeric user-group levels (UserGroupReadArBy / WriteArBy), one entry
per slot 0-15; omitted = unchanged", items: { type: object, required: [slot, read, write], properties: { slot: { type: inte
ger, minimum: 0, maximum: 15 }, read: { type: integer, minimum: 0, maximum: 255 }, write: { type: integer, minimum: 0, maxi
mum: 255 } } } }
      development_access: { type: string, nullable: true }
      sales_access: { type: string, nullable: true }
      tcc_access: { type: string, nullable: true }
      service_access: { type: string, nullable: true }
      turbine_operator_package_access: { type: string, nullable: true }
      grid_operator_access: { type: string, nullable: true }
      service_partner_access: { type: string, nullable: true }
      customer_access: { type: string, nullable: true }
      remedy_text: { type: string, nullable: true, maxLength: 4000 }
      affected_component: { type: string, nullable: true, maxLength: 200 }
      related_status_codes: { type: string, nullable: true, maxLength: 500, description: "Identifiers like StCd-01234 sep
arated by commas" }
      standard_references: { type: string, nullable: true, maxLength: 1000 }
      applies_from_version: { type: string, nullable: true, description: "Controller software version code or id" }
      applies_to_version: { type: string, nullable: true }
```

```
      planned_effective_date: { type: string, format: date, nullable: true }
      planned_deprecation_date: { type: string, format: date, nullable: true }
  StatusCodeRevisionCreate:
    allOf:
      - { $ref: '#/components/schemas/StatusCodeCreate' }
      - type: object
        properties: { justification: { type: string } }
  StatusCodeRevisionUpdate:
    type: object
    description: >
      The editable StatusCodeCreate fields (a full representation, as before), plus the optimistic
      concurrency tokens (I-01). Send `expected_row_version` (and/or `expected_updated_at`) from the revision you
      loaded: if somebody saved in between the response is 409 `conflict` with `current_row_version` and `current`
      (the stored revision) so the client can merge; nothing is overwritten. Omit both for the old last-write-wins behavi
our.
    properties:
      expected_row_version: { type: integer }
      expected_updated_at: { type: string, format: date-time }
  StatusCodeList:
    type: object
    properties:
      items: { type: array, items: { $ref: '#/components/schemas/StatusCodeRevision' } }
      total: { type: integer }
      page: { type: integer }
      page_size: { type: integer }
  RevisionDiff:
    type: object
    properties:
      status_code_id: { type: integer }
      from_revision: { type: integer }
      to_revision: { type: integer }
      field_changes:
        type: array
        items:
          type: object
          properties: { field: { type: string }, old_value: {}, new_value: {} }
  SignoffDecision:
    type: object
    required: [decision, current_password]
    properties:
      decision: { type: string, enum: [APPROVE, REJECT] }
      comment: { type: string }
      current_password: { type: string, format: password, description: "The caller's own password, re-entered for this ac
tion (I-17, SEC-026). 403 step_up_required / step_up_failed otherwise; a wrong one counts toward account lockout." }
      checklist_confirmed: { type: array, items: { type: integer }, description: "Reviewer sign-off APPROVE only: ids of
the review-checklist items confirmed (all applicable ones are required, I-23)" }
  AdminSignoffDecision:
    allOf:
      - $ref: '#/components/schemas/SignoffDecision'
      - type: object
        properties:
          access_overrides:
            type: object
            description: >
              FR-034 / SEC-013: confirm-or-override the eight Engineer-proposed audience-access
              fields. Keys are field names (development_access, sales_access, tcc_access,
              service_access, turbine_operator_package_access, grid_operator_access,
              service_partner_access, customer_access); values are lookup ids, or null to clear.
              Applied only on APPROVE; fields not named stay as proposed.
            additionalProperties: { type: integer, nullable: true }
  DomainSignoffRequest:
    type: object
    properties:
      id: { type: integer }
      change_request_id: { type: integer }
      engineering_domain: { type: string }
      requested_by: { type: integer }
      target_user_id: { type: integer, nullable: true }
      created_at: { type: string, format: date-time }
  DomainSignoff:
    type: object
    properties:
      id: { type: integer }
      engineering_domain: { type: string }
      signed_off_by: { type: integer, nullable: true }
      signed_off_at: { type: string, format: date-time, nullable: true }
      status: { type: string, enum: [REQUIRED_PENDING, SIGNED_OFF, NOT_REQUIRED] }
  ChangeRequest:
    type: object
    properties:
      id: { type: integer }
      status_code_revision_id: { type: integer }
      cr_type: { type: string, enum: [NEW, REVISION, DEPRECATION, IMPORT] }
      requested_by: { type: integer }
      justification: { type: string, nullable: true }
      import_job_id: { type: integer, nullable: true }
      state: { type: string, enum: [DRAFT, REVIEW, PENDING_APPROVAL, APPROVED, REJECTED, WITHDRAWN] }
      submitted_at: { type: string, format: date-time, nullable: true }
      chief_engineer_decided_at: { type: string, format: date-time, nullable: true }
```

```
      chief_engineer_decided_by: { type: integer, nullable: true }
      assigned_reviewer_id: { type: integer, nullable: true }
      comments: { type: array, items: { $ref: '#/components/schemas/ReviewComment' } }
      domain_signoffs: { type: array, items: { $ref: '#/components/schemas/DomainSignoff' } }
      warnings: { type: array, items: { $ref: '#/components/schemas/Warning' }, description: "Submit response only (I-04,
 I-12)" }
  ReviewComment:
    type: object
    properties:
      id: { type: integer }
      author_id: { type: integer }
      comment_text: { type: string }
      decision:
        type: string
        nullable: true
        enum: [REVIEWER_APPROVE, REVIEWER_REJECT, ADMIN_APPROVE, ADMIN_REJECT, CHIEF_ENGINEER_APPROVE, CHIEF_ENGINEER_REJ
ECT]
      created_at: { type: string, format: date-time }
  ChangeRequestList:
    type: object
    properties:
      items: { type: array, items: { $ref: '#/components/schemas/ChangeRequest' } }
      total: { type: integer }
      page: { type: integer }
      page_size: { type: integer }
  UserUsage:
    type: object
    properties:
      user_id: { type: integer }
      references: { type: object, additionalProperties: { type: integer }, description: "table.column -> rows that point
at the account" }
      reference_total: { type: integer }
      deletable: { type: boolean, description: "true only when nothing but the account's own rows exist (BR-025)" }
  MappingRow:
    type: object
    description: One row of the mapping - only what the .st programs do not carry (FR-191)
    properties:
      id: { type: integer }
      identifier: { type: string, example: "StCd-01000" }
      family: { type: string, enum: [MC, HC] }
      legacy_number: { type: integer, nullable: true }
      new_number: { type: integer }
      description: { type: string, description: "The English title" }
      functional_group: { type: string, nullable: true }
      logical_node: { type: string, nullable: true }
      status_type: { type: string, enum: [ERROR, WARNING, INFO], nullable: true }
      up_down_counter: { type: integer, enum: [0, 1], nullable: true }
      parameter_set_delay: { type: string, nullable: true, description: "The value of the parameter that sets the delay,
e.g. *5s" }
      parameter_reset_delay: { type: string, nullable: true }
      development_access: { type: string, nullable: true }
      sales_access: { type: string, nullable: true }
      tcc_access: { type: string, nullable: true }
      service_access: { type: string, nullable: true }
      turbine_operator_package_access: { type: string, nullable: true }
      grid_operator_access: { type: string, nullable: true }
      service_partner_access: { type: string, nullable: true }
      customer_access: { type: string, nullable: true }
      cause: { type: string, nullable: true }
      directive: { type: string, nullable: true }
      note: { type: string, nullable: true }
      status: { type: string, enum: [PROPOSED, FROM_PROGRAM, APPROVED, ISSUED, REJECTED] }
      source: { type: string, nullable: true }
      version: { type: integer }
      updated_at: { type: string }
  MappingPage:
    type: object
    properties:
      summary: { type: object }
      total: { type: integer }
      page: { type: integer }
      rows: { type: array, items: { $ref: '#/components/schemas/MappingRow' } }
  MappingChange:
    type: object
    properties:
      id: { type: integer }
      title: { type: string }
      reason: { type: string }
      origin: { type: string, enum: [FORM, WORKBOOK, SELECTION] }
      state: { type: string, enum: [OPEN, APPLIED, REJECTED, WITHDRAWN] }
      requested_by: { type: string }
      requested_at: { type: string }
      closed_at: { type: string, nullable: true }
      rows: { type: integer }
      approvals:
        type: array
        description: Always three entries, one per role; decision is null while the role has not decided
        items:
          type: object
```

```
          properties:
            role: { type: string, enum: [STCD_OWNER, CHIEF_ENGINEER, ADMINISTRATOR] }
            decision: { type: string, enum: [APPROVE, REJECT], nullable: true }
            by: { type: string, nullable: true }
            at: { type: string, nullable: true }
            comment: { type: string, nullable: true }
      items:
        type: array
        description: Only in the detail and after a request or a decision
        items:
          type: object
          properties:
            action: { type: string, enum: [ADD, CHANGE, CONFIRM, REJECT] }
            mapping_id: { type: integer, nullable: true }
            new_number: { type: integer }
            changed_fields: { type: array, items: { type: string } }
            after: { type: object }
  ImportJob:
    type: object
    properties:
      id: { type: integer }
      requested_by: { type: integer }
      source_format: { type: string, enum: [CSV, XLSX, ST] }
      source_file_reference: { type: string }
      row_count: { type: integer }
      status: { type: string, enum: [PENDING, VALIDATING, COMPLETE, FAILED] }
      created_count: { type: integer }
      error_count: { type: integer }
      error_report_reference: { type: string, nullable: true }
      created_at: { type: string, format: date-time }
      completed_at: { type: string, format: date-time, nullable: true }
      import_mode: { type: string, enum: [CREATE, UPDATE] }
      processed_count: { type: integer, description: "Rows handled so far (I-32)" }
      warning_count: { type: integer }
      progress_percent: { type: integer }
      error_report_xlsx_available: { type: boolean }
      change_set_id: { type: integer, nullable: true, description: "The batch change set of a status code import from .st
 programs (v4.2)" }
  StImportPreview:
    type: object
    properties:
      token: { type: string }
      options: { type: object }
      summary:
        type: object
        properties:
          files: { type: integer }
          programs: { type: integer }
          ready: { type: integer }
          changed: { type: integer }
          restorable: { type: integer }
          exists: { type: integer }
          blocked: { type: integer }
          warnings: { type: integer }
          with_legacy: { type: integer }
          without_legacy: { type: integer, description: "Programs with no legacy number - the new set is larger than the
old one" }
      items:
        type: array
        items:
          type: object
          properties:
            key: { type: string }
            file: { type: string }
            line: { type: integer }
            program_number: { type: integer, description: "StCd_<n> as written in the file (the legacy number for a Hub p
rogram)" }
            controller: { type: string, enum: [MC, HC] }
            identifier: { type: string, nullable: true, description: "The new identifier the code will be issued" }
            legacy_number: { type: integer, nullable: true }
            name: { type: string }
            group: { type: string, nullable: true }
            subgroup: { type: string, nullable: true }
            prg_idx: { type: integer, nullable: true }
            state: { type: string, enum: [ready, changed, restorable, exists, blocked], description: "ready = new; change
d = differs from the code in the catalogue (update the Draft or open a new revision); restorable = the code's change reques
t was withdrawn (open a new Draft on the same number); exists = unchanged; blocked = problems" }
            action: { type: string, enum: [create, update, revise, restore], nullable: true }
            changes: { type: array, items: { type: string }, description: "Fields in which a changed program differs from
 the catalogue" }
            issues: { type: array, items: { type: object, properties: { column: { type: string }, message: { type: string
 } } } }
            warnings: { type: array, items: { type: string } }
            existing: { type: object, nullable: true }
      file_errors: { type: array, items: { type: object } }
      notes: { type: array, items: { type: string } }
      expires_hours: { type: integer }
  LegacyMapping:
    type: object
```

```
    properties:
      summary:
        type: object
        properties:
          codes_total: { type: integer }
          with_legacy_number: { type: integer }
          new_without_legacy: { type: integer }
          by_controller: { type: object }
          identifier_issued: { type: integer }
          identifier_reserved: { type: integer }
          identifier_pending: { type: integer }
      entries:
        type: array
        items:
          type: object
          properties:
            status_code_id: { type: integer }
            controller: { type: string, enum: [MC, HC] }
            legacy_number: { type: string }
            new_identifier: { type: string, nullable: true }
            identifier_state: { type: string, enum: [issued, reserved, pending] }
            title: { type: string }
            lifecycle_state: { type: string }
            group: { type: string }
            subgroup: { type: string }
  ImportDryRun:
    type: object
    properties:
      mode: { type: string, enum: [CREATE, UPDATE] }
      source_format: { type: string }
      row_count: { type: integer }
      would_create: { type: integer }
      would_update: { type: integer }
      unchanged: { type: integer }
      error_count: { type: integer }
      warning_count: { type: integer }
      ok: { type: boolean }
      rows:
        type: array
        items:
          type: object
          properties:
            row: { type: integer }
            action: { type: string, enum: [create, update, skip, error] }
            errors: { type: array, items: { type: object, properties: { column: { type: string }, value: { type: string }
, message: { type: string }, hint: { type: string } } } }
            warnings: { type: array, items: { type: string } }
            status_code_identifier: { type: string, nullable: true }
  Release:
    type: object
    properties:
      id: { type: integer }
      name: { type: string }
      version_label: { type: string }
      description: { type: string, nullable: true }
      scope_filter: { type: object }
      status: { type: string, enum: [BUILDING, PUBLISHED] }
      created_by: { type: integer }
      created_at: { type: string, format: date-time }
      published_by: { type: integer, nullable: true }
      published_at: { type: string, format: date-time, nullable: true }
      item_count: { type: integer }
      target_software_versions: { type: array, items: { type: string }, description: "I-07" }
      warnings: { type: array, items: { $ref: '#/components/schemas/Warning' }, description: "Publish response only: plan
ned effective dates still in the future (I-27)" }
  ReleaseCreate:
    type: object
    required: [name, version_label]
    properties:
      name: { type: string }
      version_label: { type: string }
      description: { type: string }
      scope_filter: { type: object }
      target_software_versions: { type: array, items: { type: string } }
  AuditLogEntry:
    type: object
    properties:
      id: { type: integer }
      entity_type: { type: string }
      entity_id: { type: integer }
      action: { type: string }
      actor_id: { type: integer, nullable: true }
      occurred_at: { type: string, format: date-time }
      before_value: { type: object, nullable: true }
      after_value: { type: object, nullable: true }
      ip_address: { type: string, nullable: true }
      request_id: { type: string }
  # ------------------------------------------------------------------------- v4 (2026-10-22)
  Readiness:
```

```
    type: object
    properties:
      status: { type: string, enum: [ok, degraded, unavailable] }
      version: { type: string, description: "Version of the running application (FR-184)" }
      checks: { type: object, description: "database, schema, audit_chain, backup - each with a `state`" }
  SimilarCode:
    type: object
    properties:
      status_code_id: { type: integer }
      status_code_identifier: { type: string, nullable: true }
      title: { type: string }
      lifecycle_status: { $ref: '#/components/schemas/LifecycleStatus' }
      score: { type: number, description: "0..1 text similarity of the titles" }
  Warning:
    type: object
    description: A soft finding that never blocks a save - possible duplicate (I-12) or an unresolvable / deprecated refe
rence (I-04)
    properties:
      type: { type: string, enum: [POSSIBLE_DUPLICATE, UNKNOWN_REFERENCE, DEPRECATED_REFERENCE, ARCHIVED_REFERENCE, SELF_
REFERENCE, FUTURE_EFFECTIVE_DATE] }
      field: { type: string }
      message: { type: string }
  WhereUsedEntry:
    type: object
    properties:
      status_code_id: { type: integer }
      status_code_identifier: { type: string, nullable: true }
      title: { type: string }
      lifecycle_status: { $ref: '#/components/schemas/LifecycleStatus' }
      relation: { type: string, enum: [REPEATED_ERROR, SUPERSEDED_BY, RELATED, PLATFORM_VARIANT] }
  FamilyMember:
    type: object
    properties:
      status_code_id: { type: integer }
      status_code_identifier: { type: string, nullable: true }
      title: { type: string }
      turbine_platforms: { type: array, items: { type: string } }
      lifecycle_status: { $ref: '#/components/schemas/LifecycleStatus' }
      platform_variant_of_status_code_id: { type: integer, nullable: true }
  HubMappingReport:
    type: object
    properties:
      total: { type: integer }
      tally: { type: object, description: "OPEN, MAPPED_DRAFT, MAPPED_LIVE, HUB_RETIRED counts" }
      ready_to_retire: { type: boolean }
      items:
        type: array
        items:
          type: object
          properties:
            hub_status_code_id: { type: integer }
            hub_identifier: { type: string, nullable: true }
            hub_title: { type: string }
            hub_lifecycle: { type: string }
            legacy_reference_number: { type: string, nullable: true }
            mc_status_code_id: { type: integer, nullable: true }
            mc_identifier: { type: string, nullable: true }
            mc_lifecycle: { type: string, nullable: true }
            state: { type: string, enum: [OPEN, MAPPED_DRAFT, MAPPED_LIVE, HUB_RETIRED] }
  SlaInfo:
    type: object
    nullable: true
    properties:
      stage: { type: string, enum: [review, approval] }
      entered_at: { type: string, format: date-time }
      age_days: { type: integer, description: "Business days (Mon-Fri) waited at this stage" }
      level: { type: string, enum: [ok, amber, red] }
      amber_days: { type: integer }
      red_days: { type: integer }
  StalledRequest:
    allOf:
      - { $ref: '#/components/schemas/SlaInfo' }
      - type: object
        properties:
          change_request_id: { type: integer }
          title: { type: string }
          assigned_reviewer_id: { type: integer, nullable: true }
  FieldComment:
    type: object
    properties:
      id: { type: integer }
      change_request_id: { type: integer }
      field: { type: string }
      author_id: { type: integer }
      comment_text: { type: string }
      created_at: { type: string, format: date-time }
      resolved_by: { type: integer, nullable: true }
      resolved_at: { type: string, format: date-time, nullable: true }
  ChecklistItem:
```

```
    type: object
    properties:
      id: { type: integer }
      functional_system_group_id: { type: integer, nullable: true, description: "NULL = applies to every group" }
      text: { type: string }
      is_active: { type: boolean }
      sort_order: { type: integer }
  Attachment:
    type: object
    properties:
      id: { type: integer }
      change_request_id: { type: integer }
      filename: { type: string }
      content_type: { type: string }
      size_bytes: { type: integer }
      sha256: { type: string }
      description: { type: string, nullable: true }
      uploaded_by: { type: integer }
      uploaded_at: { type: string, format: date-time }
  ChangeSet:
    type: object
    properties:
      id: { type: integer }
      name: { type: string }
      description: { type: string, nullable: true }
      target_software_version: { type: string, nullable: true }
      status: { type: string, enum: [OPEN, CLOSED] }
      created_by: { type: integer }
      created_at: { type: string, format: date-time }
      closed_at: { type: string, format: date-time, nullable: true }
      change_request_ids: { type: array, items: { type: integer } }
  ChangeSetOverview:
    allOf:
      - { $ref: '#/components/schemas/ChangeSet' }
      - type: object
        properties:
          tally: { type: object, description: "CRs per lifecycle state" }
          items:
            type: array
            items:
              type: object
              properties:
                change_request_id: { type: integer }
                status_code_identifier: { type: string, nullable: true }
                title: { type: string }
                state: { type: string }
                sla: { $ref: '#/components/schemas/SlaInfo' }
  BulkDecision:
    type: object
    required: [step, decision, current_password]
    properties:
      step: { type: string, enum: [reviewer, admin, chief-engineer] }
      decision: { type: string, enum: [APPROVE, REJECT] }
      comment: { type: string }
      current_password: { type: string, format: password }
      change_request_ids: { type: array, items: { type: integer }, description: "Omit to decide every CR of the set that
waits for this step" }
  Delegation:
    type: object
    properties:
      id: { type: integer }
      delegator_id: { type: integer }
      delegate_id: { type: integer }
      role: { type: string, enum: [REVIEWER, ADMINISTRATOR, CHIEF_ENGINEER] }
      valid_from: { type: string, format: date }
      valid_to: { type: string, format: date }
      created_by: { type: integer }
      created_at: { type: string, format: date-time }
      revoked_at: { type: string, format: date-time, nullable: true }
  Notification:
    type: object
    properties:
      id: { type: integer }
      kind: { type: string, example: SLA_REMINDER }
      message: { type: string }
      link: { type: string, nullable: true }
      change_request_id: { type: integer, nullable: true }
      created_at: { type: string, format: date-time }
      read_at: { type: string, format: date-time, nullable: true }
  Setting:
    type: object
    properties:
      key: { type: string, example: 'sla.review.amber_days' }
      value: { type: string }
      default: { type: string }
      description: { type: string }
  CatalogueHealth:
    type: object
    properties:
```

```
      generated_at: { type: string }
      capacity: { type: array, items: { $ref: '#/components/schemas/CapacityRow' } }
      capacity_alerts: { type: array, items: { $ref: '#/components/schemas/CapacityRow' } }
      missing_data: { type: object }
      inconsistencies: { type: array, items: { type: object } }
      unused_numbers: { type: array, items: { type: object } }
      upcoming: { type: object }
  CapacityRow:
    type: object
    properties:
      functional_system_group: { type: string }
      controller: { type: string, enum: [MC, HC], description: "Main Controller or Hub Controller group (the health page
colours them blue and light blue)" }
      functional_subgroup: { type: string }
      label: { type: string }
      range_start: { type: integer }
      range_end: { type: integer }
      total: { type: integer }
      allocated: { type: integer }
      remaining: { type: integer }
      percent_remaining: { type: number }
      alert: { type: boolean }
      exhausted: { type: boolean }
  OccurrenceImport:
    type: object
    properties:
      id: { type: integer }
      source_label: { type: string }
      period_start: { type: string, format: date }
      period_end: { type: string, format: date }
      imported_by: { type: integer }
      imported_at: { type: string, format: date-time }
      matched_codes: { type: integer }
      unmatched_rows: { type: integer }
  AuditChainResult:
    type: object
    properties:
      ok: { type: boolean }
      checked: { type: integer }
      head_hash: { type: string, nullable: true }
      first_bad_id: { type: integer, nullable: true }
      reason: { type: string }
      verified_at: { type: string, format: date-time }
  ExportProfile:
    type: object
    properties:
      id: { type: integer }
      code: { type: string, example: SERVICE_MANUAL }
      name: { type: string }
      description: { type: string, nullable: true }
      columns:
        type: array
        items: { type: object, properties: { field: { type: string }, label: { type: string } } }
      language: { type: string, nullable: true }
      audience: { type: string, nullable: true }
      title_page: { type: boolean }
      pdf_orientation: { type: string, enum: [L, P] }
      is_active: { type: boolean }
  ExportProfileInput:
    type: object
    properties:
      code: { type: string }
      name: { type: string }
      description: { type: string }
      columns:
        type: array
        items: { type: object, required: [field, label], properties: { field: { type: string }, label: { type: string } }
 }
      language: { type: string, nullable: true }
      audience: { type: string, nullable: true }
      title_page: { type: boolean }
      pdf_orientation: { type: string, enum: [L, P] }
      is_active: { type: boolean }
  BulkSample:
    type: object
    properties:
      id: { type: integer }
      change_set_id: { type: integer }
      step: { type: string }
      domain: { type: string, nullable: true }
      status: { type: string, enum: [OPEN, PASSED, FAILED, USED, VOID] }
      population: { type: integer }
      sampled: { type: integer }
      rate_percent: { type: integer }
      drawn_by: { type: integer }
      drawn_at: { type: string }
      used_at: { type: string, nullable: true }
      items: { type: array, items: { type: object } }
```

```
    PackageJob:
      type: object
      properties:
        id: { type: integer }
        status: { type: string, enum: [PENDING, COMPLETE, FAILED] }
        done: { type: integer, description: Files written so far }
        total: { type: integer }
        error: { type: string, nullable: true }
        size_bytes: { type: integer, nullable: true }
        sha256: { type: string, nullable: true }
        created_at: { type: string }
        completed_at: { type: string, nullable: true }

    ExportRecord:
      type: object
      properties:
        id: { type: integer }
        export_type: { type: string }
        scope_type: { type: string }
        sha256: { type: string, nullable: true }
        size_bytes: { type: integer, nullable: true }
        row_count: { type: integer, nullable: true }
        profile: { type: string, nullable: true }
        audience: { type: string, nullable: true }
        language: { type: string, nullable: true }
        software_version: { type: string, nullable: true }
        requested_by: { type: integer }
        created_at: { type: string, format: date-time }
    Manifest:
      type: object
      description: "What an export is, with the SHA-256 of every file. `signature` is an HMAC-SHA-256 when MANIFEST_SIGNING
  _KEY is configured, else null."
      properties:
        manifest_version: { type: integer }
        kind: { type: string, enum: [catalogue, release-notes, offline-bundle] }
        generated_at: { type: string, format: date-time }
        generated_by: { type: object }
        release: { type: object, nullable: true }
        scope: { type: object }
        format: { type: string }
        profile: { type: string, nullable: true }
        audience: { type: string, nullable: true }
        language: { type: string, nullable: true }
        software_version: { type: string, nullable: true }
        row_count: { type: integer }
        files:
          type: array
          items: { type: object, properties: { name: { type: string }, sha256: { type: string }, size_bytes: { type: intege
  r } } }
        content_digest: { type: string, description: "SHA-256 of the data rows, independent of the file format" }
        algorithm: { type: string, example: SHA-256 }
        signature: { type: object, nullable: true }
    VerifyResult:
      type: object
      properties:
        sha256: { type: string }
        recorded: { type: boolean }
        export_job_ids: { type: array, items: { type: integer } }
        manifest:
          type: object
          properties:
            file_sha256: { type: string }
            file_listed: { type: boolean }
            file_name_matches: { type: boolean }
            signature: { type: string, enum: [valid, invalid, unsigned, no-key] }
            ok: { type: boolean }
    TranslationStatus:
      type: object
      properties:
        language: { type: string }
        label: { type: string }
        status: { type: string, enum: [MISSING, DRAFT, APPROVED] }
        title: { type: string, nullable: true }
        description: { type: string, nullable: true }
        remedy_text: { type: string, nullable: true }
        translated_by: { type: integer, nullable: true }
        approved_by: { type: integer, nullable: true }
    Translation:
      type: object
      properties:
        revision_id: { type: integer }
        language: { type: string }
        status: { type: string, enum: [DRAFT, APPROVED] }
        title: { type: string }
        description: { type: string }
        remedy_text: { type: string, nullable: true }
        translated_by: { type: integer }
        translated_at: { type: string, format: date-time }
        approved_by: { type: integer, nullable: true }
```

```
      approved_at: { type: string, format: date-time, nullable: true }
  SavedSearch:
    type: object
    properties:
      id: { type: integer }
      name: { type: string }
      filters: { type: object }
      columns: { type: array, items: { type: string } }
      query_string: { type: string, description: "Append to /status-codes to reproduce the view (shareable link)" }
      created_at: { type: string, format: date-time }
  ApiToken:
    type: object
    properties:
      id: { type: integer }
      name: { type: string }
      token_prefix: { type: string }
      scopes: { type: array, items: { type: string } }
      expires_at: { type: string, format: date-time }
      last_used_at: { type: string, format: date-time, nullable: true }
      revoked_at: { type: string, format: date-time, nullable: true }
      created_at: { type: string, format: date-time }
  ApiTokenCreated:
    allOf:
      - { $ref: '#/components/schemas/ApiToken' }
      - type: object
        properties:
          token: { type: string, description: "The secret - shown exactly once" }
  Webhook:
    type: object
    properties:
      id: { type: integer }
      name: { type: string }
      url: { type: string }
      events: { type: array, items: { type: string } }
      is_active: { type: boolean }
      last_delivery_at: { type: string, format: date-time, nullable: true }
      last_status: { type: string, nullable: true }
      created_at: { type: string, format: date-time }
  WebhookCreated:
    allOf:
      - { $ref: '#/components/schemas/Webhook' }
      - type: object
        properties:
          secret: { type: string, description: "Signing secret - shown exactly once" }
  ChangeFeed:
    type: object
    properties:
      items:
        type: array
        items:
          type: object
          properties:
            id: { type: integer }
            event: { type: string, enum: [code.released, code.deprecated, code.reinstated, code.archived] }
            occurred_at: { type: string, format: date-time }
            status_code_identifier: { type: string, nullable: true }
            title: { type: string }
            revision_number: { type: integer }
            release: { type: object, nullable: true }
      next_since_id: { type: integer }
      has_more: { type: boolean }
```

Appendix B - Database schema (SQL)

The schema every migration builds on.

```
-- ============================================================================
-- CSCM Tool ? SQLite Physical Schema (MVP) ? v4
-- Referenced by: docs/07-database-design.md, docs/06-data-dictionary.md
-- Target: SQLite 3.35+ (requires partial index / generated column support)
-- v3 changes (2026-09-17), grounded in a real export of the live status code
-- library (897 rows, 9 sheets) and the legacy tool's screenshot:
--   - Functional Subgroup renamed with real values and made MANDATORY
--     (was nullable in v2); Administrator can add new subgroups later
--   - status_category simplified to ERROR/WARNING/INFO
--   - available_group is a plain integer code, not a lettered enum
--   - alarm is a boolean, not a 4-value behaviour enum
--   - set_delay/reset_delay are free-text-with-unit, not integer seconds
--   - reset/brake/yaw/converter "program" fields are numeric passthroughs,
--     not governed enums; manual and automatic reset are separate fields
--   - access_rights_id (single enum) replaced by 8 independent
--     audience-access lookups (development/sales/tcc/service/turbine
--     operator package/grid operator/service partner/customer)
--   - new fields: up_down_counter, trigger_snapshot,
--     loadless_spinning_permitted, number_of_times_repeated,
--     repeated_over_the_course, status_code_for_repeated_error (opaque,
--     NOT a foreign key), logical_node/node_value (derived/read-only),
--     legacy_reference_number
--   - new import_job table (CSV/XLSX library import -> Draft Change
--     Requests, full governance still applies)
--   - export_job.export_type gains XLSX
--   - software_version re-added as a nullable field on status_code_revision
--     (was dropped without confirmation in an earlier pass -- open item,
--     may belong on `release` instead; see docs/06-data-dictionary.md)
--   - lookup_functional_system_group.code is NO LONGER unique on its own:
--     the Hub Controller library (range 11000-13999) reuses WTUR and WROT
--     for two new, numerically distinct groups; uniqueness moved to
--     (code, range_start). Three new groups added: WTUR (11000-11999,
--     Hub Control System), WROT (12000-12999, Hub Rotor & Pitch), WPPD
--     (13000-13999, Wind Farm) -- 11 new subgroups under them
--   - trg_prevent_group_overlap added, mirroring the existing subgroup
--     overlap trigger one level up -- Administrators can add whole new
--     Functional System Groups into the confirmed free ranges (10000-10999,
--     14000-14999, 15000-15999, 16000-16999) without a schema change
-- v4 changes (2026-10-22, improvement pack I-01..I-36, docs/21-improvement-ideas.md):
--   - status_code_revision: row_version/updated_at (optimistic concurrency, I-01);
--     remedy_text/affected_component/related_status_codes/standard_references (I-14);
--     applies_from/to_version_id (I-07); planned_effective/deprecation_date (I-27)
--   - lookup_controller_software_version, release_target_version (I-07);
--     lookup_language, revision_translation (I-15)
--   - lookup_*_access.audience_visible (I-03); lookup_functional_system_group.
--     controller_type MAIN|HUB and hub_mc_mapping (I-36)
--   - export_profile (I-02); export_job gains sha256/size/profile/audience/manifest
--     columns and wider type/scope CHECKs (I-06, I-24); import_job progress/mode (I-05, I-32)
--   - change_set (+ change_request.change_set_id, I-11); cr_attachment (I-13);
--     delegation (I-16); field_comment, review_checklist_item,
--     cr_checklist_confirmation (I-23); saved_search (I-26); notification (I-08);
--     login_attempt (I-20); app_setting; occurrence_import/code_occurrence (I-28);
--     api_token, webhook, webhook_delivery (I-29); user.ui_language (I-22)
--   - audit_log_entry.prev_hash/entry_hash: tamper-evident hash chain (I-18)
--   - the optional FTS5 search index lives in schema-fts5.sql (I-31)
-- v4.1 changes (2026-10-23, derived from 14 generated StCd_*.st programs: Main Controller
-- 1000-1010 and Hub Controller 14000/14005/14025, cross-checked against the status code workbooks;
-- docs/06-data-dictionary.md sections 1, 2c, 2d, 2f, 9d.9):
--   - status_code.prg_idx: the PrgIdx_GUi block identification, entered manually, 1..65535 (UINT),
--     unique among non-sandbox codes, frozen once a revision was released
--   - status_code_revision: 13 further Main Controller subsystem programs, 3 Hub Controller
--     programs (pitch error/warning, rotor brake), 5 Hub operation flags and the operating-mode
--     mask / task frequency / availability-calculation / zero-load parameters (all optional)
--   - lookup_user_group_slot + status_code_revision_user_group: the 16 numeric user-group slots of
--     the controller (slots 1..8 = the eight audiences) with a read and a write level per revision;
--     lookup_*_access.default_read_level / default_write_level
--   - trg_prevent_prg_idx_change_after_release, trg_require_prg_idx_for_review (inert until
--     app_setting 'status_code.require_prg_idx' = '1')
-- v4.2 changes (2026-10-23, status code import from the generated StCd_*.st programs,
-- docs/03-srs.md section 1.16, docs/06-data-dictionary.md section 5a):
--   - status_code.reserved_identifier: the number an imported program already carries; the
--     identifier issued at Submit for Review must be exactly this one, no other code may take it
--     (unique among non-sandbox codes; two triggers guard the reservation)
--   - import_job.source_format gains 'ST' and import_job.change_set_id links the batch change set
--   - import_source_item: one row per imported program (file, SHA-256, parsed attributes as JSON)
-- v4.3 changes (2026-10-23, roles and languages):
--   - role 'STCD_OWNER' (id 6): owner of the status code library; may change the role of a user
--     between VIEWER, ENGINEER and REVIEWER only, never creates users (docs/11-security-architecture.md
--     section 3, docs/03-srs.md FR-156, BR-024)
--   - lookup_language: French, Hindi and Georgian added; v4.4 adds Arabic (written right to left); every
--     name reads <country letters> <name> <flag emoji>; user.ui_language accepts the six codes
-- v4.17 changes: the legacy -> new status code mapping kept in the database and changed only with the three
--     approvals of the StCd Owner, the Chief Engineer and an Administrator (status_code_mapping, mapping_change,
```

```
--     mapping_change_item, mapping_approval)
-- v4.13 changes: the sample inspection of a bulk decision (bulk_sample, bulk_sample_item)
-- v4.9.2 changes: status_code.deleted_at (a withdrawn code is erased in place, never removed; the delete
--     trigger is strict again)
-- v4.9 changes: the technical nomenclature (terminology_term, terminology_translation); user.deleted_at (a
--     deleted account with history is erased, not removed: BR-025); status_code.withdrawn_at (a code whose
--     change request was withdrawn can be deleted by an Administrator: BR-026)
-- v4.7 changes (2026-10-24): trg_user_email_immutable (the e-mail address of a user is fixed); the tables of the
--     in-application translation review (language_reviewer, translation_review, translation_review_item,
--     translation_override)
-- v4.6 changes (2026-10-23): Spanish, Italian and Danish; Georgian moves to the end of the language list
-- v4.5 changes (2026-10-23): ten more languages (Bengali, Marathi, Telugu, Tamil, Gujarati, Urdu, Kannada, Odia,
--     Malayalam, Punjabi); user.ui_language is a two-letter code (the CHECK no longer lists them, so the next
--     language needs no table rebuild)
-- ============================================================================

PRAGMA foreign_keys = ON;

-- ----------------------------------------------------------------------------
-- Reference / lookup tables
-- ----------------------------------------------------------------------------

CREATE TABLE role (
    id          INTEGER PRIMARY KEY,
    code        TEXT NOT NULL UNIQUE,      -- ADMINISTRATOR | STCD_OWNER | ENGINEER | REVIEWER | CHIEF_ENGINEER | VIEWER
    label       TEXT NOT NULL
);
INSERT INTO role (id, code, label) VALUES
    (1, 'ADMINISTRATOR', 'Administrator (Product Owner Controls Software)'),
    (2, 'ENGINEER', 'Engineer'),
    (3, 'REVIEWER', 'Reviewer (Sub-PO for Status Codes)'),
    (4, 'CHIEF_ENGINEER', 'Chief Engineer'),
    (5, 'VIEWER', 'Viewer'),
    (6, 'STCD_OWNER', 'StCd Owner (Owner of the Status Code Library)');   -- v4.3

CREATE TABLE lookup_engineering_domain (
    id INTEGER PRIMARY KEY, code TEXT NOT NULL UNIQUE, label TEXT NOT NULL,
    description TEXT, is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0
);
INSERT INTO lookup_engineering_domain (code, label, sort_order) VALUES
    ('CONTROLS','Controls Engineering',1), ('ELECTRICAL','Electrical Engineering',2),
    ('MECHANICAL','Mechanical Engineering',3), ('GRID','Grid / Power Systems Engineering',4),
    ('SAFETY','Functional Safety Engineering',5);

-- NOTE (2026-09-17): `code` is intentionally NOT unique on its own. The Hub
-- Controller status code library (range 11000-13999) reuses the WTUR and
-- WROT prefixes for its own, numerically distinct groups (confirmed by the
-- business: same thematic prefix, different status code range) alongside a
-- new WPPD prefix for Wind Farm. Uniqueness is therefore on (code,
-- range_start), not code alone ? always resolve a group by its numeric
-- range, never by prefix alone, in application code.
CREATE TABLE lookup_functional_system_group (
    id INTEGER PRIMARY KEY, code TEXT NOT NULL, label TEXT NOT NULL,
    range_start INTEGER NOT NULL, range_end INTEGER NOT NULL,
    description TEXT, is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0,
    controller_type TEXT NOT NULL DEFAULT 'MAIN' CHECK (controller_type IN ('MAIN','HUB')),   -- I-36
    UNIQUE (code, range_start),
    CHECK (range_end > range_start)
);
INSERT INTO lookup_functional_system_group (code, label, range_start, range_end, sort_order) VALUES
    ('WCNV','Converter & Grid Interface',1000,1999,1),
    ('WGEN','Generator System',2000,2999,2),
    ('WNAC','Meteorology & Nacelle Environment',3000,3999,3),
    ('WROT','Rotor & Pitch System',4000,4999,4),
    ('WTOW','Tower & Structure',5000,5999,5),
    ('WTRF','Transformer & MV System',6000,6999,6),
    ('WTRM','Drivetrain & Gearbox',7000,7999,7),
    ('WYAW','Yaw System',8000,8999,8),
    ('WTUR','Turbine Control & Operation',9000,9999,9),
    -- Hub Controller groups (new 2026-09-17, source: "HC Status Code Number.xlsx")
    ('WTUR','Hub Controller Control System',11000,11999,10),
    ('WROT','Hub Controller Rotor & Pitch',12000,12999,11),
    ('WPPD','Wind Farm / Plant Dispatch',13000,13999,12);
UPDATE lookup_functional_system_group SET controller_type = 'HUB' WHERE range_start >= 11000;

-- Real subgroup taxonomy, corrected 2026-09-17 from the authoritative
-- row-level export (docs/06-data-dictionary.md §9a). Every hundred-block
-- (or, within WTUR's 9900-9999 span, ten-block) not listed is deliberately
-- reserved capacity for subgroups added later via lookup_functional_subgroup
-- inserts validated for non-overlap (§9a.1) -- never a schema migration.
CREATE TABLE lookup_functional_subgroup (
    id INTEGER PRIMARY KEY,
    functional_system_group_id INTEGER NOT NULL REFERENCES lookup_functional_system_group(id),
    code TEXT NOT NULL, label TEXT NOT NULL,
    sub_range_start INTEGER NOT NULL, sub_range_end INTEGER NOT NULL,
    description TEXT, is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0,
    UNIQUE (functional_system_group_id, code),
    CHECK (sub_range_end > sub_range_start)
);
```

```
-- group_range_start disambiguates WTUR/WROT, which each now identify two
-- different groups (Main Controller and Hub Controller) ? always join on
-- (code, range_start) together, never on code alone (see the note above
-- lookup_functional_system_group).
INSERT INTO lookup_functional_subgroup (functional_system_group_id, code, label, sub_range_start, sub_range_end, sort_order
  )
WITH v(group_code, group_range_start, code, label, sub_range_start, sub_range_end, sort_order) AS (
    VALUES
    ('WCNV',1000,'010xx','Converter',1000,1199,1),
    ('WCNV',1000,'012xx','Grid',1200,1299,2),
    ('WGEN',2000,'020xx','Generator',2000,2199,1),
    ('WGEN',2000,'022xx','Protection',2200,2299,2),
    ('WNAC',3000,'030xx','Meteorology',3000,3199,1),
    ('WNAC',3000,'032xx','Nacelle',3200,3299,2),
    ('WROT',4000,'040xx','Rotor',4000,4199,1),
    ('WROT',4000,'042xx','Hot Air De-icing',4200,4299,2),
    ('WROT',4000,'043xx','Lighting',4300,4399,3),
    ('WROT',4000,'044xx','Blade Sensors',4400,4499,4),
    ('WTOW',5000,'050xx','Tower',5000,5199,1),
    ('WTRF',6000,'060xx','Switches',6000,6199,1),
    ('WTRF',6000,'062xx','Transformer',6200,6299,2),
    ('WTRM',7000,'070xx','Gearbox',7000,7199,1),
    ('WTRM',7000,'072xx','Drive train',7200,7299,2),
    ('WTRM',7000,'073xx','Hydraulics',7300,7399,3),
    ('WTRM',7000,'074xx','Brake',7400,7499,4),
    ('WYAW',8000,'080xx','Yaw',8000,8199,1),
    ('WTUR',9000,'090xx','System',9000,9199,1),
    ('WTUR',9000,'092xx','Control System',9200,9299,2),
    ('WTUR',9000,'093xx','Safety',9300,9399,3),
    ('WTUR',9000,'0990x','External',9900,9929,4),
    ('WTUR',9000,'0993x','Farm',9930,9959,5),
    ('WTUR',9000,'0996x','Feedback Control',9960,9969,6),
    ('WTUR',9000,'0997x','Interface',9970,9979,7),
    ('WTUR',9000,'0998x','Power Management',9980,9989,8),
    ('WTUR',9000,'0999x','UPS',9990,9999,9),
    -- Hub Controller subgroups (new 2026-09-17, source: "HC Status Code Number.xlsx"). Bands are
    -- aligned to the hundreds digit like the Main groups (docs/05-governance-handbook.md section 2;
    -- aligned 2026-10-23, migration a1d4f6b8c203): '123xx' owns 12300-12399. A band shorter than a
    -- hundred is named by its first ten with one digit in place of the second x ('1110x' = 11100-11149,
    -- '1115x' = 11150-11199; renamed from '11-1xxA'/'11-1xxB' by migration f6c8e0a2b459).
    ('WTUR',11000,'110xx','Control System',11000,11099,1),
    ('WTUR',11000,'1110x','Field Bus',11100,11149,2),
    ('WTUR',11000,'1115x','System',11150,11199,3),
    ('WROT',12000,'1200x','Blade',12000,12049,1),
    ('WROT',12000,'1205x','Batteries',12050,12199,2),
    ('WROT',12000,'1220x','Pitch',12200,12249,3),
    ('WROT',12000,'1225x','Pitch Control',12250,12299,4),
    ('WROT',12000,'123xx','Pitch Drivers',12300,12399,5),
    ('WROT',12000,'124xx','Pitch Converter',12400,12499,6),
    ('WROT',12000,'125xx','Rotor',12500,12599,7),
    ('WPPD',13000,'130xx','Wind Farm',13000,13099,1)
)
SELECT g.id, v.code, v.label, v.sub_range_start, v.sub_range_end, v.sort_order
FROM v JOIN lookup_functional_system_group g
  ON g.code = v.group_code AND g.range_start = v.group_range_start;

CREATE TABLE lookup_turbine_platform (
    id INTEGER PRIMARY KEY, code TEXT NOT NULL UNIQUE, label TEXT NOT NULL,
    description TEXT, is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0
);
INSERT INTO lookup_turbine_platform (code, label, sort_order) VALUES
    ('2XM','2XM Platform',1), ('3XM','3XM Platform',2), ('4XM','4XM Platform',3);

-- Available Group: numeric codes only, labels TBD pending Controls
-- Engineering confirmation (docs/06-data-dictionary.md §9c). Soft reference
-- set for validation, not a hard-enforced enum, since new codes may appear.
CREATE TABLE lookup_available_group (
    id INTEGER PRIMARY KEY, code INTEGER NOT NULL UNIQUE, label TEXT,
    description TEXT, is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0
);
INSERT INTO lookup_available_group (code, sort_order)
WITH v(value) AS (
    VALUES (1),(2),(4),(5),(6),(9),(11),(12),(13),(14),(15),(16),(18),
           (20),(21),(22),(23),(24),(25),(27),(28),(30),(32)
)
SELECT value, ROW_NUMBER() OVER (ORDER BY value) FROM v;

-- Eight independent audience/access lookups, real observed values
-- (docs/06-data-dictionary.md §2d, §9d). Casing/typo variants from the raw
-- export ("See Only" vs "See only", "ProfessionaProfessional") normalized.
CREATE TABLE lookup_development_access (id INTEGER PRIMARY KEY, code TEXT NOT NULL UNIQUE, label TEXT NOT NULL, description
   TEXT, is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0, audience_visible INTEGER NOT NULL DEFAU
  LT 1 CHECK (audience_visible IN (0,1)),
    default_read_level INTEGER CHECK (default_read_level IS NULL OR default_read_level BETWEEN 0 AND 255),
    default_write_level INTEGER CHECK (default_write_level IS NULL OR default_write_level BETWEEN 0 AND 255));
INSERT INTO lookup_development_access (code, label, sort_order) VALUES
    ('SCADA_USER','SCADA User',1), ('DEPARTMENT','Department',2), ('OPERATION_CONTROL','Operation Control',3), ('SEE_ONLY',
  'See only',4);
```

```
CREATE TABLE lookup_sales_access (id INTEGER PRIMARY KEY, code TEXT NOT NULL UNIQUE, label TEXT NOT NULL, description TEXT,
   is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0, audience_visible INTEGER NOT NULL DEFAULT 1 C
  HECK (audience_visible IN (0,1)),
    default_read_level INTEGER CHECK (default_read_level IS NULL OR default_read_level BETWEEN 0 AND 255),
    default_write_level INTEGER CHECK (default_write_level IS NULL OR default_write_level BETWEEN 0 AND 255));
INSERT INTO lookup_sales_access (code, label, sort_order) VALUES
    ('SALES','Sales',1), ('COVERT','Covert',2), ('SEE_ONLY','See only',3);

CREATE TABLE lookup_tcc_access (id INTEGER PRIMARY KEY, code TEXT NOT NULL UNIQUE, label TEXT NOT NULL, description TEXT, i
  s_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0, audience_visible INTEGER NOT NULL DEFAULT 1 CHE
  CK (audience_visible IN (0,1)),
    default_read_level INTEGER CHECK (default_read_level IS NULL OR default_read_level BETWEEN 0 AND 255),
    default_write_level INTEGER CHECK (default_write_level IS NULL OR default_write_level BETWEEN 0 AND 255));
INSERT INTO lookup_tcc_access (code, label, sort_order) VALUES
    ('TELENOTDIENST','Telenotdienst',1), ('TELENOTDIENST_READ','Telenotdienst-read',2), ('PMS_DISPATCHER','PMS Dispatcher',
  3), ('SEE_ONLY','See only',4);

CREATE TABLE lookup_service_access (id INTEGER PRIMARY KEY, code TEXT NOT NULL UNIQUE, label TEXT NOT NULL, description TEX
  T, is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0, audience_visible INTEGER NOT NULL DEFAULT 1
   CHECK (audience_visible IN (0,1)),
    default_read_level INTEGER CHECK (default_read_level IS NULL OR default_read_level BETWEEN 0 AND 255),
    default_write_level INTEGER CHECK (default_write_level IS NULL OR default_write_level BETWEEN 0 AND 255));
INSERT INTO lookup_service_access (code, label, sort_order) VALUES
    ('COMMISSIONING','Commissioning',1), ('SERVICE_AND_MAINTENANCE','Service and Maintenance',2), ('SEE_ONLY','See only',3)
  ;

CREATE TABLE lookup_top_access (id INTEGER PRIMARY KEY, code TEXT NOT NULL UNIQUE, label TEXT NOT NULL, description TEXT, i
  s_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0, audience_visible INTEGER NOT NULL DEFAULT 1 CHE
  CK (audience_visible IN (0,1)),
    default_read_level INTEGER CHECK (default_read_level IS NULL OR default_read_level BETWEEN 0 AND 255),
    default_write_level INTEGER CHECK (default_write_level IS NULL OR default_write_level BETWEEN 0 AND 255));
INSERT INTO lookup_top_access (code, label, sort_order) VALUES
    ('PREVENTIVE_MAINTENANCE','Preventive Maintenance',1), ('TROUBLESHOOTING','Troubleshooting',2),
    ('TROUBLESHOOTING_ADVANCED','Troubleshooting Advanced',3), ('COVERT','Covert',4), ('SEE_ONLY','See only',5);

CREATE TABLE lookup_grid_operator_access (id INTEGER PRIMARY KEY, code TEXT NOT NULL UNIQUE, label TEXT NOT NULL, descripti
  on TEXT, is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0, audience_visible INTEGER NOT NULL DEF
  AULT 1 CHECK (audience_visible IN (0,1)),
    default_read_level INTEGER CHECK (default_read_level IS NULL OR default_read_level BETWEEN 0 AND 255),
    default_write_level INTEGER CHECK (default_write_level IS NULL OR default_write_level BETWEEN 0 AND 255));
INSERT INTO lookup_grid_operator_access (code, label, sort_order) VALUES
    ('COVERT','Covert',1), ('SEE_ONLY','See only',2);

CREATE TABLE lookup_service_partner_access (id INTEGER PRIMARY KEY, code TEXT NOT NULL UNIQUE, label TEXT NOT NULL, descrip
  tion TEXT, is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0, audience_visible INTEGER NOT NULL D
  EFAULT 1 CHECK (audience_visible IN (0,1)),
    default_read_level INTEGER CHECK (default_read_level IS NULL OR default_read_level BETWEEN 0 AND 255),
    default_write_level INTEGER CHECK (default_write_level IS NULL OR default_write_level BETWEEN 0 AND 255));
INSERT INTO lookup_service_partner_access (code, label, sort_order) VALUES
    ('SERVICE_PARTNER','Service Partner',1), ('EXT_SERVICE_PROVIDER','Ext. Service Provider',2),
    ('CUSTOMER_AFTER_WARRANTY_PROFESSIONAL','Customer after warranty Professional',3),
    ('CUSTOMER_AFTER_WARRANTY_ADVANCED','Customer after warranty Advanced',4),
    ('COVERT','Covert',5), ('SEE_ONLY','See only',6);

CREATE TABLE lookup_customer_access (id INTEGER PRIMARY KEY, code TEXT NOT NULL UNIQUE, label TEXT NOT NULL, description TE
  XT, is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0, audience_visible INTEGER NOT NULL DEFAULT
  1 CHECK (audience_visible IN (0,1)),
    default_read_level INTEGER CHECK (default_read_level IS NULL OR default_read_level BETWEEN 0 AND 255),
    default_write_level INTEGER CHECK (default_write_level IS NULL OR default_write_level BETWEEN 0 AND 255));
INSERT INTO lookup_customer_access (code, label, sort_order) VALUES
    ('STANDARD','Standard',1), ('ADVANCED','Advanced',2), ('PROFESSIONAL','Professional',3),
    ('PREMIUM','Premium',4), ('COVERT','Covert',5), ('SEE_ONLY','See only',6);

-- I-03 / decision D-02: 'Covert' is treated as "not visible to that audience"
-- (fail-closed) until the business confirms otherwise; the flag is data that
-- an Administrator can flip per value in Lookup Administration.
UPDATE lookup_development_access      SET audience_visible = 0 WHERE code = 'COVERT';
UPDATE lookup_sales_access            SET audience_visible = 0 WHERE code = 'COVERT';
UPDATE lookup_tcc_access              SET audience_visible = 0 WHERE code = 'COVERT';
UPDATE lookup_service_access          SET audience_visible = 0 WHERE code = 'COVERT';
UPDATE lookup_top_access              SET audience_visible = 0 WHERE code = 'COVERT';
UPDATE lookup_grid_operator_access    SET audience_visible = 0 WHERE code = 'COVERT';
UPDATE lookup_service_partner_access  SET audience_visible = 0 WHERE code = 'COVERT';
UPDATE lookup_customer_access         SET audience_visible = 0 WHERE code = 'COVERT';

-- v4.1: numeric user-group levels (UserGroupReadArBy / UserGroupWriteArBy, BYTE) that the controller
-- uses for each named access value. Seeded only where the generated programs showed one value for
-- every code that carries the named access value. 'Service and Maintenance' (10 or 20) and
-- 'Telenotdienst' (10 or 20) differ from code to code, so they stay NULL: the level is then
-- entered per revision in status_code_revision_user_group (docs/06-data-dictionary.md section 2d).
UPDATE lookup_development_access      SET default_read_level = 1,  default_write_level = 50 WHERE code = 'SEE_ONLY';
UPDATE lookup_development_access      SET default_read_level = 1,  default_write_level = 20 WHERE code = 'OPERATION_CONTROL
  ';
UPDATE lookup_sales_access            SET default_read_level = 1,  default_write_level = 50 WHERE code = 'SEE_ONLY';
UPDATE lookup_tcc_access              SET default_read_level = 1,  default_write_level = 50 WHERE code = 'SEE_ONLY';
UPDATE lookup_service_access          SET default_read_level = 1,  default_write_level = 50 WHERE code = 'SEE_ONLY';
UPDATE lookup_top_access              SET default_read_level = 1,  default_write_level = 50 WHERE code = 'SEE_ONLY';
UPDATE lookup_top_access              SET default_read_level = 1,  default_write_level = 10 WHERE code = 'PREVENTIVE_MAINTE
  NANCE';
```

```
UPDATE lookup_top_access              SET default_read_level = 1,  default_write_level = 15 WHERE code = 'TROUBLESHOOTING';
UPDATE lookup_grid_operator_access    SET default_read_level = 1,  default_write_level = 50 WHERE code = 'SEE_ONLY';
UPDATE lookup_service_partner_access  SET default_read_level = 1,  default_write_level = 50 WHERE code = 'SEE_ONLY';
UPDATE lookup_service_partner_access  SET default_read_level = 1,  default_write_level = 15 WHERE code = 'EXT_SERVICE_PROVI
  DER';
UPDATE lookup_service_partner_access  SET default_read_level = 1,  default_write_level = 10 WHERE code = 'CUSTOMER_AFTER_WA
  RRANTY_ADVANCED';
UPDATE lookup_customer_access         SET default_read_level = 1,  default_write_level = 50 WHERE code = 'SEE_ONLY';
UPDATE lookup_sales_access            SET default_read_level = 50, default_write_level = 50 WHERE code = 'COVERT';
UPDATE lookup_top_access              SET default_read_level = 50, default_write_level = 50 WHERE code = 'COVERT';
UPDATE lookup_grid_operator_access    SET default_read_level = 50, default_write_level = 50 WHERE code = 'COVERT';
UPDATE lookup_service_partner_access  SET default_read_level = 50, default_write_level = 50 WHERE code = 'COVERT';
UPDATE lookup_customer_access         SET default_read_level = 50, default_write_level = 50 WHERE code = 'COVERT';

-- v4.1: the 16 user-group slots of the controller (UserGroupReadArBy/WriteArBy[Idx, slot]).
-- Slots 1..8 are the eight audiences in the order of the export columns; this was verified on 14
-- programs (a 'Covert' value is read level 50 in exactly the slot of its audience). Slot 13 is
-- never written by the generator; slots 0, 9..12, 14, 15 were 50/50 in every sample (slot 9 reads
-- 1 on Main and 50 on Hub Controller codes) and stay unassigned until Controls Engineering names them.
CREATE TABLE lookup_user_group_slot (
    id INTEGER PRIMARY KEY,
    slot INTEGER NOT NULL UNIQUE CHECK (slot BETWEEN 0 AND 15),
    audience_code TEXT UNIQUE CHECK (audience_code IS NULL OR audience_code IN
        ('DEVELOPMENT','SALES','TCC','SERVICE','TURBINE_OPERATOR_PACKAGE','GRID_OPERATOR','SERVICE_PARTNER','CUSTOMER')),
    label TEXT NOT NULL, description TEXT,
    is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0
);
INSERT INTO lookup_user_group_slot (slot, audience_code, label, is_active, sort_order) VALUES
    (0,NULL,'Slot 0 (unassigned)',0,0),
    (1,'DEVELOPMENT','Development',1,1), (2,'SALES','Sales',1,2), (3,'TCC','TCC',1,3),
    (4,'SERVICE','Service',1,4), (5,'TURBINE_OPERATOR_PACKAGE','Turbine Operator Package',1,5),
    (6,'GRID_OPERATOR','Grid Operator',1,6), (7,'SERVICE_PARTNER','Service Partner',1,7),
    (8,'CUSTOMER','Customer',1,8),
    (9,NULL,'Slot 9 (unassigned)',0,9), (10,NULL,'Slot 10 (unassigned)',0,10),
    (11,NULL,'Slot 11 (unassigned)',0,11), (12,NULL,'Slot 12 (unassigned)',0,12),
    (13,NULL,'Slot 13 (never written by the generator)',0,13), (14,NULL,'Slot 14 (unassigned)',0,14),
    (15,NULL,'Slot 15 (unassigned)',0,15);

-- I-07 / decision D-01: controller software versions, ordered by sort_order.
-- A revision declares the version range it applies to; a Release targets one or more.
CREATE TABLE lookup_controller_software_version (
    id INTEGER PRIMARY KEY, code TEXT NOT NULL UNIQUE, label TEXT NOT NULL,
    description TEXT, is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0
);

-- I-15 / decision D-03: content languages. Exactly one language is the source
-- language of the catalogue (the text stored on status_code_revision).
CREATE TABLE lookup_language (
    id INTEGER PRIMARY KEY, code TEXT NOT NULL UNIQUE, label TEXT NOT NULL,
    description TEXT, is_active INTEGER NOT NULL DEFAULT 1, sort_order INTEGER NOT NULL DEFAULT 0,
    is_source INTEGER NOT NULL DEFAULT 0 CHECK (is_source IN (0,1))
);
INSERT INTO lookup_language (code, label, sort_order, is_source) VALUES
    ('en','GB English GB',1,1), ('de','DE Deutsch DE',2,0), ('fr','FR Français FR',3,0),
    ('hi','IN ?????? (Hindi) IN',4,0), ('ar','SA ??????? (Arabic) SA',6,0), ('bn','IN ????? (Bengali) IN',7,0),
    ('mr','IN ????? (Marathi) IN',8,0), ('te','IN ?????? (Telugu) IN',9,0), ('ta','IN ????? (Tamil) IN',10,0),
    ('gu','IN ??????? (Gujarati) IN',11,0), ('ur','PK ???? (Urdu) PK',12,0), ('kn','IN ????? (Kannada) IN',13,0),
    ('or','IN ????? (Odia) IN',14,0), ('ml','IN ?????? (Malayalam) IN',15,0), ('pa','IN ?????? (Punjabi) IN',16,0),
    ('es','ES Español ES',17,0), ('it','IT Italiano IT',18,0), ('da','DK Dansk DK',19,0),
    ('ka','GE ??????? (Georgian) GE',20,0);   -- v4.3-v4.6: country letters left, flag right; Georgian last

-- ---------------------------------------------------------------------------
-- Identity / access
-- ---------------------------------------------------------------------------

CREATE TABLE user (
    id                  INTEGER PRIMARY KEY,
    username            TEXT NOT NULL UNIQUE,
    email               TEXT NOT NULL UNIQUE,
    password_hash       TEXT NOT NULL,
    full_name           TEXT NOT NULL,
    role_id             INTEGER NOT NULL REFERENCES role(id),
    is_active           INTEGER NOT NULL DEFAULT 1,
    created_at          TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    last_login_at       TEXT,
    failed_login_count  INTEGER NOT NULL DEFAULT 0,
    locked_until        TEXT,
    ui_language         TEXT NOT NULL DEFAULT 'en' CHECK (length(ui_language) = 2 AND ui_language = lower(ui_language)),
  -- I-22; the application decides which codes exist (v4.5)
    deleted_at          TEXT   -- v4.9: set when an Administrator deletes an account that has history; name, address and pa
  ssword are erased, the row stays so that nothing it did loses its author
);

-- 11-security-architecture.md §4: a new password may not repeat any of the last 5
-- passwords. Holds the hashes of previous passwords (never plaintext).
CREATE TABLE user_password_history (
    id              INTEGER PRIMARY KEY,
    user_id         INTEGER NOT NULL REFERENCES user(id),
    password_hash   TEXT NOT NULL,
```

```
    retired_at      TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now'))
);
CREATE INDEX ix_pwhist_user ON user_password_history (user_id, id);

CREATE TABLE user_engineering_domain (
    id INTEGER PRIMARY KEY,
    user_id INTEGER NOT NULL REFERENCES user(id),
    engineering_domain_id INTEGER NOT NULL REFERENCES lookup_engineering_domain(id),
    UNIQUE (user_id, engineering_domain_id)
);

-- ------------------------------------------------------------------------
-- Core governed entities
-- ------------------------------------------------------------------------

CREATE TABLE status_code (
    id                                  INTEGER PRIMARY KEY,
    status_code_identifier              TEXT UNIQUE,   -- NULL until Draft->Review; StCd-XXXXX or StCd-TXXXXX
    functional_system_group_id          INTEGER NOT NULL REFERENCES lookup_functional_system_group(id),
    functional_subgroup_id              INTEGER NOT NULL REFERENCES lookup_functional_subgroup(id),  -- mandatory (v3)
    is_sandbox                          INTEGER NOT NULL DEFAULT 0 CHECK (is_sandbox IN (0,1)),
    platform_variant_of_status_code_id  INTEGER REFERENCES status_code(id),
    legacy_reference_number             TEXT,
    -- v4.1: PrgIdx_GUi, the block identification the generated program writes to the controller's
    -- global on entry. Entered by hand (the tool never allocates it), UINT range, audited.
    -- NULL only while the code is a Draft or not yet imported; required from Review on
    -- (docs/06-data-dictionary.md section 1).
    prg_idx                             INTEGER CHECK (prg_idx IS NULL OR prg_idx BETWEEN 1 AND 65535),
    -- v4.2: the number a generated StCd_<n>.st program already carries. Submit for Review issues
    -- exactly this identifier (no sequence allocation, no Administrator override), and the number
    -- allocator skips it. NULL for every code that is not imported from a program.
    reserved_identifier                 TEXT CHECK (reserved_identifier IS NULL
                                                    OR reserved_identifier GLOB 'StCd-[0-9][0-9][0-9][0-9][0-9]'),
    owner_id                            INTEGER NOT NULL REFERENCES user(id),
    created_at                          TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    -- v4.9: set when the last revision of the code was withdrawn (the code is an empty shell); only such a code can be del
  eted
    withdrawn_at                        TEXT,
    -- v4.9.2: set when an Administrator deletes a withdrawn code. The row stays (so no id is ever
    -- reused and every audit entry keeps pointing at one code) but gives up its identifier, reserved
    -- number, legacy number and PrgIdx_GUi, so the same program or row can be imported again.
    deleted_at                          TEXT,
    CHECK (status_code_identifier IS NULL
           OR status_code_identifier GLOB 'StCd-[0-9][0-9][0-9][0-9][0-9]'
           OR status_code_identifier GLOB 'StCd-T[0-9][0-9][0-9][0-9][0-9]')
);

-- v4.1: one PrgIdx_GUi per block; sandbox codes are exempt so test codes never block a real value.
CREATE UNIQUE INDEX ux_status_code_prg_idx ON status_code (prg_idx) WHERE prg_idx IS NOT NULL AND is_sandbox = 0;

-- v4.2: a number can be reserved by one code only.
CREATE UNIQUE INDEX ux_status_code_reserved_identifier ON status_code (reserved_identifier) WHERE reserved_identifier IS NO
  T NULL;

CREATE TABLE status_code_revision (
    id                              INTEGER PRIMARY KEY,
    status_code_id                  INTEGER NOT NULL REFERENCES status_code(id),
    revision_number                 INTEGER NOT NULL,
    is_current                      INTEGER NOT NULL DEFAULT 1 CHECK (is_current IN (0,1)),

    title                           TEXT NOT NULL CHECK (length(title) BETWEEN 1 AND 120),
    description                     TEXT NOT NULL CHECK (length(description) BETWEEN 10 AND 4000),
    software_version                TEXT,   -- nullable; may belong on `release` instead ? see docs/06-data-dictionary.md o
  pen item
    status_category                 TEXT NOT NULL CHECK (status_category IN ('ERROR','WARNING','INFO')),
    available_group                 INTEGER NOT NULL,
    alarm_flag                      INTEGER NOT NULL DEFAULT 0 CHECK (alarm_flag IN (0,1)),
    set_delay                       TEXT NOT NULL,
    reset_delay                     TEXT NOT NULL,
    up_down_counter                 INTEGER NOT NULL DEFAULT 0 CHECK (up_down_counter IN (0,1)),
    trigger_snapshot                INTEGER NOT NULL DEFAULT 0 CHECK (trigger_snapshot IN (0,1)),
    loadless_spinning_permitted     INTEGER NOT NULL DEFAULT 0 CHECK (loadless_spinning_permitted IN (0,1)),

    number_of_times_repeated        INTEGER CHECK (number_of_times_repeated IS NULL OR number_of_times_repeated >= 1),
    repeated_over_the_course        TEXT,
    status_code_for_repeated_error  INTEGER,   -- opaque reference, NOT a FK (docs/06-data-dictionary.md §2b)

    yaw_program                         INTEGER NOT NULL DEFAULT 0,
    manual_reset_program                INTEGER NOT NULL DEFAULT 0,
    auto_reset_program                  INTEGER NOT NULL DEFAULT 0,
    converter_reactive_power_program    INTEGER NOT NULL DEFAULT 0,
    brake_program                       INTEGER NOT NULL DEFAULT 0,
    umrichterprogramm_gsc               INTEGER NOT NULL DEFAULT 0,
    umrichterprogramm_msc               INTEGER NOT NULL DEFAULT 0,
    yaw_bearing_lubrication_program     INTEGER NOT NULL DEFAULT 0,

    -- v4.1 Main Controller subsystem programs (4XM 'Naming Conventions' sheet / generated
    -- *PrgArUs arrays). 0 = the code does not act on that subsystem. Hub Controller codes keep 0.
    converter_reset_program             INTEGER NOT NULL DEFAULT 0,   -- ConvRstLev     ResetConverterPrgArUs
```

```
  hydraulic_pump_program               INTEGER NOT NULL DEFAULT 0,   -- HyPmpLev       HydraulicPrgArUs
  gearbox_oil_pump_program             INTEGER NOT NULL DEFAULT 0,   -- GbxOilPmpLev   GearOilPumpPrgArUs
  gearbox_oil_fan_program              INTEGER NOT NULL DEFAULT 0,   -- GbxOilFanLev   GearOilCoolingPrgArUs
  gearbox_oil_heating_program          INTEGER NOT NULL DEFAULT 0,   -- GbxOilHtLev    GearHeatingPrgArUs
  generator_fan_program                INTEGER NOT NULL DEFAULT 0,   -- GnFanLev       GenFanPrgArUs
  generator_heating_program            INTEGER NOT NULL DEFAULT 0,   -- GnHtLev        GenHeatingPrgArUs
  generator_bearing_lubrication_program INTEGER NOT NULL DEFAULT 0,  -- GnBrgLuLev     GenoLubPrgArUs
  rotor_bearing_lubrication_program    INTEGER NOT NULL DEFAULT 0,   -- RotBrgLuLev    RotBearLubPrgArUs
  anemometer_heating_program           INTEGER NOT NULL DEFAULT 0,   -- AneHtLev       VaneAneHeatPrgArUs
  medium_voltage_cb_program            INTEGER NOT NULL DEFAULT 0,   -- MedCBLev       MeanVoltageCBPrgArUs ('Mean' reads
 as a mistranslation of Mittelspannung)
  tower_fan_program                    INTEGER NOT NULL DEFAULT 0,   -- TowFanLev      TowerFanPrgArUs
  transformer_oil_pump_program         INTEGER NOT NULL DEFAULT 0,   -- TrfOilPmpLev   TrafoPumpPrgArUs
  zero_load_operation                  INTEGER CHECK (zero_load_operation IS NULL OR zero_load_operation IN (0,1)),   --
ZeroLoadOperation_ArBo; Main only, NULL = not captured

  -- v4.1 Hub Controller programs and operation flags. brake_program is the Hub 'Stop Program'
  -- (same BrakePrgArUs array); ROTOR_BREAK in the Hub service export is rotor_brake_program.
  pitch_error_program                  INTEGER NOT NULL DEFAULT 0,   -- Pitch Error Program    PtErrOpPrgArUs
  pitch_warning_program                INTEGER NOT NULL DEFAULT 0,   -- Pitch Warning Program  PtWarnOpPrgArUs
  rotor_brake_program                  INTEGER NOT NULL DEFAULT 0,   -- ROTOR_BREAK            RotorBrakePrgArUs
  stop_program_40_flag                 INTEGER CHECK (stop_program_40_flag IS NULL OR stop_program_40_flag IN (0,1)),
  stop_program_80_flag                 INTEGER CHECK (stop_program_80_flag IS NULL OR stop_program_80_flag IN (0,1)),
  manual_operation_flag                INTEGER CHECK (manual_operation_flag IS NULL OR manual_operation_flag IN (0,1)),
  automatic_operation_flag             INTEGER CHECK (automatic_operation_flag IS NULL OR automatic_operation_flag IN (0,
1)),
  vane_position_flag                   INTEGER CHECK (vane_position_flag IS NULL OR vane_position_flag IN (0,1)),

  -- v4.1 parameters both controllers carry in the generated program but no workbook exports.
  -- NULL = not captured yet.
  operating_mode_mask                  INTEGER CHECK (operating_mode_mask IS NULL OR operating_mode_mask BETWEEN 0 AND 10
48575),   -- OpModeArDw: 20 bits Main, 6 bits Hub
  task_call_frequency                  INTEGER CHECK (task_call_frequency IS NULL OR task_call_frequency >= 1),
        -- TaskCallFrequencyArBy (1, 10 or 100 so far)
  availability_calc                    INTEGER CHECK (availability_calc IS NULL OR availability_calc IN (0,1)),
        -- AvailabilityCalcArBo

  development_access              INTEGER REFERENCES lookup_development_access(id),
  sales_access                    INTEGER REFERENCES lookup_sales_access(id),
  tcc_access                      INTEGER REFERENCES lookup_tcc_access(id),
  service_access                  INTEGER REFERENCES lookup_service_access(id),
  turbine_operator_package_access INTEGER REFERENCES lookup_top_access(id),
  grid_operator_access            INTEGER REFERENCES lookup_grid_operator_access(id),
  service_partner_access          INTEGER REFERENCES lookup_service_partner_access(id),
  customer_access                 INTEGER REFERENCES lookup_customer_access(id),

  -- logical_node / node_value are derived (docs/06-data-dictionary.md §2e);
  -- NOT stored as independent columns to guarantee they can never drift
  -- from the parent status_code's functional_system_group_id. Exposed via
  -- a read-only view / computed API field instead ? see 07-database-design.md.

  owner_id                        INTEGER NOT NULL REFERENCES user(id),
  lifecycle_status                TEXT NOT NULL DEFAULT 'DRAFT'
                                   CHECK (lifecycle_status IN ('DRAFT','REVIEW','PENDING_APPROVAL','APPROVED','RELEASED',
'DEPRECATED','ARCHIVED')),
  created_by                      INTEGER NOT NULL REFERENCES user(id),
  created_at                      TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
  reviewer_signoff_by             INTEGER REFERENCES user(id),
  reviewer_signoff_at             TEXT,
  admin_signoff_by                INTEGER REFERENCES user(id),
  admin_signoff_at                TEXT,
  chief_engineer_approved_by      INTEGER REFERENCES user(id),
  chief_engineer_approval_date    TEXT,
  effective_date                  TEXT,
  deprecated_reason               TEXT,
  superseded_by_status_code_id    INTEGER REFERENCES status_code(id),

  -- I-01 optimistic concurrency: row_version increments on every Draft edit.
  row_version                     INTEGER NOT NULL DEFAULT 1,
  updated_at                      TEXT,
  -- I-14 troubleshooting content (all optional)
  remedy_text                     TEXT CHECK (remedy_text IS NULL OR length(remedy_text) <= 4000),
  affected_component              TEXT CHECK (affected_component IS NULL OR length(affected_component) <= 200),
  related_status_codes            TEXT CHECK (related_status_codes IS NULL OR length(related_status_codes) <= 500),
  standard_references             TEXT CHECK (standard_references IS NULL OR length(standard_references) <= 1000),
  -- I-07 applicability (from/to inclusive; NULL = open-ended)
  applies_from_version_id         INTEGER REFERENCES lookup_controller_software_version(id),
  applies_to_version_id           INTEGER REFERENCES lookup_controller_software_version(id),
  -- I-27 planned dates (ISO yyyy-mm-dd); informative, never change the lifecycle by themselves
  planned_effective_date          TEXT,
  planned_deprecation_date        TEXT,
  UNIQUE (status_code_id, revision_number),
  CHECK (reviewer_signoff_by IS NULL OR reviewer_signoff_by <> created_by),
  CHECK (admin_signoff_by IS NULL OR admin_signoff_by <> created_by),
  CHECK (chief_engineer_approved_by IS NULL OR (
      chief_engineer_approved_by <> created_by
      AND chief_engineer_approved_by <> reviewer_signoff_by
      AND chief_engineer_approved_by <> admin_signoff_by
  ))
```

```
);

CREATE UNIQUE INDEX ux_revision_current_per_code
    ON status_code_revision (status_code_id) WHERE is_current = 1;
CREATE INDEX ix_revision_status_code ON status_code_revision (status_code_id);
CREATE INDEX ix_revision_lifecycle ON status_code_revision (lifecycle_status);
CREATE INDEX ix_revision_owner ON status_code_revision (owner_id);
CREATE INDEX ix_revision_category ON status_code_revision (status_category);
CREATE UNIQUE INDEX ux_one_open_cr_per_code
    ON status_code_revision (status_code_id) WHERE lifecycle_status IN ('DRAFT','REVIEW','PENDING_APPROVAL');

CREATE TABLE status_code_revision_platform (
    id INTEGER PRIMARY KEY,
    status_code_revision_id INTEGER NOT NULL REFERENCES status_code_revision(id),
    turbine_platform_id INTEGER NOT NULL REFERENCES lookup_turbine_platform(id),
    UNIQUE (status_code_revision_id, turbine_platform_id)
);

-- v4.1: the numeric user-group levels of one revision, one row per slot (UserGroupReadArBy /
-- UserGroupWriteArBy are BYTE arrays). Slots 1..8 mirror the eight audience fields above.
CREATE TABLE status_code_revision_user_group (
    id INTEGER PRIMARY KEY,
    status_code_revision_id INTEGER NOT NULL REFERENCES status_code_revision(id),
    user_group_slot_id INTEGER NOT NULL REFERENCES lookup_user_group_slot(id),
    read_level INTEGER NOT NULL CHECK (read_level BETWEEN 0 AND 255),
    write_level INTEGER NOT NULL CHECK (write_level BETWEEN 0 AND 255),
    UNIQUE (status_code_revision_id, user_group_slot_id)
);

-- Import (v3) ? CSV/XLSX library import, still subject to full governance
-- (docs/06-data-dictionary.md §5, 05-governance-handbook.md §6b). Created
-- before change_request so the FK below resolves at CREATE TABLE time.
CREATE TABLE import_job (
    id                      INTEGER PRIMARY KEY,
    requested_by            INTEGER NOT NULL REFERENCES user(id),
    source_format           TEXT NOT NULL CHECK (source_format IN ('CSV','XLSX','ST')),   -- 'ST' = v4.2, generated StCd pr
  ograms
    source_file_reference   TEXT NOT NULL,
    row_count               INTEGER NOT NULL,
    status                  TEXT NOT NULL DEFAULT 'PENDING' CHECK (status IN ('PENDING','VALIDATING','COMPLETE','FAILED')),
    created_count           INTEGER NOT NULL DEFAULT 0,
    error_count             INTEGER NOT NULL DEFAULT 0,
    error_report_reference  TEXT,
    created_at              TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    completed_at            TEXT,
    processed_count         INTEGER NOT NULL DEFAULT 0,   -- I-32 progress (rows handled so far)
    warning_count           INTEGER NOT NULL DEFAULT 0,   -- I-12 duplicate / I-04 reference warnings
    import_mode             TEXT NOT NULL DEFAULT 'CREATE' CHECK (import_mode IN ('CREATE','UPDATE')),   -- I-05
    error_report_xlsx_reference TEXT,
    change_set_id           INTEGER REFERENCES change_set(id)   -- v4.2: the batch change set of an ST import
);

-- v4.2: provenance of every status code created from a generated StCd_<n>.st program. The parsed
-- attributes stay as JSON, including those the catalogue has no column for yet (user-group levels,
-- the per-subsystem programs), so nothing the file said is lost and a later re-import can show what
-- changed. The file itself is kept under instance/imports/<job>/ and identified by its SHA-256.
CREATE TABLE import_source_item (
    id              INTEGER PRIMARY KEY,
    import_job_id   INTEGER NOT NULL REFERENCES import_job(id),
    status_code_id  INTEGER NOT NULL REFERENCES status_code(id),
    source_file     TEXT NOT NULL,
    source_sha256   TEXT NOT NULL CHECK (length(source_sha256) = 64),
    program_number  INTEGER NOT NULL CHECK (program_number >= 0),   -- StCd_<n> as written in the file
    controller      TEXT NOT NULL CHECK (controller IN ('MC','HC')),
    source_line     INTEGER NOT NULL DEFAULT 0,
    parsed_json     TEXT NOT NULL,
    UNIQUE (import_job_id, source_file, program_number)
);
CREATE INDEX ix_import_source_item_status_code ON import_source_item (status_code_id);

-- I-11: a change set groups the CRs that ship with one controller software
-- release. Presentation and queueing only: every CR keeps its own three decisions.
CREATE TABLE change_set (
    id                          INTEGER PRIMARY KEY,
    name                        TEXT NOT NULL UNIQUE,
    description                 TEXT,
    target_software_version_id  INTEGER REFERENCES lookup_controller_software_version(id),
    status                      TEXT NOT NULL DEFAULT 'OPEN' CHECK (status IN ('OPEN','CLOSED')),
    created_by                  INTEGER NOT NULL REFERENCES user(id),
    created_at                  TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    closed_at                   TEXT
);

CREATE TABLE change_request (
    id                      INTEGER PRIMARY KEY,
    status_code_revision_id INTEGER NOT NULL UNIQUE REFERENCES status_code_revision(id),
    cr_type                 TEXT NOT NULL CHECK (cr_type IN ('NEW','REVISION','DEPRECATION','IMPORT')),
    requested_by            INTEGER NOT NULL REFERENCES user(id),
    justification           TEXT,
```

```
    import_job_id               INTEGER REFERENCES import_job(id),
    state                       TEXT NOT NULL DEFAULT 'DRAFT'
                                 CHECK (state IN ('DRAFT','REVIEW','PENDING_APPROVAL','APPROVED','REJECTED','WITHDRAWN')),
    submitted_at                TEXT,
    chief_engineer_decided_at   TEXT,
    chief_engineer_decided_by   INTEGER REFERENCES user(id),
    assigned_reviewer_id        INTEGER REFERENCES user(id),   -- FR-039: optional Reviewer assignment, Administrator-manag
  ed
    change_set_id               INTEGER REFERENCES change_set(id),   -- I-11
    CHECK (chief_engineer_decided_by IS NULL OR chief_engineer_decided_by <> requested_by)
);
CREATE INDEX ix_cr_state ON change_request (state);
CREATE INDEX ix_cr_requested_by ON change_request (requested_by);

CREATE TABLE review_comment (
    id                  INTEGER PRIMARY KEY,
    change_request_id   INTEGER NOT NULL REFERENCES change_request(id),
    author_id           INTEGER NOT NULL REFERENCES user(id),
    comment_text        TEXT NOT NULL CHECK (length(comment_text) BETWEEN 1 AND 4000),
    decision            TEXT CHECK (decision IN (
                             'REVIEWER_APPROVE','REVIEWER_REJECT',
                             'ADMIN_APPROVE','ADMIN_REJECT',
                             'CHIEF_ENGINEER_APPROVE','CHIEF_ENGINEER_REJECT')),
    created_at          TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now'))
);
CREATE INDEX ix_comment_cr ON review_comment (change_request_id);

CREATE TABLE domain_signoff (
    id                          INTEGER PRIMARY KEY,
    change_request_id           INTEGER NOT NULL REFERENCES change_request(id),
    engineering_domain_id       INTEGER NOT NULL REFERENCES lookup_engineering_domain(id),
    signed_off_by                INTEGER NOT NULL REFERENCES user(id),
    signed_off_at                TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    UNIQUE (change_request_id, engineering_domain_id)
);

-- FR-028: the originating Engineer's explicit request for a domain sign-off,
-- optionally directed at one named Engineer holding that domain. Surfaces on
-- the target's Dashboard / My Change Requests (10-ui-ux-specification.md §6).
CREATE TABLE domain_signoff_request (
    id                          INTEGER PRIMARY KEY,
    change_request_id           INTEGER NOT NULL REFERENCES change_request(id),
    engineering_domain_id       INTEGER NOT NULL REFERENCES lookup_engineering_domain(id),
    requested_by                INTEGER NOT NULL REFERENCES user(id),
    target_user_id              INTEGER REFERENCES user(id),
    created_at                  TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    UNIQUE (change_request_id, engineering_domain_id)
);
CREATE INDEX ix_dsr_target ON domain_signoff_request (target_user_id);

CREATE TABLE release (
    id                  INTEGER PRIMARY KEY,
    name                TEXT NOT NULL,
    version_label       TEXT NOT NULL UNIQUE,
    description         TEXT,
    scope_filter        TEXT NOT NULL DEFAULT '{}',
    status              TEXT NOT NULL DEFAULT 'BUILDING' CHECK (status IN ('BUILDING','PUBLISHED')),
    created_by          INTEGER NOT NULL REFERENCES user(id),
    created_at          TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    published_by        INTEGER REFERENCES user(id),
    published_at        TEXT
);

CREATE TABLE release_item (
    id                          INTEGER PRIMARY KEY,
    release_id                  INTEGER NOT NULL REFERENCES release(id),
    status_code_revision_id     INTEGER NOT NULL UNIQUE REFERENCES status_code_revision(id)
);
CREATE INDEX ix_release_item_release ON release_item (release_id);

-- ---------------------------------------------------------------------------
-- Audit & export
-- ---------------------------------------------------------------------------

CREATE TABLE audit_log_entry (
    id              INTEGER PRIMARY KEY,
    entity_type     TEXT NOT NULL,
    entity_id       INTEGER NOT NULL,
    action          TEXT NOT NULL,
    actor_id        INTEGER REFERENCES user(id),
    occurred_at     TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    before_value    TEXT,
    after_value     TEXT,
    ip_address      TEXT,
    request_id      TEXT NOT NULL,
    prev_hash       TEXT,   -- I-18: entry_hash of the preceding row ('GENESIS' for the first)
    entry_hash      TEXT    -- I-18: SHA-256 (or HMAC-SHA-256 with AUDIT_HMAC_KEY) over this row + prev_hash
);
CREATE INDEX ix_audit_entity ON audit_log_entry (entity_type, entity_id);
```

```
CREATE INDEX ix_audit_actor ON audit_log_entry (actor_id);
CREATE INDEX ix_audit_occurred ON audit_log_entry (occurred_at);

CREATE TABLE export_job (
    id                  INTEGER PRIMARY KEY,
    requested_by        INTEGER NOT NULL REFERENCES user(id),
    export_type         TEXT NOT NULL CHECK (export_type IN ('CSV','JSON','XLSX','PDF','SQLITE_BACKUP','HTML','MD','ZIP')),
    export_scope_type   TEXT NOT NULL DEFAULT 'CATALOGUE' CHECK (export_scope_type IN ('CATALOGUE','FULL_DATABASE','RELEASE
  _NOTES','OFFLINE_BUNDLE')),
    scope               TEXT NOT NULL DEFAULT '{}',
    release_id          INTEGER REFERENCES release(id),
    status              TEXT NOT NULL DEFAULT 'PENDING' CHECK (status IN ('PENDING','COMPLETE','FAILED')),
    created_at          TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    completed_at        TEXT,
    file_reference      TEXT,
    -- I-06 integrity: what exactly left the system
    sha256              TEXT,
    size_bytes          INTEGER,
    row_count           INTEGER,
    profile_code        TEXT,
    audience            TEXT,
    language_code       TEXT,
    software_version_code TEXT,
    manifest            TEXT
);
CREATE INDEX ix_export_job_sha256 ON export_job (sha256);


-- ---------------------------------------------------------------------------
-- v4 additions (improvement pack, docs/21-improvement-ideas.md)
-- ---------------------------------------------------------------------------

-- I-07: the controller software versions a Release is published for.
CREATE TABLE release_target_version (
    id                  INTEGER PRIMARY KEY,
    release_id          INTEGER NOT NULL REFERENCES release(id),
    software_version_id INTEGER NOT NULL REFERENCES lookup_controller_software_version(id),
    UNIQUE (release_id, software_version_id)
);

-- I-02 / I-03: export profiles are Administrator-managed data, not code.
-- `columns` is a JSON array of {"field": <export column>, "label": <header>}.
CREATE TABLE export_profile (
    id              INTEGER PRIMARY KEY,
    code            TEXT NOT NULL UNIQUE,
    name            TEXT NOT NULL,
    description     TEXT,
    columns         TEXT NOT NULL,
    language_code   TEXT,                       -- NULL = the catalogue's source language
    audience        TEXT,                       -- default audience filter (I-03), NULL = none
    title_page      INTEGER NOT NULL DEFAULT 1 CHECK (title_page IN (0,1)),
    pdf_orientation TEXT NOT NULL DEFAULT 'L' CHECK (pdf_orientation IN ('L','P')),
    is_active       INTEGER NOT NULL DEFAULT 1,
    sort_order      INTEGER NOT NULL DEFAULT 0,
    created_by      INTEGER REFERENCES user(id),
    created_at      TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now'))
);
INSERT INTO export_profile (code, name, description, columns, audience, pdf_orientation, sort_order) VALUES
    ('SERVICE_MANUAL', 'Service manual',
     'Identifier, title, description, category, delays and remedy text for Technical Publications.',
     '[{"field":"status_code_identifier","label":"Status code"},{"field":"title","label":"Title"},{"field":"description","l
  abel":"Description"},{"field":"status_category","label":"Category"},{"field":"set_delay","label":"Set delay"},{"field":"res
  et_delay","label":"Reset delay"},{"field":"remedy_text","label":"Remedy"}]',
     NULL, 'L', 1),
    ('SCADA_TAGS', 'SCADA tag list',
     'Identifier, logical node, category and alarm flag for SCADA tag import.',
     '[{"field":"status_code_identifier","label":"Status code"},{"field":"logical_node","label":"Node"},{"field":"node_valu
  e","label":"Node value"},{"field":"title","label":"Tag text"},{"field":"status_category","label":"Category"},{"field":"alar
  m_flag","label":"Alarm"}]',
     NULL, 'L', 2),
    ('CUSTOMER_LIST', 'Customer list',
     'Customer-facing list; only codes the Customer audience may see (I-03).',
     '[{"field":"status_code_identifier","label":"Status code"},{"field":"title","label":"Title"},{"field":"description","l
  abel":"Description"},{"field":"status_category","label":"Category"},{"field":"remedy_text","label":"Remedy"}]',
     'customer', 'P', 3);

-- I-13: evidence on a change request (files live outside the web root; only metadata here).
CREATE TABLE cr_attachment (
    id                  INTEGER PRIMARY KEY,
    change_request_id   INTEGER NOT NULL REFERENCES change_request(id),
    filename            TEXT NOT NULL,
    content_type        TEXT NOT NULL,
    size_bytes          INTEGER NOT NULL CHECK (size_bytes >= 0),
    sha256              TEXT NOT NULL,
    storage_path        TEXT NOT NULL,
    description         TEXT,
    uploaded_by         INTEGER NOT NULL REFERENCES user(id),
    uploaded_at         TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now'))
);
```

```
CREATE INDEX ix_cr_attachment_cr ON cr_attachment (change_request_id);

-- I-15: translations of a revision's text. No row = MISSING. Approval is a light
-- review by someone other than the translator, not the three-decision cycle.
CREATE TABLE revision_translation (
    id                      INTEGER PRIMARY KEY,
    status_code_revision_id INTEGER NOT NULL REFERENCES status_code_revision(id),
    language_id             INTEGER NOT NULL REFERENCES lookup_language(id),
    title                   TEXT NOT NULL CHECK (length(title) BETWEEN 1 AND 120),
    description             TEXT NOT NULL CHECK (length(description) BETWEEN 1 AND 4000),
    remedy_text             TEXT CHECK (remedy_text IS NULL OR length(remedy_text) <= 4000),
    translation_status      TEXT NOT NULL DEFAULT 'DRAFT' CHECK (translation_status IN ('DRAFT','APPROVED')),
    translated_by           INTEGER NOT NULL REFERENCES user(id),
    translated_at           TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    approved_by             INTEGER REFERENCES user(id),
    approved_at             TEXT,
    UNIQUE (status_code_revision_id, language_id),
    CHECK (approved_by IS NULL OR approved_by <> translated_by)
);

-- I-16: delegation / out-of-office cover for the decision roles.
CREATE TABLE delegation (
    id              INTEGER PRIMARY KEY,
    delegator_id    INTEGER NOT NULL REFERENCES user(id),
    delegate_id     INTEGER NOT NULL REFERENCES user(id),
    role_code       TEXT NOT NULL CHECK (role_code IN ('REVIEWER','ADMINISTRATOR','CHIEF_ENGINEER')),
    valid_from      TEXT NOT NULL,
    valid_to        TEXT NOT NULL,
    created_by      INTEGER NOT NULL REFERENCES user(id),
    created_at      TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    revoked_at      TEXT,
    CHECK (delegator_id <> delegate_id),
    CHECK (valid_to >= valid_from)
);
CREATE INDEX ix_delegation_delegate ON delegation (delegate_id);

-- I-23: field-level review comments and per-group review checklists.
CREATE TABLE field_comment (
    id                  INTEGER PRIMARY KEY,
    change_request_id   INTEGER NOT NULL REFERENCES change_request(id),
    field_name          TEXT NOT NULL,
    author_id           INTEGER NOT NULL REFERENCES user(id),
    comment_text        TEXT NOT NULL CHECK (length(comment_text) BETWEEN 1 AND 2000),
    created_at          TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    resolved_by         INTEGER REFERENCES user(id),
    resolved_at         TEXT
);
CREATE INDEX ix_field_comment_cr ON field_comment (change_request_id);

CREATE TABLE review_checklist_item (
    id                          INTEGER PRIMARY KEY,
    functional_system_group_id  INTEGER REFERENCES lookup_functional_system_group(id),   -- NULL = every group
    text                        TEXT NOT NULL CHECK (length(text) BETWEEN 3 AND 300),
    is_active                   INTEGER NOT NULL DEFAULT 1,
    sort_order                  INTEGER NOT NULL DEFAULT 0,
    created_at                  TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now'))
);

CREATE TABLE cr_checklist_confirmation (
    id                  INTEGER PRIMARY KEY,
    change_request_id   INTEGER NOT NULL REFERENCES change_request(id),
    checklist_item_id   INTEGER NOT NULL REFERENCES review_checklist_item(id),
    item_text           TEXT NOT NULL,   -- snapshot: the wording confirmed at sign-off
    confirmed_by        INTEGER NOT NULL REFERENCES user(id),
    confirmed_at        TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    UNIQUE (change_request_id, checklist_item_id)
);

-- I-26: saved searches (definition = JSON of the list filters and visible columns).
CREATE TABLE saved_search (
    id          INTEGER PRIMARY KEY,
    user_id     INTEGER NOT NULL REFERENCES user(id),
    name        TEXT NOT NULL CHECK (length(name) BETWEEN 1 AND 80),
    definition  TEXT NOT NULL,
    created_at  TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    UNIQUE (user_id, name)
);

-- I-08: in-app notices (optionally also mailed through the organisation's own SMTP relay).
CREATE TABLE notification (
    id                  INTEGER PRIMARY KEY,
    user_id             INTEGER NOT NULL REFERENCES user(id),
    kind                TEXT NOT NULL,
    message             TEXT NOT NULL,
    link                TEXT,
    change_request_id   INTEGER REFERENCES change_request(id),
    dedupe_key          TEXT UNIQUE,
    created_at          TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    read_at             TEXT,
```

```
    emailed_at          TEXT
);
CREATE INDEX ix_notification_user ON notification (user_id, read_at);

-- I-20: login rate-limit counters live in the database so they survive a restart
-- and are shared by every worker process. attempted_at is epoch seconds.
CREATE TABLE login_attempt (
    id            INTEGER PRIMARY KEY,
    scope         TEXT NOT NULL CHECK (scope IN ('IP','USER')),
    key           TEXT NOT NULL,
    attempted_at  REAL NOT NULL
);
CREATE INDEX ix_login_attempt ON login_attempt (scope, key, attempted_at);

-- Administrator-managed settings (SLA thresholds, reviewer auto-assignment, ...).
CREATE TABLE app_setting (
    key         TEXT PRIMARY KEY,
    value       TEXT NOT NULL,
    updated_by  INTEGER REFERENCES user(id),
    updated_at  TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now'))
);

-- I-28: occurrence counts imported from SCADA/CMS (anonymised, per period).
CREATE TABLE occurrence_import (
    id              INTEGER PRIMARY KEY,
    source_label    TEXT NOT NULL,
    period_start    TEXT NOT NULL,
    period_end      TEXT NOT NULL,
    imported_by     INTEGER NOT NULL REFERENCES user(id),
    imported_at     TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    row_count       INTEGER NOT NULL DEFAULT 0,
    unmatched_count INTEGER NOT NULL DEFAULT 0,
    CHECK (period_end >= period_start)
);
CREATE TABLE code_occurrence (
    id                    INTEGER PRIMARY KEY,
    occurrence_import_id  INTEGER NOT NULL REFERENCES occurrence_import(id),
    status_code_id        INTEGER NOT NULL REFERENCES status_code(id),
    occurrences           INTEGER NOT NULL CHECK (occurrences >= 0),
    UNIQUE (occurrence_import_id, status_code_id)
);

-- I-29: scoped, hashed, expiring API tokens; webhooks for internal consumers.
CREATE TABLE api_token (
    id            INTEGER PRIMARY KEY,
    user_id       INTEGER NOT NULL REFERENCES user(id),
    name          TEXT NOT NULL,
    token_prefix  TEXT NOT NULL,
    token_hash    TEXT NOT NULL UNIQUE,
    scopes        TEXT NOT NULL,
    expires_at    TEXT NOT NULL,
    last_used_at  TEXT,
    revoked_at    TEXT,
    created_at    TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now'))
);
CREATE TABLE webhook (
    id              INTEGER PRIMARY KEY,
    name            TEXT NOT NULL,
    url             TEXT NOT NULL,
    secret          TEXT NOT NULL,
    events          TEXT NOT NULL DEFAULT 'release.published',
    is_active       INTEGER NOT NULL DEFAULT 1,
    created_by      INTEGER NOT NULL REFERENCES user(id),
    created_at      TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    last_delivery_at TEXT,
    last_status     TEXT
);
CREATE TABLE webhook_delivery (
    id            INTEGER PRIMARY KEY,
    webhook_id    INTEGER NOT NULL REFERENCES webhook(id),
    event         TEXT NOT NULL,
    payload       TEXT NOT NULL,
    http_status   INTEGER,
    ok            INTEGER NOT NULL DEFAULT 0 CHECK (ok IN (0,1)),
    error         TEXT,
    delivered_at  TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now'))
);

-- I-36: which Main Controller code replaces which Hub Controller code.
CREATE TABLE hub_mc_mapping (
    id                  INTEGER PRIMARY KEY,
    hub_status_code_id  INTEGER NOT NULL UNIQUE REFERENCES status_code(id),
    mc_status_code_id   INTEGER NOT NULL REFERENCES status_code(id),
    created_by          INTEGER NOT NULL REFERENCES user(id),
    created_at          TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    CHECK (hub_status_code_id <> mc_status_code_id)
);

-- ------------------------------------------------------------------------
```

```
-- Triggers: prevent write access to immutable states / restricted operations
-- at the database layer (defense in depth ? service layer is the primary
-- enforcement point; see docs/07-database-design.md §5)
-- ------------------------------------------------------------------------

CREATE TRIGGER trg_prevent_edit_locked_revision
BEFORE UPDATE OF title, description, status_category, available_group, alarm_flag,
    set_delay, reset_delay, brake_program, manual_reset_program, auto_reset_program,
    development_access, sales_access, tcc_access, service_access,
    turbine_operator_package_access, grid_operator_access, service_partner_access, customer_access,
    remedy_text, affected_component, related_status_codes, standard_references,
    applies_from_version_id, applies_to_version_id,
    converter_reset_program, hydraulic_pump_program, gearbox_oil_pump_program, gearbox_oil_fan_program,
    gearbox_oil_heating_program, generator_fan_program, generator_heating_program,
    generator_bearing_lubrication_program, rotor_bearing_lubrication_program, anemometer_heating_program,
    medium_voltage_cb_program, tower_fan_program, transformer_oil_pump_program, zero_load_operation,
    pitch_error_program, pitch_warning_program, rotor_brake_program,
    stop_program_40_flag, stop_program_80_flag, manual_operation_flag, automatic_operation_flag,
    vane_position_flag, operating_mode_mask, task_call_frequency, availability_calc
ON status_code_revision
WHEN OLD.lifecycle_status IN ('APPROVED','RELEASED','ARCHIVED')
BEGIN
    SELECT RAISE(ABORT, 'Cannot edit a revision in APPROVED, RELEASED or ARCHIVED state');
END;

CREATE TRIGGER trg_prevent_release_item_delete
BEFORE DELETE ON release_item
WHEN (SELECT status FROM release WHERE id = OLD.release_id) = 'PUBLISHED'
BEGIN
    SELECT RAISE(ABORT, 'Cannot remove items from a published Release');
END;

CREATE TRIGGER trg_prevent_sandbox_release
BEFORE INSERT ON release_item
WHEN (
    SELECT sc.is_sandbox FROM status_code_revision scr
    JOIN status_code sc ON sc.id = scr.status_code_id
    WHERE scr.id = NEW.status_code_revision_id
) = 1
BEGIN
    SELECT RAISE(ABORT, 'Sandbox status codes can never be added to a Release');
END;

-- FR-072/FR-073: the audit trail is append-only at the database layer.
CREATE TRIGGER trg_audit_no_update
BEFORE UPDATE ON audit_log_entry
BEGIN
    SELECT RAISE(ABORT, 'audit_log_entry is append-only');
END;

CREATE TRIGGER trg_audit_no_delete
BEFORE DELETE ON audit_log_entry
BEGIN
    SELECT RAISE(ABORT, 'audit_log_entry is append-only');
END;

CREATE TRIGGER trg_prevent_nonsandbox_delete
BEFORE DELETE ON status_code
WHEN OLD.is_sandbox = 0
BEGIN
    SELECT RAISE(ABORT, 'Only sandbox (is_sandbox=1) status codes may be deleted');
END;

-- New subgroup non-overlap check (docs/06-data-dictionary.md §9a.1): the
-- service layer is the primary enforcement point (it can produce a helpful
-- 409 with the conflicting subgroup's name); this trigger is the database-
-- layer backstop.
CREATE TRIGGER trg_prevent_subgroup_overlap
BEFORE INSERT ON lookup_functional_subgroup
WHEN EXISTS (
    SELECT 1 FROM lookup_functional_subgroup existing
    WHERE existing.functional_system_group_id = NEW.functional_system_group_id
      AND NEW.sub_range_start <= existing.sub_range_end
      AND NEW.sub_range_end >= existing.sub_range_start
)
BEGIN
    SELECT RAISE(ABORT, 'New subgroup sub-range overlaps an existing subgroup in this Functional System Group');
END;

-- New group-level non-overlap check (new 2026-09-17): an Administrator may
-- assign a brand-new Functional System Group into any free range (e.g.
-- 10000-10999, 14000-14999, 15000-15999, 16000-16999 -- docs/05-governance-
-- handbook.md §2). Mirrors trg_prevent_subgroup_overlap one level up.
CREATE TRIGGER trg_prevent_group_overlap
BEFORE INSERT ON lookup_functional_system_group
WHEN EXISTS (
    SELECT 1 FROM lookup_functional_system_group existing
    WHERE NEW.range_start <= existing.range_end
      AND NEW.range_end >= existing.range_start
```

```
)
BEGIN
    SELECT RAISE(ABORT, 'New Functional System Group range overlaps an existing group');
END;

-- v4.1: PrgIdx_GUi identifies the block in controller diagnostics, so once a revision of the code
-- has been released the number is frozen. Setting it for the first time (NULL -> value, e.g. the
-- back-fill of the legacy library) is always allowed.
CREATE TRIGGER trg_prevent_prg_idx_change_after_release
BEFORE UPDATE OF prg_idx ON status_code
WHEN OLD.prg_idx IS NOT NULL AND NEW.prg_idx IS NOT OLD.prg_idx
 AND EXISTS (SELECT 1 FROM status_code_revision r
             WHERE r.status_code_id = OLD.id AND r.lifecycle_status IN ('RELEASED','DEPRECATED','ARCHIVED'))
BEGIN
    SELECT RAISE(ABORT, 'prg_idx cannot be changed once the status code has been released');
END;

-- v4.1: database backstop for "PrgIdx_GUi is mandatory". The service layer is the primary
-- enforcement point. The trigger stays inert until app_setting 'status_code.require_prg_idx' is '1',
-- which should happen once the legacy library carries its values and the application collects the field.
CREATE TRIGGER trg_require_prg_idx_for_review
BEFORE UPDATE OF lifecycle_status ON status_code_revision
WHEN NEW.lifecycle_status IN ('REVIEW','PENDING_APPROVAL','APPROVED','RELEASED')
 AND OLD.lifecycle_status IS NOT NEW.lifecycle_status
 AND COALESCE((SELECT value FROM app_setting WHERE key = 'status_code.require_prg_idx'), '0') = '1'
 AND (SELECT is_sandbox FROM status_code WHERE id = NEW.status_code_id) = 0
 AND (SELECT prg_idx FROM status_code WHERE id = NEW.status_code_id) IS NULL
BEGIN
    SELECT RAISE(ABORT, 'A status code needs its PrgIdx_GUi before it can leave Draft');
END;

-- v4.2: an imported code can only ever be issued the number its program carries. The service layer
-- does the same check first; this is the database backstop for any other writer.
CREATE TRIGGER trg_identifier_matches_reservation
BEFORE UPDATE OF status_code_identifier ON status_code
WHEN NEW.status_code_identifier IS NOT NULL
 AND OLD.status_code_identifier IS NULL
 AND NEW.reserved_identifier IS NOT NULL
 AND NEW.status_code_identifier <> NEW.reserved_identifier
BEGIN
    SELECT RAISE(ABORT, 'The identifier of an imported code must be the number it reserved');
END;

-- v4.2: a reservation can neither be taken from an identifier already issued to another code nor
-- be made for one.
CREATE TRIGGER trg_reservation_not_issued_elsewhere
BEFORE INSERT ON status_code
WHEN NEW.reserved_identifier IS NOT NULL
 AND EXISTS (SELECT 1 FROM status_code o WHERE o.status_code_identifier = NEW.reserved_identifier)
BEGIN
    SELECT RAISE(ABORT, 'The reserved identifier is already issued to another status code');
END;

-- v4.2: and no other code can be issued a number that an imported code has reserved.
CREATE TRIGGER trg_identifier_not_reserved_elsewhere
BEFORE UPDATE OF status_code_identifier ON status_code
WHEN NEW.status_code_identifier IS NOT NULL
 AND OLD.status_code_identifier IS NULL
 AND EXISTS (SELECT 1 FROM status_code o
             WHERE o.reserved_identifier = NEW.status_code_identifier AND o.id <> NEW.id)
BEGIN
    SELECT RAISE(ABORT, 'The identifier is reserved by another imported status code');
END;

-- v4.7: the e-mail address identifies the person - also for the Windows sign-in that will replace
-- the password in a deployment - so it is fixed once the account exists. A typing error is corrected
-- by deleting the unused account and creating it again, or by deactivating it (docs/03-srs.md FR-162).
CREATE TRIGGER trg_user_email_immutable
BEFORE UPDATE OF email ON user
WHEN NEW.email <> OLD.email AND NEW.deleted_at IS NULL   -- v4.9: deleting an account erases its address
BEGIN
    SELECT RAISE(ABORT, 'The e-mail address of a user cannot be changed');
END;

-- ---------------------------------------------------------------------------
-- v4.7: translation review in the application (docs/26-translations.md, docs/03-srs.md FR-164).
-- Native speakers check the interface translations in the browser; an Administrator or StCd Owner
-- assigns them to a language and decides on their proposals; an accepted proposal becomes an
-- override that the application shows at once, without a new release.
-- ---------------------------------------------------------------------------

-- Who may review which language. Any account can be assigned (an external native speaker gets a
-- Viewer account and sees only this section).
CREATE TABLE language_reviewer (
    id              INTEGER PRIMARY KEY,
    user_id         INTEGER NOT NULL REFERENCES user(id),
    language_code   TEXT NOT NULL REFERENCES lookup_language(code),
    assigned_by     INTEGER NOT NULL REFERENCES user(id),
    assigned_at     TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
```

```
    UNIQUE (user_id, language_code)
);

-- One review of one language by one reviewer. `fingerprint` is the catalogue (shipped text plus
-- overrides) the review started from; when it differs from the catalogue later, the review shows as
-- "changed since".
CREATE TABLE translation_review (
    id              INTEGER PRIMARY KEY,
    language_code   TEXT NOT NULL REFERENCES lookup_language(code),
    reviewer_id     INTEGER NOT NULL REFERENCES user(id),
    status          TEXT NOT NULL DEFAULT 'OPEN' CHECK (status IN ('OPEN','COMPLETED')),
    started_at      TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    completed_at    TEXT,
    fingerprint     TEXT NOT NULL,
    string_count    INTEGER NOT NULL CHECK (string_count >= 0)
);
CREATE UNIQUE INDEX ux_translation_review_open ON translation_review (language_code, reviewer_id) WHERE status = 'OPEN';

-- The answers: one row per string the reviewer has answered. A proposal with no current text is a
-- translation for a string that was missing.
CREATE TABLE translation_review_item (
    id              INTEGER PRIMARY KEY,
    review_id       INTEGER NOT NULL REFERENCES translation_review(id),
    string_id       TEXT NOT NULL,
    english         TEXT NOT NULL,
    current_text    TEXT,
    verdict         TEXT CHECK (verdict IN ('OK','CHANGE','UNCLEAR')),
    proposal        TEXT CHECK (proposal IS NULL OR length(proposal) <= 1000),
    comment         TEXT CHECK (comment IS NULL OR length(comment) <= 1000),
    decision        TEXT NOT NULL DEFAULT 'PENDING' CHECK (decision IN ('PENDING','ACCEPTED','REJECTED','NOTED')),
    decided_by      INTEGER REFERENCES user(id),
    decided_at      TEXT,
    updated_at      TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    UNIQUE (review_id, string_id),
    CHECK (verdict <> 'CHANGE' OR (proposal IS NOT NULL AND length(trim(proposal)) > 0))
);

-- An accepted proposal: the text the application shows for a string of a language instead of the
-- shipped one. Removing the row brings the shipped text back.
CREATE TABLE translation_override (
    id              INTEGER PRIMARY KEY,
    language_code   TEXT NOT NULL REFERENCES lookup_language(code),
    string_id       TEXT NOT NULL,
    english         TEXT NOT NULL,
    text            TEXT NOT NULL CHECK (length(text) BETWEEN 1 AND 1000),
    review_item_id  INTEGER REFERENCES translation_review_item(id),
    created_by      INTEGER NOT NULL REFERENCES user(id),
    created_at      TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    UNIQUE (language_code, string_id)
);


-- ---------------------------------------------------------------------------
-- Technical nomenclature (v4.8, FR-166): the terms of the controller and of the workflow with their
-- translations, kept apart from the interface translations (translation_*). A translation is
-- proposed (DRAFT) by one person and approved by another (APPROVED); a term marked keep_english is
-- never translated.
-- ---------------------------------------------------------------------------
CREATE TABLE terminology_term (
    id              INTEGER PRIMARY KEY,
    code            TEXT NOT NULL UNIQUE CHECK (length(code) BETWEEN 2 AND 80),
    category        TEXT NOT NULL CHECK (category IN ('PRODUCT','DATA','STATE','PROCESS','ROLE','AUDIENCE','KEEP')),
    english         TEXT NOT NULL CHECK (length(trim(english)) BETWEEN 1 AND 200),
    definition      TEXT NOT NULL DEFAULT '' CHECK (length(definition) <= 1000),
    keep_english    INTEGER NOT NULL DEFAULT 0 CHECK (keep_english IN (0,1)),
    is_active       INTEGER NOT NULL DEFAULT 1 CHECK (is_active IN (0,1)),
    created_by      INTEGER REFERENCES user(id),
    created_at      TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    updated_at      TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now'))
);

CREATE TABLE terminology_translation (
    id              INTEGER PRIMARY KEY,
    term_id         INTEGER NOT NULL REFERENCES terminology_term(id),
    language_code   TEXT NOT NULL REFERENCES lookup_language(code),
    text            TEXT NOT NULL CHECK (length(trim(text)) BETWEEN 1 AND 200),
    status          TEXT NOT NULL DEFAULT 'DRAFT' CHECK (status IN ('DRAFT','APPROVED')),
    note            TEXT CHECK (note IS NULL OR length(note) <= 500),
    proposed_by     INTEGER REFERENCES user(id),
    proposed_at     TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    approved_by     INTEGER REFERENCES user(id),
    approved_at     TEXT,
    UNIQUE (term_id, language_code),
    CHECK ((status = 'APPROVED') = (approved_at IS NOT NULL))
);


-- v4.13: the sample inspection of a bulk decision (FR-180): which codes of a batch were drawn, who looked
CREATE TABLE bulk_sample (
```

```
    id                  INTEGER PRIMARY KEY,
    change_set_id       INTEGER NOT NULL REFERENCES change_set(id),
    step                TEXT NOT NULL CHECK (step IN ('domain','reviewer','admin','chief-engineer')),
    domain_code         TEXT CHECK (domain_code IS NULL OR length(domain_code) BETWEEN 1 AND 40),
    drawn_by            INTEGER NOT NULL REFERENCES user(id),
    drawn_at            TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    seed                TEXT NOT NULL,
    population          INTEGER NOT NULL CHECK (population >= 1),
    population_hash     TEXT NOT NULL,
    rate_percent        INTEGER NOT NULL CHECK (rate_percent BETWEEN 1 AND 100),
    status              TEXT NOT NULL DEFAULT 'OPEN' CHECK (status IN ('OPEN','PASSED','FAILED','USED','VOID')),
    used_at             TEXT,
    CHECK ((step = 'domain') = (domain_code IS NOT NULL)),
    CHECK ((status = 'USED') = (used_at IS NOT NULL))
);

CREATE TABLE bulk_sample_item (
    id                  INTEGER PRIMARY KEY,
    sample_id           INTEGER NOT NULL REFERENCES bulk_sample(id),
    change_request_id   INTEGER NOT NULL REFERENCES change_request(id),
    verdict             TEXT CHECK (verdict IN ('OK','DEFECT')),
    note                TEXT CHECK (note IS NULL OR length(note) <= 1000),
    checked_at          TEXT,
    UNIQUE (sample_id, change_request_id),
    CHECK ((verdict IS NULL) = (checked_at IS NULL))
);

CREATE INDEX ix_bulk_sample_set ON bulk_sample (change_set_id, step, status);


-- v4.17: the legacy -> new status code mapping (FR-191..FR-198). It holds only what the generated .st programs do not
-- carry: the English title, the type, the up/down counter, the named access levels, cause and directive and the
-- legacy -> new pair itself. The initial content is the workbook seed_data/legacy_mapping.xlsx (flask mapping-seed);
-- every later change is a mapping_change with the three approvals of mapping_approval.
CREATE TABLE status_code_mapping (
    id                  INTEGER PRIMARY KEY,
    family              TEXT NOT NULL CHECK (family IN ('MC','HC')),
    legacy_number       INTEGER CHECK (legacy_number IS NULL OR legacy_number >= 0),
    new_number          INTEGER NOT NULL CHECK (new_number BETWEEN 1 AND 99999),
    description         TEXT NOT NULL CHECK (length(description) BETWEEN 1 AND 120),
    functional_group    TEXT CHECK (functional_group IS NULL OR length(functional_group) <= 80),
    logical_node        TEXT CHECK (logical_node IS NULL OR length(logical_node) <= 12),
    status_type         TEXT CHECK (status_type IS NULL OR status_type IN ('ERROR','WARNING','INFO')),
    up_down_counter     INTEGER CHECK (up_down_counter IS NULL OR up_down_counter IN (0,1)),
    parameter_set_delay   TEXT CHECK (parameter_set_delay IS NULL OR length(parameter_set_delay) <= 40),
    parameter_reset_delay TEXT CHECK (parameter_reset_delay IS NULL OR length(parameter_reset_delay) <= 40),
    development_access  TEXT,
    sales_access        TEXT,
    tcc_access          TEXT,
    service_access      TEXT,
    turbine_operator_package_access TEXT,
    grid_operator_access TEXT,
    service_partner_access TEXT,
    customer_access     TEXT,
    cause               TEXT CHECK (cause IS NULL OR length(cause) <= 4000),
    directive           TEXT CHECK (directive IS NULL OR length(directive) <= 4000),
    note                TEXT CHECK (note IS NULL OR length(note) <= 2000),
    status              TEXT NOT NULL DEFAULT 'PROPOSED' CHECK (status IN ('PROPOSED','FROM_PROGRAM','APPROVED','ISSUED','R
  EJECTED')),
    source              TEXT CHECK (source IS NULL OR length(source) <= 200),
    version             INTEGER NOT NULL DEFAULT 1 CHECK (version >= 1),
    created_at          TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    updated_at          TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    UNIQUE (new_number),
    UNIQUE (family, legacy_number)
);

-- one request to change the mapping: one or several rows, decided as a whole by the three roles
CREATE TABLE mapping_change (
    id                  INTEGER PRIMARY KEY,
    title               TEXT NOT NULL CHECK (length(title) BETWEEN 1 AND 120),
    reason              TEXT NOT NULL CHECK (length(reason) BETWEEN 1 AND 2000),
    origin              TEXT NOT NULL DEFAULT 'FORM' CHECK (origin IN ('FORM','WORKBOOK','SELECTION')),
    requested_by        INTEGER NOT NULL REFERENCES user(id),
    requested_at        TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    state               TEXT NOT NULL DEFAULT 'OPEN' CHECK (state IN ('OPEN','APPLIED','REJECTED','WITHDRAWN')),
    closed_at           TEXT,
    CHECK ((state = 'OPEN') = (closed_at IS NULL))
);

CREATE TABLE mapping_change_item (
    id                  INTEGER PRIMARY KEY,
    change_id           INTEGER NOT NULL REFERENCES mapping_change(id),
    mapping_id          INTEGER REFERENCES status_code_mapping(id),
    action              TEXT NOT NULL CHECK (action IN ('ADD','CHANGE','CONFIRM','REJECT')),
    base_version        INTEGER,
    before_json         TEXT,
    after_json          TEXT NOT NULL,
    CHECK ((action = 'ADD') = (mapping_id IS NULL)),
```

```
    CHECK ((action = 'ADD') = (base_version IS NULL))
);

-- the three approvals: StCd Owner, Chief Engineer and one Administrator, each a different person who is not
-- the requester; one rejection closes the request
CREATE TABLE mapping_approval (
    id                  INTEGER PRIMARY KEY,
    change_id           INTEGER NOT NULL REFERENCES mapping_change(id),
    role_code           TEXT NOT NULL CHECK (role_code IN ('STCD_OWNER','CHIEF_ENGINEER','ADMINISTRATOR')),
    user_id             INTEGER NOT NULL REFERENCES user(id),
    decision            TEXT NOT NULL CHECK (decision IN ('APPROVE','REJECT')),
    comment             TEXT CHECK (comment IS NULL OR length(comment) <= 1000),
    decided_at          TEXT NOT NULL DEFAULT (strftime('%Y-%m-%dT%H:%M:%fZ','now')),
    UNIQUE (change_id, role_code),
    UNIQUE (change_id, user_id)
);

CREATE INDEX ix_mapping_item_change ON mapping_change_item (change_id);
CREATE INDEX ix_mapping_item_row ON mapping_change_item (mapping_id);
CREATE INDEX ix_mapping_change_state ON mapping_change (state);
CREATE INDEX ix_mapping_status ON status_code_mapping (family, status);
```

Appendix C - Search index (SQL)

The full-text search index.

```
-- ============================================================================
-- CSCM Tool ? optional full-text search index (SQLite FTS5) ? I-31
-- Applied by migration f8b2d4e6a012 after schema.sql. It is a separate file
-- because it is *derived* data: if the SQLite build has no FTS5 module the
-- migration skips it and catalogue search silently falls back to substring
-- search (LIKE); nothing else depends on it.
--
-- One row per status_code_revision (rowid = revision id), kept in step by
-- triggers. `prefix` indexes make "conv*" style matches fast.
-- ============================================================================

CREATE VIRTUAL TABLE IF NOT EXISTS status_code_fts USING fts5(
    identifier, title, description, remedy_text,
    tokenize = 'unicode61 remove_diacritics 2',
    prefix = '2 3 4'
);

-- Vocabulary of indexed terms; used for typo tolerance ("converte" ~ "converter").
CREATE VIRTUAL TABLE IF NOT EXISTS status_code_fts_vocab USING fts5vocab(status_code_fts, 'row');

CREATE TRIGGER IF NOT EXISTS trg_fts_revision_insert
AFTER INSERT ON status_code_revision
BEGIN
    INSERT INTO status_code_fts (rowid, identifier, title, description, remedy_text)
    VALUES (
        NEW.id,
        COALESCE((SELECT status_code_identifier FROM status_code WHERE id = NEW.status_code_id), ''),
        NEW.title, NEW.description, COALESCE(NEW.remedy_text, '')
    );
END;

CREATE TRIGGER IF NOT EXISTS trg_fts_revision_update
AFTER UPDATE OF title, description, remedy_text ON status_code_revision
BEGIN
    DELETE FROM status_code_fts WHERE rowid = OLD.id;
    INSERT INTO status_code_fts (rowid, identifier, title, description, remedy_text)
    VALUES (
        NEW.id,
        COALESCE((SELECT status_code_identifier FROM status_code WHERE id = NEW.status_code_id), ''),
        NEW.title, NEW.description, COALESCE(NEW.remedy_text, '')
    );
END;

CREATE TRIGGER IF NOT EXISTS trg_fts_revision_delete
AFTER DELETE ON status_code_revision
BEGIN
    DELETE FROM status_code_fts WHERE rowid = OLD.id;
END;

-- The identifier is assigned at Submit for Review, after the revision rows exist.
CREATE TRIGGER IF NOT EXISTS trg_fts_code_identifier
AFTER UPDATE OF status_code_identifier ON status_code
BEGIN
    DELETE FROM status_code_fts
     WHERE rowid IN (SELECT id FROM status_code_revision WHERE status_code_id = NEW.id);
    INSERT INTO status_code_fts (rowid, identifier, title, description, remedy_text)
    SELECT r.id, COALESCE(NEW.status_code_identifier, ''), r.title, r.description,
           COALESCE(r.remedy_text, '')
      FROM status_code_revision r
     WHERE r.status_code_id = NEW.id;
END;
```

Appendix D - Other files of the artefact folder

These files are part of the repository (docs/artifacts), not reproduced here:

- `hc-numbering-proposal.xlsx` (19 KB)
- `New Statuscode for sharing (recreated).xlsx` (31 KB)
- `test-run-history.json` (13 KB)